Open Source FEA using adapy

This report describes the main verification work of the simulation results produced by the Open Source Finite Element (FE) solvers Code Aster and Calculix.

The motivation behind producing this report is to increase confidence in the conversion of FEM models in *adapy* and also the results from the various FEA solvers.

Introduction

The results from *Code Aster* and *Calculix* and tools will be compared with proprietary FEA solvers such as;

- Abaqus
- Sestra (part of DNV's Sesam suite)

The simulations are pre- and post processed using adapy and this report is automatically generated using paradoc as part of a ci/cd pipeline within the *adapy* repositories.

For each new addition of code affecting the handling of FEM in *adapy* a new set of analyses is performed, whereas the results are gathered in this auto-generated document.

Please note! This report will not go into detail into the specifics of why the results are different between different element representations. This report will only investigate and compare the results from different FEA software.

FEA Solvers

- Calculix v2.23
- Code Aster v18.1.7
- Abaqus v2025
- Sestra v11.3.1.0

Python packages

The following python packages were instrumental in the creation of this document and the FEA results herein.

Adapy

The intention behind *adapy* is to make it easier to work with finite element models and BIM models and act as a high-level scripting library for parametric and procedural modelling techniques.

Paradoc

paradoc was created to simplify the generation of reports by creating the structure of the document and text in markdown with a string substitution scheme that lets you easily pass in tables, functions and finally be able to produce production ready documents in PDF, ODT or Microsoft Word (docx).

Summary

A summary of all performed analyses will be performed in this section and a percentage difference for each FEA solver per type of analysis will be calculated and presented.

Analyses

Eigenvalue Analysis

A simple cantilevered beam is subjected to eigenvalue analysis with a total of 11 eigenmodes requested. 49 cases are compared across solvers: abaqus, calculix, code_aster, sesam.

The beam model will be analyzed by varying the following parameters:

- Mesh type (Solid, Shell, Line)
- Mesh element types (TET, HEX, TRI, QUAD)
- Mesh element order (1st and 2nd order)
- Reduced integration elements (R)
- Different FEA solvers (Calculix, Code Aster, Abaqus, Sesam)

Model Description

Property	Value
Description	Beam(“MyBeam”, [0.0, 0.5, 0.5], [3.0, 0.5, 0.5], “IPE400”, “S420”)
Length	3.00 m
Section	IPE400
Material	S420
Young’s modulus	2.100e+11 Pa
Yield stress	4.200e+08 Pa
Density	7850 kg/m³

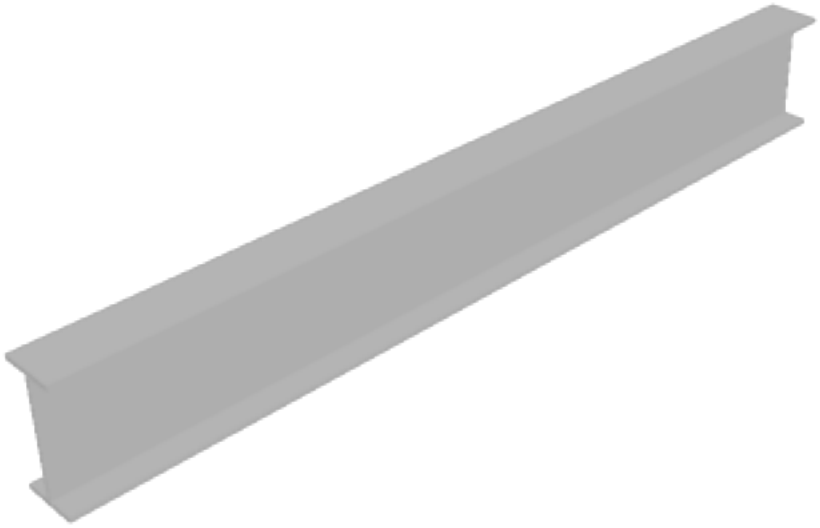


Figure 1: Cantilever beam geometry.

Summary

This section presents a comparison between the calculated results from the eigenvalue analysis of a cantilever beam using the different FEA tools.

Note! The `Sestra` and `Abaqus` analyses are not performed as part of the github actions step. The eigenmode results are kept in cached json files and imported during the compilation of the report using **paradoc**.

Eigenfrequency vs. mode number

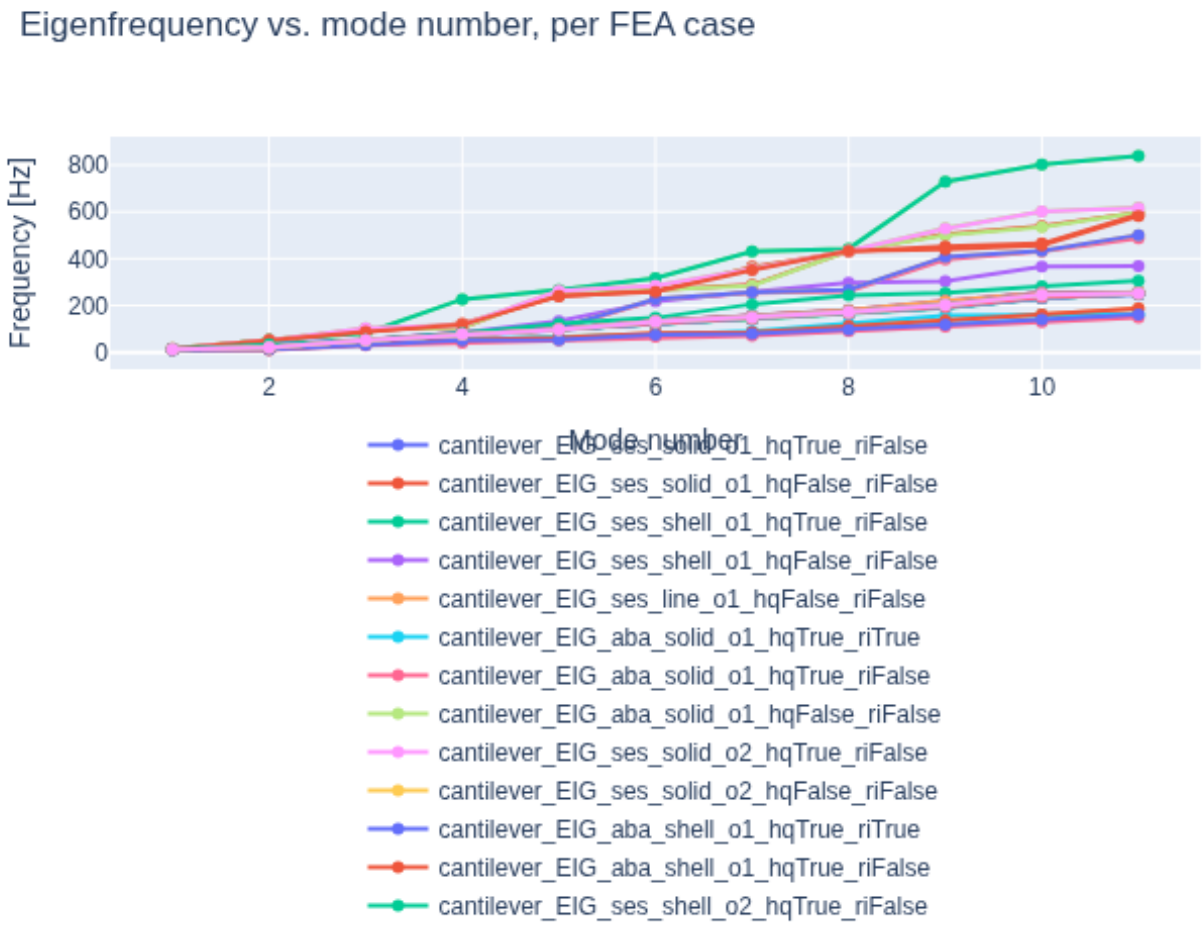


Figure 2: Eigenfrequency vs. mode number across FEA cases.

Comparison tables

Columns are sortable; the table grammar `{tbl:sortby:Mode:asc;index:no}` sets the initial sort and hides the index column.

Table 2: Eigenfrequency comparison (Hz) — solid, 1st order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9766	15.7932	12.514	13.435	15.79	15.9285	15.9277	13.6205	13.6205	14.3018	13.5291
2	20.3015	51.5418	34.157	20.731	51.532	51.5896	51.5872	21.0685	21.0685	51.6653	34.7401
3	50.6181	97.7314	49.598	50.678	97.588	102.446	102.41	50.8252	50.8252	88.2331	50.7365
4	77.3921	105.691	55.271	82.799	104.8	116.744	116.512	84.066	84.066	116.873	82.2962
5	98.1185	260.887	64.305	100.19	260.59	261.877	261.806	102.334	102.334	240.766	123.303
6	133.016	267.131	74.866	223.14	266.26	284.697	284.465	227.25	227.25	258.286	147.139
7	153.833	287.461	90.033	254.98	285.05	357.062	356.306	255.872	255.872	351.823	205.267
8	178.565	432.347	124.58	260.1	432.31	432.36	432.35	266.519	266.519	431.955	243.985
9	214.689	503.862	155.99	396.57	500.86	529.297	528.489	407.869	407.869	439.038	254.838
10	253.463	539.506	159.95	431.8	534.26	601.943	601.586	432.207	432.207	456.762	282.888
11	256.05	598.2	162.56	486.96	596.78	617.646	616.155	500.931	500.931	582.562	305.247

Table 3: Eigenfrequency comparison (Hz) — solid, 2nd order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9541	12.9537	12.943	12.955	12.954	12.9542	12.9542	12.9551	12.9551	12.9428	12.9427
2	20.0445	20.2945	20.029	20.15	20.294	20.2876	20.2877	20.1503	20.1503	20.0288	20.0288
3	50.5342	50.5401	50.498	50.541	50.54	50.5401	50.5402	50.5412	50.5412	50.4977	50.4977
4	76.7467	77.1278	76.687	76.89	77.128	77.2297	77.2297	76.8903	76.8903	76.6876	76.6875
5	97.4723	97.8296	97.391	97.594	97.83	97.8506	97.8505	97.5939	97.5939	97.3913	97.3912
6	130.865	132.478	130.81	131.37	132.48	133.015	133.014	131.365	131.365	130.814	130.809

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
7	148.496	153.631	148.47	149.96	153.63	154.804	154.802	149.958	149.958	148.478	148.474
8	172.243	178.114	172.17	173.92	178.12	181.029	181.026	173.925	173.925	172.177	172.171
9	201.904	215.815	201.84	205.29	215.82	219.501	219.497	205.292	205.292	201.846	201.839
10	246.353	253.203	246.17	251.35	253.2	253.193	253.193	251.347	251.347	246.177	246.165
11	252.742	254.466	252.57	252.92	254.47	254.685	254.684	252.917	252.917	252.573	252.573

Table 4: Eigenfrequency comparison (Hz) — shell, 1st order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRIR	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7628	13.6414	12.373	12.78	13.6869	13.7254	13.642	12.7639	13.1463	14.2558
2	19.3943	20.1983	19.23	19.593	20.3643	36.6896	20.4181	19.6304	19.9719	53.2618
3	51.9912	52.1922	51.849	51.998	52.1874	52.2485	52.1927	51.9997	52.0595	87.9153
4	75.1918	79.5226	73.558	75.683	80.2058	83.7093	79.6596	75.4574	77.2361	121.85
5	95.9297	101.756	93.709	96.393	102.043	132.928	102.01	96.2959	98.7619	239.804
6	125.116	125.093	126.87	126.96	124.093	218.167	125.459	125.898	125.442	262.83
7	141.556	143.164	144.58	145.57	145.486	259.245	144.895	145.282	143.864	365.1
8	165.03	166.668	168.38	169.39	168.62	297.856	168.951	169.441	167.974	431.774
9	190.099	191.533	197.51	198.75	199.719	304.464	195.184	200.494	193.96	451.745
10	229.46	230.489	241.27	242.55	242.995	365.943	235.115	247.276	233.817	464.44
11	250.786	257.362	245.07	252.49	257.447	368.764	257.409	251.554	257.105	589.24

Table 5: Eigenfrequency comparison (Hz) — shell, 2nd order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7554	12.7558	12.758	12.758	12.756	12.7562	12.7566	12.7576	12.7561	12.7559
2	19.4731	19.5048	19.486	19.486	19.567	19.5254	19.4509	19.5638	19.4736	19.5021
3	51.9702	51.9704	51.998	51.998	51.969	51.974	51.97	51.9801	51.97	51.9741
4	75.1245	75.1716	75.149	75.149	75.145	75.1848	75.1443	75.2685	75.1252	75.1399
5	95.8886	95.9337	95.92	95.92	96.001	95.9713	95.8707	96.0372	95.8893	95.9349
6	124.987	125.173	125	125	125.21	125.465	125.087	125.889	124.987	125.257
7	142.234	142.768	142.32	142.32	142.77	142.995	142.359	143.72	142.233	142.497
8	165.718	166.334	165.85	165.85	166.6	166.579	165.79	167.347	165.719	165.981
9	191.999	193.308	192.18	192.18	192.84	194.014	192.406	195.577	191.998	192.87
10	232.626	234.613	232.87	232.87	233.36	236.042	233.389	238.389	232.628	234.347
11	250.095	250.24	250.17	250.17	250.13	250.287	250.175	250.521	250.096	250.147

Table 6: Eigenfrequency comparison (Hz) — line, 1st order.

Mode	ses	aba	ccx	ca	ccxR
1	12.2454	10.785	12.9734	12.2454	12.9734
2	12.9431	12.928	52.5867	12.9793	52.5867
3	36.7525	32.342	81.1511	36.7525	81.1511
4	50.5542	50.543	226.646	52.9559	226.646
5	61.3087	53.855	267.32	61.3087	267.32
6	79.793	75.296	316.915	81.3401	316.915
7	85.9466	79.59	431.04	85.9466	431.04
8	110.699	96.637	442.61	110.699	442.61
9	135.6	117.85	728.596	135.6	728.596
10	160.681	138.9	802.316	160.681	802.316
11	185.976	159.77	838.75	185.976	838.75

Table 7: Eigenfrequency comparison (Hz) — line, 2nd order.

Mode	ses	aba
1	10.039	10.786
2	10.6144	12.932
3	30.1078	32.358
4	41.2358	50.557
5	50.1488	53.93
6	65.3265	75.502
7	70.1429	79.684
8	90.0695	97.074
9	109.907	118.65
10	129.631	140.22

Mode	ses	aba
11	149.217	161.8

Effective modal mass

Effective modal mass [kg] per case, summed over the captured modes in the global X / Y / Z directions. Every solver reports it; Code Aster and Sesam give translational mass only. Summing over enough modes approaches the structure's total mass in each direction.

Table 8: Effective modal mass [kg] per case, summed over the captured modes in the global X/Y/Z directions. Code_Aster and Sesam report translational effective mass only.

Case	Modes	ΣMeff X	ΣMeff Y	ΣMeff Z
aba_line_o1	11	0	152.8	118.2
aba_line_o2	11	0	152.8	118.2
aba_shell_o1_QUAD	11	0	168	122
aba_shell_o1_QUADR	11	0	168.4	122
aba_shell_o2_QUAD	11	0	167.8	122
aba_shell_o2_QUADR	11	0	167.8	122
aba_shell_o2_TRI	11	0	167.9	122
aba_solid_o1_HEX	11	153.5	170.5	158.2
aba_solid_o1_HEXR	11	0	154.3	119
aba_solid_o1_TET	11	153.4	172	170.9
aba_solid_o2_HEX	11	0	163.3	158.6
aba_solid_o2_HEXR	11	0	163.2	158.7
aba_solid_o2_TET	11	0	162	158.6
ca_line_o1	11	0	152.3	116.5
ca_shell_o1_QUAD	11	0	167.6	163.7
ca_shell_o1_TRI	11	0	167.3	163.7
ca_shell_o2_QUAD	11	0	167.8	122
ca_shell_o2_TRI	11	0	167.8	122
ca_solid_o1_HEX	11	153.5	170.8	158.3
ca_solid_o1_HEXR	11	0	164.7	158.5
ca_solid_o1_TET	11	153.4	172.6	171
ca_solid_o2_HEX	11	0	163.3	158.6
ca_solid_o2_HEXR	11	0	163.2	158.7
ca_solid_o2_TET	11	0	162.2	158.6
ccx_line_o1	11	153.9	174.4	161.2
ccx_line_o1R	11	153.9	174.4	161.2
ccx_shell_o1_QUAD	11	0	168.1	122
ccx_shell_o1_QUADR	11	158	176.8	176.5
ccx_shell_o1_TRI	11	0	174.8	163.6
ccx_shell_o2_QUAD	11	0	167.8	122
ccx_shell_o2_QUADR	11	0	167.8	122
ccx_shell_o2_TRI	11	0	167.8	122
ccx_solid_o1_HEX	11	153.4	170.7	158.2
ccx_solid_o1_HEXR	11	153.5	171.8	171.3
ccx_solid_o1_TET	11	153.4	172.6	171
ccx_solid_o2_HEX	11	0	163.3	158.6
ccx_solid_o2_HEXR	11	0	163.2	158.7
ccx_solid_o2_TET	11	0	162.2	158.6
ses_line_o1	11	0	153	120.2
ses_line_o2	11	0	231.8	180
ses_shell_o1_QUAD	11	0	167.6	122
ses_shell_o1_TRI	11	0	167.1	163.6
ses_shell_o2_QUAD	11	0	167.8	122
ses_shell_o2_TRI	11	0	167.9	122
ses_solid_o1_HEX	11	0	161.7	158.6
ses_solid_o1_TET	11	153.4	172.1	170.9
ses_solid_o2_HEX	11	0	163.1	158.7
ses_solid_o2_TET	11	0	162	158.6

The per-mode effective masses and participation factors, compared across solvers for every mesh configuration, are in the appendix.

Short description:

Fem formats

- ccx: Calculix

- ca: Code Aster
- ses: Sesam
- aba: Abaqus

Element Types

- TET: Tetrahedrons
- HEX: Hexahedrons
- TRI: Triangle
- QUAD: Quadrilateral

Reduced Integration

All element types ending with capital R (QUAD**R**, HEX**R** etc.) are reduced integration elements.

Nonlinear Analysis

A simple cantilever beam with a series of cutouts is subjugated to a large vertical acceleration field. The level of plastic strains are monitored for comparison across different FE tools.

Model Description

Summary

Plate Analysis

The analyses above are one beam, in three representations. This section is the plate case: an `ada.Plate` object, meshed into shell elements by adapy's own meshing module, written out as an input deck per solver, and checked against a closed form as well as against the other solvers.

Nothing here drives a solver through its API. Each solver reads a file adapy wrote — an `.inp`, a `.fem`, a `.comm`/`.med` — which is the same route a user's own model takes, so what this section verifies is the thing adapy actually ships.

9 cases ran on the machine that built this report, using: calculix, code_aster. The remaining columns come from cached results measured elsewhere, the same arrangement the eigenvalue section uses for Abaqus and Sestra.

Model Description

A rectangular strip, simply supported on its two short edges and held in cylindrical bending on the two long ones, under a uniform surface pressure.

Property	Value
Span	4.00 m
Width	0.50 m
Thickness	0.010 m
Pressure	1000 Pa
Material	S355 (E = 210 GPa, ν = 0.3, ρ = 7850 kg/m^3)



Figure 3: Plate strip geometry.

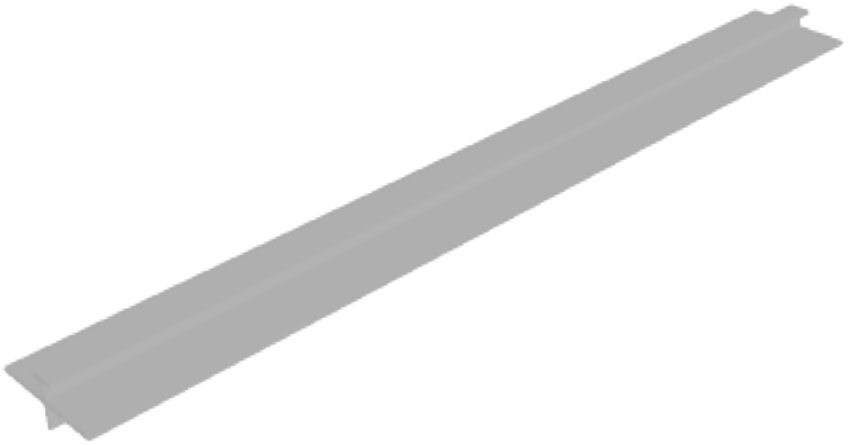


Figure 4: Plate strip with a T-profile stiffener on its centreline.

The span-to-thickness ratio is 400, so transverse shear sits several orders below the discretisation error and the thin-plate closed forms are the right reference rather than an approximation of one.

The supports are written as four node sets rather than one per edge. Simple support and cylindrical bending overlap at the four corners, and the obvious spelling constrains `u2` twice there: Abaqus and Sestra accept that, while Code_Aster refuses it outright (`<ASSEMBLA_26> … est bloqué plusieurs fois`). A model that only some solvers will read is not a cross-solver case, so the corner nodes carry only the rotation the long edges add.

Static analysis under pressure

The static case is the only one that reaches the distributed-load records — Abaqus' `*Dsload`, Calculix' `*DLOAD`, Code_Aster's `FORCE_COQUE` and Sesam's `BEUSLO`. The eigenvalue analyses reach the mass and stiffness matrices and nothing else, so a pressure written wrongly is invisible to them.

Mid-span deflection against the closed form $5\,q\,L^4 / (384\,D)$, with $D = E\,t^3 / (12\,(1-\nu^2))$ — which evaluates to 0.173333 m for this strip. Halving the seed should quarter the remaining error; the three rows are a factor of two apart so the order is readable off the table rather than asserted.

Table 10: Plate strip mid-span deflection [m] under 1 kPa, by solver and mesh seed. The closed form is 5 q L^4 / (384 D) for a strip in cylindrical bending.

Seed [m]	aba	ca	ccx	ses	closed form
0.125	0.173065	0.173198	0.173065	0.173198	0.173333
0.0625	0.173269	0.173299	0.173268	0.173299	0.173333
0.03125	0.173319	0.173325	0.173319	0.173325	0.173333

The same strip with a T-profile stiffener along its centreline, meshed as line elements sharing the shells' nodes. There is no closed form for it — a parallel-spring estimate is not one — so this is an agreement case between solvers, and it is what carries a T section through the deck writers.

Table 11: The same strip with a T-profile stiffener on its centreline [m]. No closed form: this is a cross-solver agreement case.

Seed [m]	aba_st	ca_st	ses_st
0.125	0.000584531	0.000710295	0.000720708
0.0625	0.000585478	0.000710711	0.000720976
0.03125	0.000585996	0.000710815	0.000720795

Sign convention

Every writer above places a positive pressure so that it pushes **into** the face it names. That is Abaqus' convention, and it has to be shared, because a load direction that depends on the output format makes a cross-solver comparison meaningless.

It is worth stating which face that is, because it is not a global direction: gmsh winds every element of this strip with its normal along −z, so the positive face looks *down* and a positive pressure on it moves the strip *up*. The deflections above are therefore positive. `tests/fem/test_pressure_load_cross_format.py` pins this per format.

Eigenvalue analysis

The same plate, as an eigenvalue analysis, against `f_n = n² π / (2 L²) √(D / (ρ t))` — the simply supported beam frequencies with the beam's `EI / (ρ A)` replaced by the strip's `D / (ρ t)`. Valid for the cylindrical modes, which are the ones the long-edge constraint admits.

Table 12: Plate strip eigenfrequencies [Hz], by solver, at a 0.0625 m seed. The closed form is f_n = n^2 pi / (2 L^2) sqrt(D / (rho t)), valid for the cylindrical modes the long-edge constraint admits.

Mode	aba	ca	ccx	ses	closed form
1	1.5371	1.53661	1.53675	1.53661	1.53661
2	6.1536	6.14644	6.14861	6.14644	6.14644
3	13.866	13.8295	13.8405	13.8295	13.8295
4	24.7	24.5858	24.6205	24.5858	24.5857
5	38.695	38.4153	38.5002	38.4153	38.4152
6	55.9	55.3181	55.4943	55.3182	55.3179

Short description of the solver column names:

- ccx: Calculix
- ca: Code Aster
- ses: Sesam
- aba: Abaqus

A `_st` suffix marks the stiffened variant.

Detailed results

Mode shapes and deflected shapes from every format that ran on the machine that built this report, laid out so the formats can be read against each other: one section per mesh configuration, one heading per mode, and the formats side by side under it. A swapped mode pair or a differently deflected shape shows up as two pictures that do not match.

Abaqus and Sesam cannot run where this report is built. Their mode shapes are the raw results a licensed machine committed beside its cached frequencies, and the figures are rendered from them here; a case with no committed results has no figures. Solver versions: Abaqus v2025, Calculix v2.23, Code Aster v18.1.7, Sesam v11.3.1.0.

Cantilever modal participation

Per-mode effective modal mass and participation factors in the global X / Y / Z directions, for every mesh configuration, with the formats side by side. Code Aster and Sesam report the translational directions only.

A participation factor Γ is relative to how the solver normalised its eigenvectors: Abaqus scales the largest displacement to 1, the others scale to unit generalised mass. Raw factors from different solvers therefore do not compare. The tables give the factor scaled to unit generalised mass, sign(Γ)·√Meff, which is what the others report as-is. Its sign is the solver's choice of eigenvector sign and may differ from mode to mode. The effective mass Meff = $\Gamma^2 \cdot m_{gen}$ does not depend on the normalisation.

Solid, 1st order

Table 13: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.4	117.5	116.9	116.4	117.5	118.2	118.2	116.4	116.4	116.5	116.4
1	Z	0	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	118.3	0	0	118.2	118.4	118.4	0	0	118.5	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	34.9	0	0	34.9	35.5	35.5	0	0	36.2	0
3	Z	118.3	0	119	118.2	0	0	0	118.2	118.2	0	118.3
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	0	2.1	36	0	0	0	36	36.1	0	36.1
4	Z	0	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	12.6	0
5	Z	0	39.8	0	0	39.7	39.9	39.9	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	13.3	31.4	12.4	13.2	12.8	12.8	12.5	12.5	0	0
6	Z	0	0	0	0	0.1	0	0	0	0	40.2	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.2	0	2.8	0	0	0	0	0	0	0	8.9
7	Z	0	0	0	40	0	0	0	40	40.1	0	0
8	X	0	153.4	0	0	153.4	153.4	153.4	0	0	153.5	0
8	Y	5.7	0	0	0	0	0	0	0	0	0	3.1
8	Z	0	0	0	0	0	0	0	0	0	0	0.3
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.4	6.3	1.1	5.7	6.3	6.1	6.1	5.9	5.9	0.4	0
9	Z	0	0	0	0	0	0	0	0	0	0	39.9
10	X	0	0	0	153.5	0	0	0	153.4	153.5	0	0
10	Y	0	0	0	0	0	0	0	0	0	6.1	0
10	Z	40.3	0	0	0	0	12.8	12.8	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	0	12.8	0	0	12.8	0	0	0	0	12.7	0

Table 14: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	-0	0	0	-0	0	-0	0	-0	0	0	0
1	Y	10.79	-10.841	10.813	10.788	10.838	-10.871	10.871	10.787	10.787	10.794	10.791
1	Z	0	0.017	-0	-0	-0.017	0.019	-0.019	-0	-0	0	-0.001
2	X	-0	-0	0	0	0	0	0	-0	-0	-0	0.001
2	Y	-0	-0.028	0	0	0.028	0.03	0.03	0	0	0	0.022
2	Z	0	-10.876	-0	-0	10.874	10.88	10.88	-0	0	-10.884	-0
3	X	-0	0	-0	-0	-0	0	-0	-0	-0	0	0
3	Y	-0	5.911	0	0	-5.909	5.961	-5.961	0	0	-6.013	0
3	Z	10.876	-0.019	10.907	10.872	0.019	-0.02	0.02	10.871	10.872	-0	10.876
4	X	-0	-0.003	-0	-0	0.003	0.003	0.003	0	0	0	-0
4	Y	5.995	0.069	-1.456	-6.002	-0.075	-0.108	-0.11	6.004	-6.004	-0	-6.01
4	Z	-0	-0.003	0	0	0.003	0.007	0.008	0	-0	-0	-0.004
5	X	0	-0.001	-0	0	-0.001	0	-0	-0	-0	-0	-0.002
5	Y	-0	0.122	-0	-0	0.132	-0.028	0.029	0	-0	-3.549	-0.084
5	Z	0	-6.305	0	0	-6.305	6.313	-6.315	-0	-0	0	0.006
6	X	-0	0	0	-0	0	0	-0.001	0	-0	0	-0.001
6	Y	-0.114	3.642	-5.603	3.522	3.638	-3.576	3.577	3.53	3.532	-0	0.192
6	Z	0	0.207	-0	0	0.226	-0.056	0.057	0	-0	-6.34	-0.015
7	X	-0	0	-0	-0	0	-0.021	-0.021	0	0	0	0
7	Y	-1.788	0.027	-1.671	0	0.03	0.04	0.04	-0	-0	-0	2.99
7	Z	0	-0.033	0	-6.326	-0.035	-0.02	-0.02	6.327	-6.329	-0	0.071
8	X	0	12.384	-0	-0	12.385	12.384	12.386	0	-0	12.389	0.001
8	Y	2.39	-0.001	-0	0	-0.001	0	0	0	0	0	1.76
8	Z	0	0	-0	0	0	-0	-0	-0	-0	0	-0.503
9	X	-0	-0.005	-0	0	-0.006	0.001	0.001	-0	-0	0	0.001
9	Y	0.662	-2.518	-1.056	-2.385	-2.515	-2.462	-2.464	2.426	-2.43	-0.619	-0.115
9	Z	-0	0.007	-0	0	0.007	0.064	0.064	0	0	-0	-6.318
10	X	-0	-0.015	0.019	12.391	-0.016	0.001	-0.001	-12.385	12.388	-0	0.006
10	Y	0	-0.015	-0	0	-0.017	-0.045	0.045	-0	-0	2.478	0.086
10	Z	-6.346	0.003	0	-0	0.003	-3.572	3.576	0	0	-0	-0.006
11	X	0	0.001	-0	0	-0.001	-0.012	0.012	0	0	-0	0.005
11	Y	-0	0	0	0	-0	0.016	-0.016	-0	0	-0	0.009
11	Z	0	-3.578	-0	0	3.578	0.029	-0.031	0	-0	3.558	-0.001

Solid, 2nd order

Table 15: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.5	116.5	116.6	116.5	116.5	116.5	116.5	116.5	116.5	116.6	116.6
1	Z	0	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0	0
3	Z	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	36	35.9	35.9	36	36	36	35.9	35.9	35.9	35.9
4	Z	0	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.4	3.1	3.4	3.3	3.1	3.2	3.2	3.3	3.3	3.4	3.4
7	Z	0	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0	0
8	Y	5.3	5.8	5.3	5.4	5.8	5.9	5.9	5.4	5.4	5.3	5.3
8	Z	0	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.5	0.6	0.5	0.5	0.5	0.5	0.6	0.6
9	Z	0	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0	0
10	Y	1.5	0	1.5	1.6	0	0	0	1.6	1.6	1.5	1.5

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
10	Z	0	40.2	0	0	40.2	40.2	40.2	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	40.3	0	40.3	40.3	0	0	0	40.3	40.3	40.3	40.3

Table 16: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	-0	-0	0	0	-0	0	-0	-0	0	-0
1	Y	10.794	-10.794	10.796	10.794	10.794	-10.794	10.794	10.794	10.794	10.796	10.796
1	Z	-0	-0	0	0	0	-0	0	-0	0	0	-0
2	X	0	0	-0	-0	0	-0	0	-0	-0	-0	-0
2	Y	-0	-0	0	0	-0	-0	0	0	0	-0	0
2	Z	0	0	-0	0	0	-0	0	0	-0	0	0
3	X	0	0	-0	-0	0	-0	-0	-0	-0	-0	-0
3	Y	0	-0	-0	-0	-0	-0	-0	0	-0	-0	0
3	Z	10.88	10.879	10.882	10.879	10.879	10.879	10.879	10.879	10.879	10.882	10.882
4	X	-0	0	-0	-0	0	-0	-0	0	0	0	-0
4	Y	-5.991	-5.997	-5.992	-5.993	-5.997	-6.004	-6.004	-5.993	-5.993	-5.992	-5.992
4	Z	0	-0	-0	-0	-0	0	0	0	-0	-0	-0
5	X	-0	0	0	0	-0	-0	0	0	-0	-0	-0
5	Y	0	0.001	-0	-0	-0.001	0.001	-0.001	0	-0	-0	-0
5	Z	-0	-0	-0	-0	0	-0	0	-0	0	0	0
6	X	0	-0	-0	0	0	0	0	0	-0	-0	0
6	Y	-0.096	-0.119	-0.097	-0.103	0.119	-0.093	-0.093	0.103	-0.103	0.096	-0.097
6	Z	0	0	0	-0	-0	0	0	0	-0	0	-0
7	X	0	0	-0	0	-0	0	0	-0	0	0	-0
7	Y	-1.839	-1.768	1.84	1.818	1.768	1.799	1.799	1.818	1.818	1.84	1.84
7	Z	0	0	-0	0	-0	0	0	0	0	0	0
8	X	-0	0	0	0	-0	0	0	-0	-0	0	-0
8	Y	-2.305	-2.406	2.304	2.328	2.405	2.436	2.435	2.328	2.328	2.304	2.304
8	Z	-0	0	0	-0	-0	-0	-0	-0	-0	0	-0
9	X	-0	0	-0	0	-0	0	0	-0	0	-0	-0
9	Y	-0.746	-0.774	-0.747	-0.719	0.774	0.69	0.69	0.72	-0.719	0.747	-0.747
9	Z	0	-0	0	0	0	-0	-0	0	-0	0	-0
10	X	-0	-0	0	0	-0	-0	0	-0	-0	-0	0
10	Y	1.211	0	1.212	1.267	0	-0	0	-1.267	1.267	-1.213	1.212
10	Z	0	-6.344	-0	-0	-6.344	6.344	-6.344	0	-0	0	-0
11	X	-0	-0	0	0	-0	-0	0	-0	-0	0	-0
11	Y	0	-0.001	0	0	-0.001	-0.008	0.008	0	-0	-0	-0
11	Z	6.346	-0.006	-6.345	-6.344	-0.006	0.004	-0.004	6.345	-6.344	6.345	-6.345

Shell, 1st order

Table 17: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	120	119.9	119.7	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	122.1
3	X	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	37.2
3	Z	122	122	122	122	122.1	122	122	122	0
4	X	0	0	0	0	0	0	0	0	0
4	Y	36.8	36.6	37	36.9	37.3	36.6	36.9	36.8	0
4	Z	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	13
5	Z	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0
6	Y	0	0.1	0	0	12.7	0.1	0	0	0
6	Z	0	0	0	0	0	0	0	0	41.5
7	X	0	0	0	0	0	0	0	0	0
7	Y	3.5	2.7	3.8	3.3	0	2.7	3.3	3.2	0

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
7	Z	0	0	0	0	41.6	0	0	0	0
8	X	0	0	0	0	0	0	0	0	158
8	Y	5.4	5.9	5.4	5.7	0	6	5.8	5.7	0
8	Z	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.6	0	0.6	0.6	0.6	5.7
9	Z	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0
10	Y	1.3	1.3	1.6	1.5	1.5	1.3	1.7	1.3	1
10	Z	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	3.6	0	0	0	0
11	Z	0	41.6	0	0	0	41.6	0	41.6	12.9

Table 18: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	-0	-0	0	-0	0	0	-0
1	Y	10.95	-10.951	10.952	10.951	10.941	10.951	10.95	10.951	10.95
1	Z	-0	-0	-0	-0	0	0	0	-0	0
2	X	-0	0	-0	-0	-0	0	0	-0	-0
2	Y	-0	-0	0	0	0.002	-0	0	-0	0
2	Z	-0	0	0	0	-0	0	0	0	-11.051
3	X	0	-0	-0	-0	-0	0	-0	-0	-0
3	Y	-0	0	0	0	0	-0	0	0	6.101
3	Z	-11.046	-11.047	11.046	11.046	-11.048	11.047	-11.045	11.046	0
4	X	-0	0	-0	-0	-0	0	-0	-0	0
4	Y	-6.067	-6.048	-6.081	-6.076	-6.104	-6.053	-6.072	6.065	0
4	Z	-0	-0	-0	-0	0	-0	0	0	0
5	X	0	-0	0	-0	0	0	-0	-0	-0
5	Y	0	0	-0	-0	-0.005	-0	-0	0	-3.601
5	Z	-0	-0	0	-0	-0	-0	0	-0	0
6	X	-0	-0	0	0	-0	-0	0	-0	0
6	Y	0.134	-0.252	-0.078	-0.132	-3.561	0.263	-0.174	-0.194	-0
6	Z	-0	0	-0	-0	-0	-0	-0	-0	-6.443
7	X	0	-0	-0	-0	-0	-0	-0	0	-0
7	Y	-1.86	-1.651	1.947	1.829	-0	-1.652	1.805	1.776	-0
7	Z	0	0	0	0	-6.446	0	0	0	0
8	X	0	-0	0	-0	0	0	-0	-0	12.568
8	Y	2.334	2.438	2.332	2.383	0.003	-2.455	-2.406	2.398	0
8	Z	0	0	-0	-0	0	-0	0	-0	0
9	X	-0	-0	-0	0	-0	0	-0	-0	-0
9	Y	-0.798	-0.781	-0.786	-0.771	-0.156	0.788	-0.76	-0.795	-2.388
9	Z	0	-0	0	0	0	-0	0	0	-0
10	X	0	0	-0	0	-0	0	-0	-0	-0
10	Y	-1.15	-1.128	1.269	1.232	1.218	-1.148	1.31	1.152	1.006
10	Z	0	-0	0	0	-0	0	0	-0	-0
11	X	0	-0	-0	0	-0	0	0	0	-0
11	Y	-0	-0	0	0	-1.9	-0	0	0	0
11	Z	0	-6.451	0	0	0	6.453	0	6.453	3.589

Shell, 2nd order

Table 19: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0
3	Z	122	122	122	122	122	122	122	122	122	122
4	X	0	0	0	0	0	0	0	0	0	0
4	Y	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
4	Z	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0
7	Y	3.5	3.4	3.5	3.5	3.5	3.4	3.4	3.4	3.5	3.4
7	Z	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0
8	Y	5.5	5.6	5.5	5.5	5.5	5.6	5.5	5.6	5.5	5.5
8	Z	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0
9	Y	0.7	0.6	0.7	0.7	0.7	0.6	0.6	0.6	0.7	0.6
9	Z	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0
10	Y	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
10	Z	0	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0
11	Z	0	0	0	0	0	0	0	0	0	0

Table 20: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	0	0	0	-0	0	0	-0	-0
1	Y	-10.952	10.952	10.951	10.951	10.952	10.952	10.952	10.951	10.952	10.951
1	Z	-0	-0	0	0	0	0	-0	0	0	0
2	X	-0	0	0	0	0	-0	-0	0	0	0
2	Y	0	0	0	0	0	0	-0	0	-0	0
2	Z	0	0	0	0	-0	-0	0	-0	-0	-0
3	X	-0	-0	0	0	0	-0	0	0	0	0
3	Y	-0	-0	0	0	0	0	0	0	0	0
3	Z	11.047	-11.047	11.044	11.044	11.047	-11.046	11.047	-11.046	11.047	-11.046
4	X	-0	-0	-0	-0	0	-0	-0	-0	0	0
4	Y	-6.071	6.072	-6.071	-6.071	-6.073	6.072	6.071	6.073	-6.071	6.071
4	Z	0	-0	0	0	0	-0	-0	-0	0	-0
5	X	0	-0	-0	-0	-0	0	-0	-0	0	-0
5	Y	0	0	-0	-0	-0	0	-0	-0	-0	-0
5	Z	-0	0	-0	-0	0	-0	-0	-0	-0	0
6	X	-0	0	-0	-0	-0	-0	0	-0	0	0
6	Y	-0.137	-0.144	-0.137	-0.137	-0.138	-0.144	0.142	0.141	-0.137	0.138
6	Z	0	0	0	0	-0	-0	-0	0	0	0
7	X	0	-0	-0	-0	-0	-0	-0	-0	-0	-0
7	Y	1.859	-1.85	1.858	1.858	1.88	1.848	-1.849	1.839	-1.859	1.852
7	Z	-0	0	-0	-0	0	0	0	0	0	0
8	X	-0	0	-0	-0	-0	0	-0	-0	-0	0
8	Y	-2.35	-2.358	2.351	2.351	2.349	2.356	2.353	-2.362	2.35	-2.35
8	Z	-0	-0	0	0	0	-0	0	0	0	0
9	X	0	0	-0	-0	0	-0	-0	0	-0	0
9	Y	-0.809	-0.801	-0.808	-0.808	-0.818	0.793	0.801	0.781	-0.809	0.799
9	Z	0	0	0	0	-0	-0	-0	-0	-0	-0
10	X	-0	-0	-0	-0	-0	-0	0	-0	-0	-0
10	Y	-1.171	-1.185	1.17	1.17	1.175	1.193	1.178	-1.201	1.17	1.178
10	Z	-0	-0	-0	-0	0.001	0	0	-0	0	0
11	X	-0	-0	0	0	0	-0	-0	-0	-0	-0
11	Y	-0	0.001	0	0	0	0.001	-0.001	0	0	0
11	Z	0	-0	-0	-0	0	-0	-0	0	-0	-0

Line, 1st order

Table 21: Effective modal mass [kg] per mode, global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	0	0	0	0
1	Y	0	0	116.4	0	116.4

Mode	Direction	ses	aba	ccx	ca	ccxR
1	Z	0	0	0	0	0
2	X	0	0	0	0	0
2	Y	116.7	116.6	0	116.5	0
2	Z	0	0	114.8	0	114.8
3	X	0	0	0	0	0
3	Y	0	0	35.7	0	35.7
3	Z	0	0	0	0	0
4	X	0	0	0	0	0
4	Y	0	0	12.3	0	12.3
4	Z	120.2	118.2	0	116.5	0
5	X	0	0	0	0	0
5	Y	0	0	0	0	0
5	Z	0	0	0	0	0
6	X	0	0	0	0	0
6	Y	36.3	0	0	35.8	0
6	Z	0	0	35	0	35
7	X	0	0	153.9	0	153.9
7	Y	0	36.2	0	0	0
7	Z	0	0	0	0	0
8	X	0	0	0	0	0
8	Y	0	0	6.2	0	6.2
8	Z	0	0	0	0	0
9	X	0	0	0	0	0
9	Y	0	0	3.8	0	3.8
9	Z	0	0	0	0	0
10	X	0	0	0	0	0
10	Y	0	0	0	0	0
10	Z	0	0	0	0	0
11	X	0	0	0	0	0
11	Y	0	0	0	0	0
11	Z	0	0	11.4	0	11.4

Table 22: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	-0	0	0	0
1	Y	-0	-0	-10.789	-0	-10.789
1	Z	-0	0	0	0	0
2	X	0	-0	0	0	0
2	Y	10.803	10.797	-0	10.793	-0
2	Z	-0	-0	-10.716	0	-10.716
3	X	0	0	-0	0	-0
3	Y	-0	-0	5.979	-0	5.979
3	Z	-0	0	-0	-0	-0
4	X	0	0	-0	-0	-0
4	Y	-0	-0	3.5	0	3.5
4	Z	-10.963	10.87	-0	10.793	-0
5	X	-0	-0	0	0	0
5	Y	-0	0	-0	0	-0
5	Z	0	0	0	0	0
6	X	0	-0	-0	0	-0
6	Y	-6.024	-0	-0	-5.981	-0
6	Z	0	0	-5.913	-0	-5.913
7	X	-0	0	12.407	0	12.407
7	Y	-0	-6.016	-0	0	-0
7	Z	-0	-0	-0	-0	-0
8	X	0	-0	0	0	0
8	Y	-0	-0	2.498	0	2.498
8	Z	-0	0	0	-0	0
9	X	-0	0	0	-0	0
9	Y	0	0	-1.937	0	-1.937
9	Z	0	0	-0	-0	-0
10	X	-0	0	-0	-0	-0
10	Y	-0	0	0	0	0
10	Z	-0	-0	0	0	0
11	X	-0	0	0	0	0
11	Y	-0	-0	0	-0	0
11	Z	-0	0	-3.374	0	-3.374

Line, 2nd order

Table 23: Effective modal mass [kg] per mode, global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	0	0
2	X	0	0
2	Y	176.8	116.6
2	Z	0	0
3	X	0	0
3	Y	0	0
3	Z	0	0
4	X	0	0
4	Y	0	0
4	Z	180	118.2
5	X	0	0
5	Y	0	0
5	Z	0	0
6	X	0	0
6	Y	55	0
6	Z	0	0
7	X	0	0
7	Y	0	36.2
7	Z	0	0
8	X	0	0
8	Y	0	0
8	Z	0	0
9	X	0	0
9	Y	0	0
9	Z	0	0
10	X	0	0
10	Y	0	0
10	Z	0	0
11	X	0	0
11	Y	0	0
11	Z	0	0

Table 24: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	-0	-0
2	X	-0	-0
2	Y	13.297	10.798
2	Z	0	-0
3	X	-0	0
3	Y	-0	-0
3	Z	-0	-0
4	X	0	-0
4	Y	-0	-0
4	Z	-13.418	10.871
5	X	0	-0
5	Y	0	0
5	Z	-0	0
6	X	0	-0
6	Y	-7.413	0
6	Z	0	-0
7	X	-0	0
7	Y	-0	-6.015
7	Z	-0	-0
8	X	-0	0
8	Y	-0	0
8	Z	-0	-0
9	X	0	0
9	Y	0	0
9	Z	0	-0
10	X	0	-0
10	Y	0	0
10	Z	-0	0
11	X	0	-0

Mode	Direction	ses	aba
11	Y	0	-0
11	Z	-0	0

Cantilever eigenvalue analysis

Line, 1st order

Mode 1

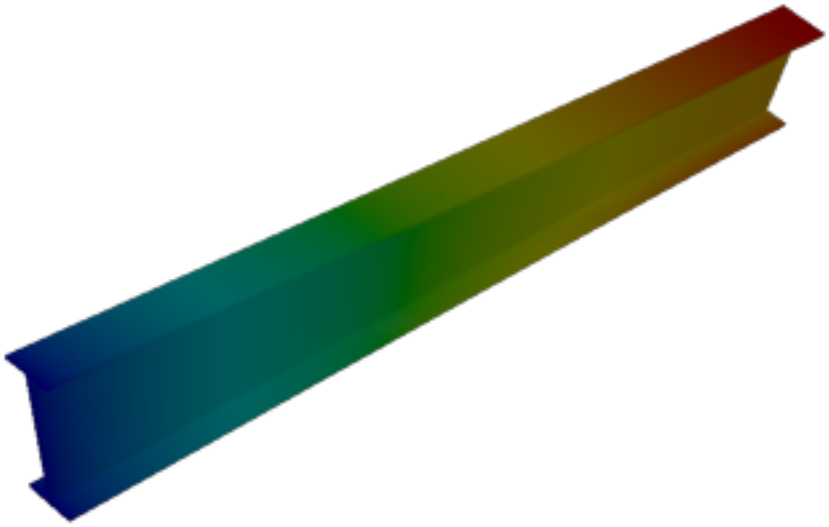


Figure 5: Abaqus — mode 1, 10.785 Hz

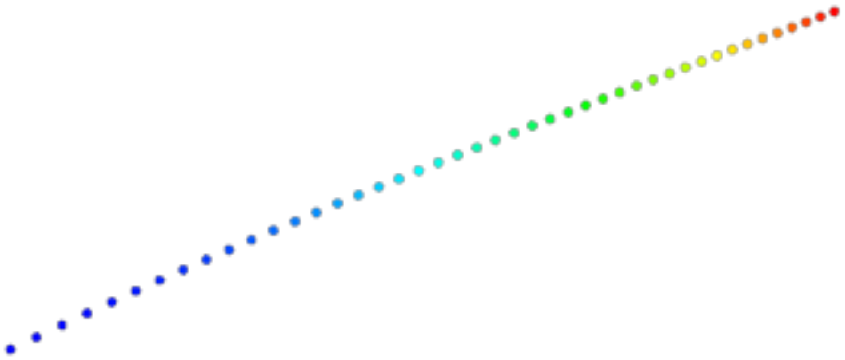


Figure 6: Calculix — mode 1, 12.973 Hz

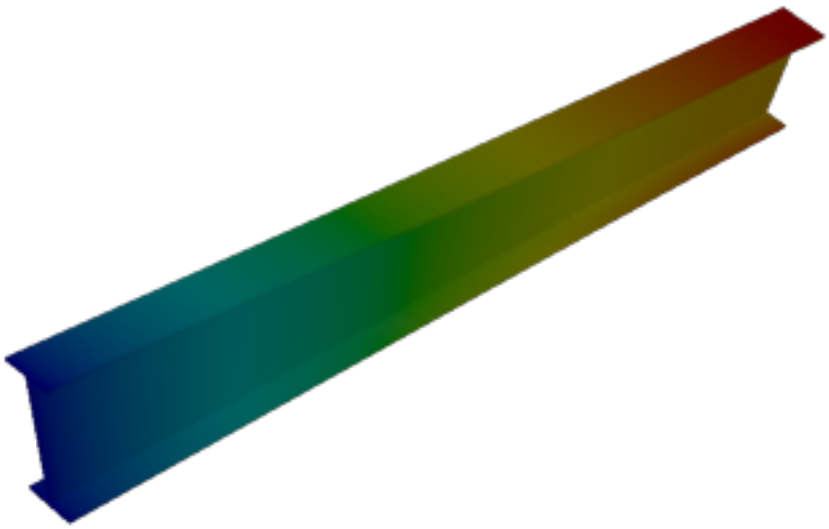


Figure 7: Code_Aster — mode 1, 12.245 Hz

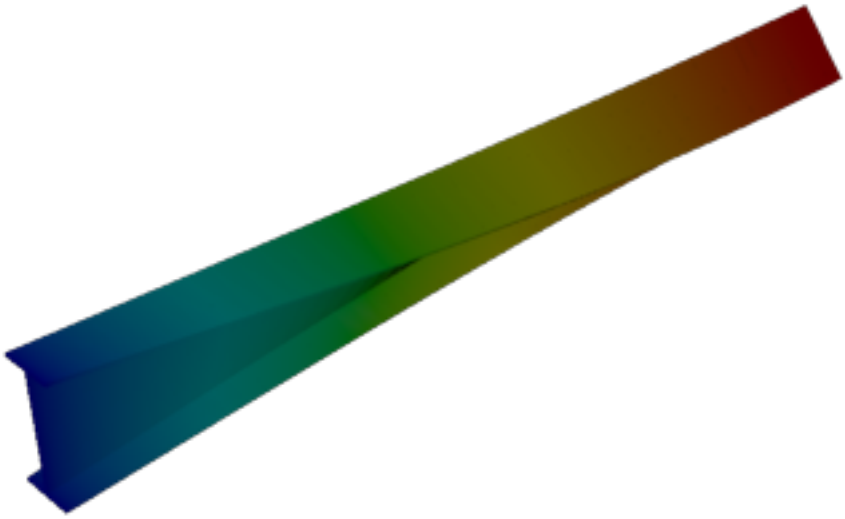


Figure 8: Sesam — mode 1, 12.245 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	10.785	-10.59	4592.3	4.83e-34	1.00e-29	1.05e-31	-2.20e-17	-3.17e-15	3.24e-16
Calculix	12.973	+7.55	6644.56	1.74e-37	116.40	9.37e-33	4.17e-19	-10.789	9.68e-17
Code_Aster	12.245	+1.52	5919.77	2.44e-45	7.41e-28	2.59e-35	4.94e-23	-2.72e-14	5.09e-18
Sesam	12.245	+1.52	5919.77	1.37e-35	1.81e-28	4.01e-31	3.70e-18	-1.35e-14	-6.33e-16

Mode 2

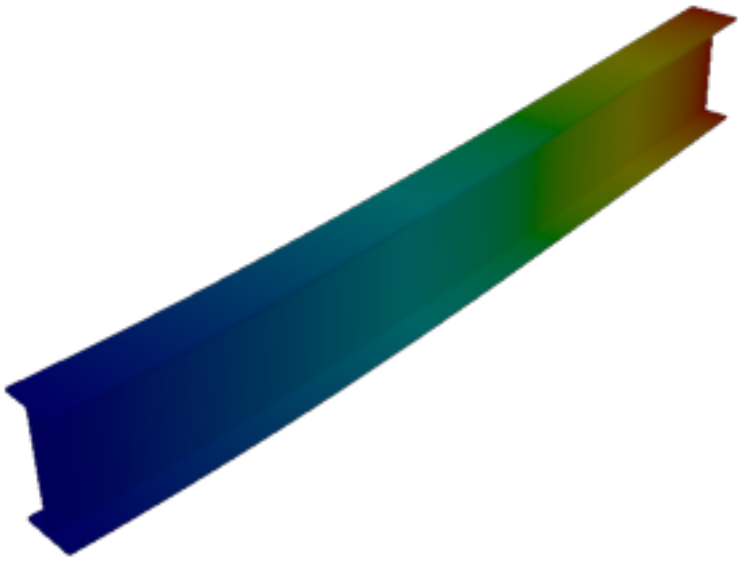


Figure 9: Abaqus — mode 2, 12.928 Hz

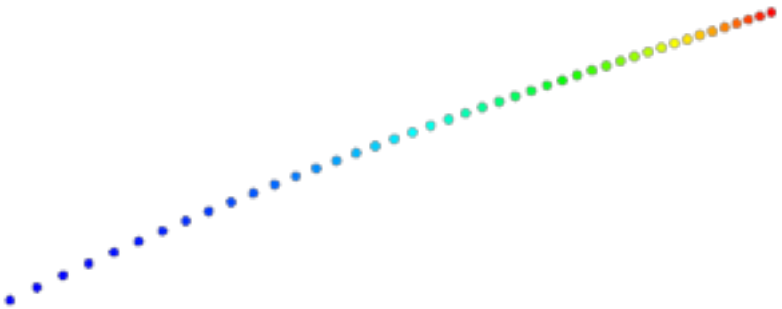


Figure 10: Calculix — mode 2, 52.587 Hz

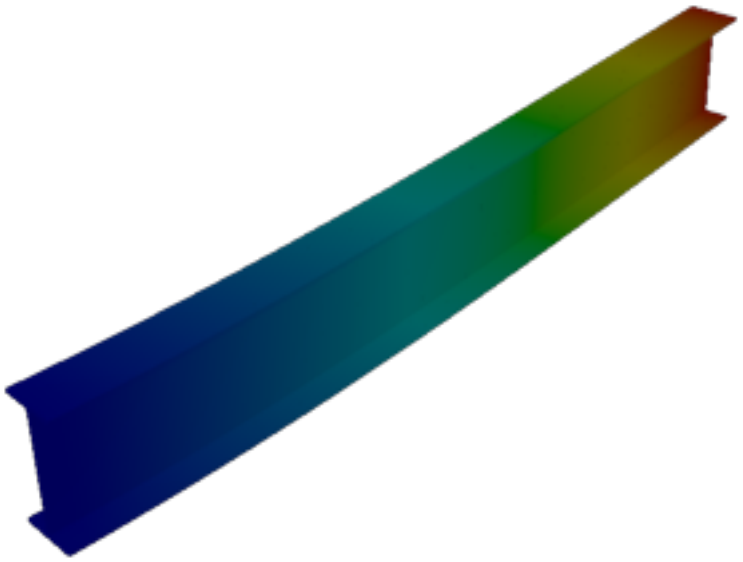


Figure 11: Code_Aster — mode 2, 12.979 Hz

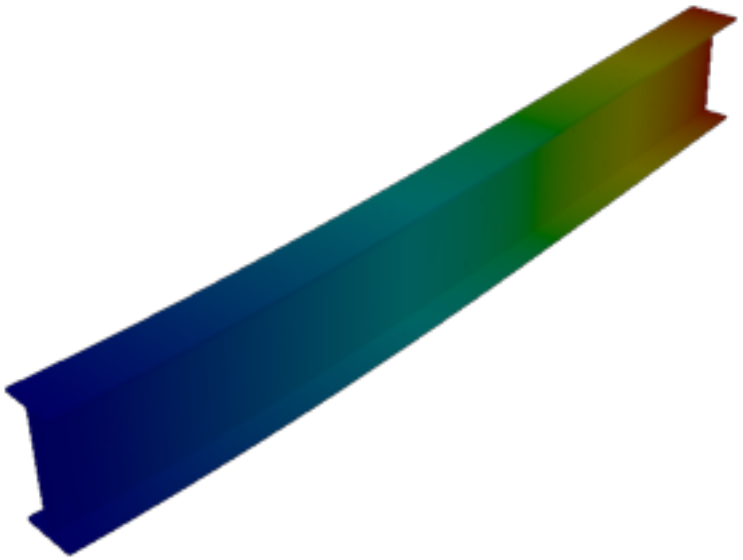


Figure 12: Sesam — mode 2, 12.943 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.928	-43.45	6598.4	6.75e-31	116.57	3.27e-31	-8.21e-16	10.797	-5.72e-16
Calculix	52.587	+130.05	109172	2.95e-34	6.28e-31	114.82	1.72e-17	-7.92e-16	-10.716
Code_Aster	12.979	-43.22	6650.66	2.90e-44	116.48	3.86e-32	1.70e-22	10.793	1.96e-16
Sesam	12.943	-43.38	6613.55	4.26e-36	116.71	1.68e-31	2.06e-18	10.803	-4.10e-16

Mode 3

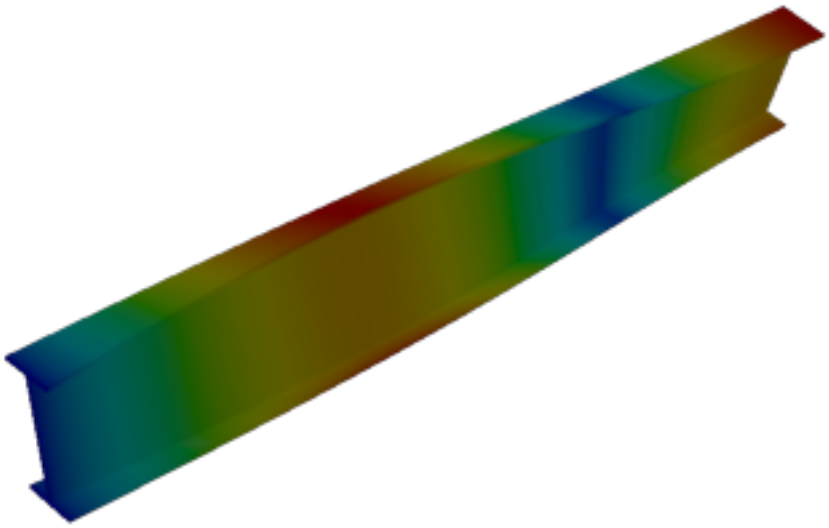


Figure 13: Abaqus — mode 3, 32.342 Hz

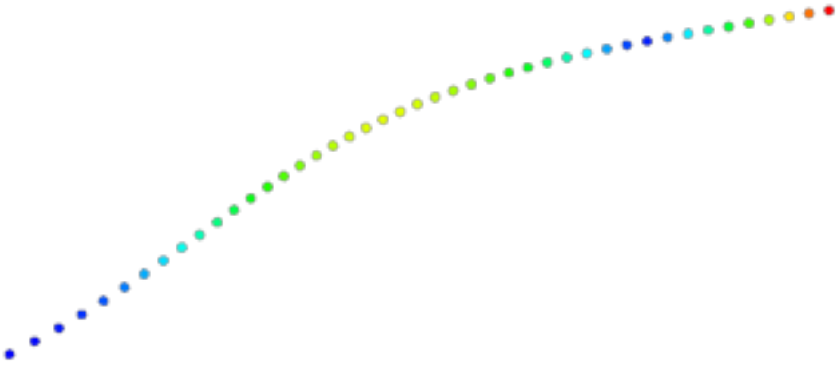


Figure 14: Calculix — mode 3, 81.151 Hz

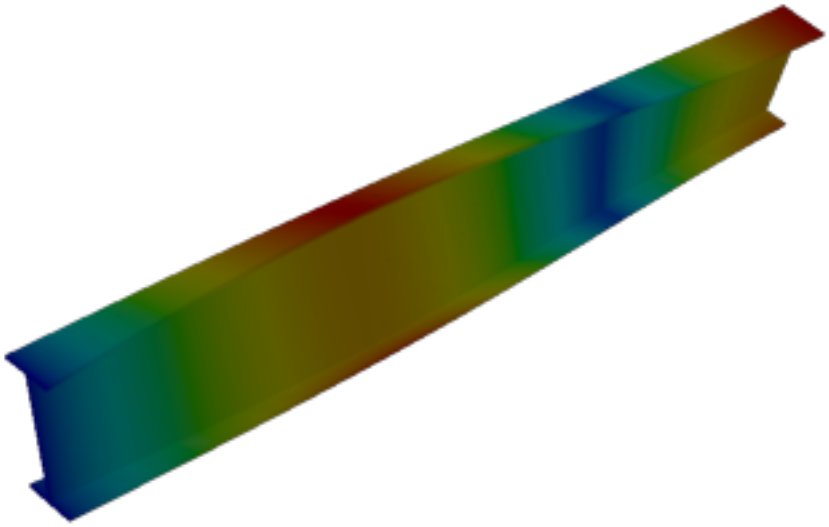


Figure 15: Code_Aster — mode 3, 36.752 Hz

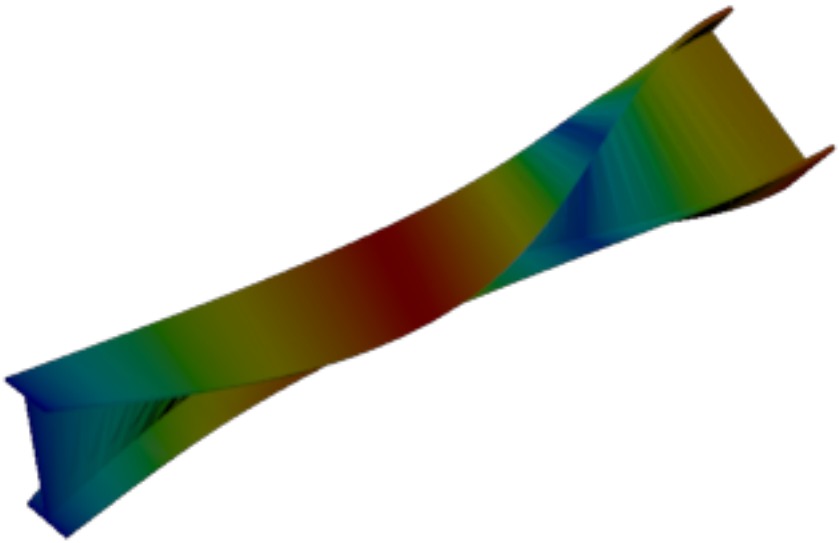


Figure 16: Sesam — mode 3, 36.752 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	32.342	-30.82	41294	7.85e-31	8.23e-32	1.14e-28	8.86e-16	-2.87e-16	1.07e-14
Calculix	81.151	+73.59	259985	4.06e-33	35.74	4.38e-30	-6.37e-17	5.979	-2.09e-15
Code_Aster	36.752	-21.38	53325.3	9.37e-41	1.13e-28	6.11e-33	9.68e-21	-1.07e-14	-7.82e-17
Sesam	36.752	-21.38	53325.3	9.01e-34	4.75e-30	1.52e-28	3.00e-17	-2.18e-15	-1.23e-14

Mode 4

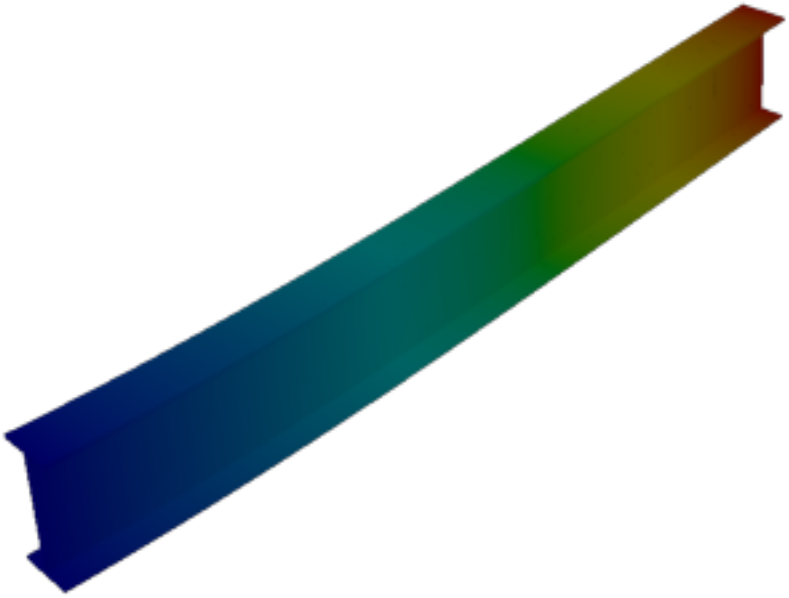


Figure 17: Abaqus — mode 4, 50.543 Hz

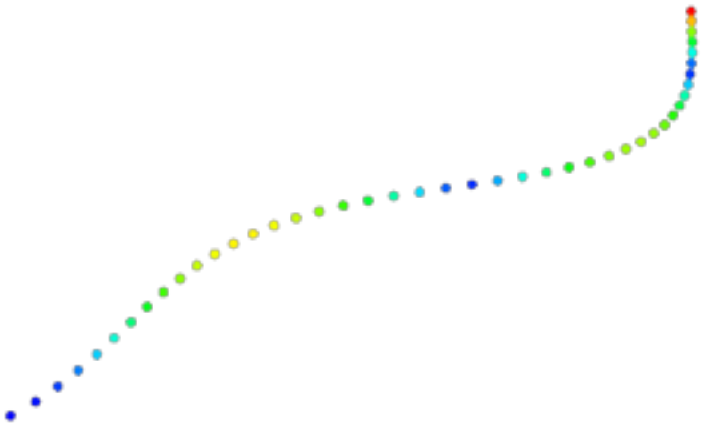


Figure 18: Calculix — mode 4, 226.646 Hz

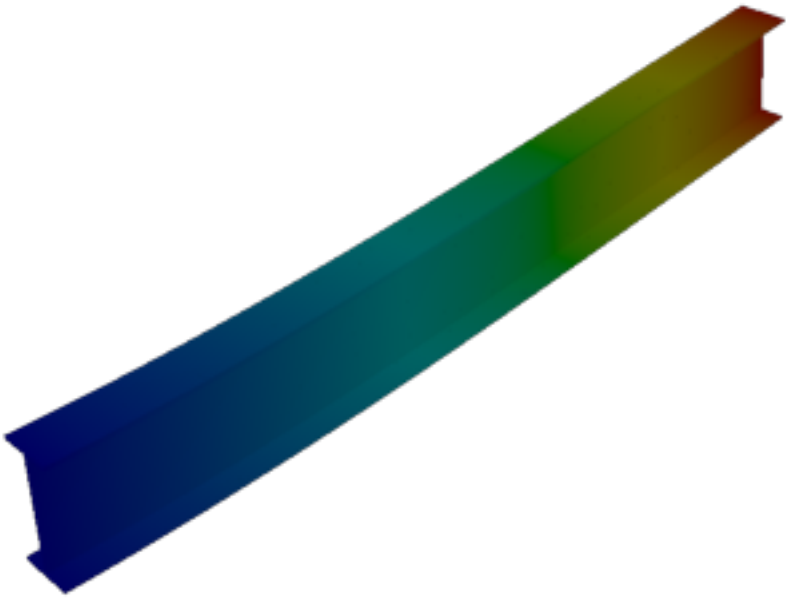


Figure 19: Code_Aster — mode 4, 52.956 Hz

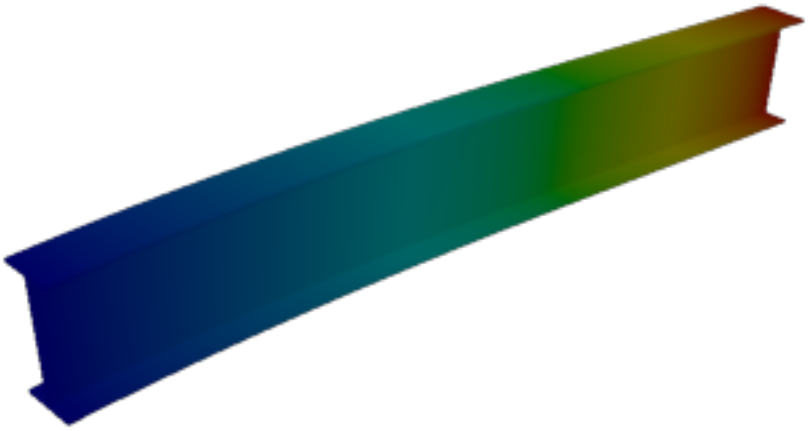


Figure 20: Sesam — mode 4, 50.554 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.543	-46.89	100852	2.51e-30	1.42e-29	118.16	1.58e-15	-3.76e-15	10.870
Calculix	226.646	+138.14	2.02794e+06	1.30e-30	12.25	2.11e-29	-1.14e-15	3.500	-4.59e-15
Code_Aster	52.956	-44.36	110710	7.80e-39	1.34e-32	116.48	-8.83e-20	1.16e-16	10.793
Sesam	50.554	-46.88	100896	1.69e-33	1.87e-30	120.18	4.11e-17	-1.37e-15	-10.963

Mode 5

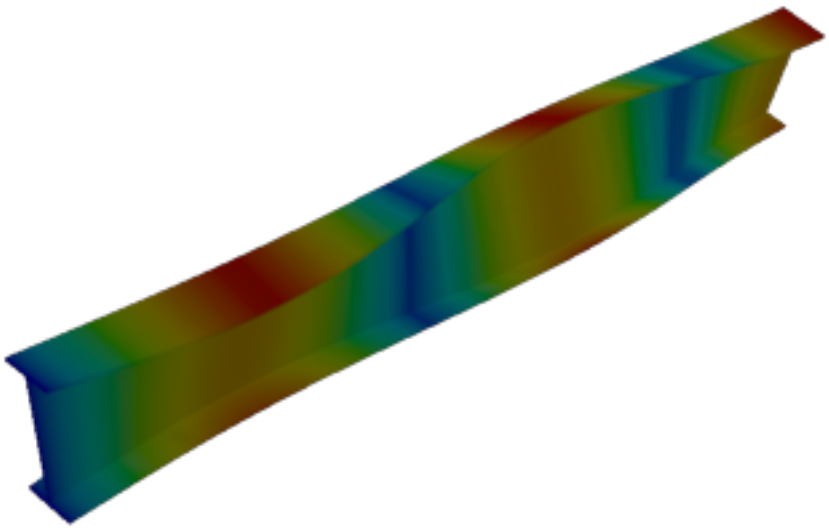


Figure 21: Abaqus — mode 5, 53.855 Hz

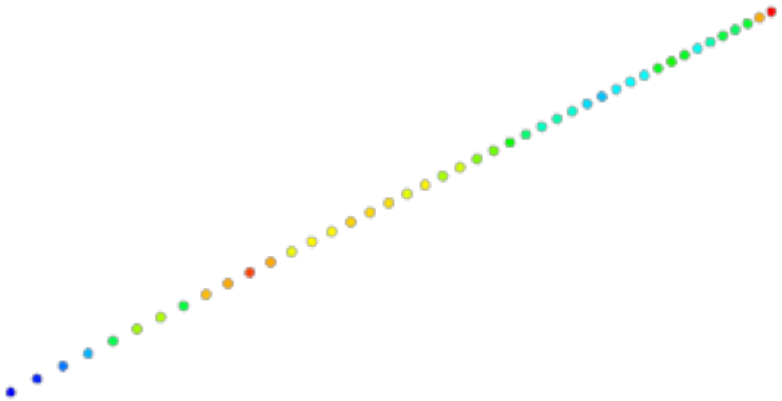


Figure 22: Calculix — mode 5, 267.320 Hz

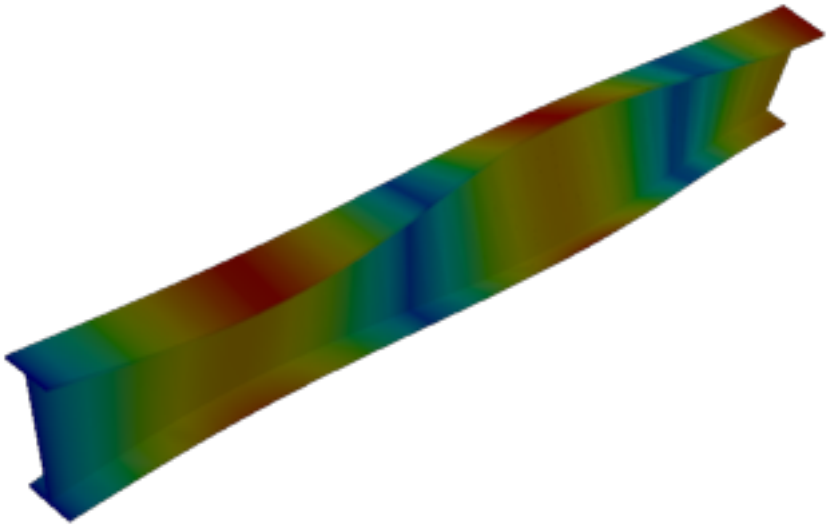


Figure 23: Code_Aster — mode 5, 61.309 Hz

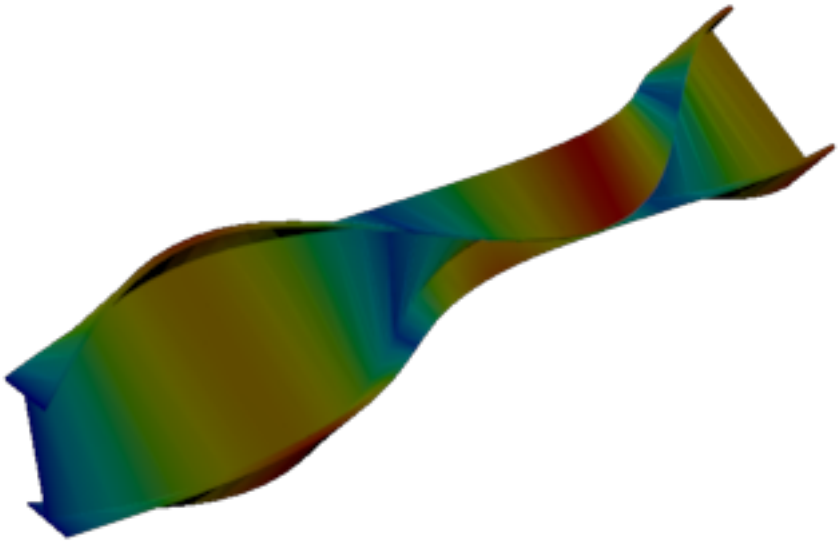


Figure 24: Sesam — mode 5, 61.309 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	53.855	-51.46	114501	1.75e-29	8.90e-30	2.61e-27	-4.18e-15	2.98e-15	5.11e-14
Calculix	267.320	+140.94	2.82112e+06	3.28e-30	4.98e-30	2.38e-29	1.81e-15	-2.23e-15	4.88e-15
Code_Aster	61.309	-44.74	148390	3.69e-37	5.51e-31	1.71e-26	6.08e-19	7.42e-16	1.31e-13
Sesam	61.309	-44.74	148390	5.33e-36	2.55e-29	4.34e-29	-2.31e-18	-5.05e-15	6.59e-15

Mode 6

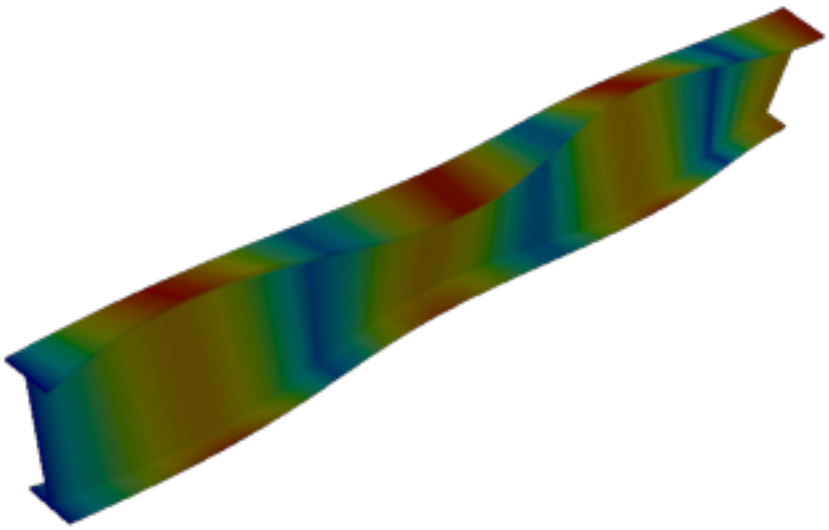


Figure 25: Abaqus — mode 6, 75.296 Hz

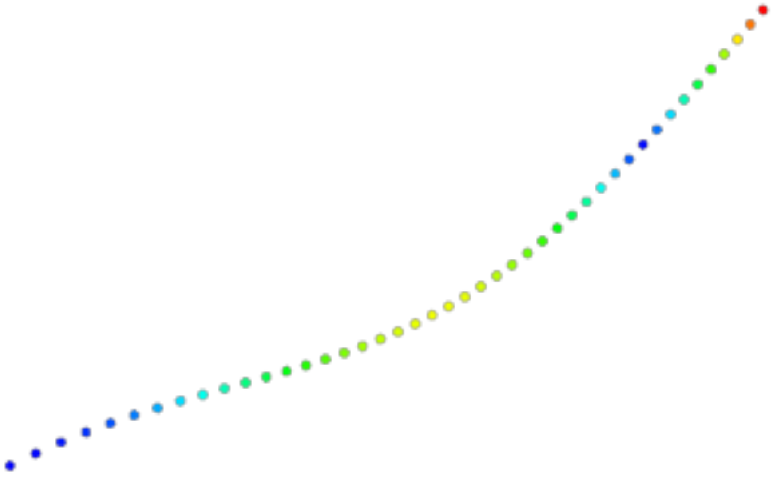


Figure 26: Calculix — mode 6, 316.915 Hz

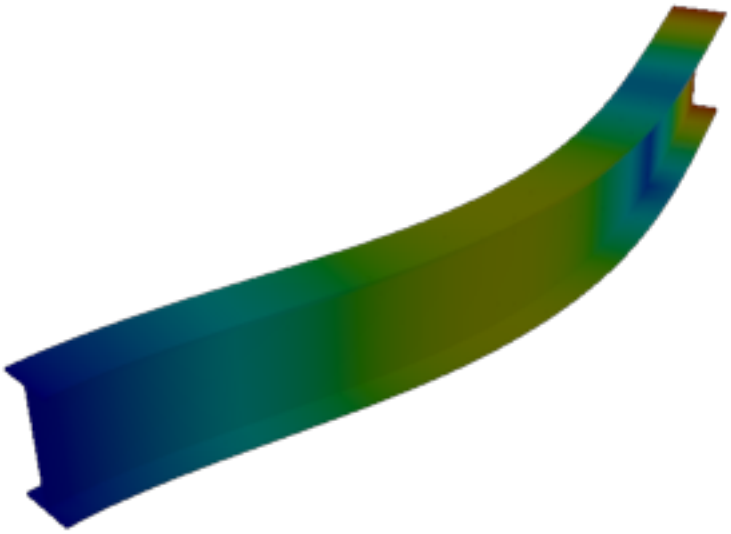


Figure 27: Code_Aster — mode 6, 81.340 Hz

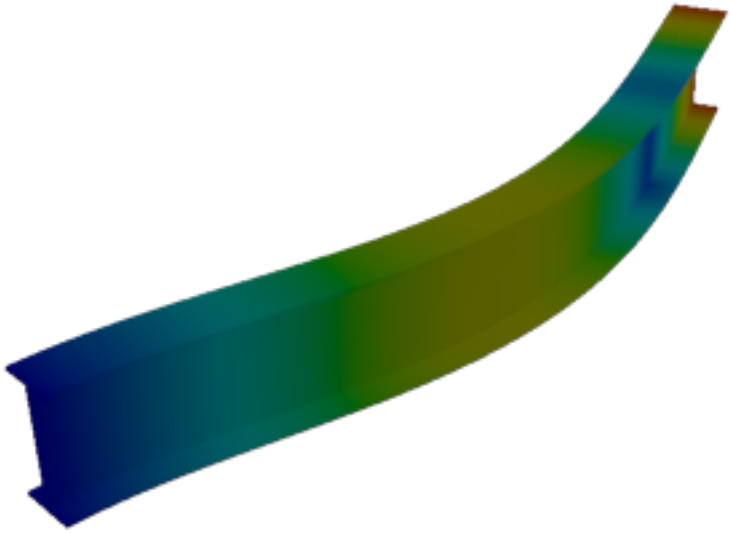


Figure 28: Sesam — mode 6, 79.793 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.296	-45.57	223823	3.82e-29	2.17e-28	3.67e-28	-6.18e-15	-1.47e-14	1.92e-14
Calculix	316.915	+129.09	3.96503e+06	1.31e-33	7.09e-30	34.97	-3.62e-17	-2.66e-15	-5.913
Code_Aster	81.340	-41.20	261198	1.94e-38	35.78	6.32e-30	1.39e-19	-5.981	-2.51e-15
Sesam	79.793	-42.32	251356	6.55e-33	36.29	2.49e-29	8.09e-17	-6.024	4.99e-15

Mode 7

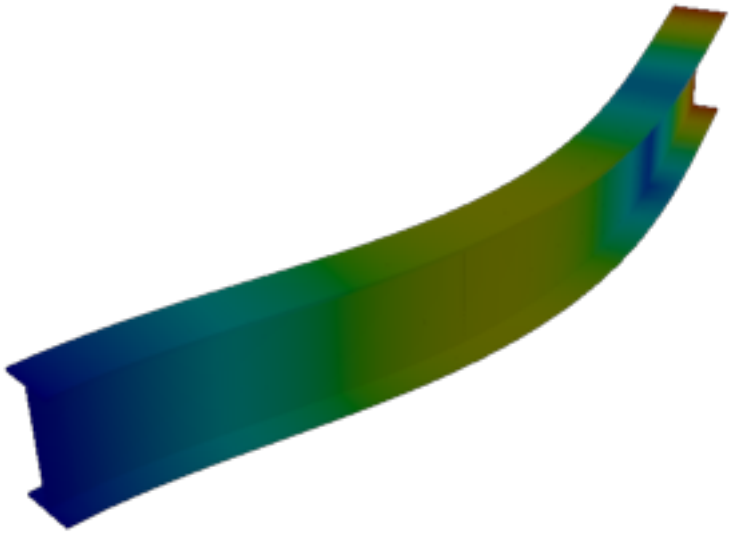


Figure 29: Abaqus — mode 7, 79.590 Hz

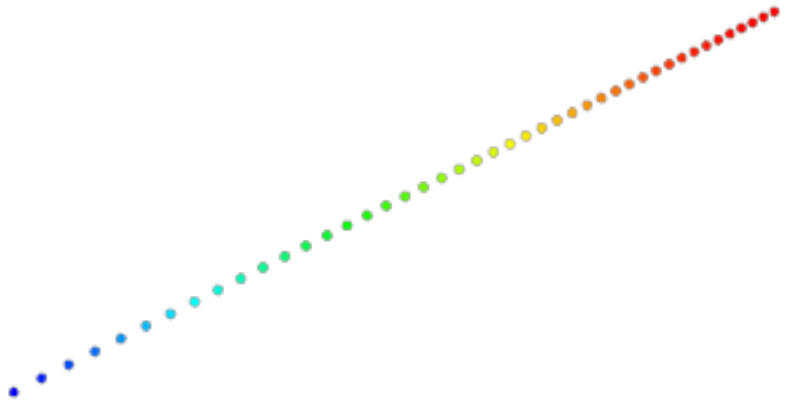


Figure 30: Calculix — mode 7, 431.040 Hz

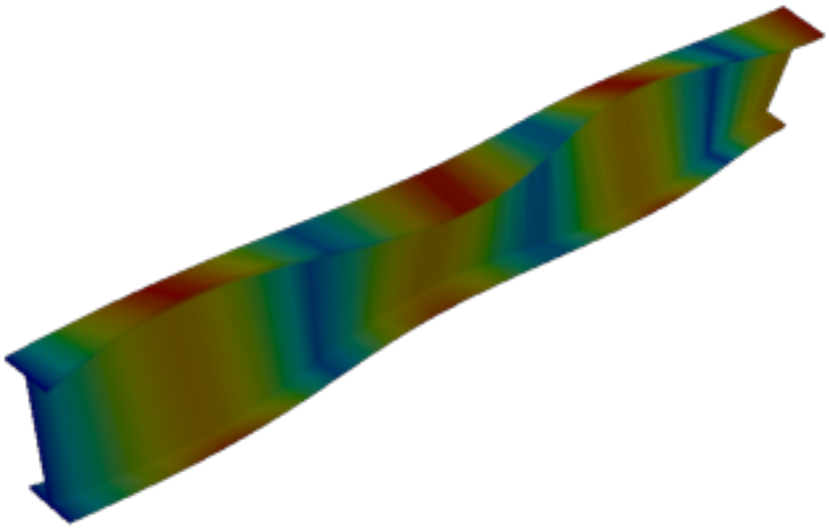


Figure 31: Code_Aster — mode 7, 85.947 Hz

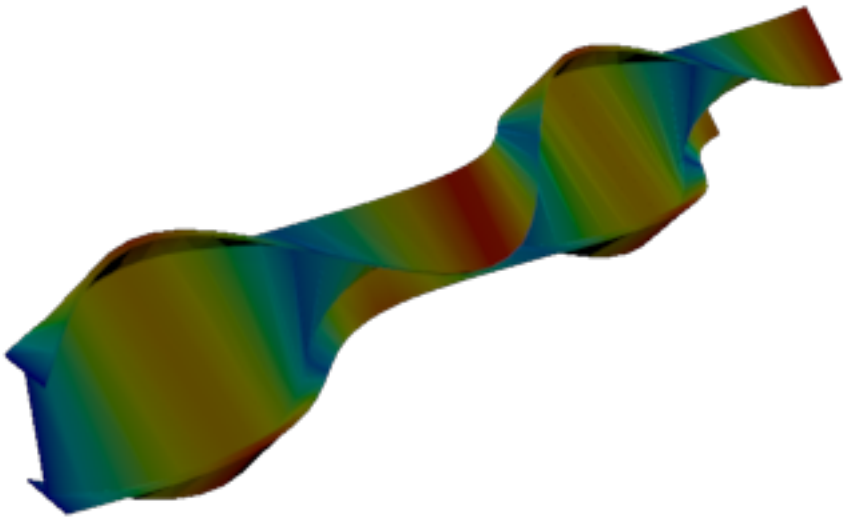


Figure 32: Sesam — mode 7, 85.947 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	79.590	-53.36	250078	1.01e-30	36.20	1.38e-29	1.01e-15	-6.016	-3.72e-15
Calculix	431.040	+152.62	7.33492e+06	153.94	1.29e-27	4.83e-31	12.407	-3.60e-14	-6.95e-16
Code_Aster	85.947	-49.63	291620	4.21e-36	2.16e-27	1.62e-29	2.05e-18	4.64e-14	-4.03e-15
Sesam	85.947	-49.63	291620	1.01e-32	4.87e-31	2.32e-31	-1.01e-16	-6.98e-16	-4.81e-16

Mode 8

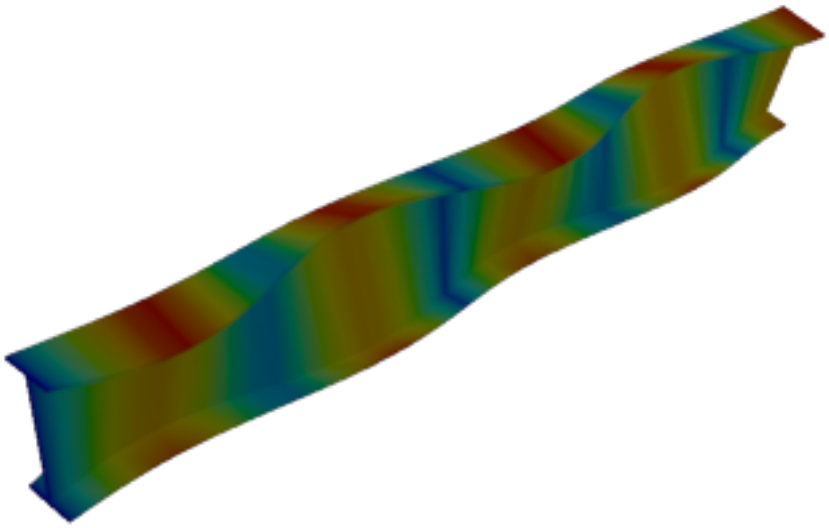


Figure 33: Abaqus — mode 8, 96.637 Hz

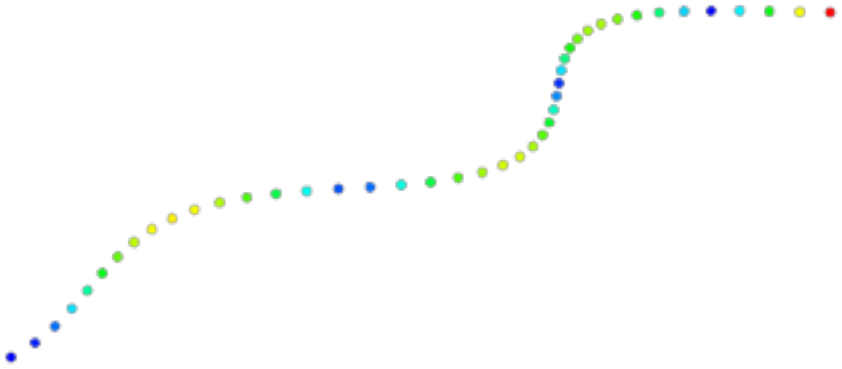


Figure 34: Calculix — mode 8, 442.610 Hz

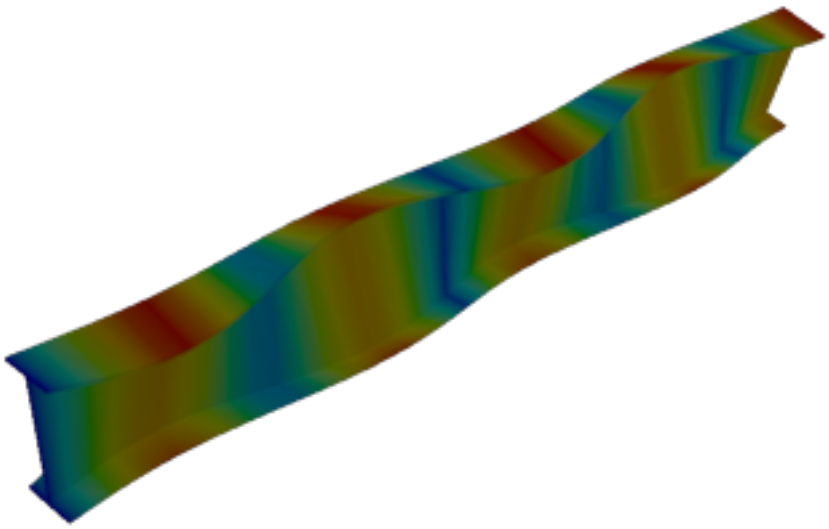


Figure 35: Code_Aster — mode 8, 110.699 Hz

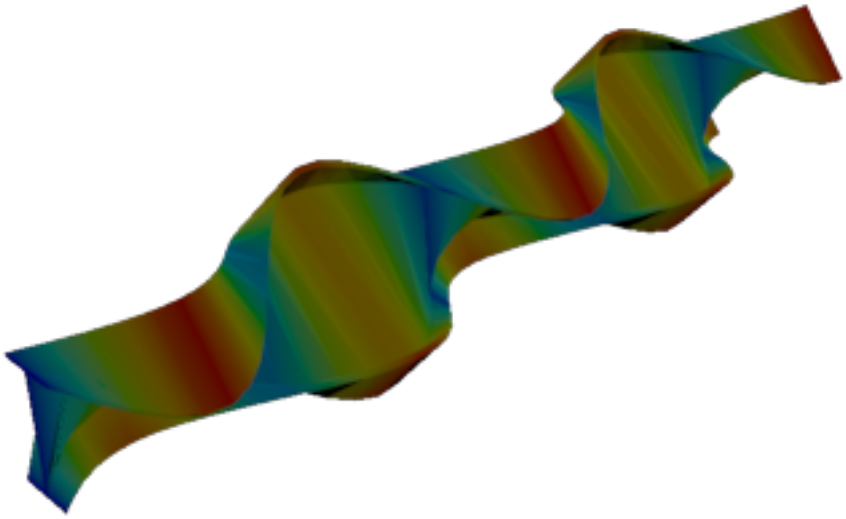


Figure 36: Sesam — mode 8, 110.699 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.637	-49.18	368677	4.04e-26	1.03e-27	4.21e-27	-2.01e-13	-3.20e-14	6.49e-14
Calculix	442.610	+132.75	7.73397e+06	3.15e-26	6.24	6.00e-30	1.77e-13	2.498	2.45e-15
Code_Aster	110.699	-41.79	483782	5.52e-35	2.21e-28	3.97e-30	7.43e-18	1.49e-14	-1.99e-15
Sesam	110.699	-41.79	483782	5.75e-32	4.96e-30	6.35e-30	2.40e-16	-2.23e-15	-2.52e-15

Mode 9

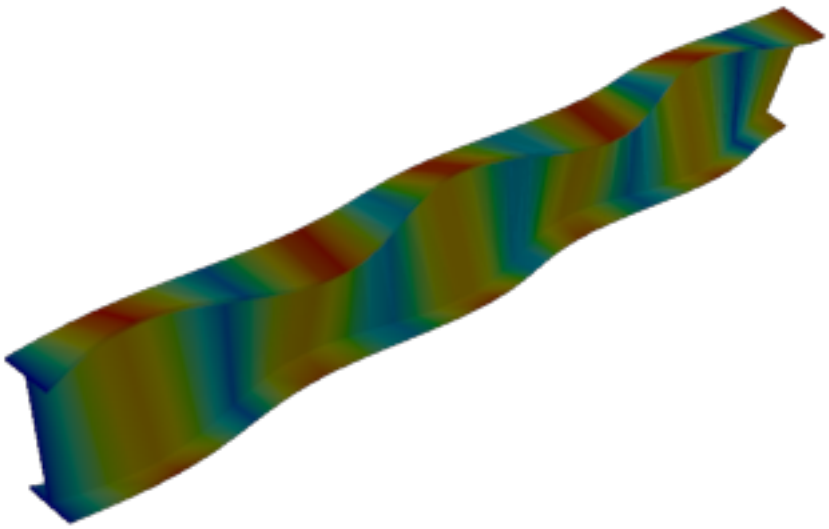


Figure 37: Abaqus — mode 9, 117.849 Hz

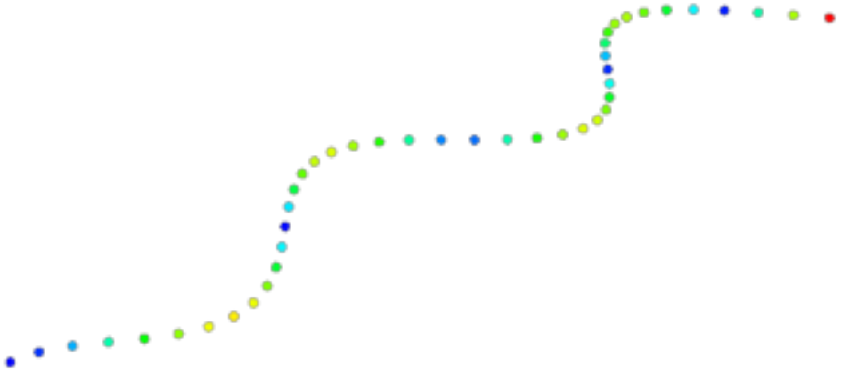


Figure 38: Calculix — mode 9, 728.596 Hz

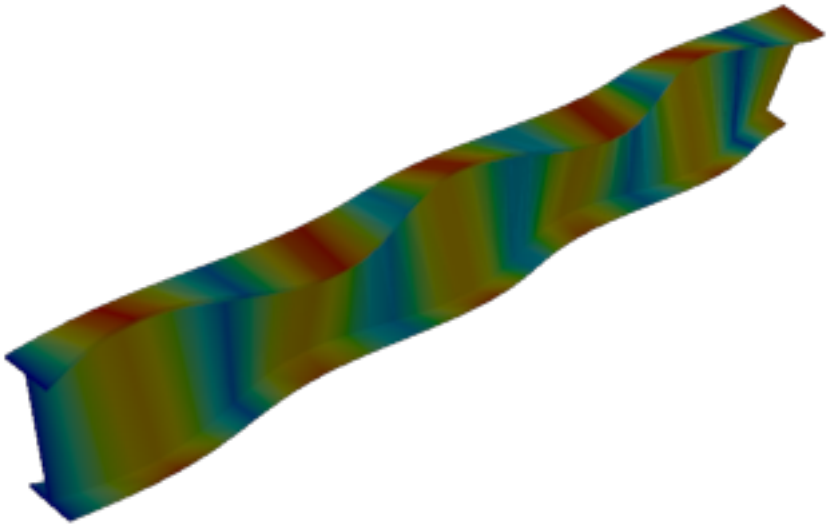


Figure 39: Code_Aster — mode 9, 135.600 Hz

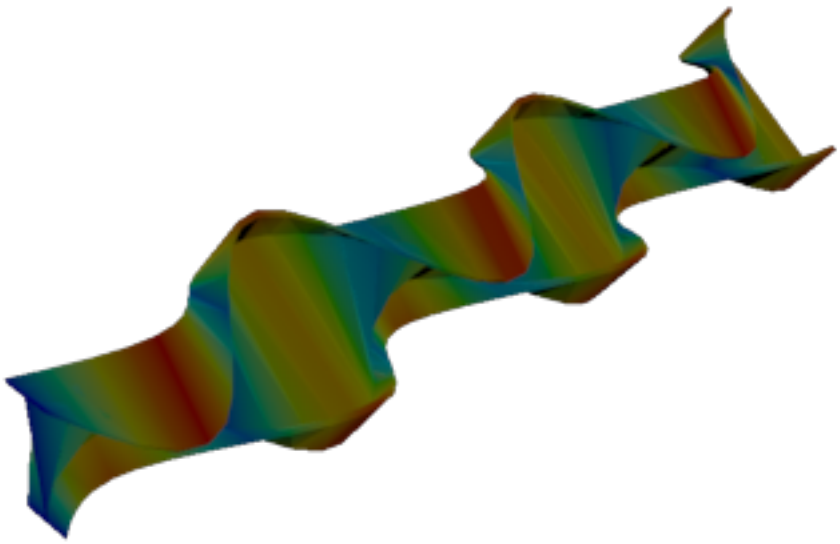


Figure 40: Sesam — mode 9, 135.600 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	117.850	-57.82	548290	9.24e-23	1.72e-25	7.37e-24	9.61e-12	4.15e-13	2.71e-12
Calculix	728.596	+160.76	2.09572e+07	1.49e-29	3.75	1.24e-28	3.85e-15	-1.937	-1.11e-14
Code_Aster	135.600	-51.47	725900	5.68e-35	4.70e-28	8.29e-30	-7.54e-18	2.17e-14	-2.88e-15
Sesam	135.600	-51.47	725900	8.08e-31	4.13e-29	6.76e-29	-8.99e-16	6.43e-15	8.22e-15

Mode 10

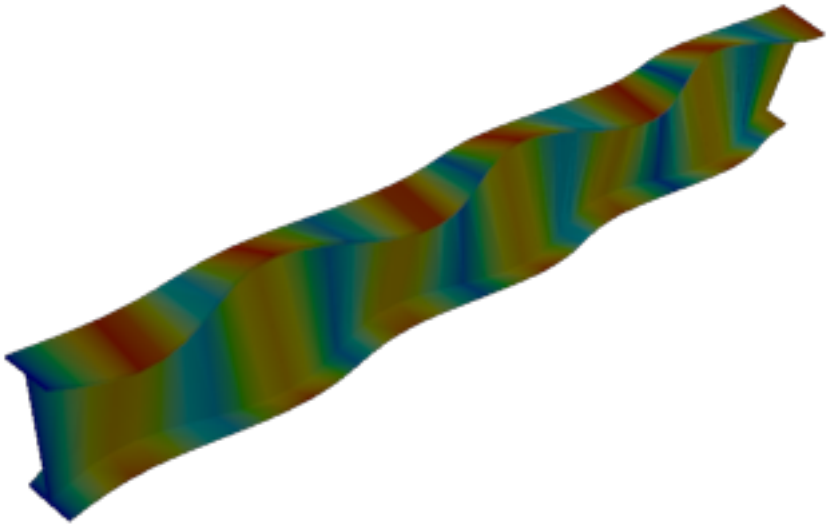


Figure 41: Abaqus — mode 10, 138.904 Hz

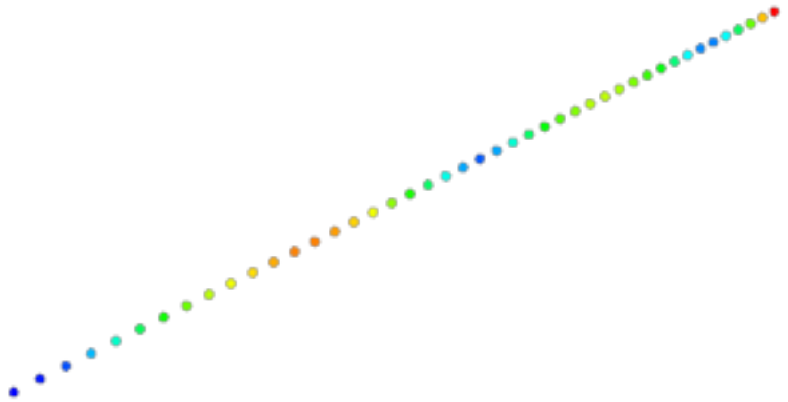


Figure 42: Calculix — mode 10, 802.316 Hz

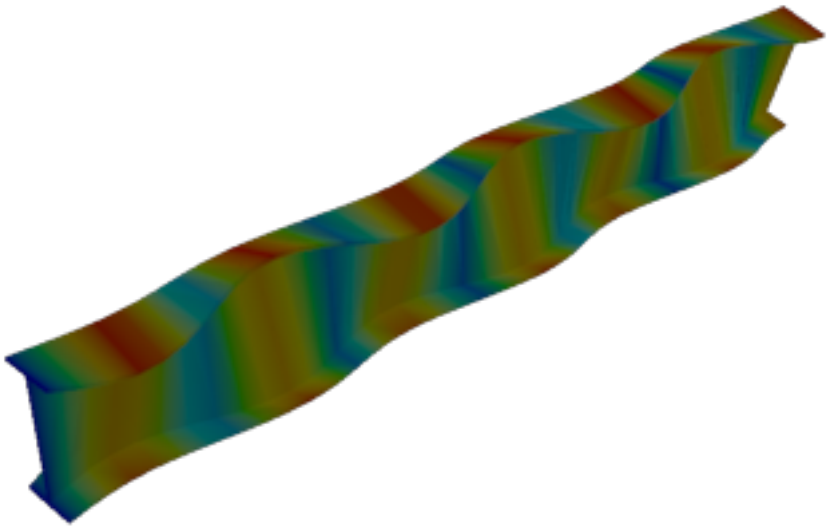


Figure 43: Code_Aster — mode 10, 160.681 Hz

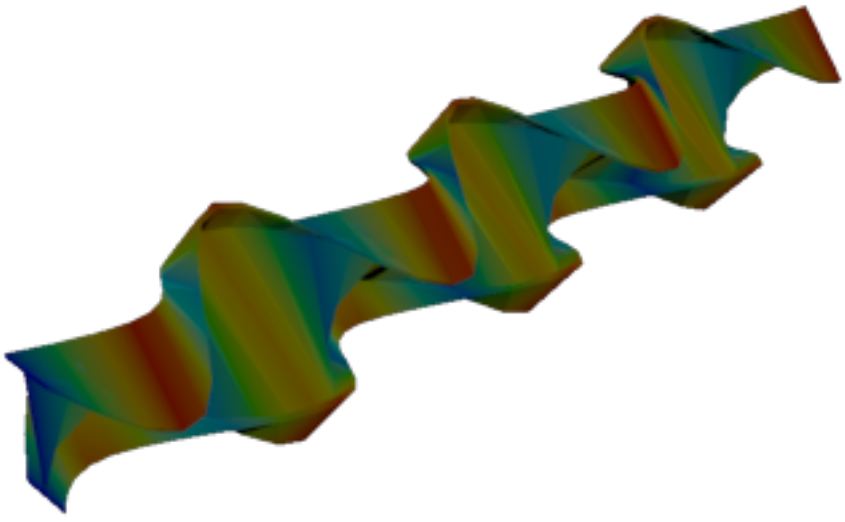


Figure 44: Sesam — mode 10, 160.681 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	138.900	-55.99	761704	1.00e-20	2.09e-22	2.61e-21	1.00e-10	1.45e-11	-5.11e-11
Calculix	802.316	+154.18	2.54127e+07	3.17e-30	9.74e-30	3.78e-28	-1.78e-15	3.12e-15	1.94e-14
Code_Aster	160.681	-49.09	1.01927e+06	2.47e-34	1.70e-27	7.63e-30	-1.57e-17	4.12e-14	2.76e-15
Sesam	160.681	-49.09	1.01927e+06	1.48e-24	9.00e-28	2.96e-25	-1.22e-12	-3.00e-14	-5.44e-13

Mode 11

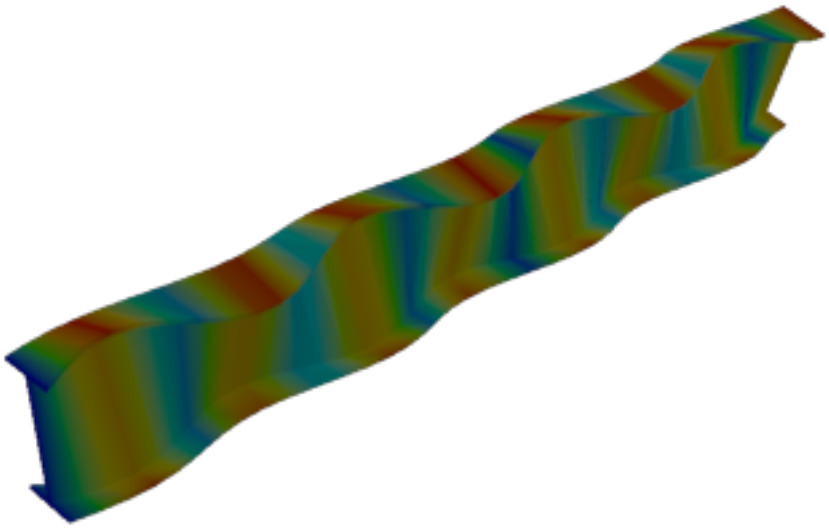


Figure 45: Abaqus — mode 11, 159.773 Hz

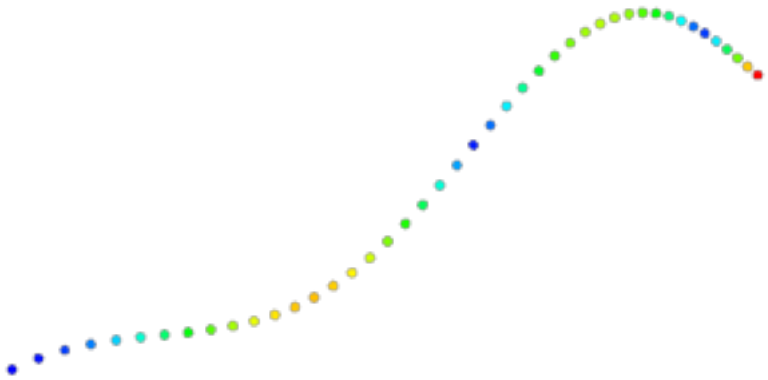


Figure 46: Calculix — mode 11, 838.750 Hz

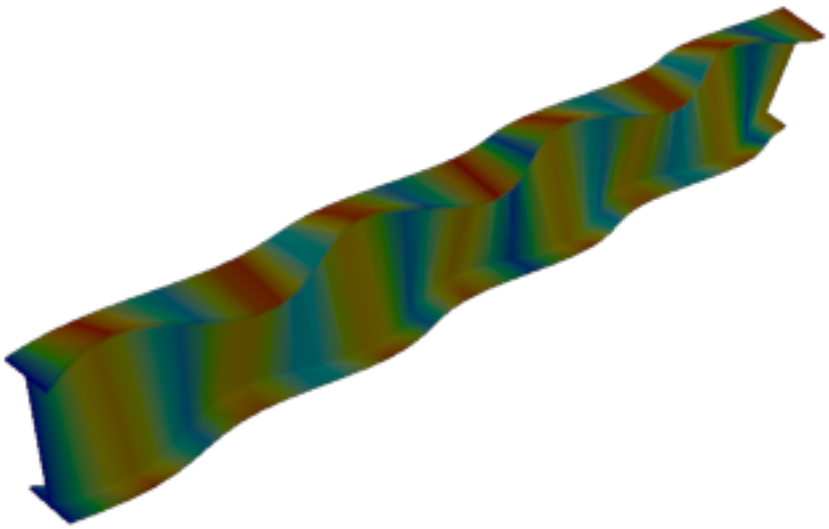


Figure 47: Code_Aster — mode 11, 185.976 Hz

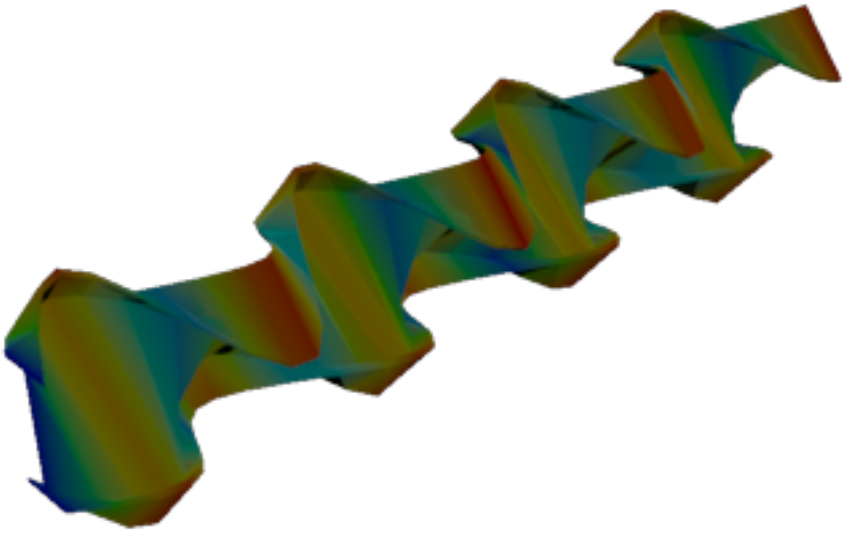


Figure 48: Sesam — mode 11, 185.976 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	159.770	-53.37	1.00778e+06	1.05e-18	1.96e-22	2.87e-21	1.02e-09	-1.40e-11	5.35e-11
Calculix	838.750	+144.81	2.77731e+07	1.58e-31	5.81e-29	11.38	3.98e-16	7.62e-15	-3.374
Code_Aster	185.976	-45.72	1.36544e+06	1.30e-35	5.33e-28	6.00e-29	3.61e-18	-2.31e-14	7.75e-15
Sesam	185.976	-45.72	1.36544e+06	4.16e-15	1.83e-18	8.10e-16	-6.45e-08	-1.35e-09	-2.85e-08

Line, 1st order, reduced integration

Mode 1

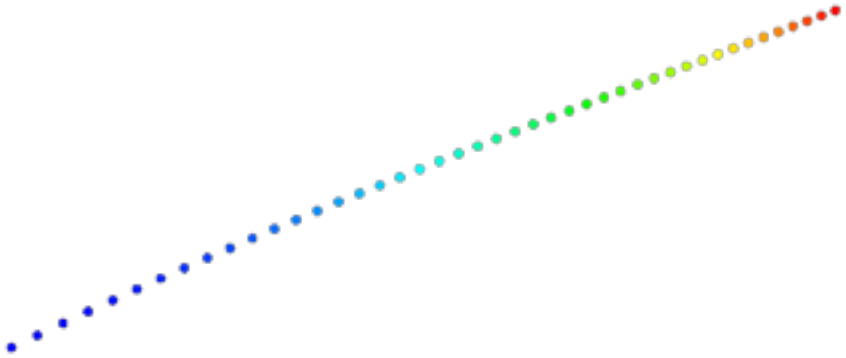


Figure 49: Calculix — mode 1, 12.973 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	12.973	6644.56	1.74e-37	116.40	9.37e-33	4.17e-19	-10.789	9.68e-17

Mode 2

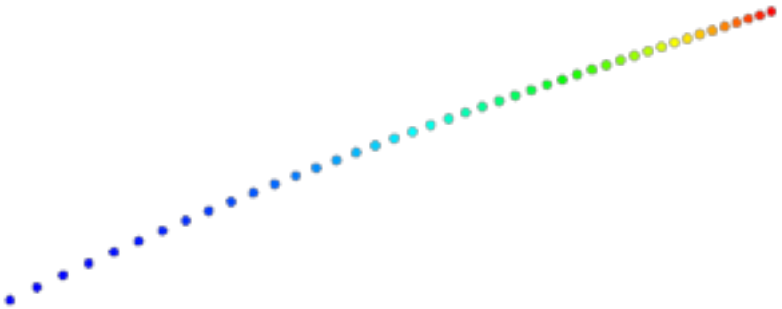


Figure 50: Calculix — mode 2, 52.587 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	52.587	109172	2.95e-34	6.28e-31	114.82	1.72e-17	-7.92e-16	-10.716

Mode 3

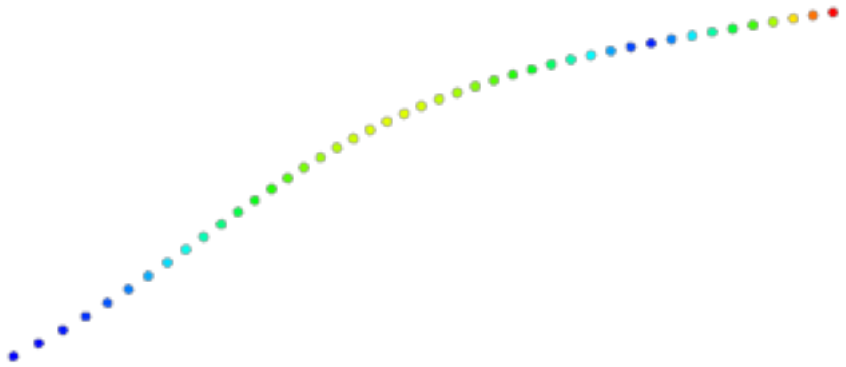


Figure 51: Calculix — mode 3, 81.151 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	81.151	259985	4.06e-33	35.74	4.38e-30	-6.37e-17	5.979	-2.09e-15

Mode 4

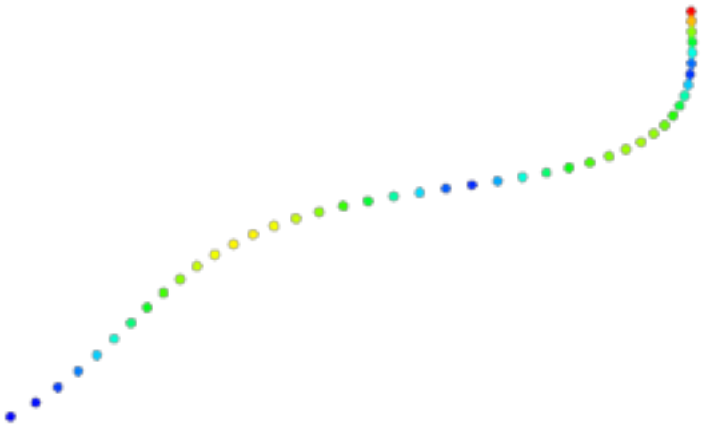


Figure 52: Calculix — mode 4, 226.646 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	226.646	2.02794e+06	1.30e-30	12.25	2.11e-29	-1.14e-15	3.500	-4.59e-15

Mode 5

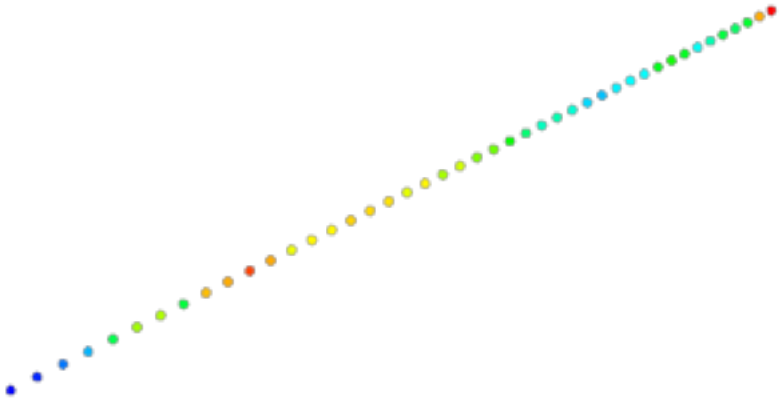


Figure 53: Calculix — mode 5, 267.320 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	267.320	2.82112e+06	3.28e-30	4.98e-30	2.38e-29	1.81e-15	-2.23e-15	4.88e-15

Mode 6

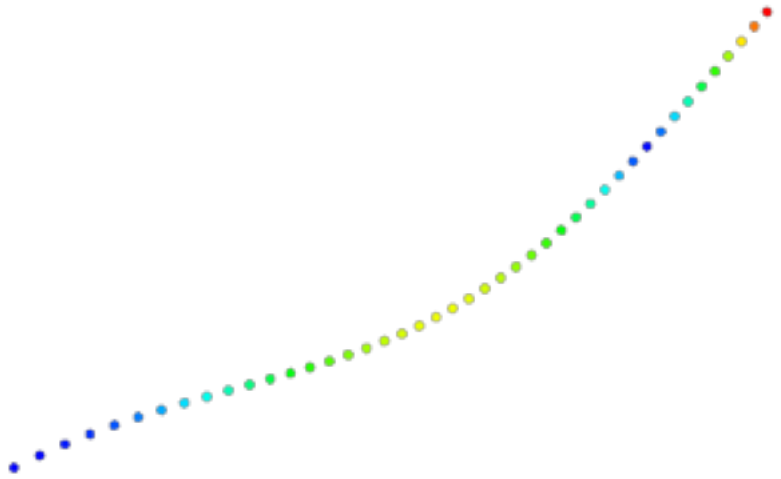


Figure 54: Calculix — mode 6, 316.915 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	316.915	3.96503e+06	1.31e-33	7.09e-30	34.97	-3.62e-17	-2.66e-15	-5.913

Mode 7

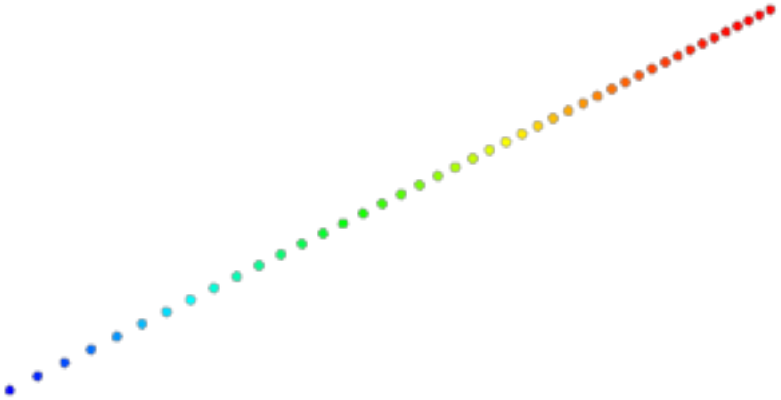


Figure 55: Calculix — mode 7, 431.040 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	431.040	7.33492e+06	153.94	1.29e-27	4.83e-31	12.407	-3.60e-14	-6.95e-16

Mode 8

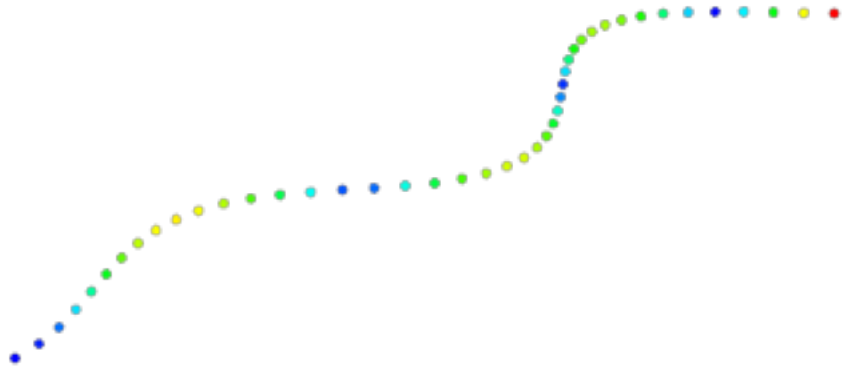


Figure 56: Calculix — mode 8, 442.610 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	442.610	7.73397e+06	3.15e-26	6.24	6.00e-30	1.77e-13	2.498	2.45e-15

Mode 9

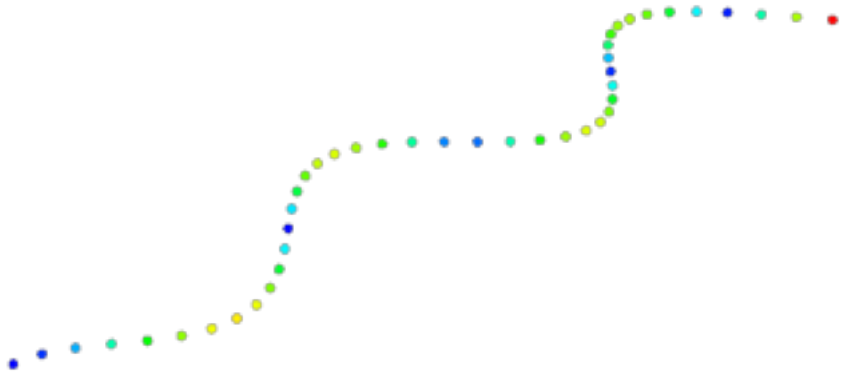


Figure 57: Calculix — mode 9, 728.596 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	728.596	2.09572e+07	1.49e-29	3.75	1.24e-28	3.85e-15	-1.937	-1.11e-14

Mode 10

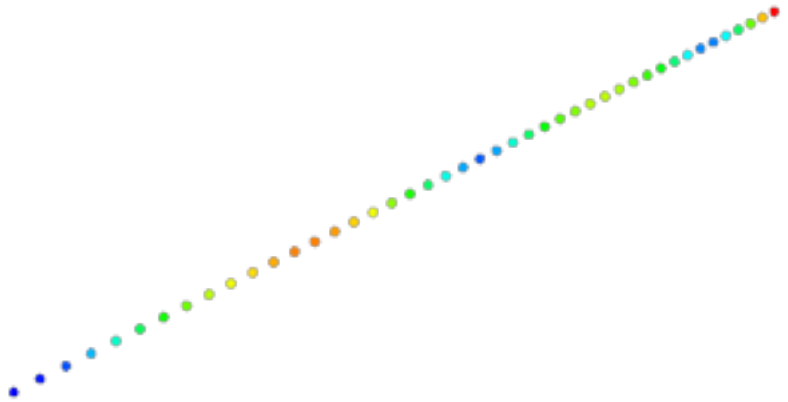


Figure 58: Calculix — mode 10, 802.316 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	802.316	2.54127e+07	3.17e-30	9.74e-30	3.78e-28	-1.78e-15	3.12e-15	1.94e-14

Mode 11

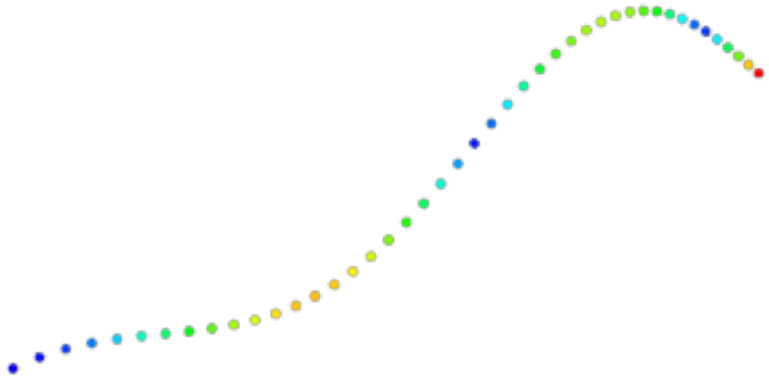


Figure 59: Calculix — mode 11, 838.750 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	838.750	2.77731e+07	1.58e-31	5.81e-29	11.38	3.98e-16	7.62e-15	-3.374

Line, 2nd order

Mode 1

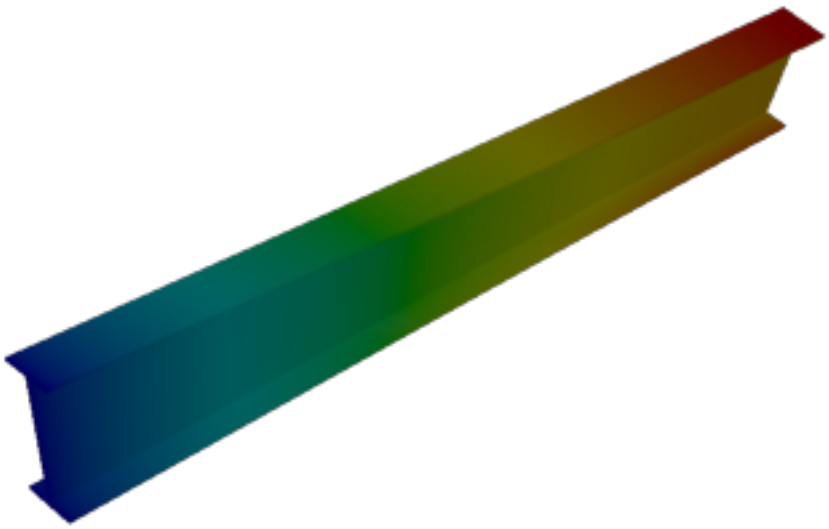


Figure 60: Abaqus — mode 1, 10.786 Hz

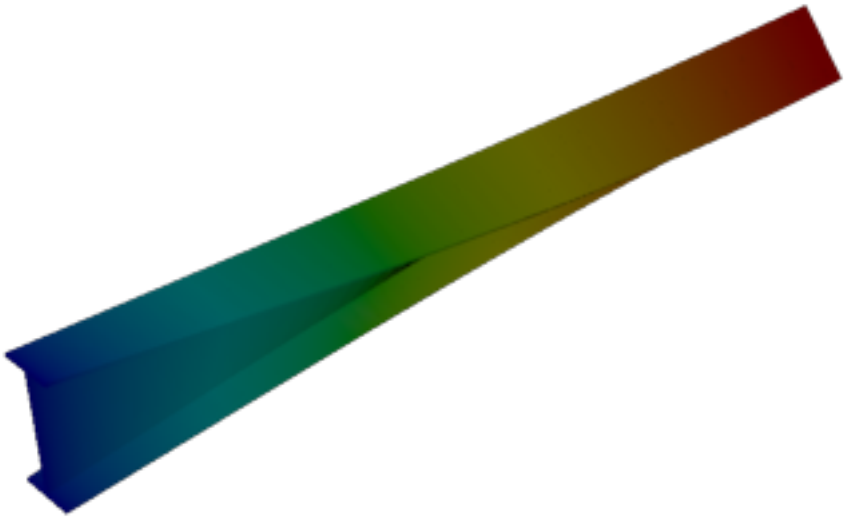


Figure 61: Sesam — mode 1, 10.039 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	10.786	+3.59	4592.8	3.02e-31	1.31e-30	3.09e-30	5.50e-16	1.14e-15	-1.76e-15
Sesam	10.039	-3.59	3978.68	6.58e-40	1.55e-27	2.99e-29	2.57e-20	3.94e-14	-5.47e-15

Mode 2

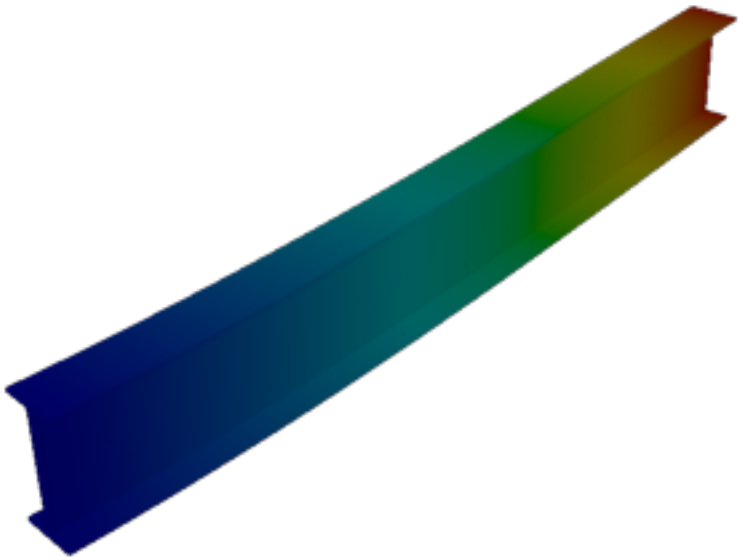


Figure 62: Abaqus — mode 2, 12.932 Hz

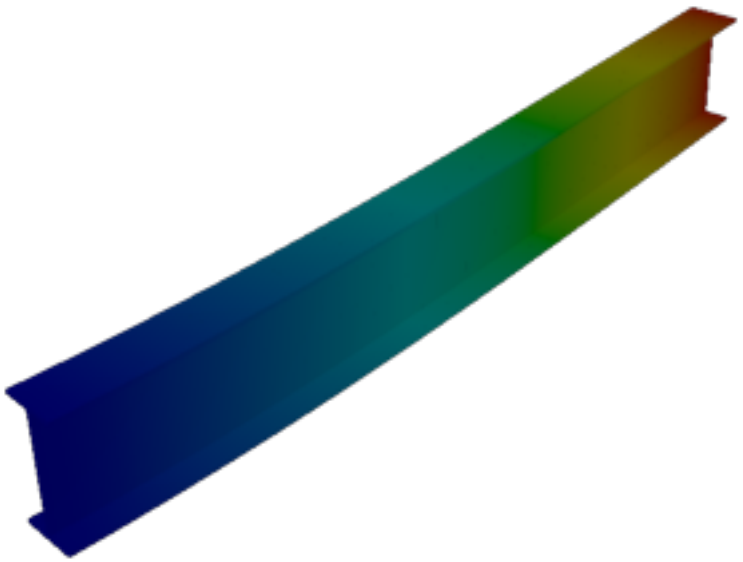


Figure 63: Sesam — mode 2, 10.614 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.932	+9.84	6602.4	1.50e-32	116.59	2.60e-32	-1.22e-16	10.798	-1.61e-16
Sesam	10.614	-9.84	4447.85	3.31e-28	176.82	4.08e-30	-1.82e-14	13.297	2.02e-15

Mode 3

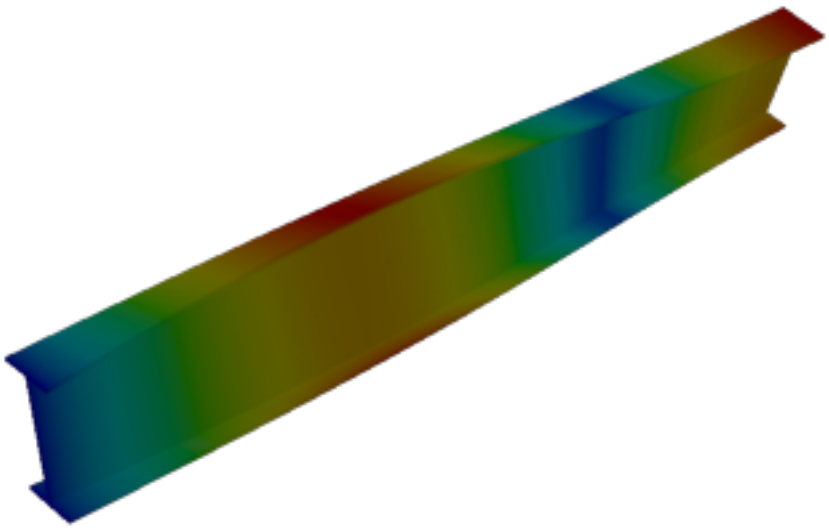


Figure 64: Abaqus — mode 3, 32.358 Hz

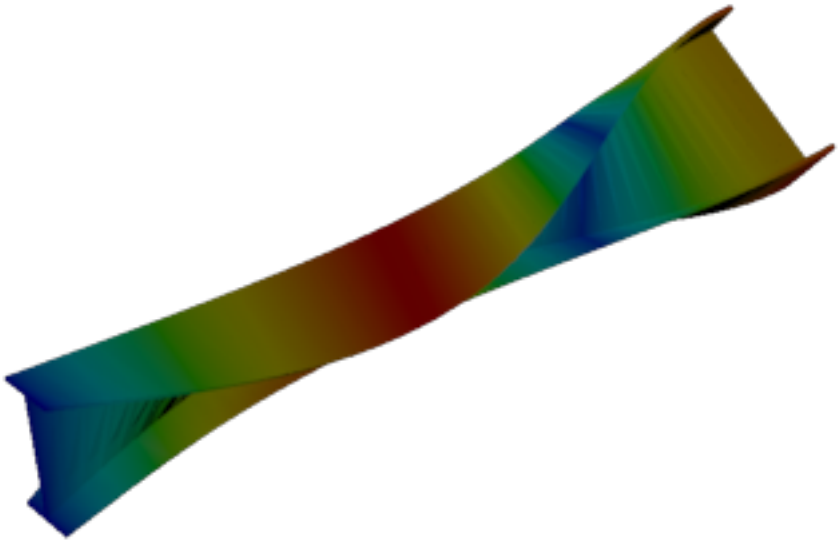


Figure 65: Sesam — mode 3, 30.108 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	32.358	+3.60	41335	2.64e-31	5.69e-29	8.91e-29	5.14e-16	-7.54e-15	-9.44e-15
Sesam	30.108	-3.60	35786.3	8.89e-37	8.24e-31	2.35e-28	-9.43e-19	-9.08e-16	-1.53e-14

Mode 4

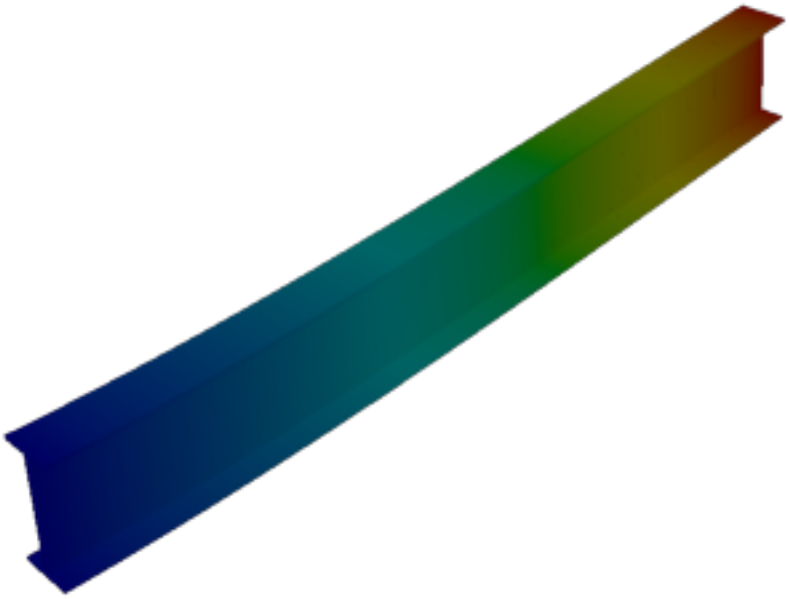


Figure 66: Abaqus — mode 4, 50.557 Hz

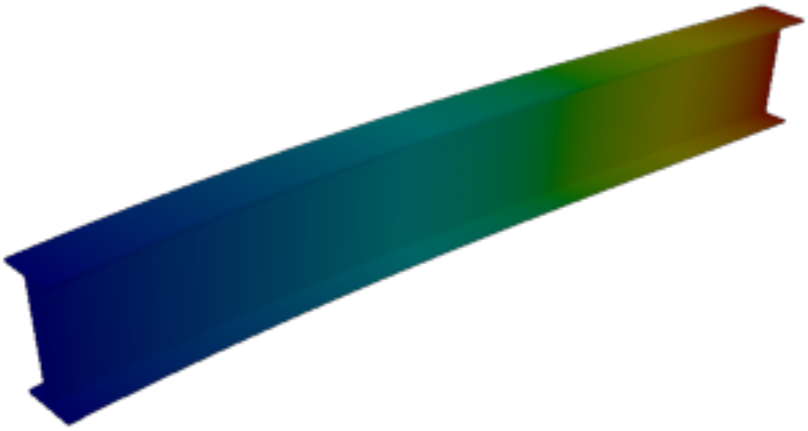


Figure 67: Sesam — mode 4, 41.236 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.557	+10.15	100907	6.29e-36	9.63e-28	118.18	-2.51e-18	-3.10e-14	10.871
Sesam	41.236	-10.15	67128.9	3.40e-28	1.54e-29	180.03	1.84e-14	-3.93e-15	-13.418

Mode 5

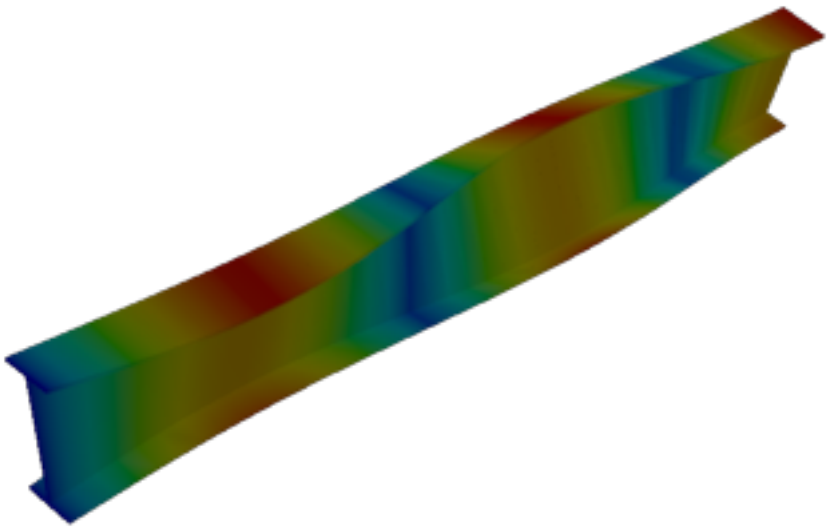


Figure 68: Abaqus — mode 5, 53.930 Hz

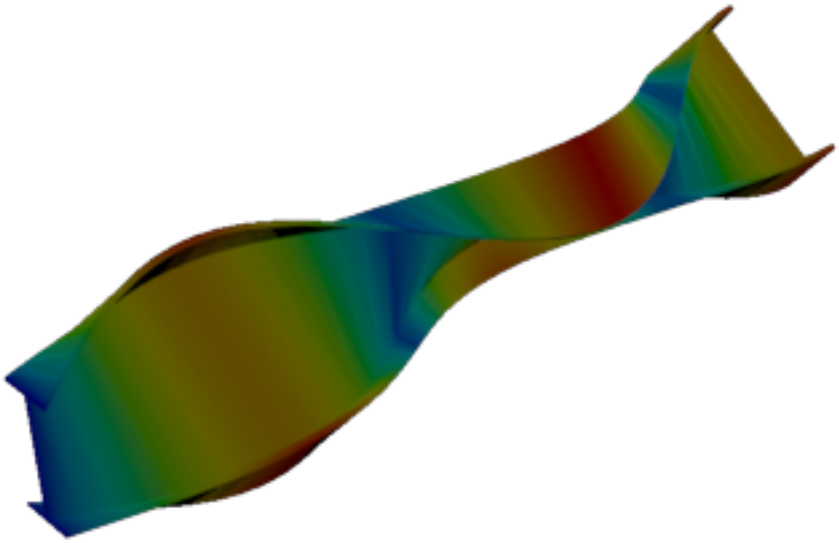


Figure 69: Sesam — mode 5, 50.149 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	53.930	+3.63	114820	1.13e-29	6.97e-31	2.94e-28	-3.36e-15	8.35e-16	1.72e-14
Sesam	50.149	-3.63	99284.4	9.63e-36	8.79e-31	5.87e-31	3.10e-18	9.37e-16	-7.66e-16

Mode 6

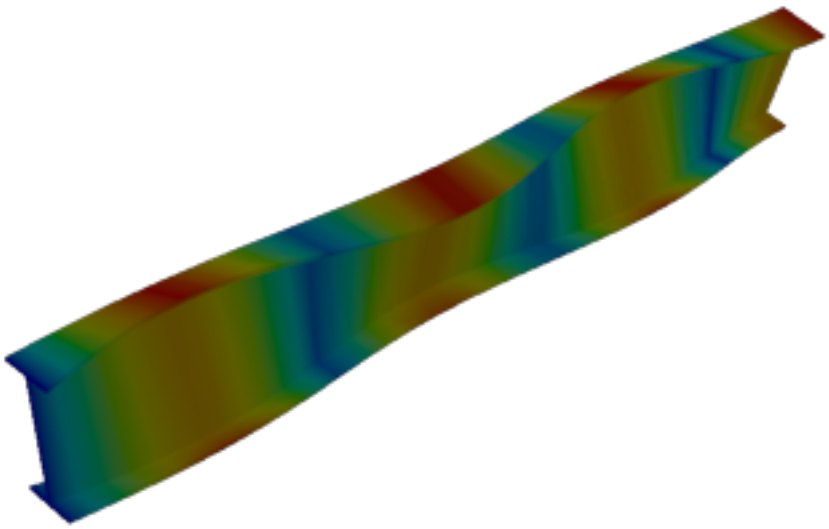


Figure 70: Abaqus — mode 6, 75.502 Hz

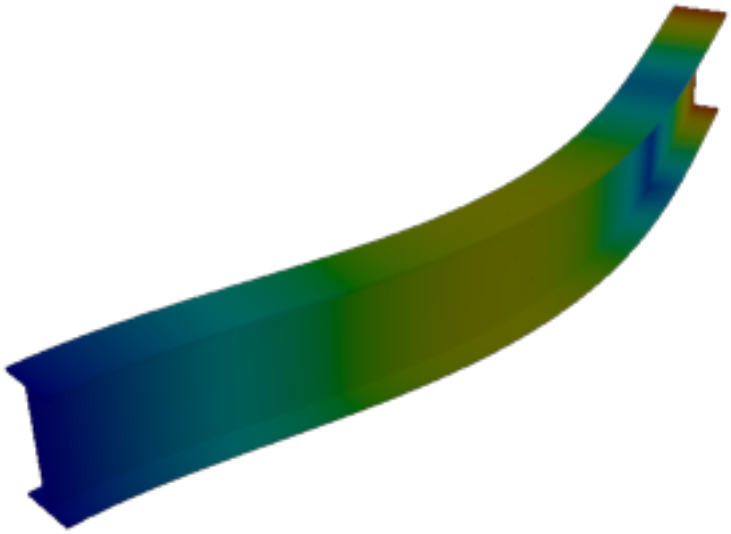


Figure 71: Sesam — mode 6, 65.327 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.502	+7.23	225048	6.44e-34	3.07e-27	1.85e-29	-2.54e-17	5.54e-14	-4.30e-15
Sesam	65.327	-7.23	168476	1.08e-28	54.95	9.03e-29	1.04e-14	-7.413	9.50e-15

Mode 7

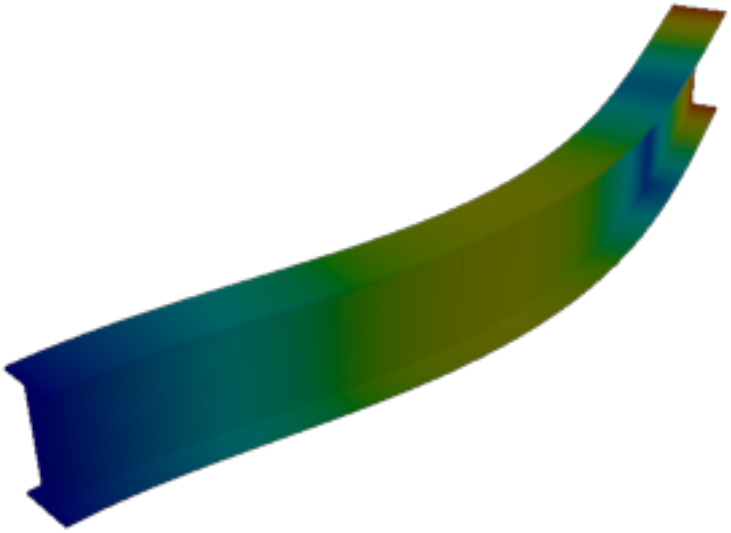


Figure 72: Abaqus — mode 7, 79.684 Hz

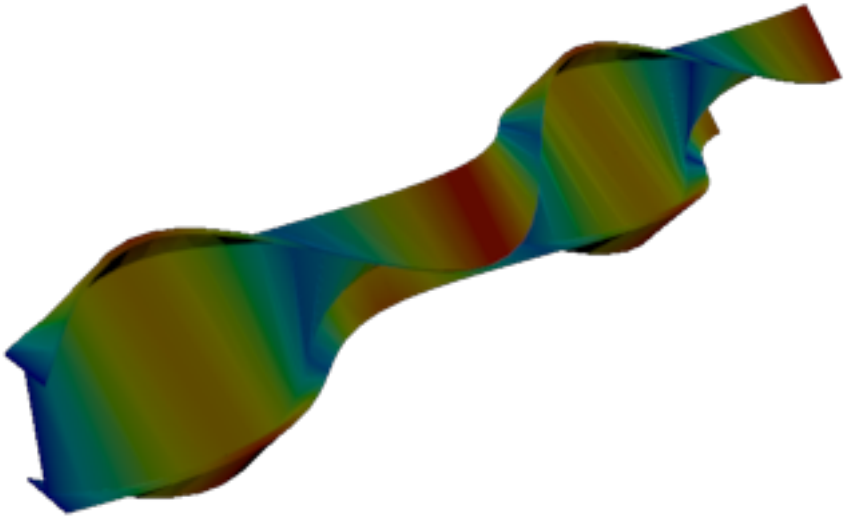


Figure 73: Sesam — mode 7, 70.143 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	79.684	+6.37	250668	9.57e-31	36.19	1.97e-27	9.78e-16	-6.015	-4.44e-14
Sesam	70.143	-6.37	194235	8.79e-34	2.31e-28	4.09e-29	-2.97e-17	-1.52e-14	-6.40e-15

Mode 8

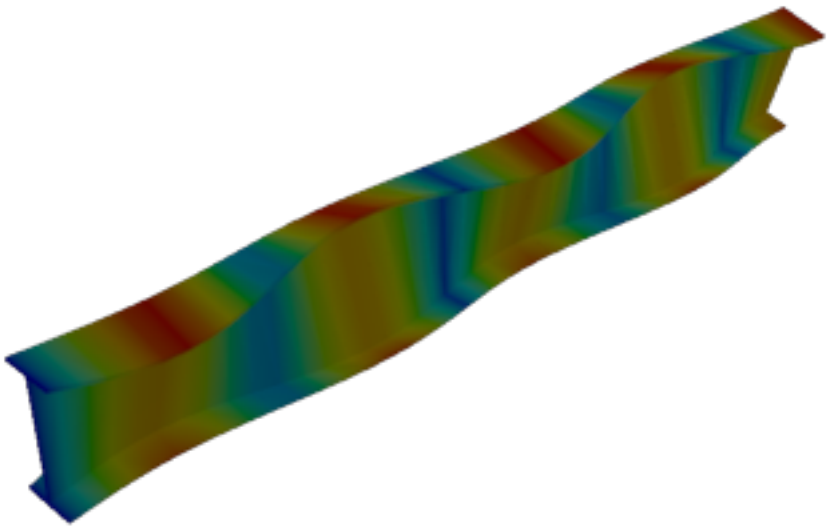


Figure 74: Abaqus — mode 8, 97.074 Hz

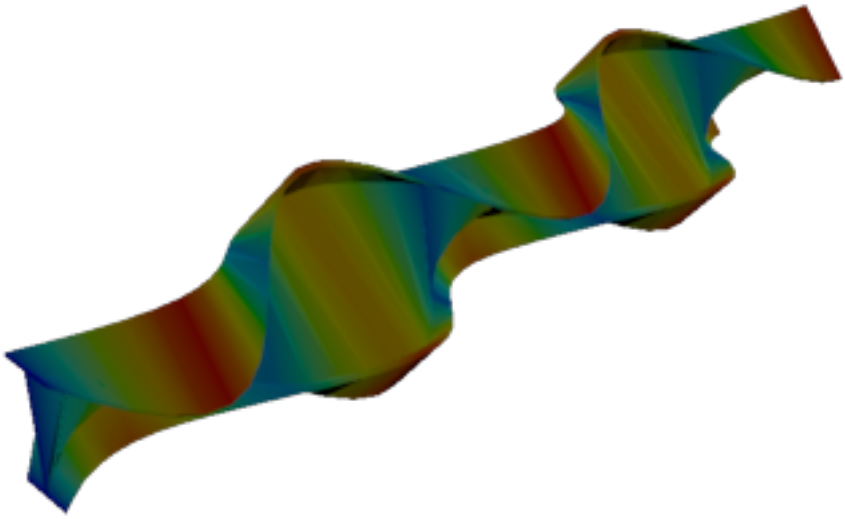


Figure 75: Sesam — mode 8, 90.070 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.074	+3.74	372022	4.46e-28	1.86e-30	1.09e-27	2.11e-14	1.36e-15	-3.31e-14
Sesam	90.070	-3.74	320270	4.07e-34	1.53e-30	2.45e-29	-2.02e-17	-1.24e-15	-4.95e-15

Mode 9

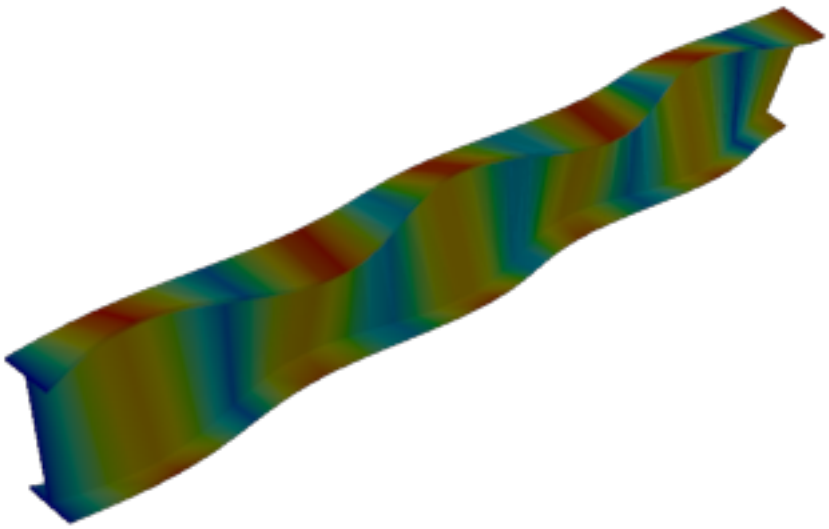


Figure 76: Abaqus — mode 9, 118.648 Hz

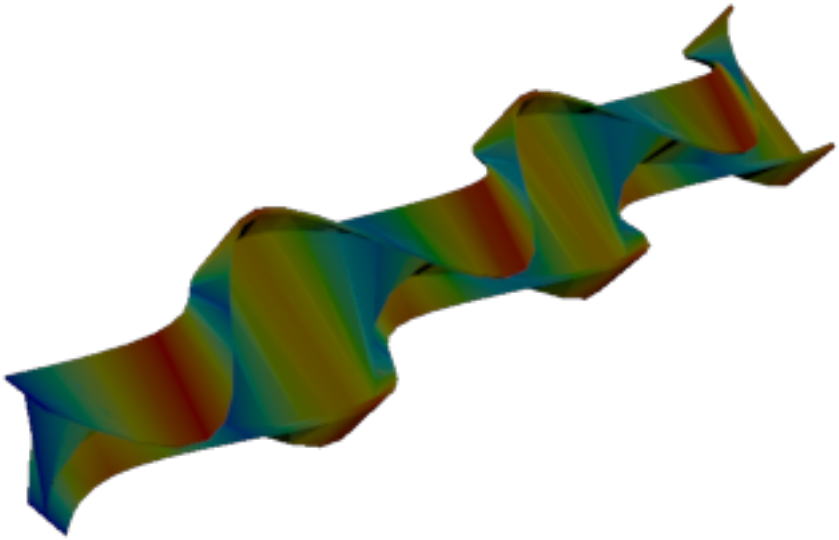


Figure 77: Sesam — mode 9, 109.907 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	118.650	+3.83	555748	3.94e-25	6.15e-26	2.26e-25	6.28e-13	2.48e-13	-4.75e-13
Sesam	109.907	-3.83	476881	1.08e-30	1.18e-29	9.75e-30	1.04e-15	3.44e-15	3.12e-15

Mode 10

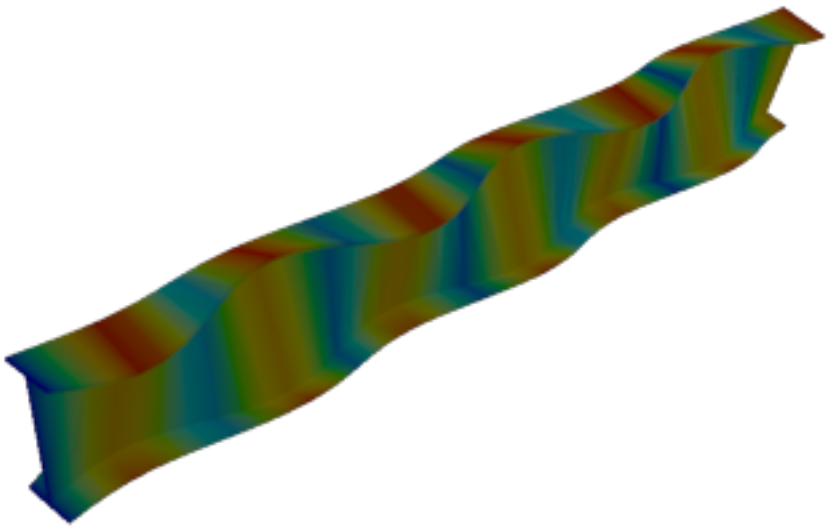


Figure 78: Abaqus — mode 10, 140.222 Hz

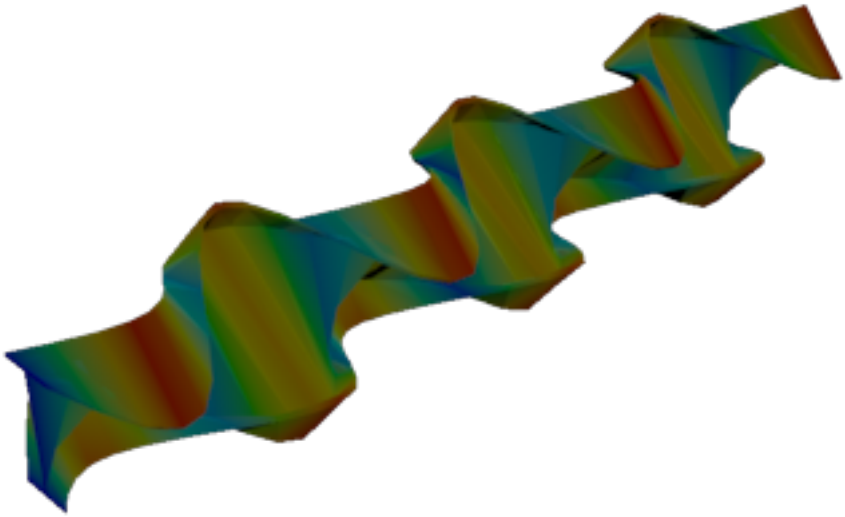


Figure 79: Sesam — mode 10, 129.631 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	140.220	+3.92	776236	2.58e-24	4.27e-23	1.99e-23	-1.61e-12	6.54e-12	4.46e-12
Sesam	129.631	-3.92	663407	4.27e-23	7.83e-25	9.59e-23	6.53e-12	8.85e-13	-9.79e-12

Mode 11

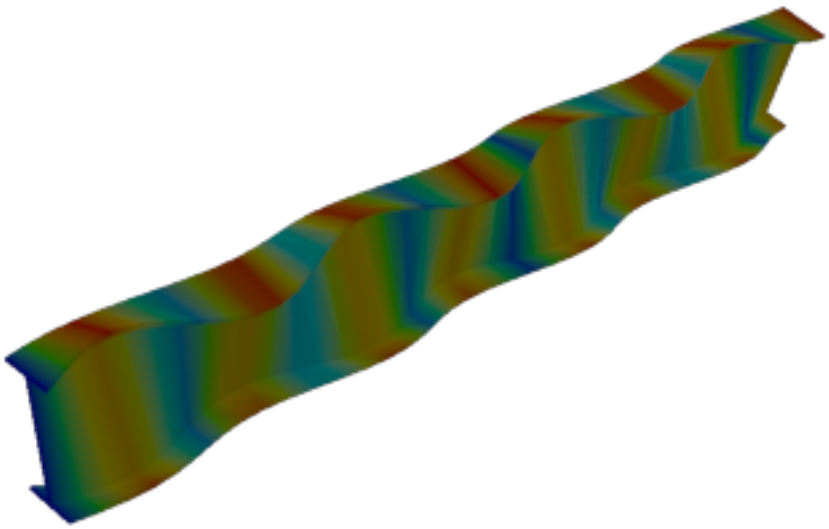


Figure 80: Abaqus — mode 11, 161.799 Hz

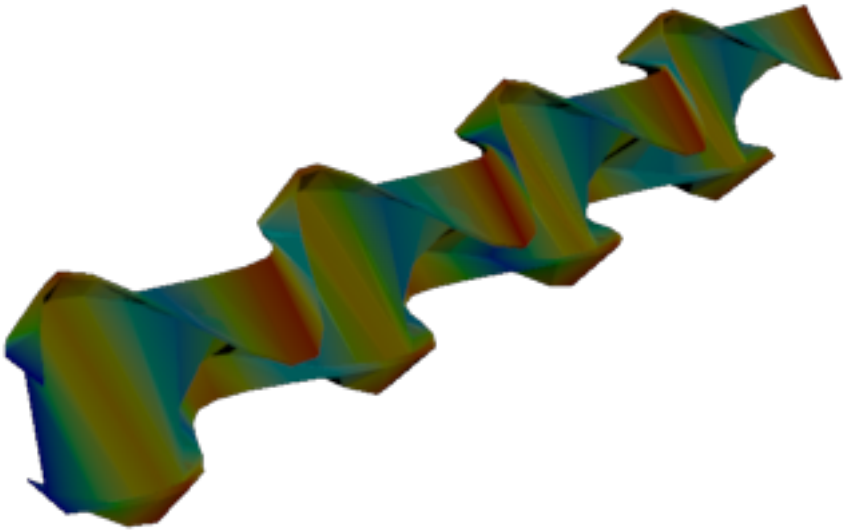


Figure 81: Sesam — mode 11, 149.217 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	161.800	+4.05	1.03351e+06	2.30e-19	1.18e-19	3.67e-20	-4.80e-10	-3.43e-10	1.92e-10
Sesam	149.217	-4.05	879011	9.89e-16	1.84e-17	2.21e-15	3.14e-08	4.28e-09	-4.70e-08

Shell, 1st order, TRI

Mode 1

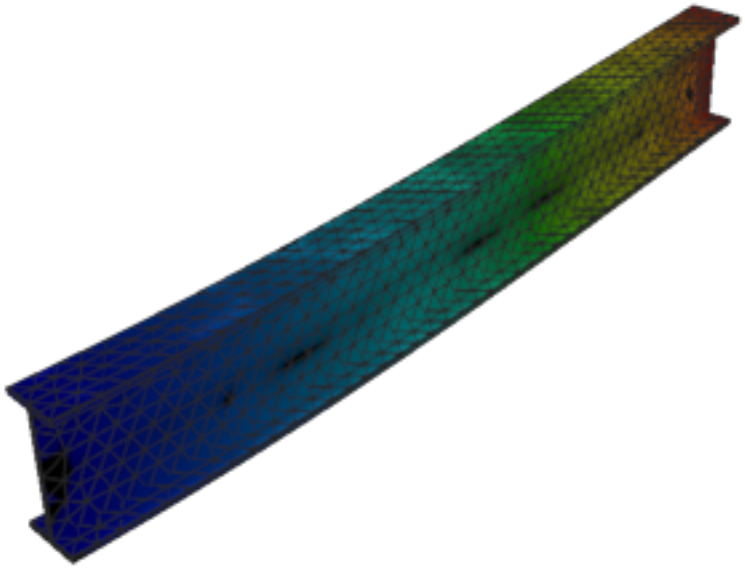


Figure 82: Calculix — mode 1, 13.725 Hz

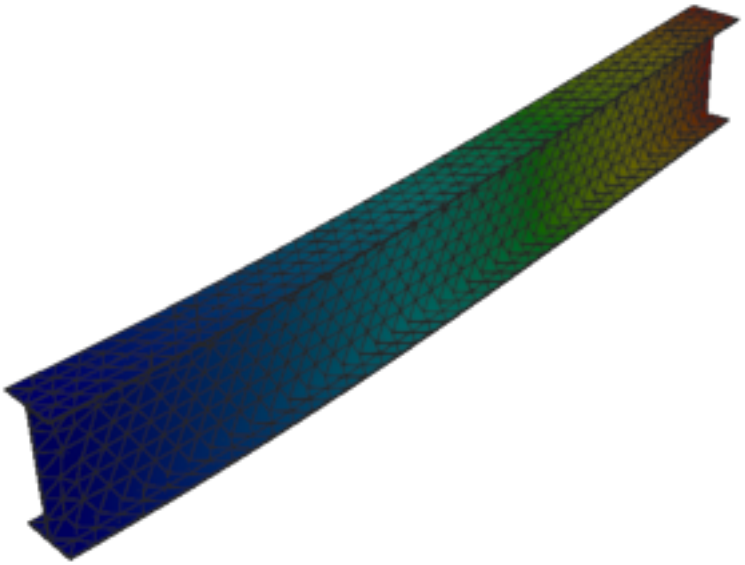


Figure 83: Code_Aster — mode 1, 13.642 Hz

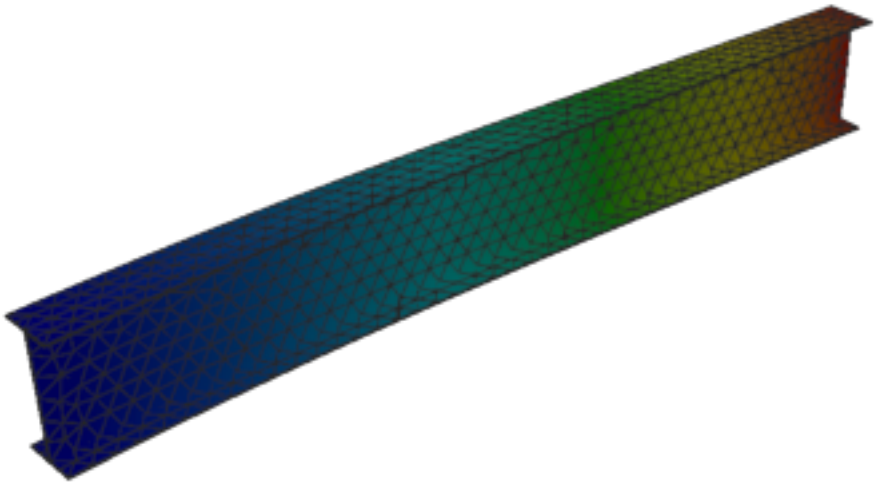


Figure 84: Sesam — mode 1, 13.641 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	13.725	+0.41	7437.24	9.89e-19	119.69	1.11e-11	9.95e-10	10.941	3.34e-06
Code_Aster	13.642	-0.20	7347.12	1.48e-27	119.93	1.17e-25	-3.85e-14	10.951	3.42e-13
Sesam	13.641	-0.21	7346.4	2.31e-27	119.93	9.89e-25	-4.81e-14	-10.951	-9.94e-13

Mode 2

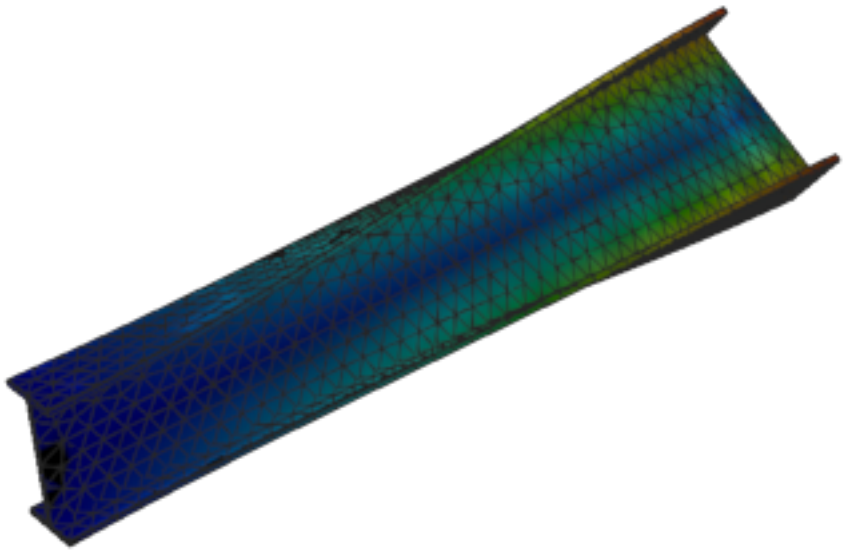


Figure 85: Calculix — mode 2, 36.690 Hz

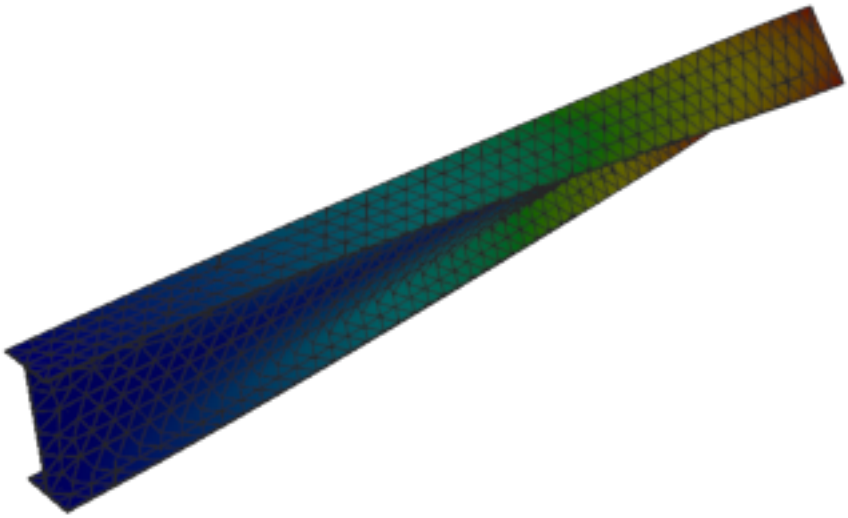


Figure 86: Code_Aster — mode 2, 20.418 Hz

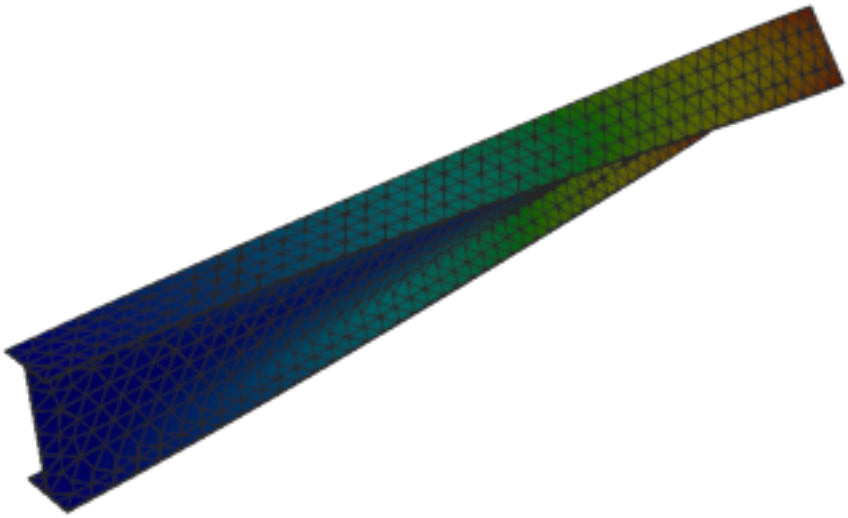


Figure 87: Sesam — mode 2, 20.198 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	36.690	+42.38	53142.9	2.89e-08	4.24e-06	5.81e-14	-1.70e-04	0.002	-2.41e-07
Code_Aster	20.418	-20.76	16458.5	4.91e-27	1.63e-08	7.48e-26	7.01e-14	-1.28e-04	2.73e-13
Sesam	20.198	-21.62	16106.1	2.48e-26	3.02e-11	1.06e-25	1.57e-13	-5.49e-06	3.25e-13

Mode 3

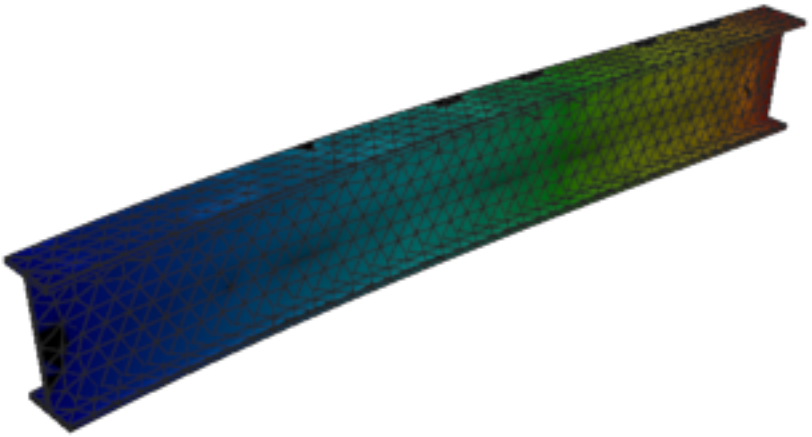


Figure 88: Calculix — mode 3, 52.249 Hz

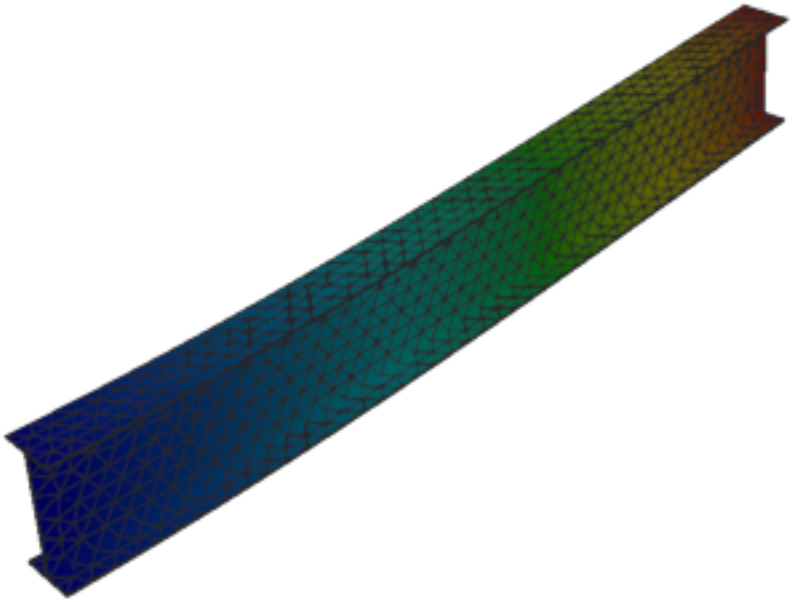


Figure 89: Code_Aster — mode 3, 52.193 Hz

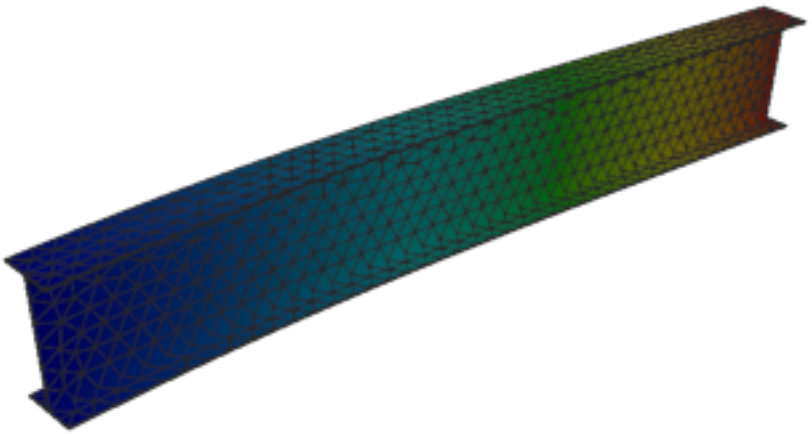


Figure 90: Sesam — mode 3, 52.192 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	52.249	+0.07	107772	1.80e-10	4.08e-10	122.06	-1.34e-05	2.02e-05	-11.048
Code_Aster	52.193	-0.04	107542	1.54e-10	3.07e-25	122.03	1.24e-05	-5.54e-13	11.047
Sesam	52.192	-0.04	107540	4.79e-10	1.53e-24	122.03	-2.19e-05	1.24e-12	-11.047

Mode 4

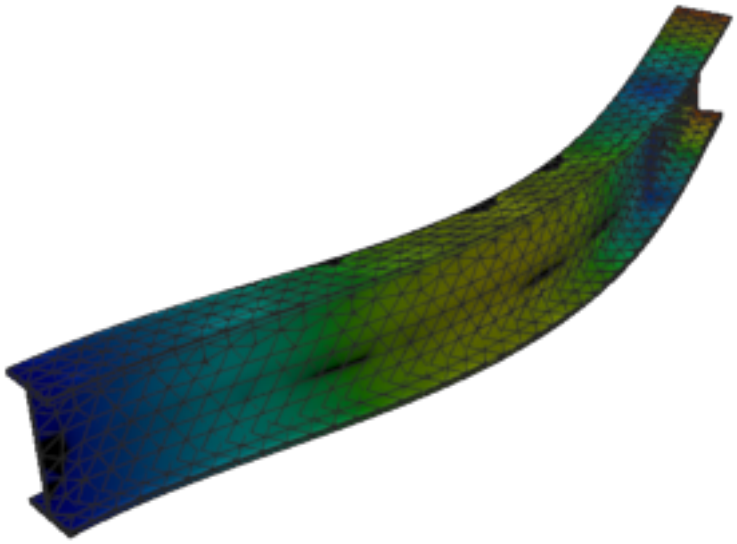


Figure 91: Calculix — mode 4, 83.709 Hz

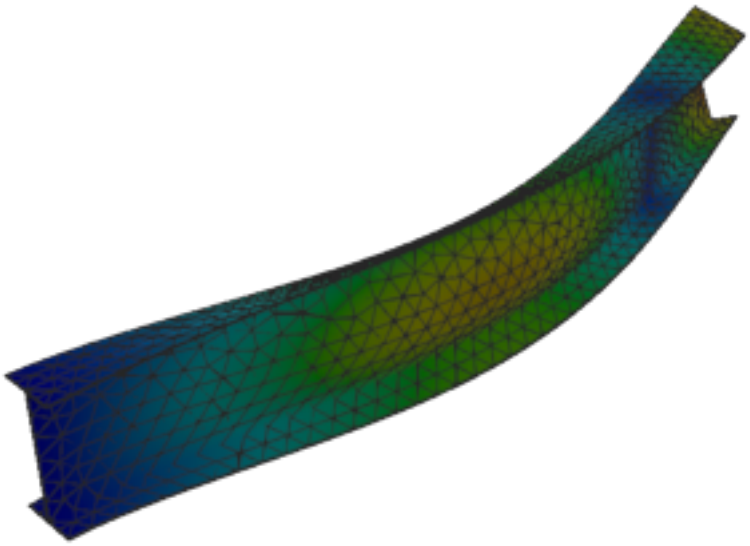


Figure 92: Code_Aster — mode 4, 79.660 Hz

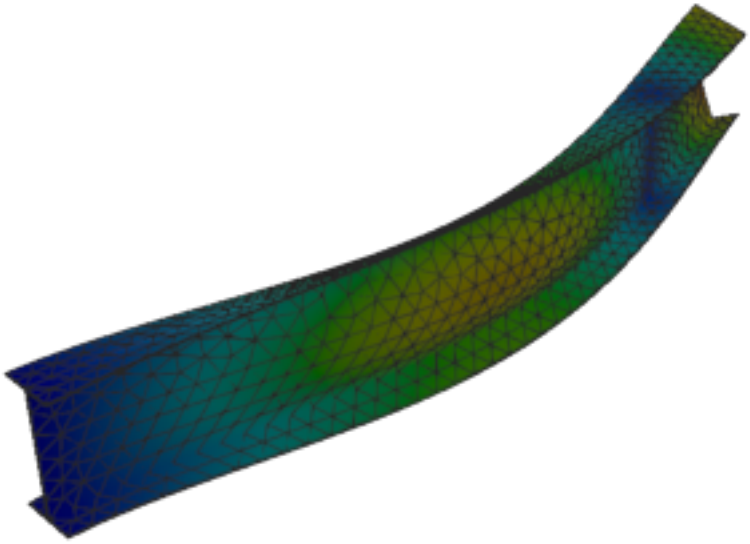


Figure 93: Sesam — mode 4, 79.523 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	83.709	+3.39	276635	1.88e-12	37.26	1.12e-11	-1.37e-06	-6.104	3.35e-06
Code_Aster	79.660	-1.61	250516	1.53e-26	36.64	1.82e-25	1.24e-13	-6.053	-4.26e-13
Sesam	79.523	-1.78	249655	1.60e-28	36.58	1.25e-25	1.26e-14	-6.048	-3.54e-13

Mode 5

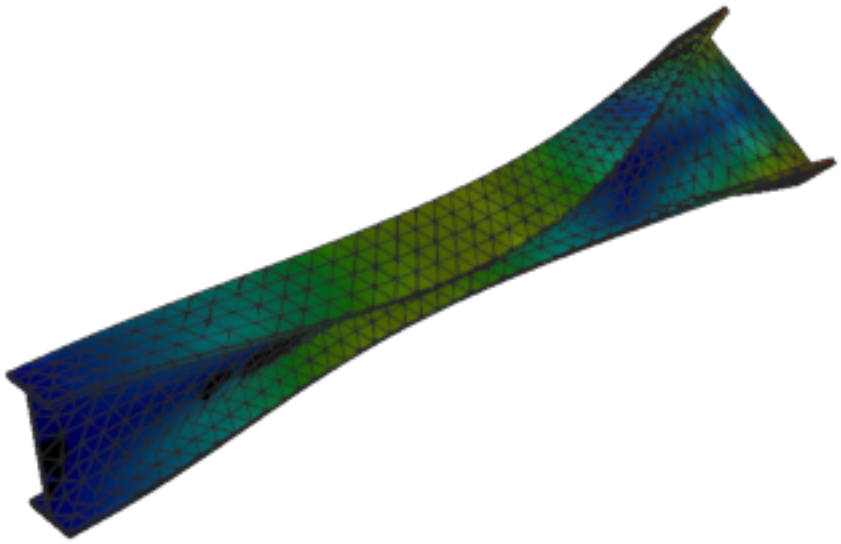


Figure 94: Calculix — mode 5, 132.927 Hz

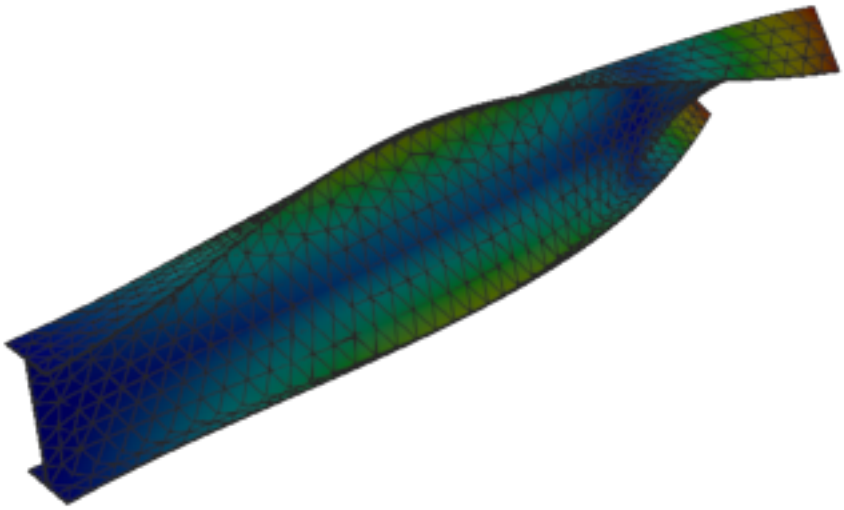


Figure 95: Code_Aster — mode 5, 102.010 Hz

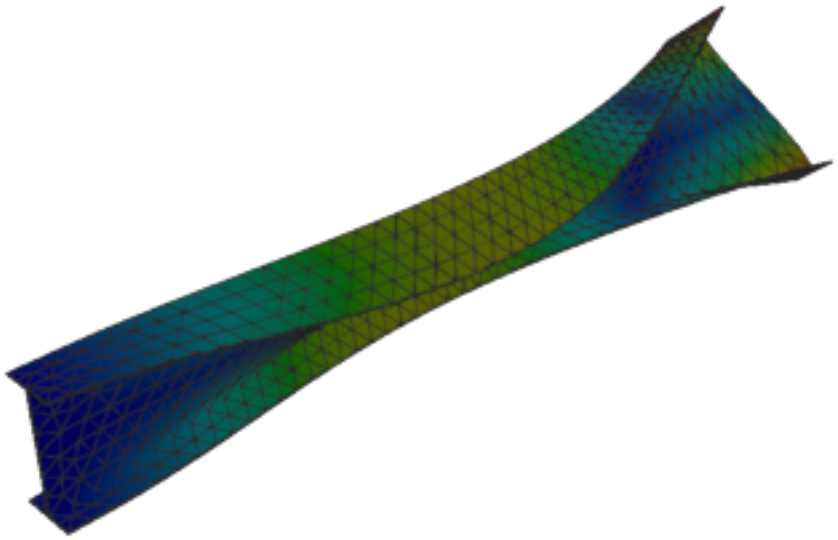


Figure 96: Sesam — mode 5, 101.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	132.928	+18.44	697572	4.77e-09	2.89e-05	2.99e-14	6.91e-05	-0.005	-1.73e-07
Code_Aster	102.010	-9.11	410817	1.16e-26	3.22e-09	2.68e-25	1.08e-13	-5.68e-05	-5.18e-13
Sesam	101.756	-9.33	408770	4.49e-27	4.27e-09	6.32e-27	-6.70e-14	6.54e-05	-7.95e-14

Mode 6

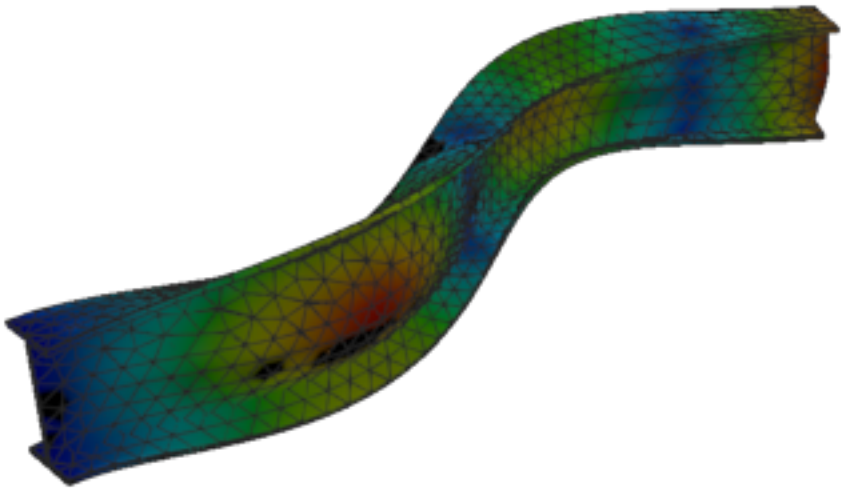


Figure 97: Calculix — mode 6, 218.167 Hz

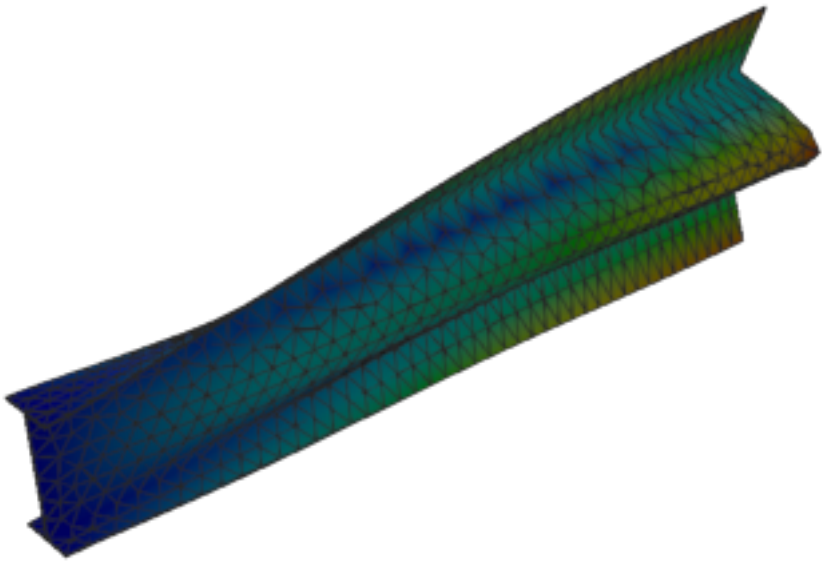


Figure 98: Code_Aster — mode 6, 125.459 Hz

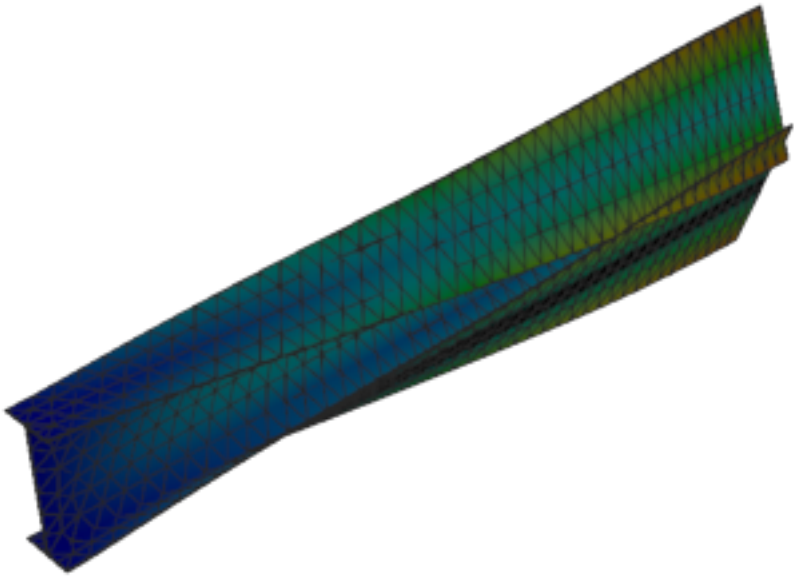


Figure 99: Sesam — mode 6, 125.093 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	218.167	+39.64	1.87905e+06	1.01e-11	12.68	1.01e-11	-3.18e-06	-3.561	-3.17e-06
Code_Aster	125.459	-19.70	621390	5.78e-28	0.07	7.34e-25	-2.40e-14	0.263	-8.57e-13
Sesam	125.093	-19.94	617772	7.12e-28	0.06	1.10e-26	-2.67e-14	-0.252	1.05e-13

Mode 7

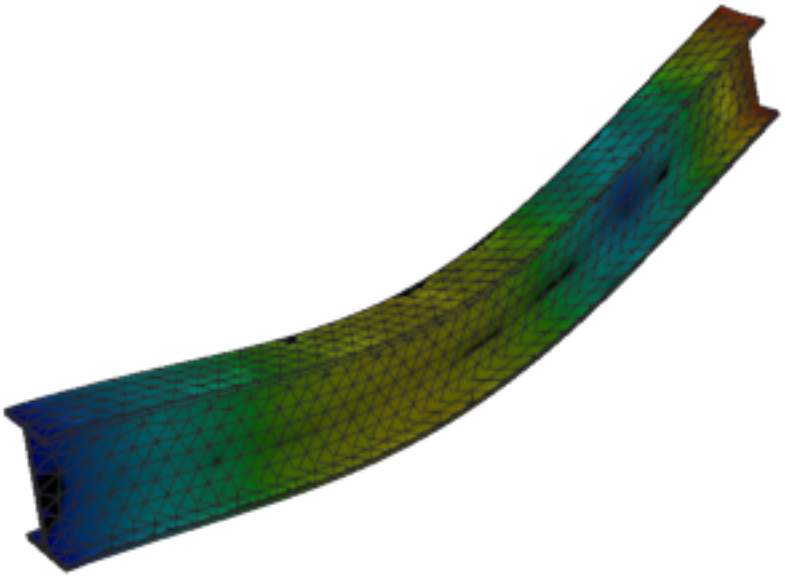


Figure 100: Calculix — mode 7, 259.245 Hz

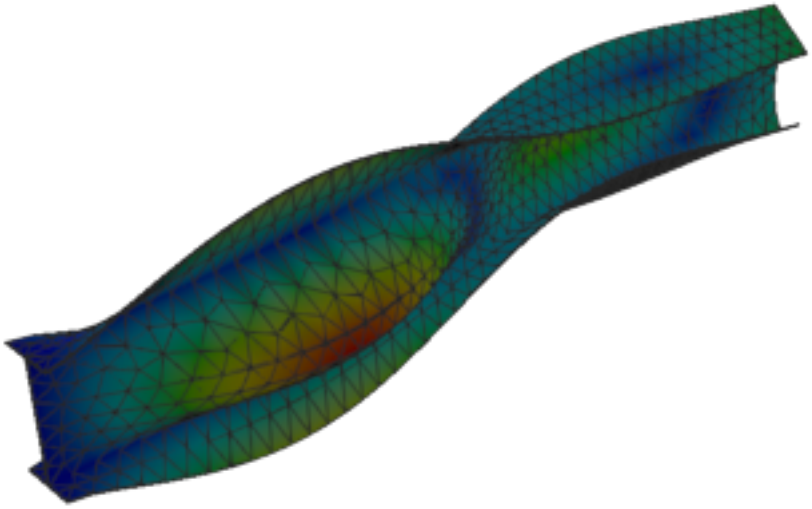


Figure 101: Code_Aster — mode 7, 144.895 Hz

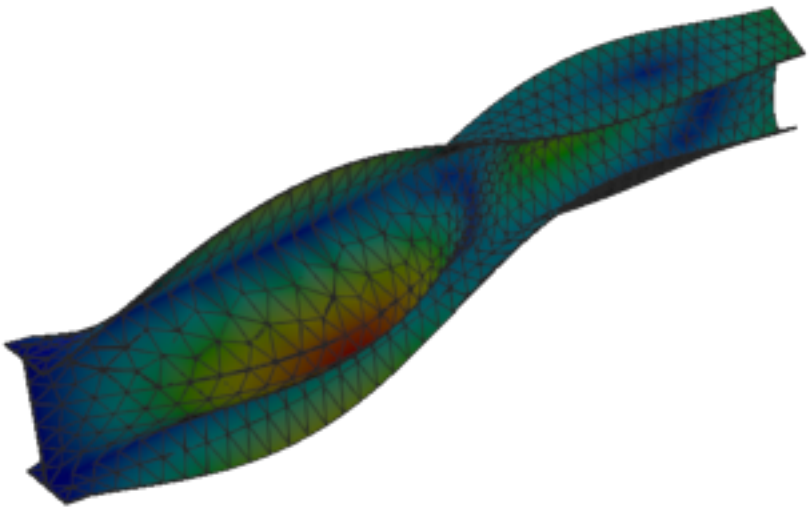


Figure 102: Sesam — mode 7, 143.164 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	259.245	+42.10	2.65327e+06	6.03e-11	3.23e-10	41.55	-7.77e-06	-1.80e-05	-6.446
Code_Aster	144.895	-20.58	828828	1.71e-27	2.73	9.34e-26	-4.13e-14	-1.652	3.06e-13
Sesam	143.164	-21.53	809150	7.95e-30	2.72	2.49e-27	-2.82e-15	-1.651	4.99e-14

Mode 8

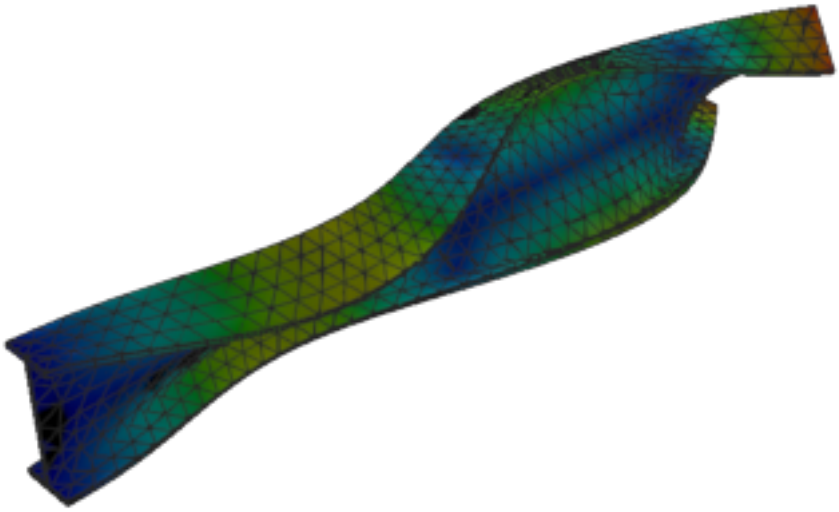


Figure 103: Calculix — mode 8, 297.856 Hz

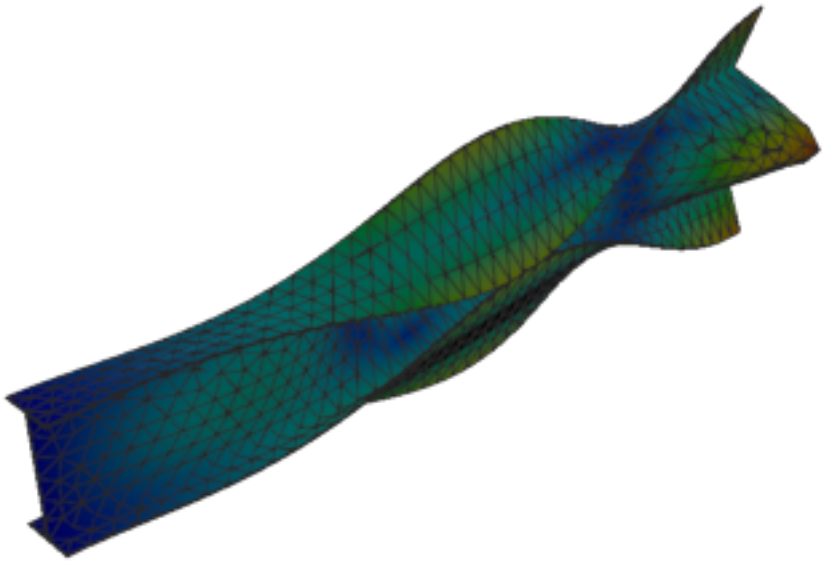


Figure 104: Code_Aster — mode 8, 168.951 Hz

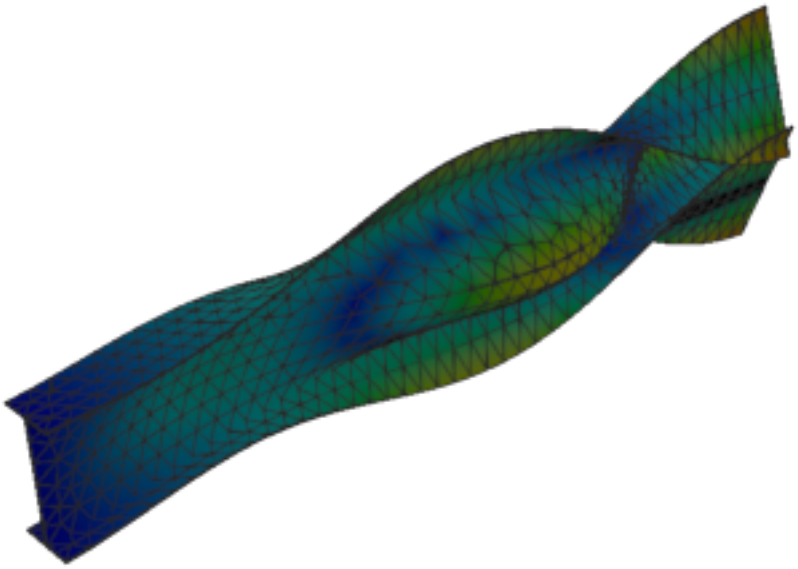


Figure 105: Sesam — mode 8, 166.668 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	297.856	+41.06	3.50246e+06	2.44e-09	7.20e-06	2.53e-14	4.94e-05	0.003	1.59e-07
Code_Aster	168.951	-19.99	1.12689e+06	5.89e-27	6.03	3.09e-26	7.67e-14	-2.455	-1.76e-13
Sesam	166.668	-21.07	1.09664e+06	4.67e-28	5.94	7.43e-31	-2.16e-14	2.438	8.62e-16

Mode 9

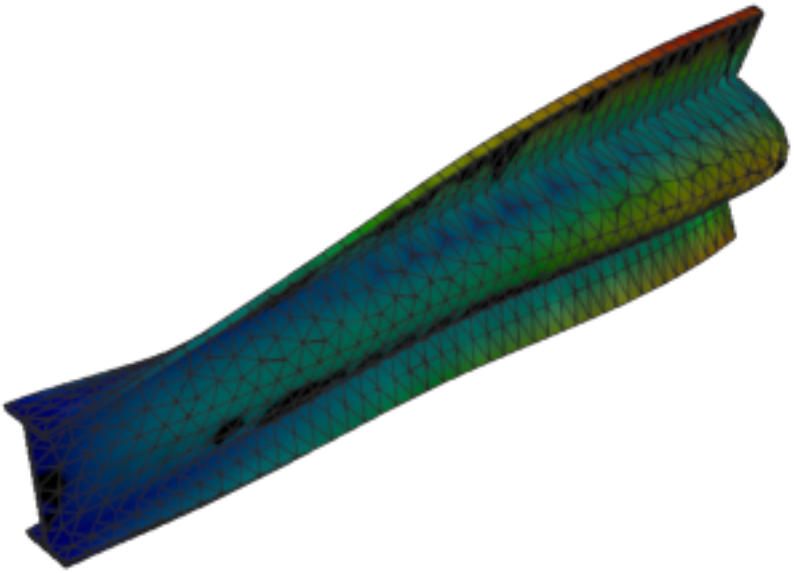


Figure 106: Calculix — mode 9, 304.464 Hz

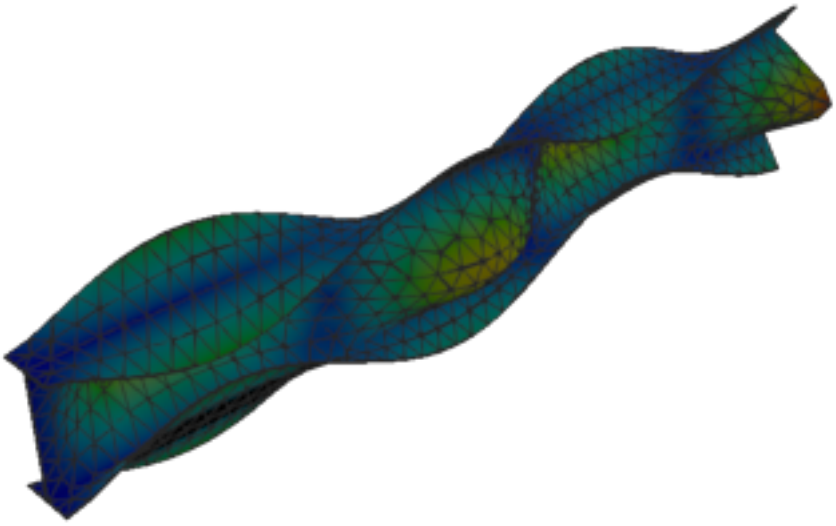


Figure 107: Code_Aster — mode 9, 195.184 Hz

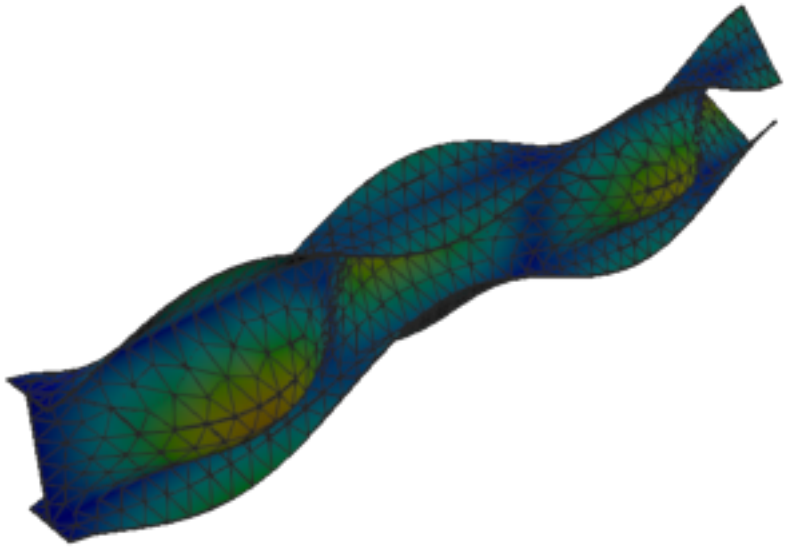


Figure 108: Sesam — mode 9, 191.533 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	304.464	+32.15	3.65958e+06	2.70e-09	0.02	8.66e-13	-5.19e-05	-0.156	9.30e-07
Code_Aster	195.184	-15.28	1.504e+06	5.26e-28	0.62	7.49e-27	2.29e-14	0.788	-8.66e-14
Sesam	191.533	-16.87	1.44826e+06	2.49e-28	0.61	5.28e-28	-1.58e-14	-0.781	-2.30e-14

Mode 10

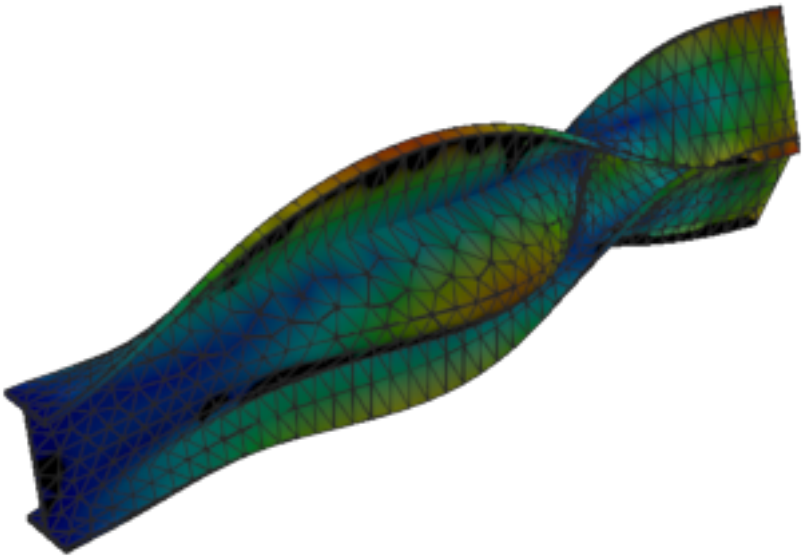


Figure 109: Calculix — mode 10, 365.943 Hz

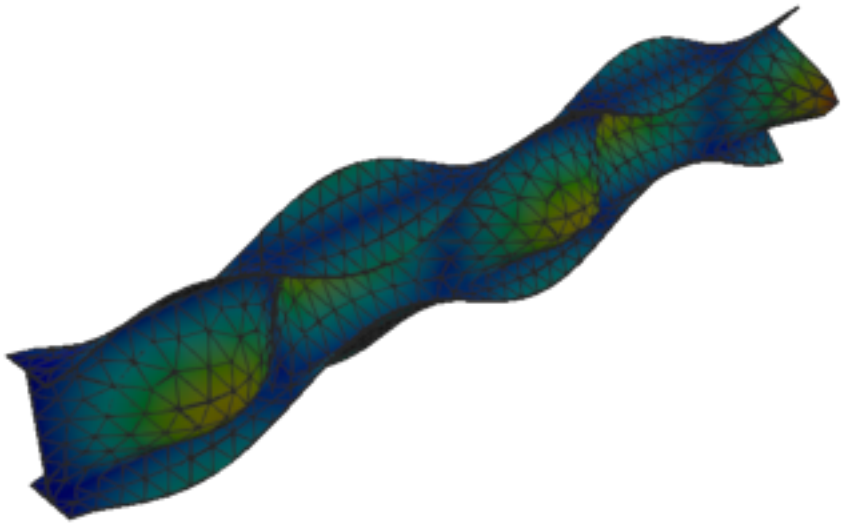


Figure 110: Code_Aster — mode 10, 235.115 Hz

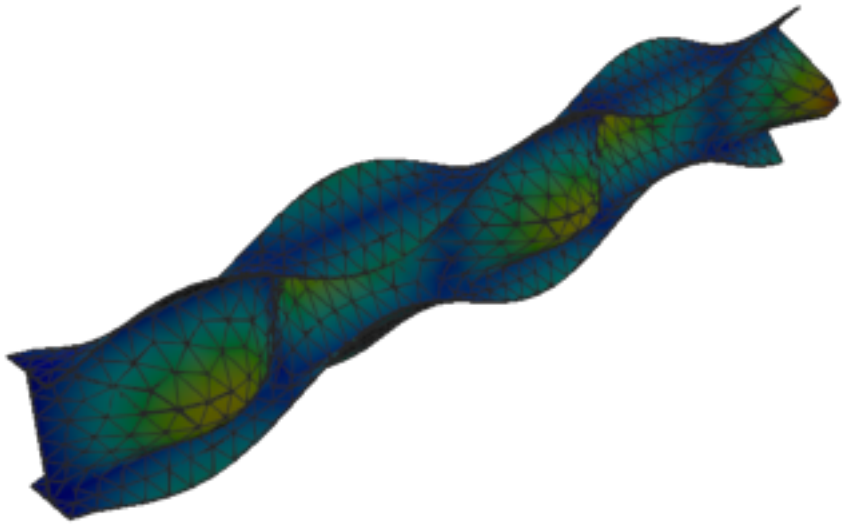


Figure 111: Sesam — mode 10, 230.489 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	365.943	+32.02	5.28673e+06	1.72e-09	1.48	3.23e-12	-4.15e-05	1.218	-1.80e-06
Code_Aster	235.115	-15.18	2.18233e+06	2.11e-27	1.32	4.83e-27	4.59e-14	-1.148	6.95e-14
Sesam	230.489	-16.85	2.09731e+06	1.88e-28	1.27	4.80e-28	1.37e-14	-1.128	-2.19e-14

Mode 11

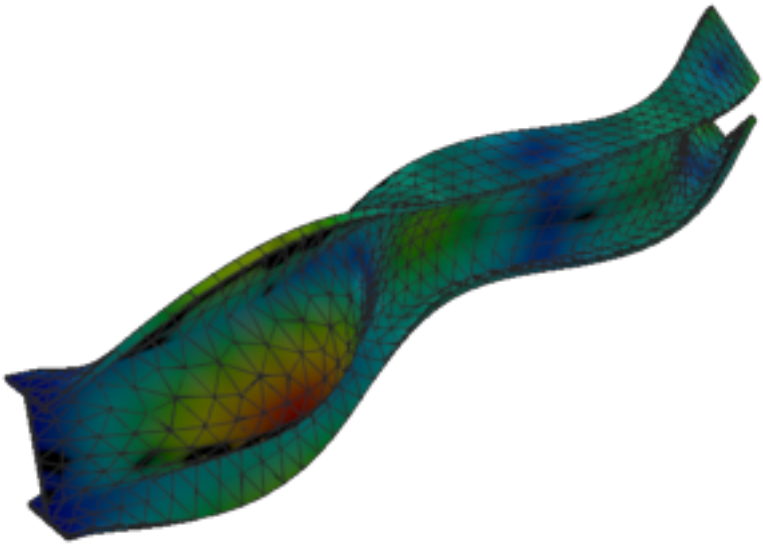


Figure 112: Calculix — mode 11, 368.764 Hz

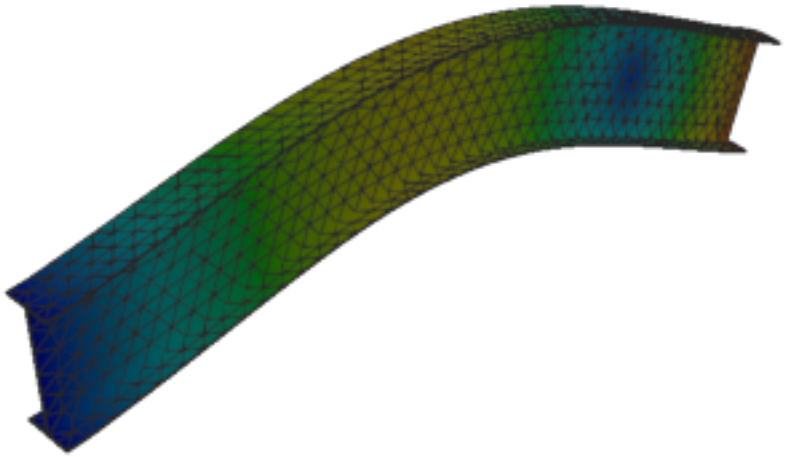


Figure 113: Code_Aster — mode 11, 257.409 Hz

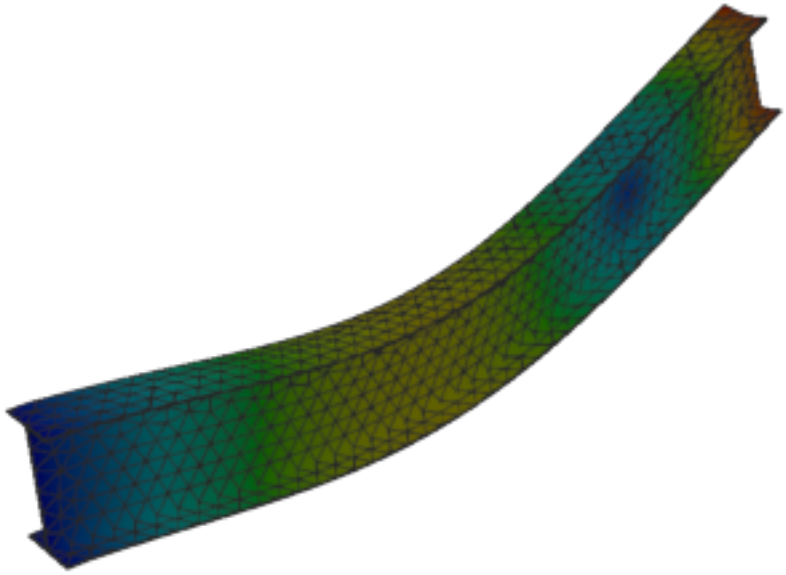


Figure 114: Sesam — mode 11, 257.362 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	368.764	+25.21	5.36853e+06	1.81e-10	3.61	4.88e-12	-1.34e-05	-1.900	2.21e-06
Code_Aster	257.409	-12.60	2.61582e+06	1.20e-09	1.38e-26	41.64	3.47e-05	-1.17e-13	6.453
Sesam	257.362	-12.61	2.61487e+06	6.51e-09	3.57e-22	41.62	-8.07e-05	-1.89e-11	-6.451

Shell, 1st order, QUAD

Mode 1

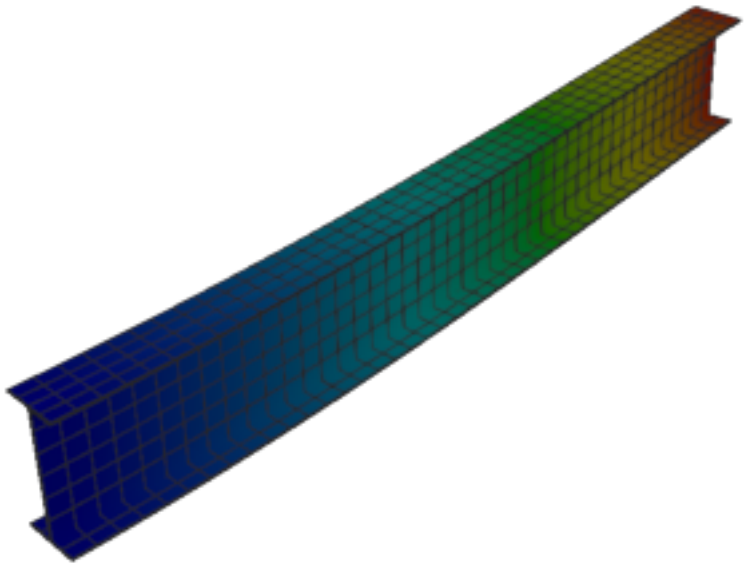


Figure 115: Abaqus — mode 1, 12.780 Hz

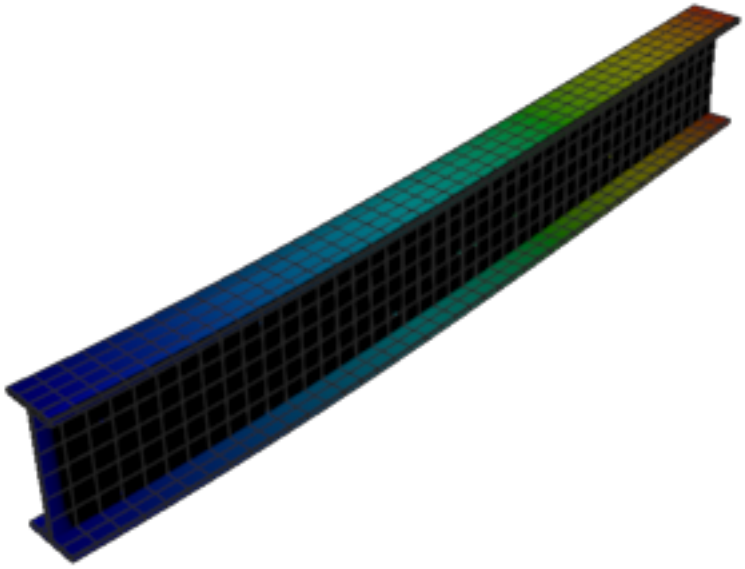


Figure 116: Calculix — mode 1, 12.764 Hz

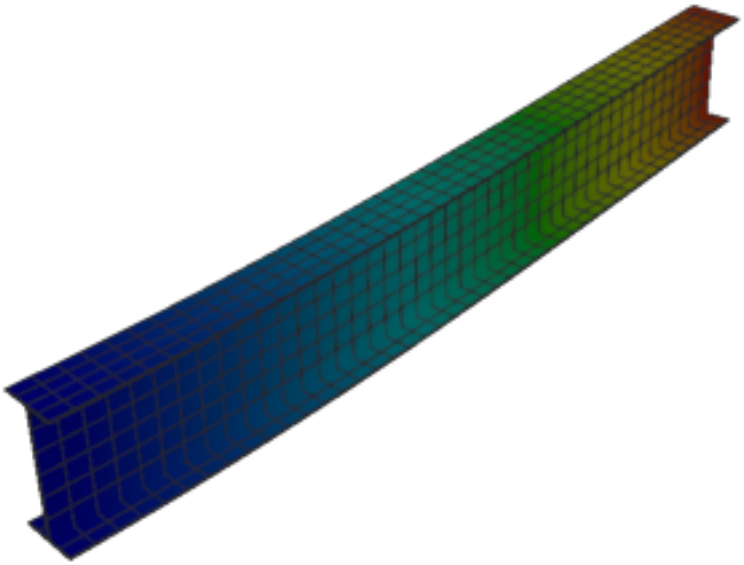


Figure 117: Code_Aster — mode 1, 13.146 Hz

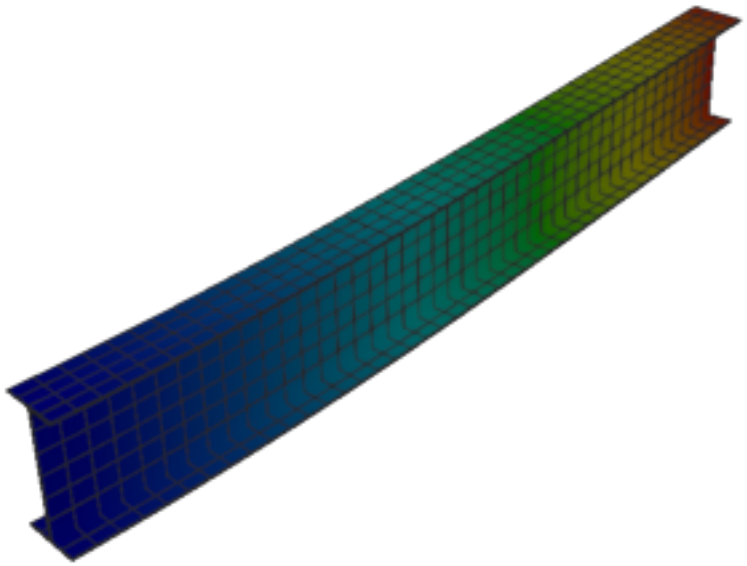


Figure 118: Sesam — mode 1, 12.763 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.780	-0.65	6448.4	7.41e-27	119.92	1.38e-25	-8.61e-14	10.951	-3.71e-13
Calculix	12.764	-0.77	6431.71	1.18e-24	119.90	1.74e-12	1.09e-12	10.950	1.32e-06
Code_Aster	13.146	+2.20	6822.87	5.73e-28	119.92	3.53e-25	2.39e-14	10.951	-5.94e-13
Sesam	12.763	-0.78	6430.64	2.41e-28	119.90	3.46e-25	-1.55e-14	10.950	-5.88e-13

Mode 2

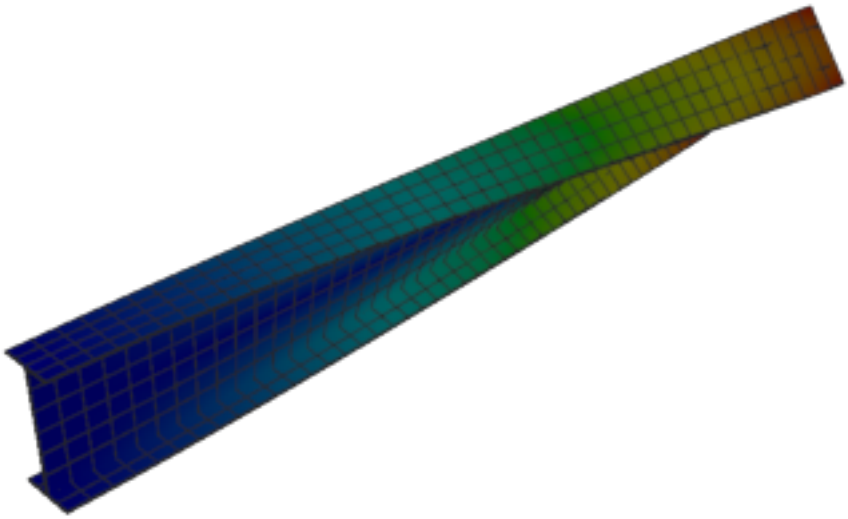


Figure 119: Abaqus — mode 2, 19.593 Hz

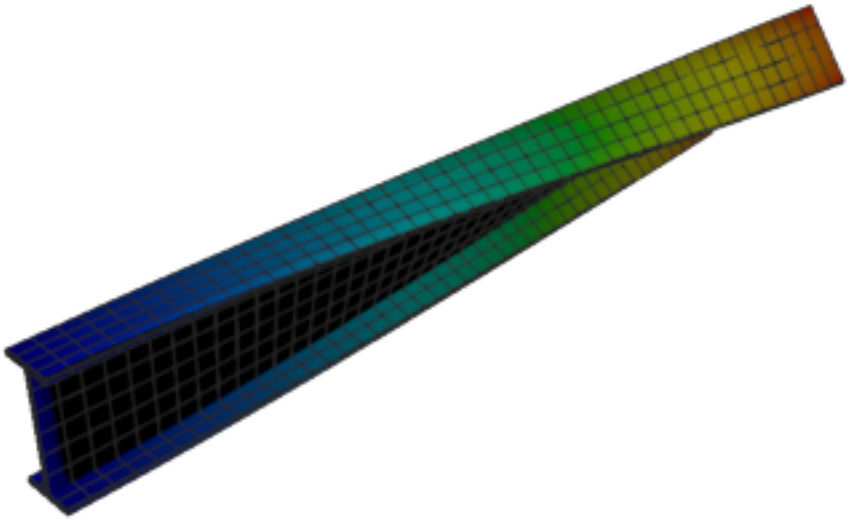


Figure 120: Calculix — mode 2, 19.630 Hz

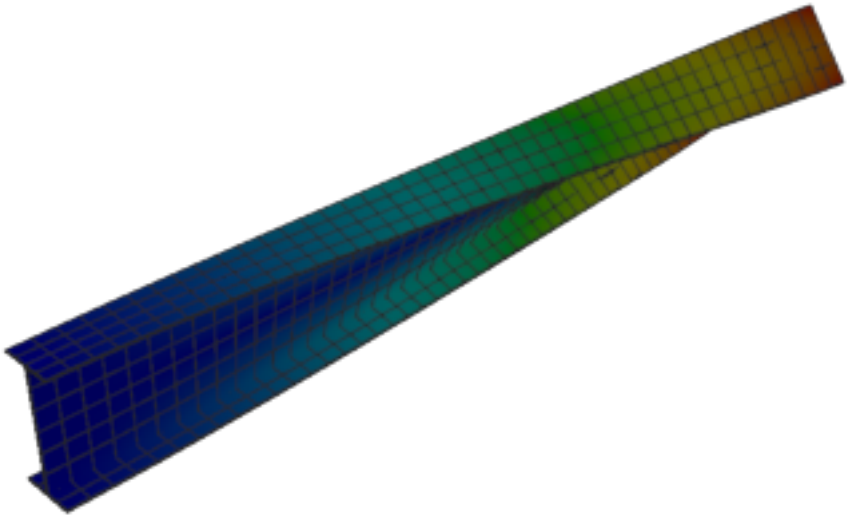


Figure 121: Code_Aster — mode 2, 19.972 Hz

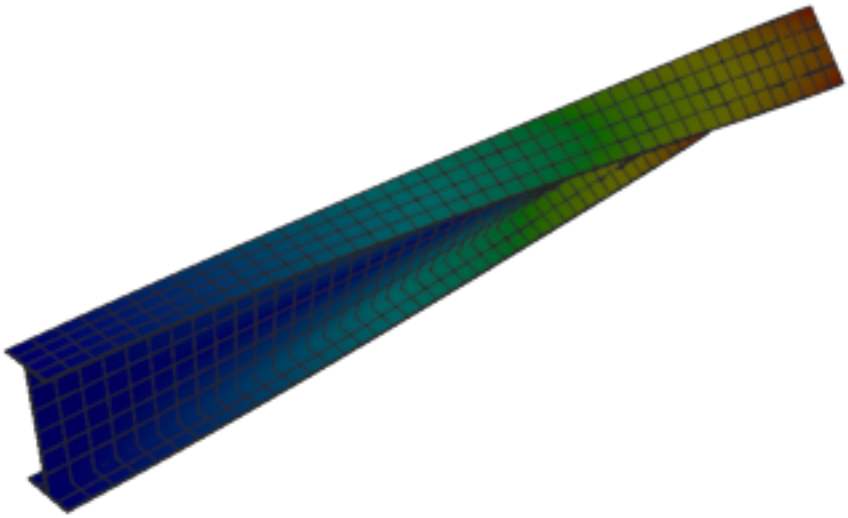


Figure 122: Sesam — mode 2, 19.394 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.593	-0.28	15155	1.46e-27	1.08e-19	7.67e-28	-3.83e-14	3.28e-10	2.77e-14
Calculix	19.630	-0.09	15213	3.11e-09	1.86e-17	2.79e-20	5.58e-05	4.31e-09	1.67e-10
Code_Aster	19.972	+1.65	15747.1	8.93e-27	4.15e-19	1.50e-24	-9.45e-14	-6.44e-10	1.23e-12
Sesam	19.394	-1.29	14849.3	5.82e-27	6.29e-19	1.98e-24	-7.63e-14	-7.93e-10	-1.41e-12

Mode 3

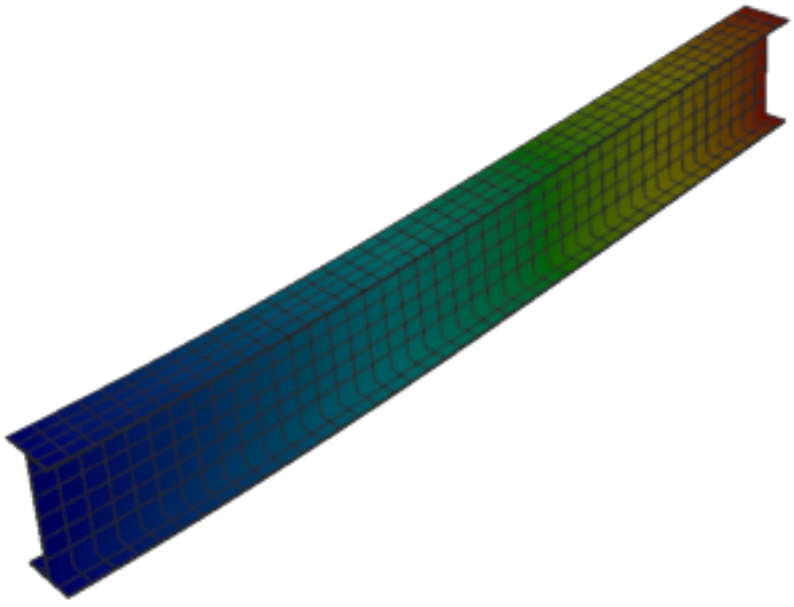


Figure 123: Abaqus — mode 3, 51.998 Hz

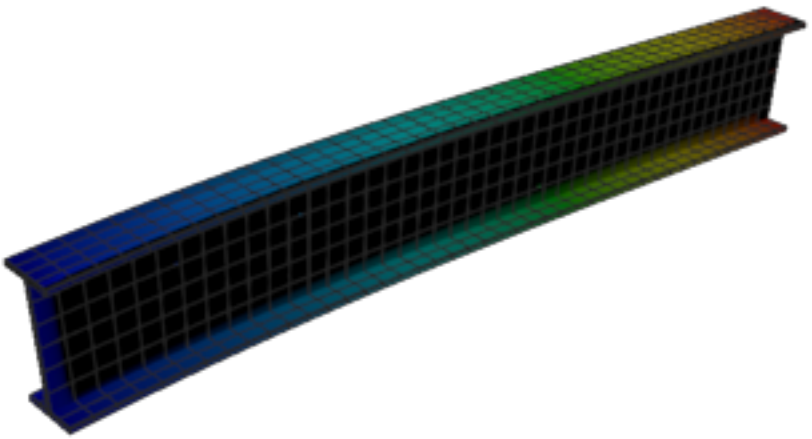


Figure 124: Calculix — mode 3, 52.000 Hz

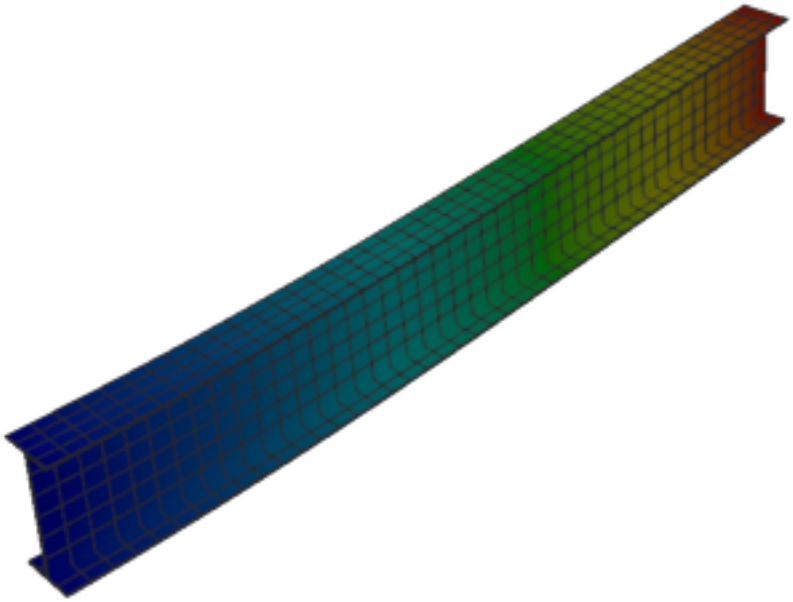


Figure 125: Code_Aster — mode 3, 52.060 Hz

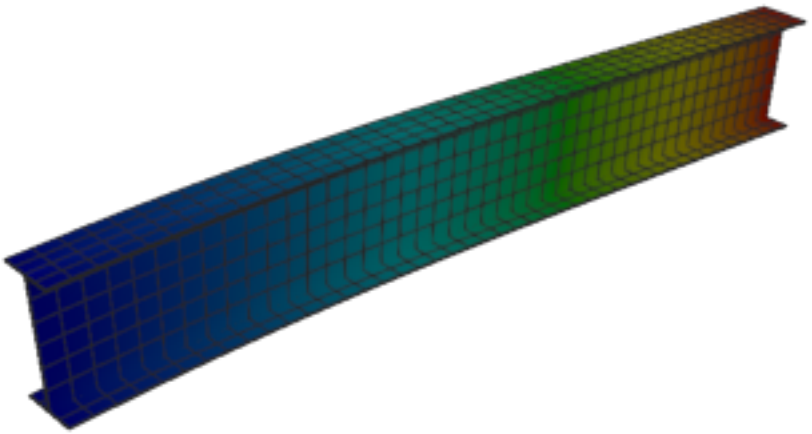


Figure 126: Sesam — mode 3, 51.991 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	-0.03	106741	1.35e-26	1.05e-26	122.01	-1.16e-13	1.03e-13	11.046
Calculix	52.000	-0.02	106749	9.22e-26	4.50e-11	121.98	-3.04e-13	6.71e-06	-11.045
Code_Aster	52.060	+0.09	106994	9.26e-28	2.72e-25	122.02	-3.04e-14	5.21e-13	11.046
Sesam	51.991	-0.04	106714	3.83e-27	1.19e-25	122.00	6.19e-14	-3.45e-13	-11.046

Mode 4

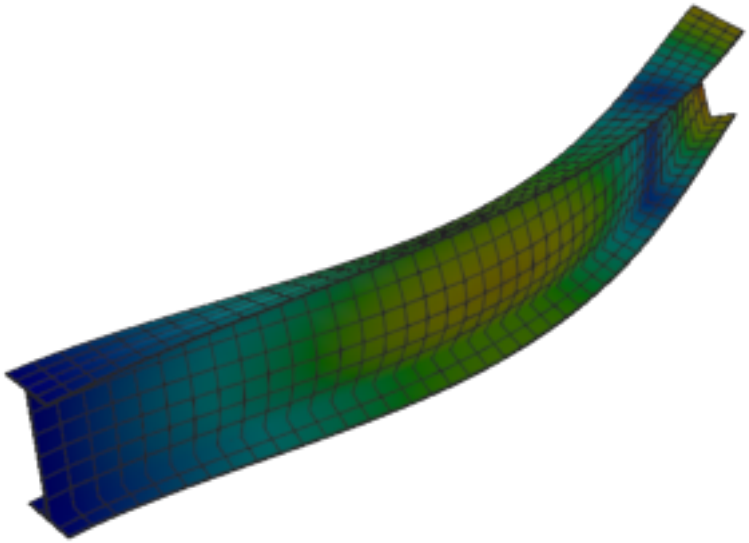


Figure 127: Abaqus — mode 4, 75.683 Hz

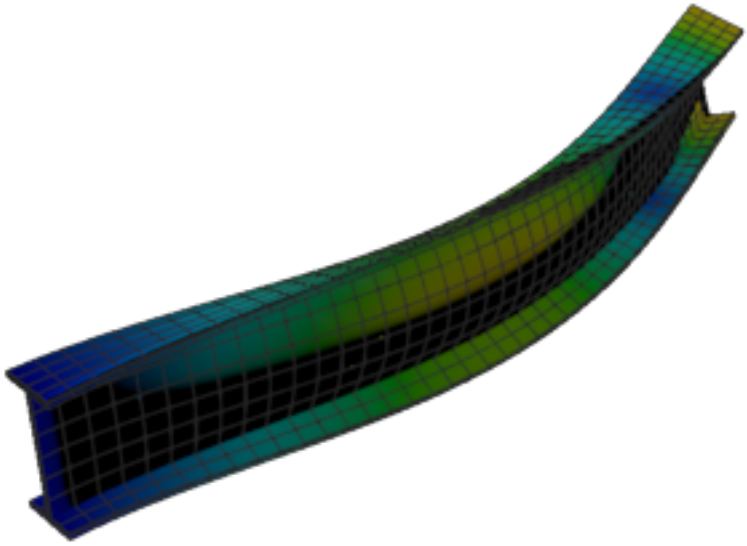


Figure 128: Calculix — mode 4, 75.457 Hz

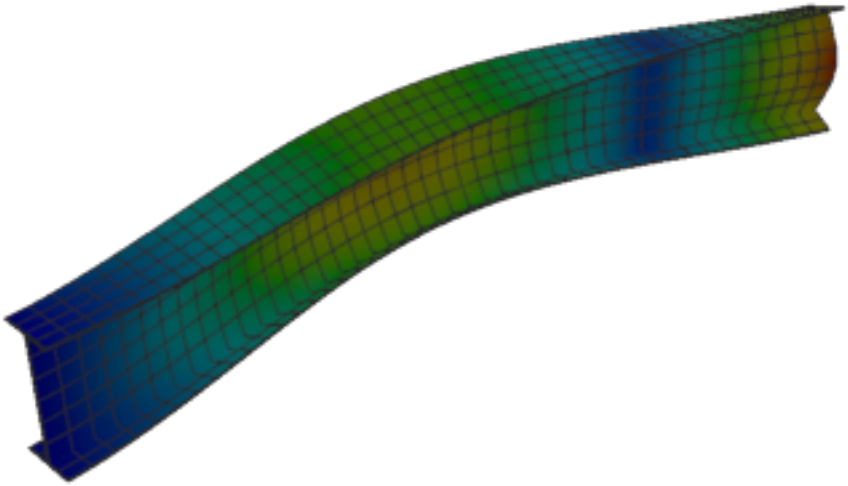


Figure 129: Code_Aster — mode 4, 77.236 Hz

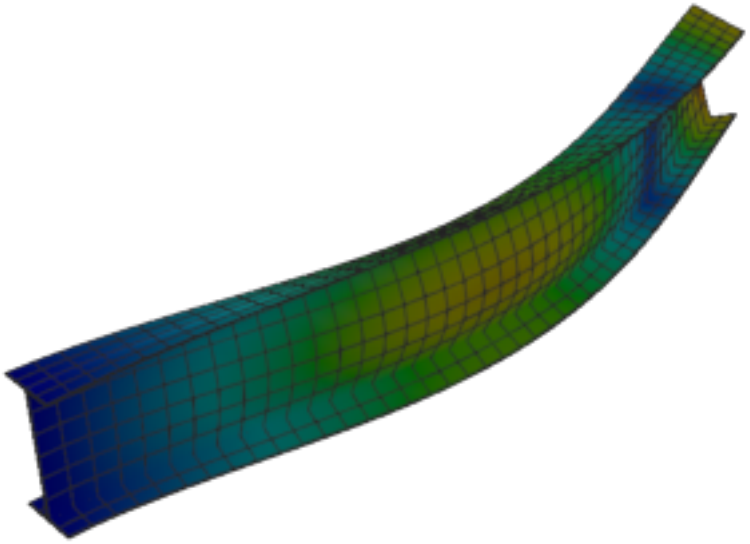


Figure 130: Sesam — mode 4, 75.192 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.683	-0.28	226132	1.88e-28	36.92	2.68e-25	-1.37e-14	-6.076	-5.18e-13
Calculix	75.457	-0.57	224783	1.01e-25	36.87	1.53e-12	-3.18e-13	-6.072	1.24e-06
Code_Aster	77.236	+1.77	235505	3.33e-27	36.78	3.14e-26	-5.77e-14	6.065	1.77e-13
Sesam	75.192	-0.92	223203	1.83e-27	36.80	1.95e-25	-4.28e-14	-6.067	-4.41e-13

Mode 5

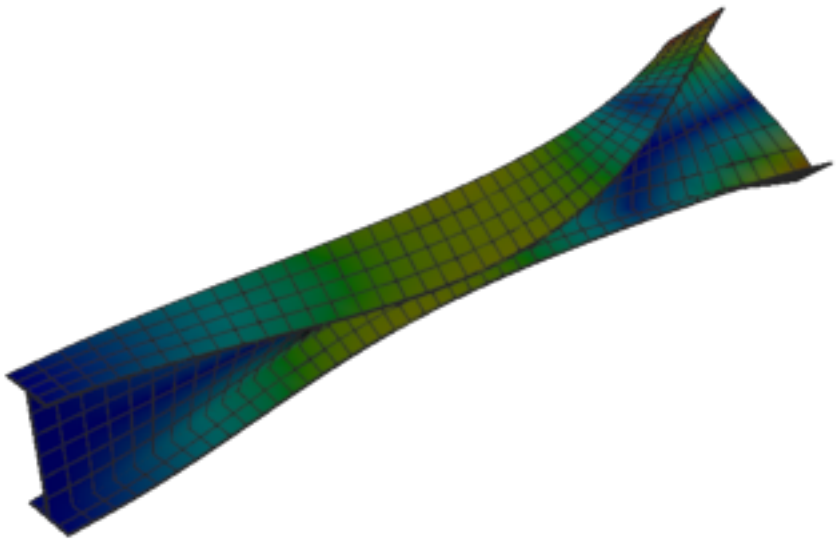


Figure 131: Abaqus — mode 5, 96.393 Hz

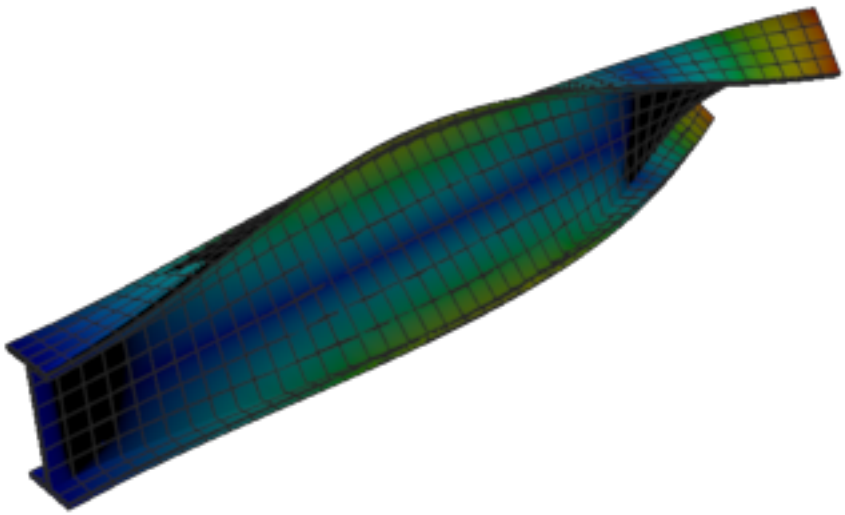


Figure 132: Calculix — mode 5, 96.296 Hz

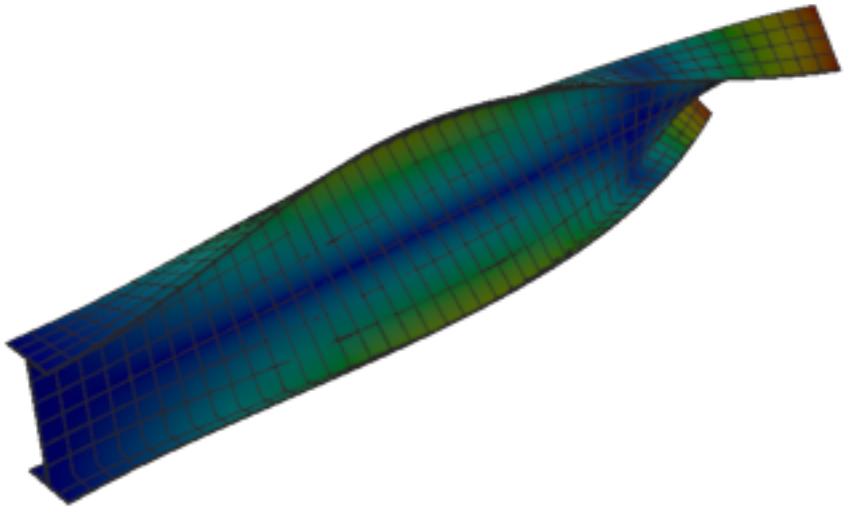


Figure 133: Code_Aster — mode 5, 98.762 Hz

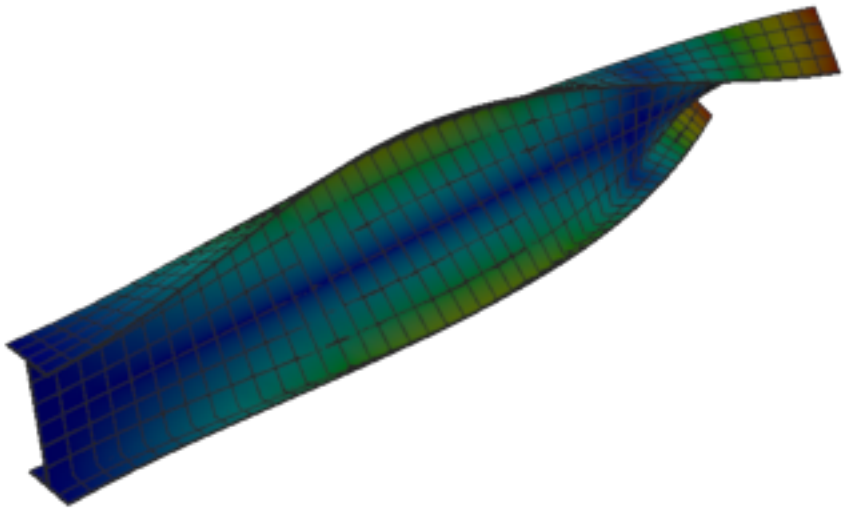


Figure 134: Sesam — mode 5, 95.930 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.393	-0.47	366818	7.25e-28	1.11e-22	7.70e-26	-2.69e-14	-1.05e-11	-2.77e-13
Calculix	96.296	-0.57	366079	8.46e-10	5.61e-21	1.48e-22	-2.91e-05	-7.49e-11	1.22e-11
Code_Aster	98.762	+1.98	385069	6.28e-29	9.74e-23	7.49e-26	-7.92e-15	9.87e-12	-2.74e-13
Sesam	95.930	-0.95	363300	3.32e-27	2.02e-23	9.04e-27	5.76e-14	4.49e-12	-9.51e-14

Mode 6

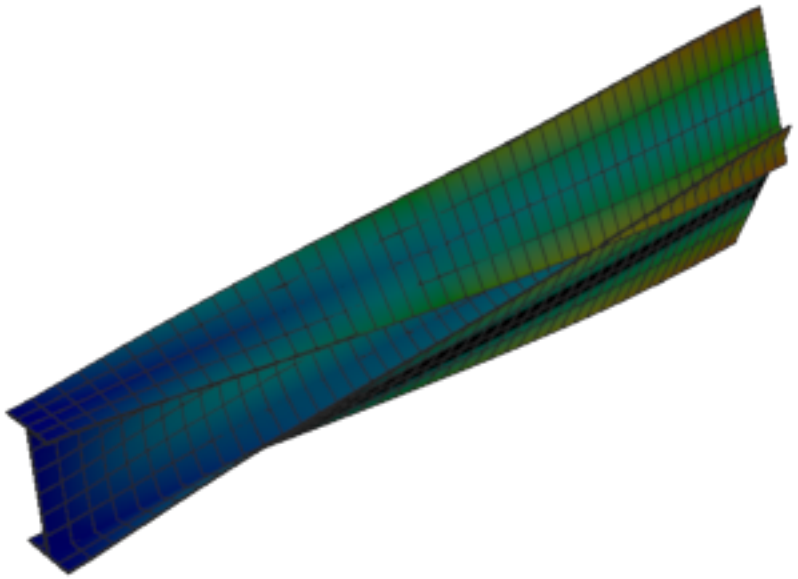


Figure 135: Abaqus — mode 6, 126.961 Hz

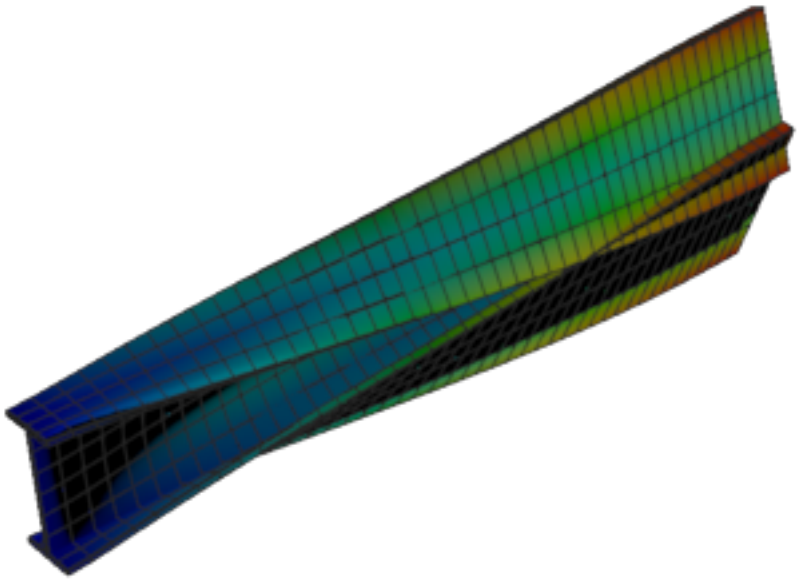


Figure 136: Calculix — mode 6, 125.898 Hz

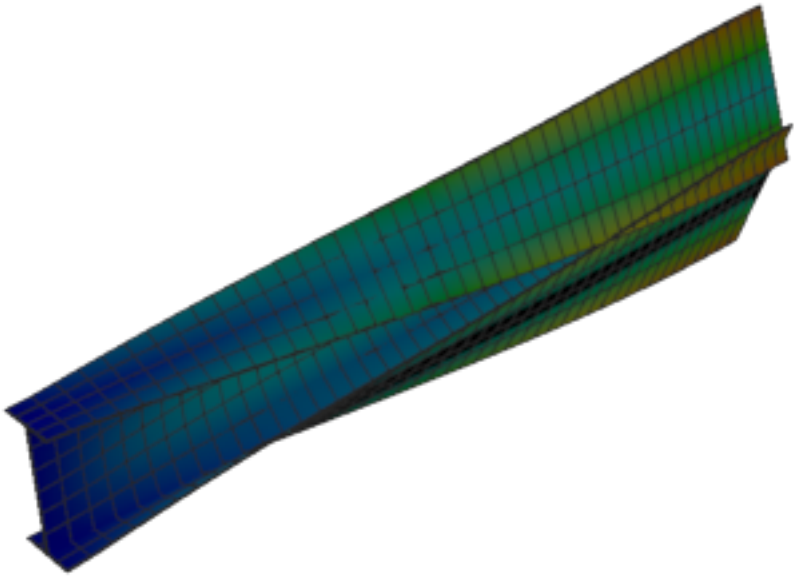


Figure 137: Code_Aster — mode 6, 125.442 Hz

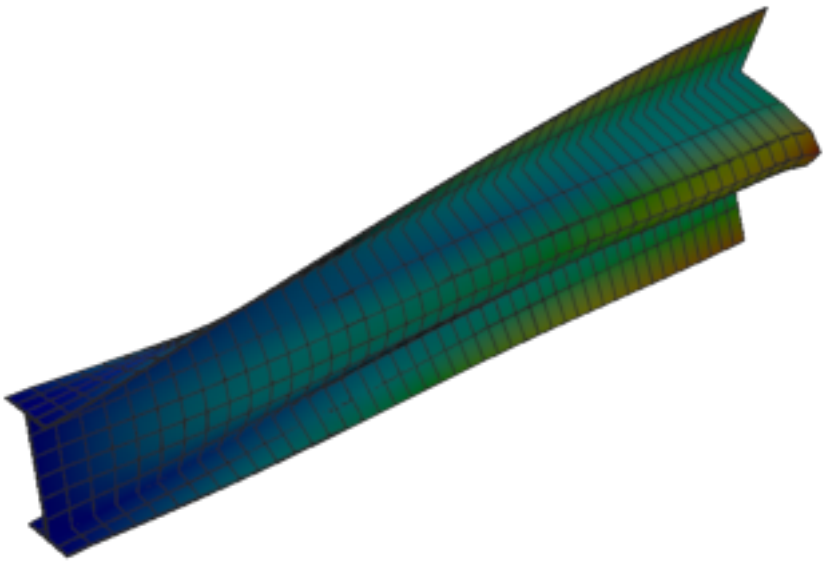


Figure 138: Sesam — mode 6, 125.116 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	126.960	+0.88	636351	1.52e-27	0.02	2.42e-26	3.90e-14	-0.132	-1.56e-13
Calculix	125.898	+0.03	625743	4.03e-28	0.03	4.13e-15	2.01e-14	-0.174	-6.43e-08
Code_Aster	125.442	-0.33	621220	2.55e-28	0.04	5.76e-25	-1.60e-14	-0.194	-7.59e-13
Sesam	125.116	-0.59	617993	9.40e-30	0.02	2.19e-27	-3.07e-15	0.134	-4.68e-14

Mode 7

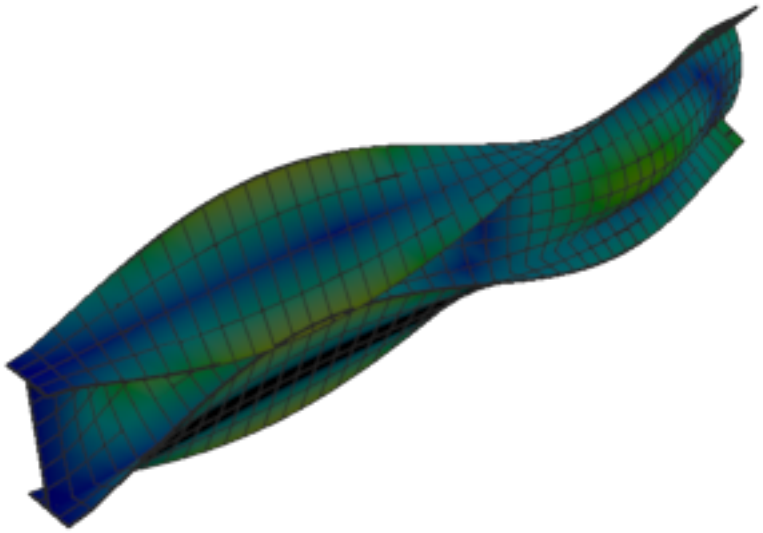


Figure 139: Abaqus — mode 7, 145.566 Hz

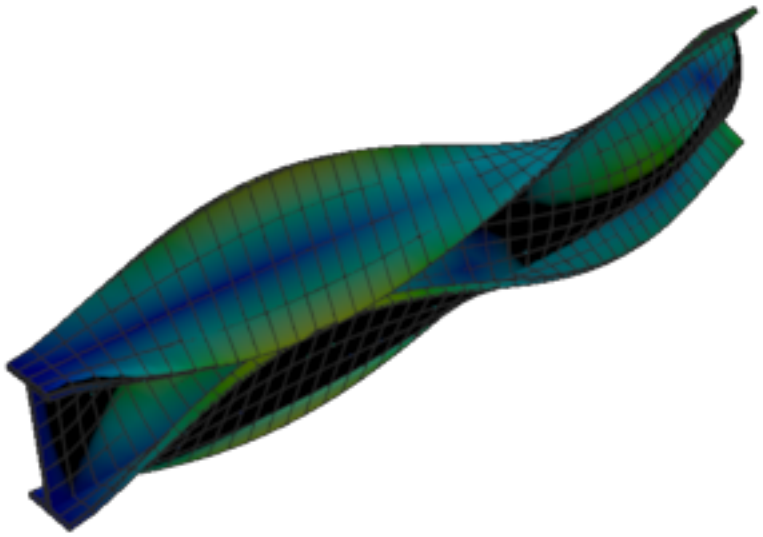


Figure 140: Calculix — mode 7, 145.283 Hz

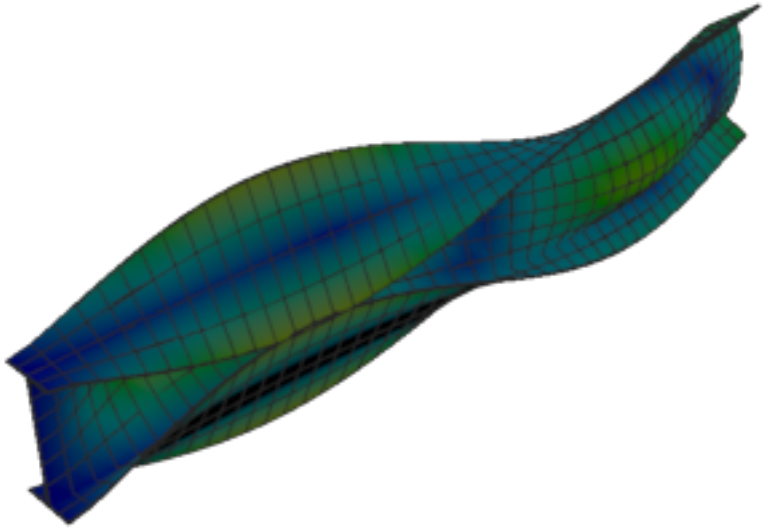


Figure 141: Code_Aster — mode 7, 143.864 Hz

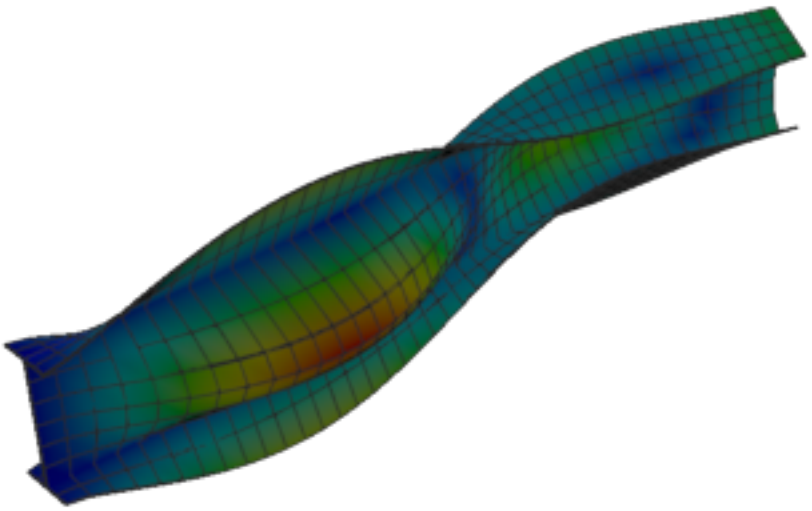


Figure 142: Sesam — mode 7, 141.556 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	145.570	+1.04	836523	2.58e-27	3.34	1.12e-26	-5.08e-14	1.829	1.06e-13
Calculix	145.282	+0.84	833272	1.59e-26	3.26	6.68e-13	-1.26e-13	1.805	8.17e-07
Code_Aster	143.864	-0.14	817074	5.14e-28	3.15	8.15e-26	2.27e-14	1.776	2.86e-13
Sesam	141.556	-1.74	791068	4.10e-28	3.46	8.46e-30	2.02e-14	-1.860	2.91e-15

Mode 8

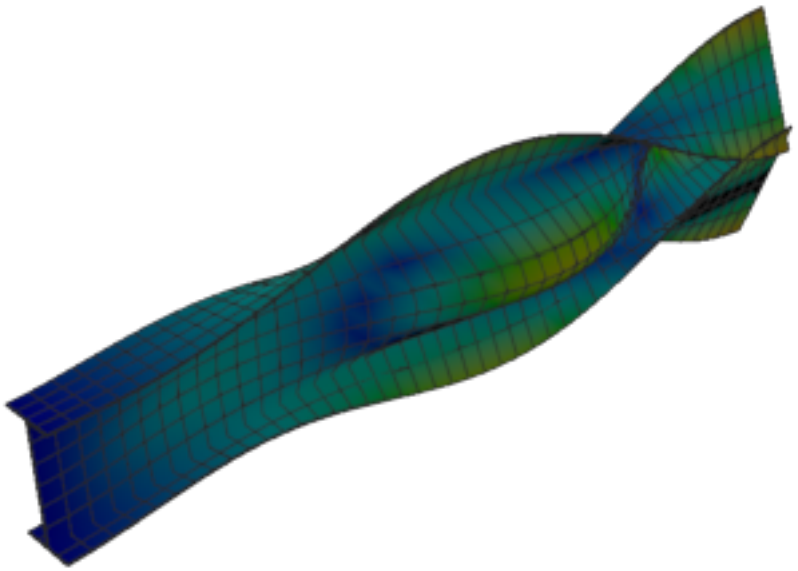


Figure 143: Abaqus — mode 8, 169.389 Hz

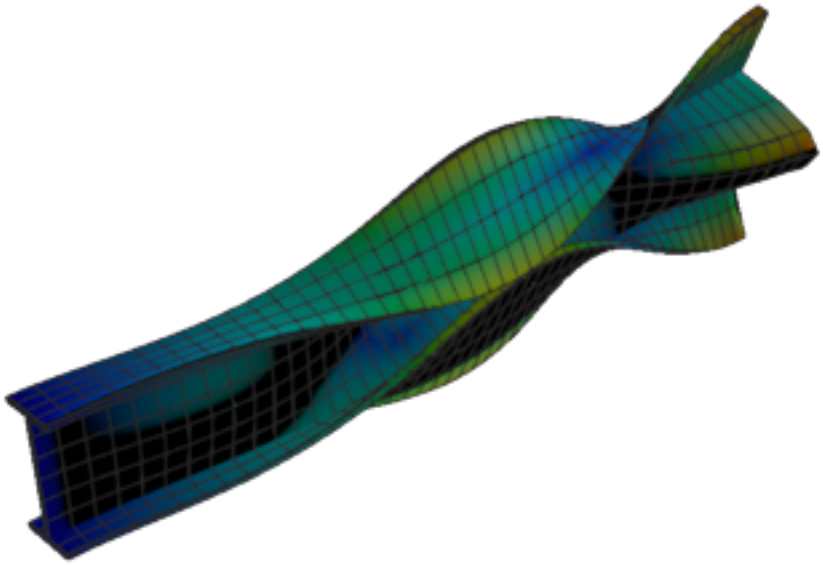


Figure 144: Calculix — mode 8, 169.441 Hz

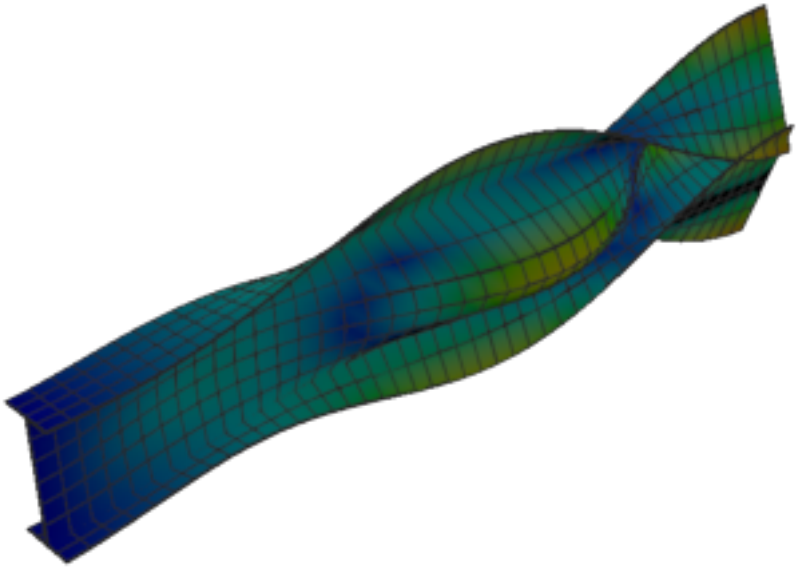


Figure 145: Code_Aster — mode 8, 167.974 Hz

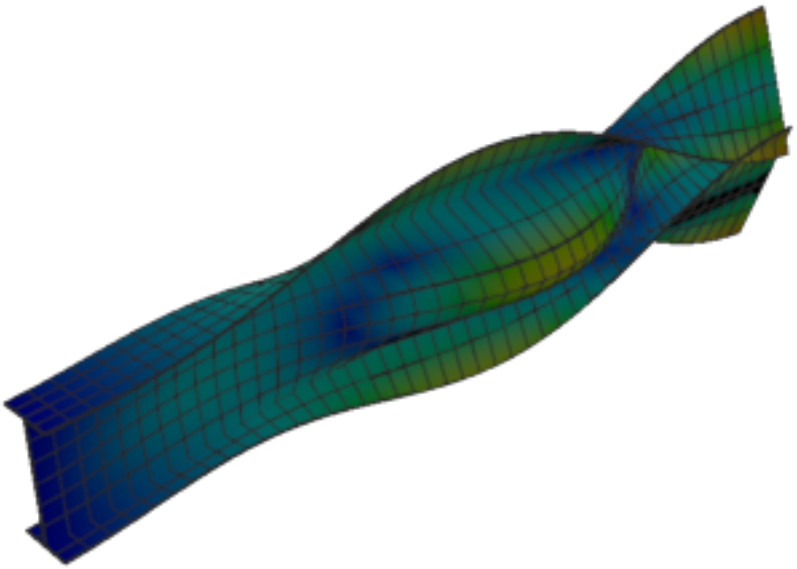


Figure 146: Sesam — mode 8, 165.030 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	169.390	+0.85	1.13275e+06	2.66e-25	5.68	6.71e-24	-5.16e-13	2.383	-2.59e-12
Calculix	169.441	+0.88	1.13343e+06	1.08e-25	5.79	1.37e-14	-3.29e-13	-2.406	1.17e-07
Code_Aster	167.974	+0.01	1.11389e+06	3.09e-27	5.75	3.62e-25	-5.56e-14	2.398	-6.01e-13
Sesam	165.030	-1.74	1.07519e+06	8.05e-28	5.45	1.35e-26	2.84e-14	2.334	1.16e-13

Mode 9

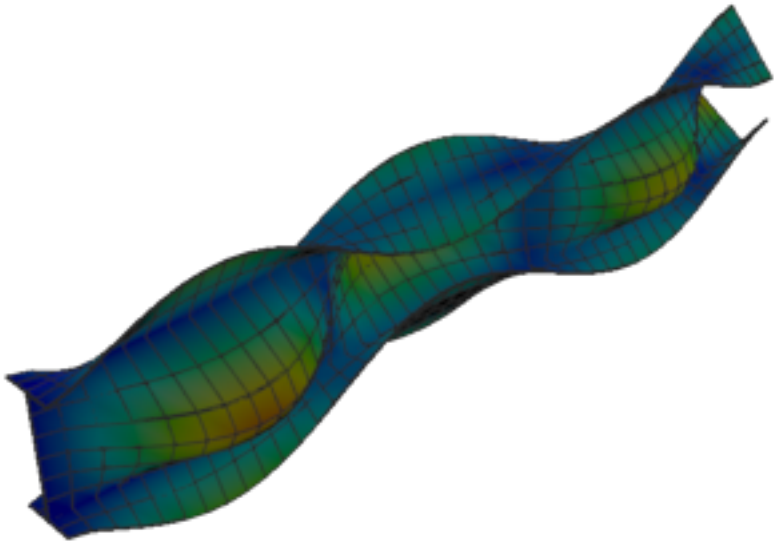


Figure 147: Abaqus — mode 9, 198.748 Hz

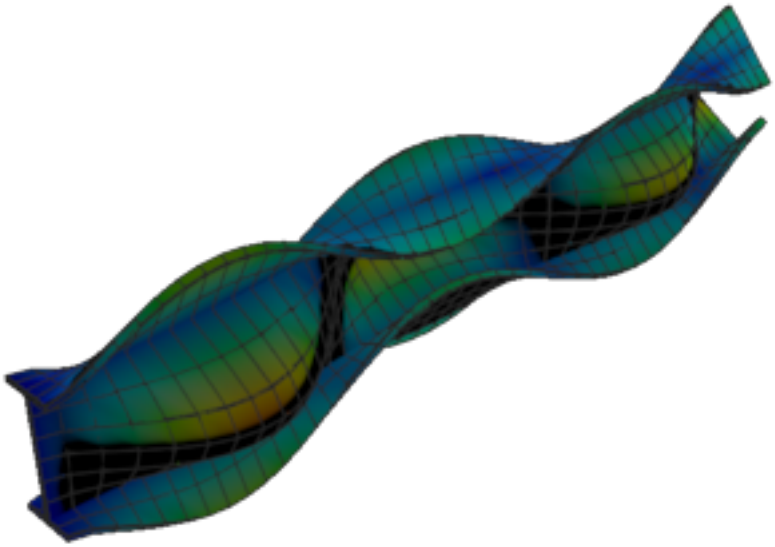


Figure 148: Calculix — mode 9, 200.495 Hz

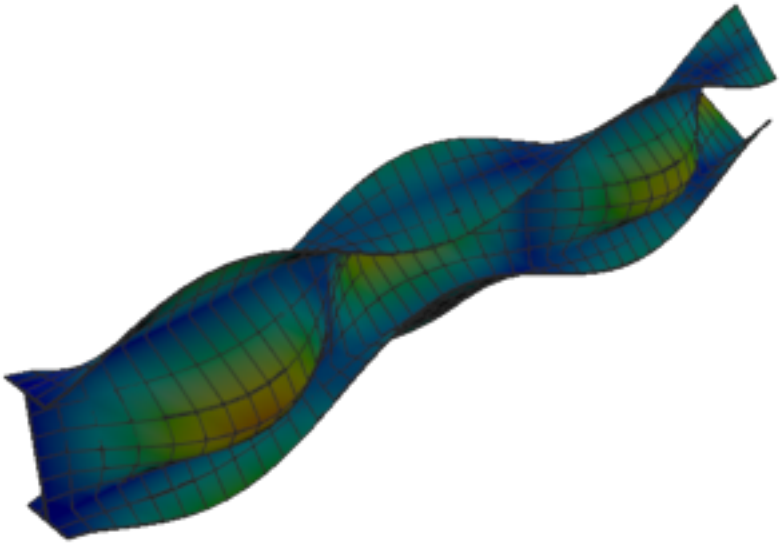


Figure 149: Code_Aster — mode 9, 193.960 Hz

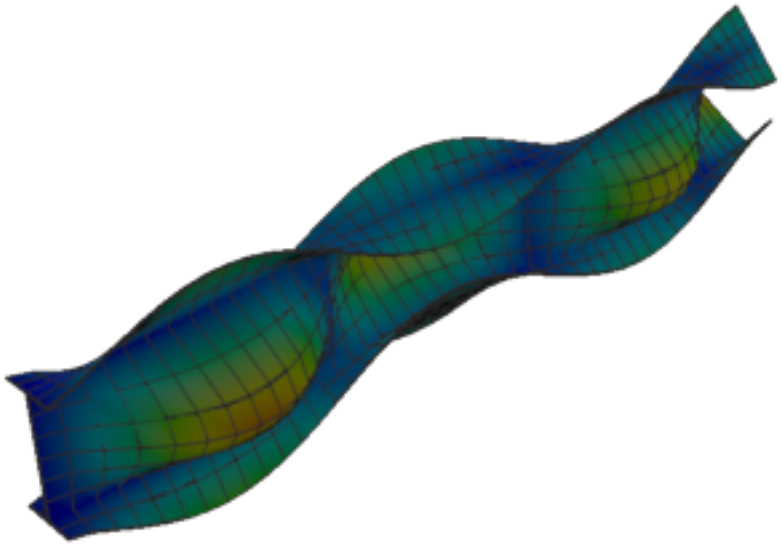


Figure 150: Sesam — mode 9, 190.099 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	198.750	+1.49	1.55943e+06	6.12e-22	0.59	1.15e-20	2.47e-11	-0.771	1.07e-10
Calculix	200.494	+2.38	1.58696e+06	1.04e-25	0.58	3.83e-13	-3.23e-13	-0.760	6.19e-07
Code_Aster	193.960	-0.95	1.48519e+06	2.45e-27	0.63	5.78e-29	-4.95e-14	-0.795	7.61e-15
Sesam	190.099	-2.92	1.42665e+06	1.48e-29	0.64	2.81e-27	-3.85e-15	-0.798	5.30e-14

Mode 10

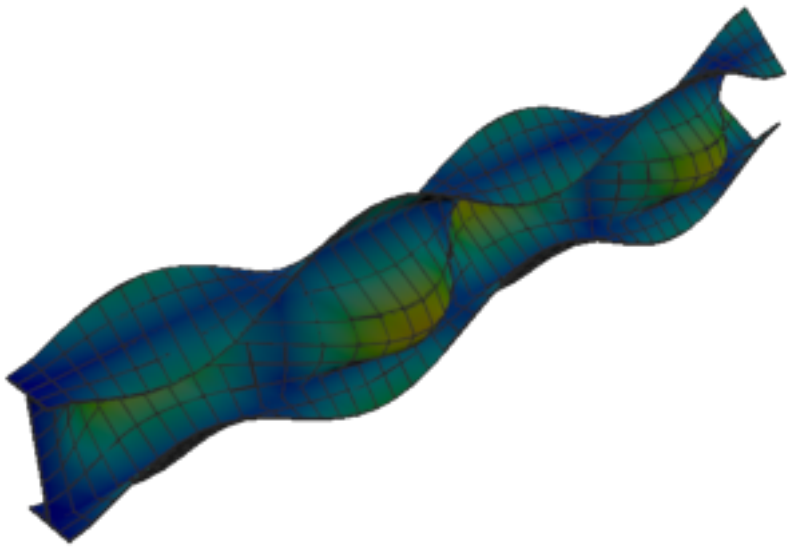


Figure 151: Abaqus — mode 10, 242.545 Hz

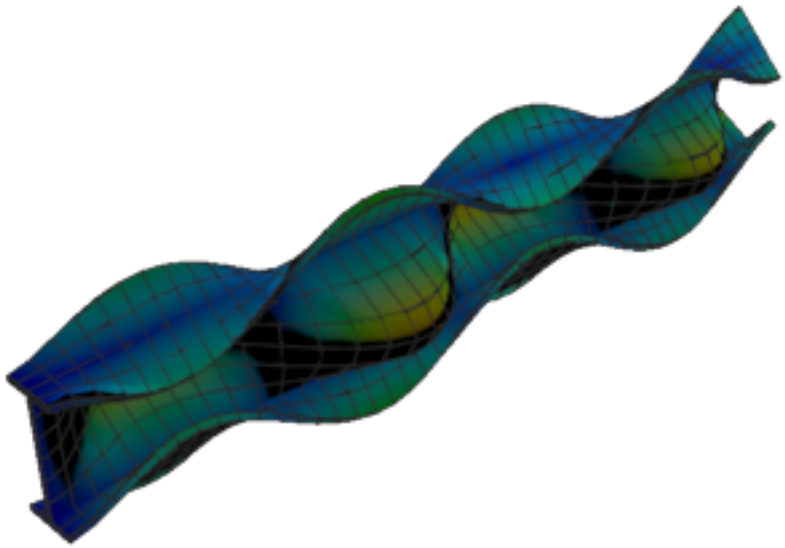


Figure 152: Calculix — mode 10, 247.276 Hz

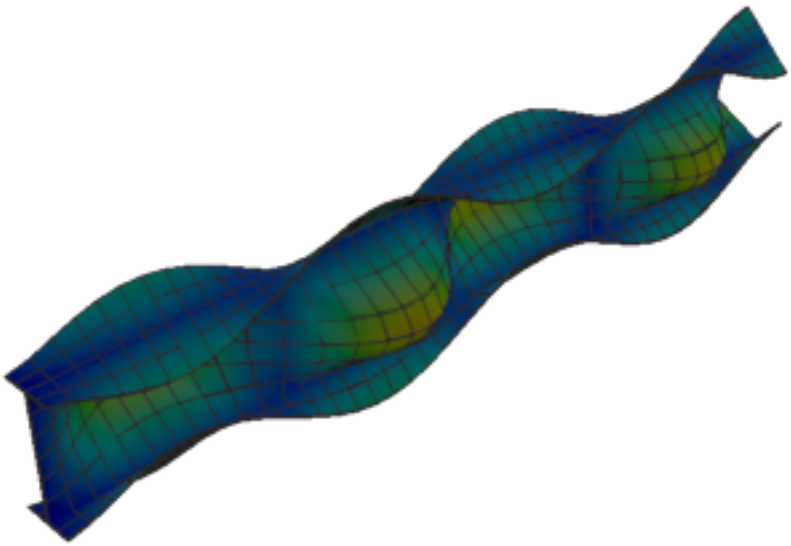


Figure 153: Code_Aster — mode 10, 233.817 Hz

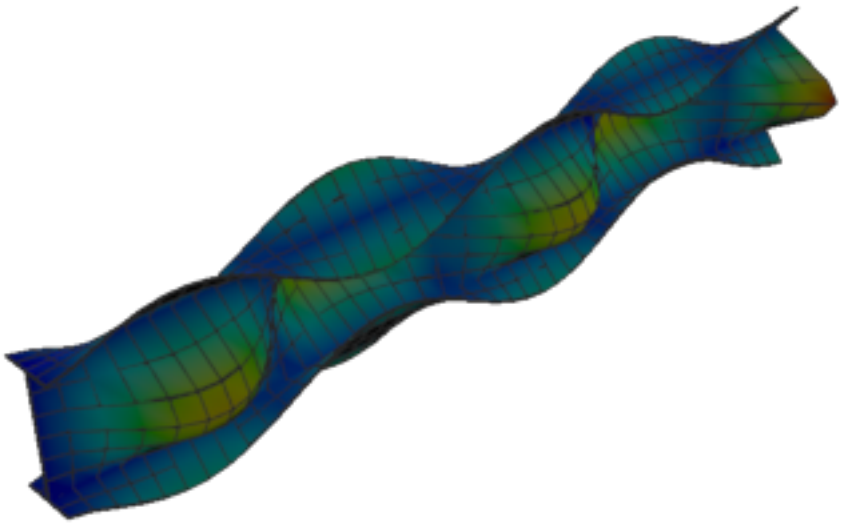


Figure 154: Sesam — mode 10, 229.460 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	242.550	+1.79	2.32244e+06	3.62e-17	1.52	5.68e-16	6.02e-09	1.232	2.38e-08
Calculix	247.276	+3.78	2.41392e+06	6.50e-25	1.71	2.90e-13	-8.06e-13	1.310	5.39e-07
Code_Aster	233.817	-1.87	2.1583e+06	3.50e-27	1.33	2.79e-26	-5.92e-14	1.152	-1.67e-13
Sesam	229.460	-3.70	2.07861e+06	5.01e-29	1.32	1.10e-28	7.08e-15	-1.150	1.05e-14

Mode 11

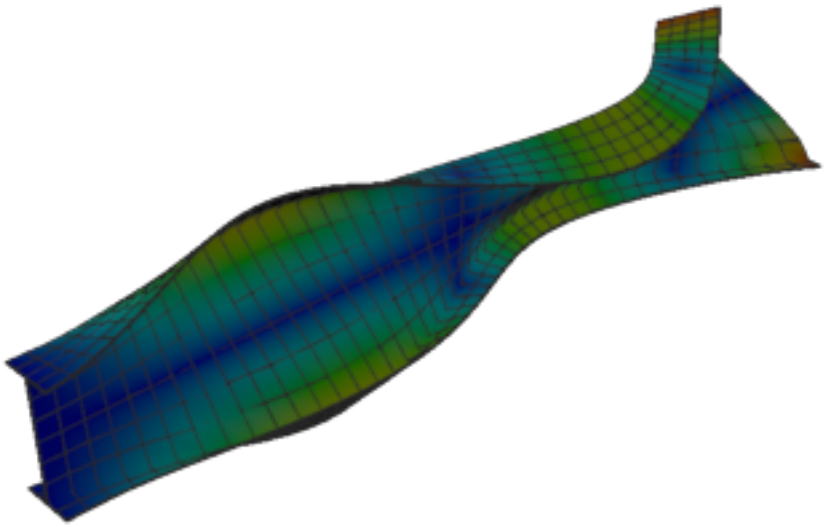


Figure 155: Abaqus — mode 11, 252.486 Hz

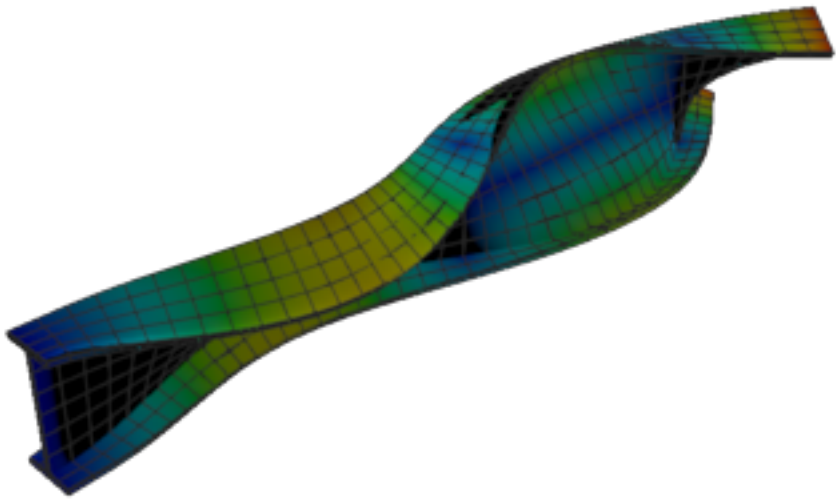


Figure 156: Calculix — mode 11, 251.554 Hz

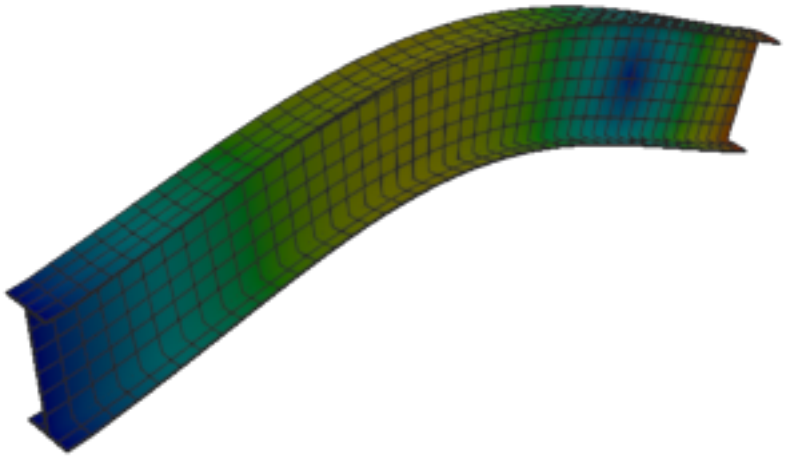


Figure 157: Code_Aster — mode 11, 257.105 Hz

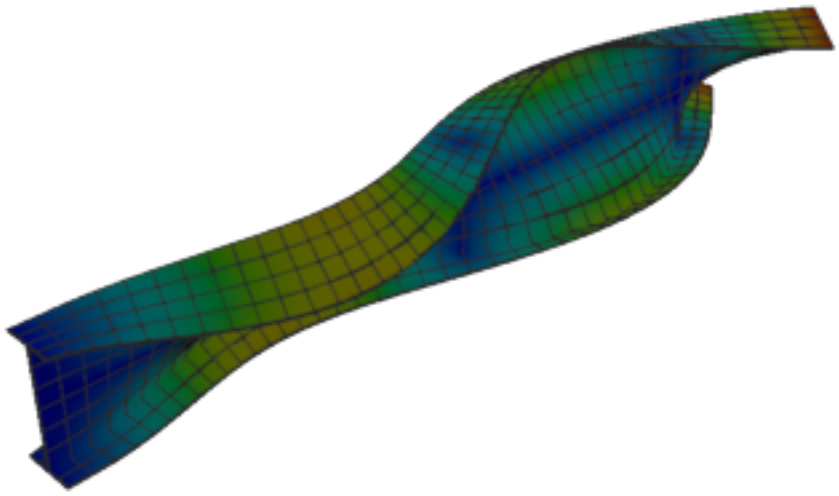


Figure 158: Sesam — mode 11, 250.786 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.490	-0.20	2.51672e+06	4.88e-17	7.27e-17	7.10e-16	6.99e-09	8.53e-09	2.67e-08
Calculix	251.554	-0.57	2.49818e+06	2.66e-10	6.32e-23	1.13e-21	1.63e-05	7.95e-12	3.37e-11
Code_Aster	257.105	+1.63	2.60965e+06	1.80e-26	2.95e-31	41.64	1.34e-13	5.43e-16	6.453
Sesam	250.786	-0.87	2.48294e+06	6.32e-21	7.50e-20	8.72e-20	7.95e-11	-2.74e-10	2.95e-10

Shell, 1st order, QUAD, reduced integration

Mode 1

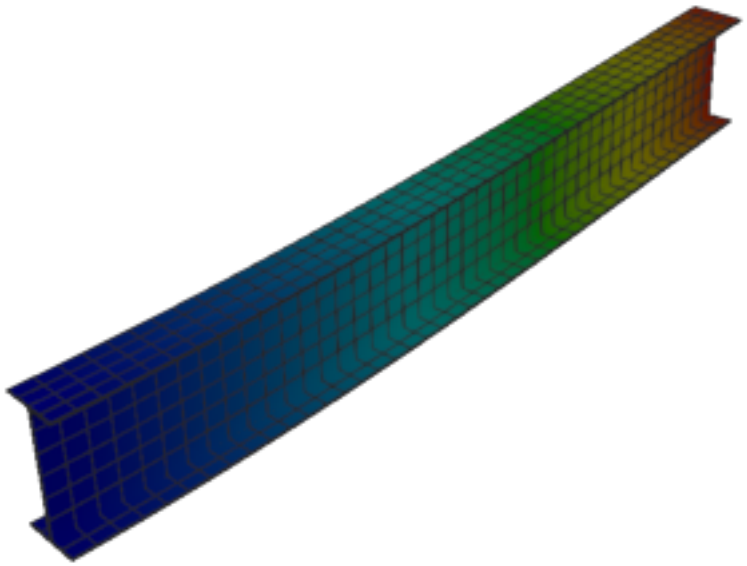


Figure 159: Abaqus — mode 1, 12.373 Hz

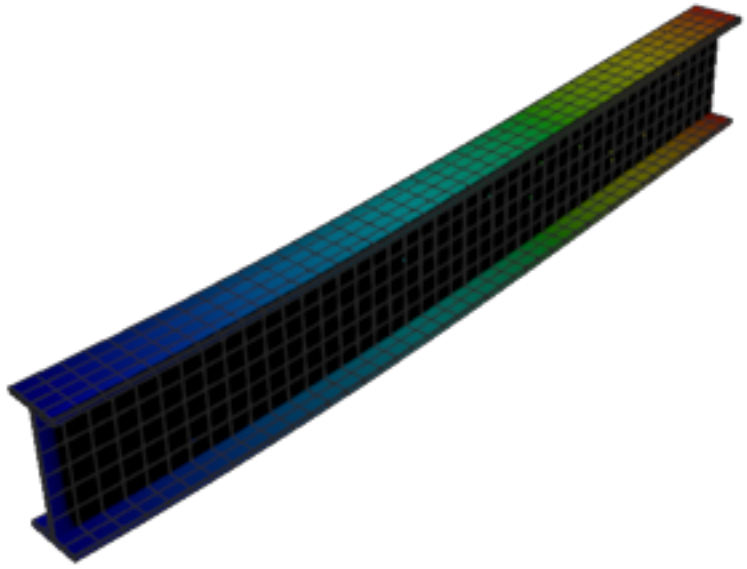


Figure 160: Calculix — mode 1, 14.256 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.373	-7.07	6043.5	1.78e-28	119.95	5.35e-25	-1.33e-14	10.952	-7.31e-13
Calculix	14.256	+7.07	8023.13	1.87e-26	119.90	6.12e-22	-1.37e-13	10.950	2.47e-11

Mode 2

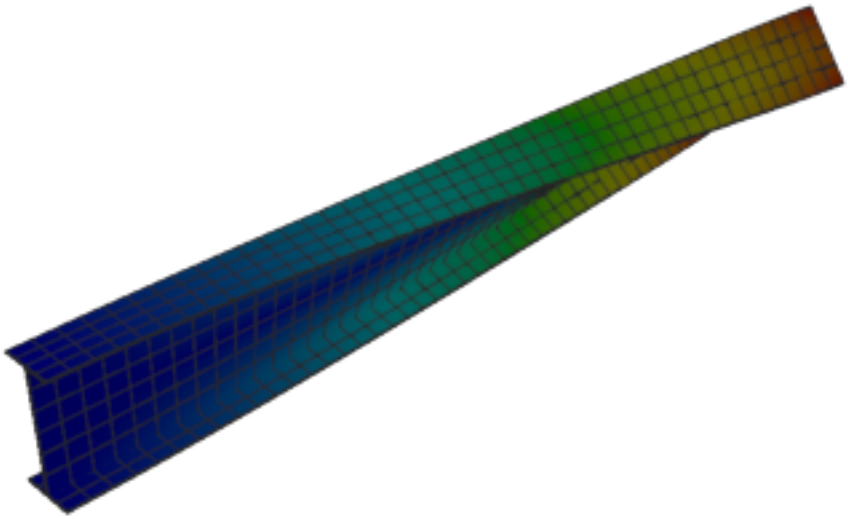


Figure 161: Abaqus — mode 2, 19.230 Hz

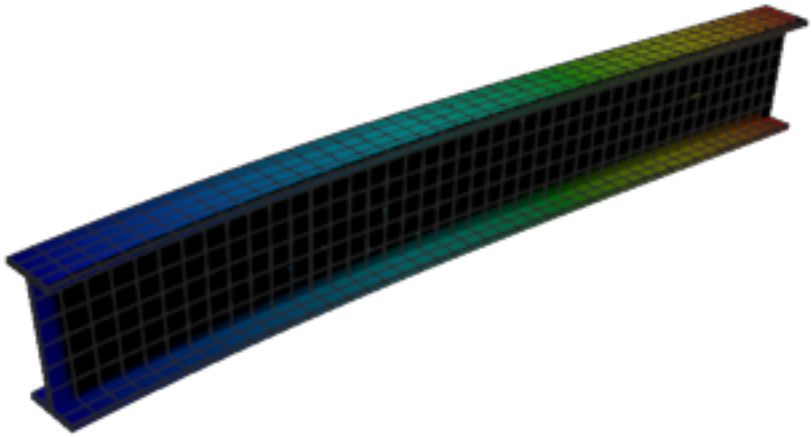


Figure 162: Calculix — mode 2, 53.262 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.230	-46.95	14599	1.25e-26	4.60e-21	3.10e-26	-1.12e-13	6.78e-11	1.76e-13
Calculix	53.262	+46.95	111993	4.93e-27	8.03e-22	122.13	-7.02e-14	2.83e-11	-11.051

Mode 3

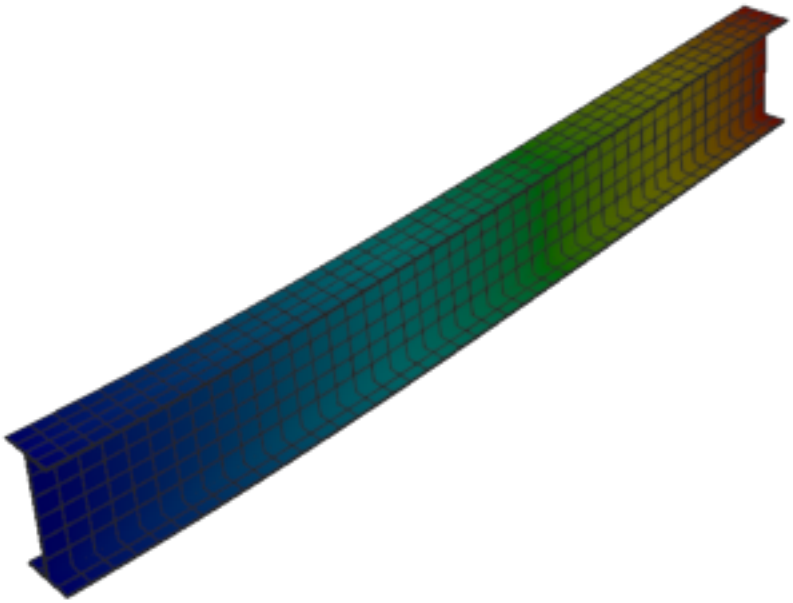


Figure 163: Abaqus — mode 3, 51.849 Hz

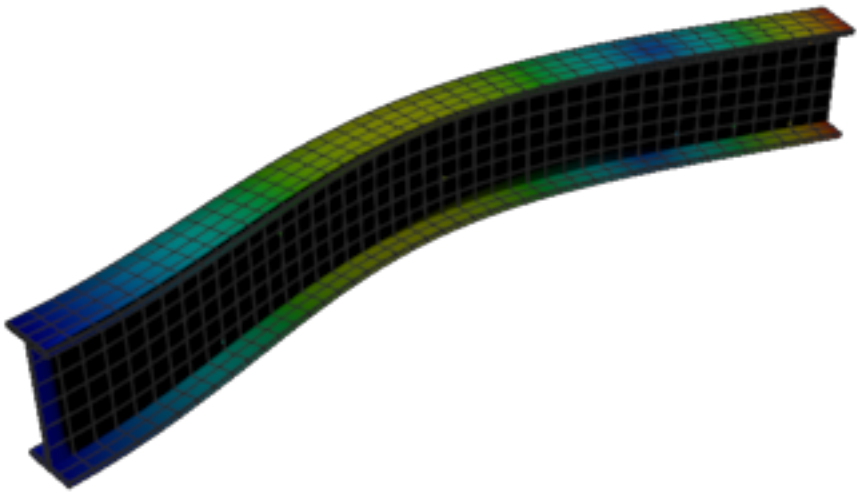


Figure 164: Calculix — mode 3, 87.915 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.849	-25.81	106129	2.48e-26	1.19e-25	122.02	-1.58e-13	3.45e-13	11.046
Calculix	87.915	+25.81	305133	8.49e-26	37.23	4.65e-23	-2.91e-13	6.101	6.82e-12

Mode 4

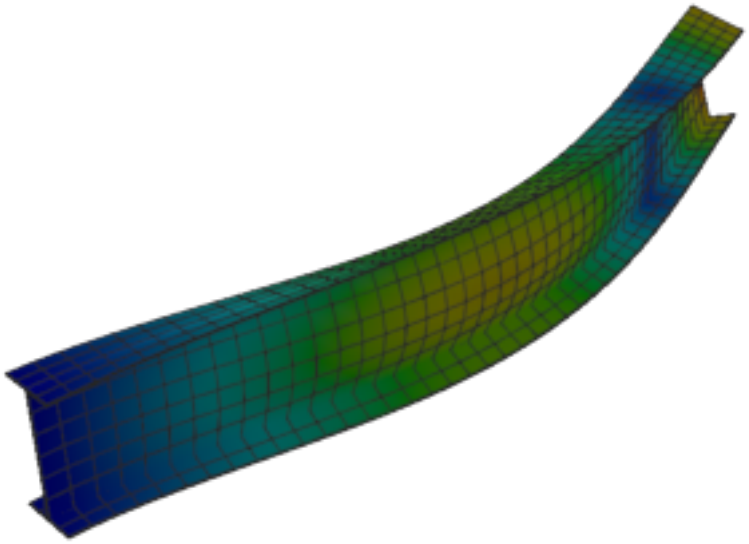


Figure 165: Abaqus — mode 4, 73.558 Hz

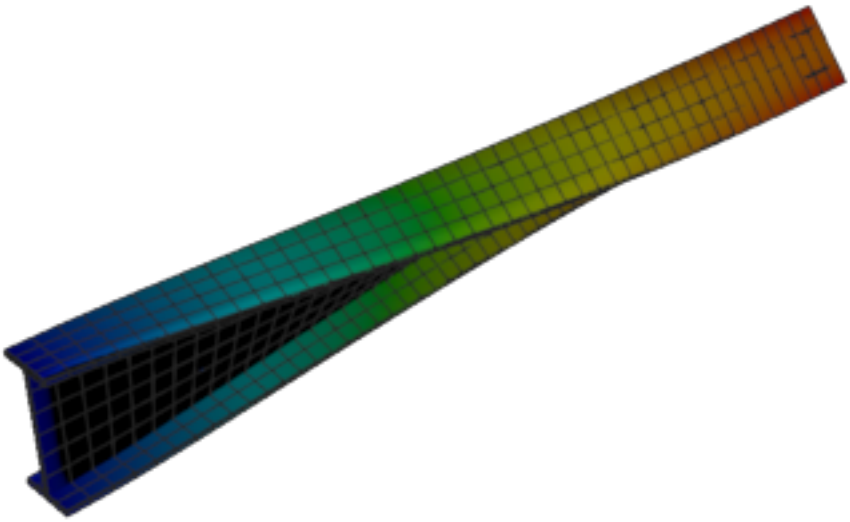


Figure 166: Calculix — mode 4, 121.850 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	73.558	-24.71	213611	2.90e-28	36.98	6.94e-25	-1.70e-14	-6.081	-8.33e-13
Calculix	121.850	+24.71	586156	6.61e-28	4.01e-22	9.84e-24	2.57e-14	2.00e-11	3.14e-12

Mode 5

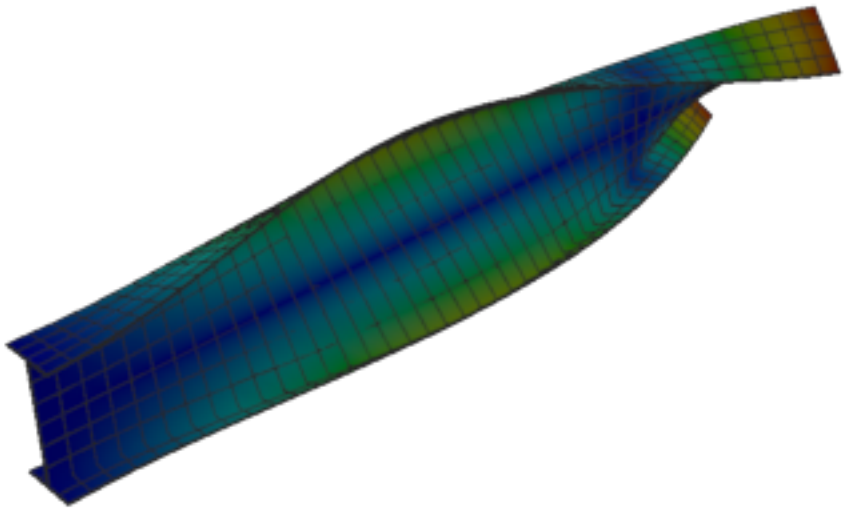


Figure 167: Abaqus — mode 5, 93.709 Hz

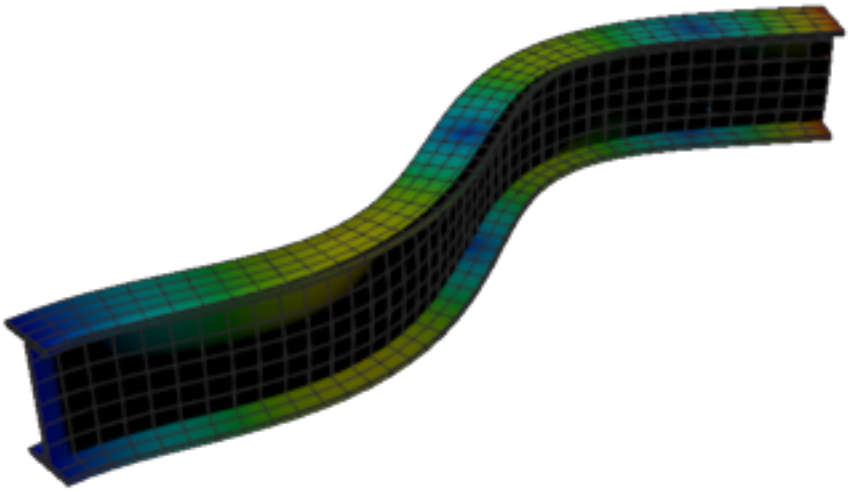


Figure 168: Calculix — mode 5, 239.804 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	93.709	-43.80	346673	9.12e-28	8.02e-23	5.37e-26	3.02e-14	-8.96e-12	2.32e-13
Calculix	239.804	+43.80	2.27025e+06	1.53e-26	12.96	3.22e-24	-1.24e-13	-3.601	1.80e-12

Mode 6

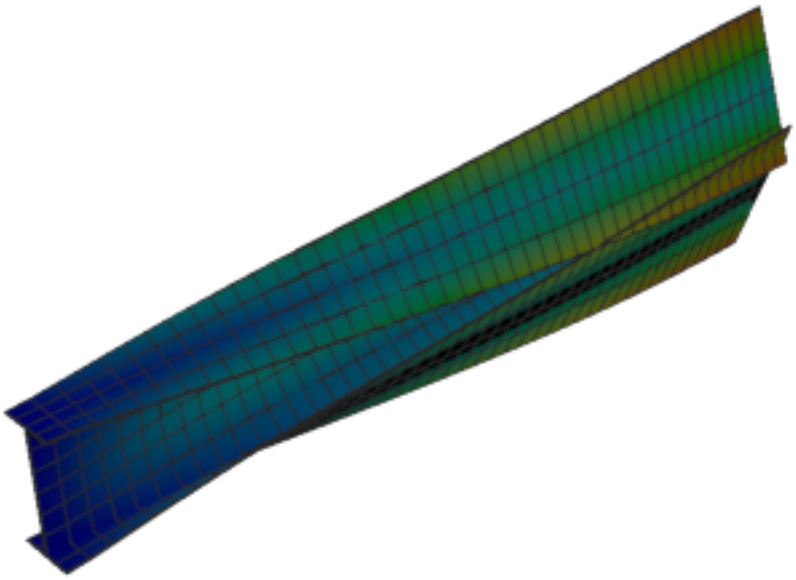


Figure 169: Abaqus — mode 6, 126.870 Hz

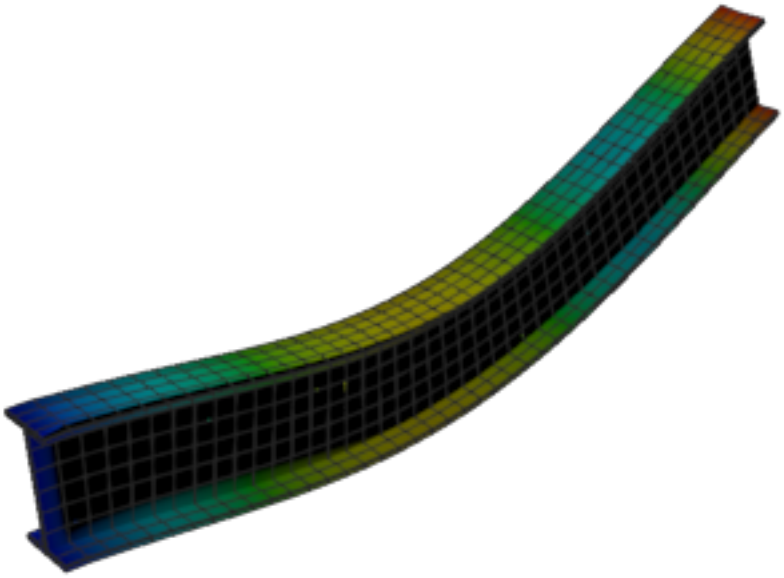


Figure 170: Calculix — mode 6, 262.830 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	126.870	-34.89	635447	1.80e-30	0.01	1.84e-25	1.34e-15	-0.078	-4.29e-13
Calculix	262.830	+34.89	2.72716e+06	2.80e-26	1.10e-24	41.52	1.67e-13	-1.05e-12	-6.443

Mode 7

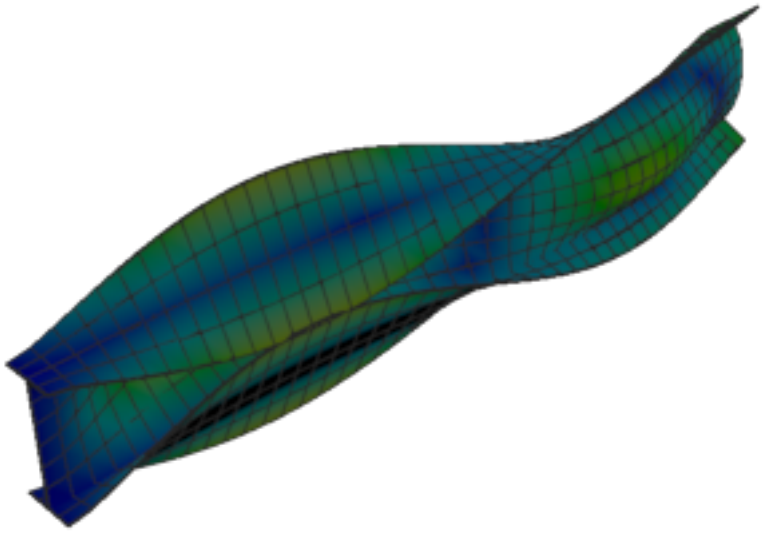


Figure 171: Abaqus — mode 7, 144.579 Hz

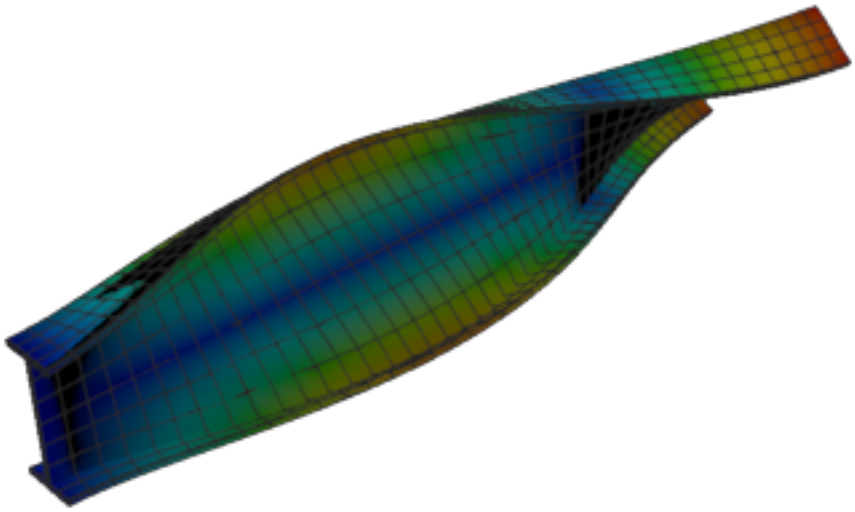


Figure 172: Calculix — mode 7, 365.100 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	144.580	-43.27	825226	1.99e-29	3.79	2.06e-26	-4.46e-15	1.947	1.43e-13
Calculix	365.100	+43.27	5.26239e+06	1.49e-25	6.79e-24	3.09e-25	-3.86e-13	-2.61e-12	5.56e-13

Mode 8

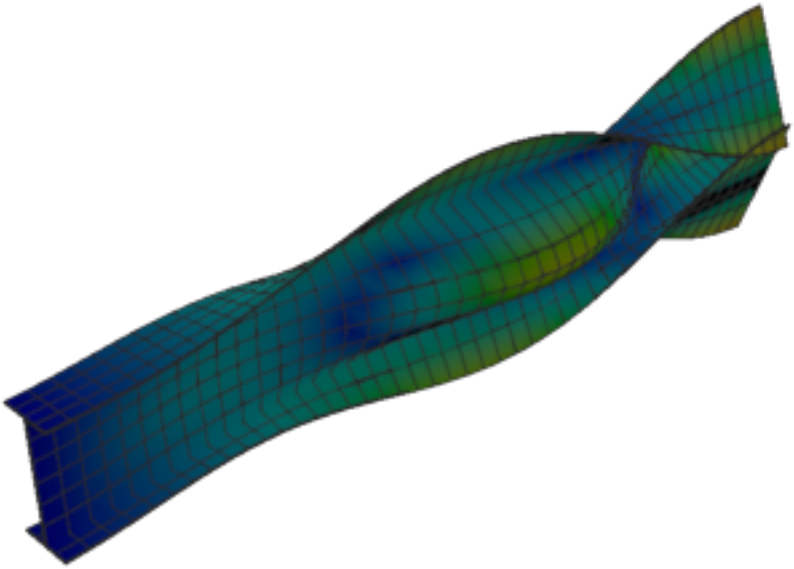


Figure 173: Abaqus — mode 8, 168.379 Hz

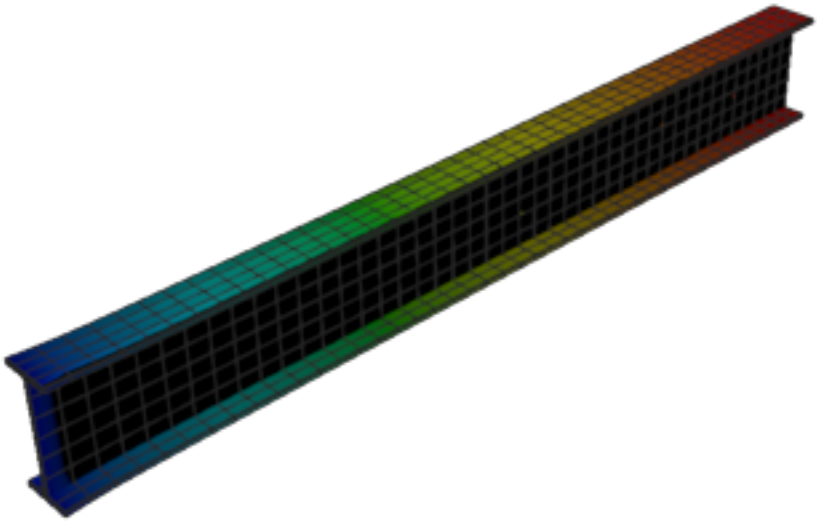


Figure 174: Calculix — mode 8, 431.774 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	168.380	-43.89	1.11927e+06	2.51e-28	5.44	1.97e-25	1.58e-14	2.332	-4.43e-13
Calculix	431.774	+43.89	7.35992e+06	157.96	2.40e-26	6.49e-27	12.568	1.55e-13	8.05e-14

Mode 9

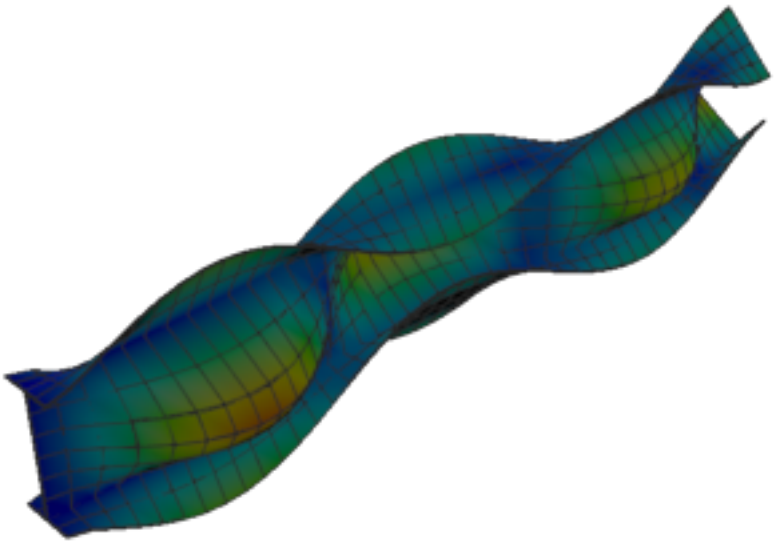


Figure 175: Abaqus — mode 9, 197.515 Hz

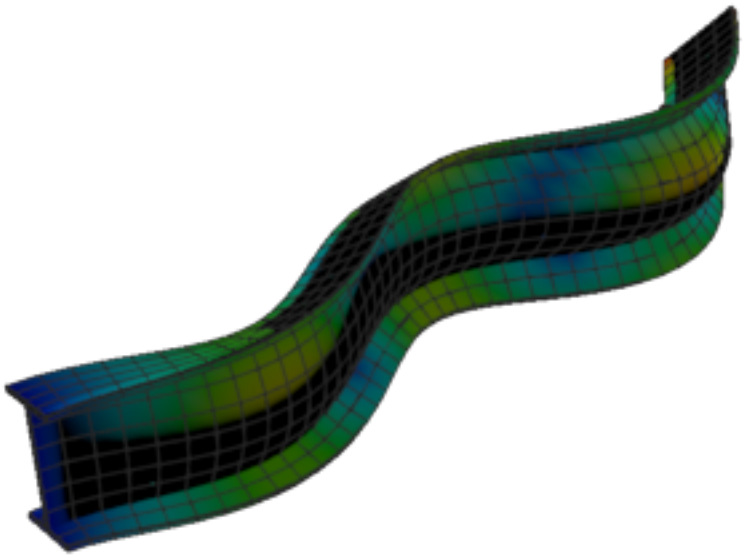


Figure 176: Calculix — mode 9, 451.745 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	197.510	-39.16	1.54014e+06	1.19e-26	0.62	1.84e-25	-1.09e-13	-0.786	4.29e-13
Calculix	451.745	+39.16	8.0565e+06	7.61e-25	5.70	3.22e-25	-8.72e-13	-2.388	-5.67e-13

Mode 10

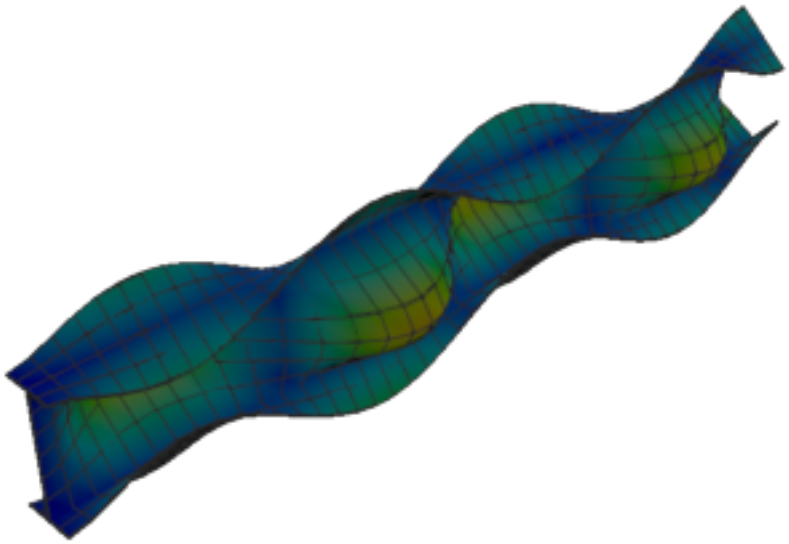


Figure 177: Abaqus — mode 10, 241.275 Hz

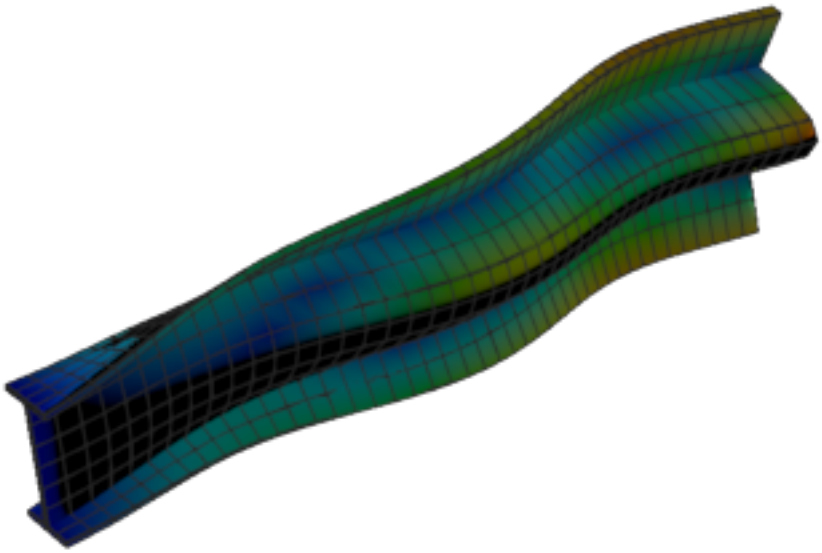


Figure 178: Calculix — mode 10, 464.440 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	241.270	-31.62	2.29818e+06	2.67e-22	1.61	1.69e-20	-1.63e-11	1.269	1.30e-10
Calculix	464.440	+31.62	8.51568e+06	1.15e-24	1.01	2.87e-24	-1.07e-12	1.006	-1.69e-12

Mode 11

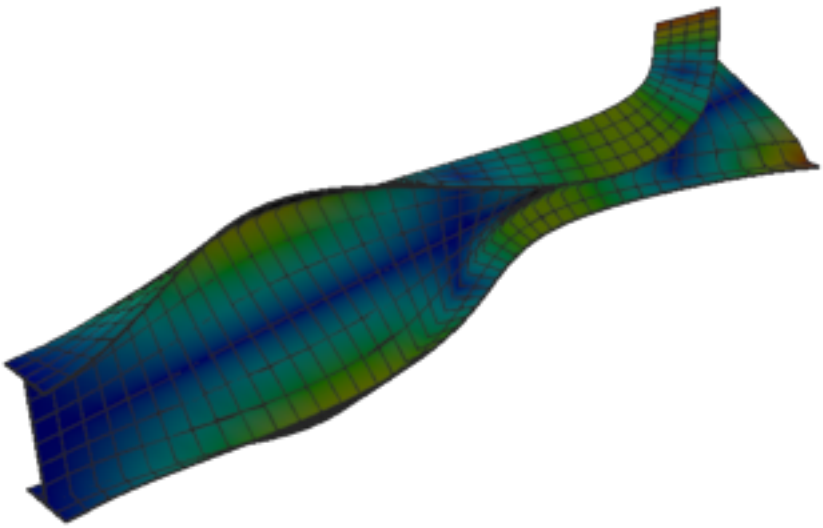


Figure 179: Abaqus — mode 11, 245.071 Hz

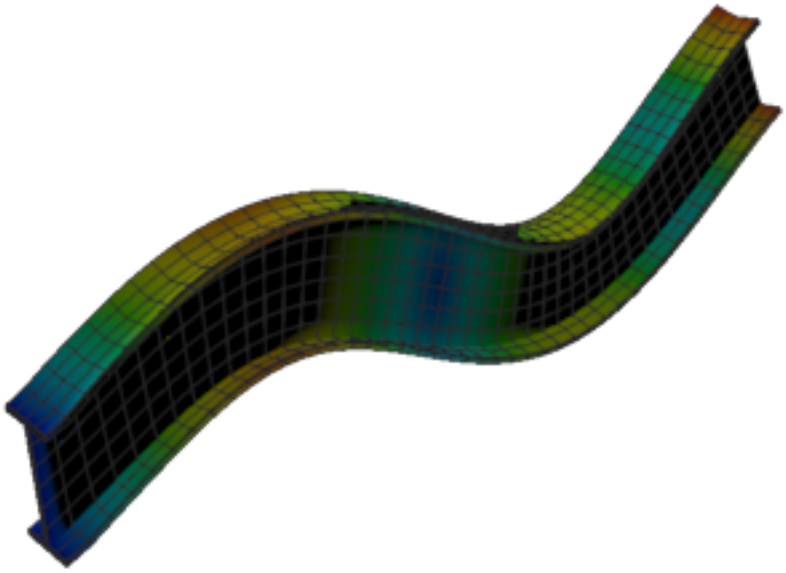


Figure 180: Calculix — mode 11, 589.240 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	245.070	-41.25	2.37106e+06	3.55e-23	8.35e-21	8.47e-21	-5.95e-12	9.14e-11	9.20e-11
Calculix	589.240	+41.25	1.3707e+07	3.93e-26	1.86e-26	12.88	-1.98e-13	1.36e-13	3.589

Shell, 2nd order, TRI

Mode 1

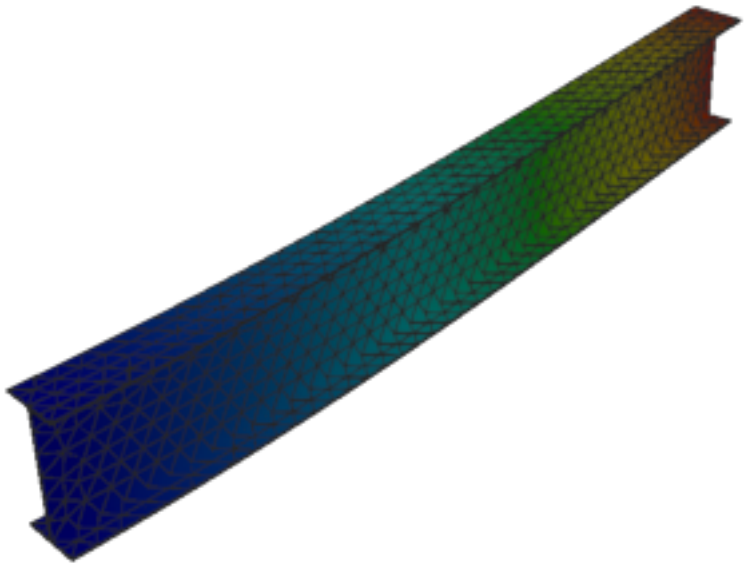


Figure 181: Abaqus — mode 1, 12.756 Hz

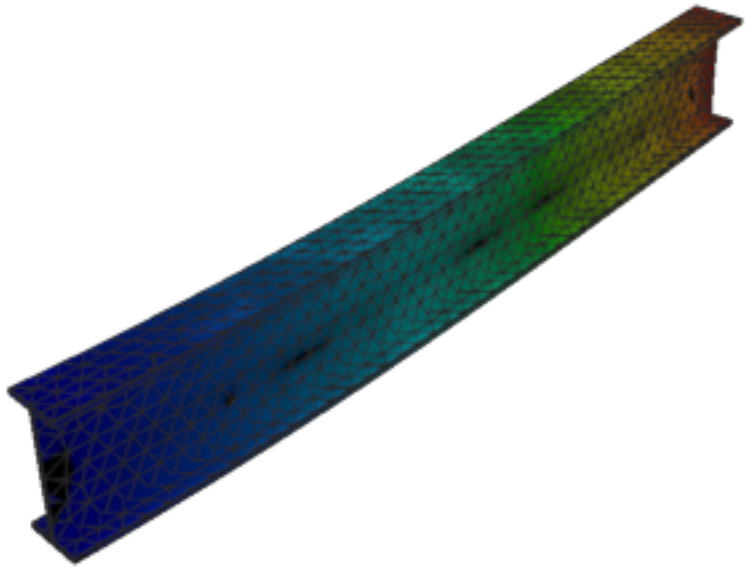


Figure 182: Calculix — mode 1, 12.756 Hz

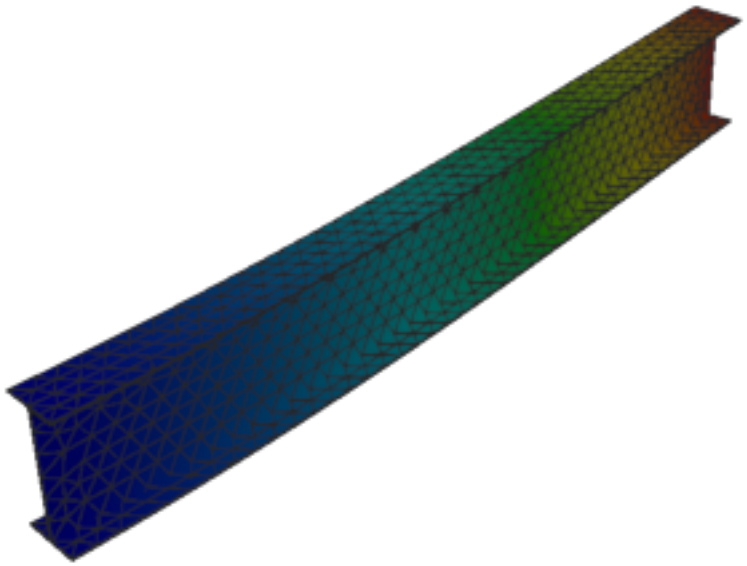


Figure 183: Code_Aster — mode 1, 12.757 Hz

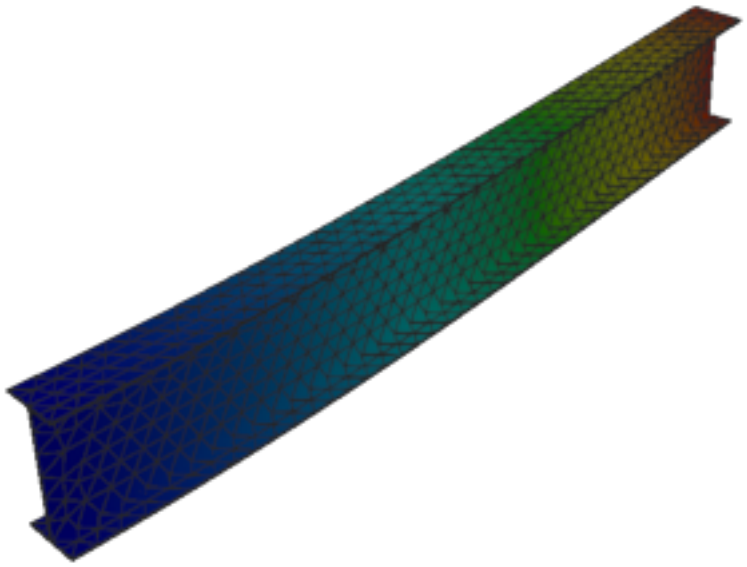


Figure 184: Sesam — mode 1, 12.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.756	-1.04e-03	6423.3	2.07e-18	119.94	3.02e-12	1.44e-09	10.952	1.74e-06
Calculix	12.756	4.50e-04	6423.95	1.62e-16	119.94	1.22e-12	-1.27e-08	10.952	1.11e-06
Code_Aster	12.757	3.43e-03	6424.33	2.51e-26	119.94	4.99e-26	1.59e-13	10.952	-2.23e-13
Sesam	12.756	-2.84e-03	6423.52	2.49e-28	119.94	4.86e-24	-1.58e-14	10.952	-2.20e-12

Mode 2

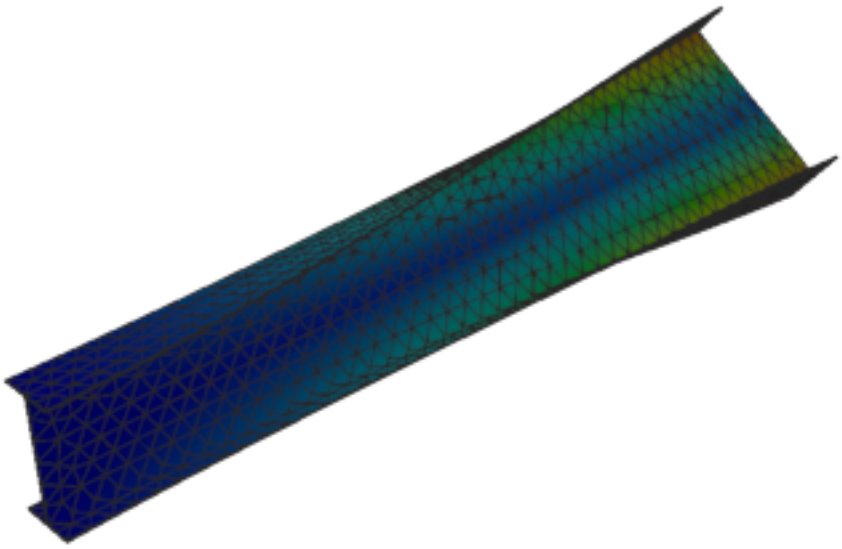


Figure 185: Abaqus — mode 2, 19.567 Hz

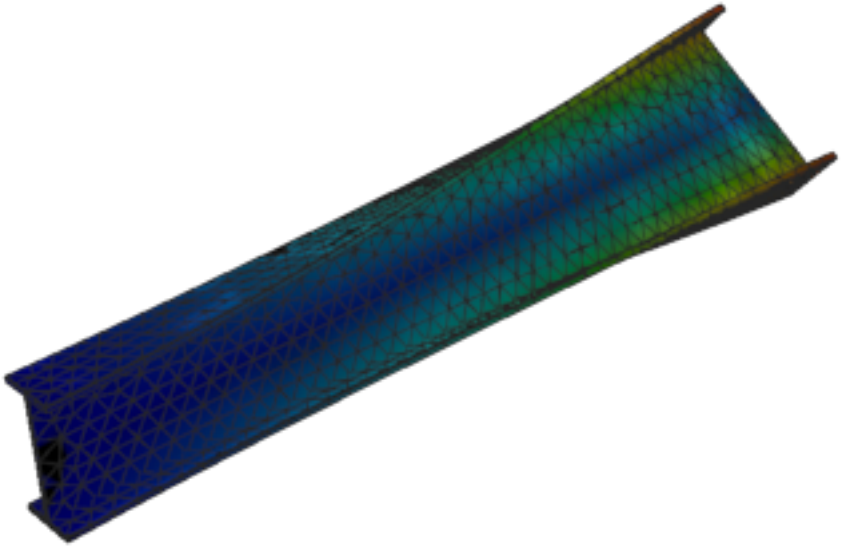


Figure 186: Calculix — mode 2, 19.525 Hz

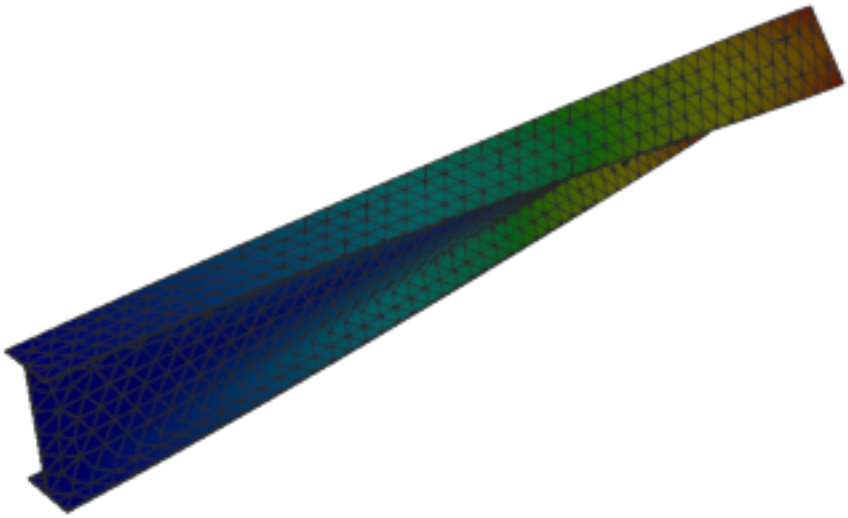


Figure 187: Code_Aster — mode 2, 19.451 Hz

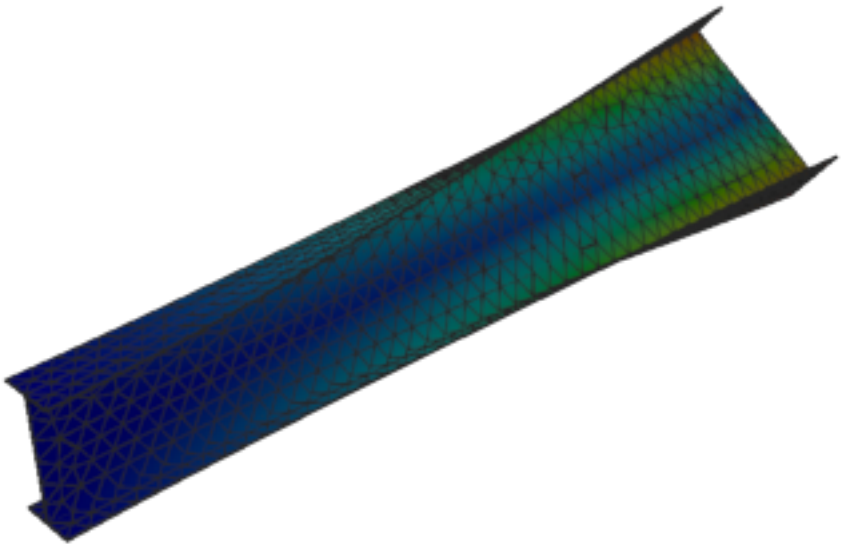


Figure 188: Sesam — mode 2, 19.505 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.567	+0.28	15115	3.71e-13	3.08e-10	4.41e-12	6.09e-07	1.75e-05	-2.10e-06
Calculix	19.525	+0.07	15050.8	2.96e-09	2.59e-09	1.08e-14	-5.44e-05	5.09e-05	-1.04e-07
Code_Aster	19.451	-0.31	14936.2	1.12e-26	4.57e-09	2.69e-23	-1.06e-13	-6.76e-05	5.19e-12
Sesam	19.505	-0.04	15019.1	1.90e-25	1.78e-08	2.63e-25	4.36e-13	1.33e-04	5.13e-13

Mode 3

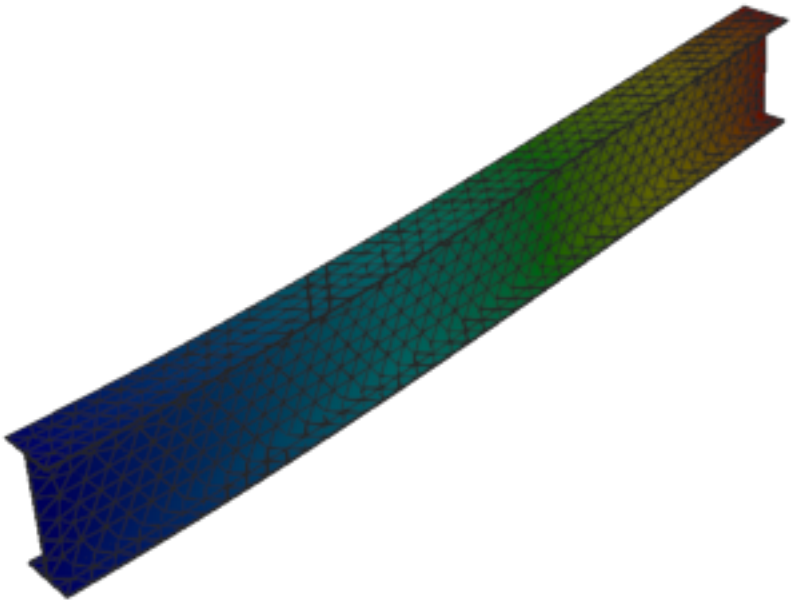


Figure 189: Abaqus — mode 3, 51.969 Hz

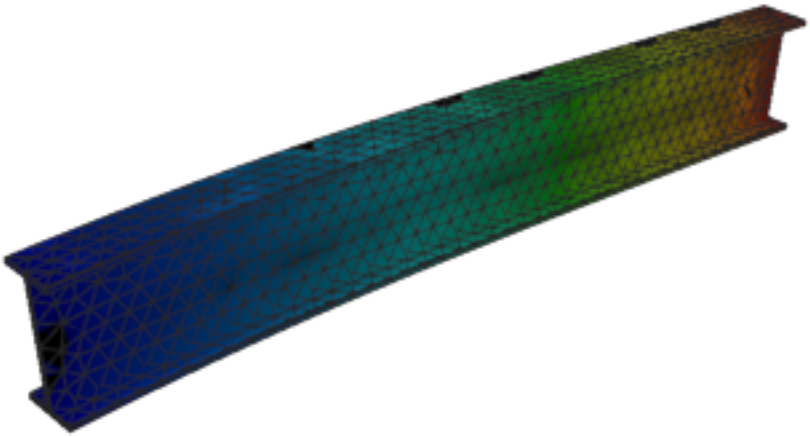


Figure 190: Calculix — mode 3, 51.974 Hz

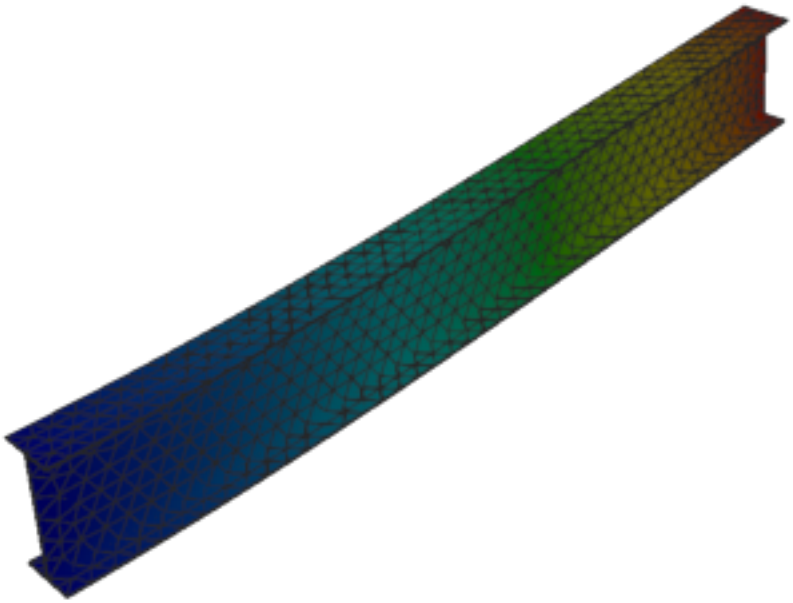


Figure 191: Code_Aster — mode 3, 51.970 Hz

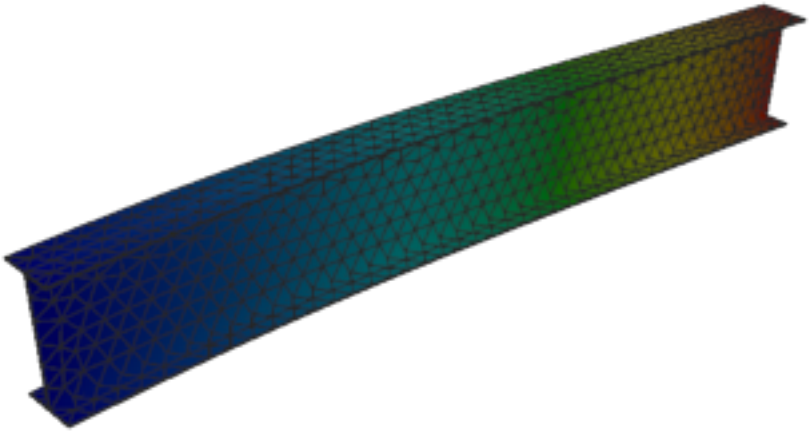


Figure 192: Sesam — mode 3, 51.970 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.969	-3.58e-03	106622	4.50e-14	9.00e-10	122.03	2.12e-07	3.00e-05	11.047
Calculix	51.974	+0.01	106643	5.80e-14	4.56e-11	122.02	-2.41e-07	6.75e-06	-11.046
Code_Aster	51.970	-1.63e-03	106627	3.55e-14	1.48e-24	122.03	1.89e-07	1.21e-12	11.047
Sesam	51.970	-9.06e-04	106628	6.59e-14	1.01e-23	122.03	-2.57e-07	-3.18e-12	-11.047

Mode 4

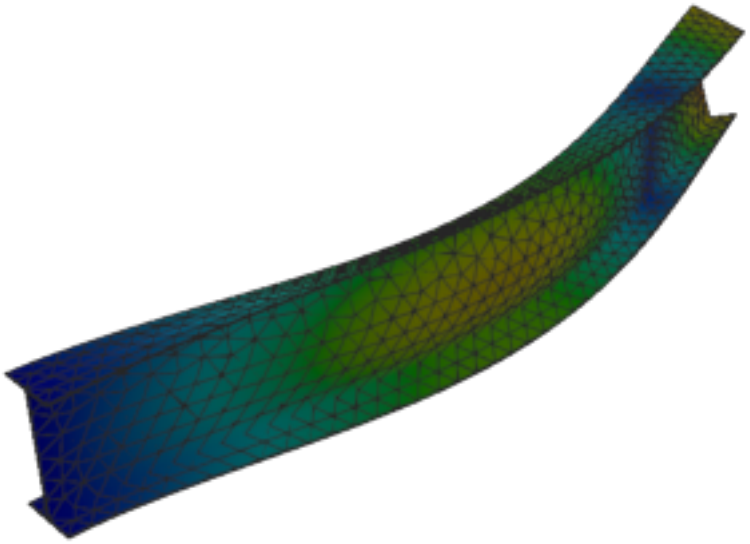


Figure 193: Abaqus — mode 4, 75.145 Hz

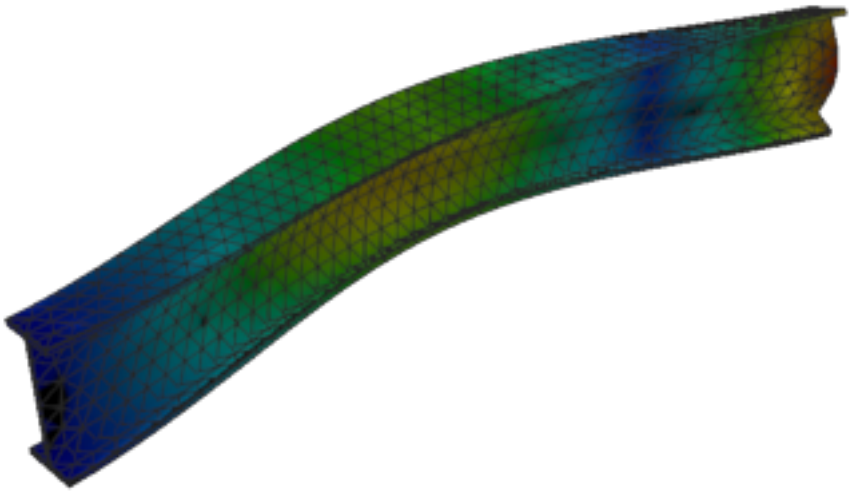


Figure 194: Calculix — mode 4, 75.185 Hz

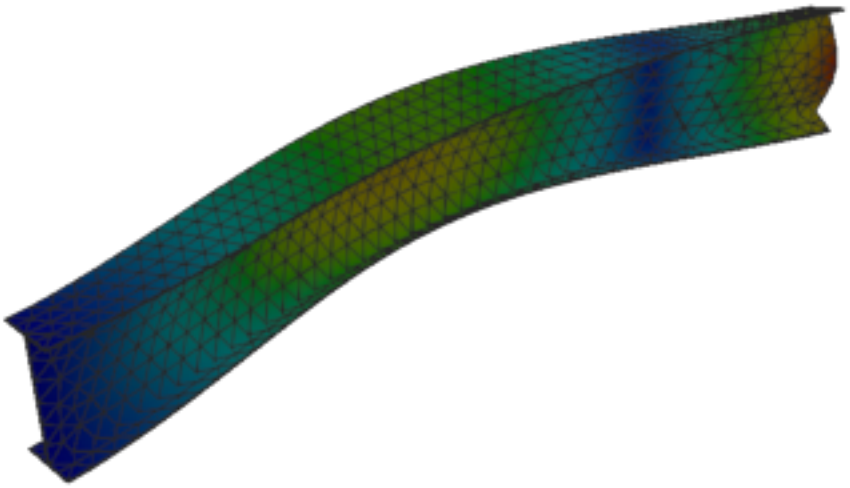


Figure 195: Code_Aster — mode 4, 75.144 Hz

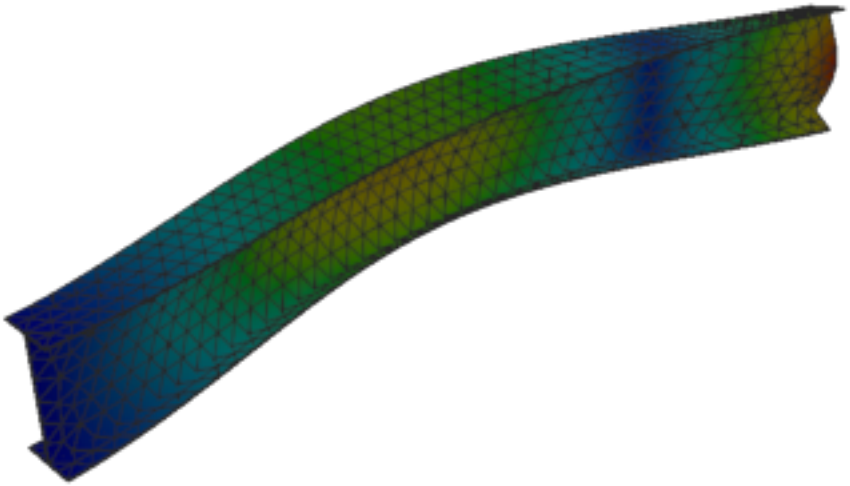


Figure 196: Sesam — mode 4, 75.172 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.145	-0.02	222925	1.24e-16	36.88	1.80e-09	1.11e-08	-6.073	4.24e-05
Calculix	75.185	+0.03	223162	9.99e-14	36.87	1.02e-12	-3.16e-07	6.072	-1.01e-06
Code_Aster	75.144	-0.02	222921	1.20e-26	36.86	3.28e-24	-1.10e-13	6.071	-1.81e-12
Sesam	75.172	+0.01	223083	1.70e-25	36.87	2.56e-24	-4.12e-13	6.072	-1.60e-12

Mode 5

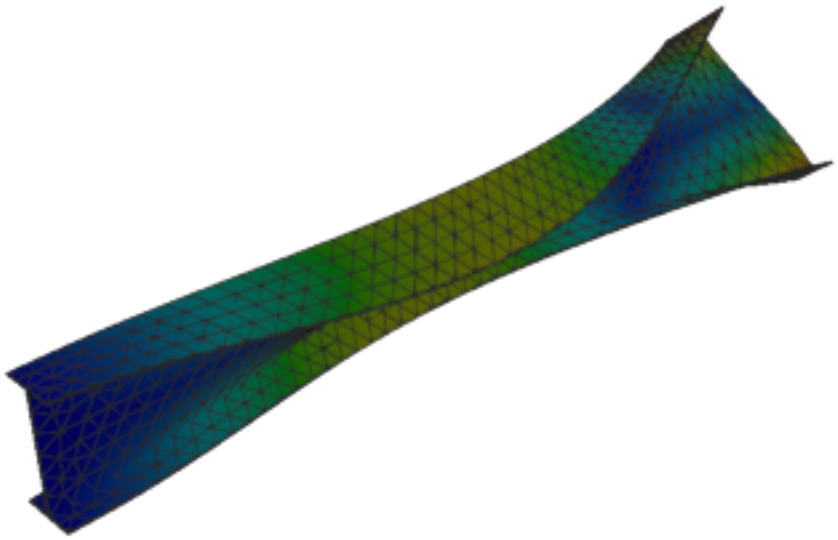


Figure 197: Abaqus — mode 5, 96.001 Hz

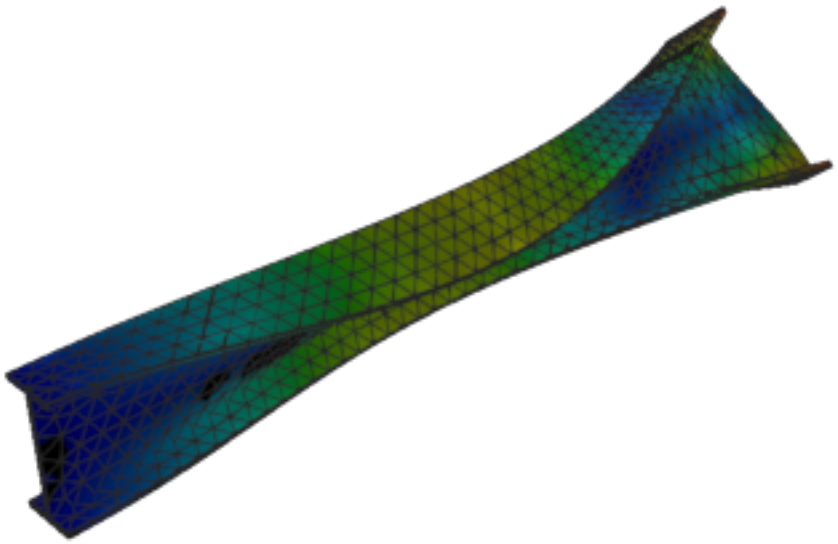


Figure 198: Calculix — mode 5, 95.971 Hz

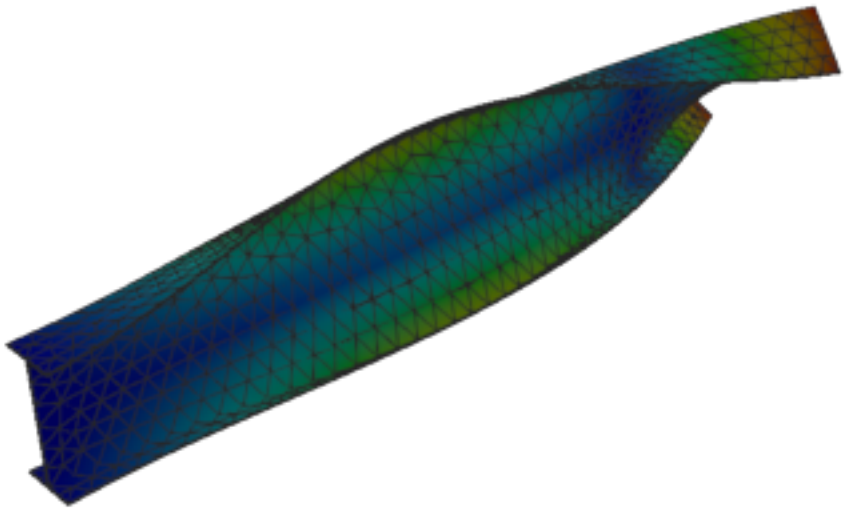


Figure 199: Code_Aster — mode 5, 95.871 Hz

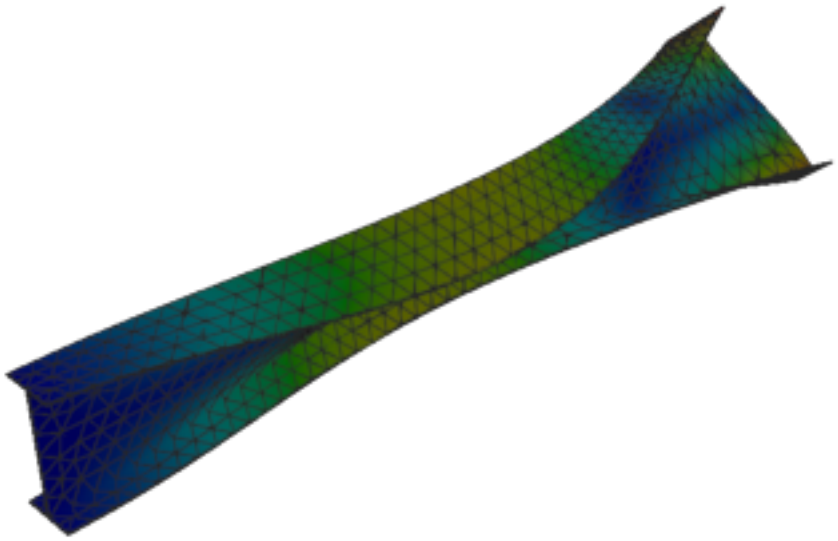


Figure 200: Sesam — mode 5, 95.934 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.001	+0.06	363843	8.08e-12	6.17e-09	5.02e-12	-2.84e-06	-7.85e-05	2.24e-06
Calculix	95.971	+0.03	363616	8.10e-10	1.26e-10	3.55e-15	2.85e-05	1.12e-05	-5.96e-08
Code_Aster	95.871	-0.08	362854	5.98e-26	1.61e-08	1.06e-24	-2.45e-13	-1.27e-04	-1.03e-12
Sesam	95.934	-0.01	363330	5.45e-26	1.06e-09	5.71e-24	-2.34e-13	3.26e-05	2.39e-12

Mode 6

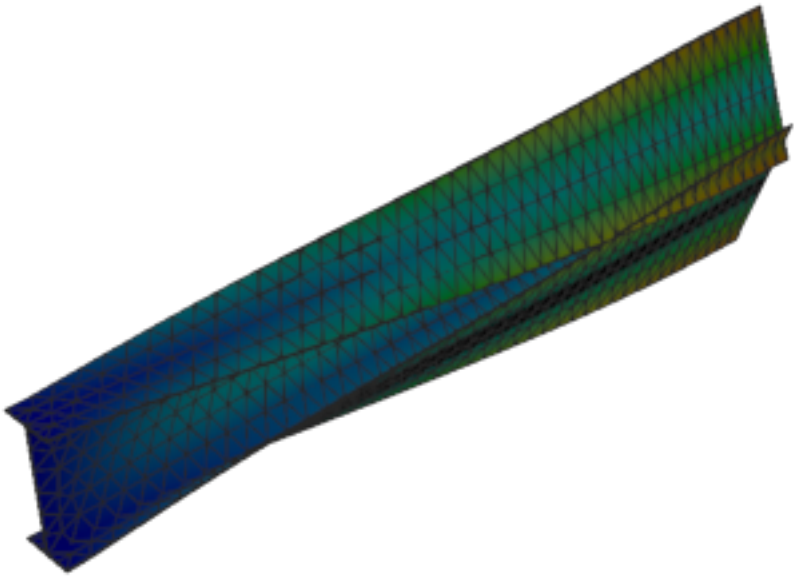


Figure 201: Abaqus — mode 6, 125.207 Hz

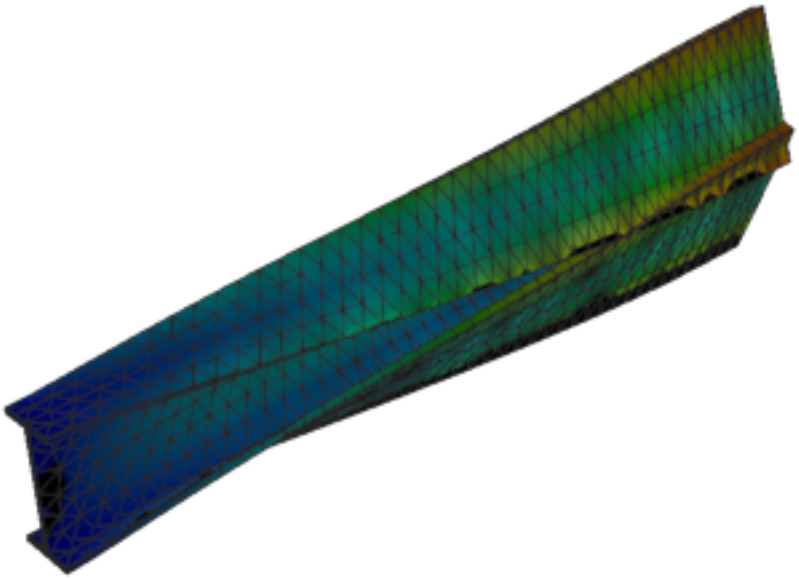


Figure 202: Calculix — mode 6, 125.465 Hz

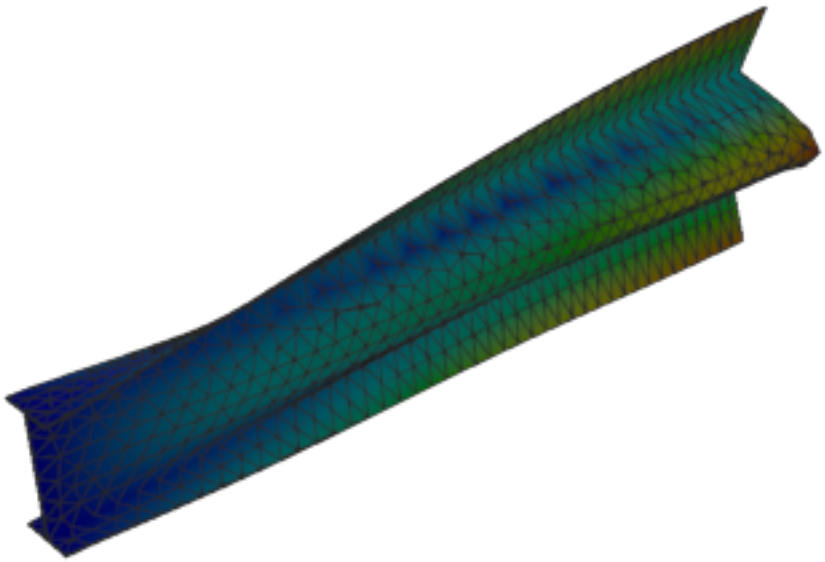


Figure 203: Code_Aster — mode 6, 125.087 Hz

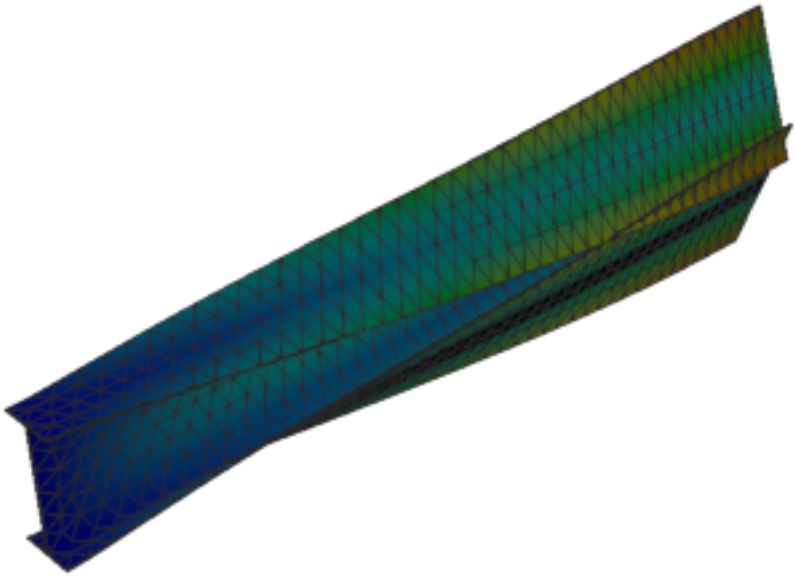


Figure 204: Sesam — mode 6, 125.173 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.210	-0.02	618895	4.62e-16	0.02	1.76e-10	-2.15e-08	-0.138	-1.33e-05
Calculix	125.465	+0.18	621448	3.80e-12	0.02	1.15e-13	-1.95e-06	-0.144	-3.39e-07
Code_Aster	125.087	-0.12	617709	1.54e-26	0.02	4.07e-24	1.24e-13	0.142	-2.02e-12
Sesam	125.173	-0.05	618555	1.15e-26	0.02	8.07e-23	1.07e-13	-0.144	8.98e-12

Mode 7

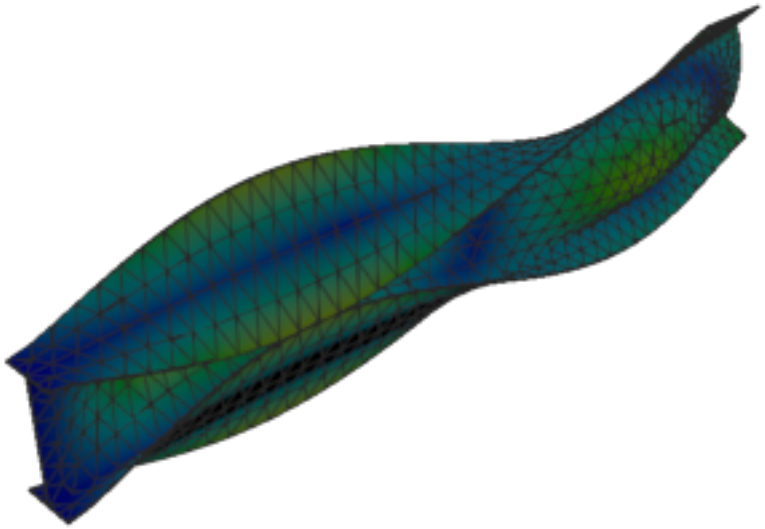


Figure 205: Abaqus — mode 7, 142.773 Hz

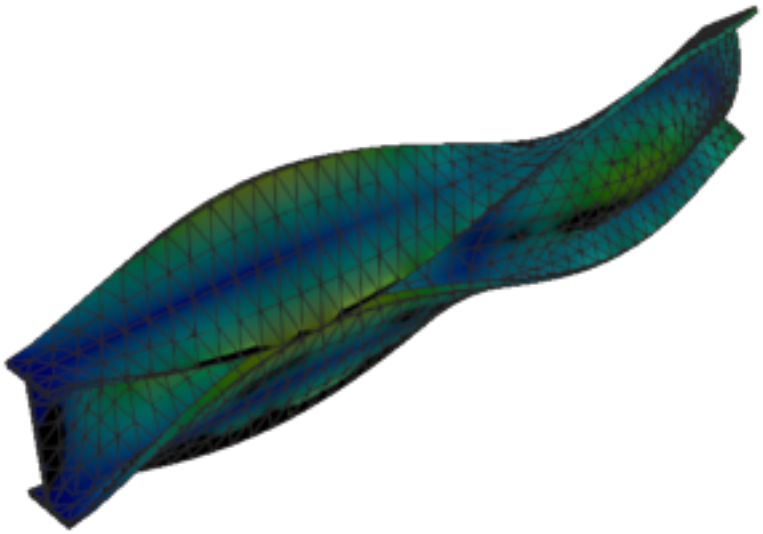


Figure 206: Calculix — mode 7, 142.995 Hz

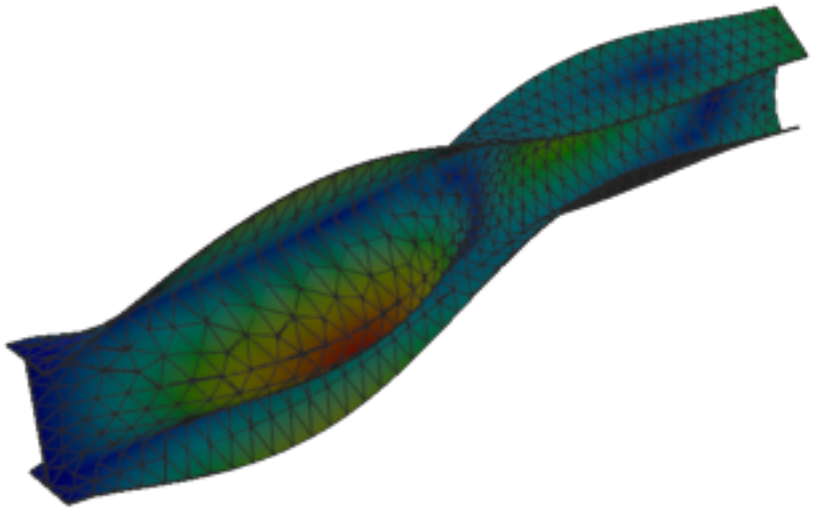


Figure 207: Code_Aster — mode 7, 142.359 Hz

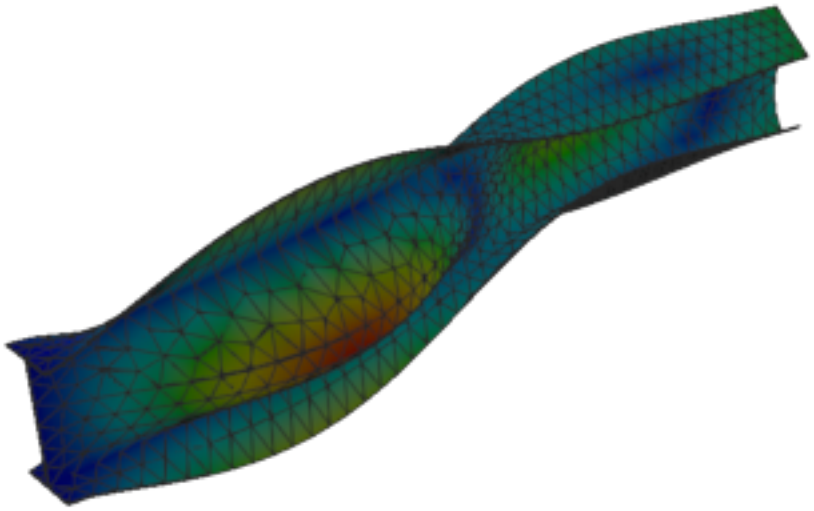


Figure 208: Sesam — mode 7, 142.768 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.770	+0.03	804737	9.41e-15	3.54	1.08e-09	-9.70e-08	1.880	3.28e-05
Calculix	142.995	+0.19	807237	7.68e-13	3.41	5.68e-13	-8.76e-07	1.848	7.54e-07
Code_Aster	142.359	-0.26	800074	2.54e-26	3.42	3.14e-25	-1.59e-13	-1.849	5.60e-13
Sesam	142.768	+0.03	804681	4.61e-26	3.42	1.06e-23	-2.15e-13	-1.850	3.26e-12

Mode 8

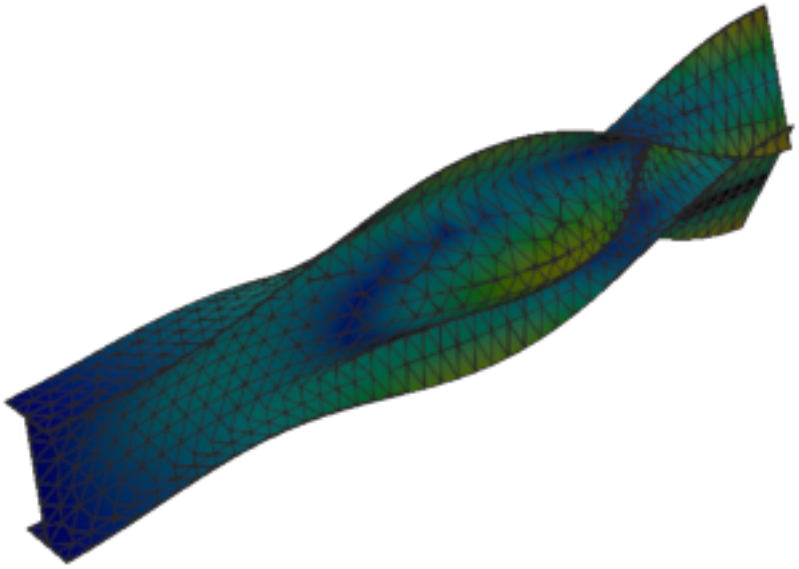


Figure 209: Abaqus — mode 8, 166.598 Hz

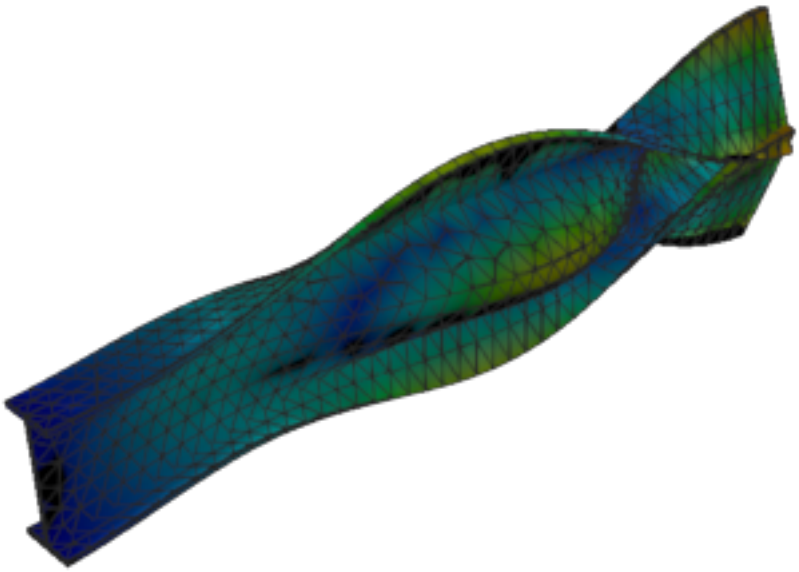


Figure 210: Calculix — mode 8, 166.579 Hz

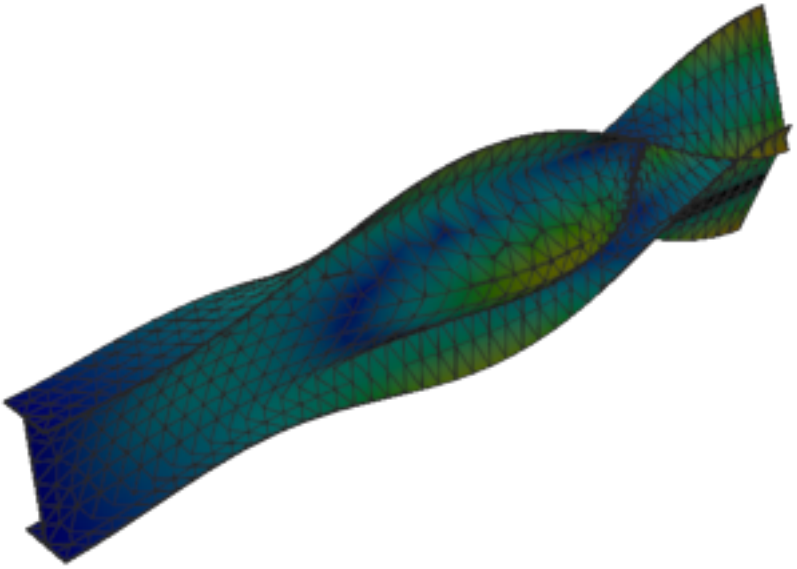


Figure 211: Code_Aster — mode 8, 165.790 Hz

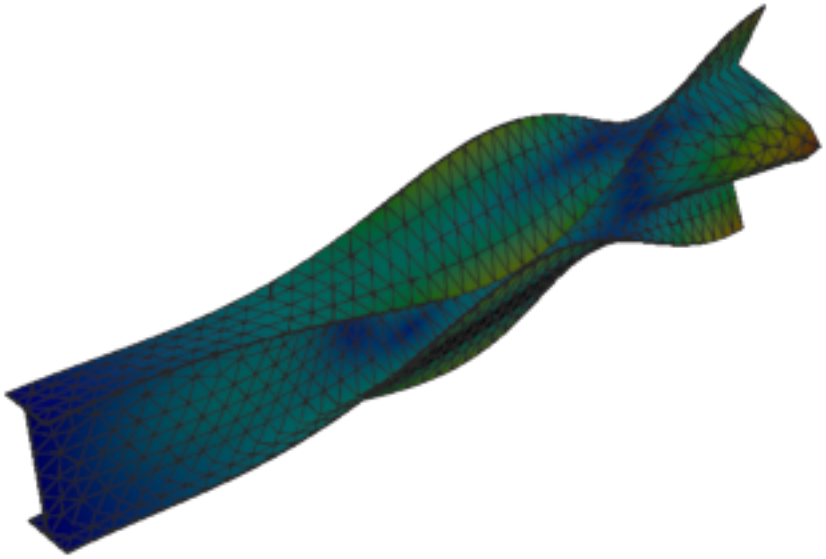


Figure 212: Sesam — mode 8, 166.334 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	166.600	+0.17	1.09572e+06	1.06e-14	5.52	1.13e-08	-1.03e-07	2.349	1.06e-04
Calculix	166.579	+0.15	1.09547e+06	5.11e-13	5.55	2.25e-13	7.15e-07	2.356	-4.75e-07
Code_Aster	165.790	-0.32	1.08511e+06	1.18e-26	5.54	3.14e-25	-1.09e-13	2.353	5.60e-13
Sesam	166.334	4.85e-03	1.09224e+06	7.25e-26	5.56	9.43e-24	2.69e-13	-2.358	-3.07e-12

Mode 9

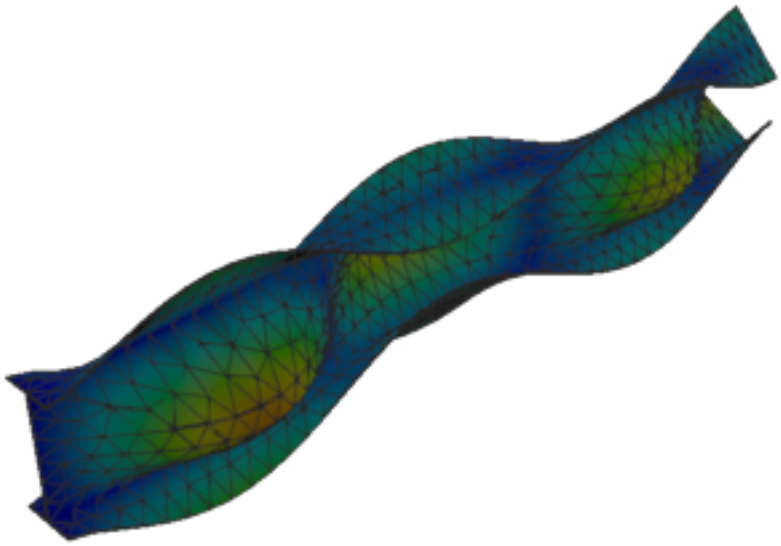


Figure 213: Abaqus — mode 9, 192.840 Hz

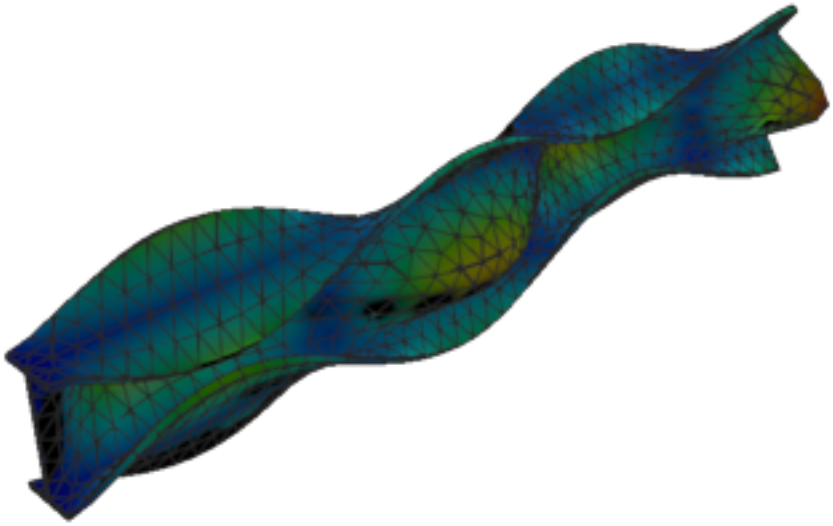


Figure 214: Calculix — mode 9, 194.014 Hz

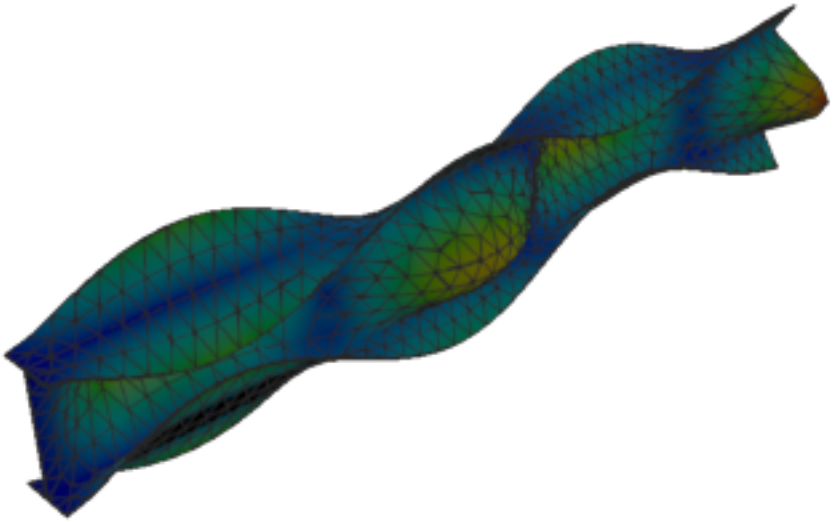


Figure 215: Code_Aster — mode 9, 192.406 Hz

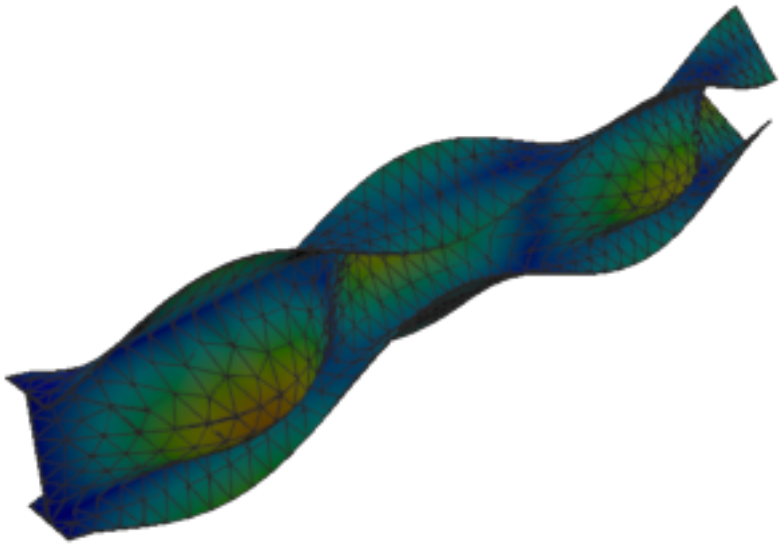


Figure 216: Sesam — mode 9, 193.308 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.840	-0.16	1.4681e+06	5.91e-14	0.67	5.79e-09	2.43e-07	-0.818	-7.61e-05
Calculix	194.014	+0.45	1.48603e+06	4.33e-13	0.63	7.51e-14	-6.58e-07	0.793	-2.74e-07
Code_Aster	192.406	-0.38	1.46149e+06	2.47e-27	0.64	8.63e-26	-4.97e-14	0.801	-2.94e-13
Sesam	193.308	+0.09	1.47524e+06	1.19e-27	0.64	3.81e-25	3.44e-14	-0.801	6.18e-13

Mode 10

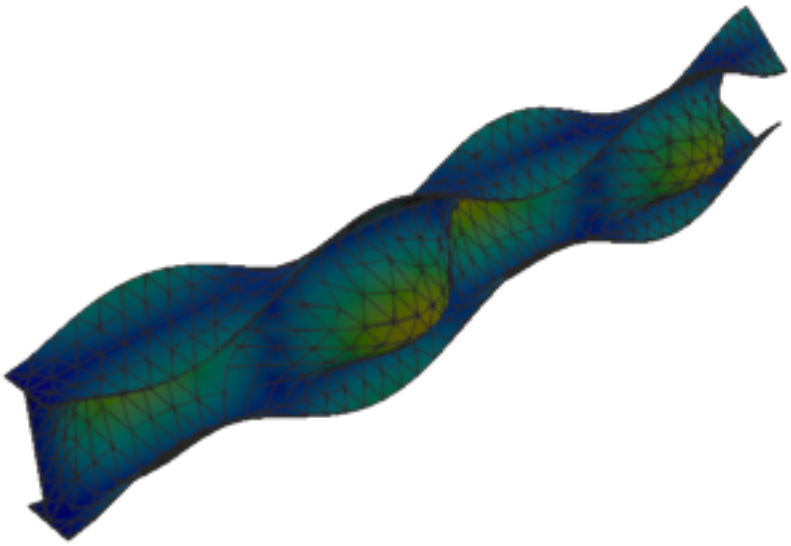


Figure 217: Abaqus — mode 10, 233.355 Hz

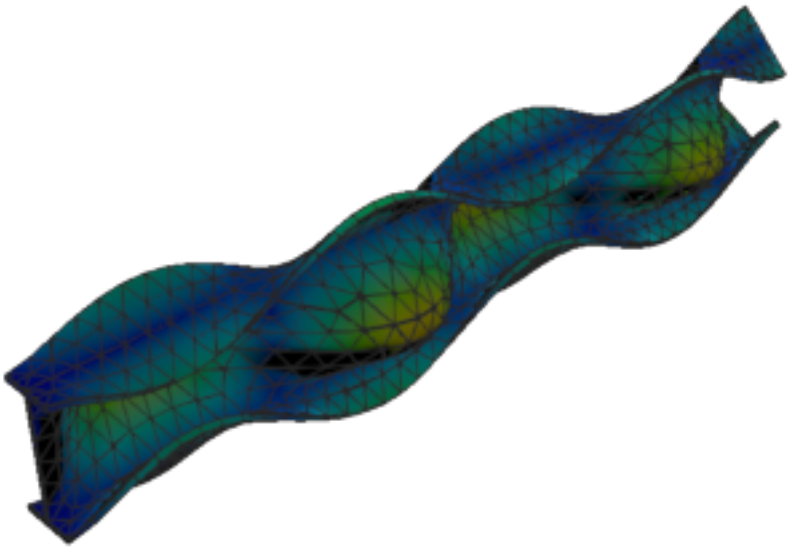


Figure 218: Calculix — mode 10, 236.042 Hz

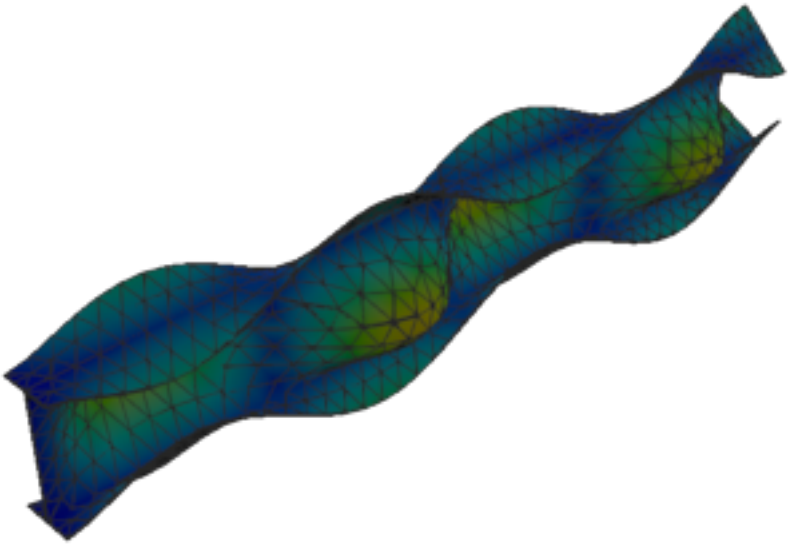


Figure 219: Code_Aster — mode 10, 233.389 Hz

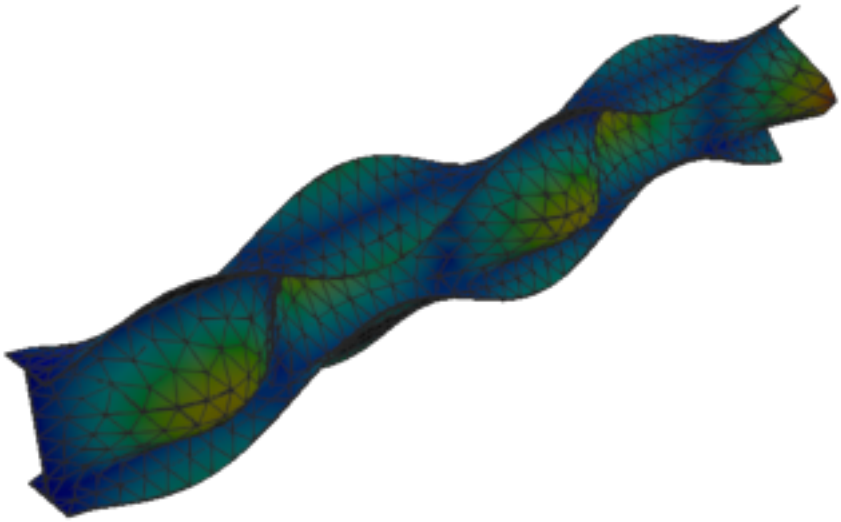


Figure 220: Sesam — mode 10, 234.613 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	233.360	-0.42	2.14979e+06	3.28e-15	1.38	3.26e-07	-5.73e-08	1.175	0.001
Calculix	236.042	+0.72	2.19956e+06	1.09e-14	1.42	2.89e-14	-1.04e-07	1.193	1.70e-07
Code_Aster	233.389	-0.41	2.15041e+06	2.31e-29	1.39	6.58e-25	4.80e-15	1.178	8.11e-13
Sesam	234.613	+0.11	2.17302e+06	1.23e-18	1.40	2.14e-20	-1.11e-09	-1.185	-1.46e-10

Mode 11

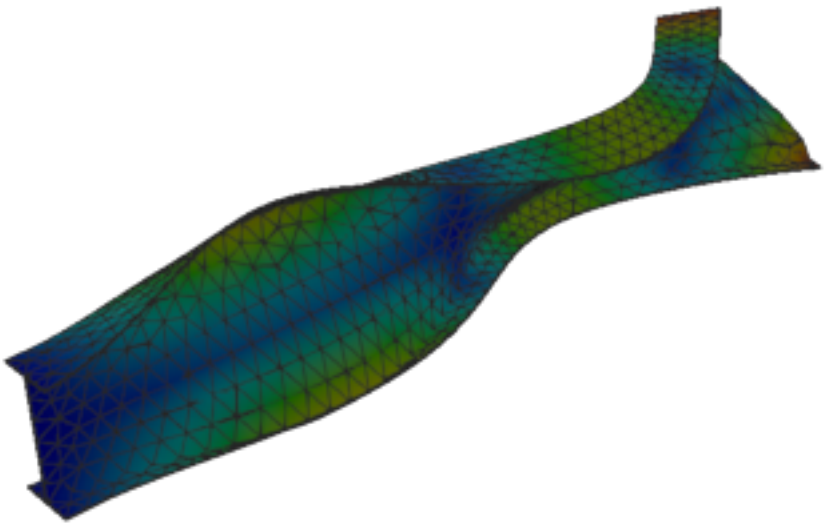


Figure 221: Abaqus — mode 11, 250.125 Hz

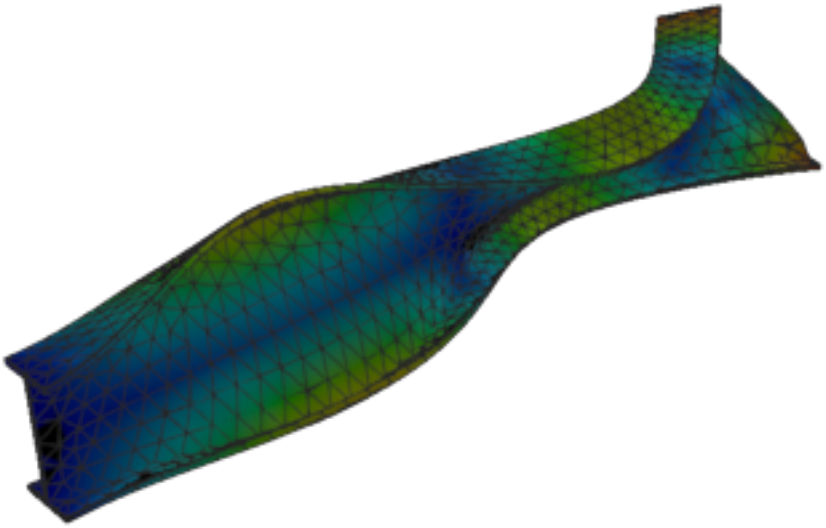


Figure 222: Calculix — mode 11, 250.287 Hz

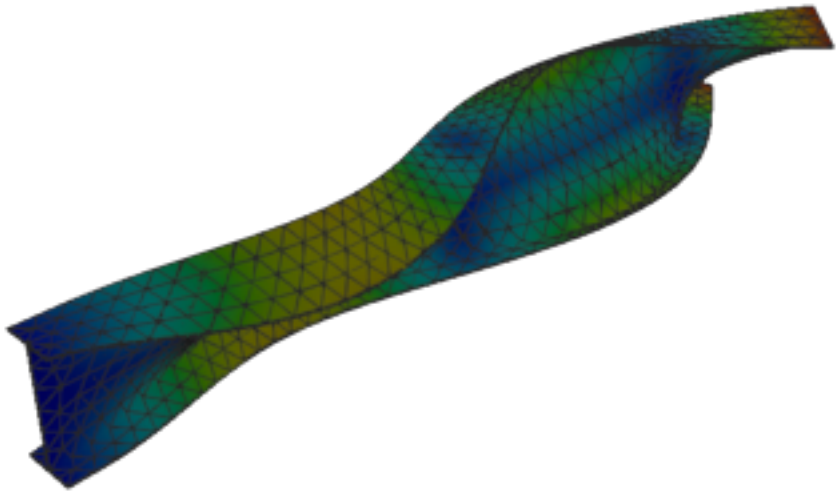


Figure 223: Code_Aster — mode 11, 250.175 Hz

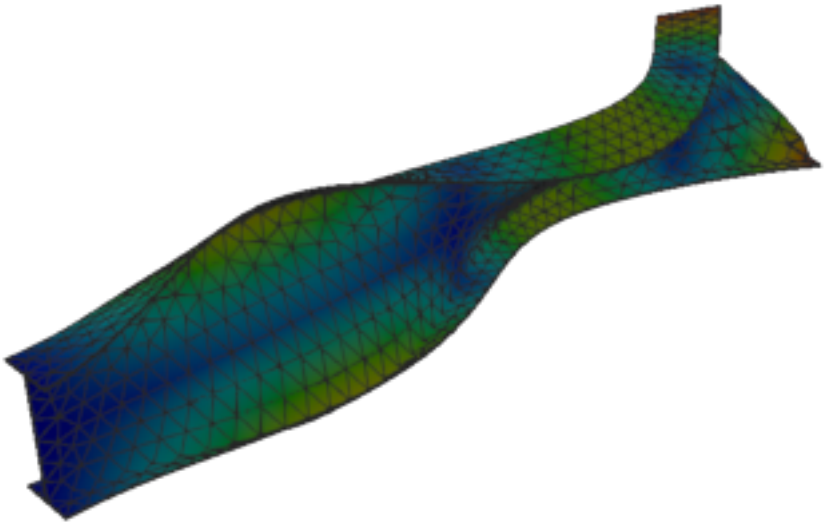


Figure 224: Sesam — mode 11, 250.240 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.130	-0.03	2.46987e+06	9.03e-11	3.18e-08	4.03e-09	9.50e-06	1.78e-04	6.35e-05
Calculix	250.287	+0.03	2.47306e+06	2.16e-10	5.05e-07	3.53e-15	-1.47e-05	0.001	-5.94e-08
Code_Aster	250.175	-0.01	2.47085e+06	1.31e-27	5.41e-07	2.47e-24	-3.62e-14	-0.001	-1.57e-12
Sesam	250.240	+0.01	2.47215e+06	1.60e-13	1.65e-06	2.77e-15	-4.01e-07	0.001	-5.26e-08

Shell, 2nd order, QUAD

Mode 1

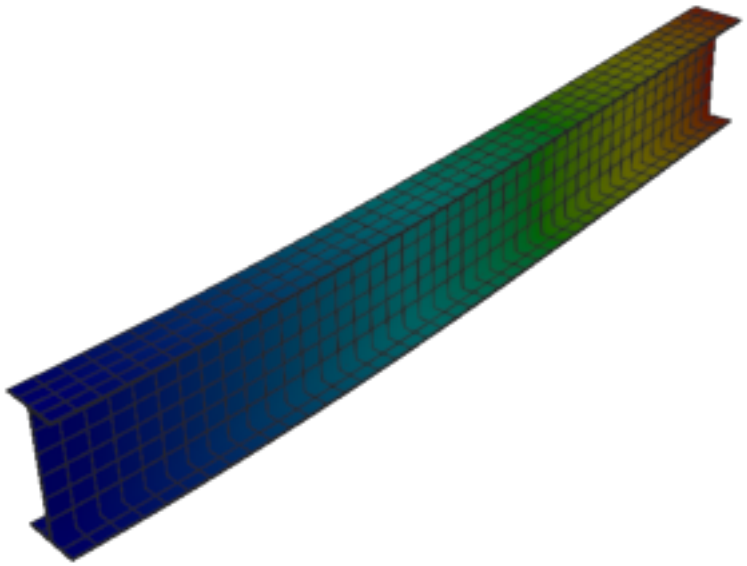


Figure 225: Abaqus — mode 1, 12.758 Hz

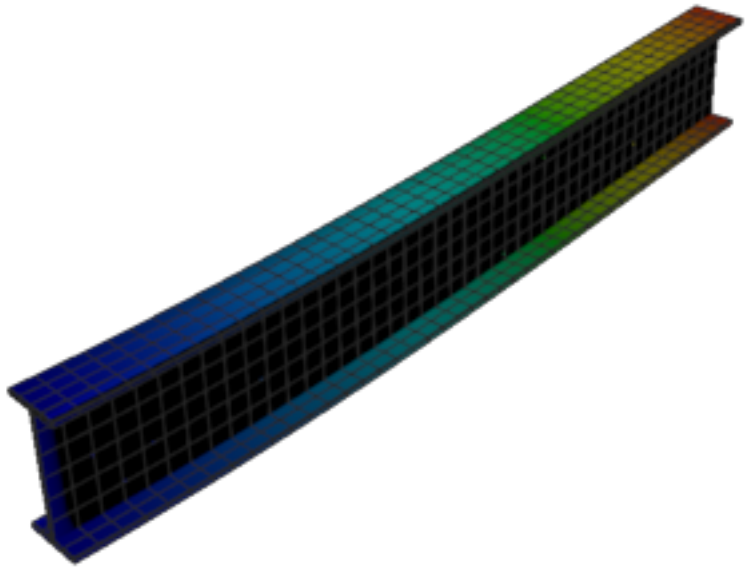


Figure 226: Calculix — mode 1, 12.758 Hz

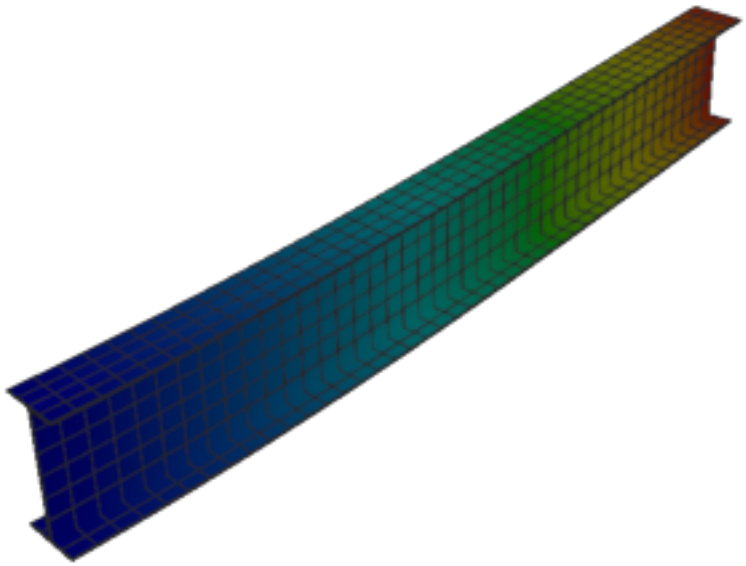


Figure 227: Code_Aster — mode 1, 12.756 Hz

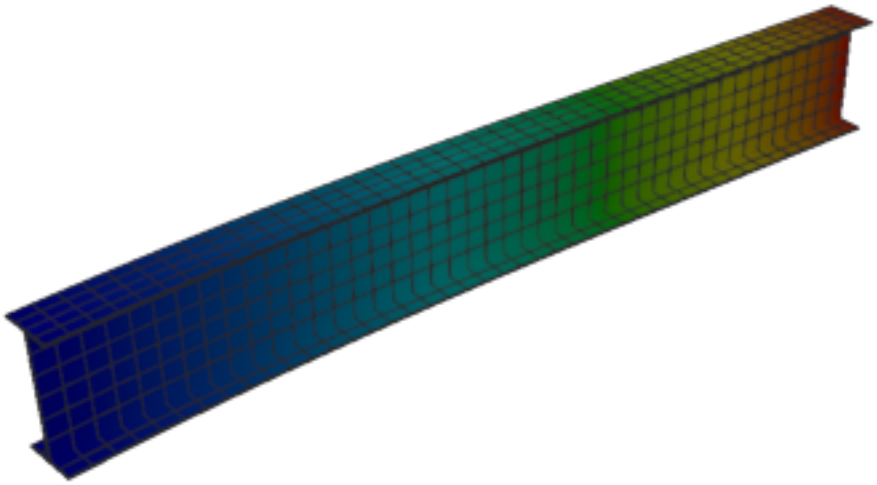


Figure 228: Sesam — mode 1, 12.755 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.758	+0.01	6426.1	1.23e-25	119.92	9.18e-25	3.51e-13	10.951	9.58e-13
Calculix	12.758	+0.01	6425.39	1.29e-23	119.93	1.74e-12	3.59e-12	10.951	1.32e-06
Code_Aster	12.756	-0.01	6423.86	3.33e-27	119.94	1.40e-25	-5.77e-14	10.952	3.74e-13
Sesam	12.755	-0.01	6423.15	3.17e-26	119.94	2.23e-24	-1.78e-13	-10.952	-1.49e-12

Mode 2

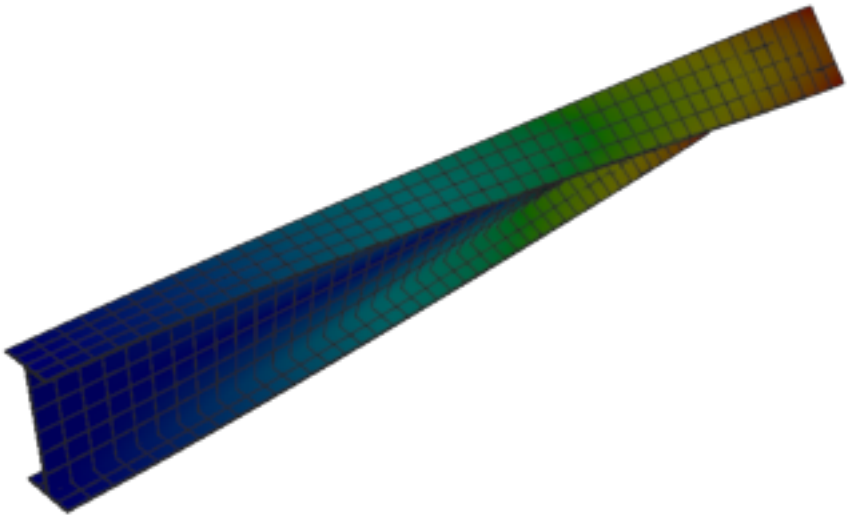


Figure 229: Abaqus — mode 2, 19.486 Hz

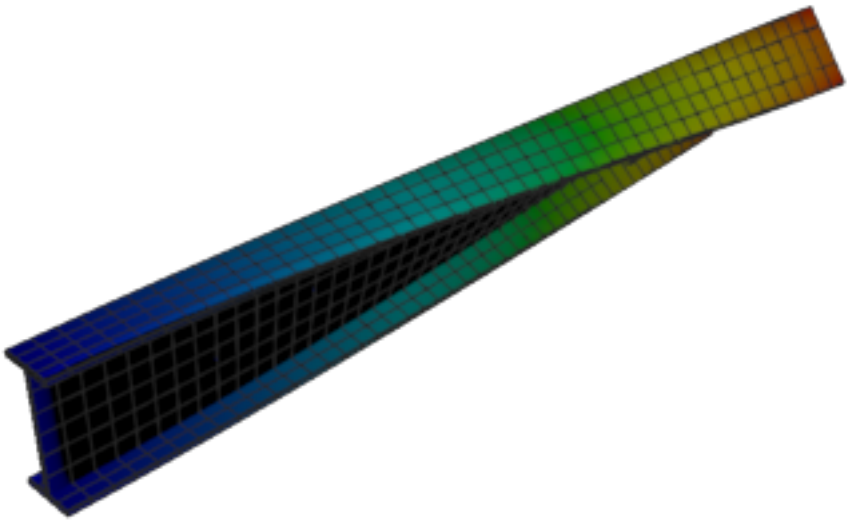


Figure 230: Calculix — mode 2, 19.564 Hz

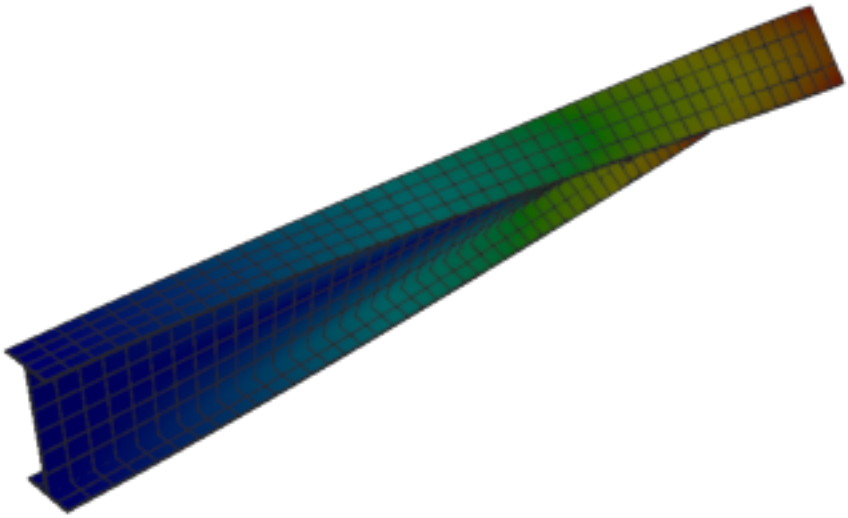


Figure 231: Code_Aster — mode 2, 19.474 Hz

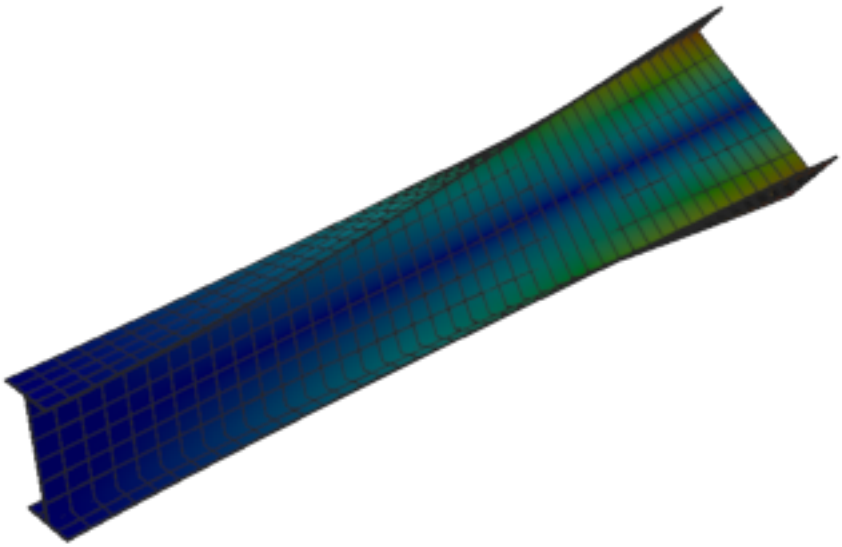


Figure 232: Sesam — mode 2, 19.473 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.486	-0.07	14990	1.04e-26	4.65e-20	2.40e-23	1.02e-13	2.16e-10	4.90e-12
Calculix	19.564	+0.33	15110.1	3.11e-09	1.08e-15	6.33e-19	5.58e-05	3.28e-08	-7.95e-10
Code_Aster	19.474	-0.13	14971	1.19e-26	2.41e-18	1.48e-22	1.09e-13	-1.55e-09	-1.22e-11
Sesam	19.473	-0.13	14970.3	4.18e-26	1.44e-18	3.35e-23	-2.04e-13	1.20e-09	5.79e-12

Mode 3

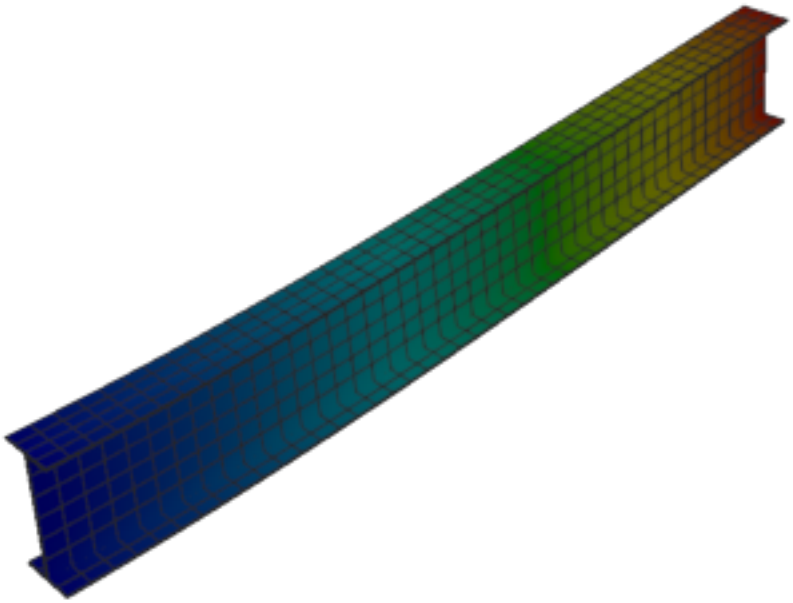


Figure 233: Abaqus — mode 3, 51.998 Hz

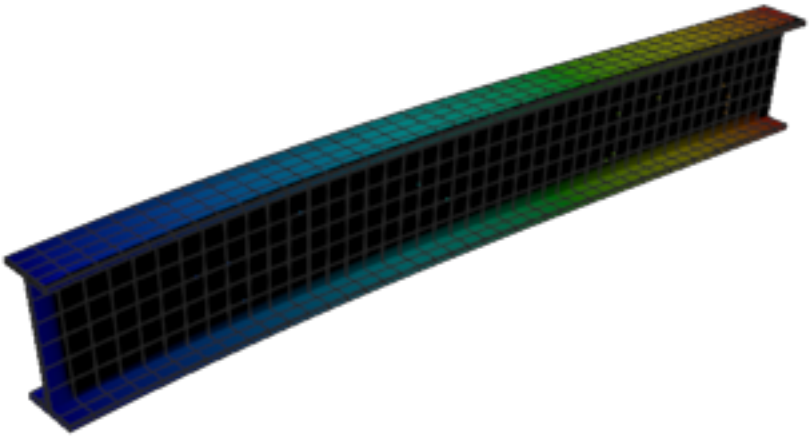


Figure 234: Calculix — mode 3, 51.980 Hz

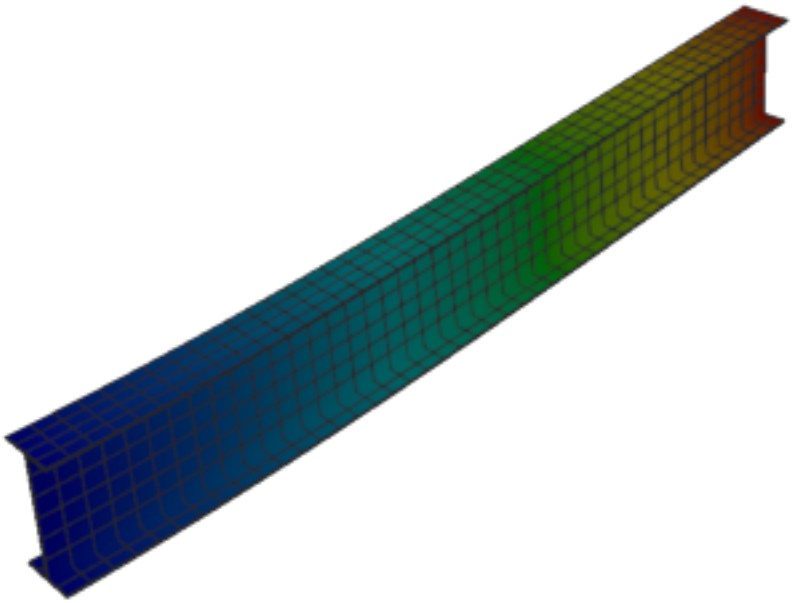


Figure 235: Code_Aster — mode 3, 51.970 Hz

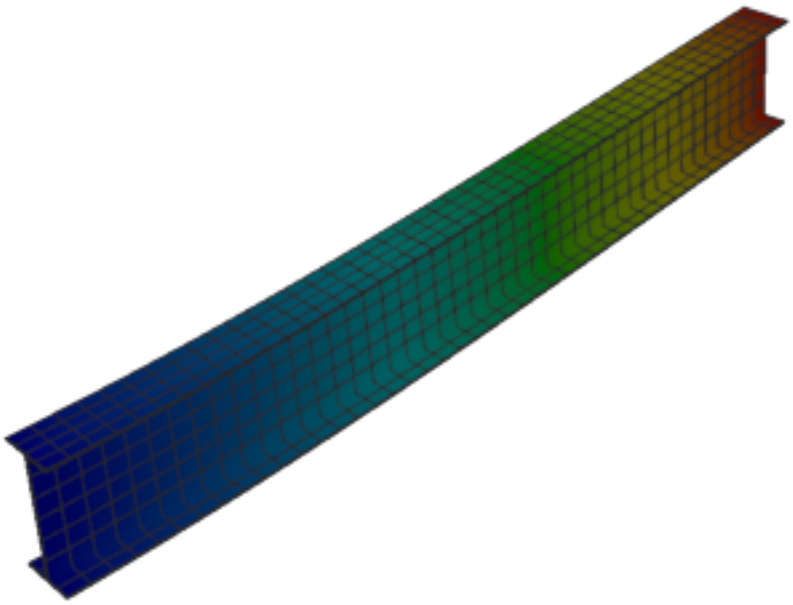


Figure 236: Sesam — mode 3, 51.970 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	+0.04	106740	5.15e-27	9.66e-26	121.97	7.18e-14	3.11e-13	11.044
Calculix	51.980	1.01e-03	106668	1.38e-23	4.52e-11	122.01	3.72e-12	6.73e-06	-11.046
Code_Aster	51.970	-0.02	106626	1.82e-26	1.94e-26	122.03	1.35e-13	1.39e-13	11.047
Sesam	51.970	-0.02	106628	4.55e-26	4.93e-25	122.03	-2.13e-13	-7.02e-13	11.047

Mode 4

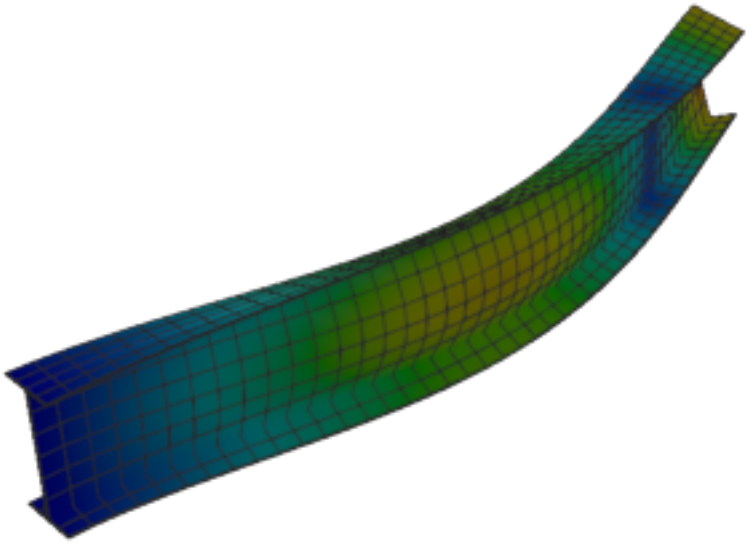


Figure 237: Abaqus — mode 4, 75.149 Hz

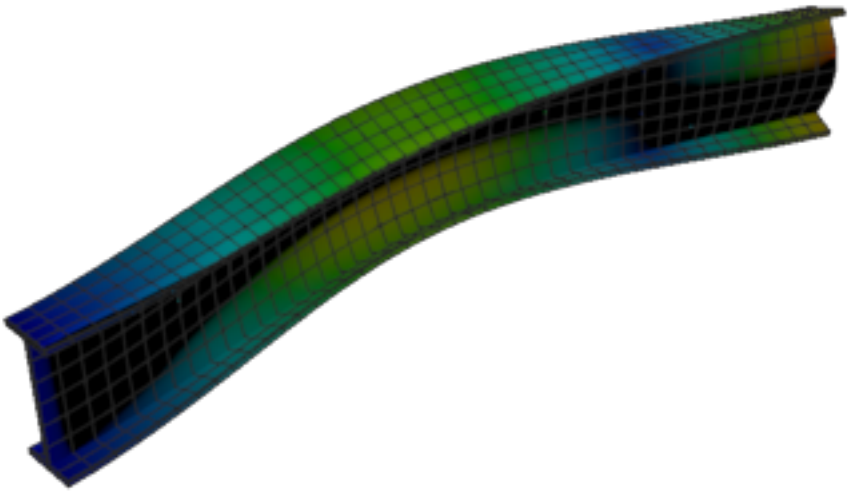


Figure 238: Calculix — mode 4, 75.269 Hz

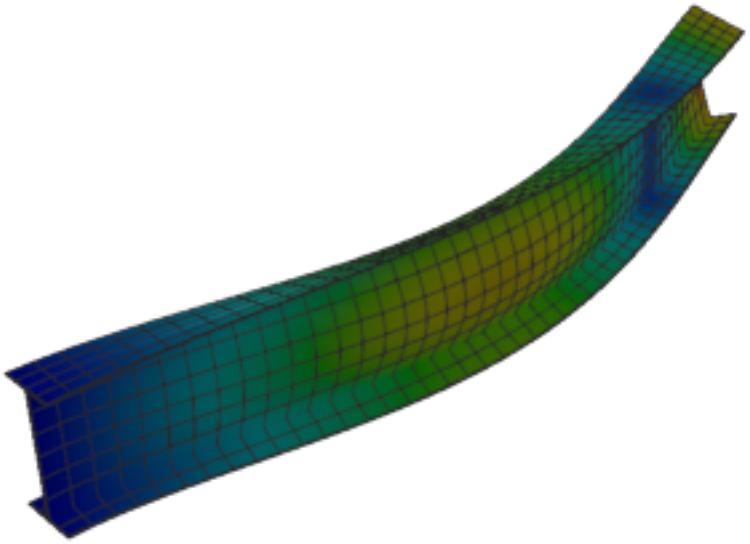


Figure 239: Code_Aster — mode 4, 75.125 Hz

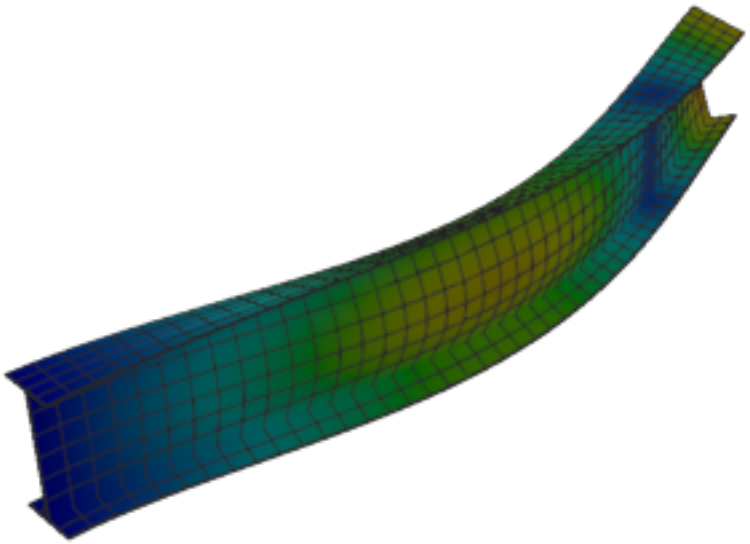


Figure 240: Sesam — mode 4, 75.125 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.149	-0.02	222949	1.40e-27	36.85	6.29e-24	-3.75e-14	-6.071	2.51e-12
Calculix	75.269	+0.14	223659	1.71e-24	36.88	1.39e-12	-1.31e-12	6.073	-1.18e-06
Code_Aster	75.125	-0.06	222808	2.82e-26	36.86	8.50e-25	1.68e-13	-6.071	9.22e-13
Sesam	75.125	-0.06	222804	3.41e-28	36.86	2.78e-24	-1.85e-14	-6.071	1.67e-12

Mode 5

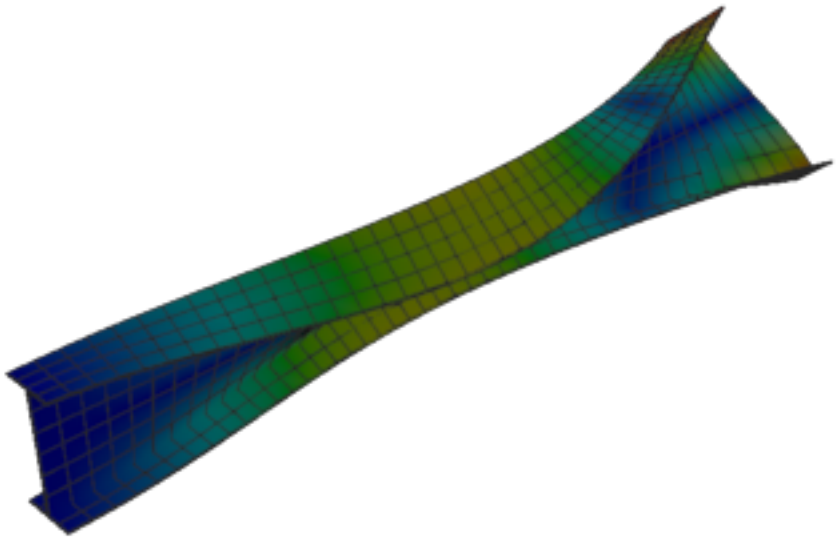


Figure 241: Abaqus — mode 5, 95.920 Hz

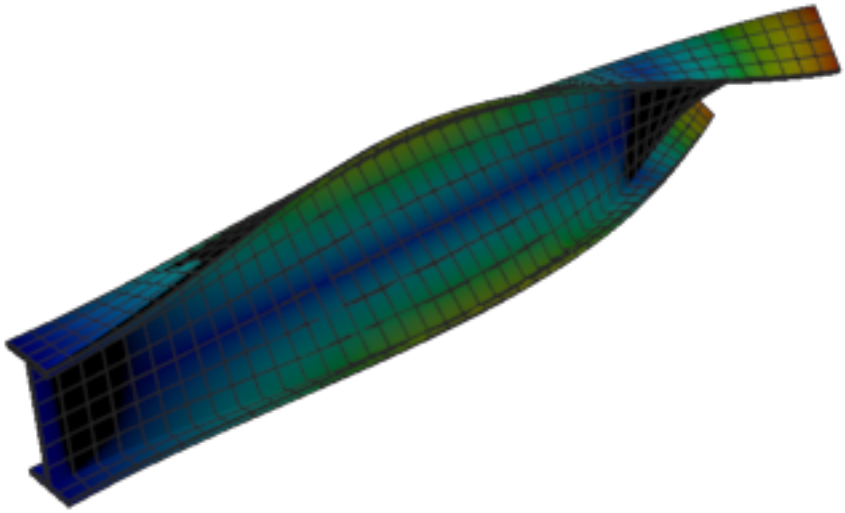


Figure 242: Calculix — mode 5, 96.037 Hz

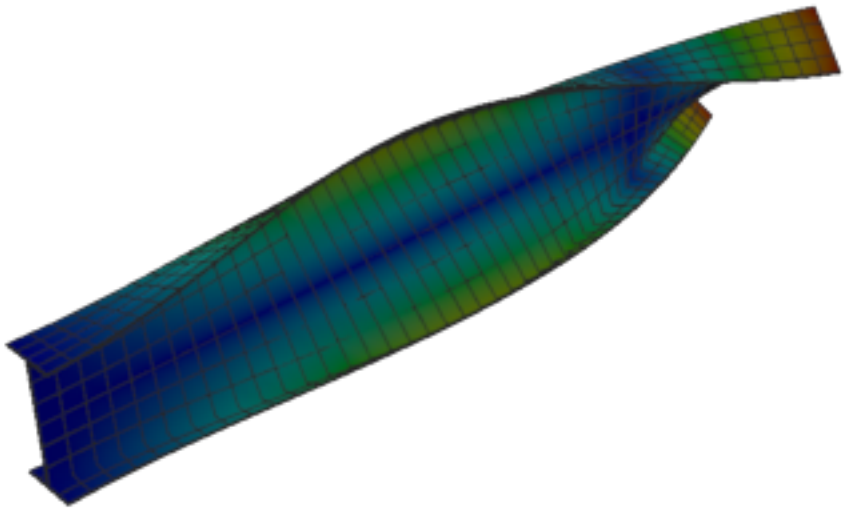


Figure 243: Code_Aster — mode 5, 95.889 Hz

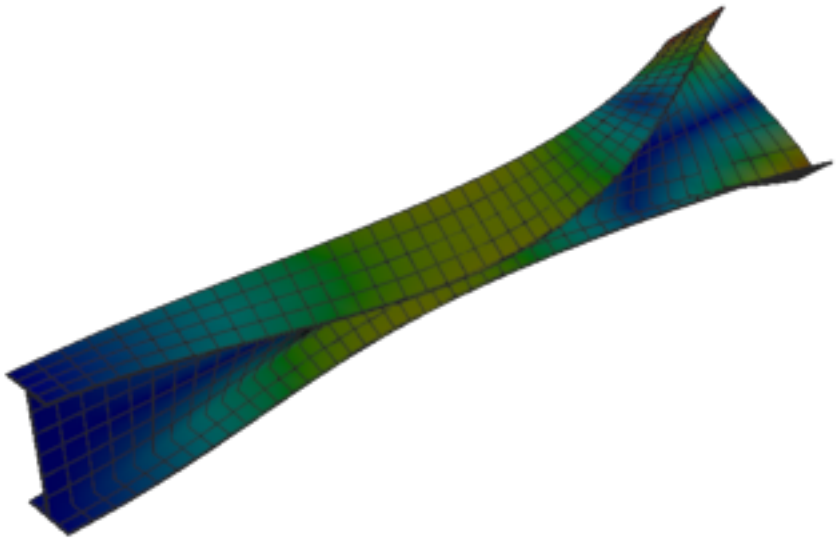


Figure 244: Sesam — mode 5, 95.889 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	95.920	-0.01	363224	7.50e-27	2.54e-21	8.04e-25	-8.66e-14	-5.04e-11	-8.97e-13
Calculix	96.037	+0.11	364115	8.58e-10	7.52e-19	4.09e-21	-2.93e-05	-8.67e-10	-6.40e-11
Code_Aster	95.889	-0.05	362994	7.82e-27	1.77e-21	5.02e-24	8.84e-14	-4.21e-11	-2.24e-12
Sesam	95.889	-0.05	362989	3.04e-26	1.19e-21	4.30e-24	1.74e-13	3.45e-11	-2.07e-12

Mode 6

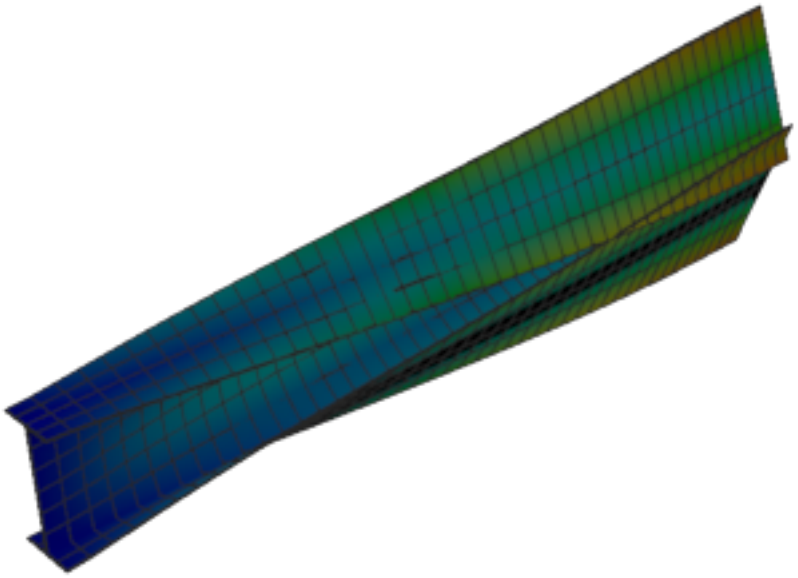


Figure 245: Abaqus — mode 6, 125.001 Hz

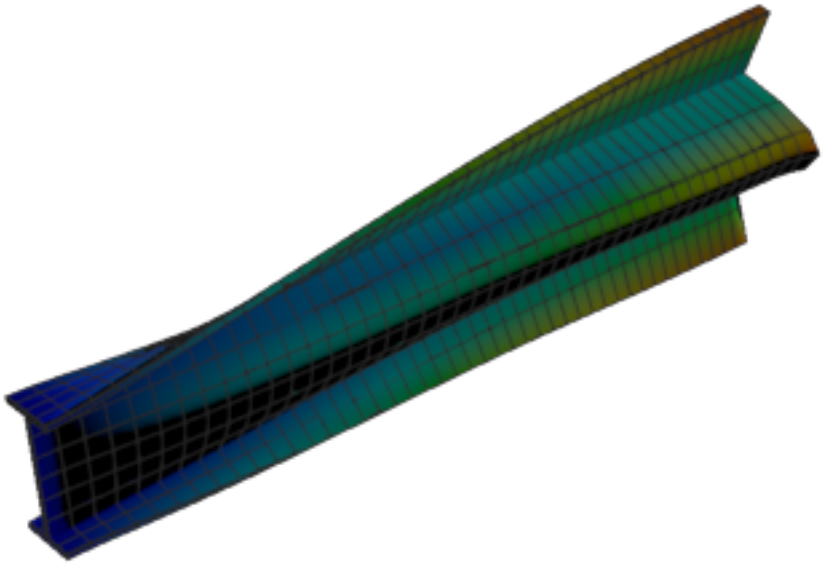


Figure 246: Calculix — mode 6, 125.889 Hz

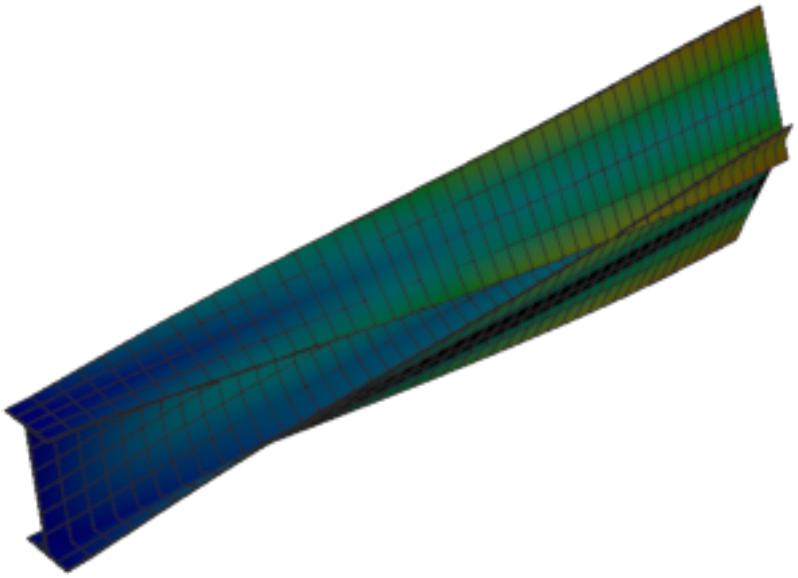


Figure 247: Code_Aster — mode 6, 124.987 Hz

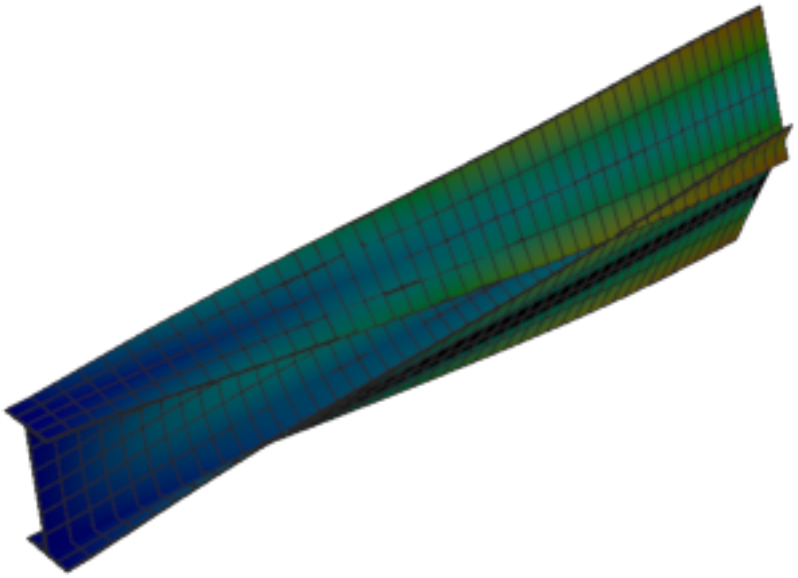


Figure 248: Sesam — mode 6, 124.987 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.000	-0.17	616856	5.44e-26	0.02	4.78e-23	-2.33e-13	-0.137	6.92e-12
Calculix	125.889	+0.54	625652	1.05e-24	0.02	1.64e-13	-1.03e-12	0.141	4.05e-07
Code_Aster	124.987	-0.18	616718	4.03e-27	0.02	3.10e-23	6.35e-14	-0.137	5.57e-12
Sesam	124.987	-0.18	616722	1.24e-26	0.02	1.35e-24	-1.11e-13	-0.137	1.16e-12

Mode 7

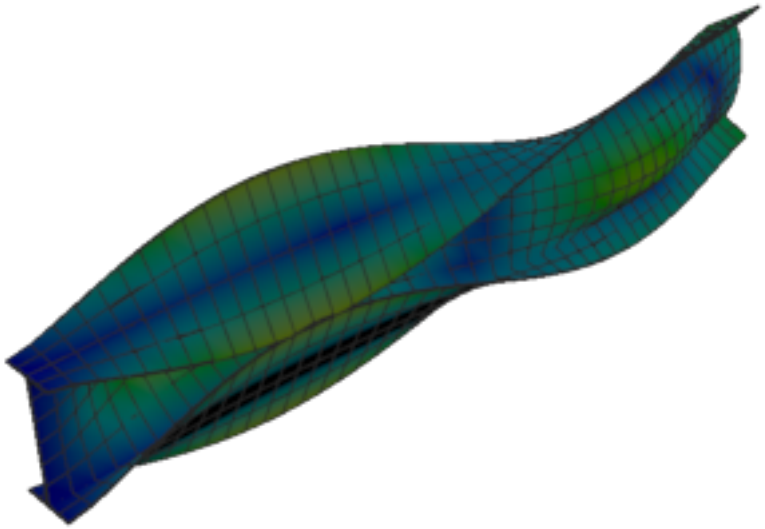


Figure 249: Abaqus — mode 7, 142.325 Hz

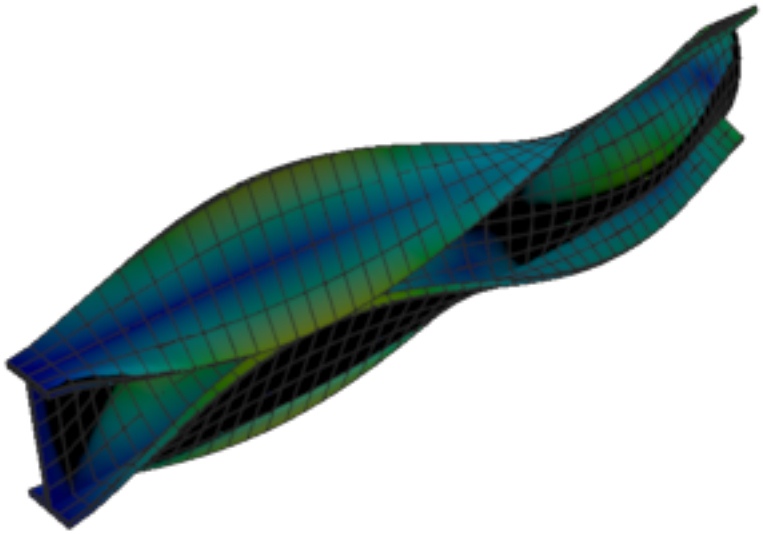


Figure 250: Calculix — mode 7, 143.720 Hz

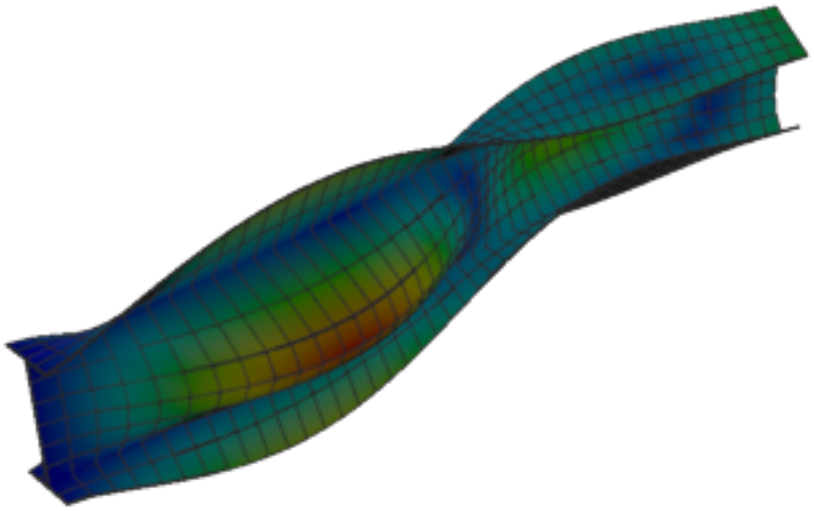


Figure 251: Code_Aster — mode 7, 142.233 Hz

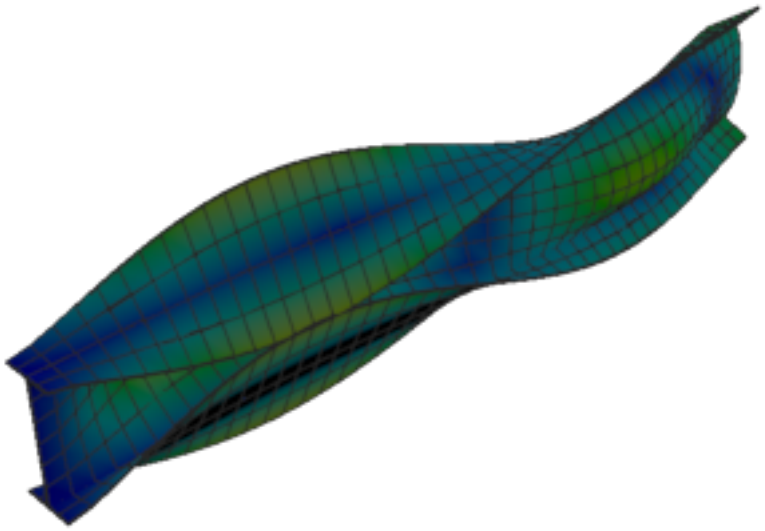


Figure 252: Sesam — mode 7, 142.234 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.320	-0.21	799690	2.49e-27	3.45	4.42e-24	-4.99e-14	1.858	-2.10e-12
Calculix	143.720	+0.77	815440	6.94e-26	3.38	8.05e-13	-2.63e-13	1.839	8.97e-07
Code_Aster	142.233	-0.28	798656	2.75e-27	3.45	2.74e-24	-5.24e-14	-1.859	1.66e-12
Sesam	142.234	-0.28	798665	3.85e-28	3.45	6.92e-26	1.96e-14	1.859	-2.63e-13

Mode 8

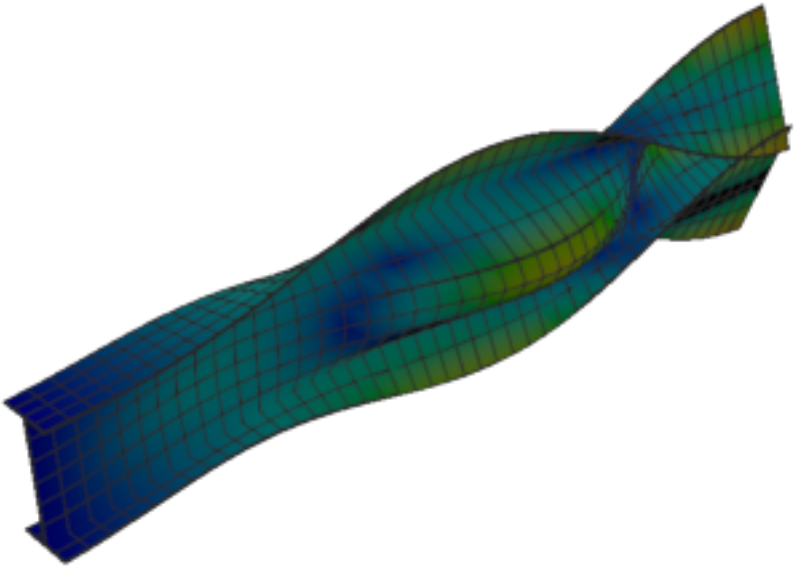


Figure 253: Abaqus — mode 8, 165.849 Hz

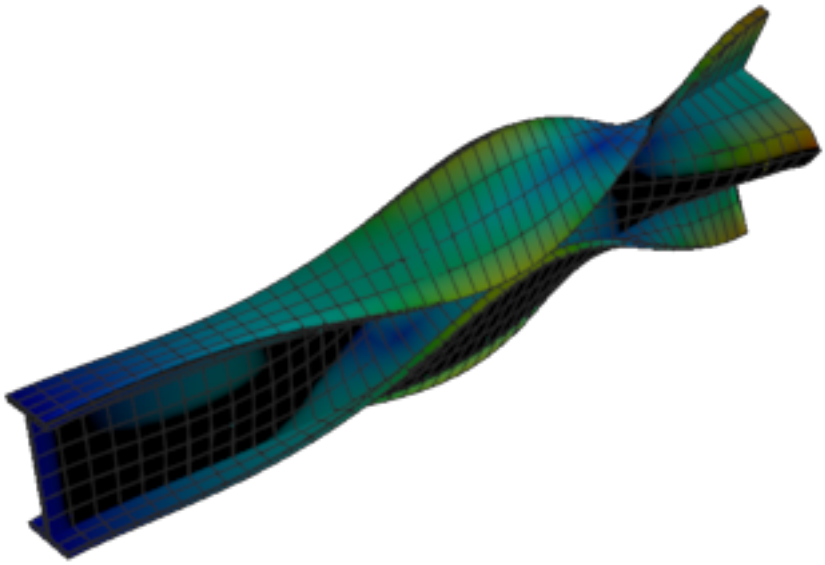


Figure 254: Calculix — mode 8, 167.347 Hz

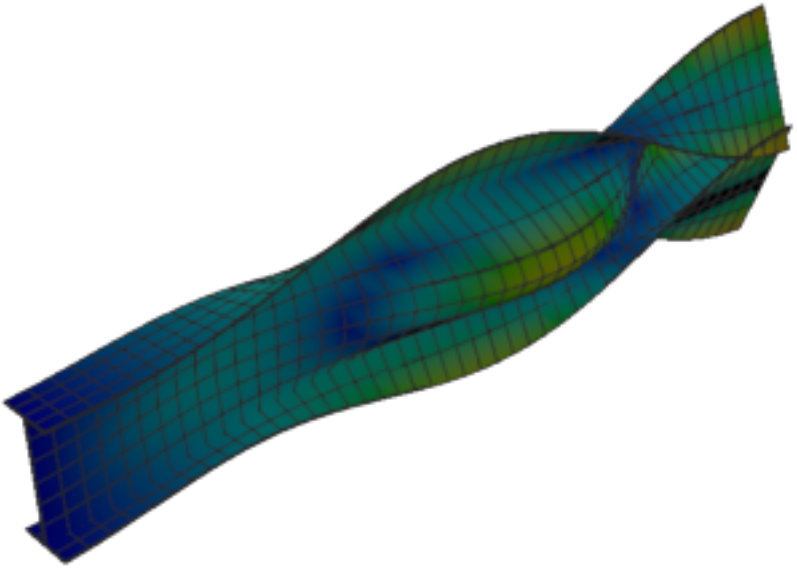


Figure 255: Code_Aster — mode 8, 165.719 Hz

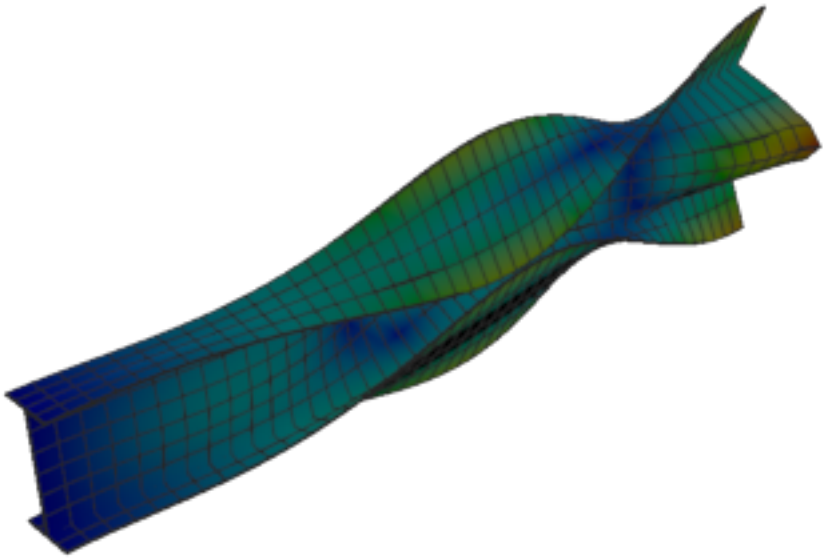


Figure 256: Sesam — mode 8, 165.718 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	165.850	-0.19	1.08589e+06	2.43e-27	5.53	1.06e-23	-4.93e-14	2.351	3.25e-12
Calculix	167.347	+0.72	1.10559e+06	2.91e-24	5.58	2.92e-13	-1.71e-12	-2.362	5.40e-07
Code_Aster	165.719	-0.26	1.08419e+06	2.11e-26	5.52	3.83e-24	-1.45e-13	2.350	1.96e-12
Sesam	165.718	-0.26	1.08418e+06	3.12e-28	5.52	6.16e-25	-1.77e-14	-2.350	-7.85e-13

Mode 9

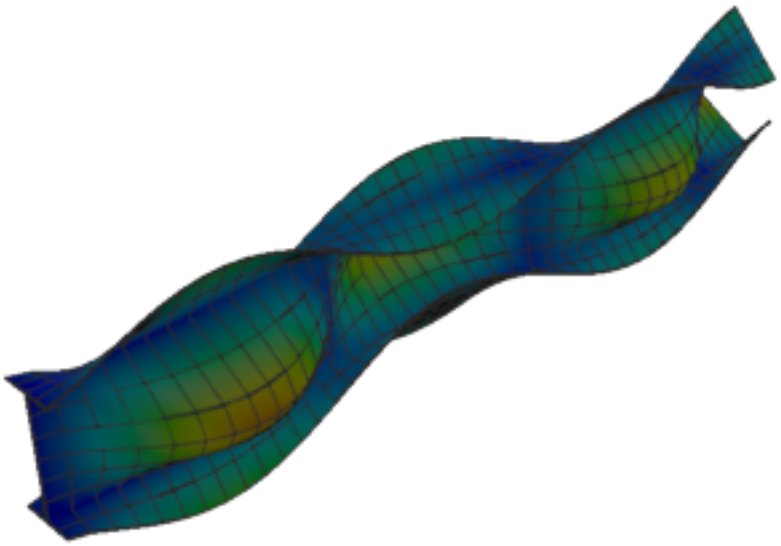


Figure 257: Abaqus — mode 9, 192.184 Hz

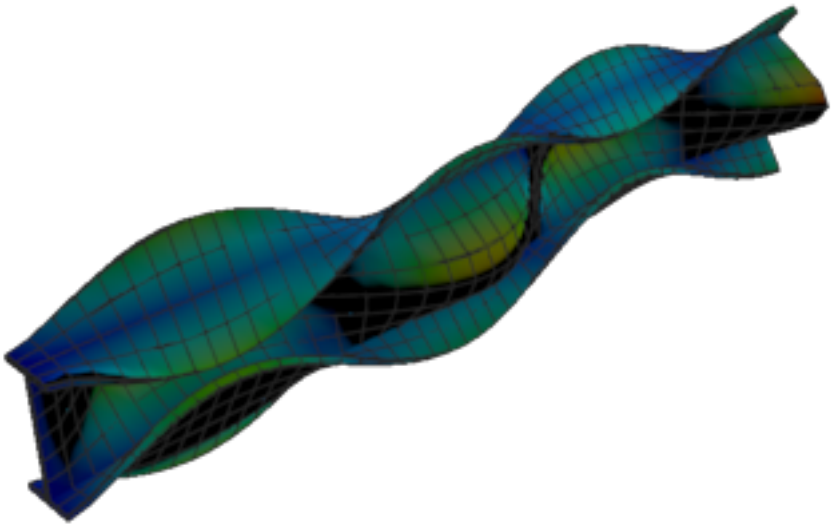


Figure 258: Calculix — mode 9, 195.577 Hz

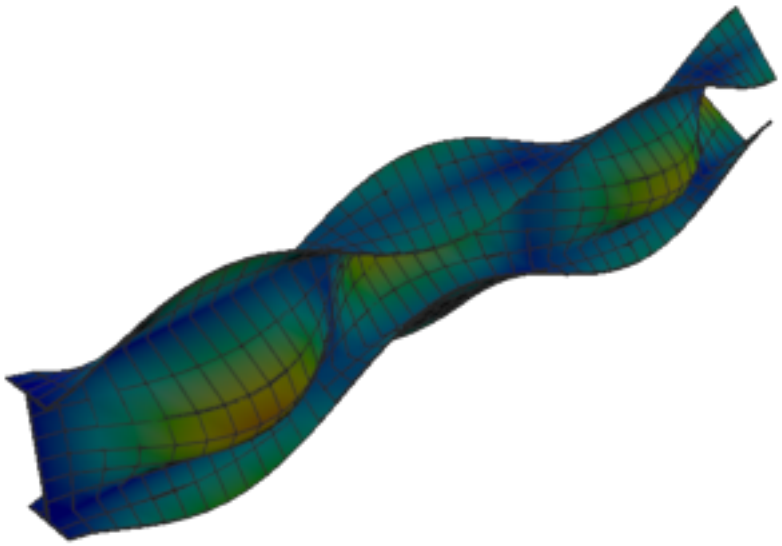


Figure 259: Code_Aster — mode 9, 191.998 Hz

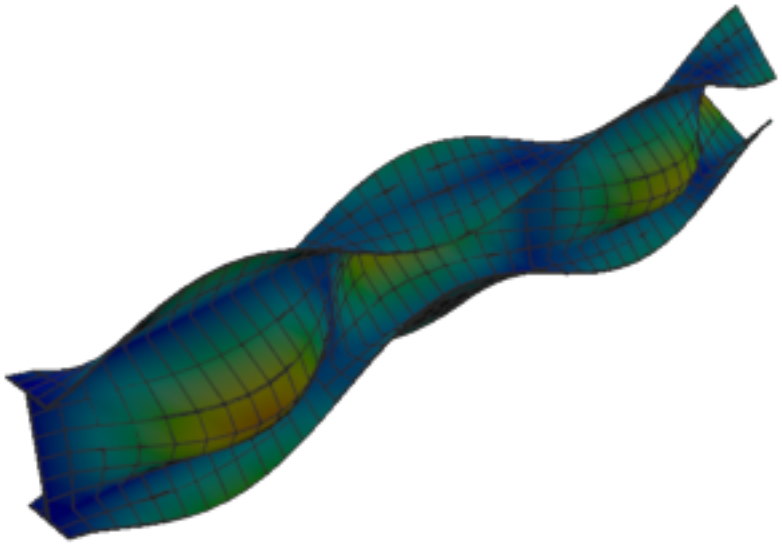


Figure 260: Sesam — mode 9, 191.999 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.180	-0.39	1.45812e+06	1.15e-27	0.65	1.20e-25	-3.40e-14	-0.808	3.47e-13
Calculix	195.577	+1.37	1.51006e+06	3.54e-25	0.61	9.24e-14	5.95e-13	0.781	-3.04e-07
Code_Aster	191.998	-0.49	1.45531e+06	4.91e-28	0.65	4.66e-25	-2.22e-14	-0.809	-6.82e-13
Sesam	191.999	-0.49	1.45531e+06	2.21e-28	0.65	1.82e-28	1.49e-14	-0.809	1.35e-14

Mode 10

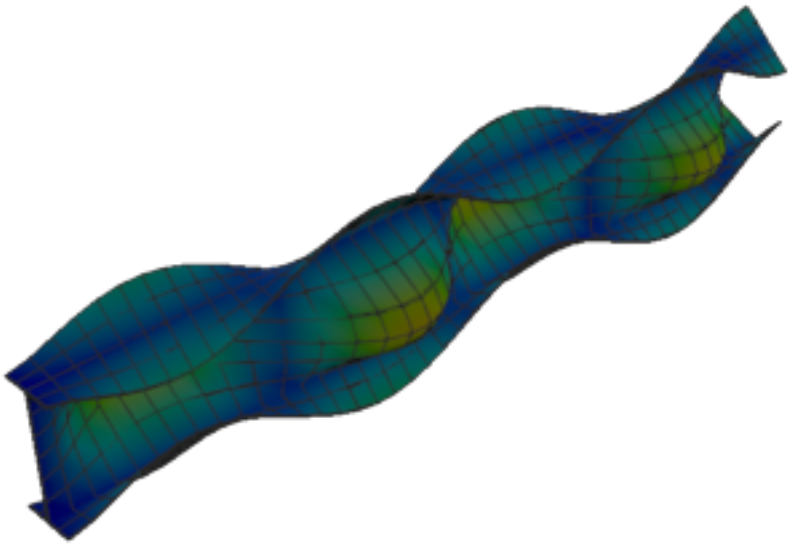


Figure 261: Abaqus — mode 10, 232.867 Hz

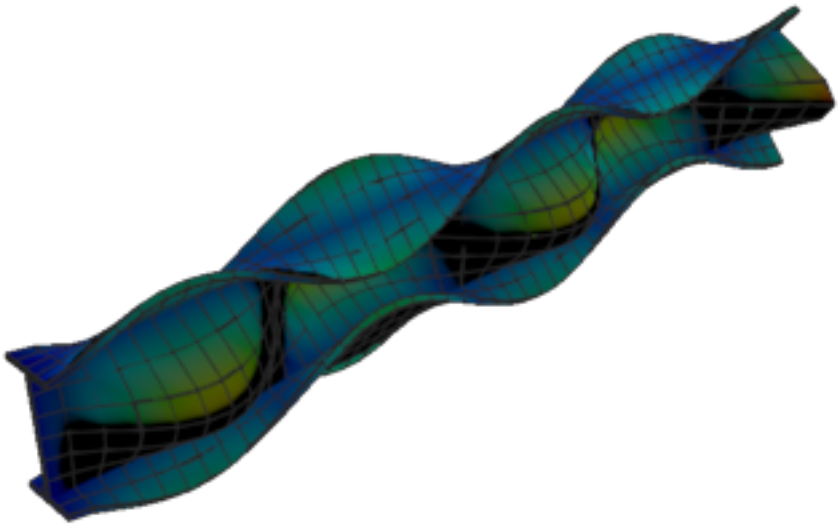


Figure 262: Calculix — mode 10, 238.389 Hz

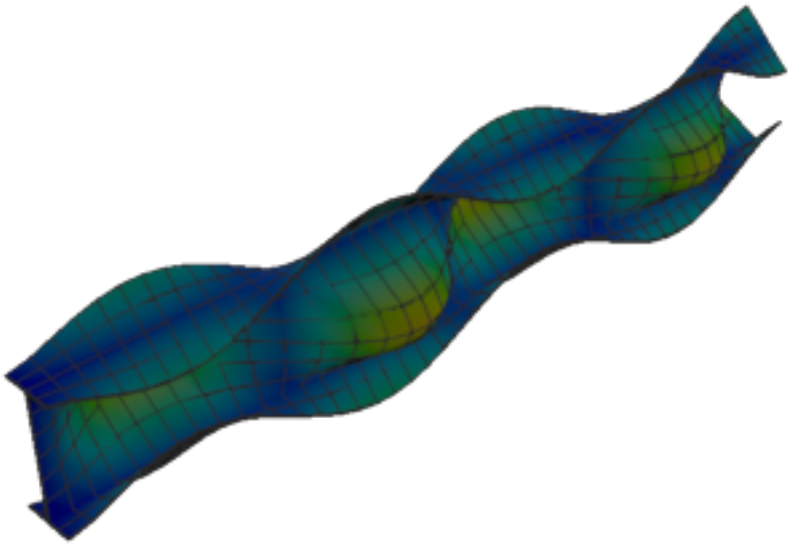


Figure 263: Code_Aster — mode 10, 232.628 Hz

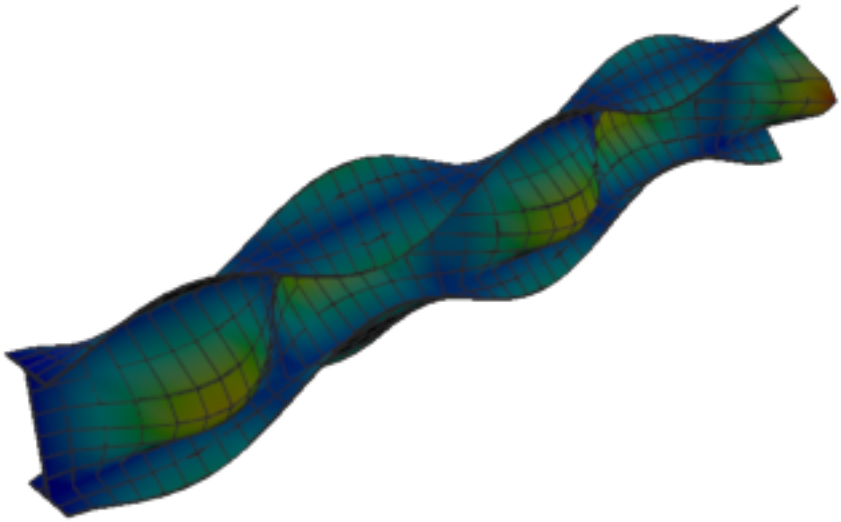


Figure 264: Sesam — mode 10, 232.626 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	232.870	-0.54	2.14079e+06	4.88e-26	1.37	1.33e-23	-2.21e-13	1.170	-3.65e-12
Calculix	238.389	+1.82	2.24354e+06	7.93e-24	1.44	5.13e-14	-2.82e-12	-1.201	-2.27e-07
Code_Aster	232.628	-0.64	2.13641e+06	3.71e-27	1.37	1.27e-24	-6.09e-14	1.170	1.13e-12
Sesam	232.626	-0.64	2.13637e+06	1.17e-26	1.37	1.70e-25	-1.08e-13	-1.171	-4.12e-13

Mode 11

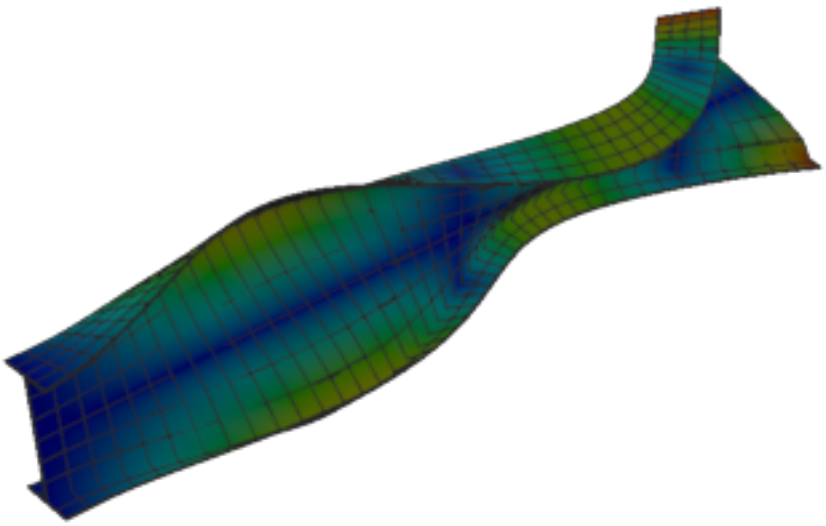


Figure 265: Abaqus — mode 11, 250.173 Hz

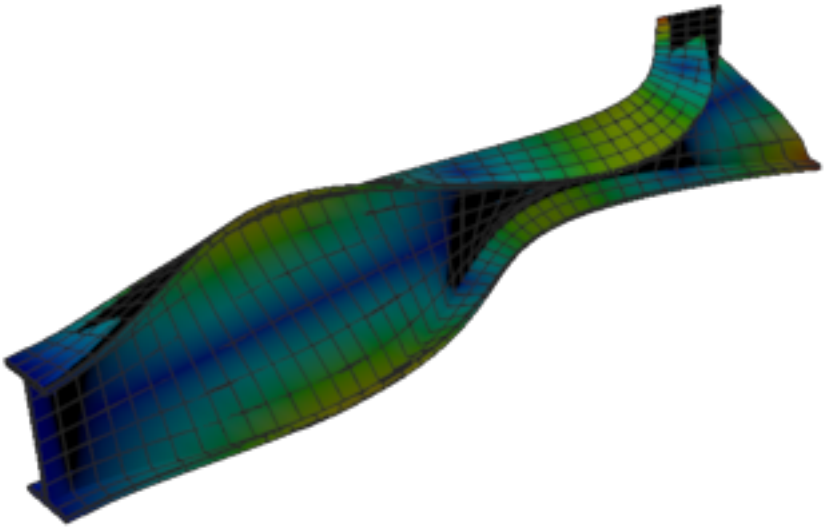


Figure 266: Calculix — mode 11, 250.521 Hz

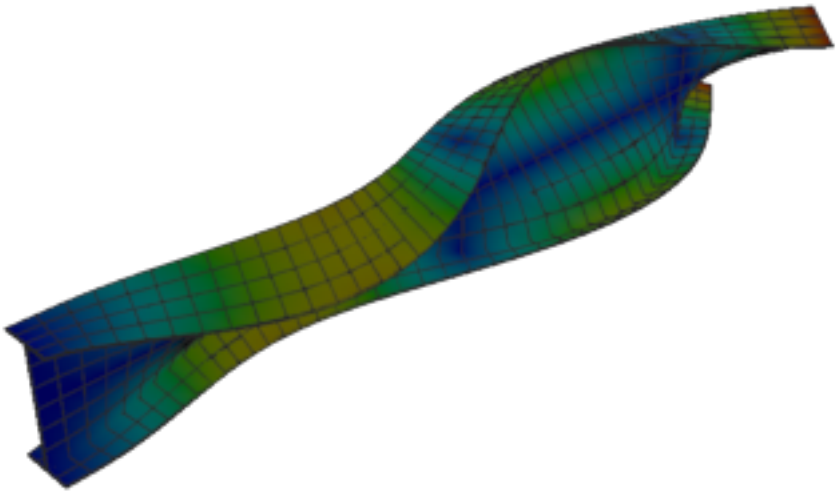


Figure 267: Code_Aster — mode 11, 250.096 Hz

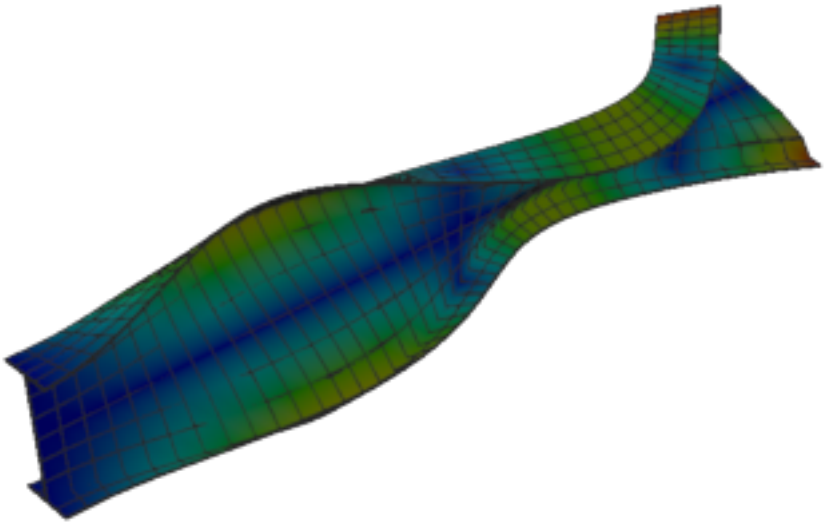


Figure 268: Sesam — mode 11, 250.095 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.170	-0.02	2.47082e+06	3.28e-21	3.98e-19	6.25e-19	5.72e-11	6.31e-10	-7.90e-10
Calculix	250.521	+0.12	2.47769e+06	2.31e-10	2.30e-22	4.17e-20	-1.52e-05	1.52e-11	2.04e-10
Code_Aster	250.096	-0.05	2.46929e+06	6.87e-28	1.30e-24	7.49e-25	-2.62e-14	1.14e-12	-8.65e-13
Sesam	250.095	-0.05	2.46928e+06	1.10e-19	4.31e-22	2.22e-21	-3.32e-10	-2.08e-11	4.71e-11

Shell, 2nd order, QUAD, reduced integration

Mode 1

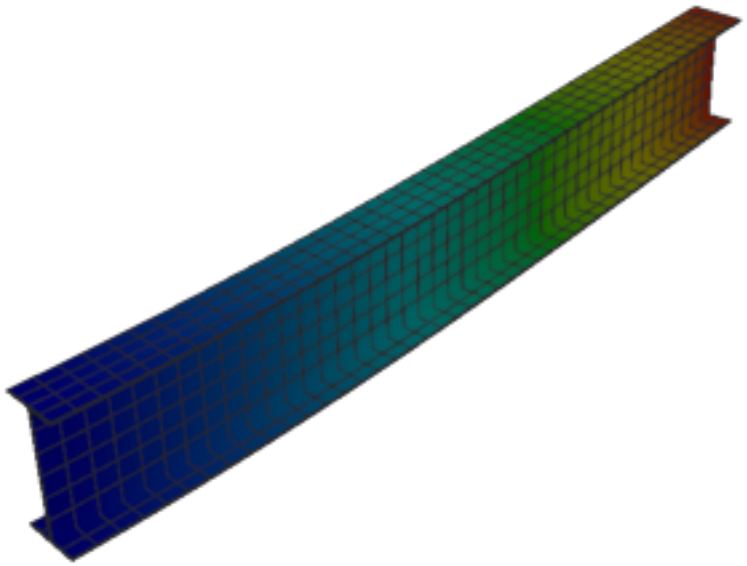


Figure 269: Abaqus — mode 1, 12.758 Hz

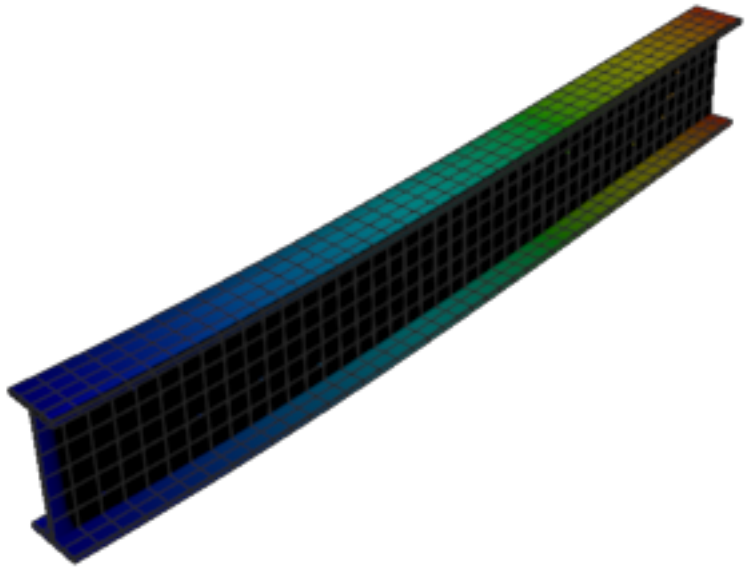


Figure 270: Calculix — mode 1, 12.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.758	+0.01	6426.1	1.23e-25	119.92	9.18e-25	3.51e-13	10.951	9.58e-13
Calculix	12.756	-0.01	6423.65	4.95e-25	119.93	1.74e-12	-7.04e-13	10.951	1.32e-06

Mode 2

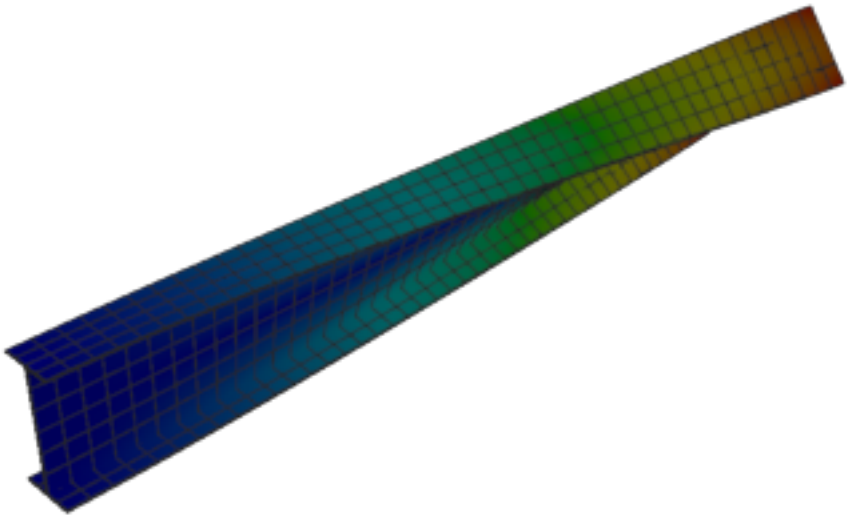


Figure 271: Abaqus — mode 2, 19.486 Hz

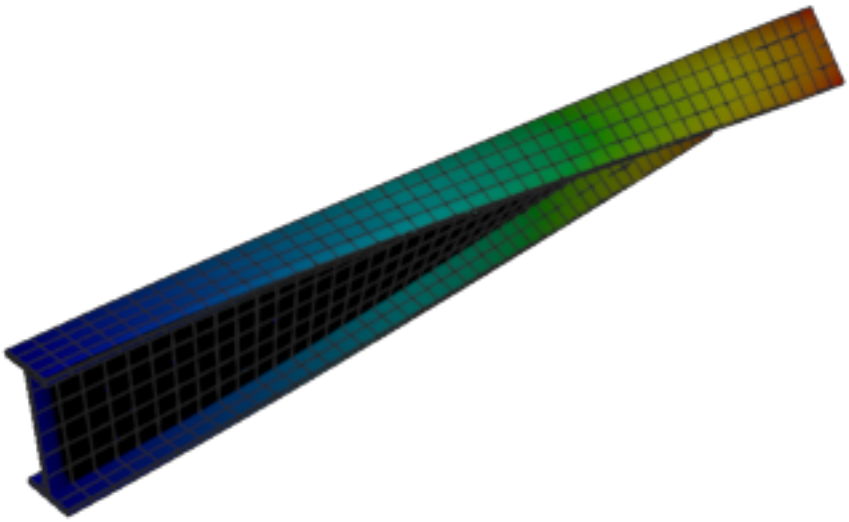


Figure 272: Calculix — mode 2, 19.502 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.486	-0.04	14990	1.04e-26	4.65e-20	2.40e-23	1.02e-13	2.16e-10	4.90e-12
Calculix	19.502	+0.04	15015	3.11e-09	2.97e-16	8.11e-20	5.58e-05	1.72e-08	-2.85e-10

Mode 3

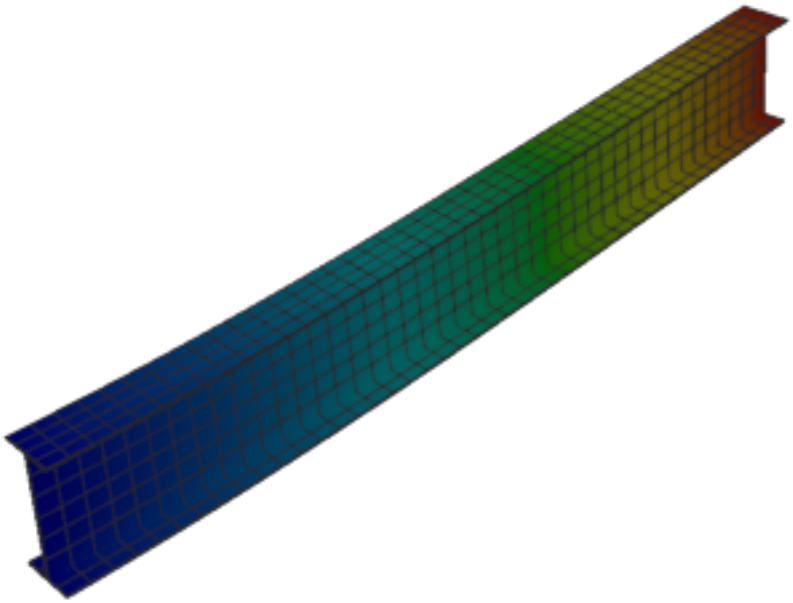


Figure 273: Abaqus — mode 3, 51.998 Hz

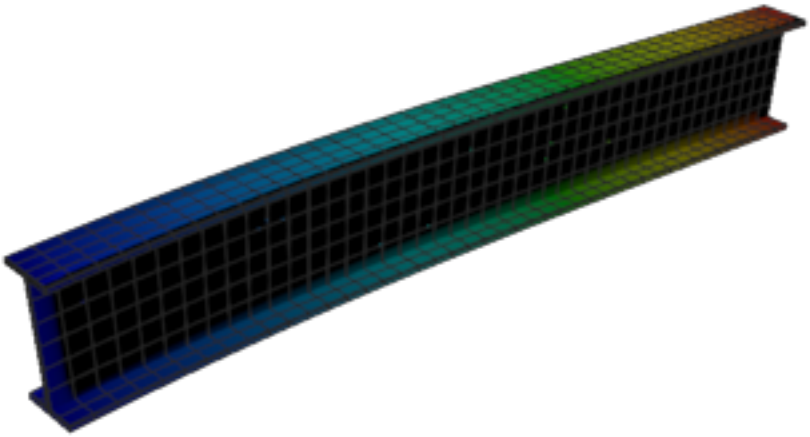


Figure 274: Calculix — mode 3, 51.974 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	+0.02	106740	5.15e-27	9.66e-26	121.97	7.18e-14	3.11e-13	11.044
Calculix	51.974	-0.02	106644	2.79e-24	4.52e-11	122.02	1.67e-12	6.73e-06	-11.046

Mode 4

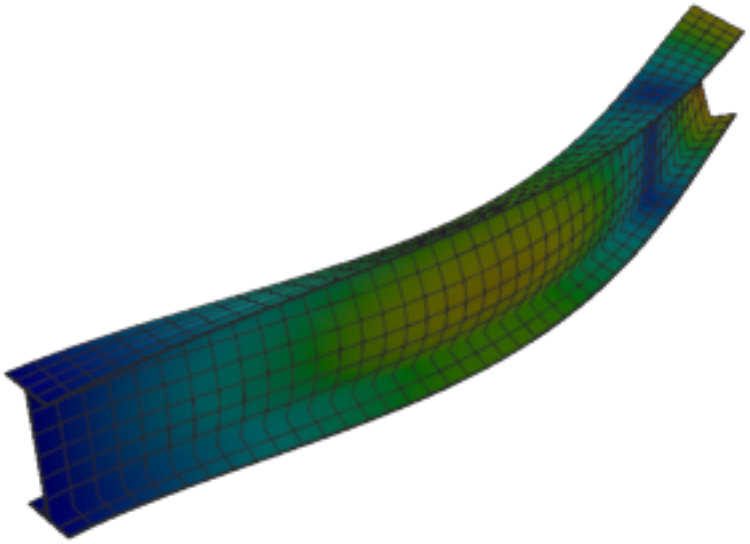


Figure 275: Abaqus — mode 4, 75.149 Hz

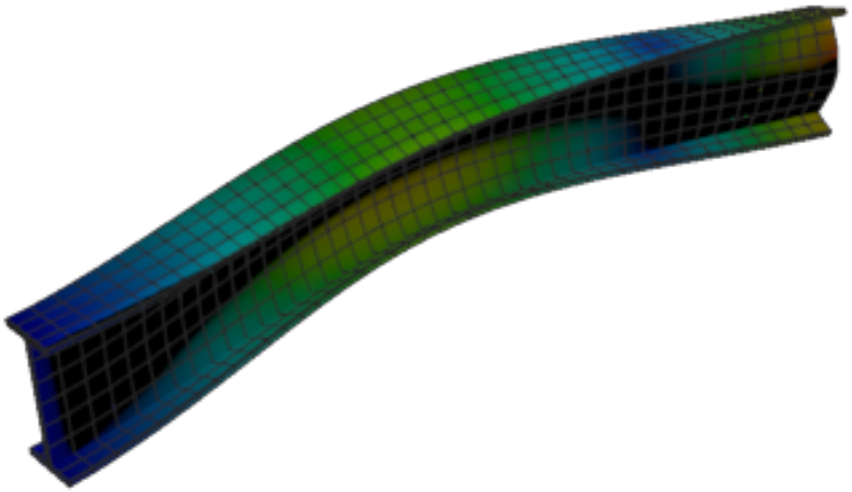


Figure 276: Calculix — mode 4, 75.140 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.149	+0.01	222949	1.40e-27	36.85	6.29e-24	-3.75e-14	-6.071	2.51e-12
Calculix	75.140	-0.01	222895	2.73e-24	36.85	1.38e-12	1.65e-12	6.071	-1.18e-06

Mode 5

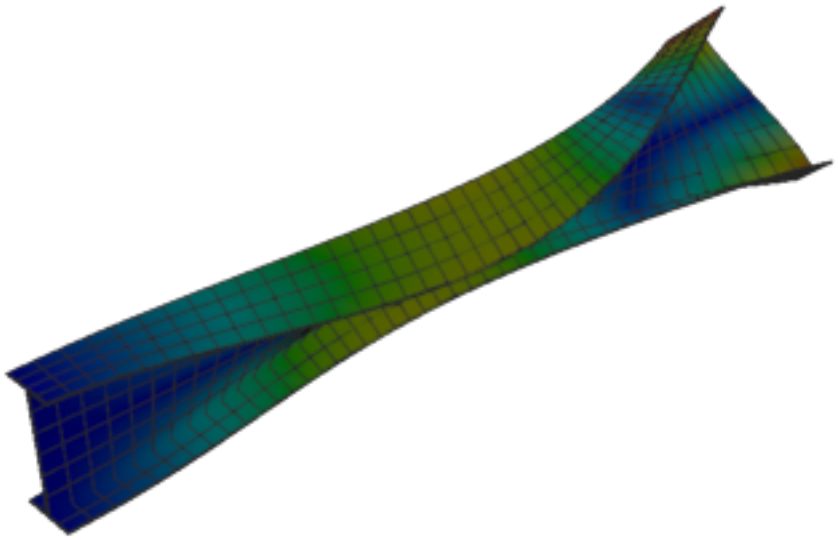


Figure 277: Abaqus — mode 5, 95.920 Hz

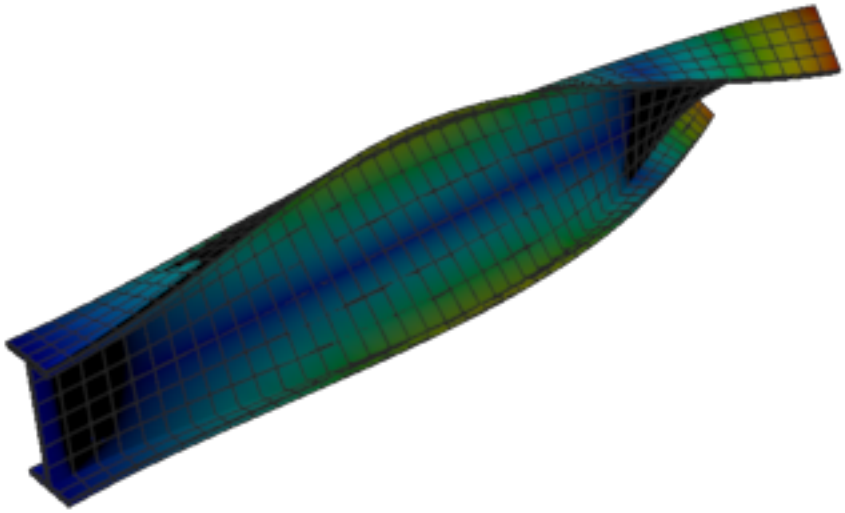


Figure 278: Calculix — mode 5, 95.935 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	95.920	-0.01	363224	7.50e-27	2.54e-21	8.04e-25	-8.66e-14	-5.04e-11	-8.97e-13
Calculix	95.935	+0.01	363340	8.60e-10	2.62e-19	7.05e-22	-2.93e-05	-5.11e-10	2.66e-11

Mode 6

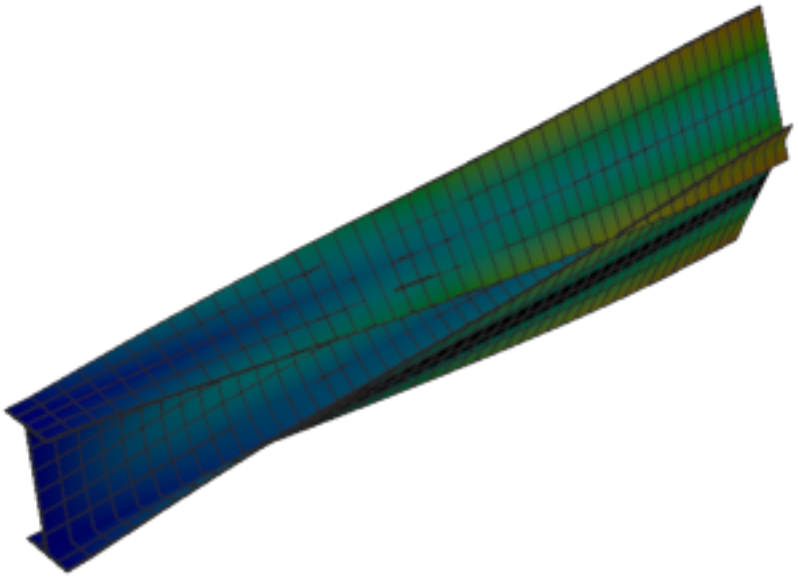


Figure 279: Abaqus — mode 6, 125.001 Hz

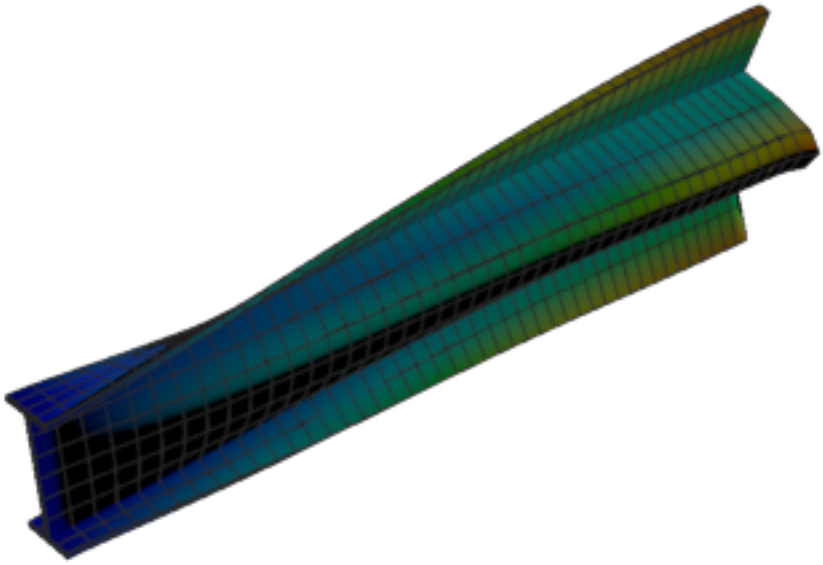


Figure 280: Calculix — mode 6, 125.257 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.000	-0.10	616856	5.44e-26	0.02	4.78e-23	-2.33e-13	-0.137	6.92e-12
Calculix	125.257	+0.10	619385	5.54e-25	0.02	1.65e-13	7.44e-13	0.138	4.06e-07

Mode 7

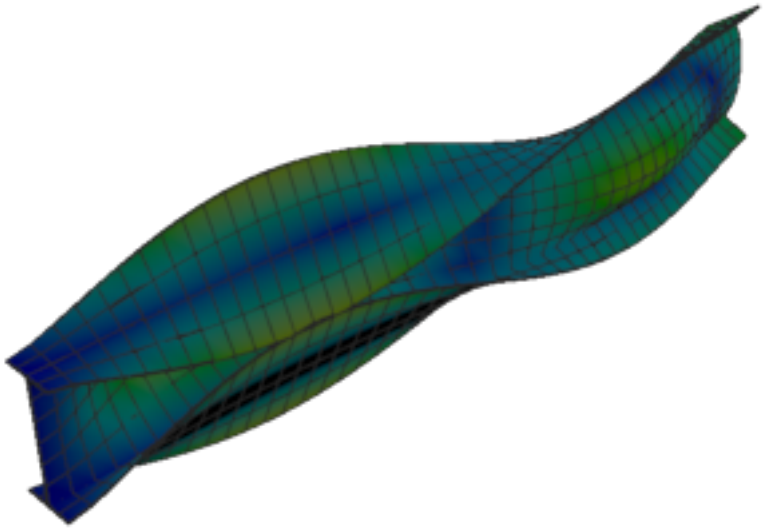


Figure 281: Abaqus — mode 7, 142.325 Hz

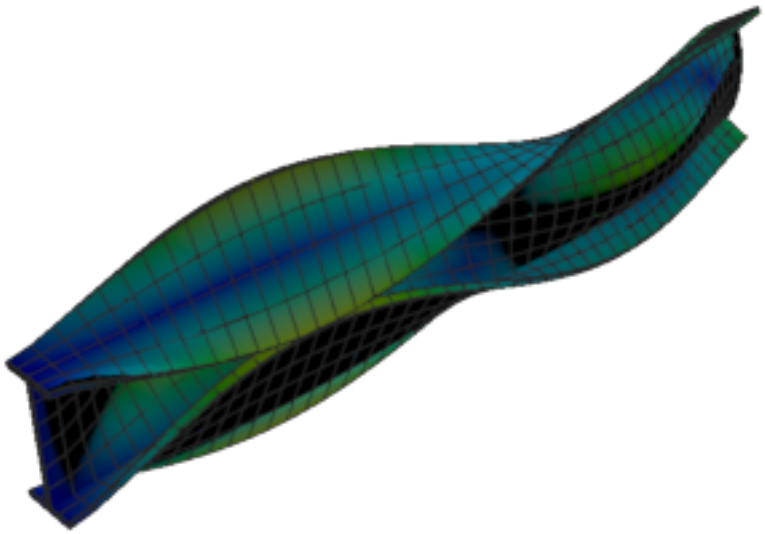


Figure 282: Calculix — mode 7, 142.497 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.320	-0.06	799690	2.49e-27	3.45	4.42e-24	-4.99e-14	1.858	-2.10e-12
Calculix	142.497	+0.06	801628	6.39e-25	3.43	7.82e-13	-7.99e-13	1.852	8.84e-07

Mode 8

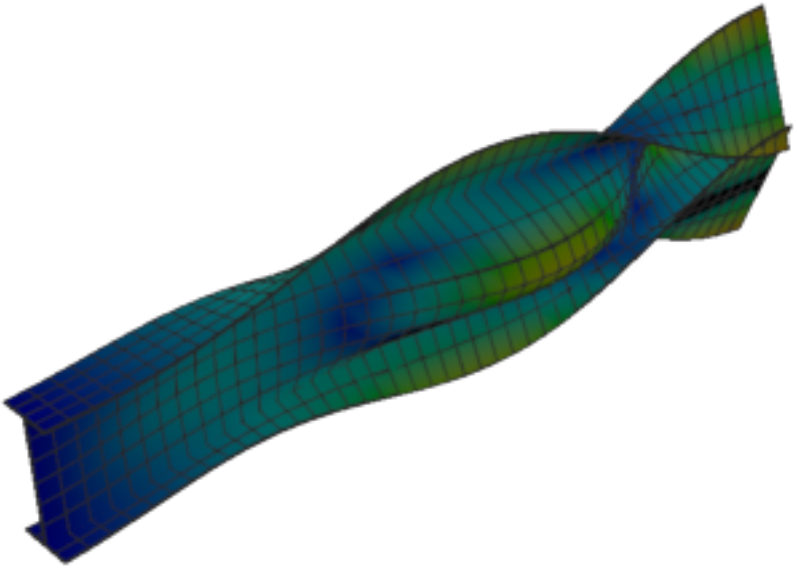


Figure 283: Abaqus — mode 8, 165.849 Hz

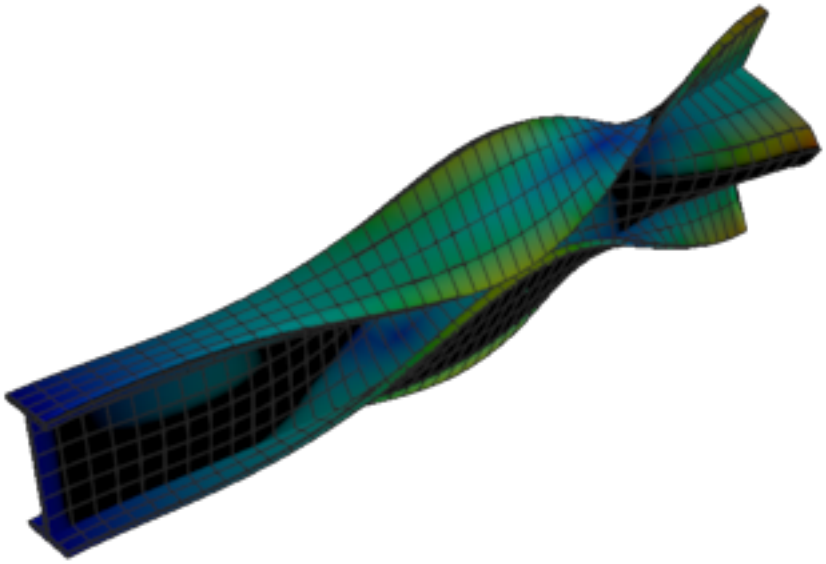


Figure 284: Calculix — mode 8, 165.981 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	165.850	-0.04	1.08589e+06	2.43e-27	5.53	1.06e-23	-4.93e-14	2.351	3.25e-12
Calculix	165.981	+0.04	1.08762e+06	9.68e-25	5.52	3.21e-13	9.84e-13	-2.350	5.66e-07

Mode 9

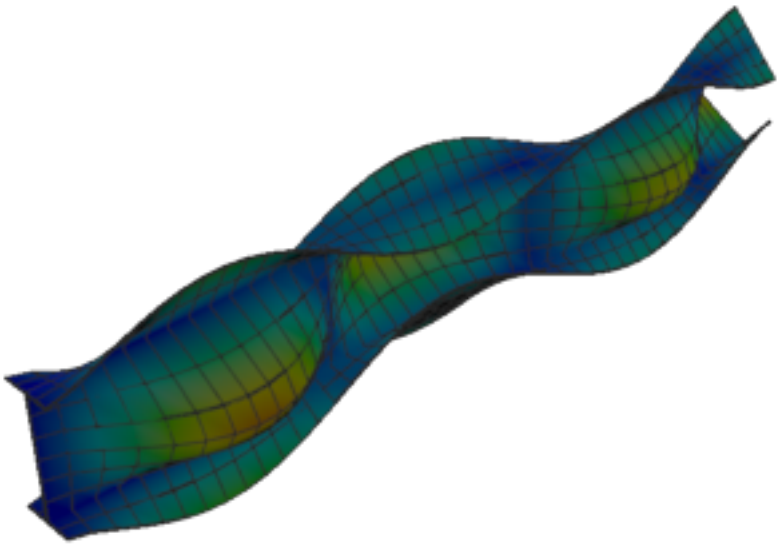


Figure 285: Abaqus — mode 9, 192.184 Hz

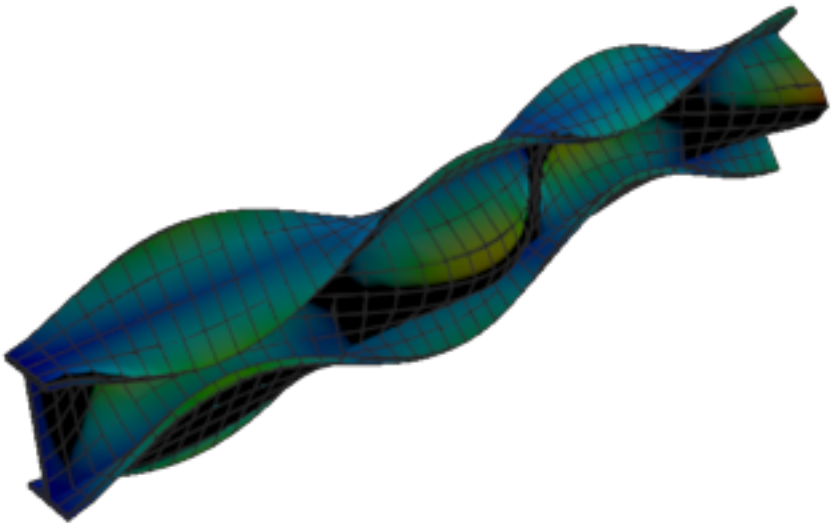


Figure 286: Calculix — mode 9, 192.870 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.180	-0.18	1.45812e+06	1.15e-27	0.65	1.20e-25	-3.40e-14	-0.808	3.47e-13
Calculix	192.870	+0.18	1.46855e+06	3.29e-24	0.64	7.71e-14	1.81e-12	0.799	-2.78e-07

Mode 10

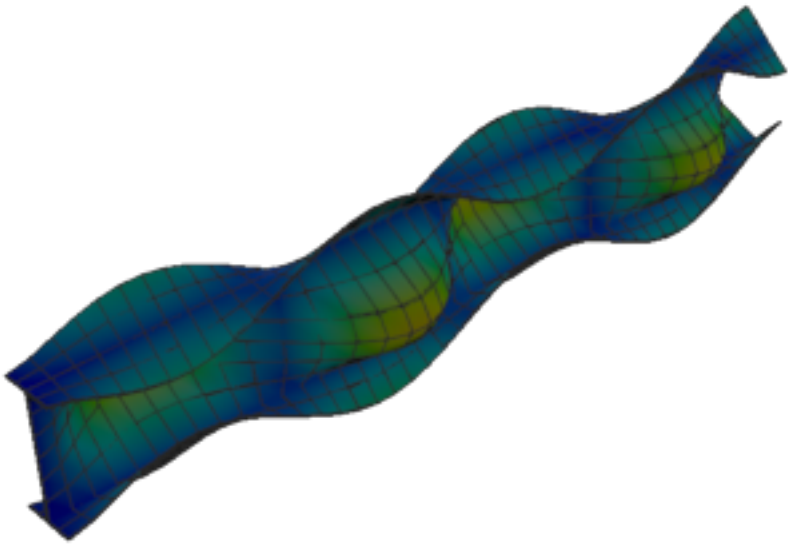


Figure 287: Abaqus — mode 10, 232.867 Hz

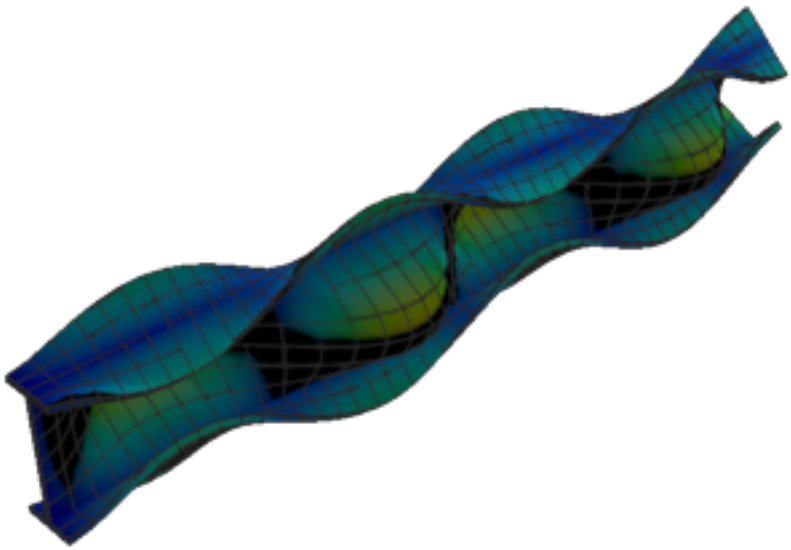


Figure 288: Calculix — mode 10, 234.347 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	232.870	-0.32	2.14079e+06	4.88e-26	1.37	1.33e-23	-2.21e-13	1.170	-3.65e-12
Calculix	234.347	+0.32	2.1681e+06	5.08e-24	1.39	3.57e-14	-2.25e-12	1.178	1.89e-07

Mode 11

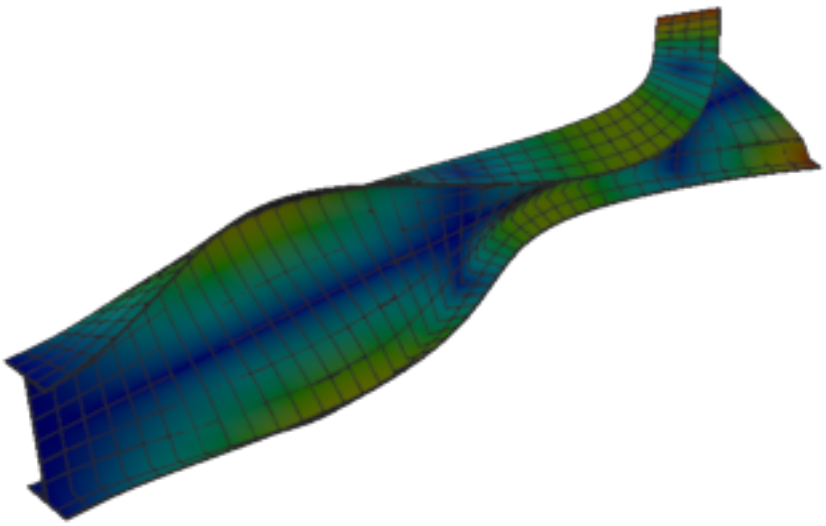


Figure 289: Abaqus — mode 11, 250.173 Hz

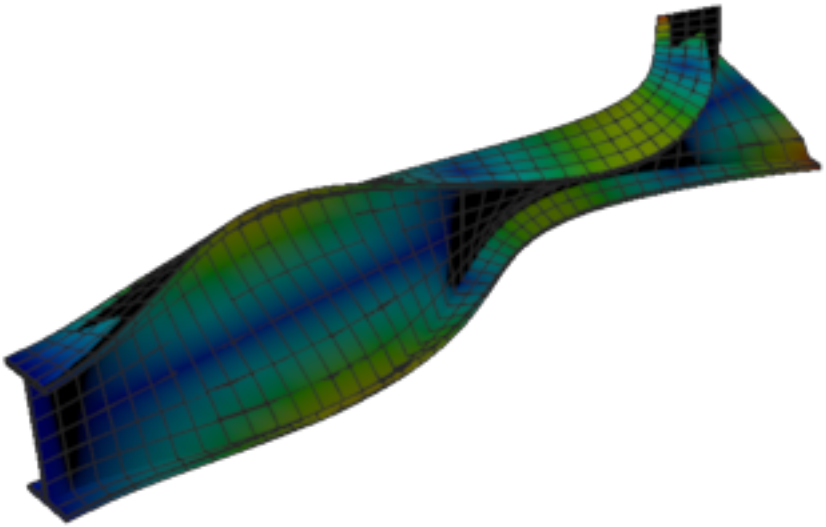


Figure 290: Calculix — mode 11, 250.147 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.170	4.62e-03	2.47082e+06	3.28e-21	3.98e-19	6.25e-19	5.72e-11	6.31e-10	-7.90e-10
Calculix	250.147	-4.62e-03	2.4703e+06	2.11e-10	3.23e-21	1.17e-22	-1.45e-05	5.68e-11	-1.08e-11

Solid, 1st order, TET

Mode 1

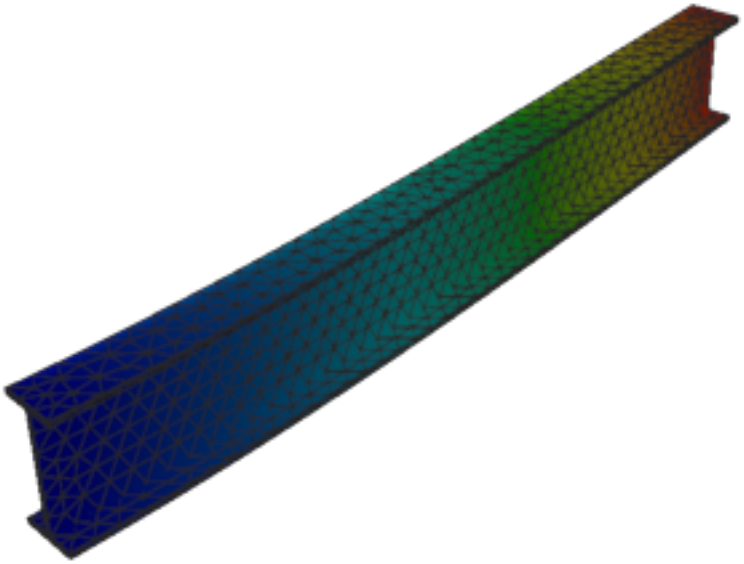


Figure 291: Abaqus — mode 1, 15.790 Hz

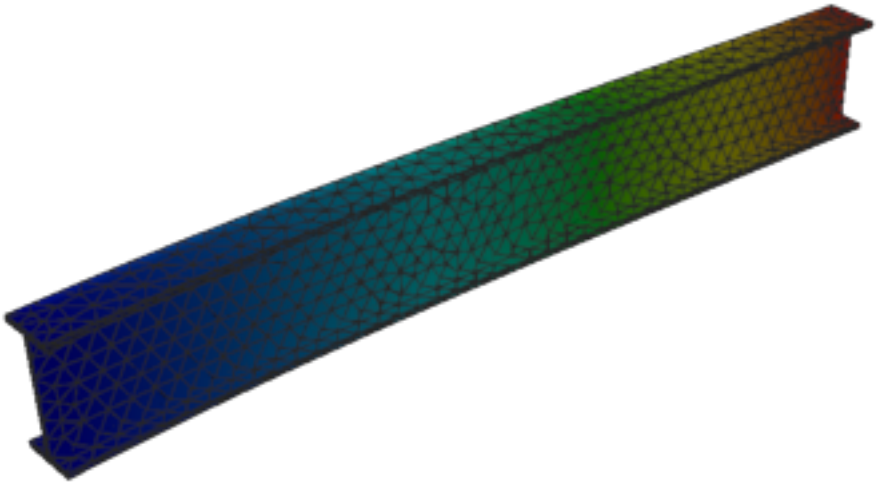


Figure 292: Calculix — mode 1, 15.928 Hz

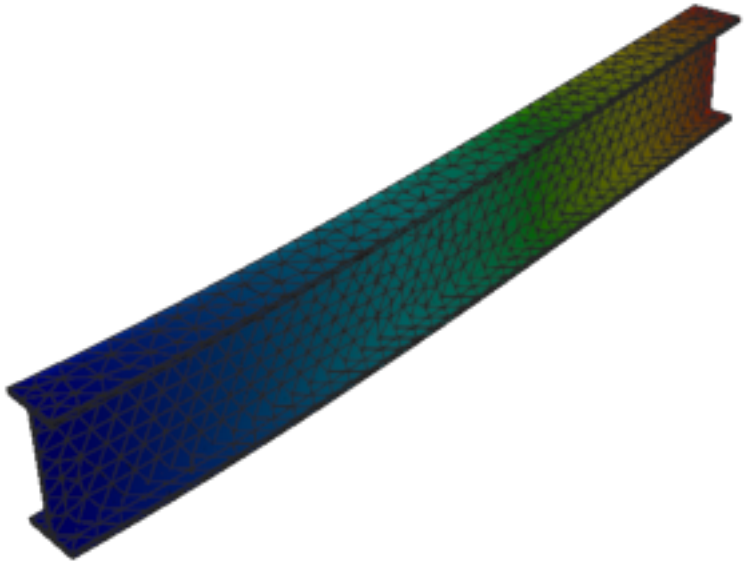


Figure 293: Code_Aster — mode 1, 15.928 Hz

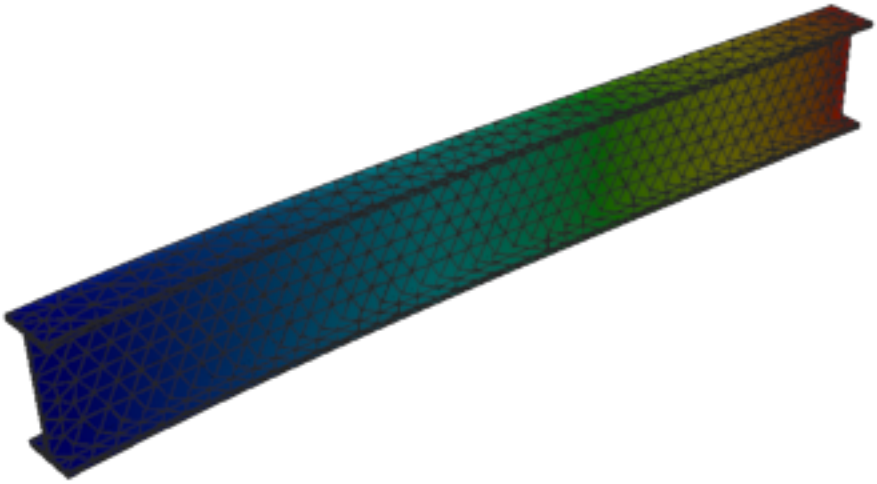


Figure 294: Sesam — mode 1, 15.793 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	15.790	-0.44	9842.9	2.50e-14	117.47	2.82e-04	1.58e-07	10.838	-0.017
Calculix	15.928	+0.43	10016.3	1.60e-09	118.18	3.43e-04	-4.00e-05	-10.871	0.019
Code_Aster	15.928	+0.43	10015.3	1.52e-09	118.17	3.43e-04	3.89e-05	10.871	-0.019
Sesam	15.793	-0.42	9846.92	5.82e-13	117.52	2.82e-04	7.63e-07	-10.841	0.017

Mode 2

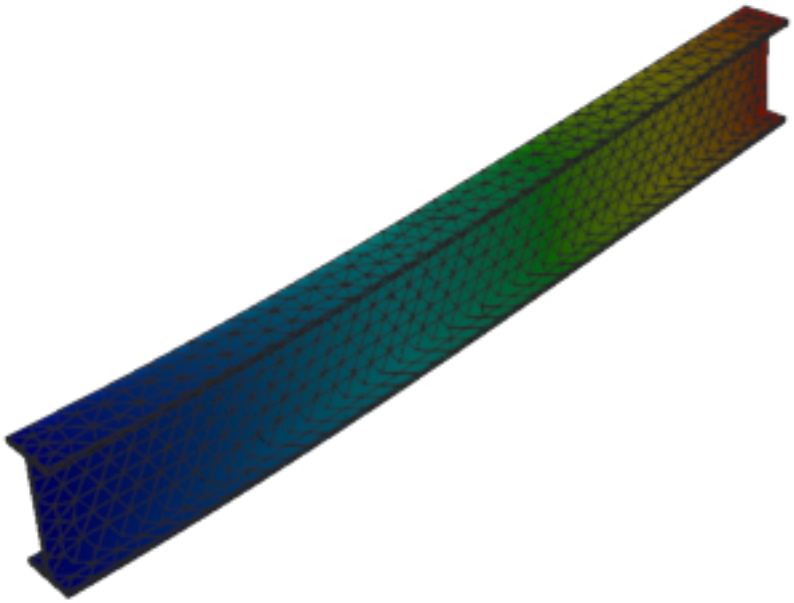


Figure 295: Abaqus — mode 2, 51.532 Hz

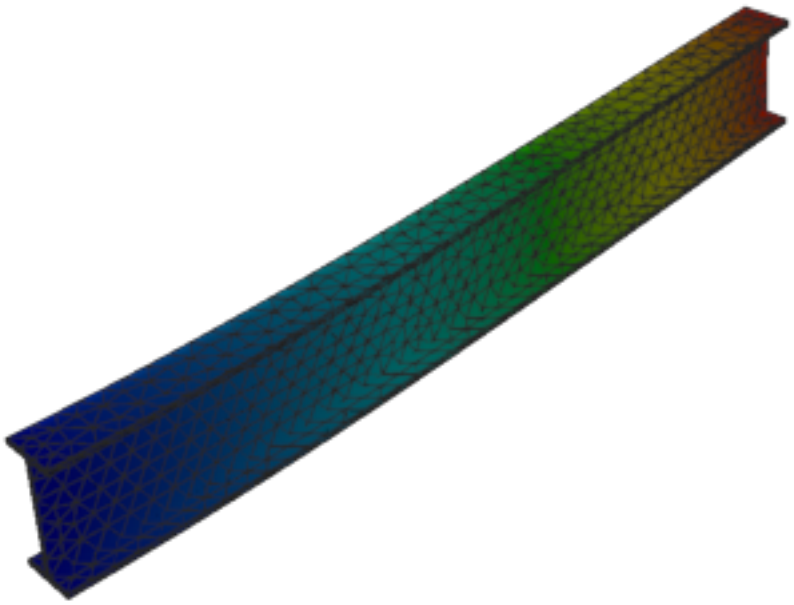


Figure 296: Calculix — mode 2, 51.590 Hz

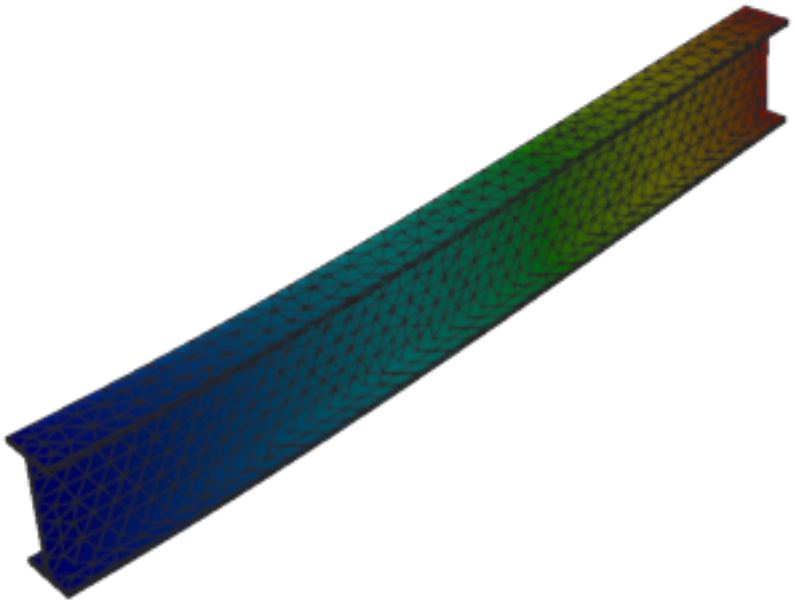


Figure 297: Code_Aster — mode 2, 51.587 Hz

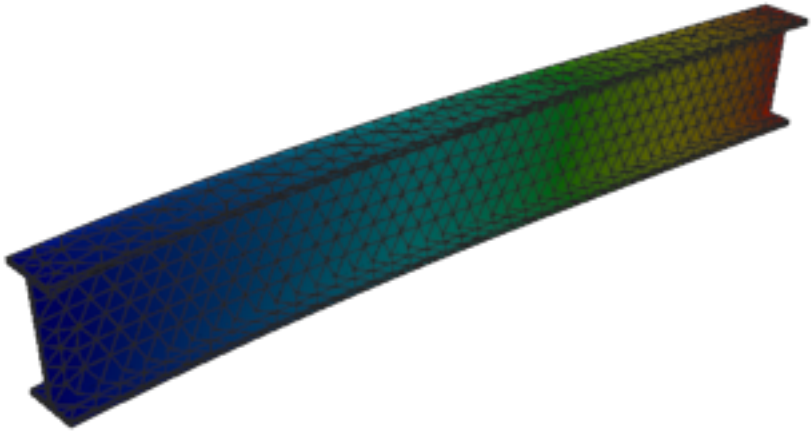


Figure 298: Sesam — mode 2, 51.542 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.532	-0.06	104836	1.44e-08	7.90e-04	118.25	1.20e-04	0.028	10.874
Calculix	51.590	+0.05	105071	6.78e-09	9.26e-04	118.38	8.23e-05	0.030	10.880
Code_Aster	51.587	+0.05	105061	7.19e-09	9.26e-04	118.37	8.48e-05	0.030	10.880
Sesam	51.542	-0.04	104877	1.53e-08	7.86e-04	118.29	-1.24e-04	-0.028	-10.876

Mode 3

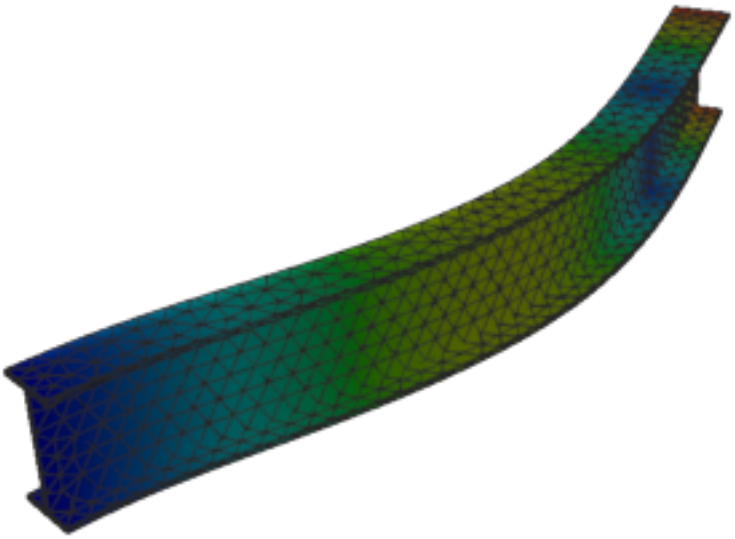


Figure 299: Abaqus — mode 3, 97.588 Hz

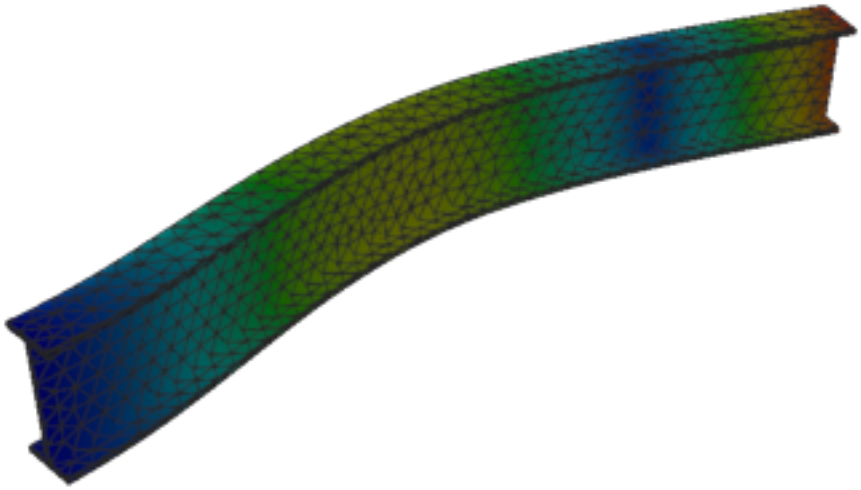


Figure 300: Calculix — mode 3, 102.446 Hz

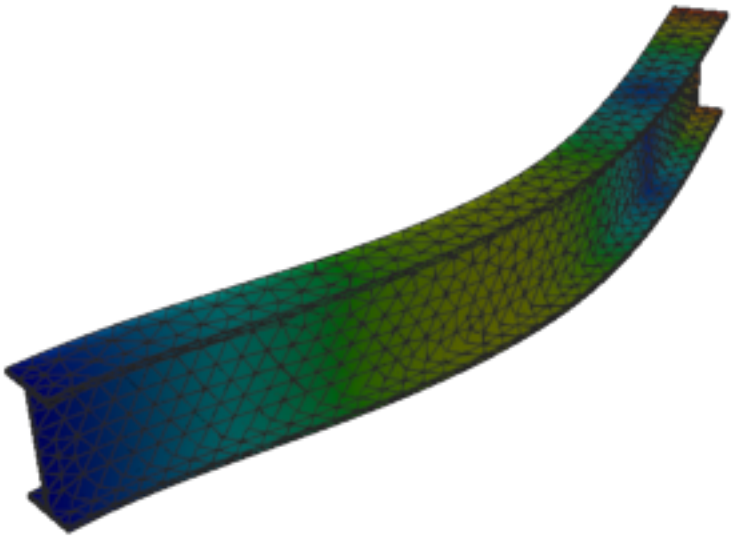


Figure 301: Code_Aster — mode 3, 102.410 Hz

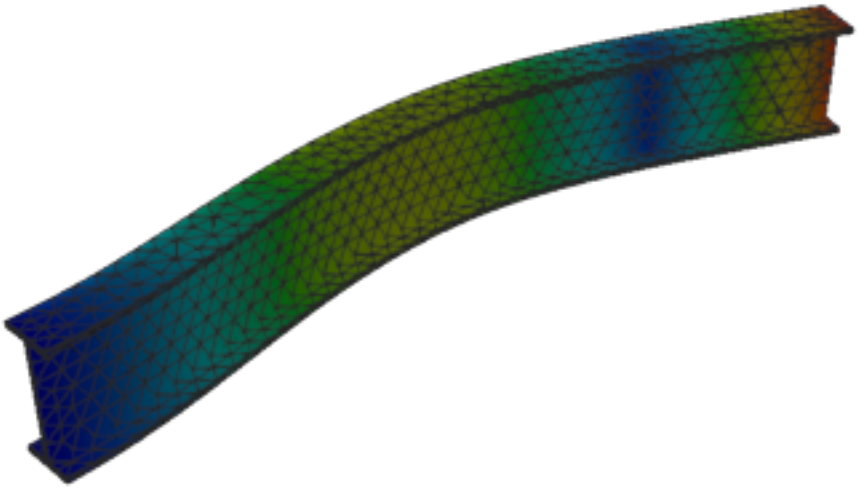


Figure 302: Sesam — mode 3, 97.731 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.588	-2.45	375967	4.61e-08	34.91	3.71e-04	-2.15e-04	-5.909	0.019
Calculix	102.446	+2.40	414334	1.14e-07	35.54	4.01e-04	3.38e-04	5.961	-0.020
Code_Aster	102.410	+2.37	414043	1.10e-07	35.54	3.99e-04	-3.32e-04	-5.961	0.020
Sesam	97.731	-2.31	377075	4.30e-08	34.94	3.65e-04	2.07e-04	5.911	-0.019

Mode 4

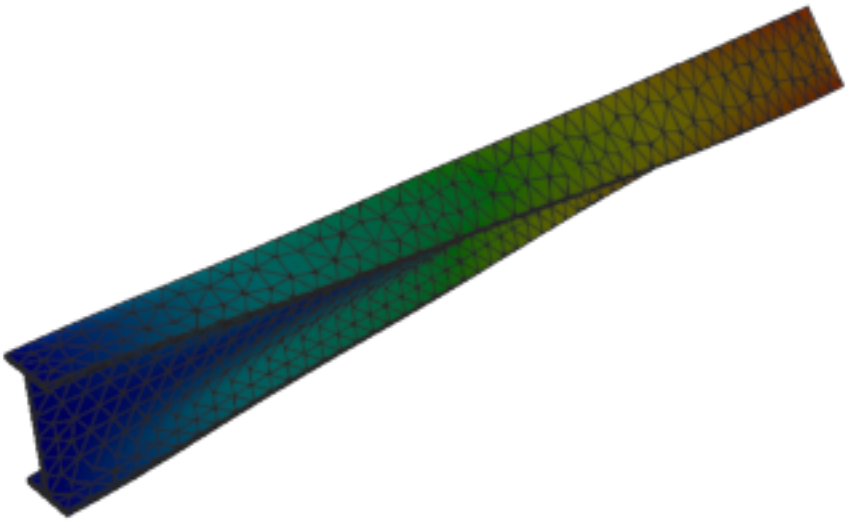


Figure 303: Abaqus — mode 4, 104.803 Hz

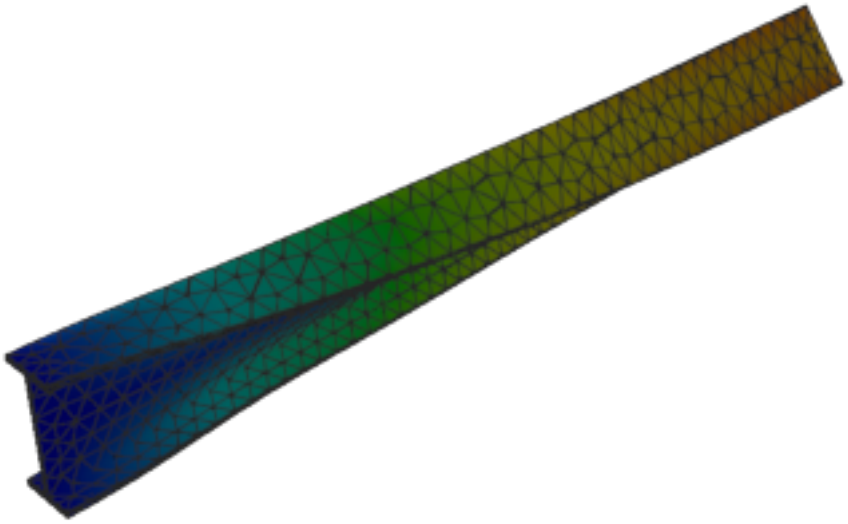


Figure 304: Calculix — mode 4, 116.744 Hz

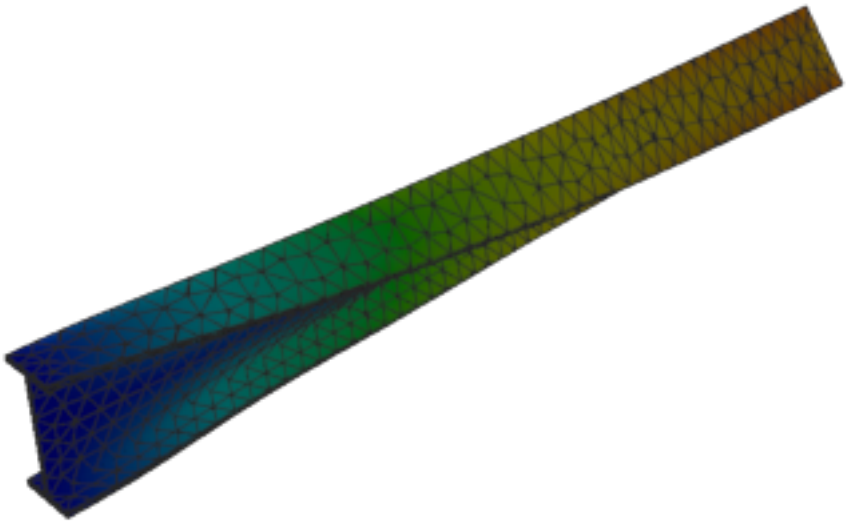


Figure 305: Code_Aster — mode 4, 116.512 Hz

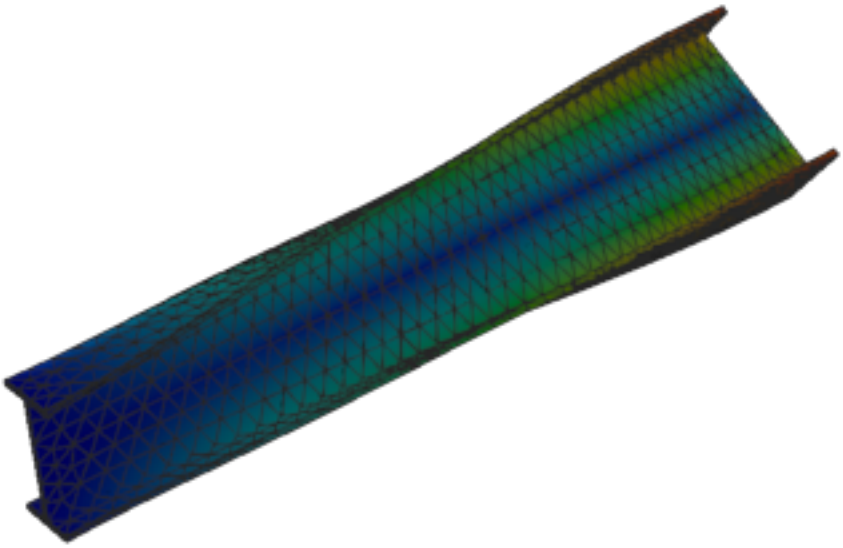


Figure 306: Sesam — mode 4, 105.691 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	104.800	-5.53	433617	6.83e-06	0.01	1.07e-05	0.003	-0.075	0.003
Calculix	116.744	+5.24	538061	9.63e-06	0.01	5.55e-05	0.003	-0.108	0.007
Code_Aster	116.512	+5.03	535924	9.53e-06	0.01	5.63e-05	0.003	-0.110	0.008
Sesam	105.691	-4.73	440993	6.99e-06	4.82e-03	1.08e-05	-0.003	0.069	-0.003

Mode 5

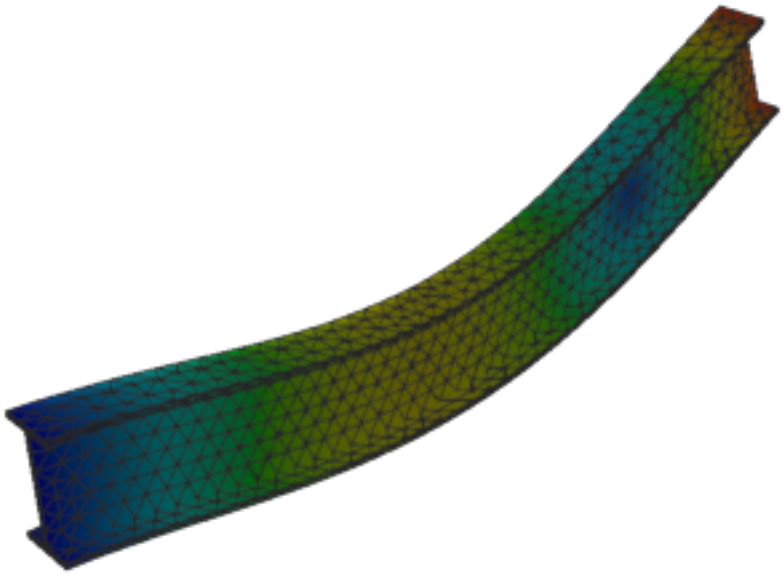


Figure 307: Abaqus — mode 5, 260.590 Hz

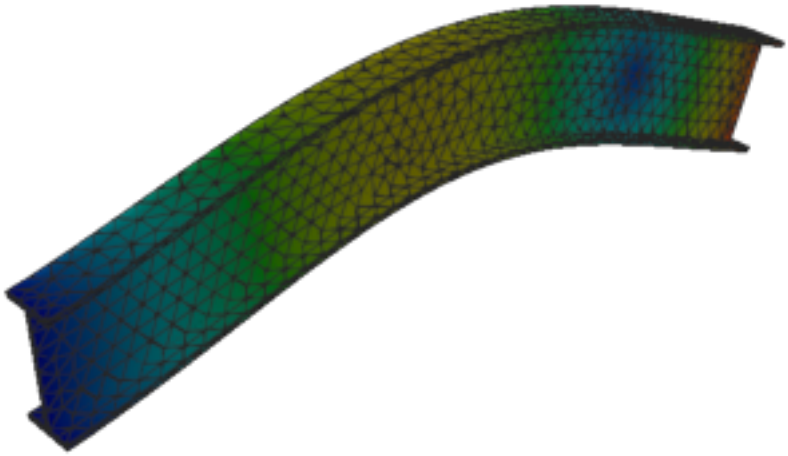


Figure 308: Calculix — mode 5, 261.877 Hz

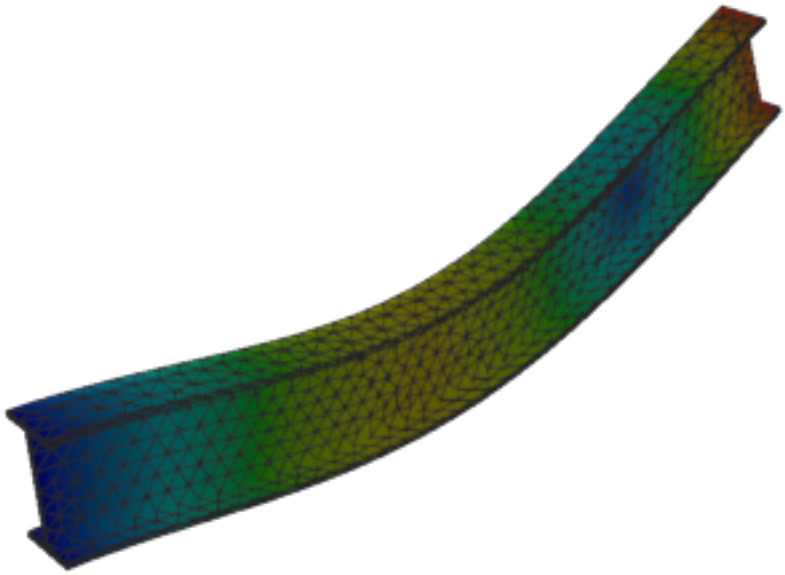


Figure 309: Code_Aster — mode 5, 261.806 Hz

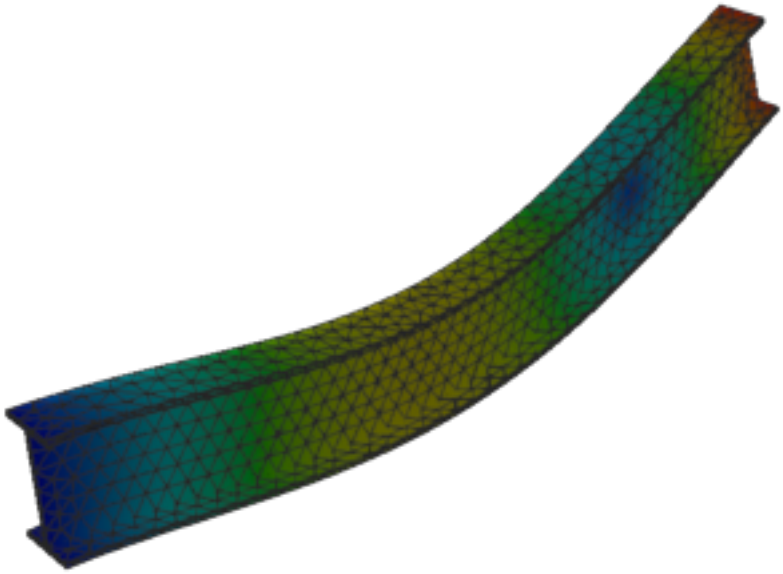


Figure 310: Sesam — mode 5, 260.887 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	260.590	-0.27	2.68086e+06	3.30e-07	0.02	39.75	-0.001	0.132	-6.305
Calculix	261.877	+0.22	2.70742e+06	9.91e-08	7.99e-04	39.86	3.15e-04	-0.028	6.313
Code_Aster	261.806	+0.20	2.70594e+06	1.05e-07	8.24e-04	39.88	-3.24e-04	0.029	-6.315
Sesam	260.887	-0.15	2.68699e+06	3.61e-07	0.01	39.76	-0.001	0.122	-6.305

Mode 6

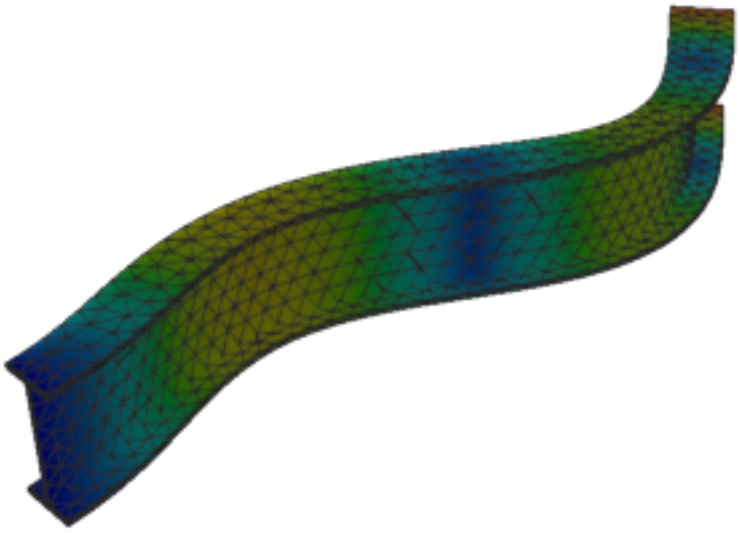


Figure 311: Abaqus — mode 6, 266.263 Hz

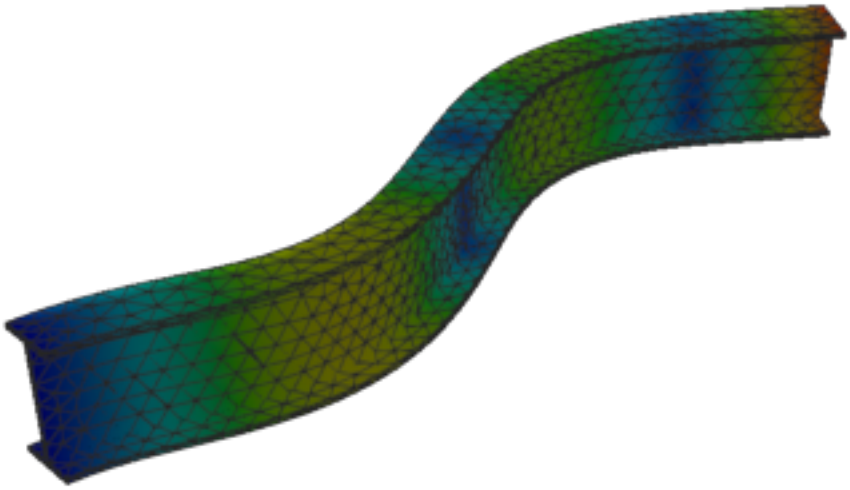


Figure 312: Calculix — mode 6, 284.697 Hz

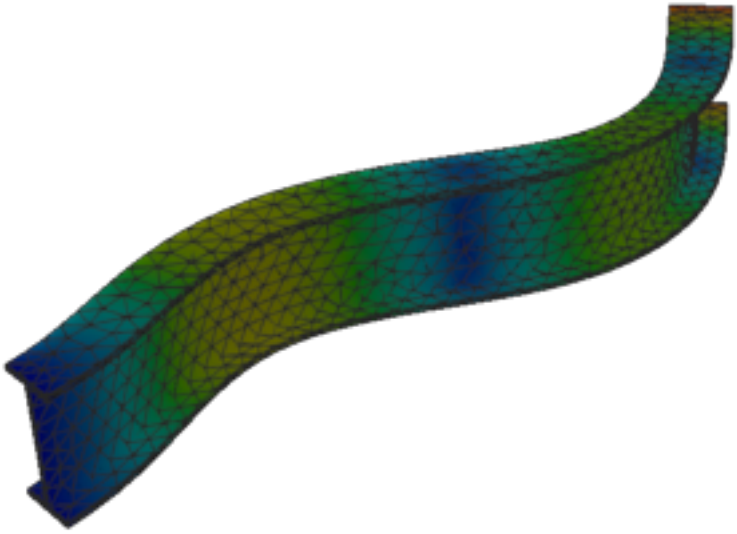


Figure 313: Code_Aster — mode 6, 284.465 Hz

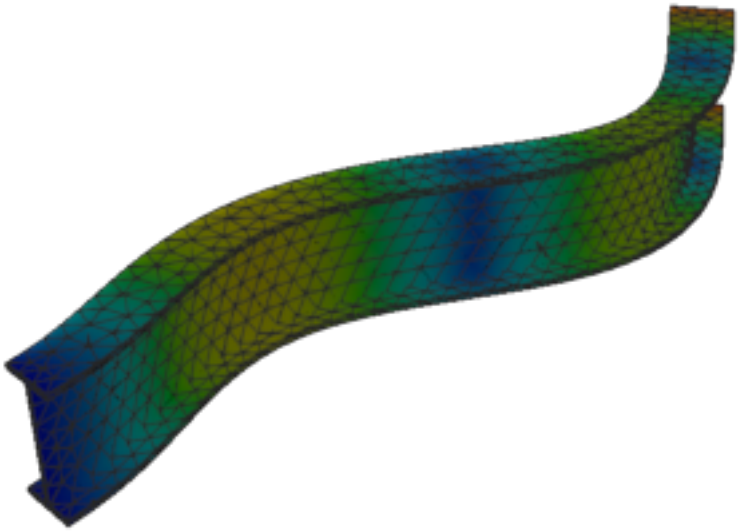


Figure 314: Sesam — mode 6, 267.131 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	266.260	-3.40	2.79886e+06	1.72e-08	13.24	0.05	1.31e-04	3.638	0.226
Calculix	284.697	+3.29	3.19983e+06	2.41e-07	12.79	3.17e-03	4.91e-04	-3.576	-0.056
Code_Aster	284.465	+3.20	3.19461e+06	2.62e-07	12.80	3.27e-03	-0.001	3.577	0.057
Sesam	267.131	-3.09	2.81714e+06	1.49e-08	13.26	0.04	1.22e-04	3.642	0.207

Mode 7

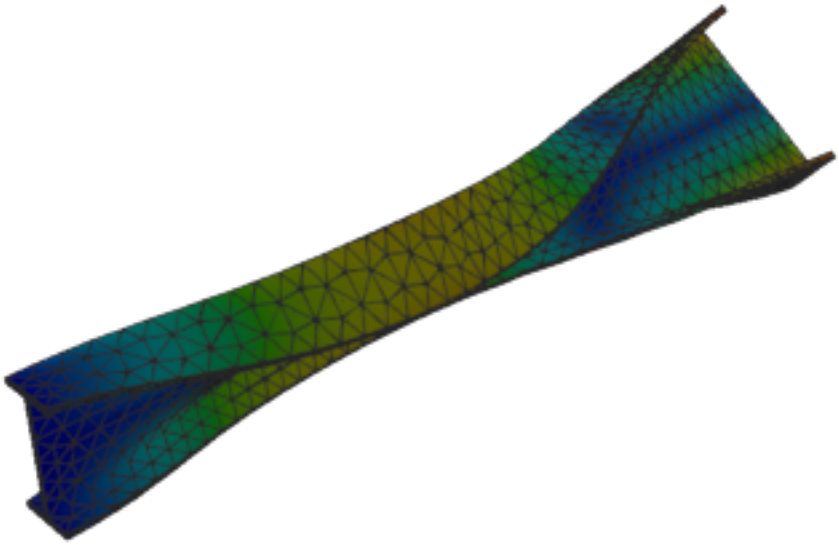


Figure 315: Abaqus — mode 7, 285.051 Hz

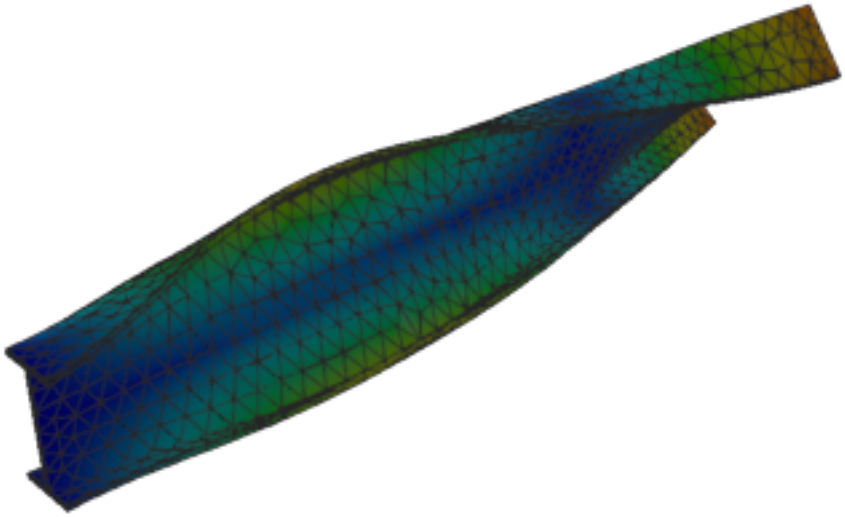


Figure 316: Calculix — mode 7, 357.062 Hz

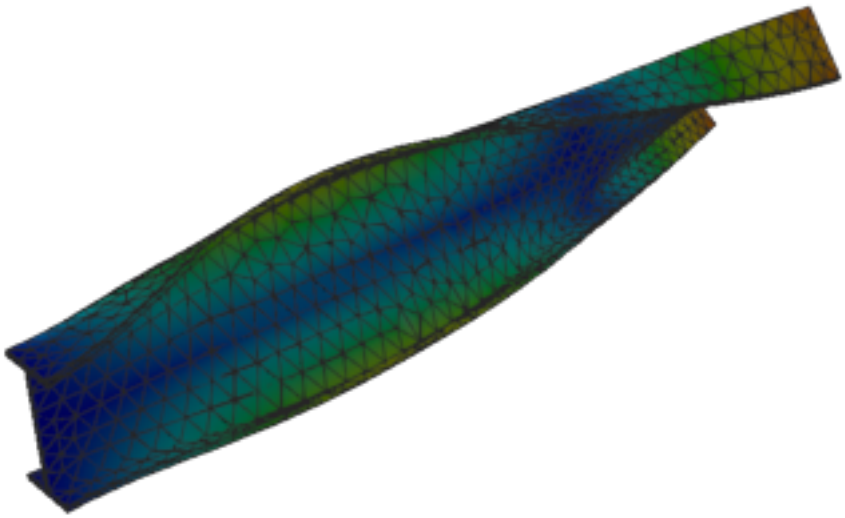


Figure 317: Code_Aster — mode 7, 356.306 Hz

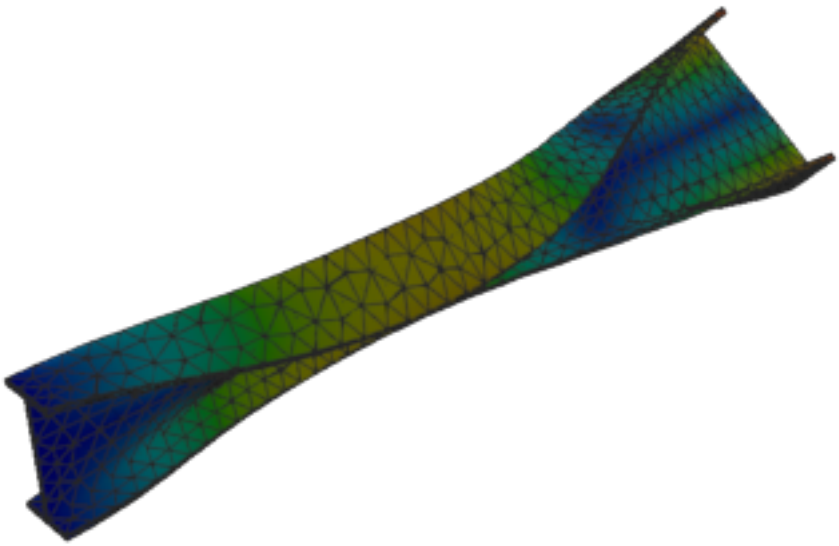


Figure 318: Sesam — mode 7, 287.461 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	285.050	-11.33	3.20779e+06	1.44e-07	9.28e-04	1.25e-03	3.79e-04	0.030	-0.035
Calculix	357.062	+11.07	5.03322e+06	4.30e-04	1.56e-03	3.86e-04	-0.021	0.040	-0.020
Code_Aster	356.306	+10.84	5.01194e+06	4.21e-04	1.60e-03	3.90e-04	-0.021	0.040	-0.020
Sesam	287.461	-10.58	3.26225e+06	1.26e-07	7.47e-04	1.07e-03	3.54e-04	0.027	-0.033

Mode 8

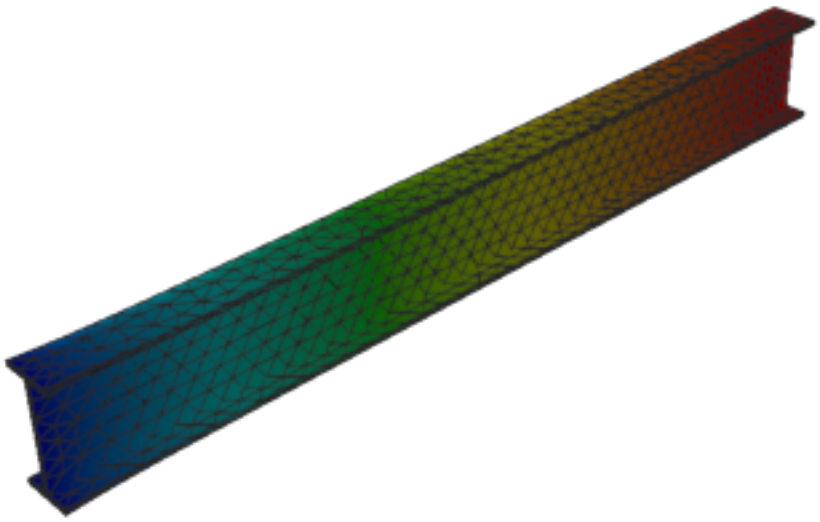


Figure 319: Abaqus — mode 8, 432.310 Hz

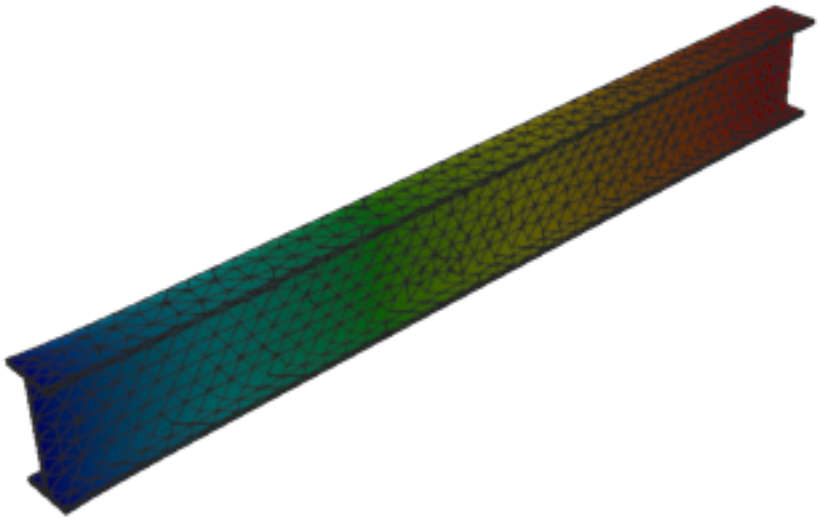


Figure 320: Calculix — mode 8, 432.360 Hz

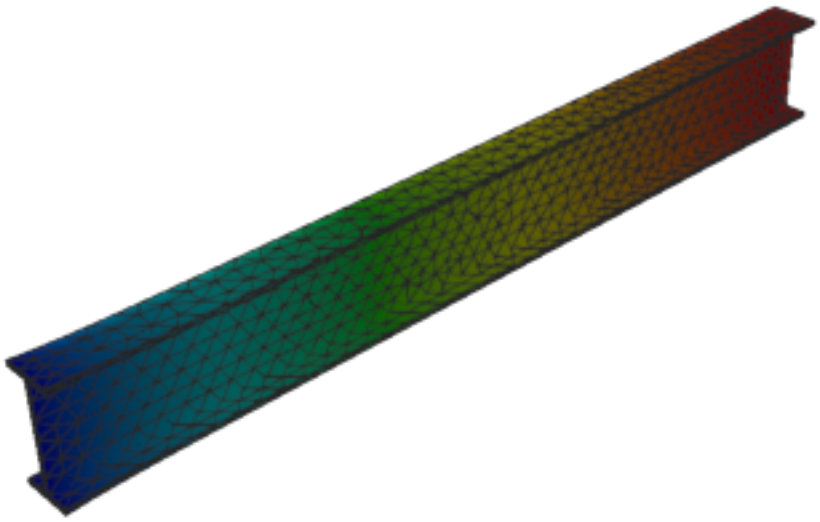


Figure 321: Code_Aster — mode 8, 432.350 Hz

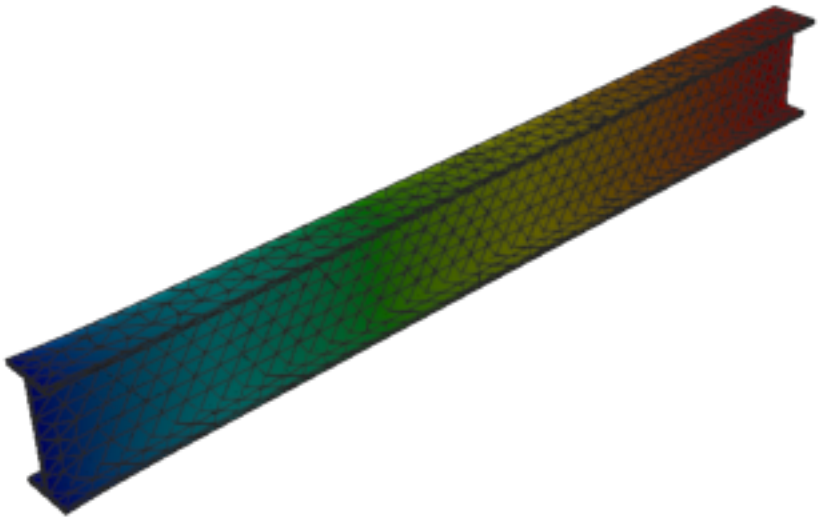


Figure 322: Sesam — mode 8, 432.347 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	432.310	-0.01	7.37819e+06	153.38	2.06e-06	3.31e-08	12.385	-0.001	1.82e-04
Calculix	432.360	4.12e-03	7.37989e+06	153.38	9.52e-08	2.76e-08	12.384	3.09e-04	-1.66e-04
Code_Aster	432.350	1.96e-03	7.37957e+06	153.41	9.68e-08	2.69e-08	12.386	3.11e-04	-1.64e-04
Sesam	432.347	1.27e-03	7.37947e+06	153.37	1.91e-06	2.88e-08	12.384	-0.001	1.70e-04

Mode 9

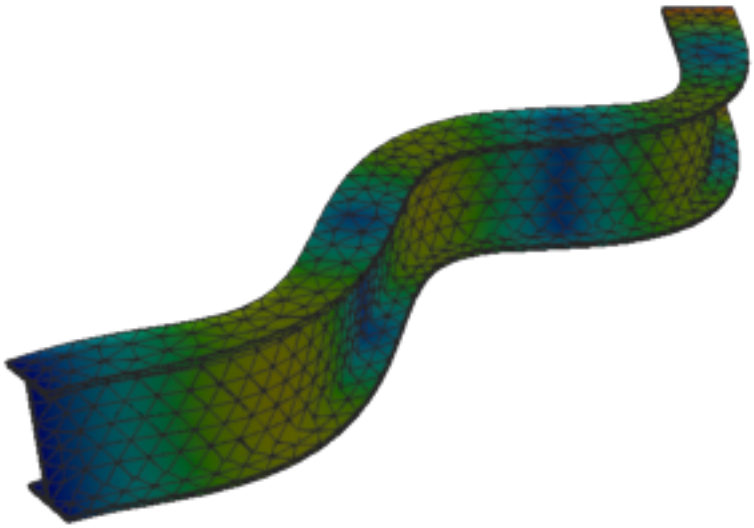


Figure 323: Abaqus — mode 9, 500.864 Hz

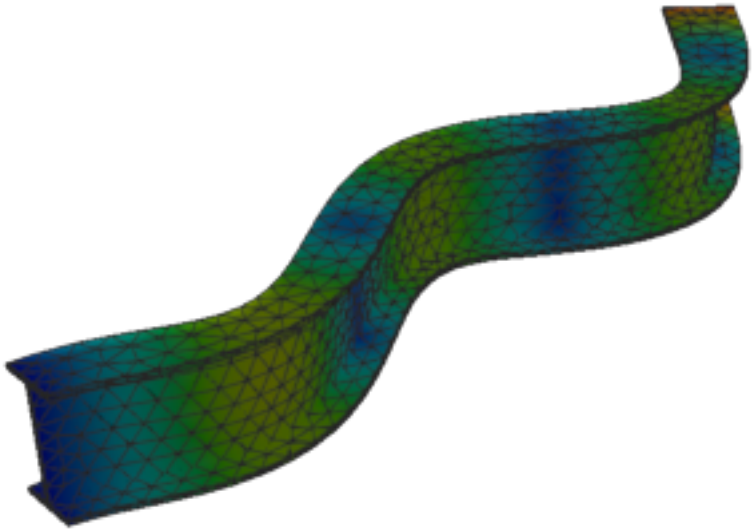


Figure 324: Calculix — mode 9, 529.297 Hz

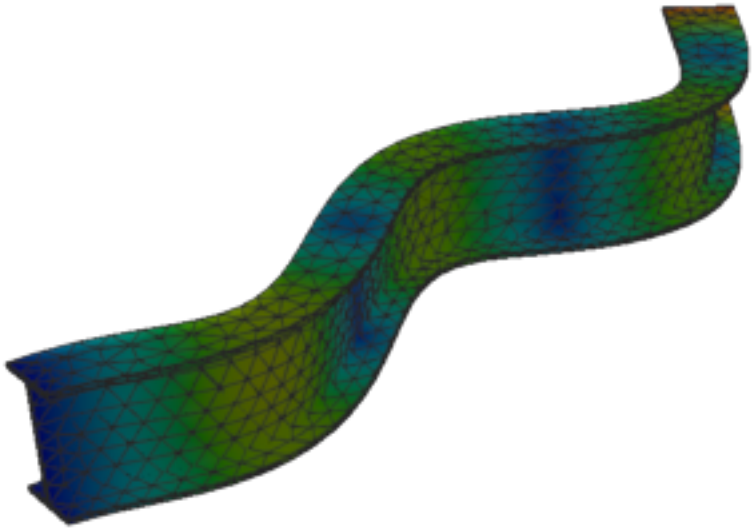


Figure 325: Code_Aster — mode 9, 528.489 Hz

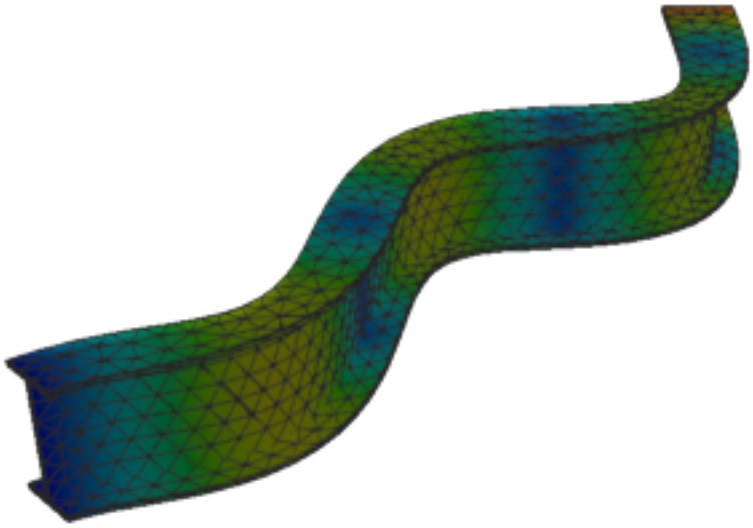


Figure 326: Sesam — mode 9, 503.862 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	500.860	-2.86	9.90373e+06	3.39e-05	6.33	4.73e-05	-0.006	-2.515	0.007
Calculix	529.297	+2.65	1.10601e+07	8.02e-07	6.06	4.08e-03	0.001	-2.462	0.064
Code_Aster	528.489	+2.49	1.10263e+07	7.98e-07	6.07	4.07e-03	0.001	-2.464	0.064
Sesam	503.862	-2.28	1.00226e+07	3.01e-05	6.34	4.92e-05	-0.005	-2.518	0.007

Mode 10

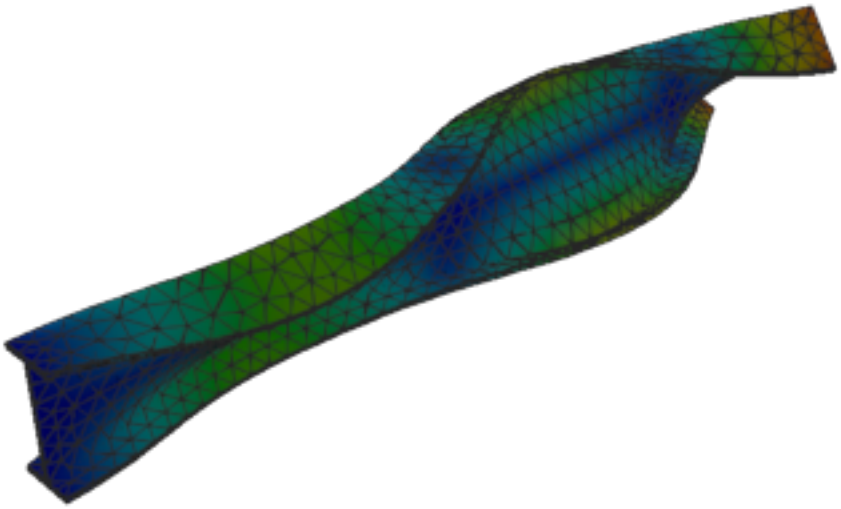


Figure 327: Abaqus — mode 10, 534.263 Hz

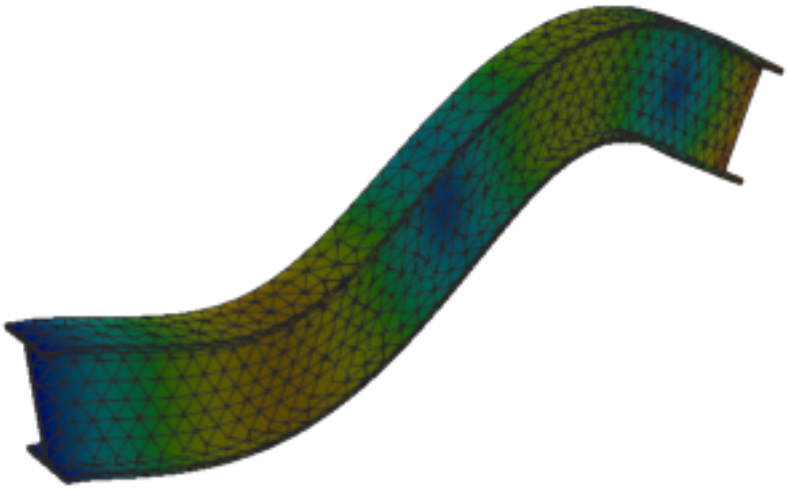


Figure 328: Calculix — mode 10, 601.943 Hz

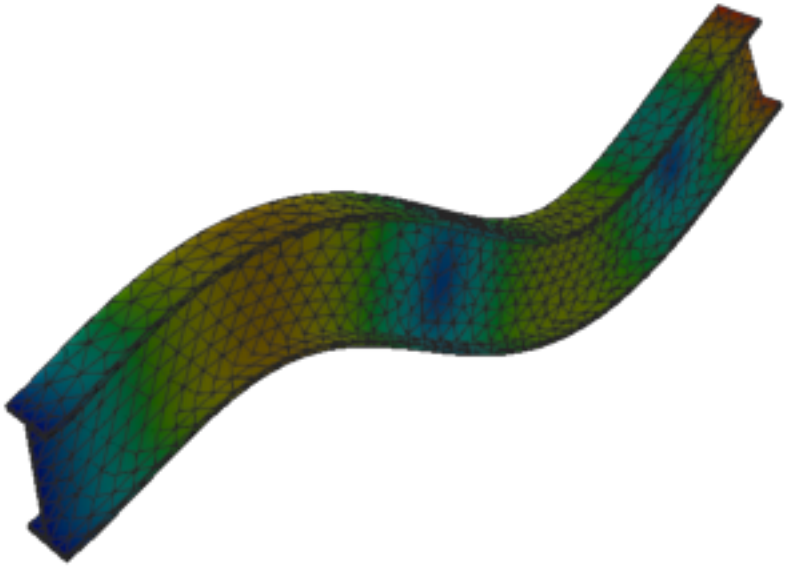


Figure 329: Code_Aster — mode 10, 601.586 Hz

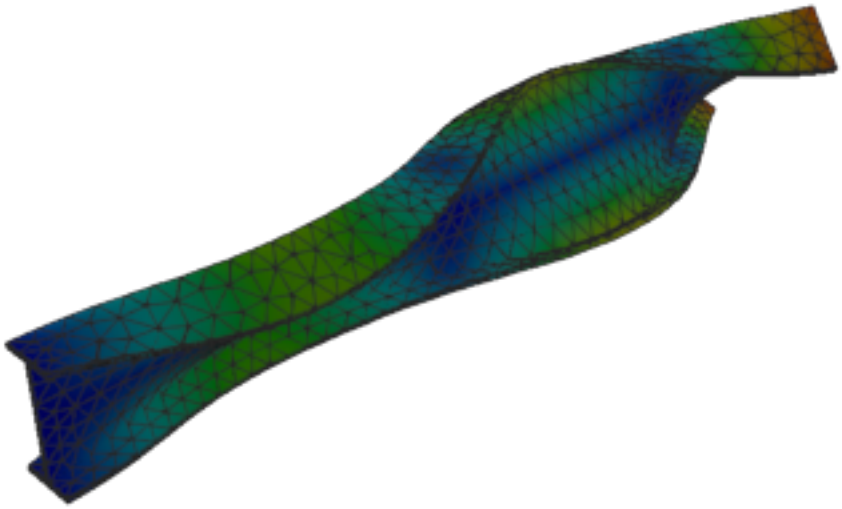


Figure 330: Sesam — mode 10, 539.506 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	534.260	-6.16	1.12686e+07	2.60e-04	2.77e-04	6.63e-06	-0.016	-0.017	0.003
Calculix	601.943	+5.73	1.43044e+07	3.09e-07	2.06e-03	12.76	0.001	-0.045	-3.572
Code_Aster	601.586	+5.67	1.42875e+07	3.19e-07	2.07e-03	12.79	-0.001	0.045	3.576
Sesam	539.506	-5.24	1.14909e+07	2.36e-04	2.31e-04	6.55e-06	-0.015	-0.015	0.003

Mode 11

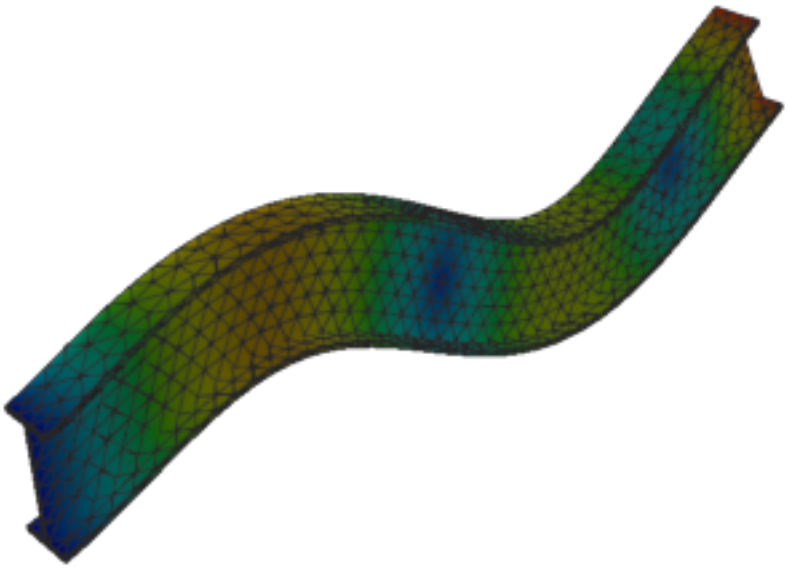


Figure 331: Abaqus — mode 11, 596.780 Hz

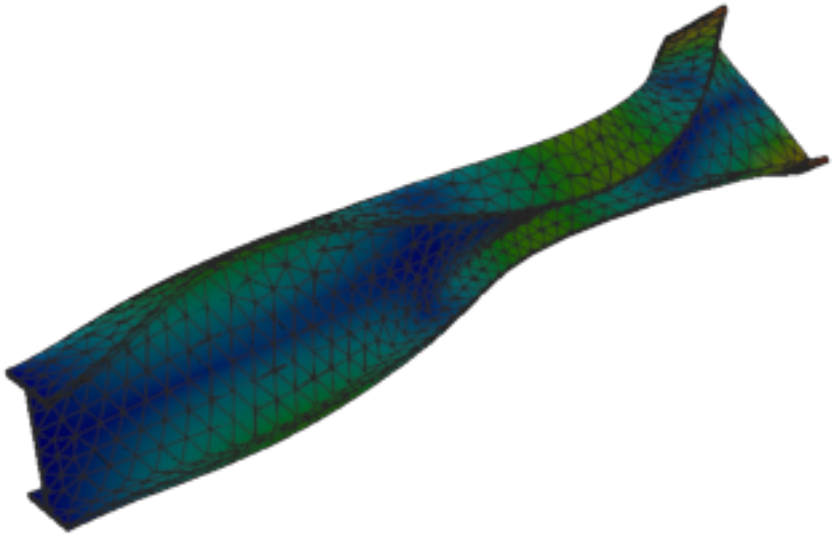


Figure 332: Calculix — mode 11, 617.646 Hz

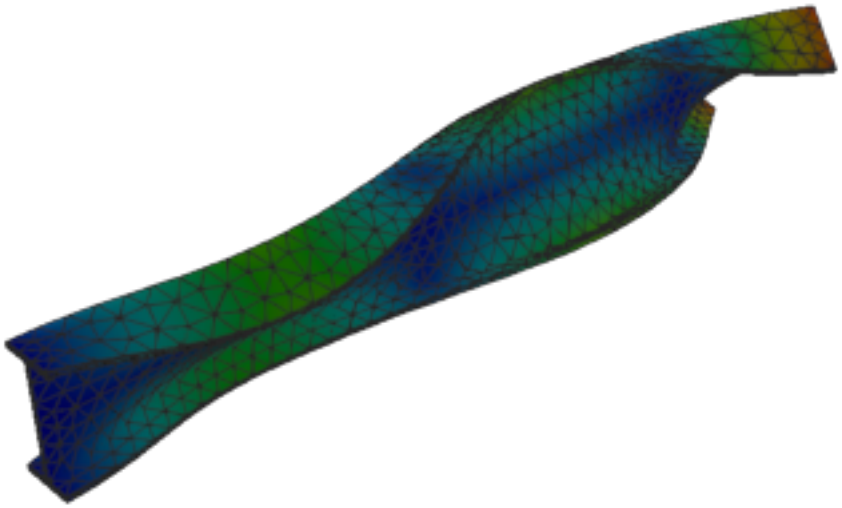


Figure 333: Code_Aster — mode 11, 616.155 Hz

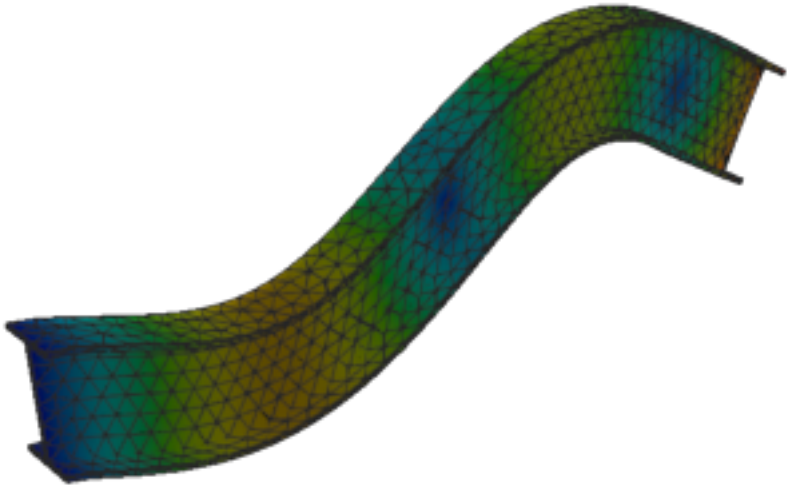


Figure 334: Sesam — mode 11, 598.200 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	596.780	-1.72	1.40601e+07	4.55e-07	1.33e-07	12.80	-0.001	-3.65e-04	3.578
Calculix	617.646	+1.72	1.50605e+07	1.44e-04	2.59e-04	8.16e-04	-0.012	0.016	0.029
Code_Aster	616.155	+1.48	1.49879e+07	1.45e-04	2.65e-04	9.64e-04	0.012	-0.016	-0.031
Sesam	598.200	-1.48	1.41271e+07	4.36e-07	3.62e-08	12.80	0.001	1.90e-04	-3.578

Solid, 1st order, HEX

Mode 1

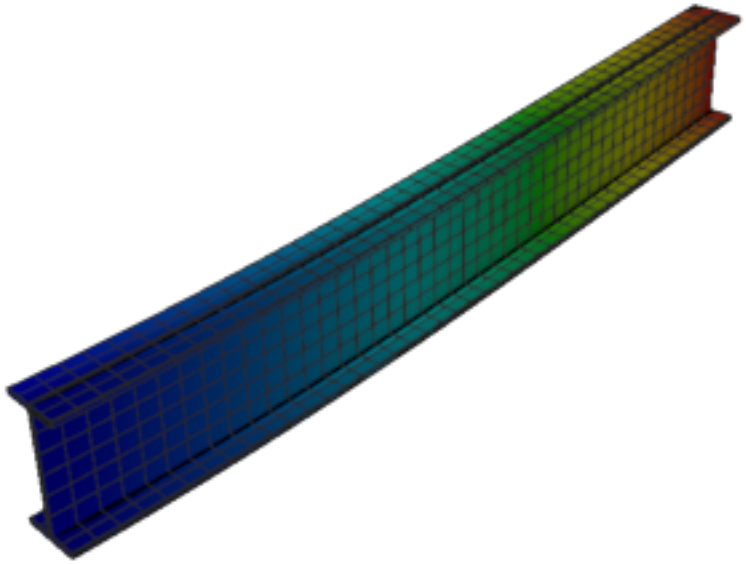


Figure 335: Abaqus — mode 1, 13.435 Hz

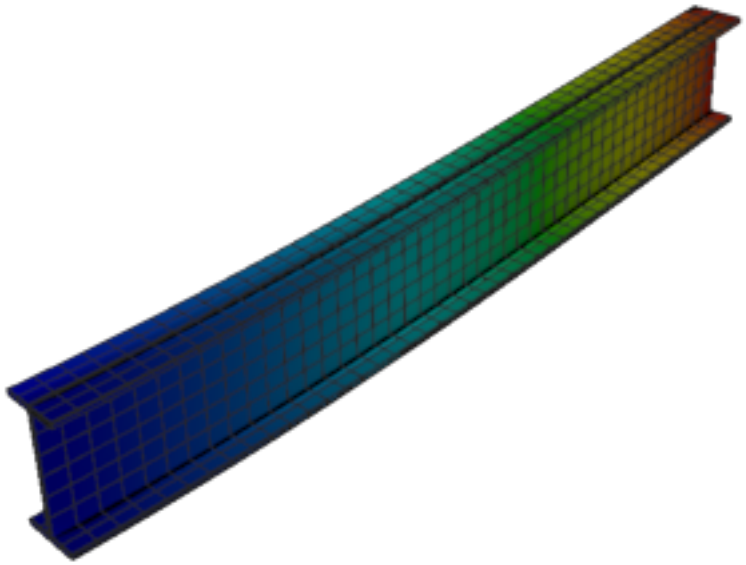


Figure 336: Calculix — mode 1, 13.620 Hz

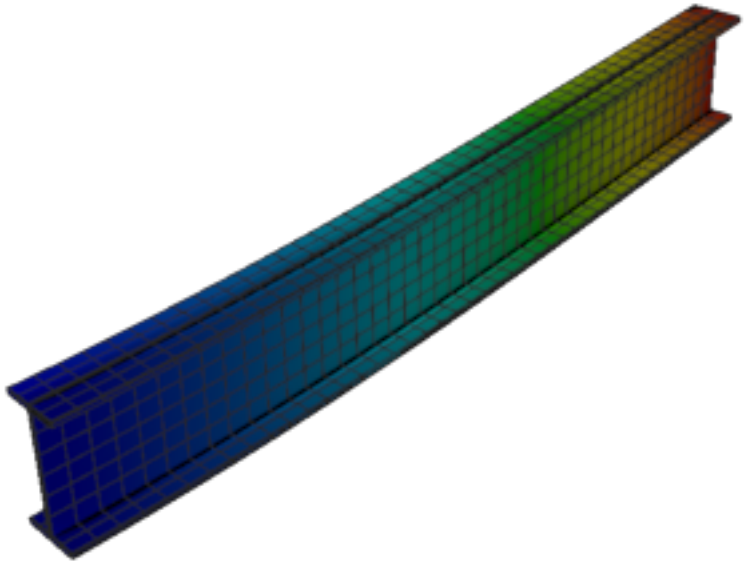


Figure 337: Code_Aster — mode 1, 13.620 Hz

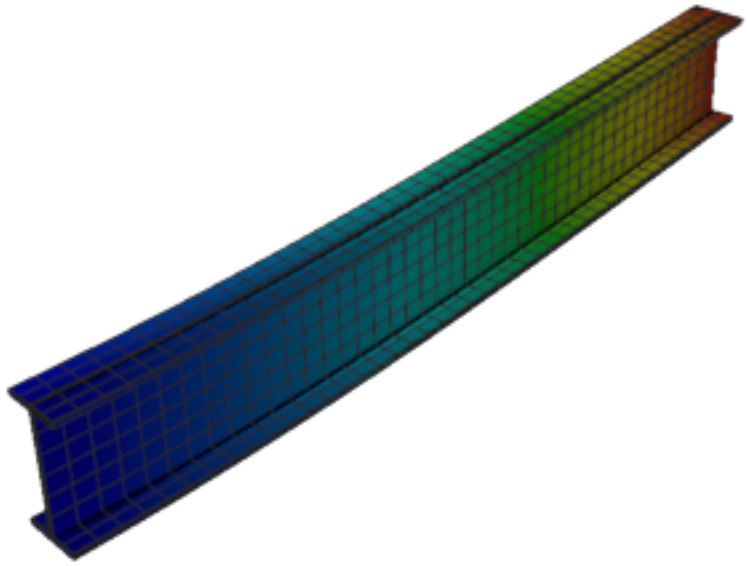


Figure 338: Sesam — mode 1, 12.977 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	13.435	+0.16	7125.8	4.92e-29	116.39	5.47e-23	-7.01e-15	10.788	-7.39e-12
Calculix	13.620	+1.55	7323.93	3.07e-25	116.36	7.93e-23	-5.54e-13	10.787	-8.91e-12
Code_Aster	13.620	+1.55	7323.93	3.34e-26	116.36	1.98e-22	1.83e-13	10.787	-1.41e-11
Sesam	12.977	-3.25	6647.9	9.71e-25	116.43	6.43e-21	-9.85e-13	10.790	8.02e-11

Mode 2

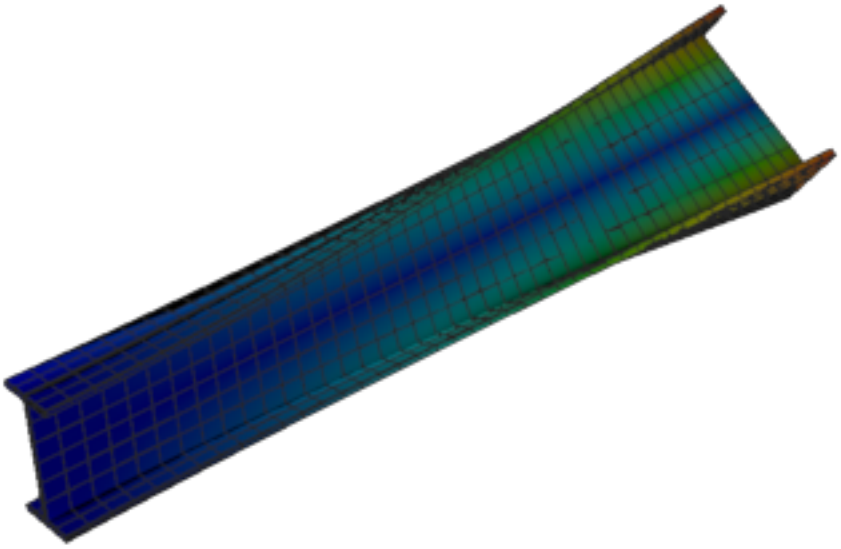


Figure 339: Abaqus — mode 2, 20.731 Hz

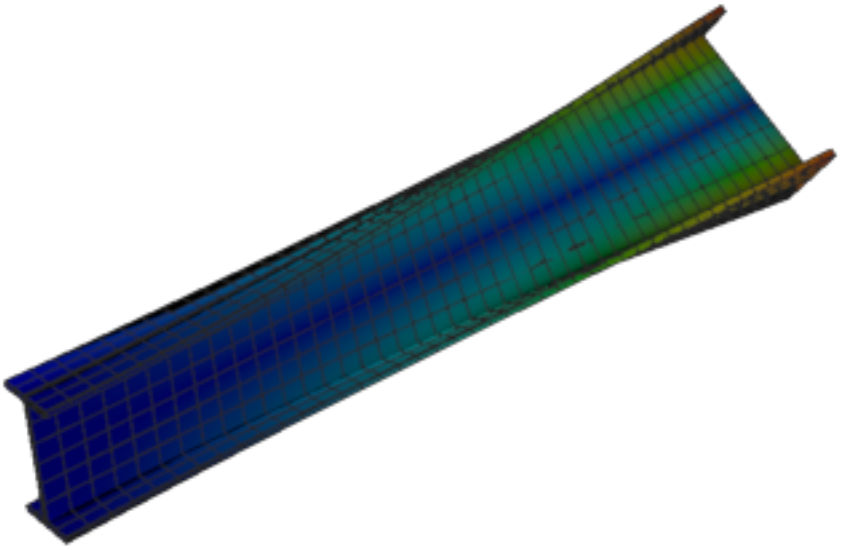


Figure 340: Calculix — mode 2, 21.069 Hz

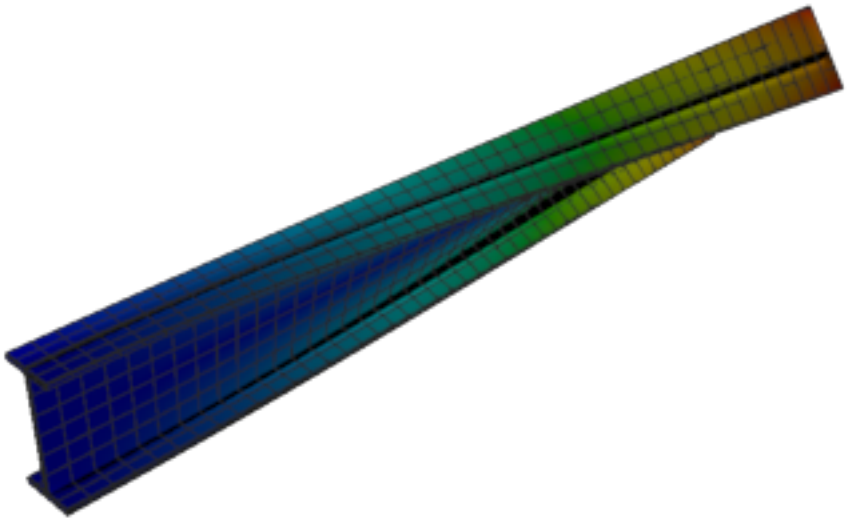


Figure 341: Code_Aster — mode 2, 21.069 Hz

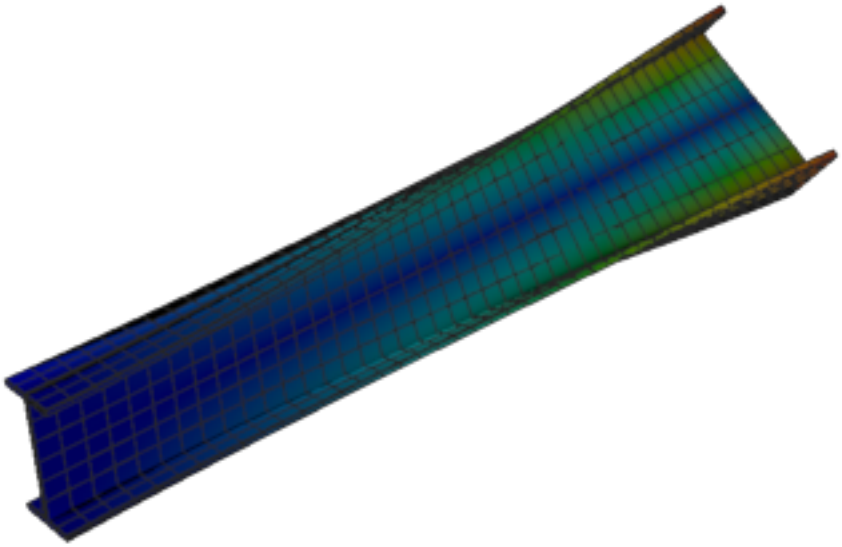


Figure 342: Sesam — mode 2, 20.302 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.731	-0.30	16967	7.03e-26	1.48e-18	2.87e-22	2.65e-13	1.22e-09	-1.69e-11
Calculix	21.069	+1.33	17523.8	2.84e-30	5.85e-19	9.43e-21	-1.68e-15	7.65e-10	-9.71e-11
Code_Aster	21.069	+1.33	17523.8	1.97e-28	1.10e-17	3.14e-22	-1.40e-14	3.32e-09	1.77e-11
Sesam	20.302	-2.36	16271.1	1.70e-26	3.10e-18	2.42e-20	-1.31e-13	-1.76e-09	1.56e-10

Mode 3

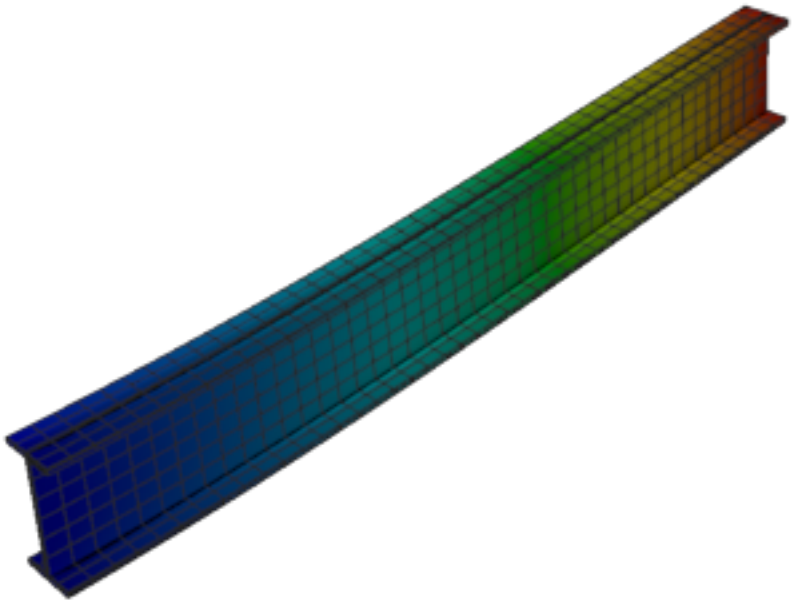


Figure 343: Abaqus — mode 3, 50.678 Hz

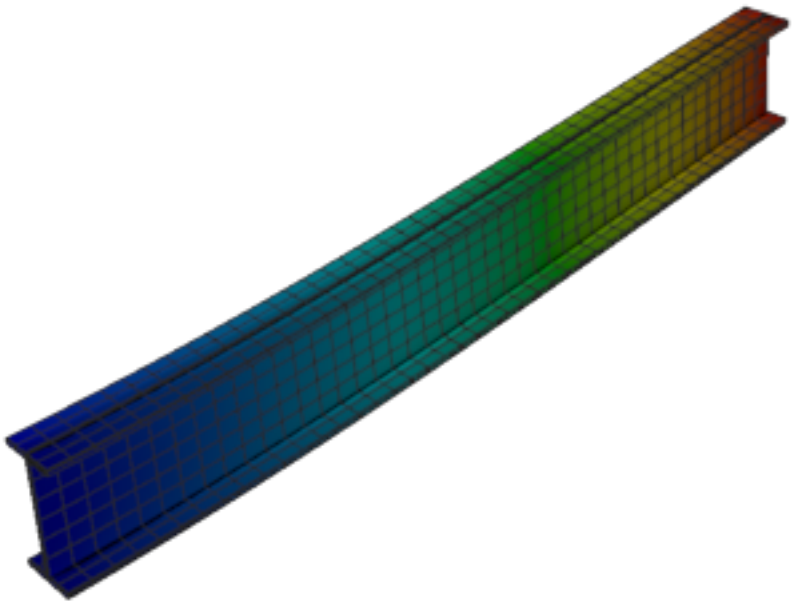


Figure 344: Calculix — mode 3, 50.825 Hz

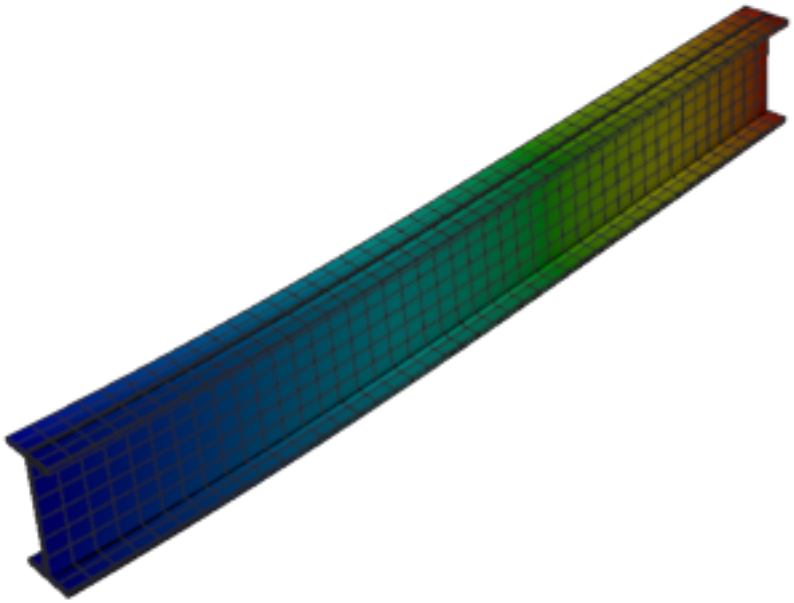


Figure 345: Code_Aster — mode 3, 50.825 Hz

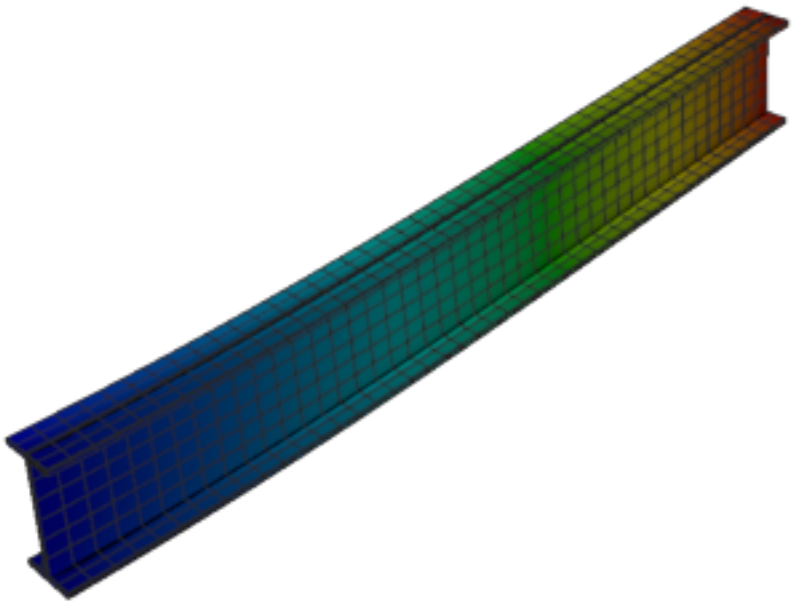


Figure 346: Sesam — mode 3, 50.618 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.678	-0.12	101392	1.09e-25	2.57e-22	118.20	-3.30e-13	1.60e-11	10.872
Calculix	50.825	+0.17	101981	3.20e-26	2.09e-23	118.19	-1.79e-13	4.57e-12	10.871
Code_Aster	50.825	+0.17	101981	3.12e-25	3.28e-23	118.19	-5.58e-13	5.73e-12	10.872
Sesam	50.618	-0.23	101151	1.20e-24	4.37e-21	118.29	-1.10e-12	-6.61e-11	10.876

Mode 4

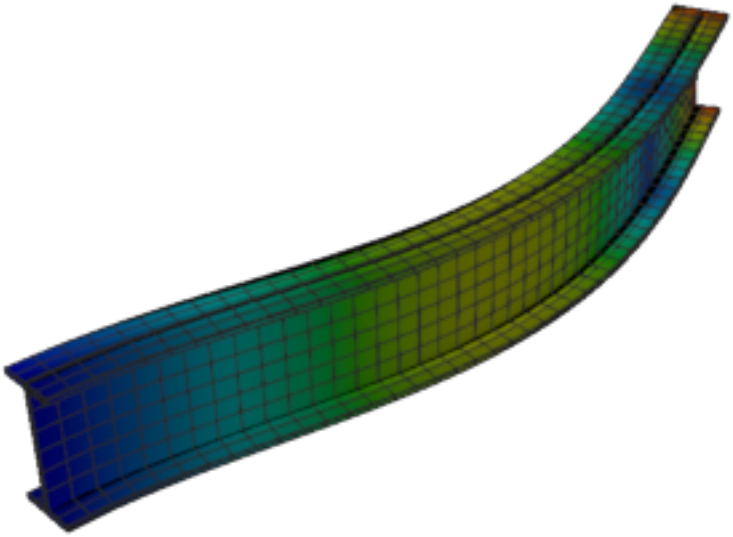


Figure 347: Abaqus — mode 4, 82.799 Hz

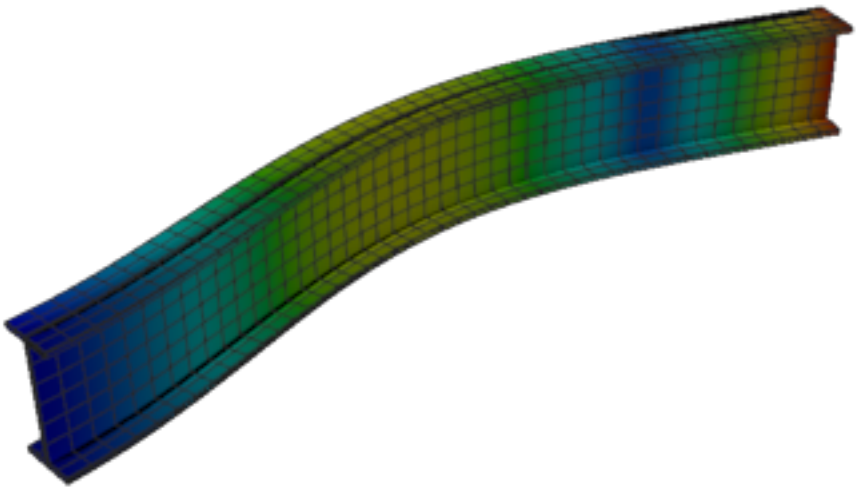


Figure 348: Calculix — mode 4, 84.066 Hz

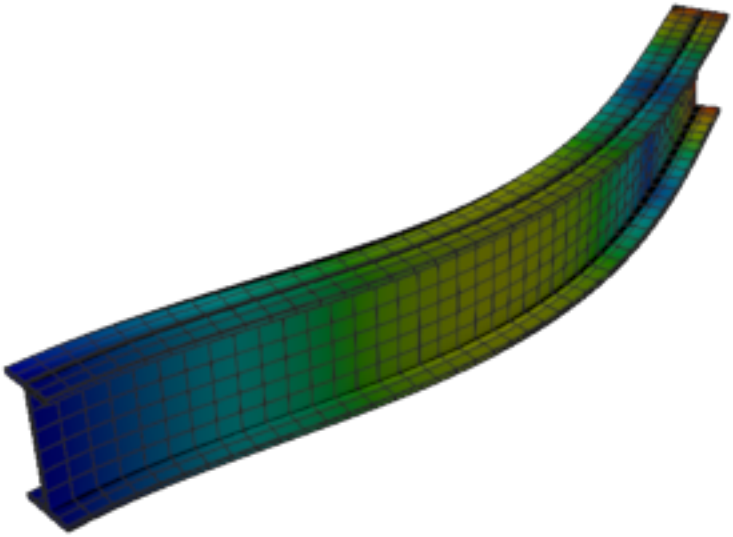


Figure 349: Code_Aster — mode 4, 84.066 Hz

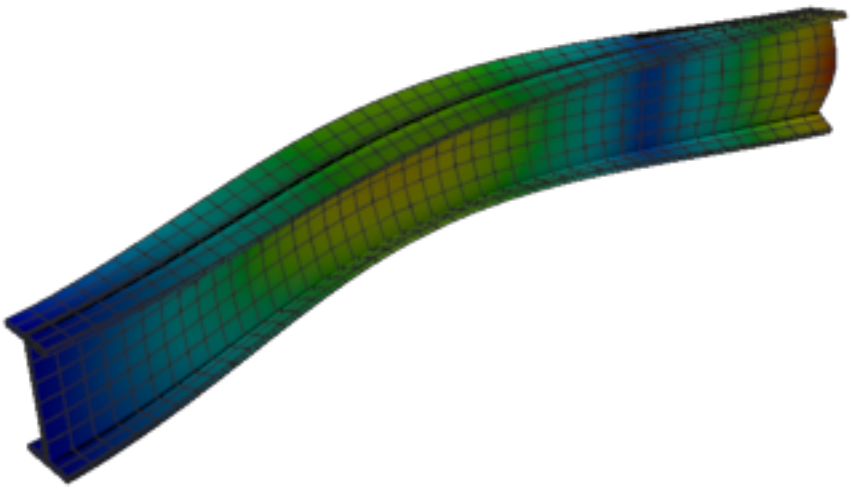


Figure 350: Sesam — mode 4, 77.392 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	82.799	+0.87	270652	7.52e-26	36.02	2.70e-22	-2.74e-13	-6.002	1.64e-11
Calculix	84.066	+2.42	278998	4.64e-26	36.05	4.98e-23	2.15e-13	6.004	7.06e-12
Code_Aster	84.066	+2.42	278998	2.15e-26	36.05	2.12e-22	1.47e-13	-6.004	-1.46e-11
Sesam	77.392	-5.71	236458	2.29e-26	35.94	6.77e-22	-1.51e-13	5.995	-2.60e-11

Mode 5

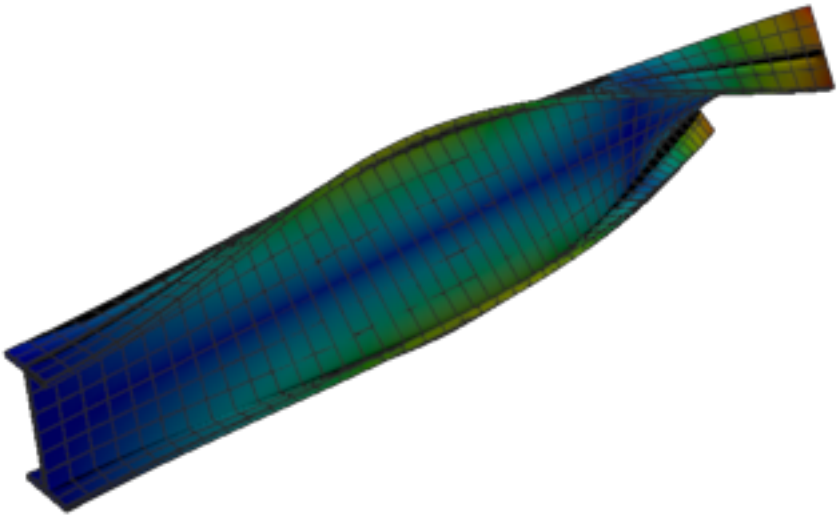


Figure 351: Abaqus — mode 5, 100.195 Hz

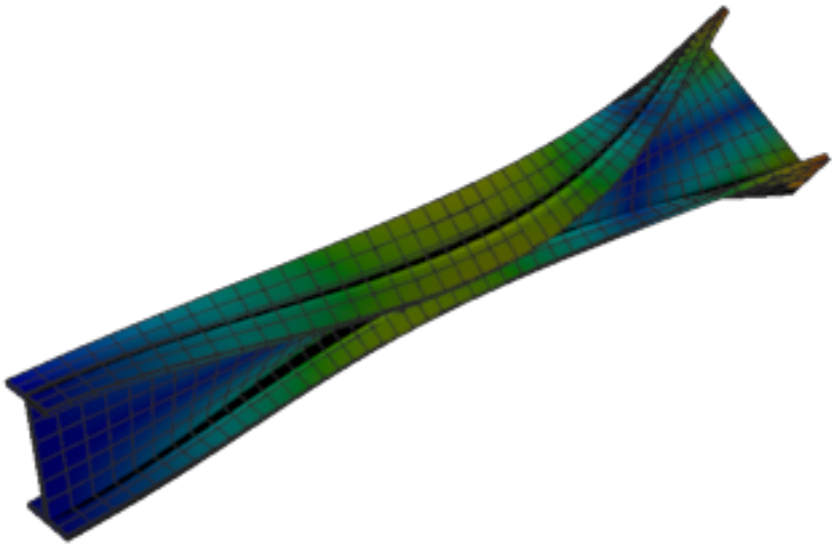


Figure 352: Calculix — mode 5, 102.334 Hz

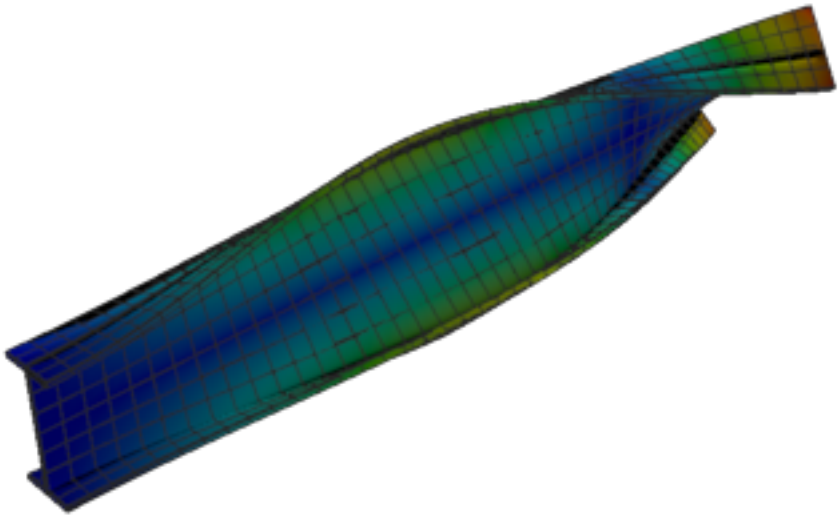


Figure 353: Code_Aster — mode 5, 102.334 Hz

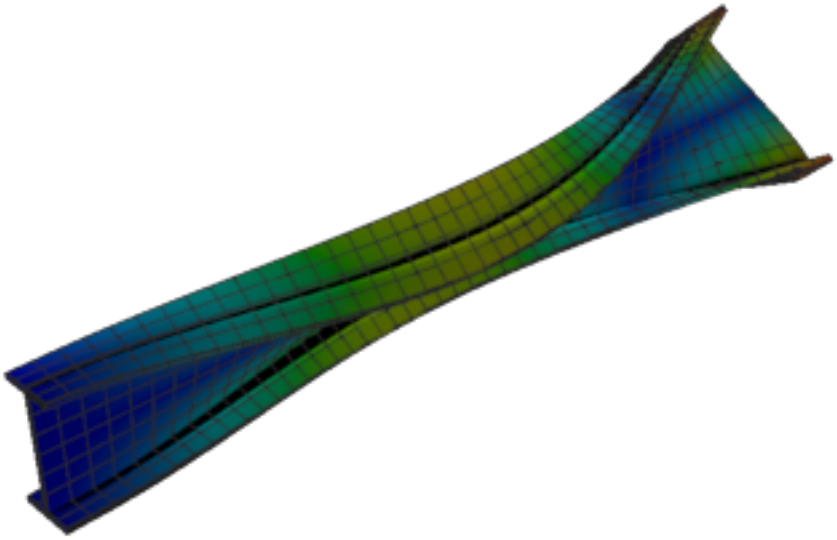


Figure 354: Sesam — mode 5, 98.119 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	100.190	-0.55	396325	4.10e-26	6.87e-21	7.52e-23	2.03e-13	-8.29e-11	8.67e-12
Calculix	102.334	+1.58	413428	1.70e-27	6.42e-21	4.55e-23	-4.12e-14	8.01e-11	-6.74e-12
Code_Aster	102.334	+1.58	413428	9.62e-27	3.93e-20	3.02e-24	-9.81e-14	-1.98e-10	-1.74e-12
Sesam	98.119	-2.61	380068	1.41e-25	3.09e-21	8.31e-23	3.76e-13	-5.56e-11	9.12e-12

Mode 6

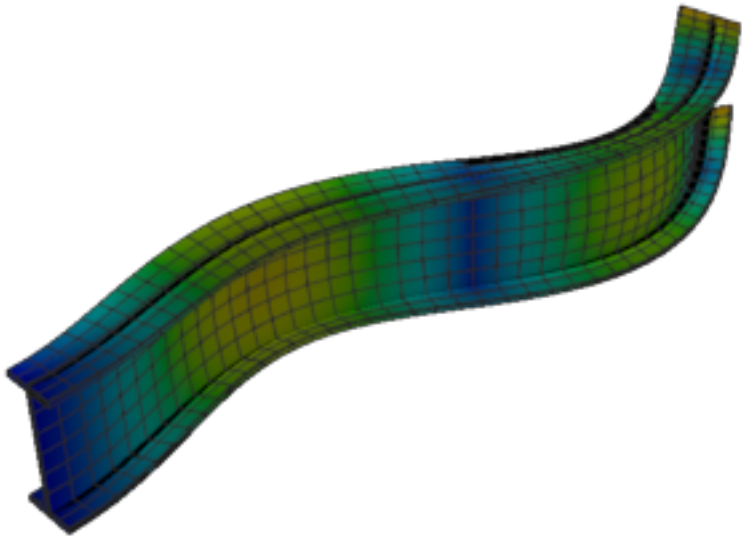


Figure 355: Abaqus — mode 6, 223.139 Hz

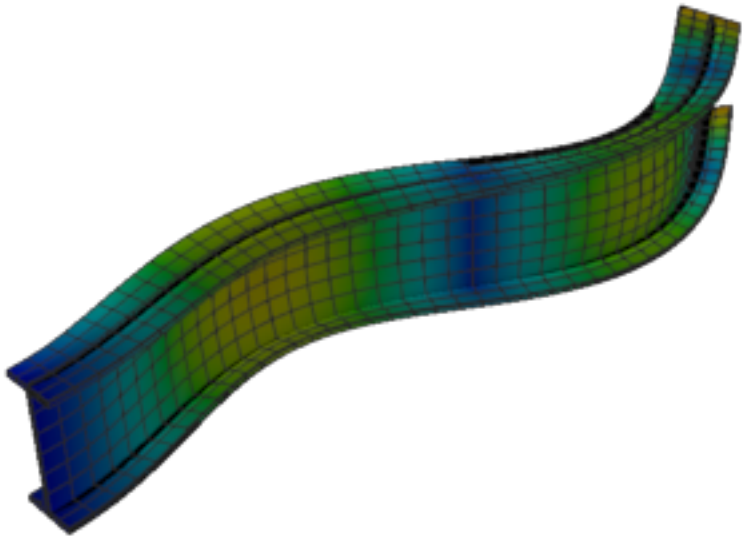


Figure 356: Calculix — mode 6, 227.250 Hz

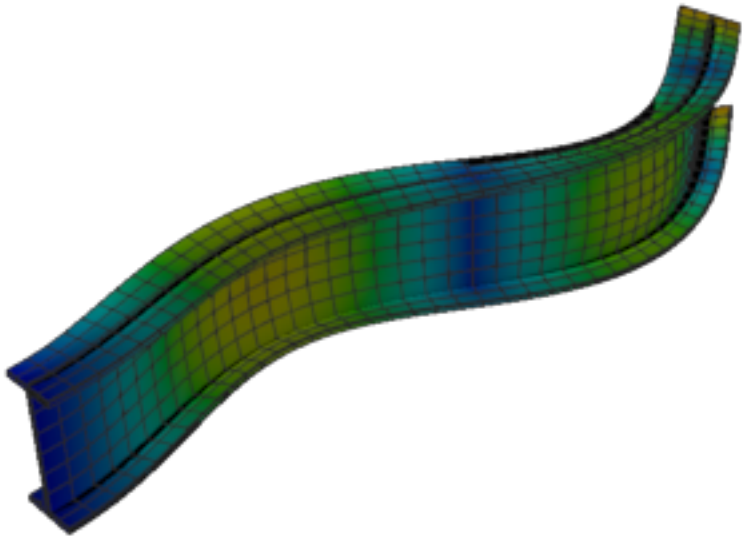


Figure 357: Code_Aster — mode 6, 227.250 Hz

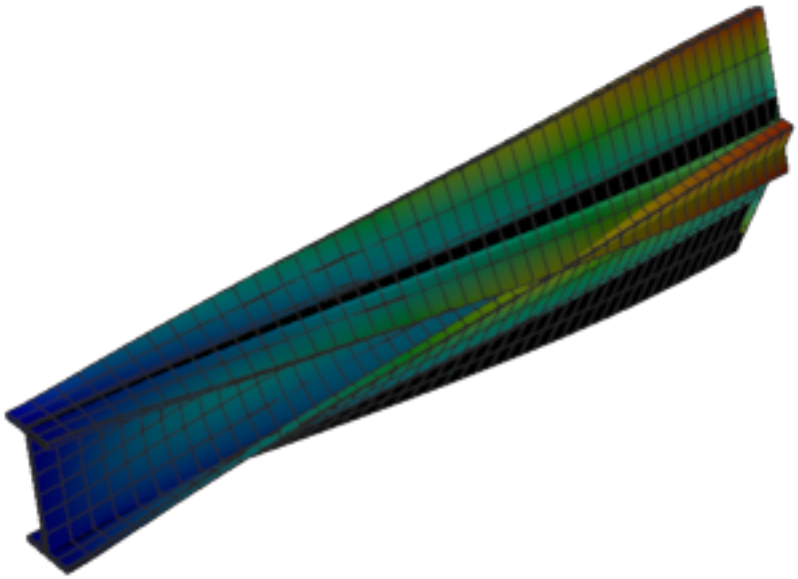


Figure 358: Sesam — mode 6, 133.016 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	223.140	+10.10	1.96567e+06	2.33e-25	12.40	1.64e-23	-4.83e-13	3.522	4.05e-12
Calculix	227.250	+12.13	2.03876e+06	1.37e-26	12.46	3.21e-24	1.17e-13	3.530	1.79e-12
Code_Aster	227.250	+12.13	2.03876e+06	4.29e-26	12.47	5.64e-25	-2.07e-13	3.532	-7.51e-13
Sesam	133.016	-34.37	698501	5.75e-26	0.01	1.11e-20	-2.40e-13	-0.114	1.05e-10

Mode 7

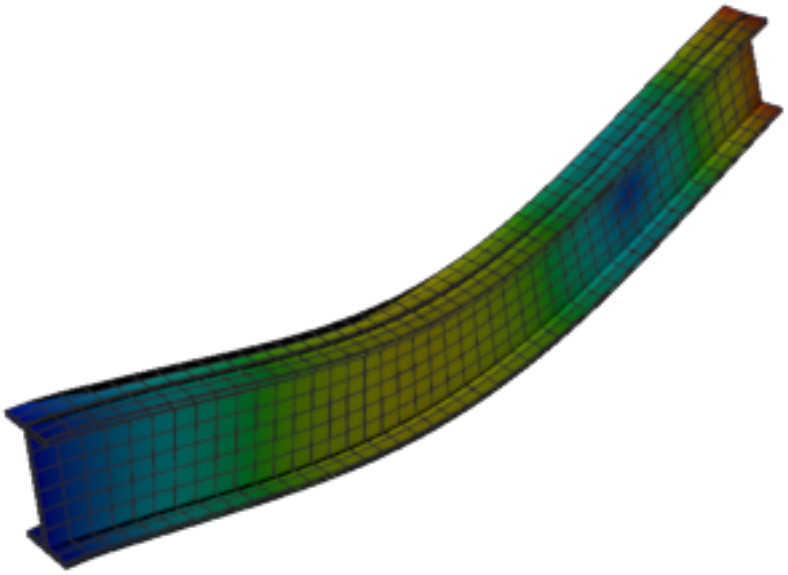


Figure 359: Abaqus — mode 7, 254.984 Hz

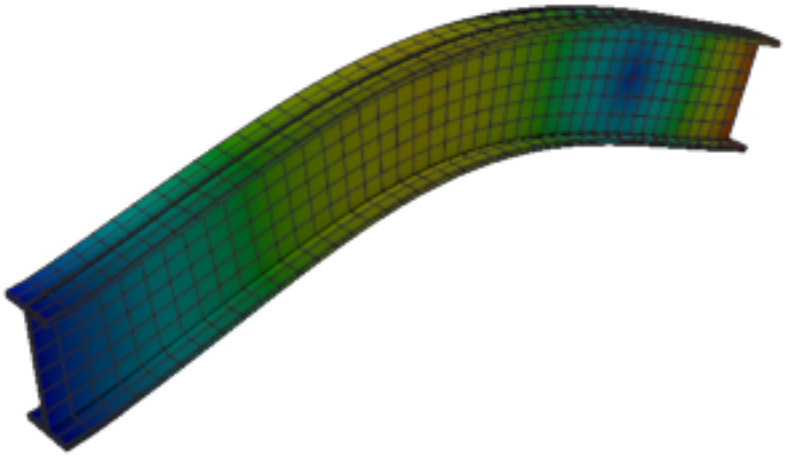


Figure 360: Calculix — mode 7, 255.872 Hz

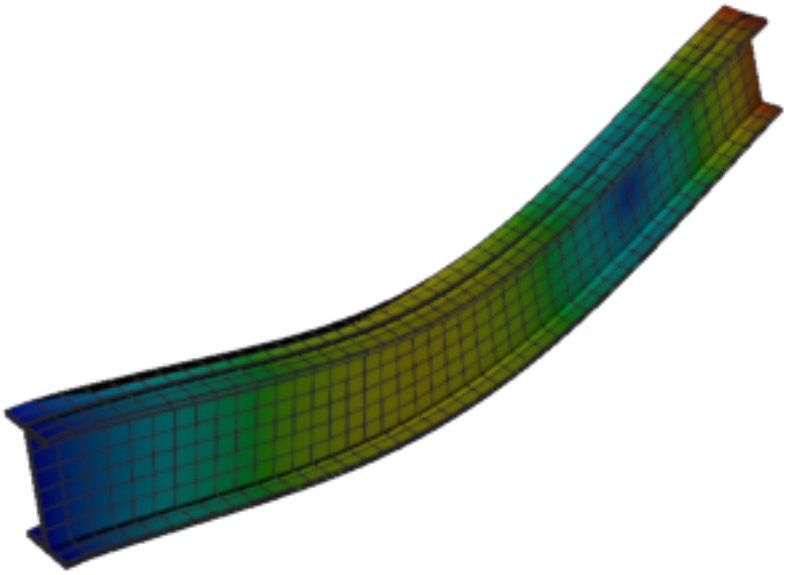


Figure 361: Code_Aster — mode 7, 255.872 Hz

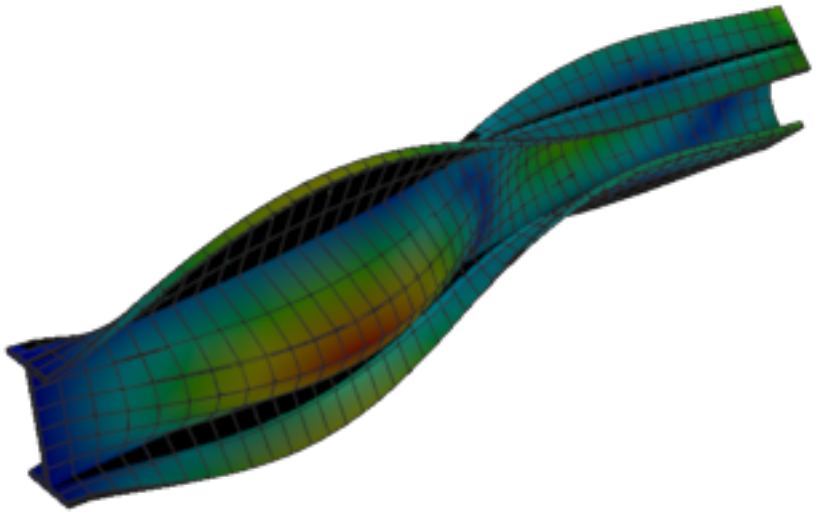


Figure 362: Sesam — mode 7, 153.833 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	254.980	+10.79	2.56676e+06	2.24e-25	2.85e-24	40.01	-4.74e-13	1.69e-12	-6.326
Calculix	255.872	+11.18	2.58466e+06	7.68e-25	2.60e-25	40.03	8.76e-13	-5.10e-13	6.327
Code_Aster	255.872	+11.18	2.58466e+06	7.10e-26	5.53e-25	40.06	2.66e-13	-7.43e-13	-6.329
Sesam	153.833	-33.16	934235	1.46e-25	3.20	1.07e-21	-3.82e-13	-1.788	3.27e-11

Mode 8

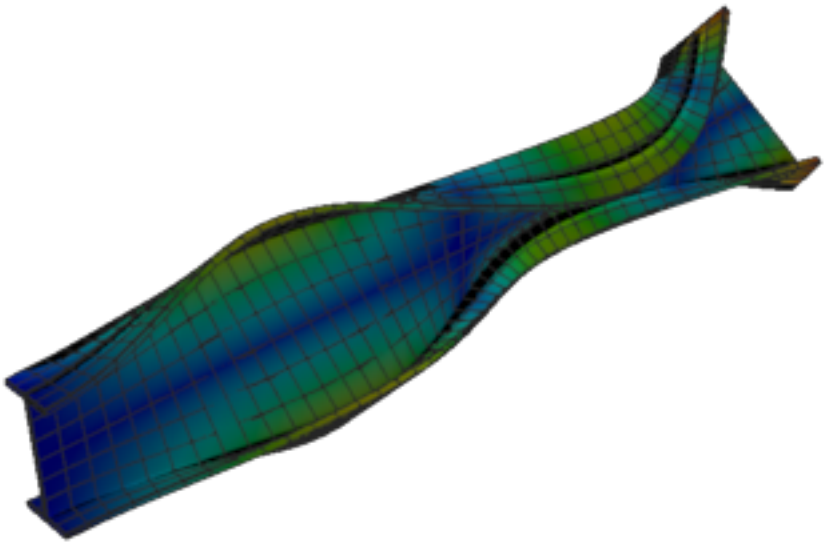


Figure 363: Abaqus — mode 8, 260.095 Hz

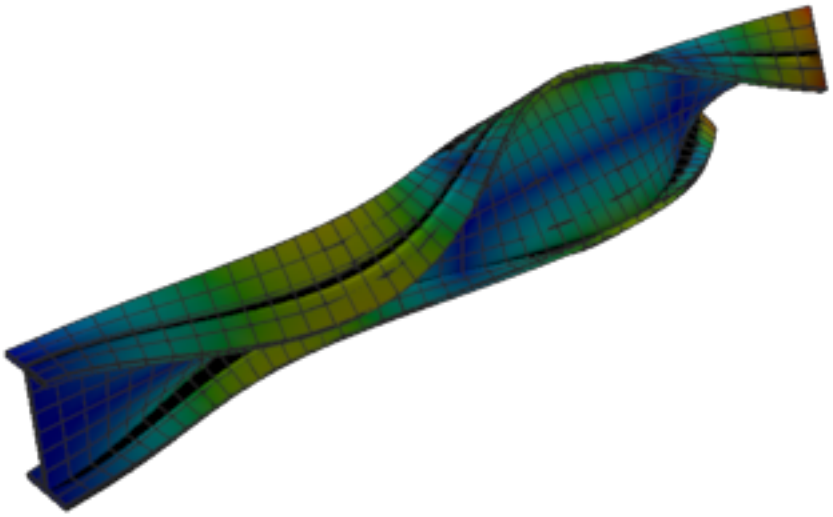


Figure 364: Calculix — mode 8, 266.519 Hz

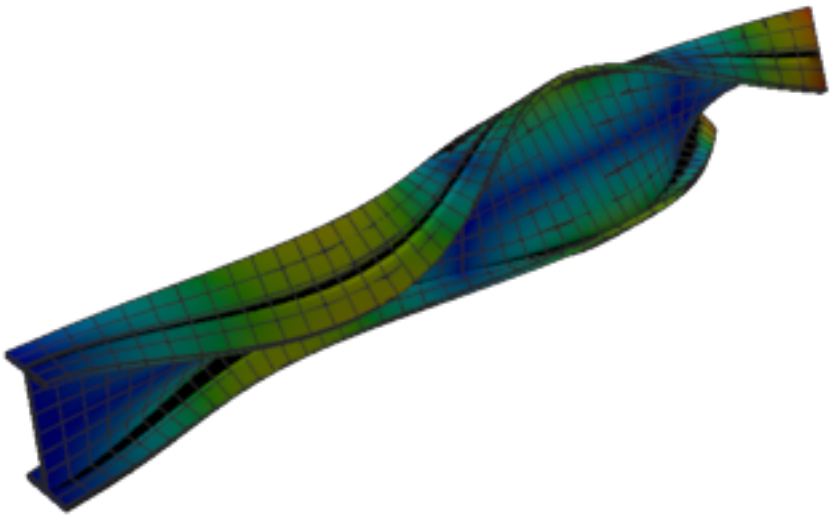


Figure 365: Code_Aster — mode 8, 266.519 Hz

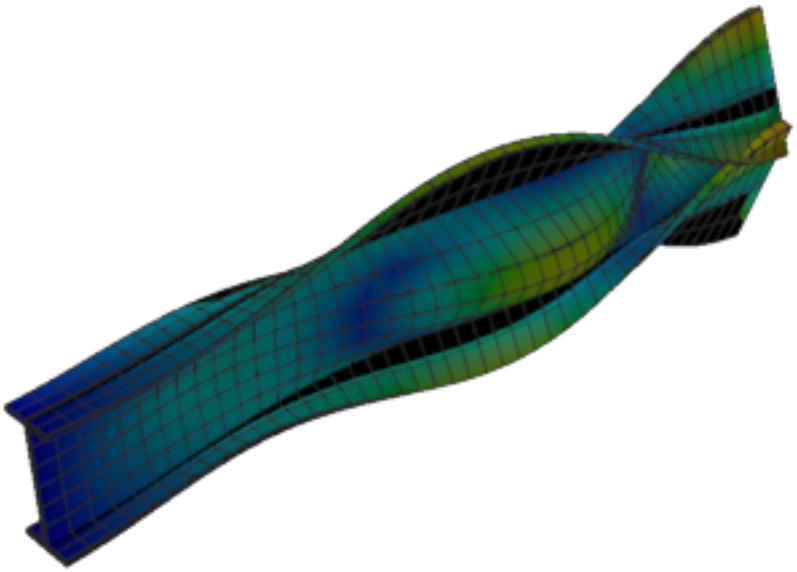


Figure 366: Sesam — mode 8, 178.565 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	260.100	+7.07	2.6707e+06	2.66e-26	6.24e-23	3.82e-21	-1.63e-13	7.90e-12	6.18e-11
Calculix	266.519	+9.71	2.80424e+06	5.05e-27	2.40e-22	2.23e-22	7.10e-14	1.55e-11	-1.49e-11
Code_Aster	266.519	+9.71	2.80424e+06	3.00e-26	3.09e-23	8.16e-23	-1.73e-13	5.56e-12	-9.03e-12
Sesam	178.565	-26.49	1.25879e+06	2.86e-28	5.71	2.39e-21	1.69e-14	2.390	4.89e-11

Mode 9

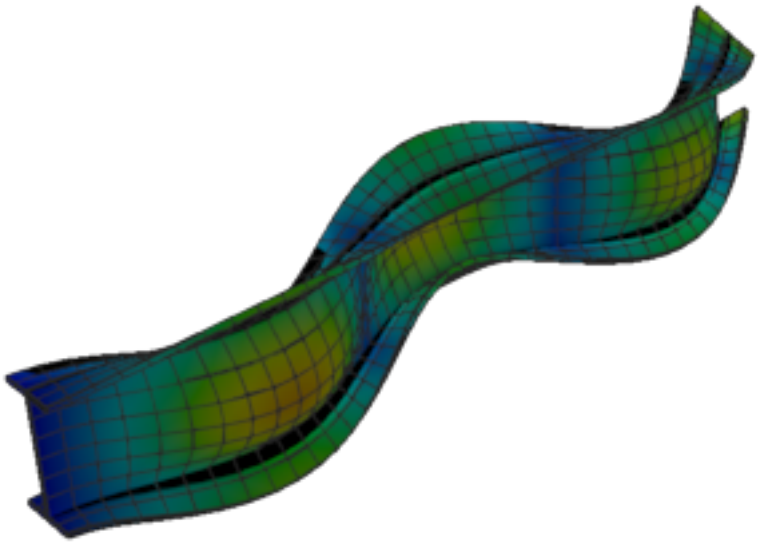


Figure 367: Abaqus — mode 9, 396.567 Hz

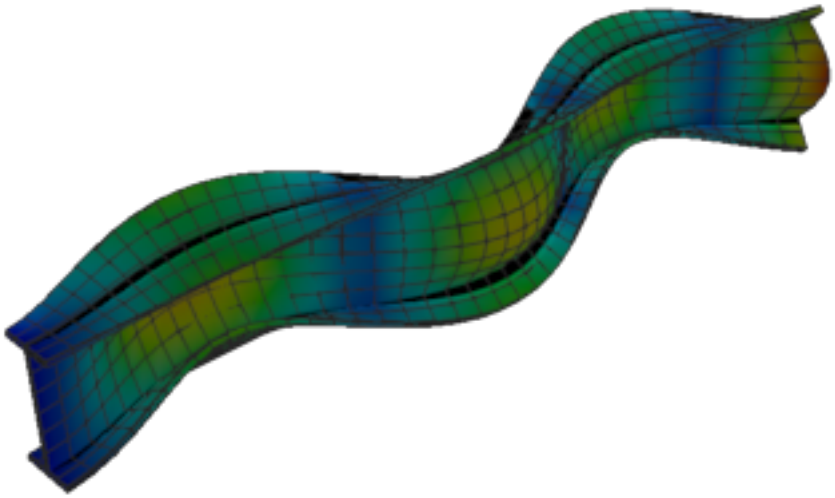


Figure 368: Calculix — mode 9, 407.869 Hz

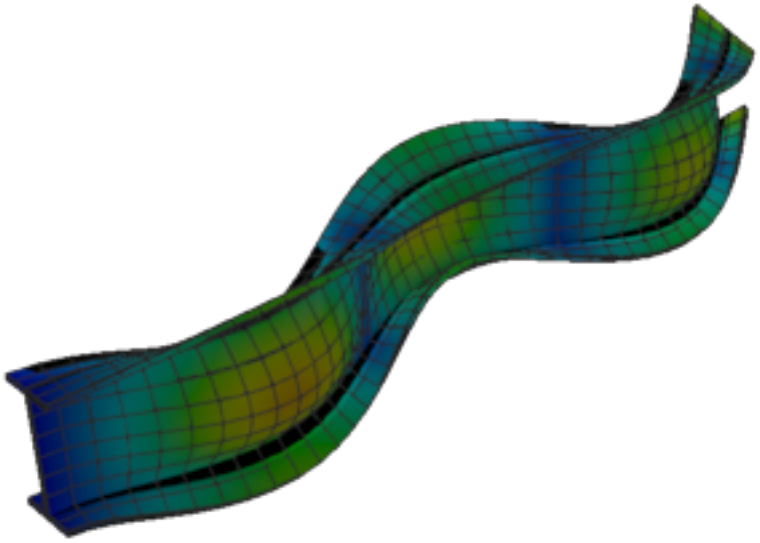


Figure 369: Code_Aster — mode 9, 407.869 Hz

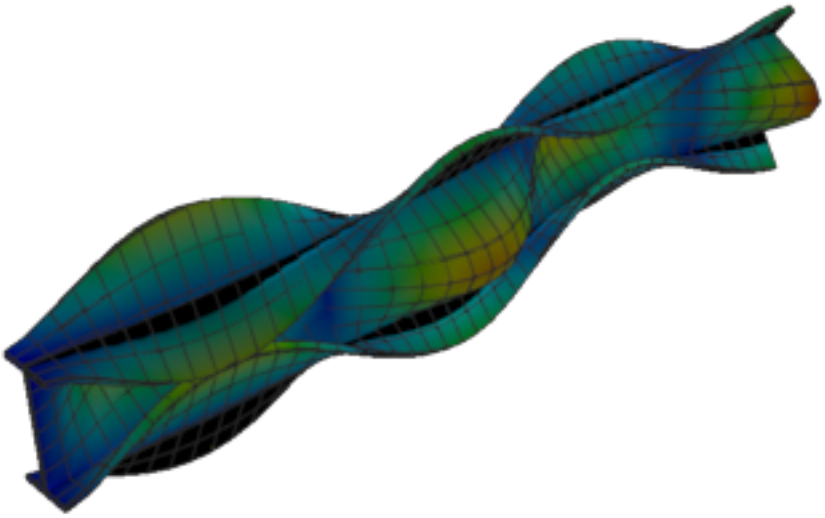


Figure 370: Sesam — mode 9, 214.689 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	396.570	+11.16	6.20858e+06	2.52e-22	5.69	2.13e-23	1.59e-11	-2.385	4.61e-12
Calculix	407.869	+14.33	6.56752e+06	2.22e-24	5.88	4.86e-25	-1.49e-12	2.426	6.97e-13
Code_Aster	407.869	+14.33	6.56753e+06	1.12e-23	5.90	3.42e-25	-3.34e-12	-2.430	5.84e-13
Sesam	214.689	-39.82	1.81962e+06	3.48e-26	0.44	2.56e-22	-1.87e-13	0.662	-1.60e-11

Mode 10

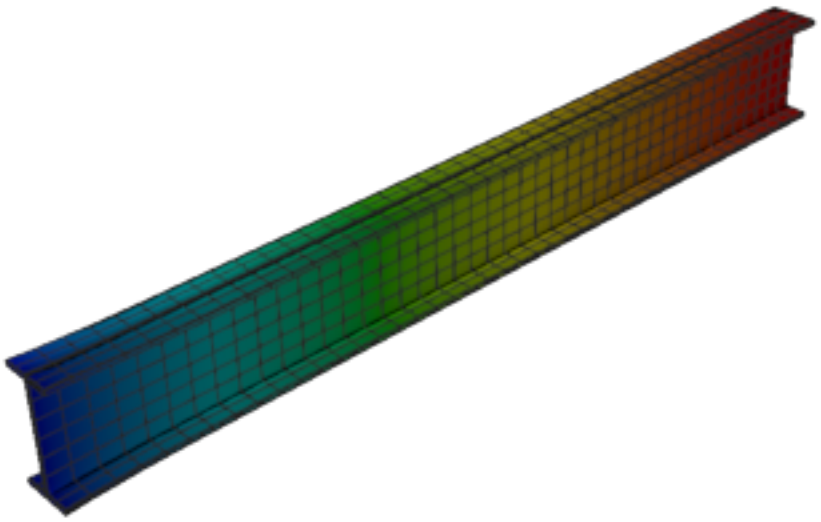


Figure 371: Abaqus — mode 10, 431.802 Hz

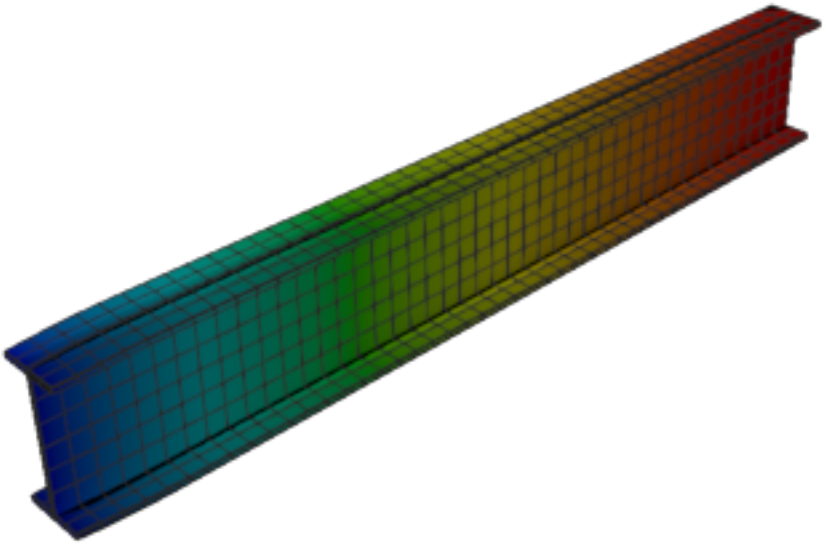


Figure 372: Calculix — mode 10, 432.207 Hz

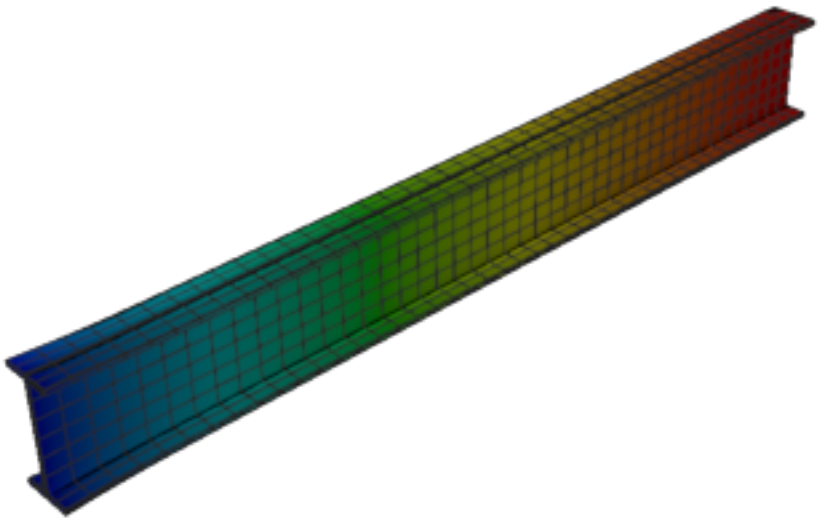


Figure 373: Code_Aster — mode 10, 432.207 Hz

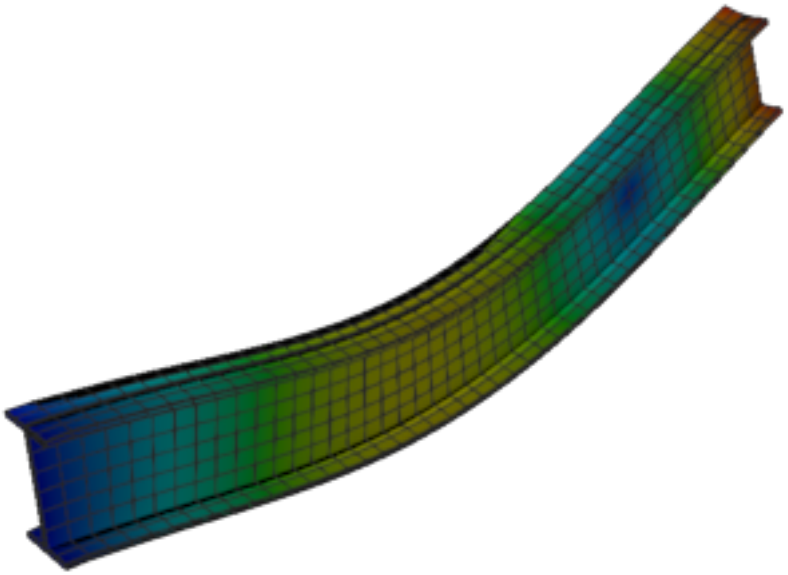


Figure 374: Sesam — mode 10, 253.463 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	431.800	+11.46	7.36085e+06	153.54	1.07e-21	9.26e-21	12.391	3.27e-11	-9.62e-11
Calculix	432.207	+11.56	7.3747e+06	153.40	3.75e-25	1.02e-25	-12.385	-6.13e-13	3.19e-13
Code_Aster	432.207	+11.56	7.3747e+06	153.46	4.59e-25	3.59e-25	12.388	-6.77e-13	5.99e-13
Sesam	253.463	-34.58	2.53623e+06	1.48e-19	5.08e-20	40.27	-3.85e-10	2.25e-10	-6.346

Mode 11

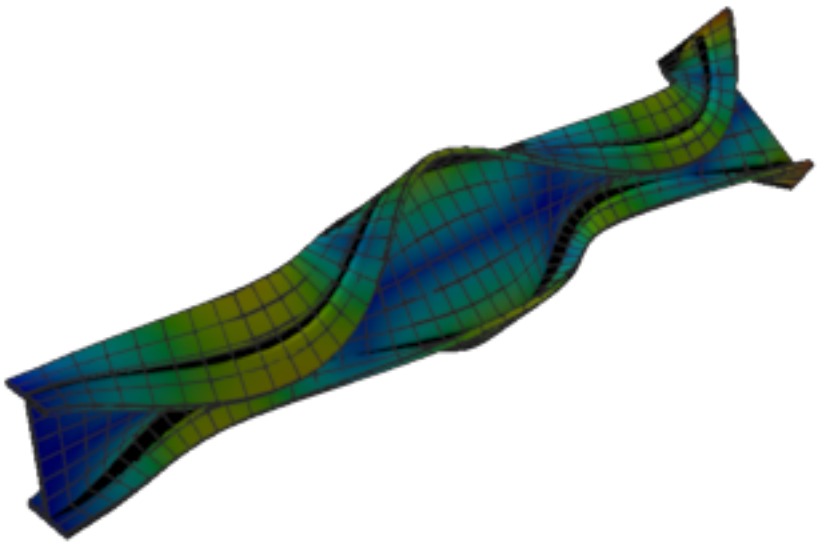


Figure 375: Abaqus — mode 11, 486.965 Hz

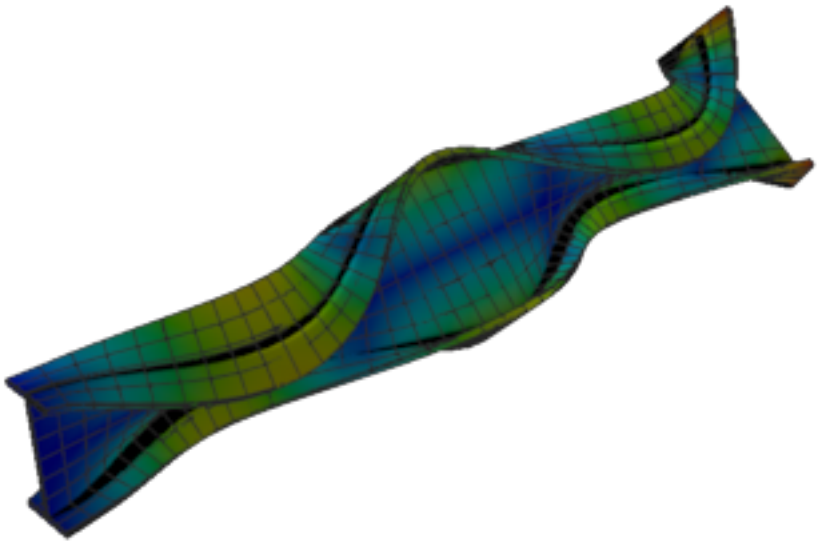


Figure 376: Calculix — mode 11, 500.931 Hz

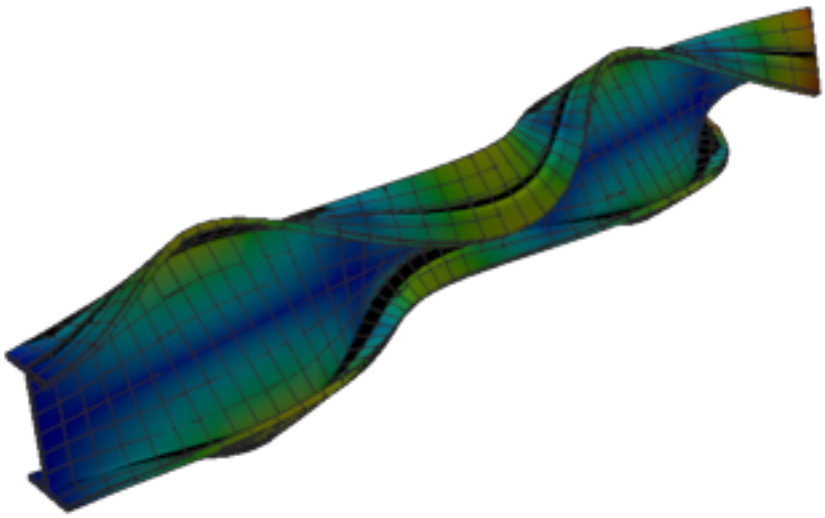


Figure 377: Code_Aster — mode 11, 500.931 Hz

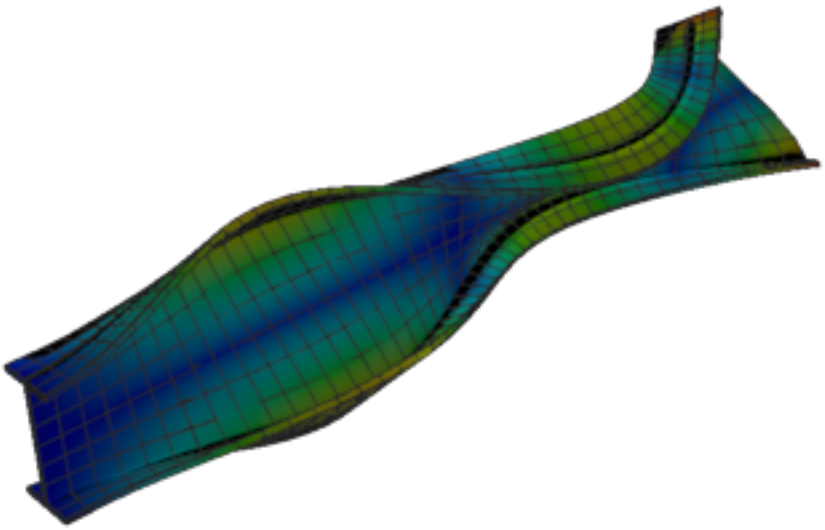


Figure 378: Sesam — mode 11, 256.050 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	486.960	+11.63	9.3617e+06	1.09e-16	2.28e-18	1.17e-18	1.04e-08	1.51e-09	1.08e-09
Calculix	500.931	+14.84	9.90641e+06	2.23e-25	1.21e-24	2.22e-24	4.72e-13	-1.10e-12	1.49e-12
Code_Aster	500.931	+14.84	9.90641e+06	2.35e-25	2.66e-25	5.46e-25	4.85e-13	5.16e-13	-7.39e-13
Sesam	256.050	-41.30	2.58827e+06	4.62e-20	1.26e-20	3.01e-23	2.15e-10	-1.12e-10	5.49e-12

Solid, 1st order, HEX, reduced integration

Mode 1

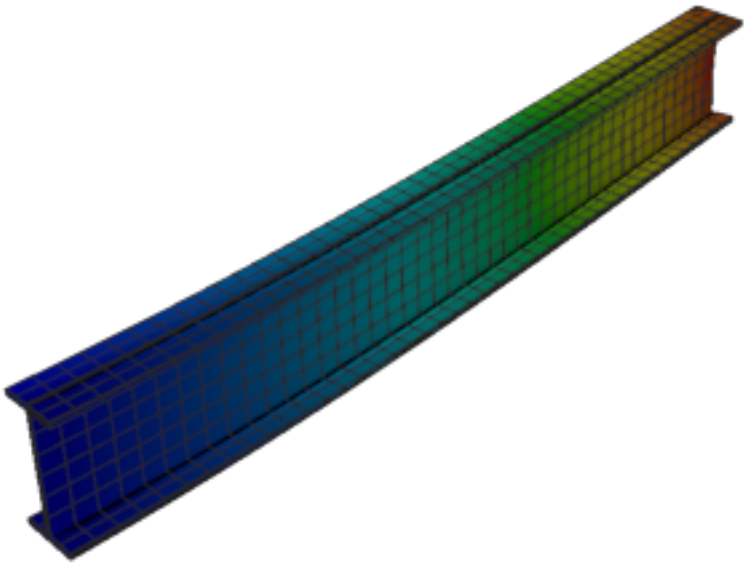


Figure 379: Abaqus — mode 1, 12.514 Hz

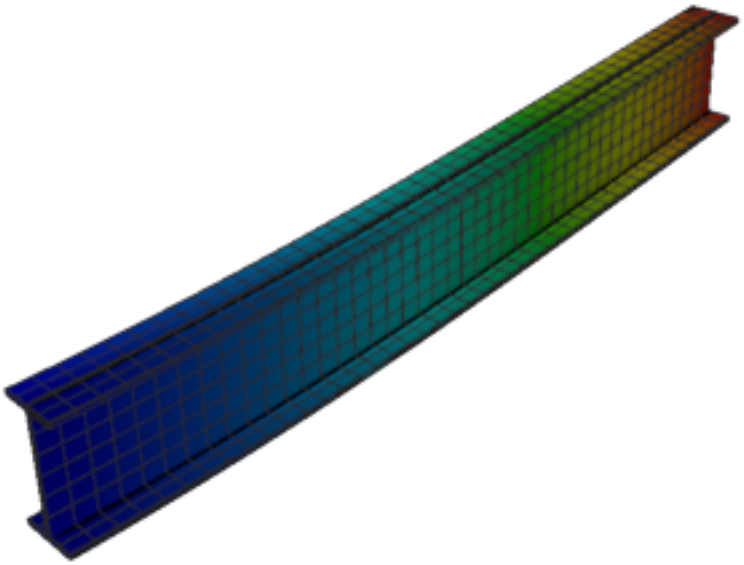


Figure 380: Calculix — mode 1, 14.302 Hz

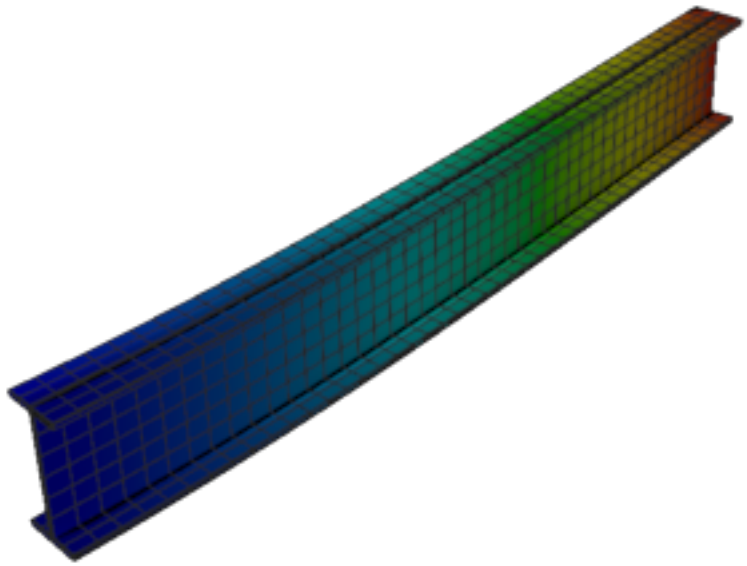


Figure 381: Code_Aster — mode 1, 13.529 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.514	-6.95	6182	1.92e-26	116.92	5.56e-23	1.39e-13	10.813	-7.45e-12
Calculix	14.302	+6.35	8074.92	5.38e-26	116.50	6.38e-22	2.32e-13	10.794	2.53e-11
Code_Aster	13.529	+0.60	7225.96	1.74e-09	116.44	1.64e-06	4.17e-05	10.791	-0.001

Mode 2

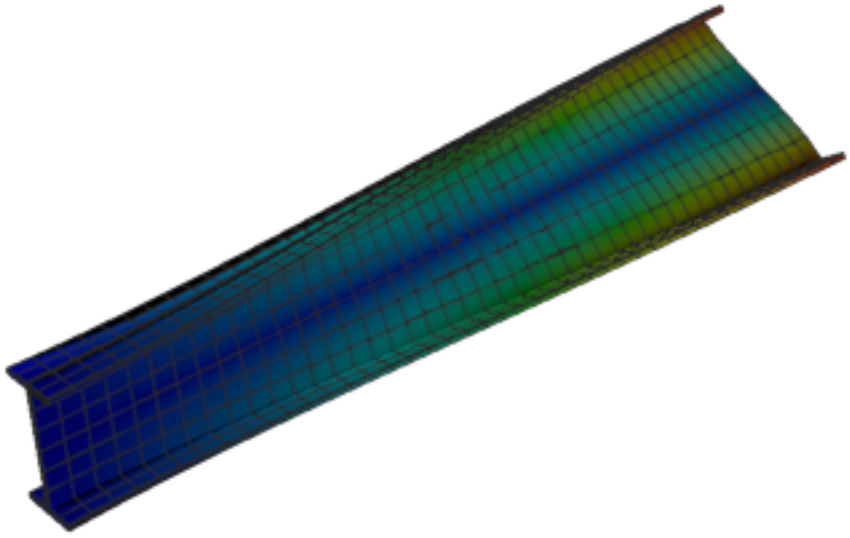


Figure 382: Abaqus — mode 2, 34.157 Hz

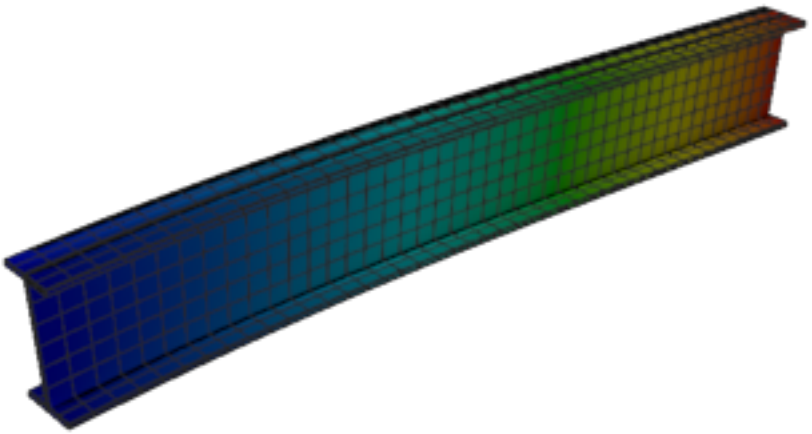


Figure 383: Calculix — mode 2, 51.665 Hz

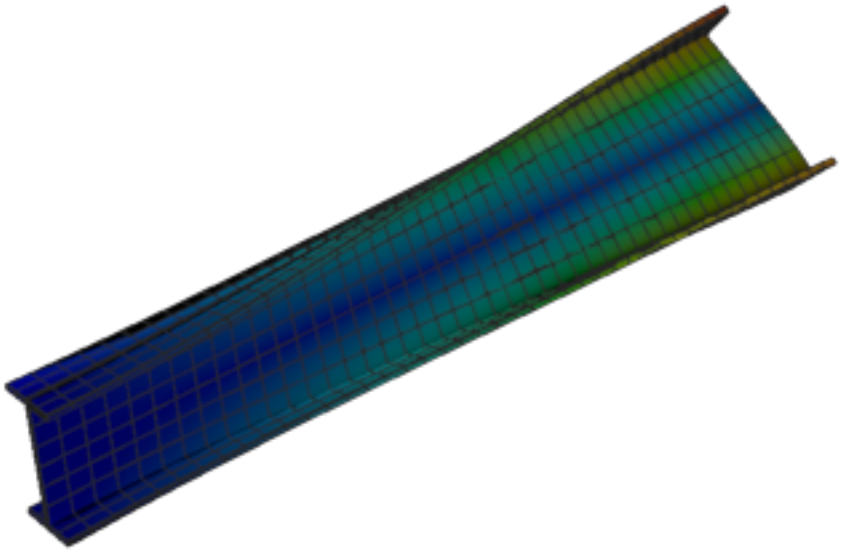


Figure 384: Code_Aster — mode 2, 34.740 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	34.157	-15.01	46060	4.54e-27	1.11e-19	8.80e-22	6.74e-14	3.33e-10	-2.97e-11
Calculix	51.665	+28.56	105380	6.55e-26	8.26e-22	118.46	-2.56e-13	2.87e-11	-10.884
Code_Aster	34.740	-13.55	47645.5	3.35e-07	5.01e-04	1.48e-07	0.001	0.022	-3.84e-04

Mode 3

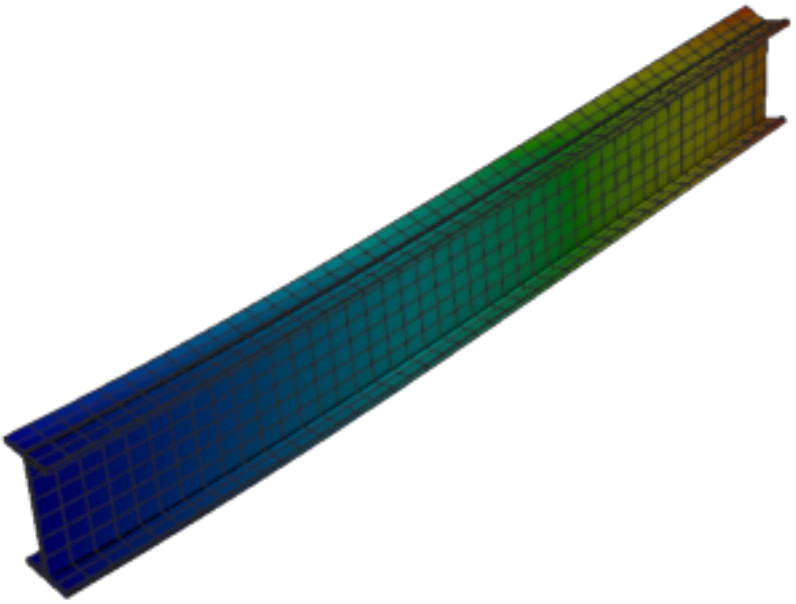


Figure 385: Abaqus — mode 3, 49.598 Hz

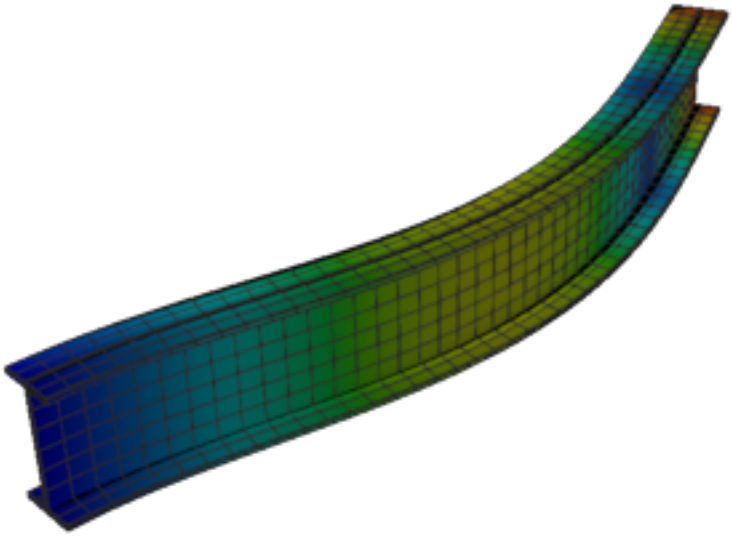


Figure 386: Calculix — mode 3, 88.233 Hz

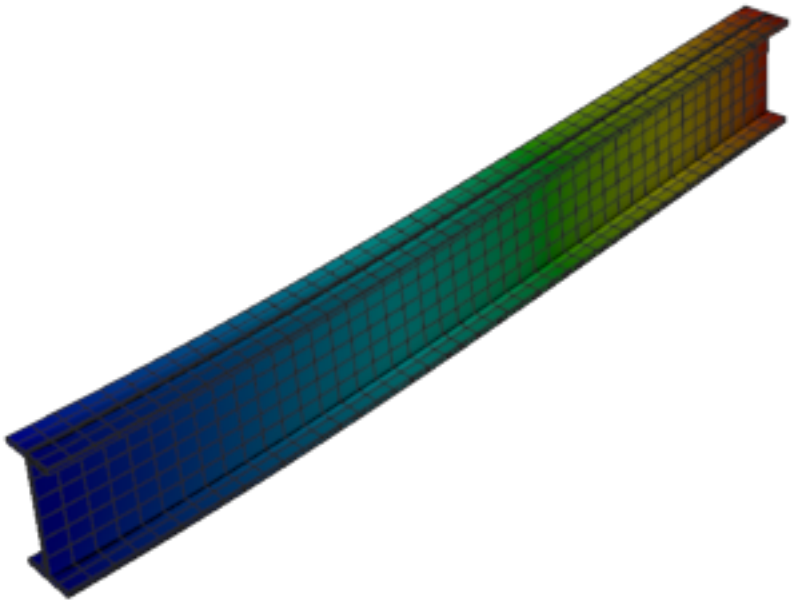


Figure 387: Code_Aster — mode 3, 50.737 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	49.598	-21.09	97114	1.57e-26	1.07e-20	118.97	-1.25e-13	1.03e-10	10.907
Calculix	88.233	+40.37	307343	1.15e-26	36.16	4.83e-23	1.07e-13	-6.013	-6.95e-12
Code_Aster	50.737	-19.28	101625	5.88e-09	8.10e-08	118.29	7.67e-05	2.85e-04	10.876

Mode 4

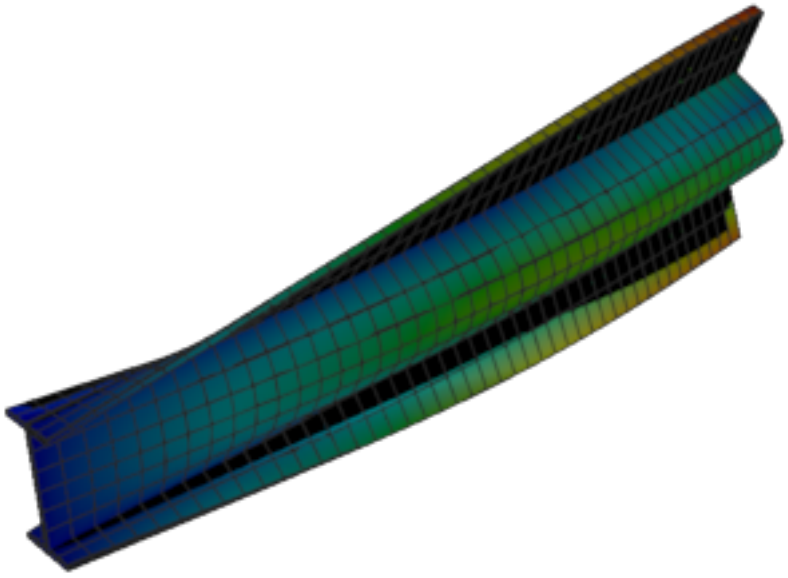


Figure 388: Abaqus — mode 4, 55.271 Hz

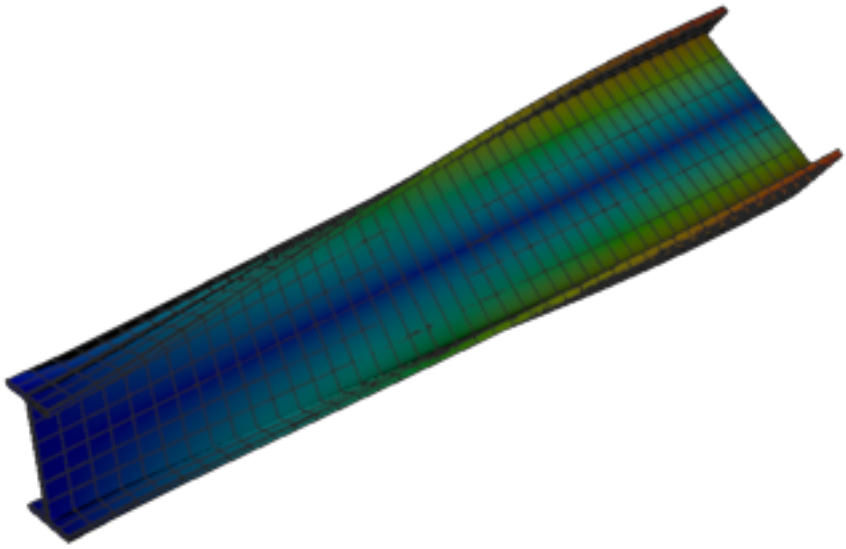


Figure 389: Calculix — mode 4, 116.873 Hz

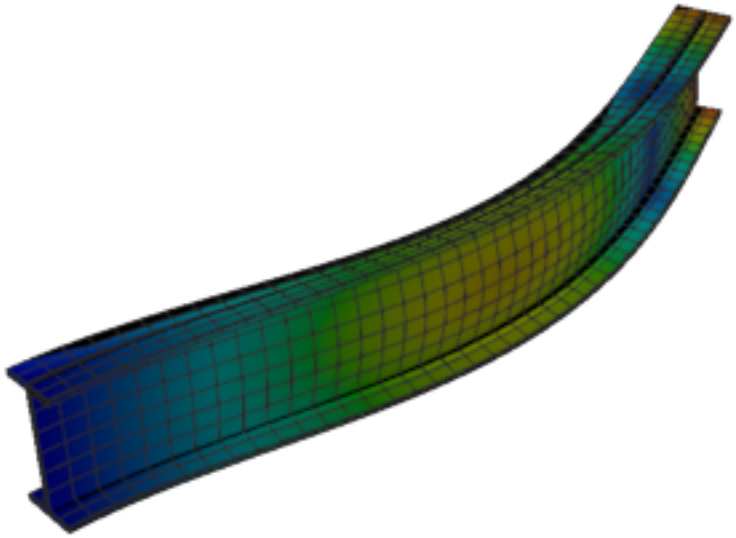


Figure 390: Code_Aster — mode 4, 82.296 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	55.271	-34.83	120602	1.14e-25	2.12	1.05e-18	-3.37e-13	-1.456	1.02e-09
Calculix	116.873	+37.80	539244	3.53e-27	9.54e-22	1.90e-26	5.94e-14	-3.09e-11	-1.38e-13
Code_Aster	82.296	-2.97	267374	6.08e-08	36.12	1.81e-05	-2.47e-04	-6.010	-0.004

Mode 5

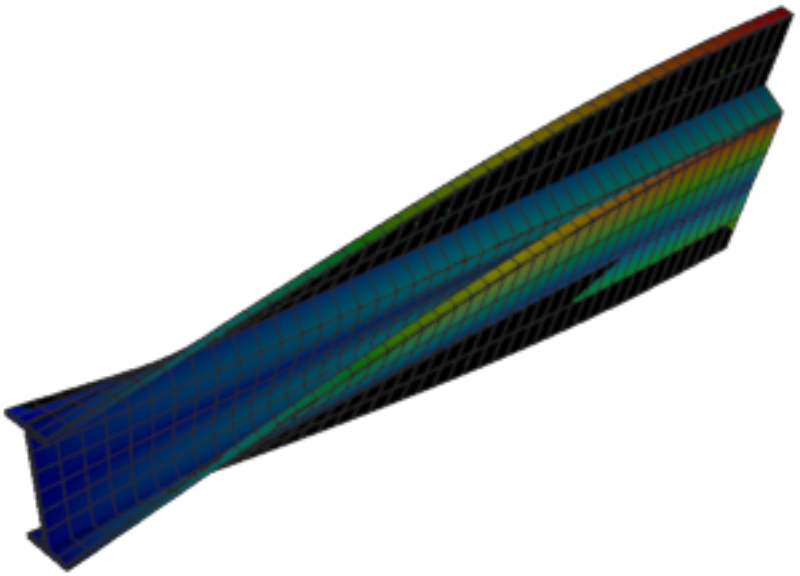


Figure 391: Abaqus — mode 5, 64.305 Hz

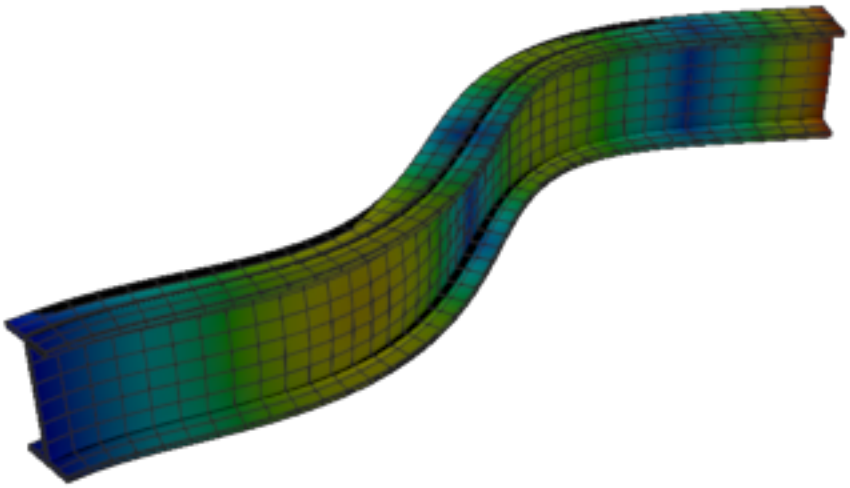


Figure 392: Calculix — mode 5, 240.766 Hz

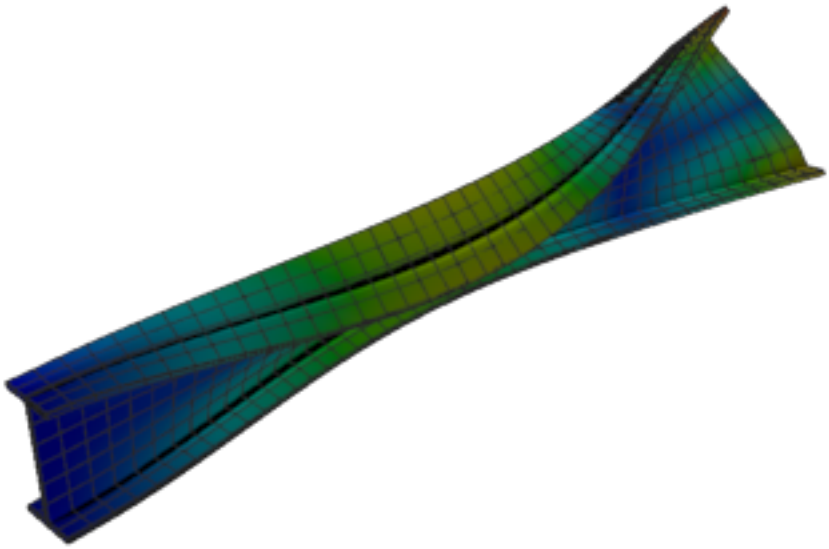


Figure 393: Code_Aster — mode 5, 123.303 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	64.305	-54.97	163249	1.13e-26	8.64e-20	1.24e-20	-1.06e-13	-2.94e-10	1.11e-10
Calculix	240.766	+68.61	2.2885e+06	3.27e-26	12.59	3.11e-25	-1.81e-13	-3.549	5.58e-13
Code_Aster	123.303	-13.65	600212	3.19e-06	0.01	4.14e-05	-0.002	-0.084	0.006

Mode 6

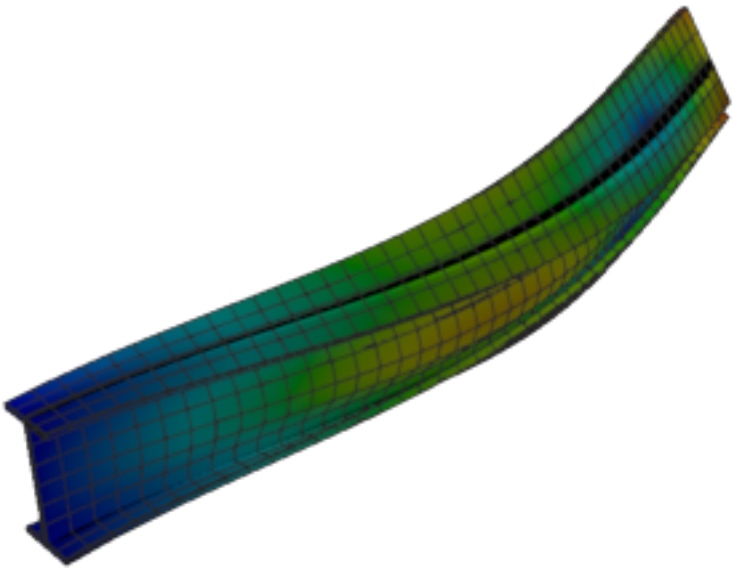


Figure 394: Abaqus — mode 6, 74.866 Hz

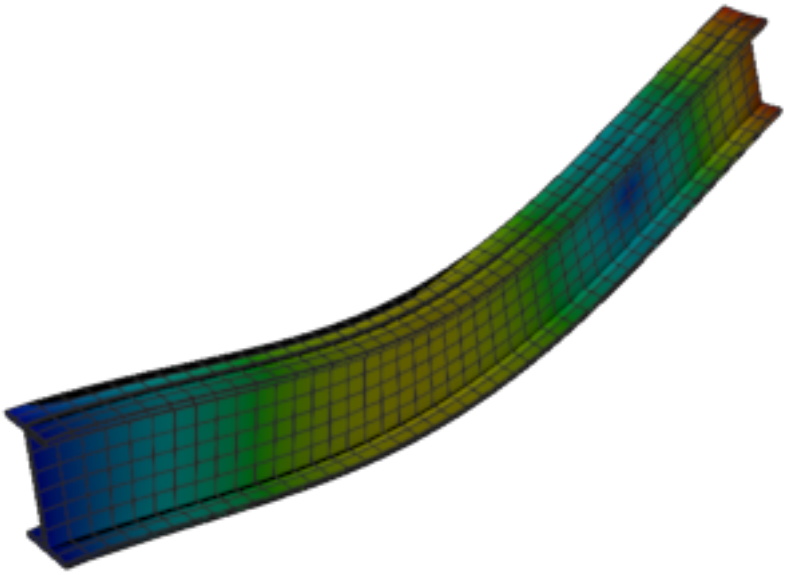


Figure 395: Calculix — mode 6, 258.286 Hz

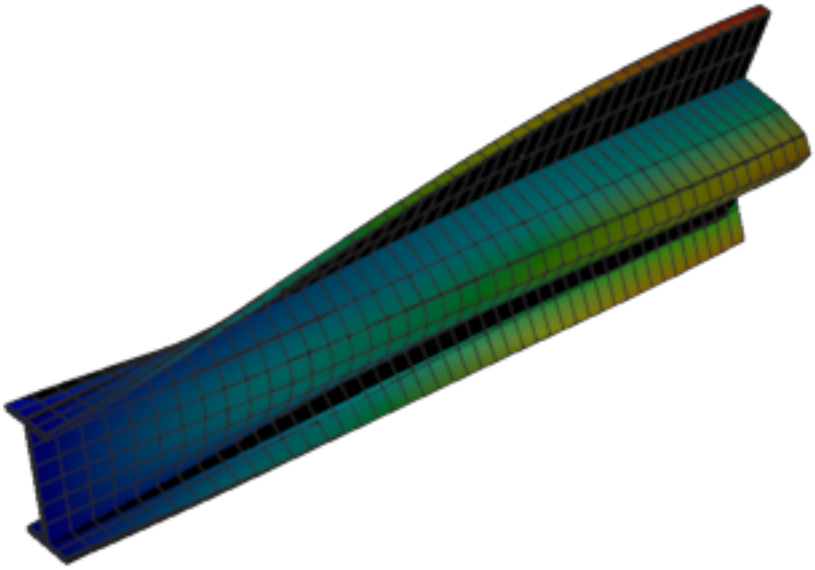


Figure 396: Code_Aster — mode 6, 147.139 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	74.866	-53.24	221271	1.12e-26	31.40	1.02e-20	1.06e-13	-5.603	-1.01e-10
Calculix	258.286	+61.33	2.63368e+06	2.30e-25	2.45e-26	40.19	4.80e-13	-1.56e-13	-6.340
Code_Aster	147.139	-8.09	854706	3.65e-07	0.04	2.32e-04	-0.001	0.192	-0.015

Mode 7

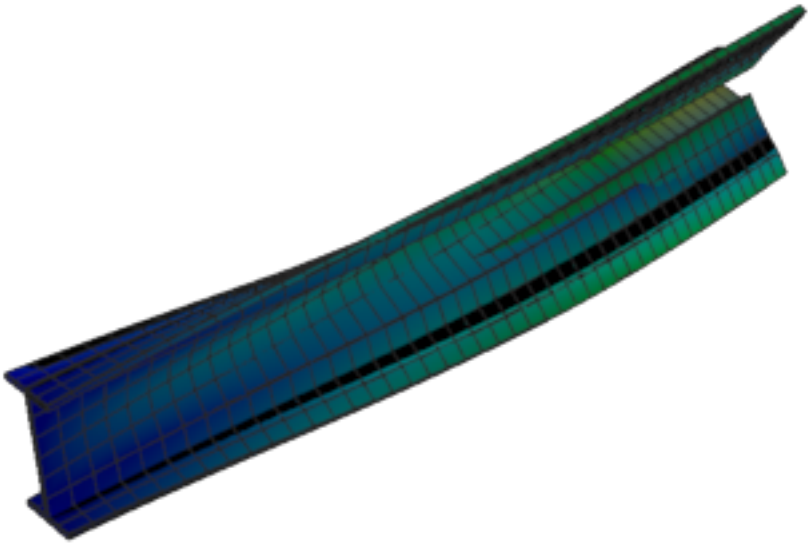


Figure 397: Abaqus — mode 7, 90.033 Hz

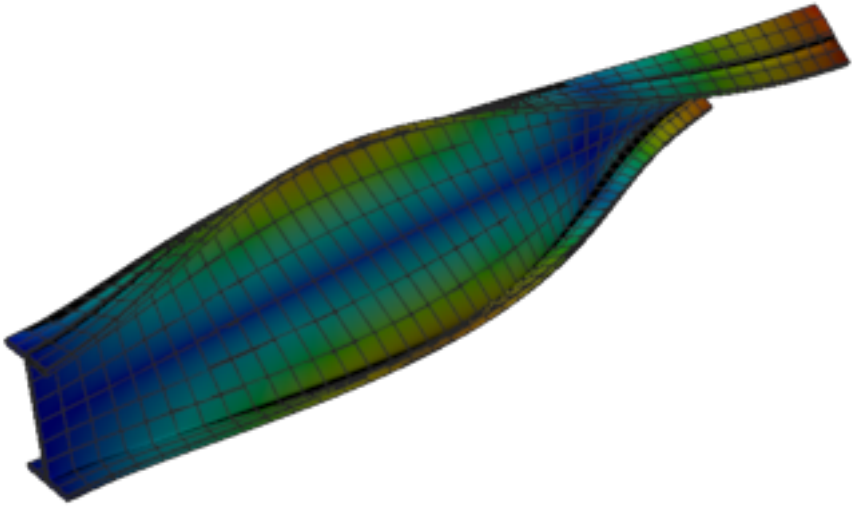


Figure 398: Calculix — mode 7, 351.823 Hz

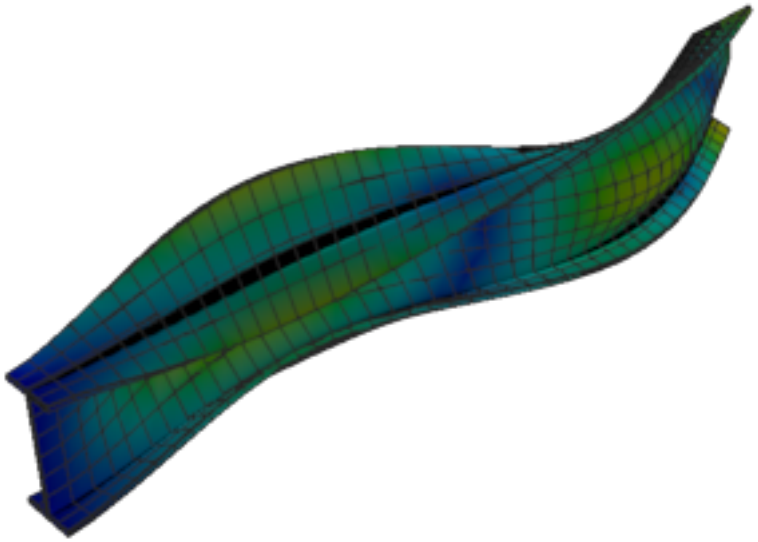


Figure 399: Code_Aster — mode 7, 205.267 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	90.033	-58.26	320010	1.17e-28	2.79	5.27e-21	-1.08e-14	-1.671	7.26e-11
Calculix	351.823	+63.10	4.88661e+06	7.91e-27	2.11e-24	1.60e-27	8.89e-14	-1.45e-12	-4.00e-14
Code_Aster	205.267	-4.84	1.66341e+06	2.28e-09	8.94	0.01	4.77e-05	2.990	0.071

Mode 8

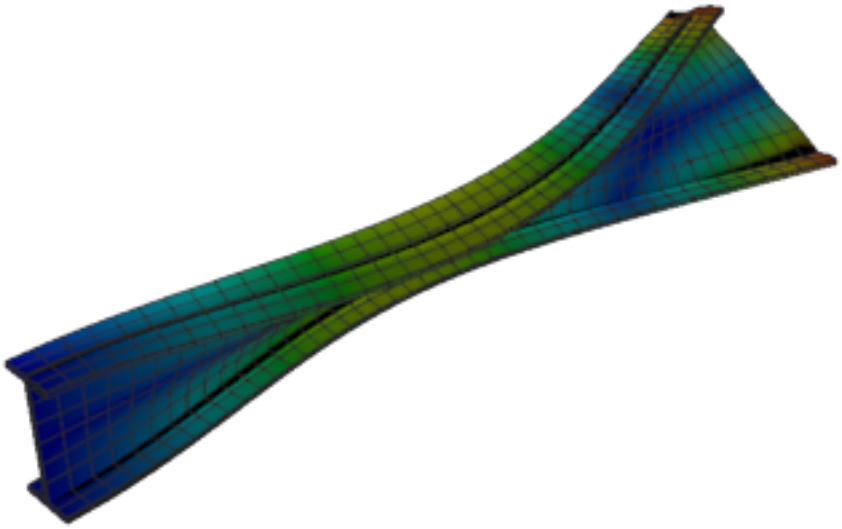


Figure 400: Abaqus — mode 8, 124.578 Hz

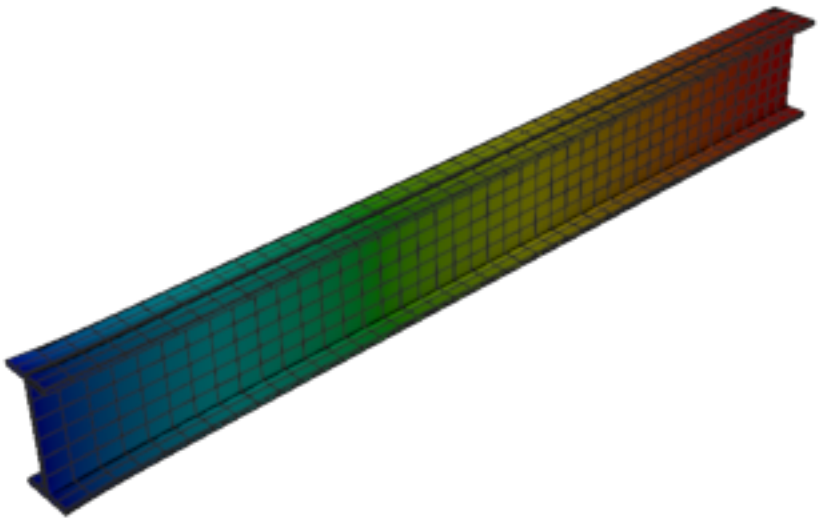


Figure 401: Calculix — mode 8, 431.955 Hz

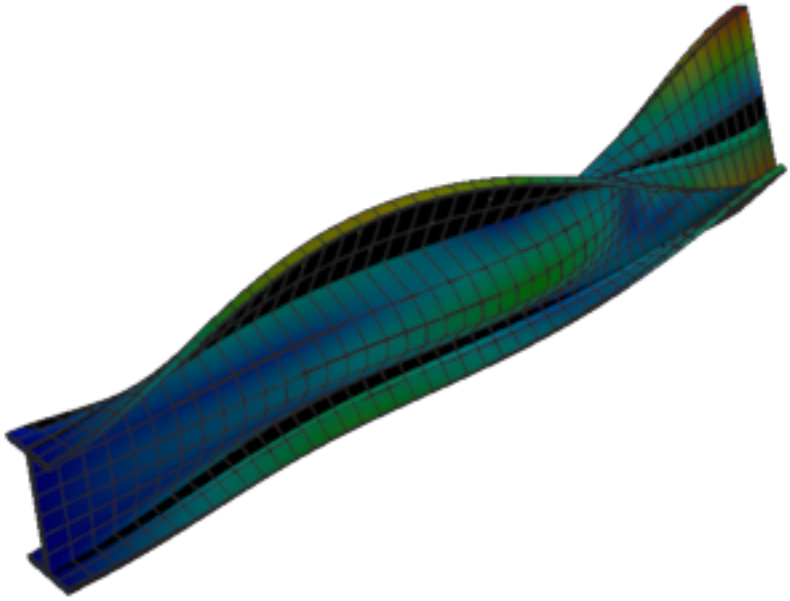


Figure 402: Code_Aster — mode 8, 243.985 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	124.580	-53.31	612693	3.87e-24	3.38e-21	2.90e-24	-1.97e-12	-5.82e-11	-1.70e-12
Calculix	431.955	+61.88	7.36609e+06	153.48	1.64e-27	2.24e-26	12.389	4.05e-14	1.50e-13
Code_Aster	243.985	-8.57	2.3501e+06	6.66e-07	3.10	0.25	0.001	1.760	-0.503

Mode 9

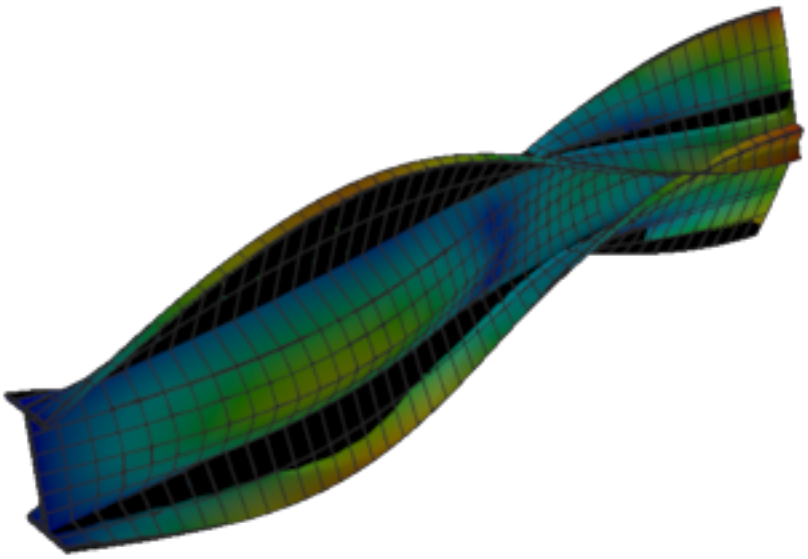


Figure 403: Abaqus — mode 9, 155.992 Hz

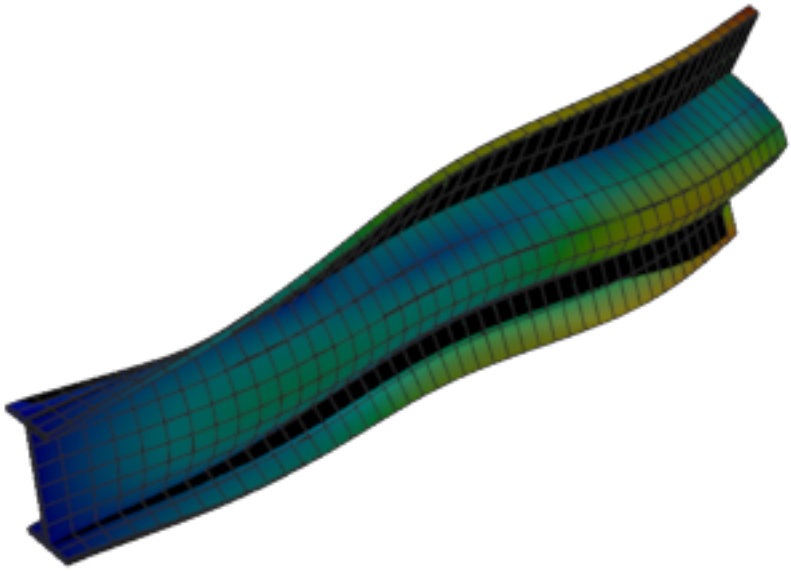


Figure 404: Calculix — mode 9, 439.038 Hz

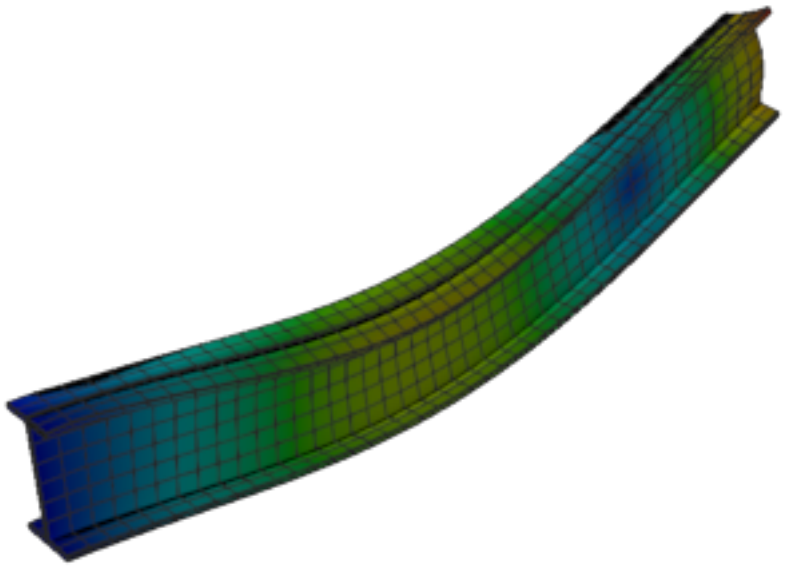


Figure 405: Code_Aster — mode 9, 254.838 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	155.990	-44.94	960645	7.43e-21	1.12	2.07e-20	-8.62e-11	-1.056	-1.44e-10
Calculix	439.038	+54.98	7.60964e+06	1.40e-23	0.38	1.42e-27	3.74e-12	-0.619	-3.77e-14
Code_Aster	254.838	-10.04	2.56382e+06	1.17e-06	0.01	39.92	0.001	-0.115	-6.318

Mode 10

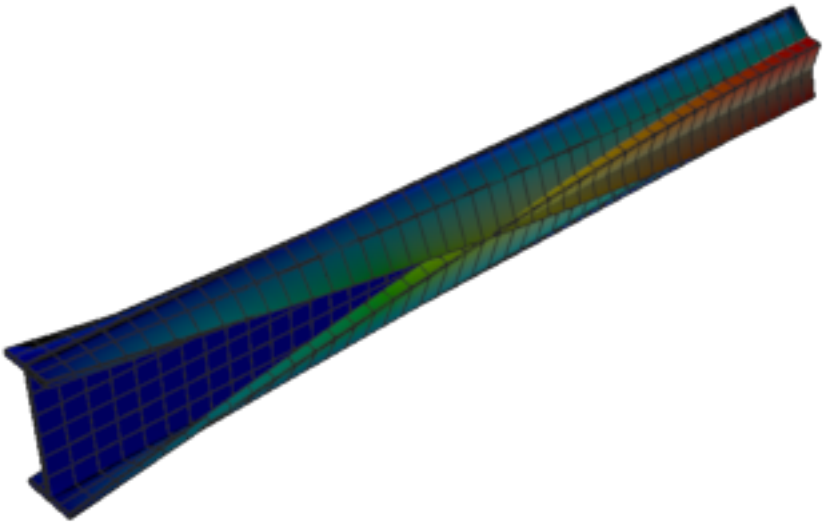


Figure 406: Abaqus — mode 10, 159.952 Hz

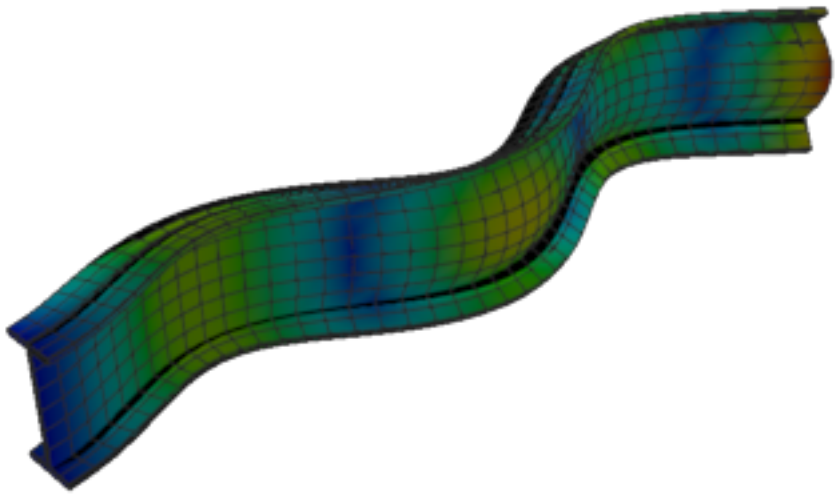


Figure 407: Calculix — mode 10, 456.762 Hz

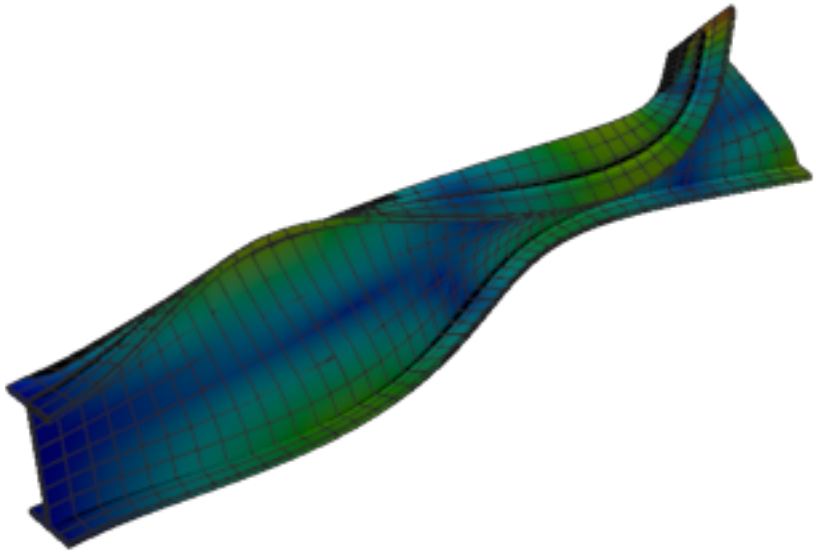


Figure 408: Code_Aster — mode 10, 282.888 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	159.950	-46.66	1.01004e+06	3.65e-04	8.80e-19	3.04e-19	0.019	-9.38e-10	5.51e-10
Calculix	456.762	+52.32	8.23646e+06	1.29e-25	6.14	7.58e-30	-3.59e-13	2.478	-2.75e-15
Code_Aster	282.888	-5.66	3.15929e+06	3.28e-05	0.01	3.08e-05	0.006	0.086	-0.006

Mode 11

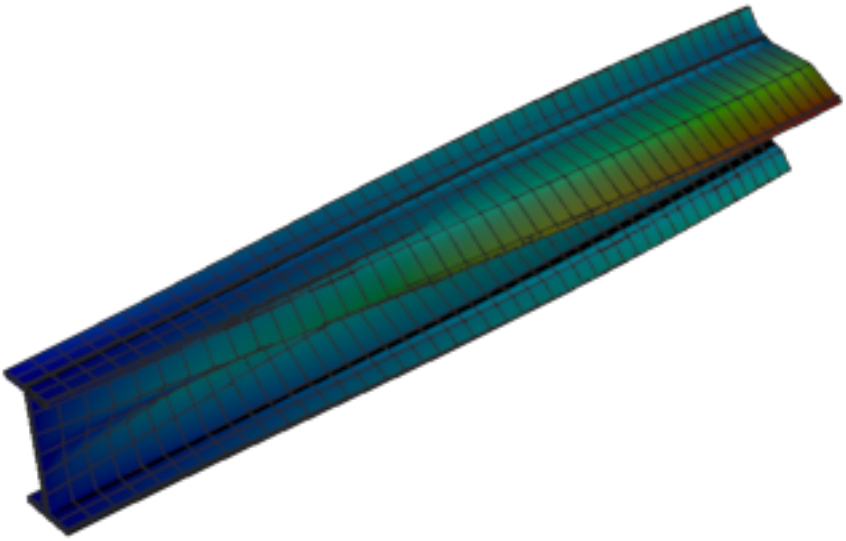


Figure 409: Abaqus — mode 11, 162.557 Hz

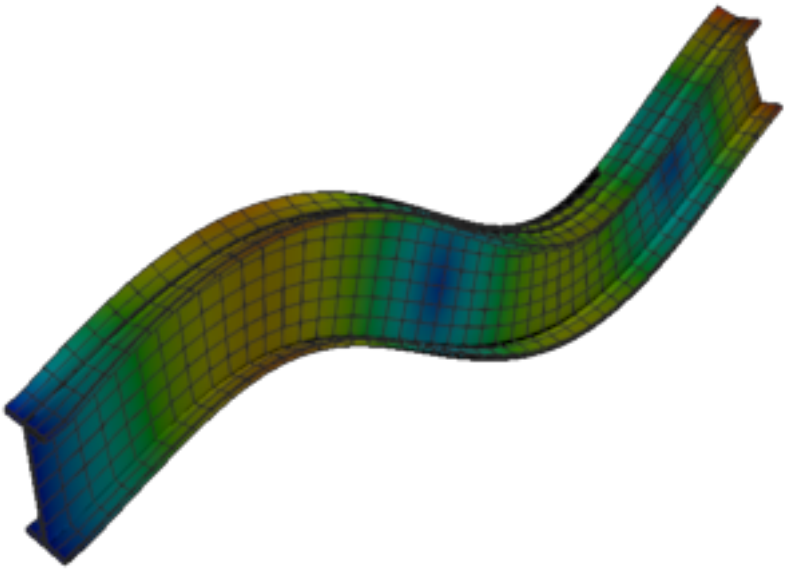


Figure 410: Calculix — mode 11, 582.562 Hz

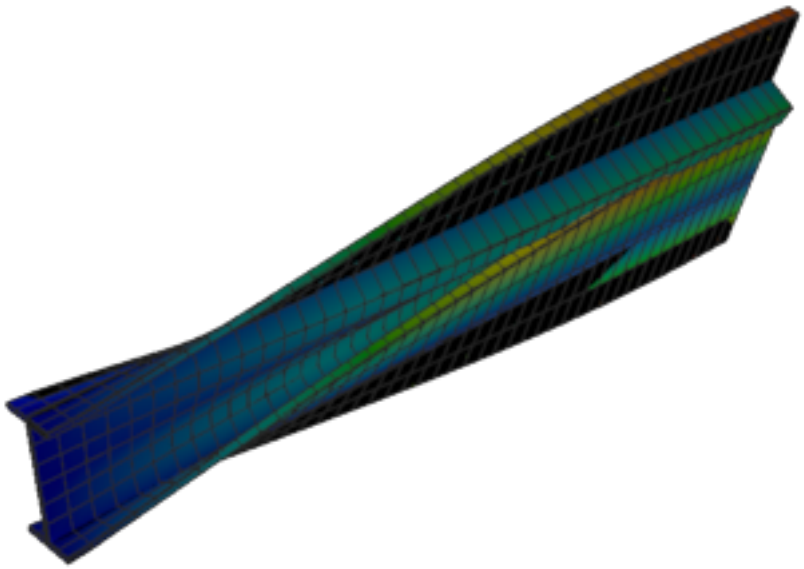


Figure 411: Code_Aster — mode 11, 305.247 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	162.560	-53.57	1.04321e+06	4.88e-19	1.57e-19	3.29e-20	-6.98e-10	3.96e-10	-1.81e-10
Calculix	582.562	+66.39	1.33981e+07	2.76e-25	1.21e-26	12.66	-5.25e-13	-1.10e-13	3.558
Code_Aster	305.247	-12.82	3.67844e+06	2.38e-05	8.68e-05	3.72e-07	0.005	0.009	-0.001

Solid, 2nd order, TET

Mode 1

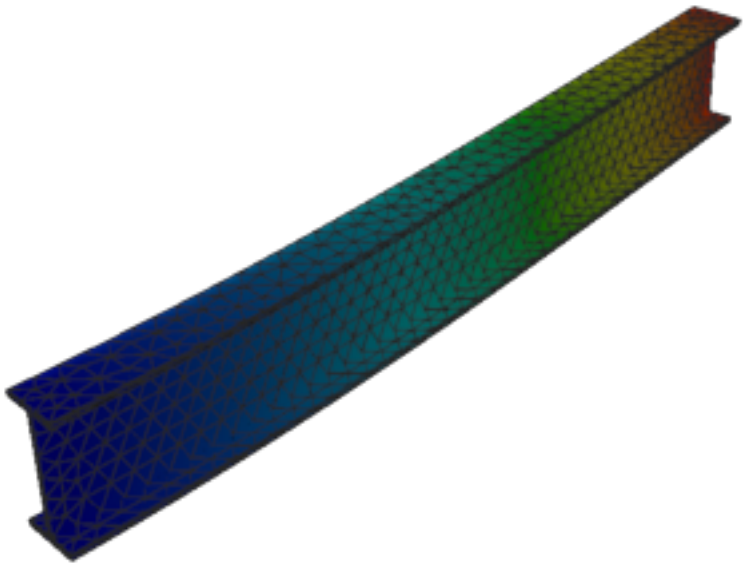


Figure 412: Abaqus — mode 1, 12.954 Hz

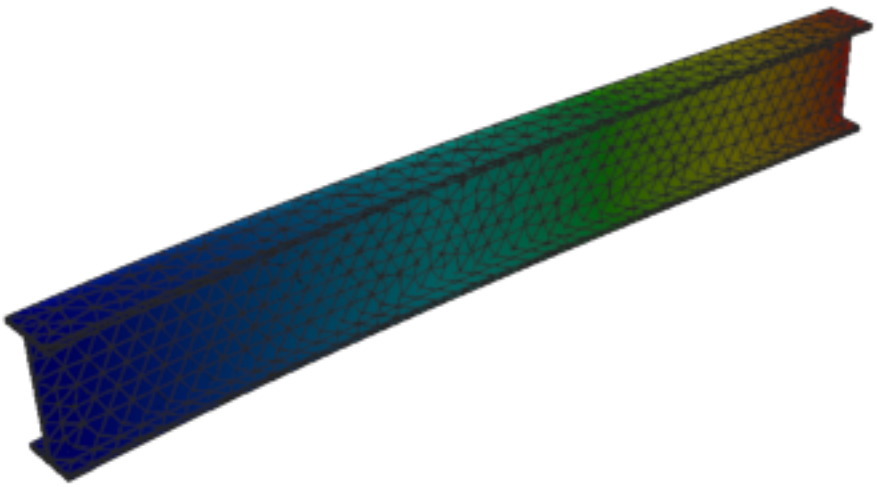


Figure 413: Calculix — mode 1, 12.954 Hz

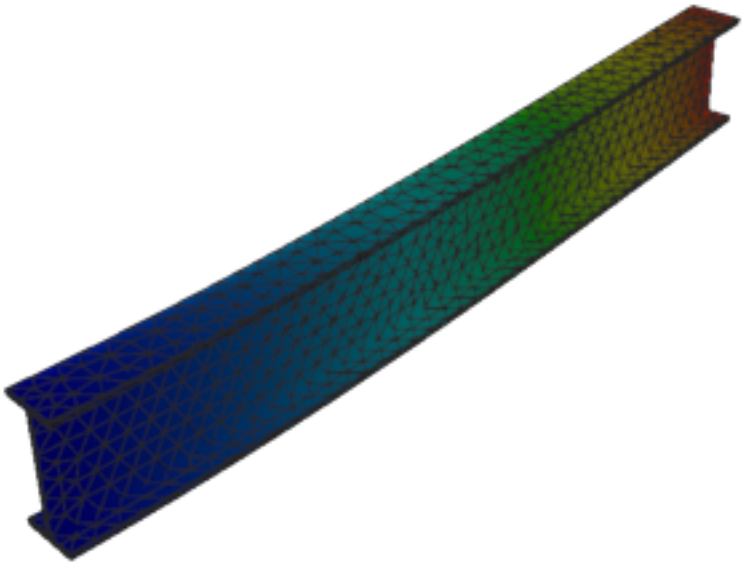


Figure 414: Code_Aster — mode 1, 12.954 Hz

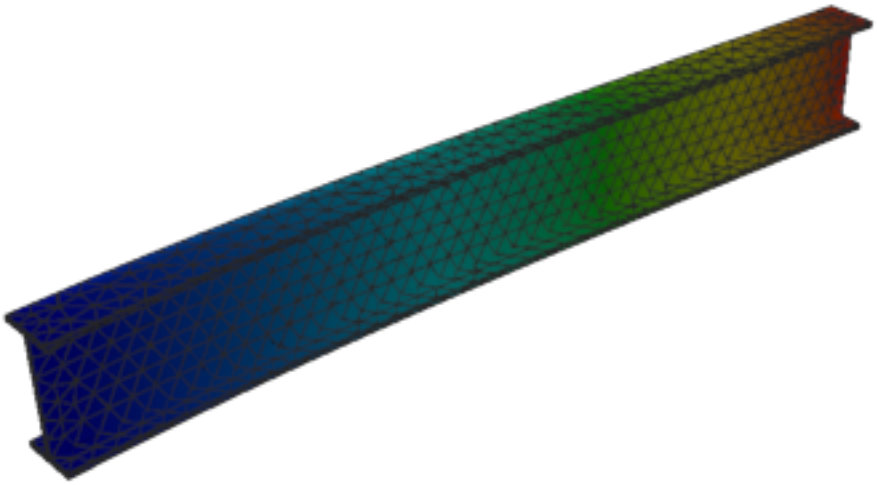


Figure 415: Sesam — mode 1, 12.954 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.954	-1.55e-04	6624.5	3.65e-13	116.51	2.81e-11	6.04e-07	10.794	5.30e-06
Calculix	12.954	1.16e-03	6624.89	4.94e-13	116.51	2.88e-10	-7.03e-07	-10.794	-1.70e-05
Code_Aster	12.954	1.16e-03	6624.89	5.04e-13	116.51	2.91e-10	7.10e-07	10.794	1.71e-05
Sesam	12.954	-2.16e-03	6624.46	3.01e-13	116.51	2.70e-11	-5.49e-07	-10.794	-5.20e-06

Mode 2

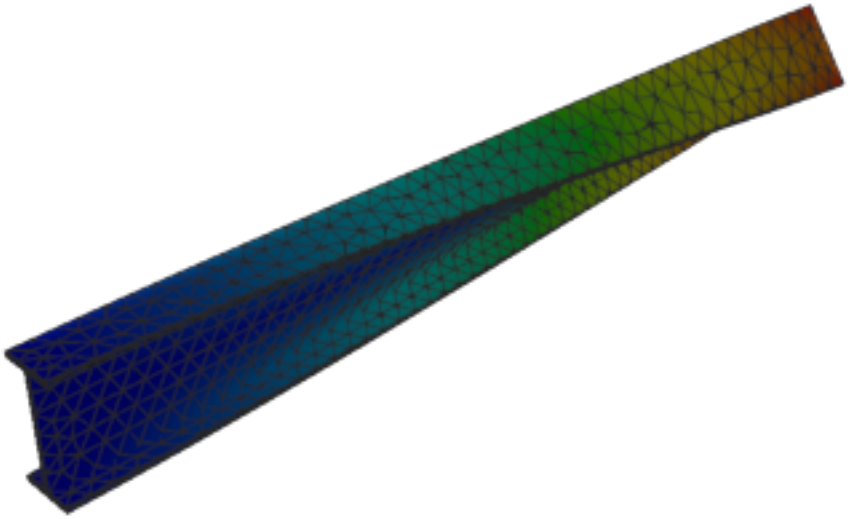


Figure 416: Abaqus — mode 2, 20.294 Hz

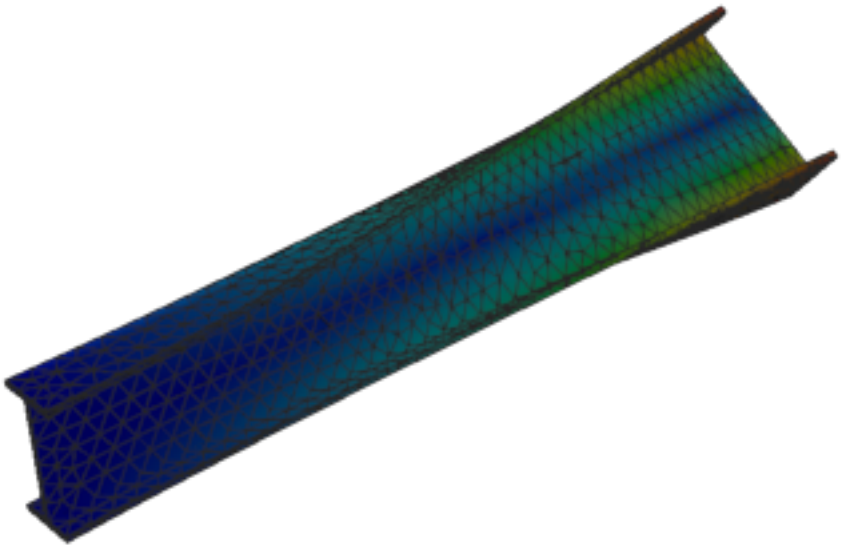


Figure 417: Calculix — mode 2, 20.288 Hz

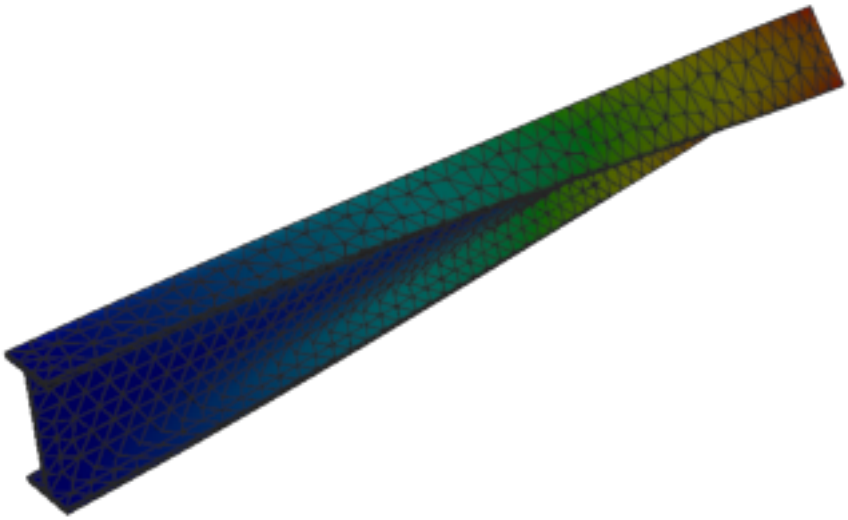


Figure 418: Code_Aster — mode 2, 20.288 Hz

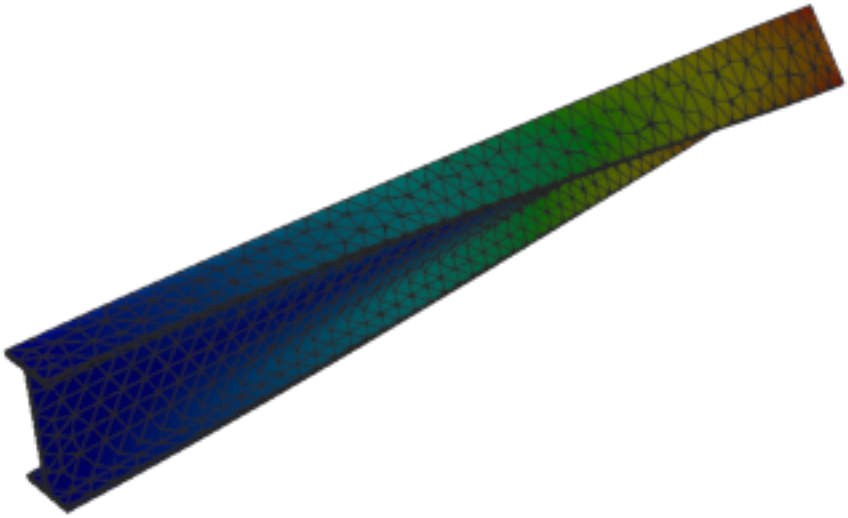


Figure 419: Sesam — mode 2, 20.294 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.294	+0.02	16260	8.19e-12	5.86e-08	1.99e-09	2.86e-06	-2.42e-04	4.47e-05
Calculix	20.288	-0.02	16248.9	3.07e-11	1.64e-07	3.90e-10	-5.54e-06	-4.05e-04	-1.98e-05
Code_Aster	20.288	-0.02	16248.9	3.26e-11	1.64e-07	3.90e-10	5.71e-06	4.05e-04	1.98e-05
Sesam	20.294	+0.02	16259.8	8.59e-12	5.86e-08	1.98e-09	2.93e-06	-2.42e-04	4.45e-05

Mode 3

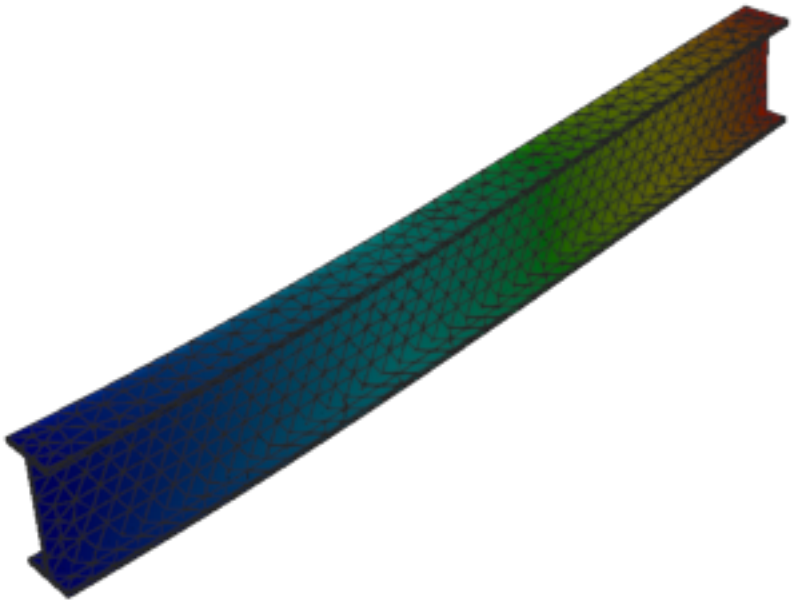


Figure 420: Abaqus — mode 3, 50.540 Hz

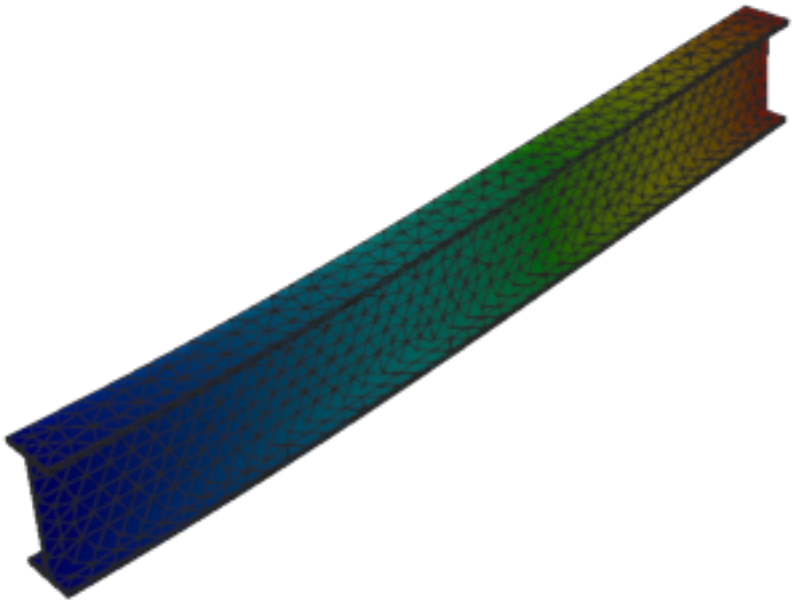


Figure 421: Calculix — mode 3, 50.540 Hz

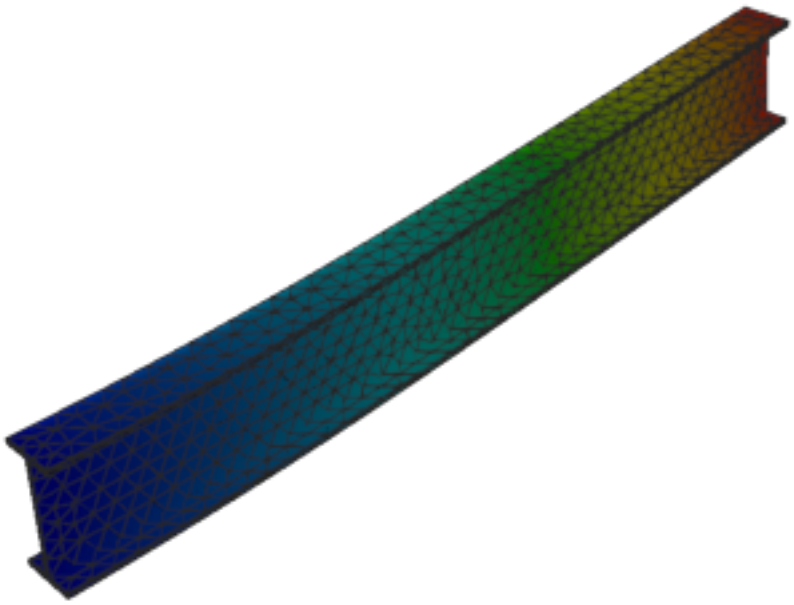


Figure 422: Code_Aster — mode 3, 50.540 Hz

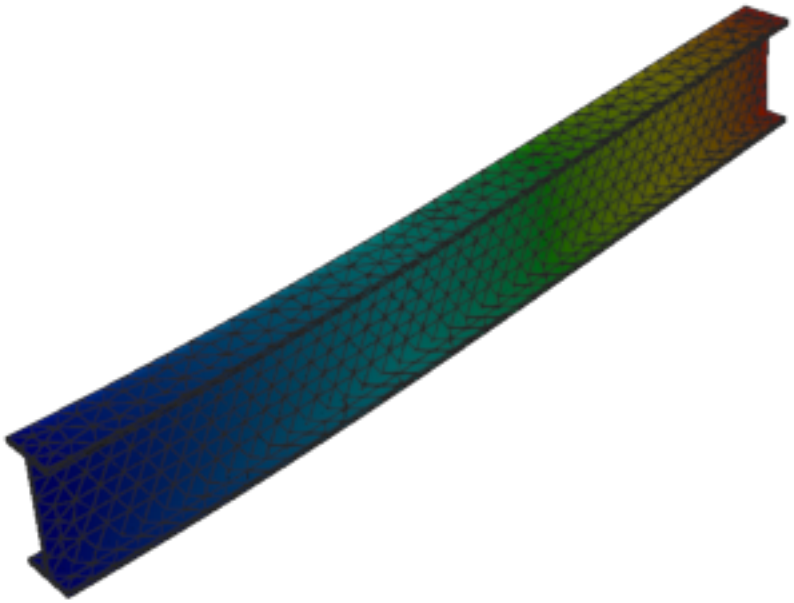


Figure 423: Sesam — mode 3, 50.540 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.540	-1.75e-04	100840	3.09e-10	1.03e-08	118.35	1.76e-05	-1.01e-04	10.879
Calculix	50.540	1.22e-04	100840	4.32e-10	6.11e-09	118.35	-2.08e-05	-7.82e-05	10.879
Code_Aster	50.540	1.29e-04	100840	4.42e-10	6.08e-09	118.35	-2.10e-05	-7.80e-05	10.879
Sesam	50.540	-7.59e-05	100840	3.10e-10	1.03e-08	118.35	1.76e-05	-1.01e-04	10.879

Mode 4

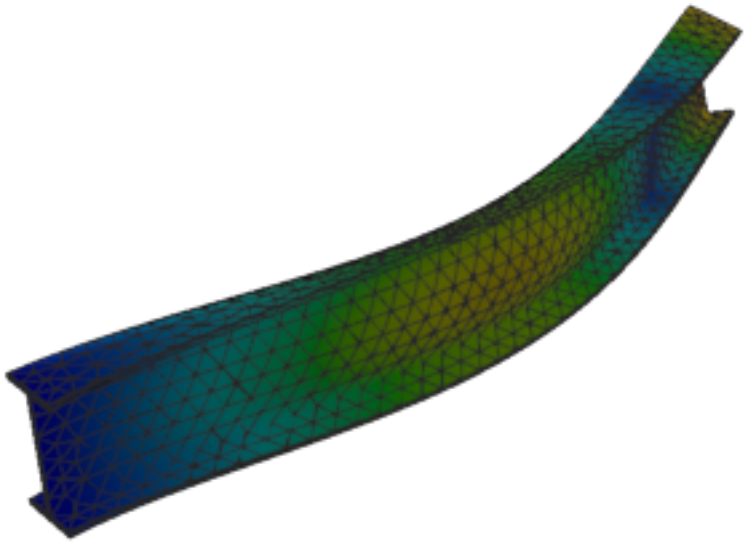


Figure 424: Abaqus — mode 4, 77.128 Hz

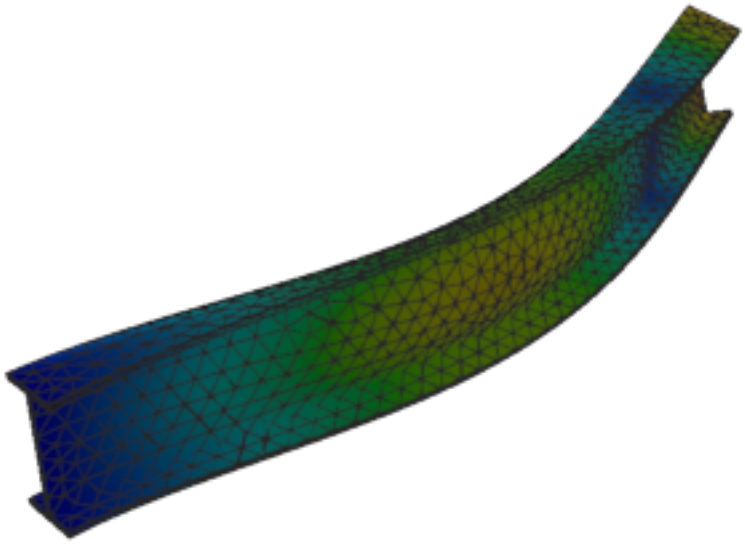


Figure 425: Calculix — mode 4, 77.230 Hz

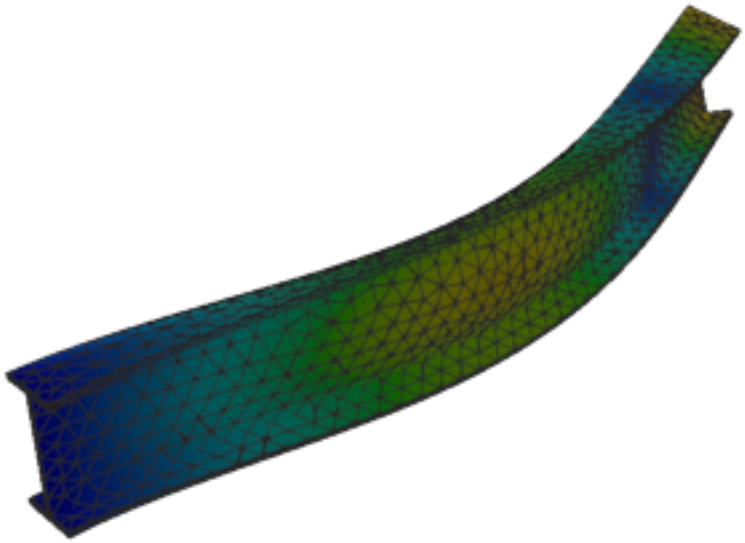


Figure 426: Code_Aster — mode 4, 77.230 Hz

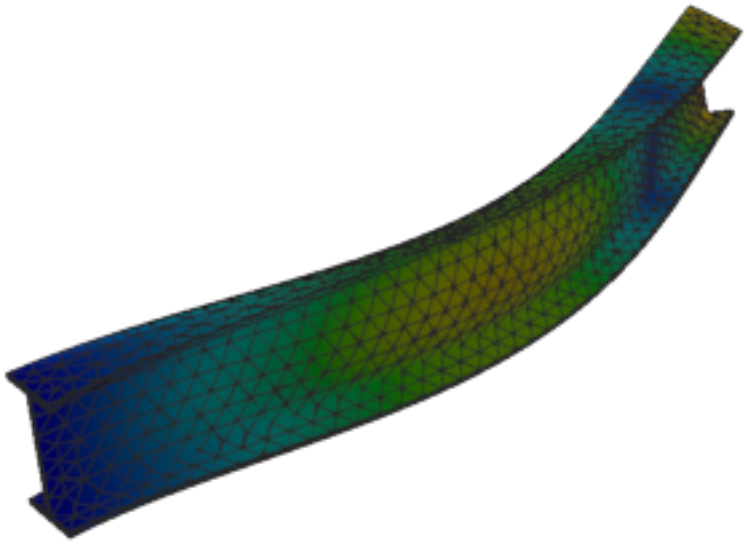


Figure 427: Sesam — mode 4, 77.128 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	77.128	-0.07	234845	6.40e-12	35.97	1.94e-09	2.53e-06	-5.997	-4.41e-05
Calculix	77.230	+0.07	235466	4.79e-11	36.04	7.89e-10	-6.92e-06	-6.004	2.81e-05
Code_Aster	77.230	+0.07	235466	4.90e-11	36.04	7.59e-10	-7.00e-06	-6.004	2.75e-05
Sesam	77.128	-0.07	234845	1.04e-11	35.97	1.89e-09	3.22e-06	-5.997	-4.35e-05

Mode 5

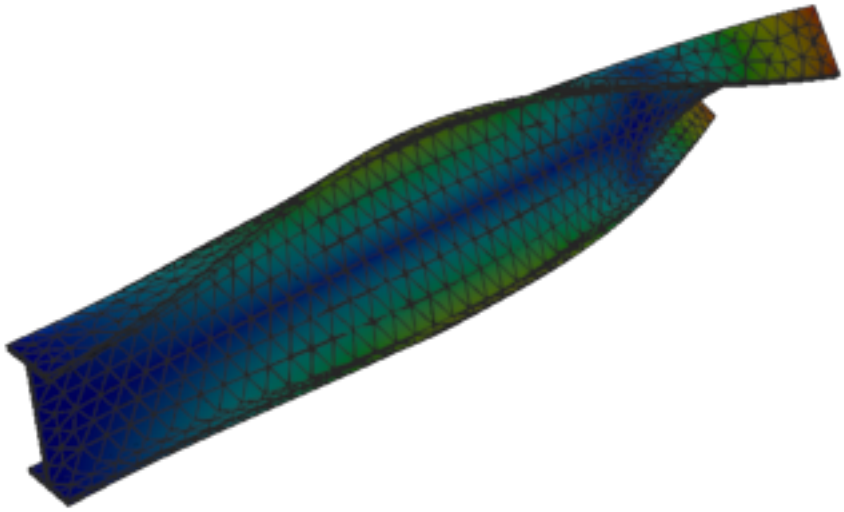


Figure 428: Abaqus — mode 5, 97.830 Hz

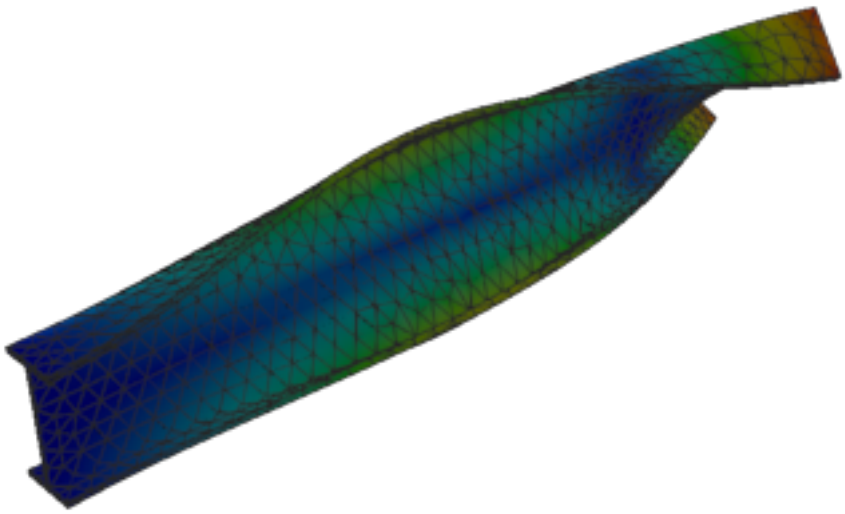


Figure 429: Calculix — mode 5, 97.851 Hz

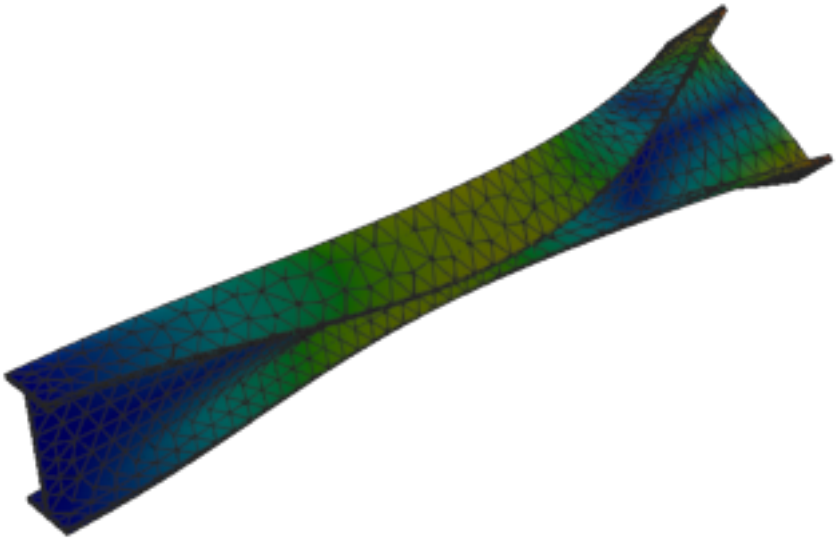


Figure 430: Code_Aster — mode 5, 97.851 Hz

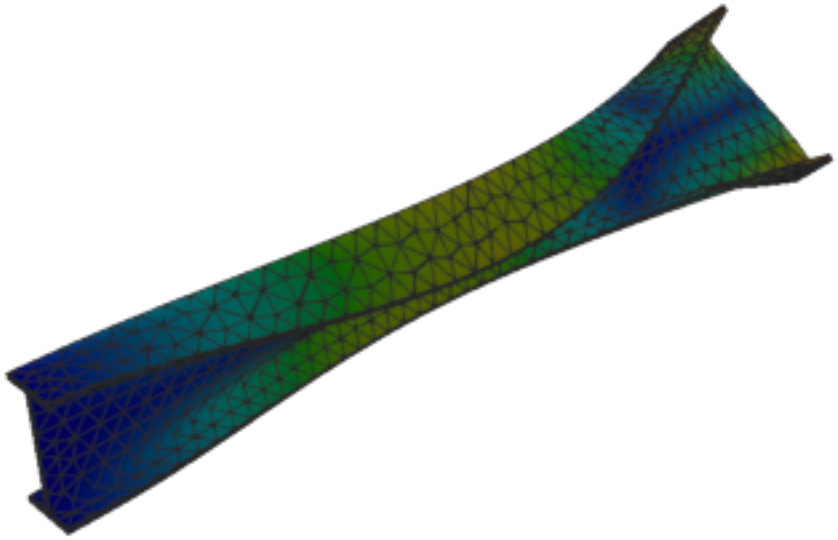


Figure 431: Sesam — mode 5, 97.830 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.830	-0.01	377834	1.93e-10	3.14e-07	6.49e-09	-1.39e-05	-0.001	8.06e-05
Calculix	97.851	+0.01	377996	7.91e-10	6.66e-07	5.29e-10	-2.81e-05	0.001	-2.30e-05
Code_Aster	97.851	+0.01	377995	8.39e-10	6.65e-07	5.24e-10	2.90e-05	-0.001	2.29e-05
Sesam	97.830	-0.01	377833	1.99e-10	3.14e-07	6.59e-09	1.41e-05	0.001	-8.12e-05

Mode 6

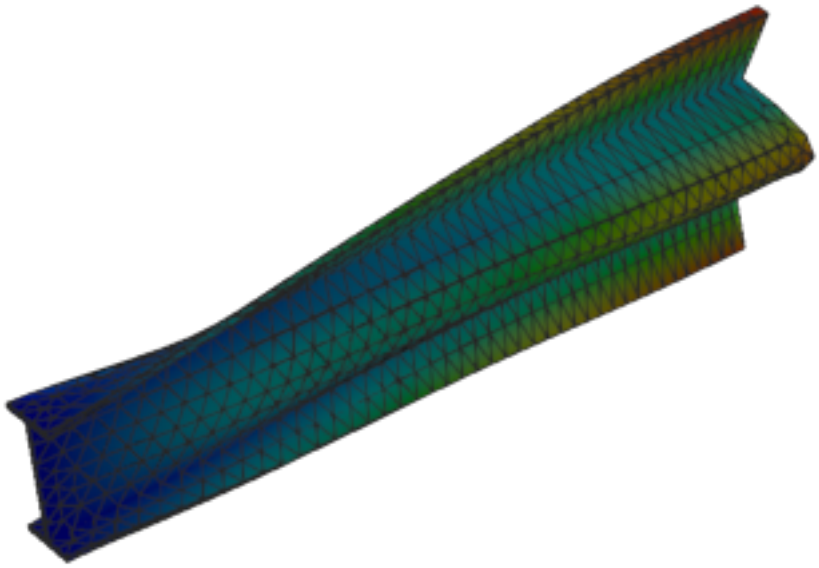


Figure 432: Abaqus — mode 6, 132.482 Hz

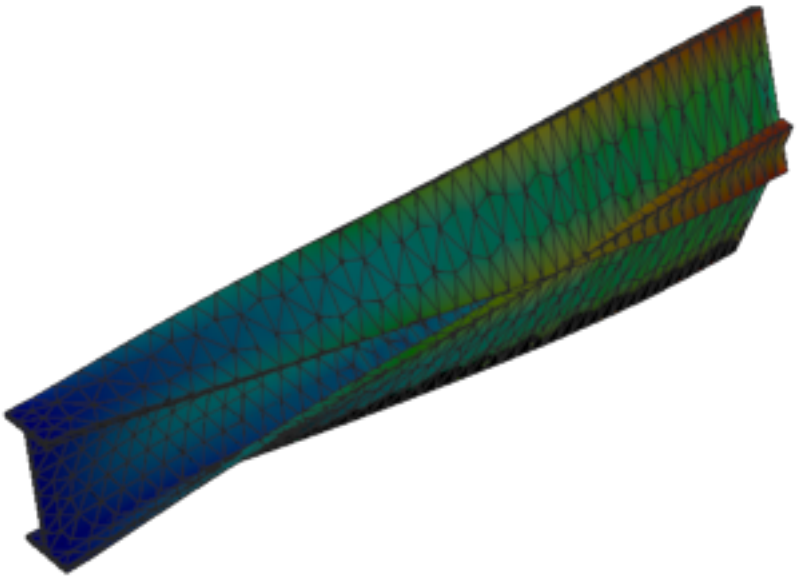


Figure 433: Calculix — mode 6, 133.015 Hz

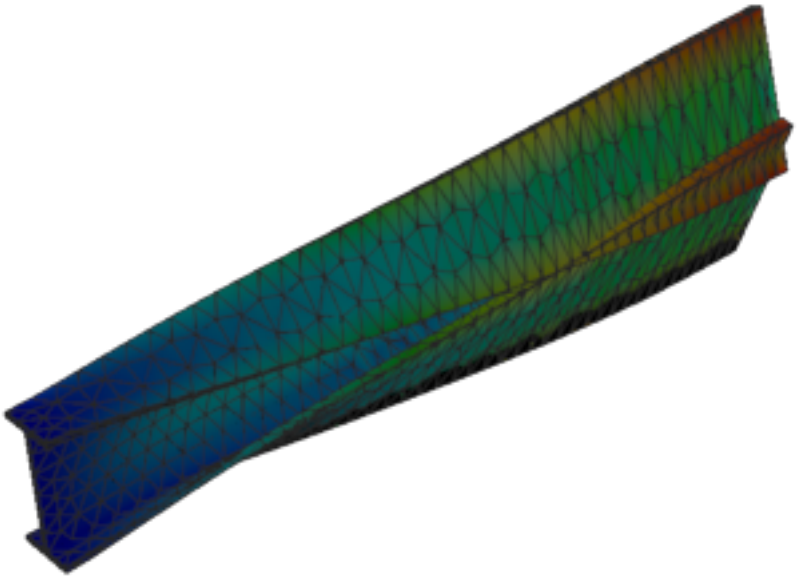


Figure 434: Code_Aster — mode 6, 133.014 Hz

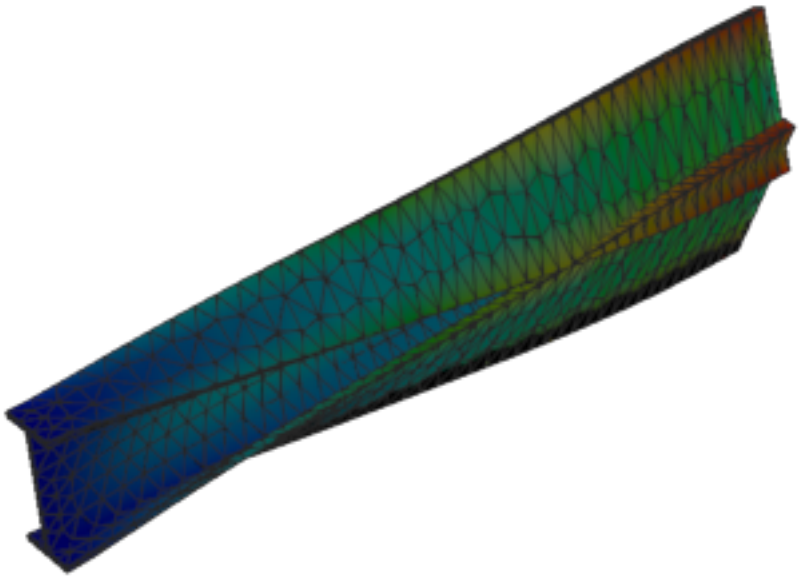


Figure 435: Sesam — mode 6, 132.478 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	132.480	-0.20	692901	1.05e-09	0.01	2.04e-09	3.23e-05	0.119	-4.51e-05
Calculix	133.015	+0.20	698496	3.94e-10	0.01	2.21e-09	1.98e-05	-0.093	4.70e-05
Code_Aster	133.014	+0.20	698479	4.00e-10	0.01	2.27e-09	2.00e-05	-0.093	4.76e-05
Sesam	132.478	-0.20	692867	1.14e-09	0.01	2.04e-09	-3.38e-05	-0.119	4.52e-05

Mode 7

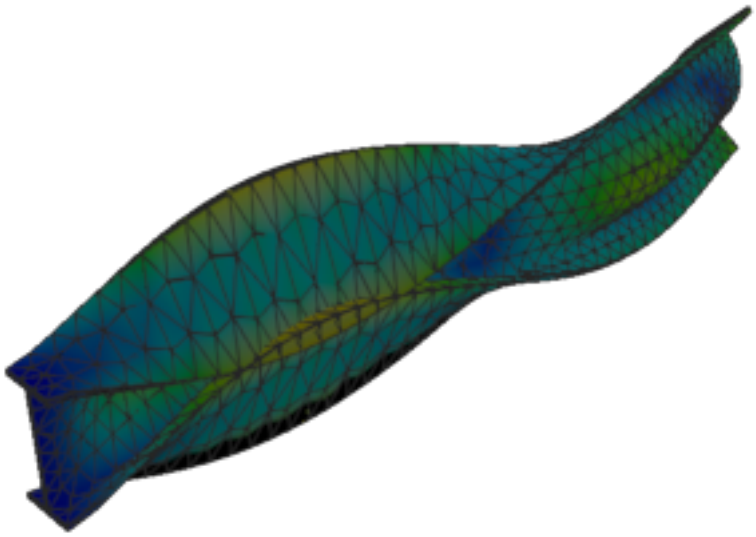


Figure 436: Abaqus — mode 7, 153.634 Hz

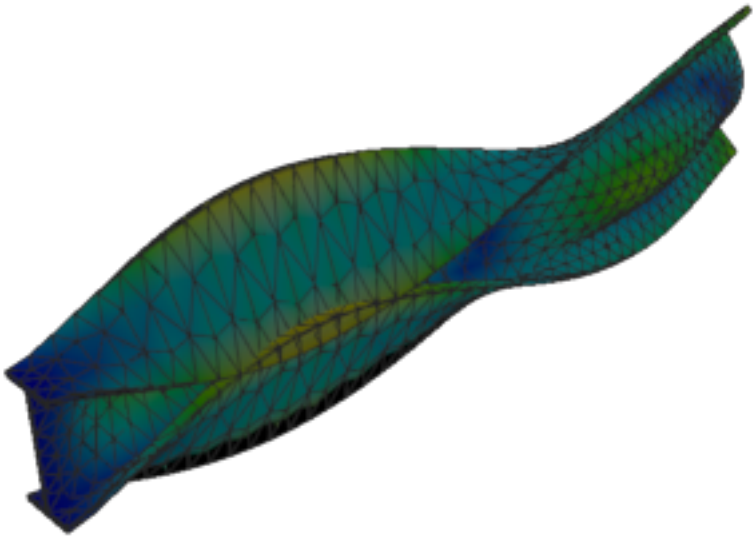


Figure 437: Calculix — mode 7, 154.803 Hz

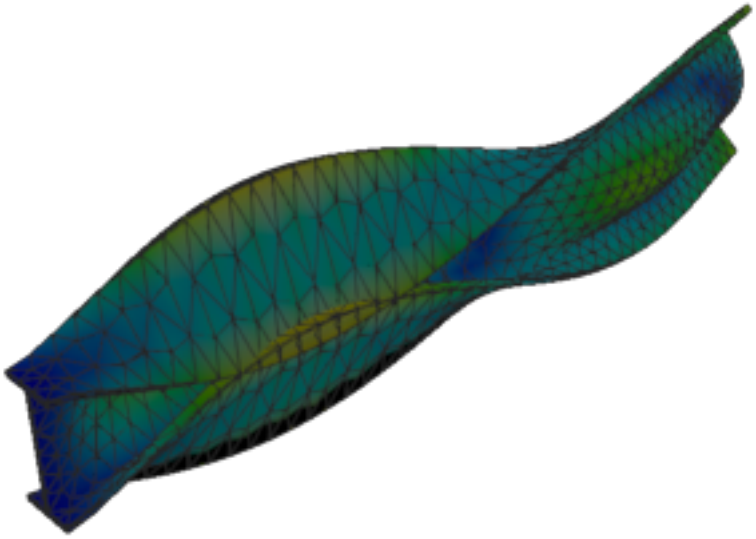


Figure 438: Code_Aster — mode 7, 154.802 Hz

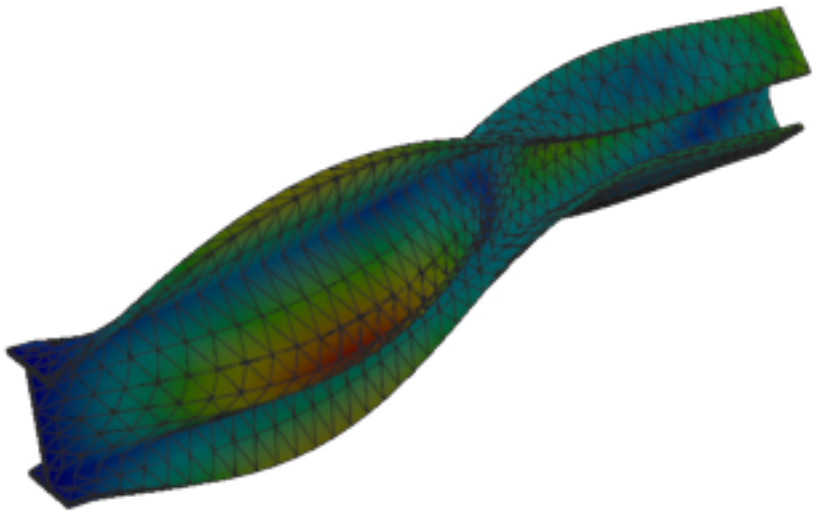


Figure 439: Sesam — mode 7, 153.631 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	153.630	-0.38	931828	1.43e-08	3.12	2.11e-09	-1.19e-04	1.768	-4.59e-05
Calculix	154.804	+0.38	946065	6.91e-09	3.24	7.22e-08	8.31e-05	1.799	2.69e-04
Code_Aster	154.802	+0.38	946046	7.01e-09	3.23	7.26e-08	8.37e-05	1.799	2.69e-04
Sesam	153.631	-0.38	931794	1.56e-08	3.12	2.15e-09	1.25e-04	-1.768	4.63e-05

Mode 8

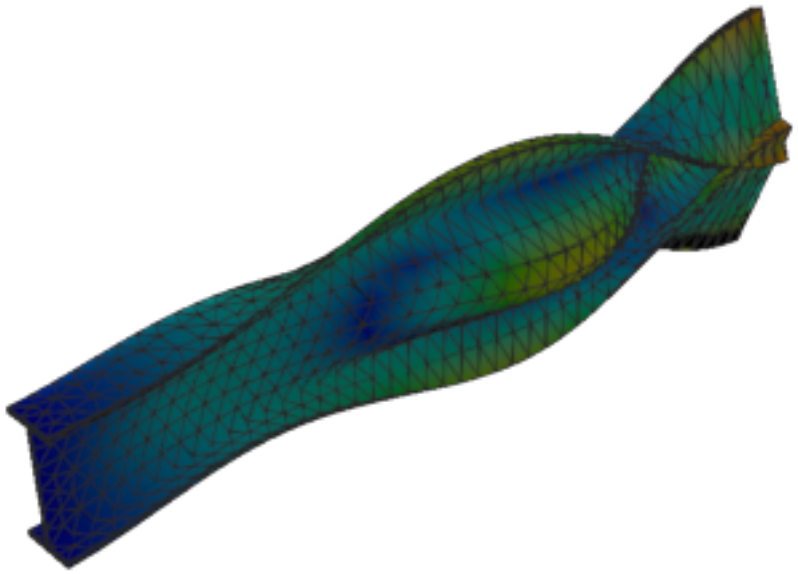


Figure 440: Abaqus — mode 8, 178.118 Hz

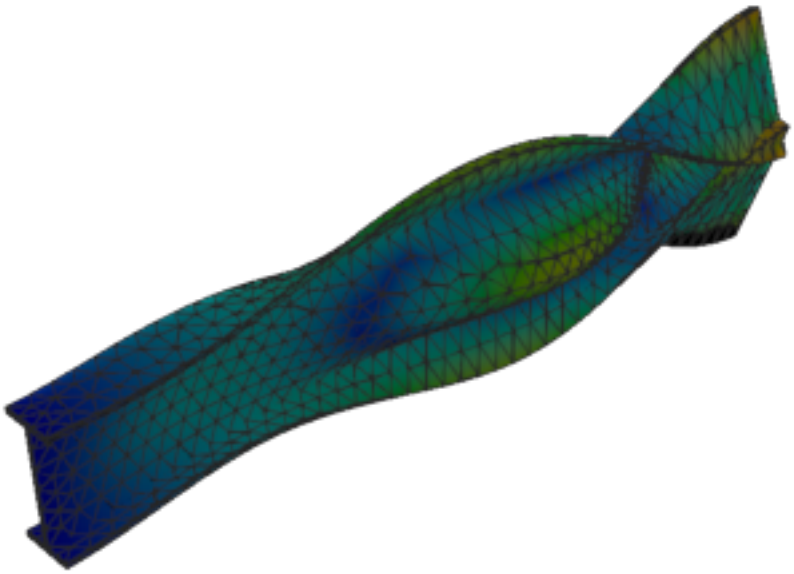


Figure 441: Calculix — mode 8, 181.029 Hz

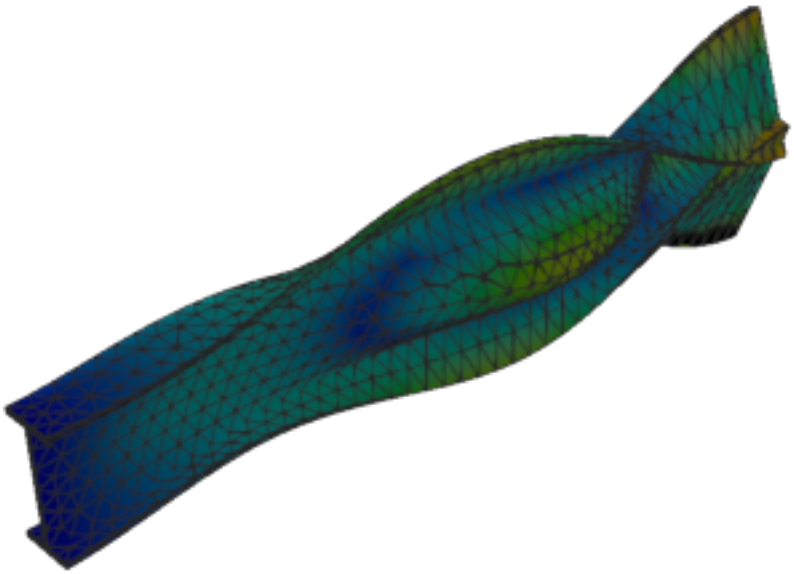


Figure 442: Code_Aster — mode 8, 181.026 Hz

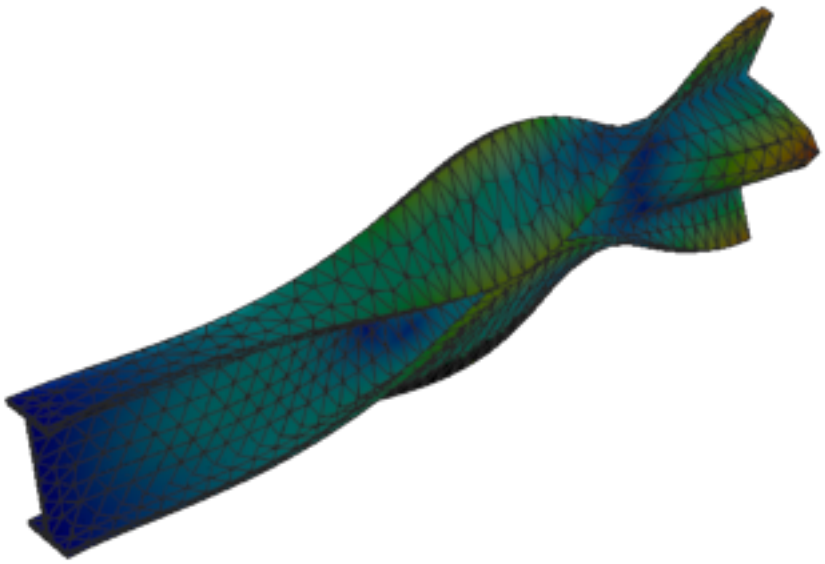


Figure 443: Sesam — mode 8, 178.114 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	178.120	-0.81	1.2525e+06	2.65e-10	5.79	5.99e-10	-1.63e-05	2.405	-2.45e-05
Calculix	181.029	+0.81	1.29376e+06	6.86e-10	5.93	6.55e-09	2.62e-05	2.436	-8.09e-05
Code_Aster	181.026	+0.81	1.29373e+06	6.94e-10	5.93	6.39e-09	2.63e-05	2.435	-7.99e-05
Sesam	178.114	-0.81	1.25243e+06	3.03e-10	5.79	6.40e-10	1.74e-05	-2.406	2.53e-05

Mode 9

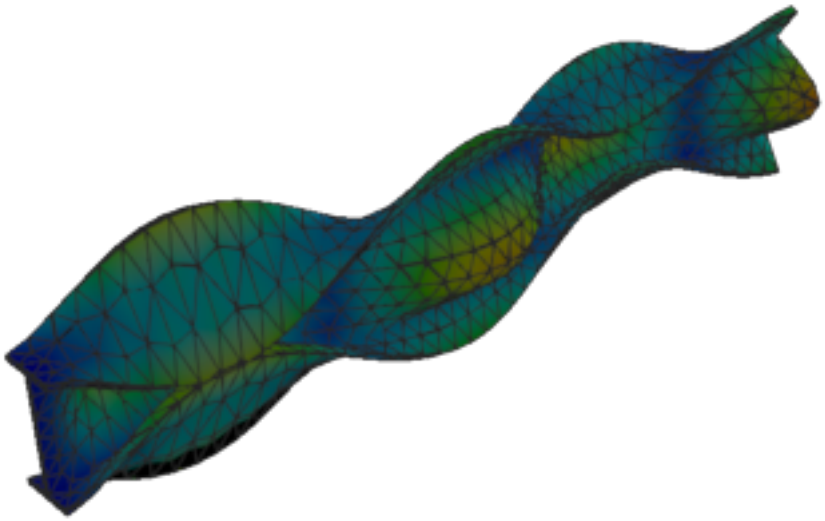


Figure 444: Abaqus — mode 9, 215.822 Hz

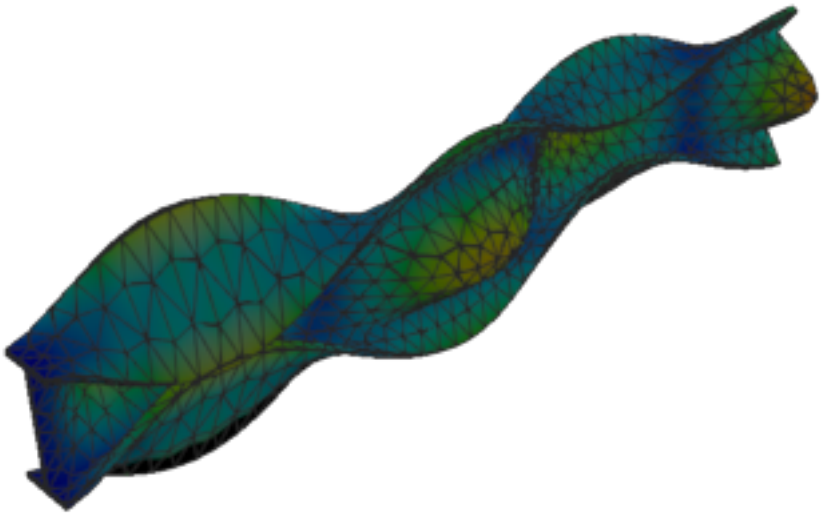


Figure 445: Calculix — mode 9, 219.501 Hz

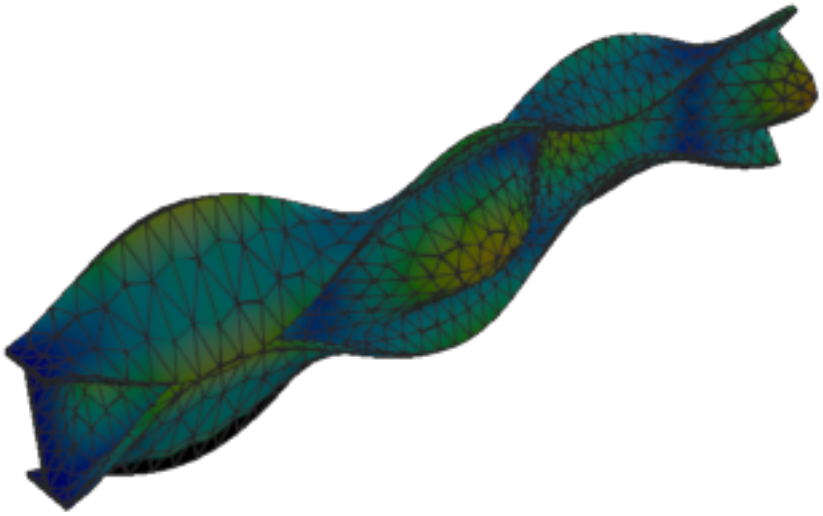


Figure 446: Code_Aster — mode 9, 219.497 Hz

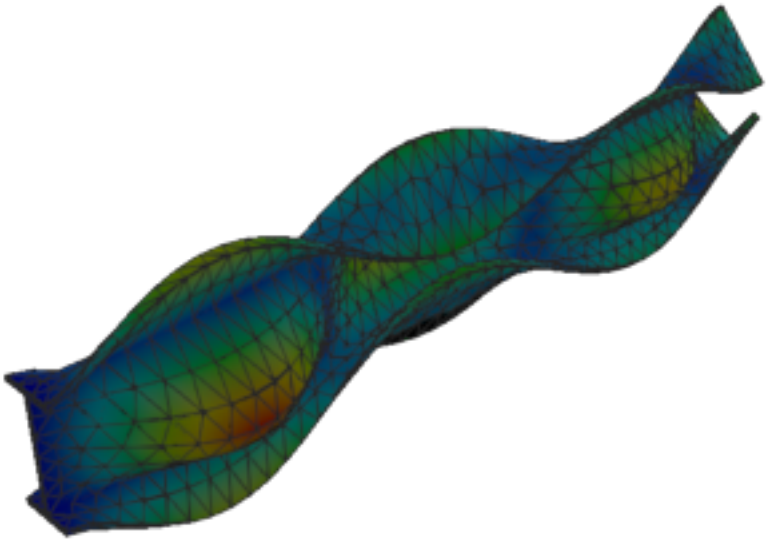


Figure 447: Sesam — mode 9, 215.815 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	215.820	-0.84	1.83887e+06	8.48e-08	0.60	8.50e-08	-2.91e-04	0.774	2.91e-04
Calculix	219.501	+0.85	1.9021e+06	5.19e-08	0.48	8.50e-08	2.28e-04	0.690	-2.92e-04
Code_Aster	219.497	+0.84	1.90203e+06	5.25e-08	0.48	8.41e-08	2.29e-04	0.690	-2.90e-04
Sesam	215.815	-0.85	1.83875e+06	9.03e-08	0.60	8.47e-08	3.00e-04	-0.774	-2.91e-04

Mode 10

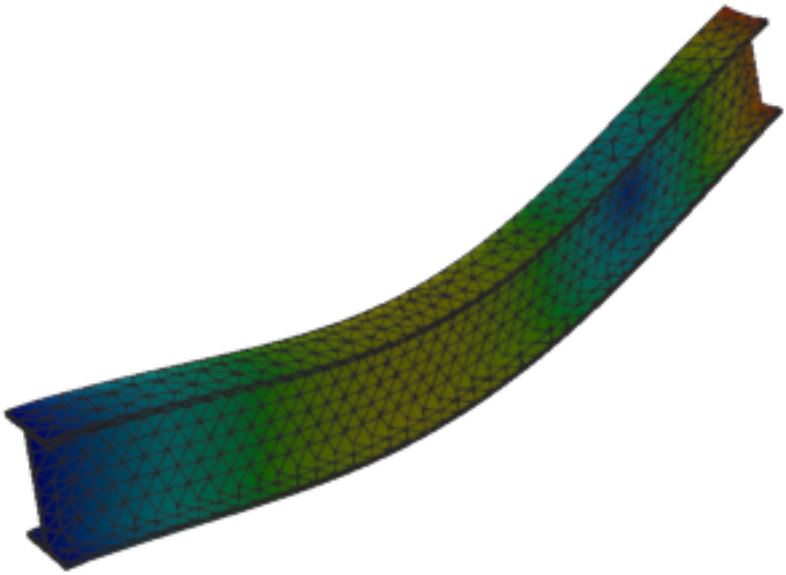


Figure 448: Abaqus — mode 10, 253.203 Hz

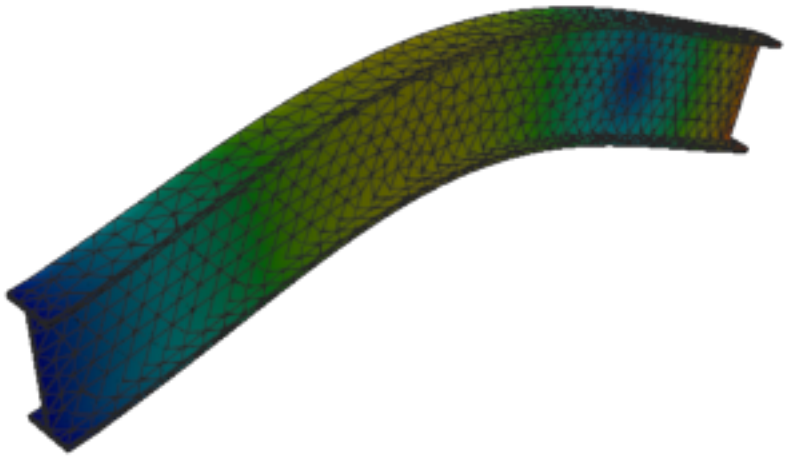


Figure 449: Calculix — mode 10, 253.193 Hz

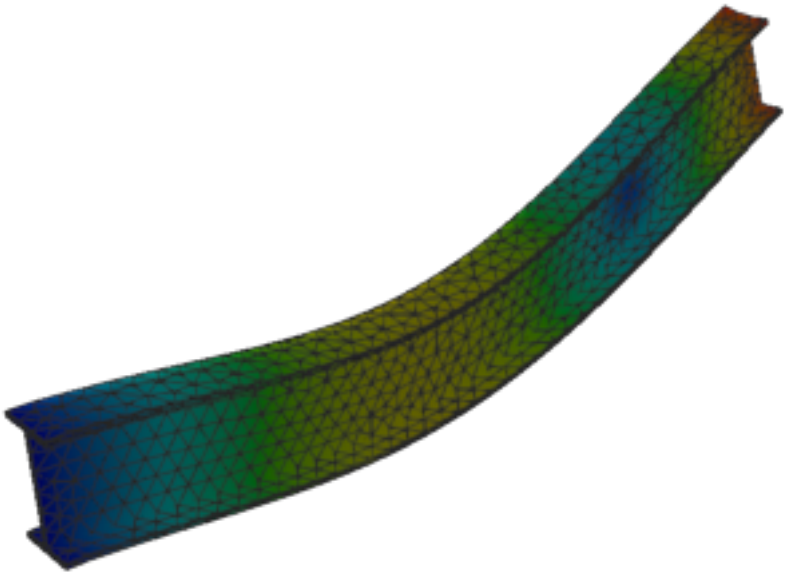


Figure 450: Code_Aster — mode 10, 253.193 Hz

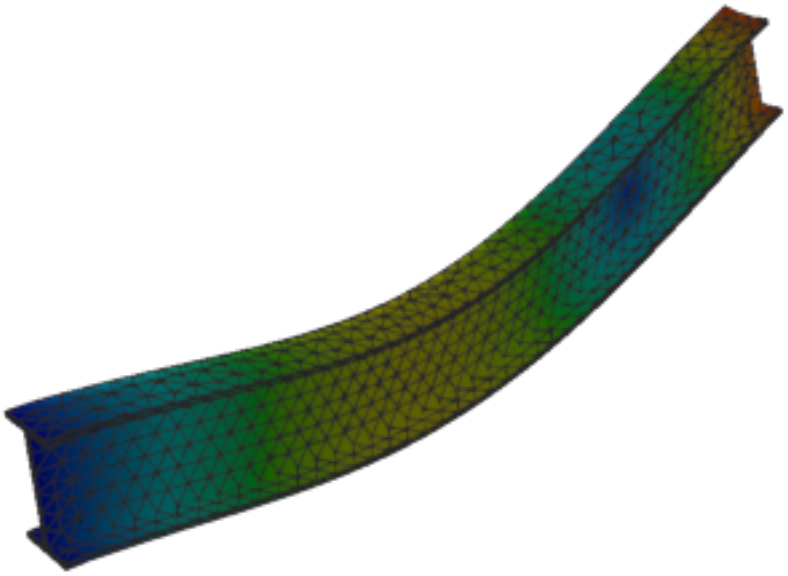


Figure 451: Sesam — mode 10, 253.203 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	253.200	1.16e-03	2.53103e+06	8.86e-09	1.97e-08	40.25	-9.41e-05	1.40e-04	-6.344
Calculix	253.193	-1.72e-03	2.53082e+06	1.34e-08	1.74e-07	40.25	-1.16e-04	-4.17e-04	6.344
Code_Aster	253.193	-1.74e-03	2.53082e+06	1.36e-08	1.74e-07	40.25	1.17e-04	4.17e-04	-6.344
Sesam	253.203	2.31e-03	2.53103e+06	9.13e-09	1.99e-08	40.25	-9.56e-05	1.41e-04	-6.344

Mode 11

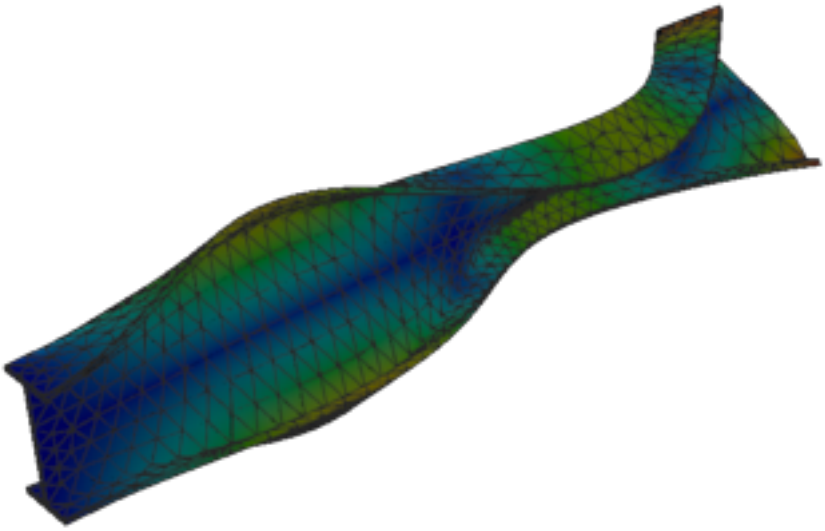


Figure 452: Abaqus — mode 11, 254.466 Hz

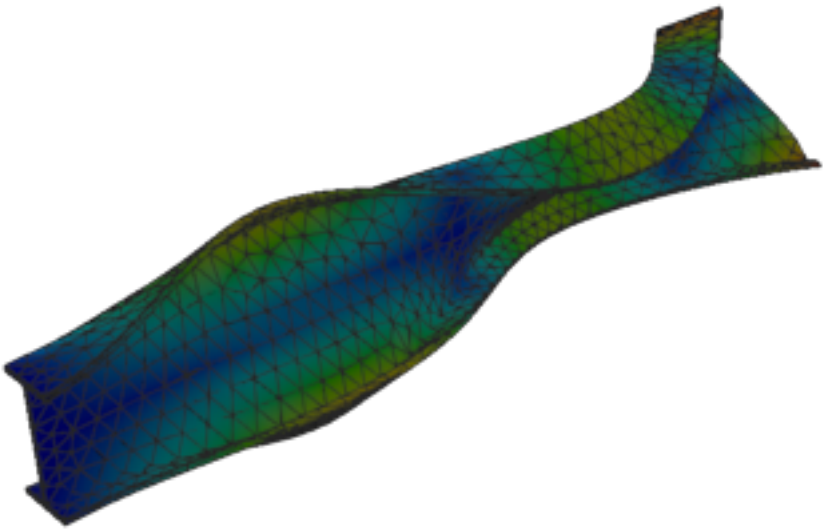


Figure 453: Calculix — mode 11, 254.685 Hz

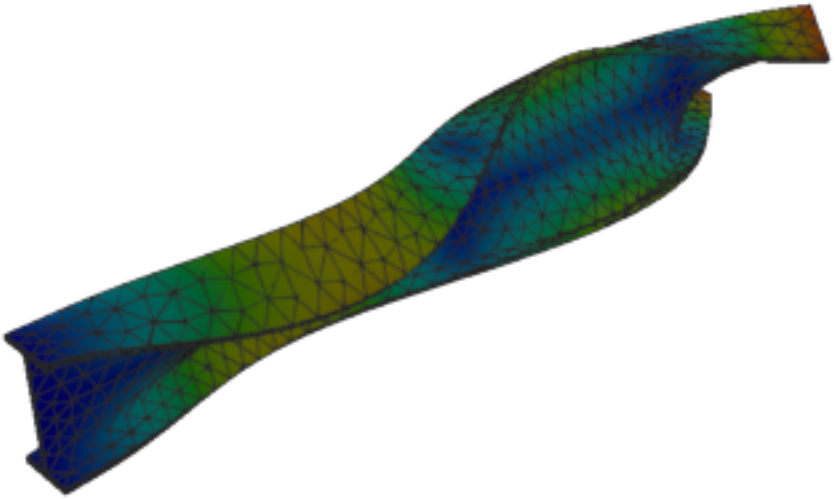


Figure 454: Code_Aster — mode 11, 254.684 Hz

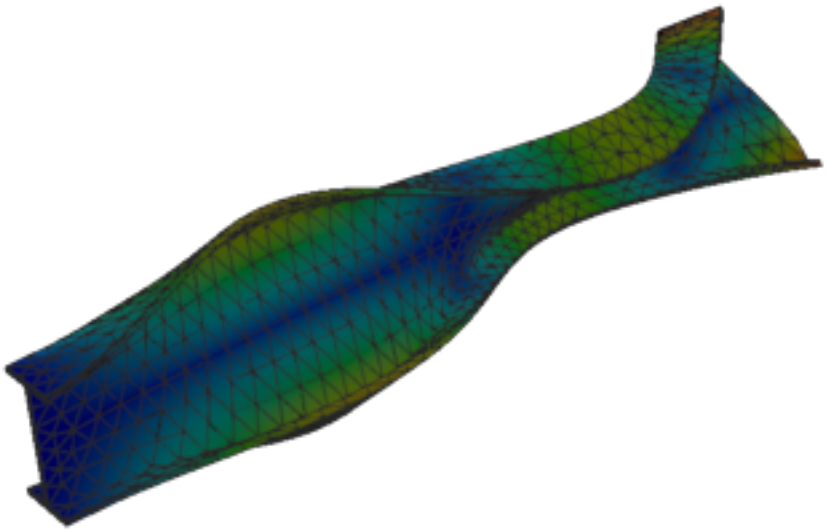


Figure 455: Sesam — mode 11, 254.466 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	254.470	-0.04	2.55635e+06	1.63e-09	6.96e-07	4.16e-05	-4.04e-05	-0.001	-0.006
Calculix	254.685	+0.04	2.56074e+06	9.09e-09	6.08e-05	1.61e-05	-9.53e-05	-0.008	0.004
Code_Aster	254.684	+0.04	2.56073e+06	9.51e-09	6.08e-05	1.60e-05	9.75e-05	0.008	-0.004
Sesam	254.466	-0.04	2.55634e+06	1.66e-09	6.95e-07	4.17e-05	-4.07e-05	-0.001	-0.006

Solid, 2nd order, HEX

Mode 1

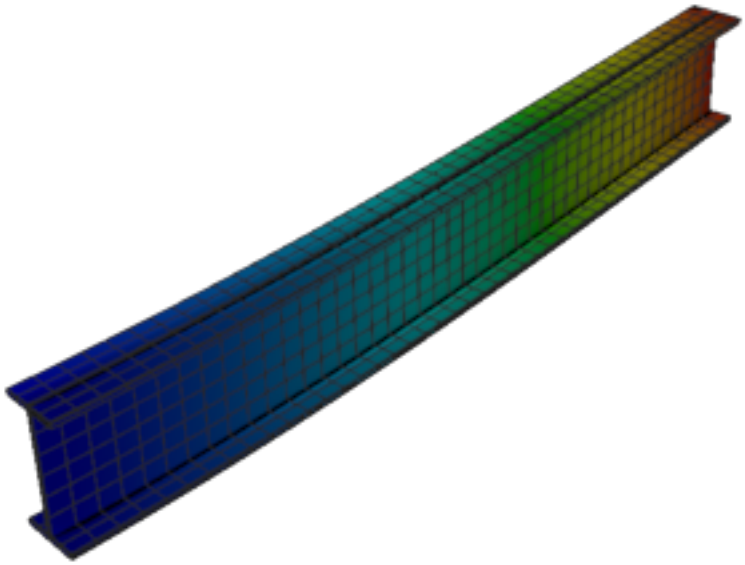


Figure 456: Abaqus — mode 1, 12.955 Hz

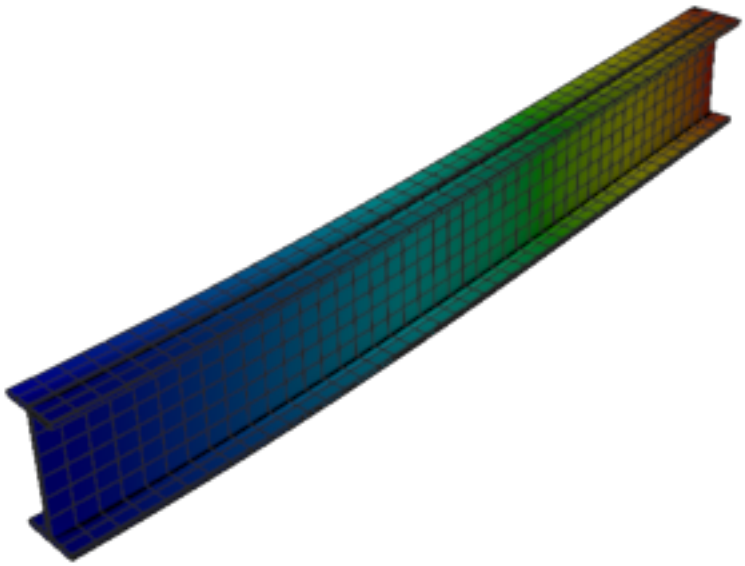


Figure 457: Calculix — mode 1, 12.955 Hz

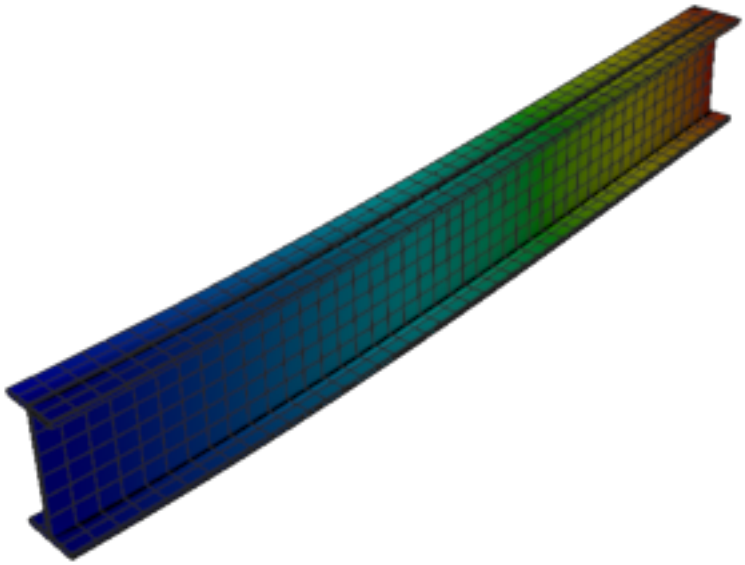


Figure 458: Code_Aster — mode 1, 12.955 Hz

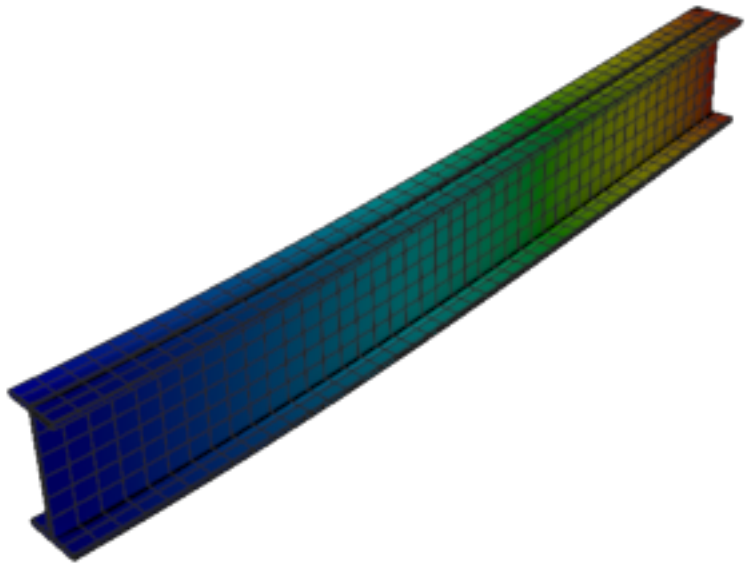


Figure 459: Sesam — mode 1, 12.954 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.955	1.39e-03	6625.8	3.08e-24	116.50	5.64e-20	1.76e-12	10.794	2.37e-10
Calculix	12.955	1.93e-03	6625.81	2.50e-24	116.50	6.61e-20	-1.58e-12	10.794	-2.57e-10
Code_Aster	12.955	1.93e-03	6625.81	1.58e-24	116.50	2.27e-21	-1.26e-12	10.794	4.77e-11
Sesam	12.954	-0.01	6624.86	4.71e-24	116.51	4.62e-21	2.17e-12	10.794	-6.80e-11

Mode 2

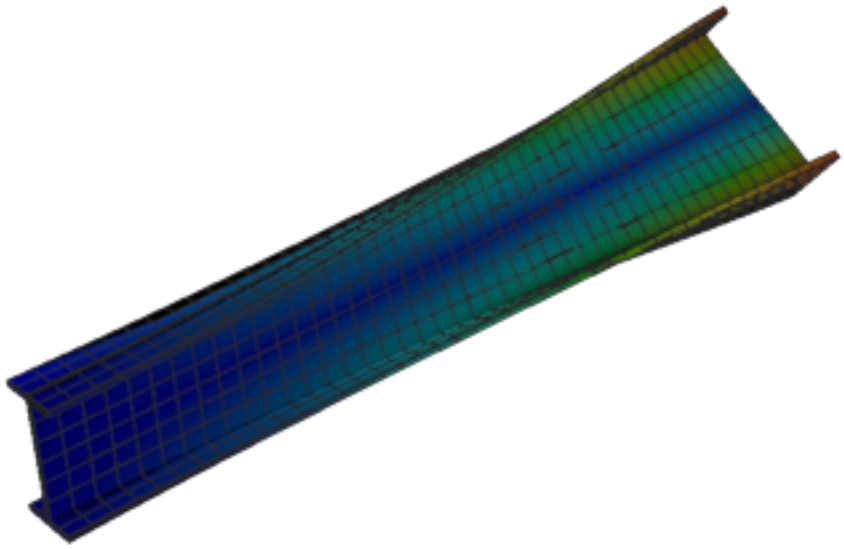


Figure 460: Abaqus — mode 2, 20.150 Hz

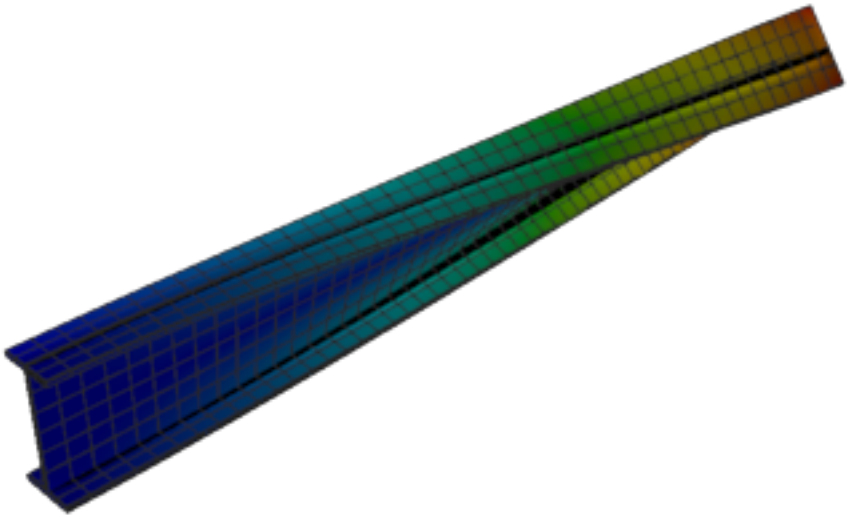


Figure 461: Calculix — mode 2, 20.150 Hz

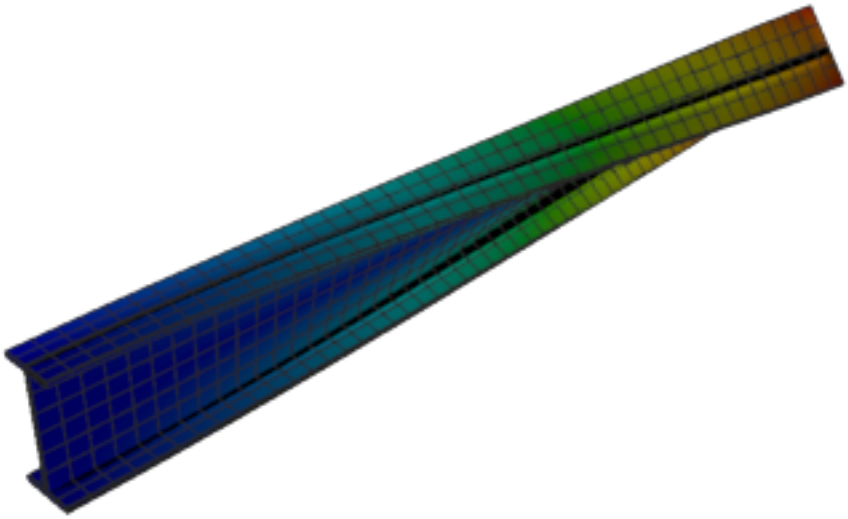


Figure 462: Code_Aster — mode 2, 20.150 Hz

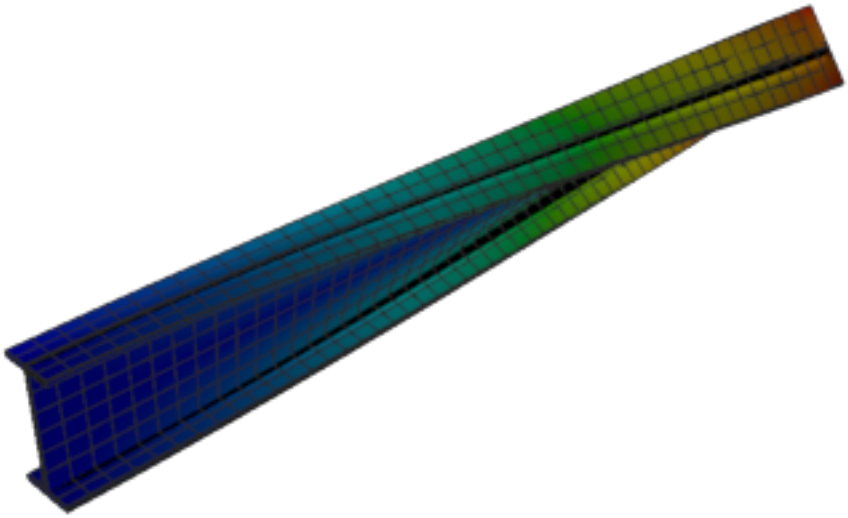


Figure 463: Sesam — mode 2, 20.044 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.150	+0.13	16030	2.26e-25	4.71e-16	2.32e-18	-4.76e-13	2.17e-08	1.52e-09
Calculix	20.150	+0.13	16029.6	7.08e-26	6.44e-16	1.72e-18	-2.66e-13	2.54e-08	1.31e-09
Code_Aster	20.150	+0.13	16029.6	1.72e-24	2.30e-17	2.99e-21	-1.31e-12	4.79e-09	-5.46e-11
Sesam	20.044	-0.39	15861.7	2.58e-24	5.48e-15	8.69e-19	1.61e-12	-7.40e-08	9.32e-10

Mode 3

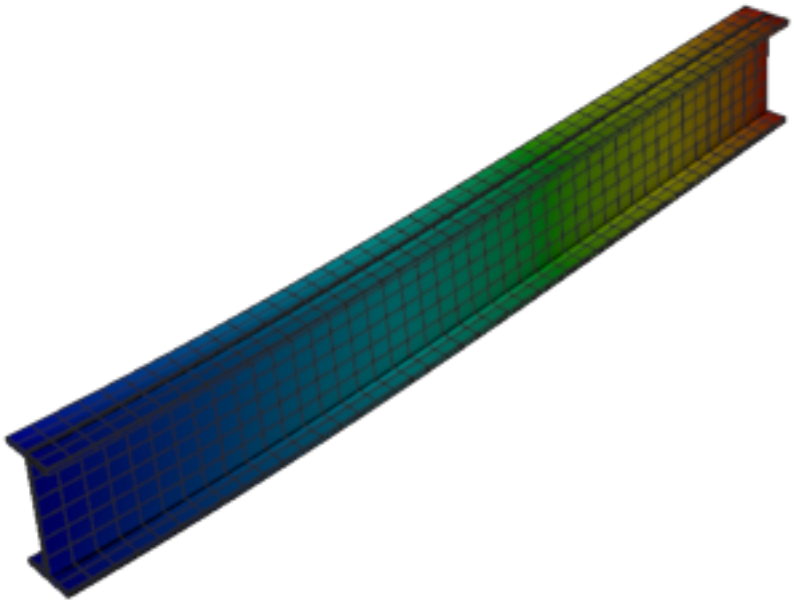


Figure 464: Abaqus — mode 3, 50.541 Hz

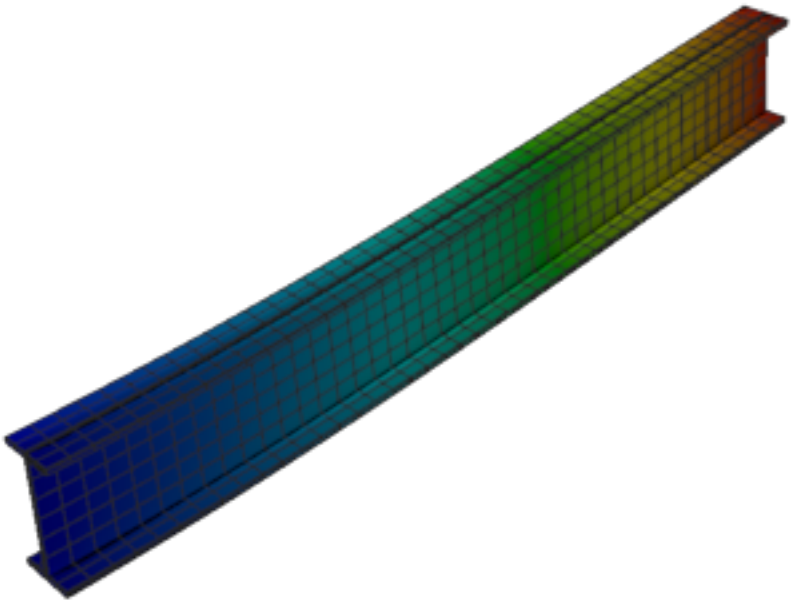


Figure 465: Calculix — mode 3, 50.541 Hz

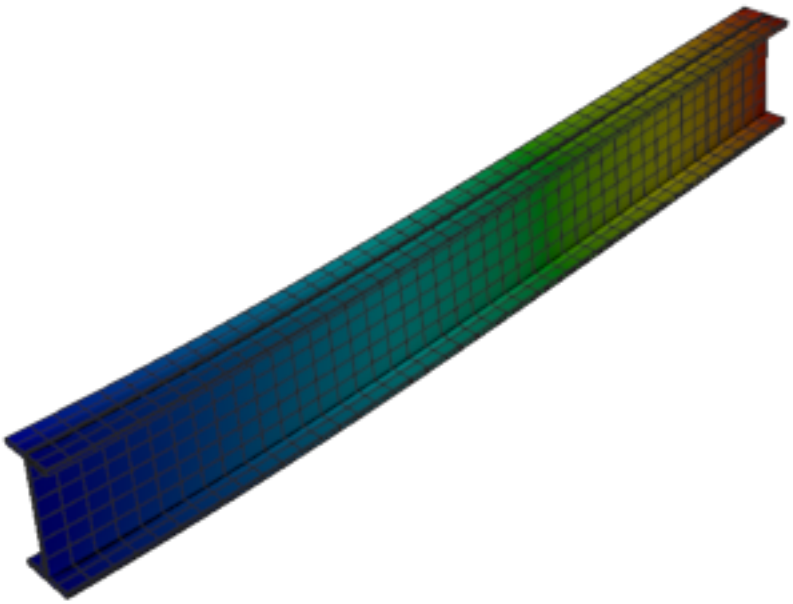


Figure 466: Code_Aster — mode 3, 50.541 Hz

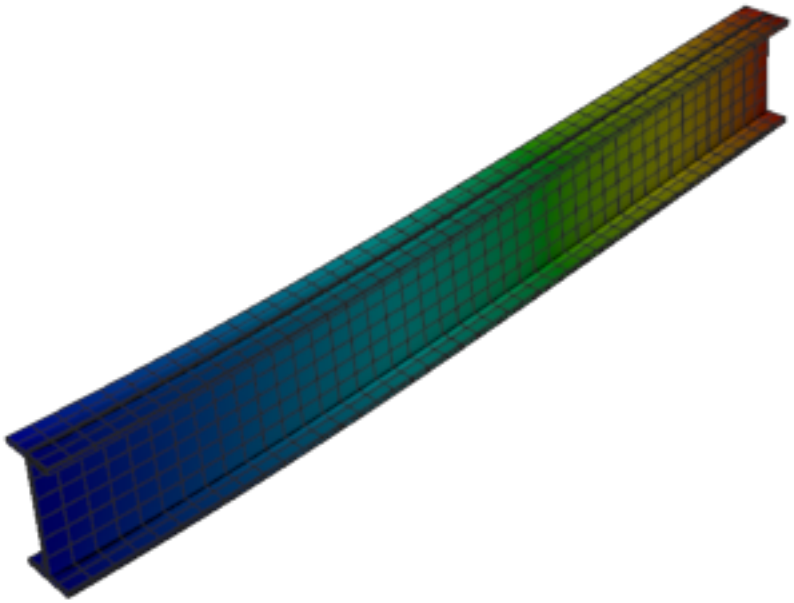


Figure 467: Sesam — mode 3, 50.534 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.541	3.14e-03	100844	1.71e-23	8.62e-20	118.36	-4.13e-12	-2.94e-10	10.879
Calculix	50.541	3.61e-03	100844	1.81e-24	1.19e-19	118.36	-1.35e-12	3.44e-10	10.879
Code_Aster	50.541	3.62e-03	100844	3.72e-24	1.03e-20	118.36	-1.93e-12	-1.01e-10	10.879
Sesam	50.534	-0.01	100816	6.44e-24	9.81e-21	118.37	2.54e-12	9.90e-11	10.880

Mode 4

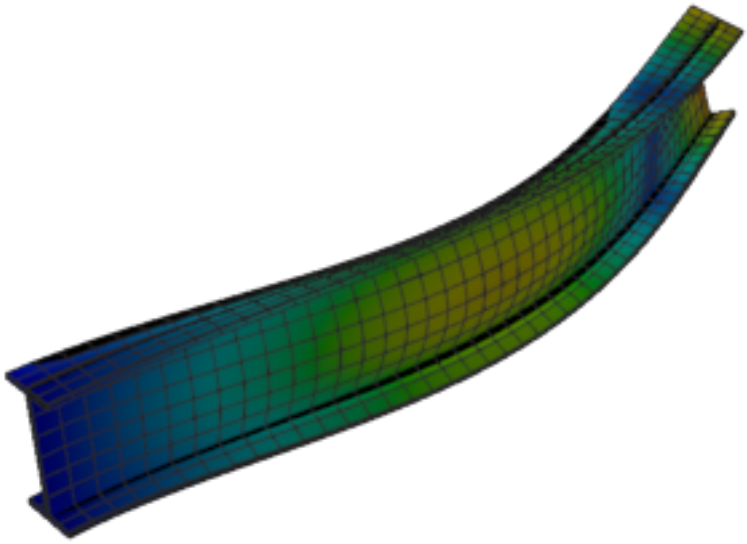


Figure 468: Abaqus — mode 4, 76.890 Hz

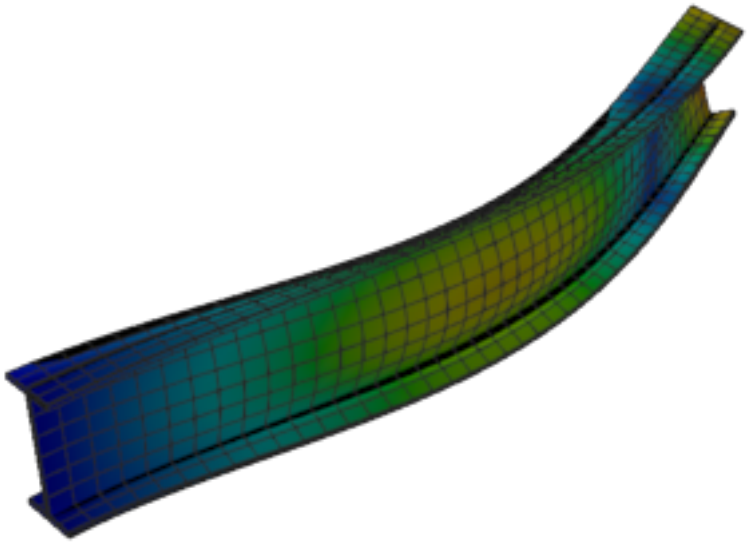


Figure 469: Calculix — mode 4, 76.890 Hz

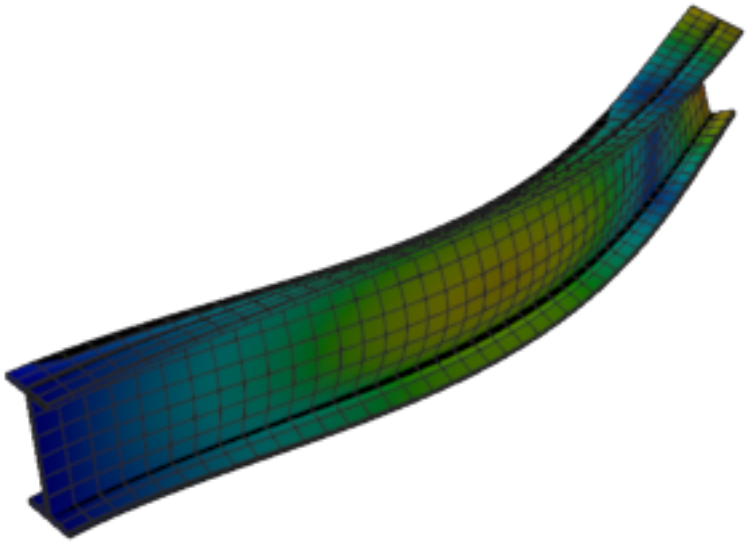


Figure 470: Code_Aster — mode 4, 76.890 Hz

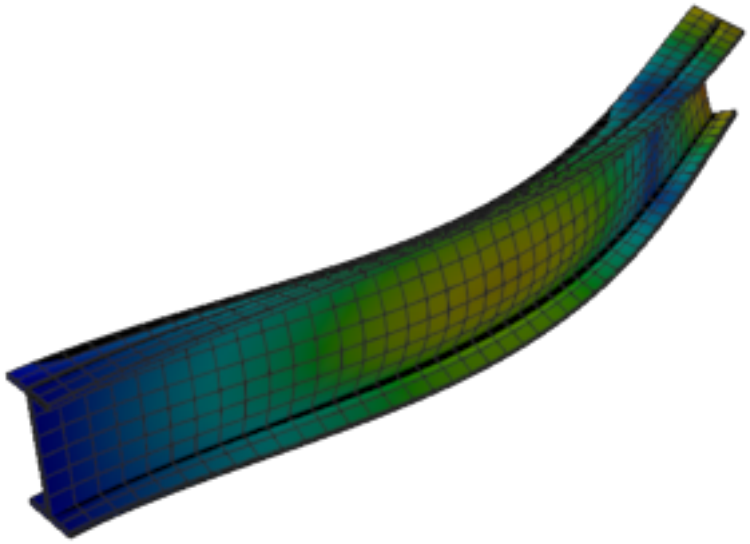


Figure 471: Sesam — mode 4, 76.747 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	76.890	+0.05	233401	9.37e-28	35.91	1.14e-20	-3.06e-14	-5.993	-1.07e-10
Calculix	76.890	+0.05	233401	4.29e-24	35.91	2.88e-20	2.07e-12	-5.993	1.70e-10
Code_Aster	76.890	+0.05	233401	1.25e-25	35.91	7.59e-21	3.54e-13	-5.993	-8.71e-11
Sesam	76.747	-0.14	232530	2.14e-25	35.89	4.07e-21	-4.63e-13	-5.991	6.38e-11

Mode 5

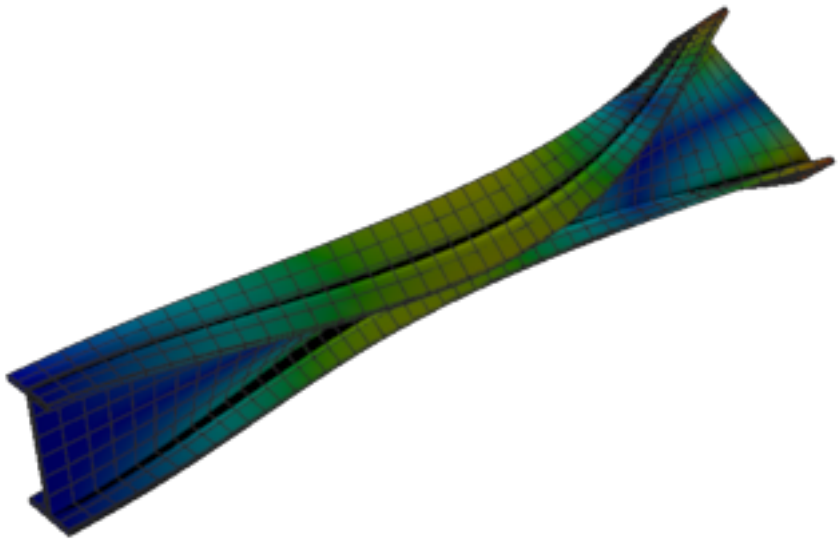


Figure 472: Abaqus — mode 5, 97.594 Hz

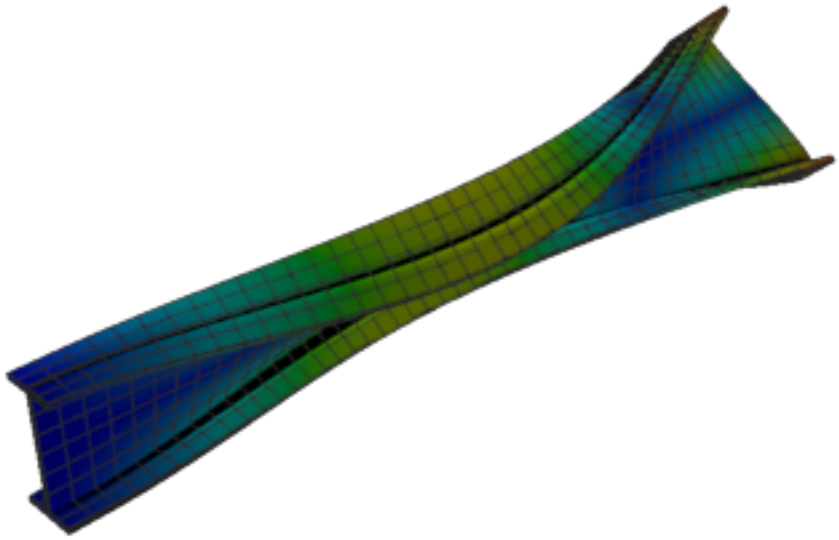


Figure 473: Calculix — mode 5, 97.594 Hz

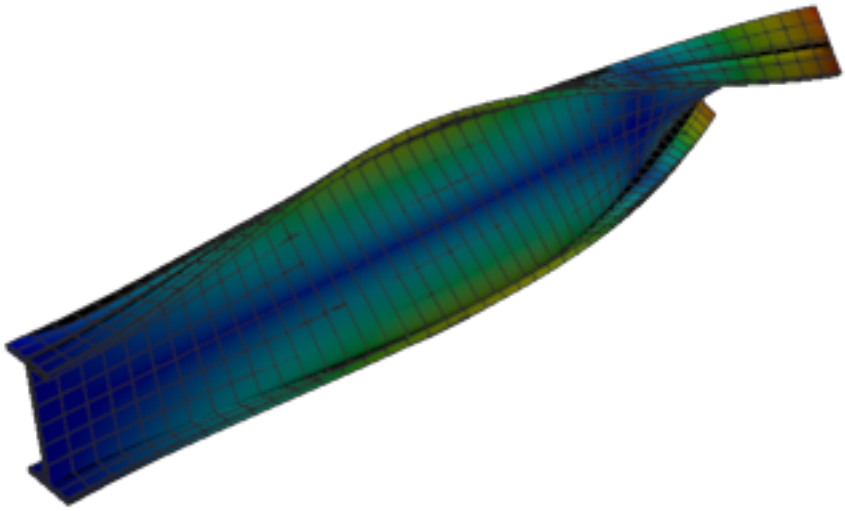


Figure 474: Code_Aster — mode 5, 97.594 Hz

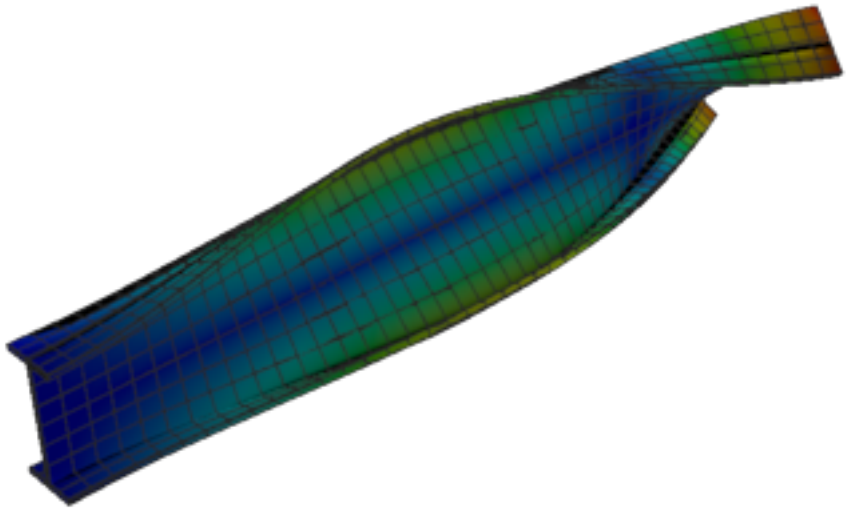


Figure 475: Sesam — mode 5, 97.472 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.594	+0.03	376015	1.21e-24	3.12e-19	1.62e-21	1.10e-12	-5.59e-10	-4.02e-11
Calculix	97.594	+0.03	376015	2.79e-24	6.84e-19	2.66e-21	1.67e-12	8.27e-10	-5.15e-11
Code_Aster	97.594	+0.03	376015	2.71e-27	7.32e-20	4.63e-21	-5.21e-14	-2.71e-10	6.80e-11
Sesam	97.472	-0.09	375078	2.82e-24	3.90e-18	4.03e-21	-1.68e-12	1.97e-09	-6.35e-11

Mode 6

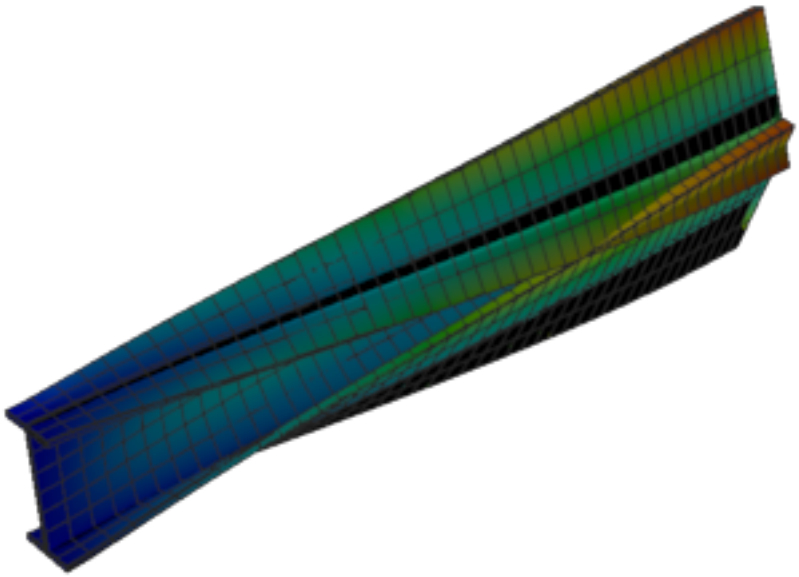


Figure 476: Abaqus — mode 6, 131.365 Hz

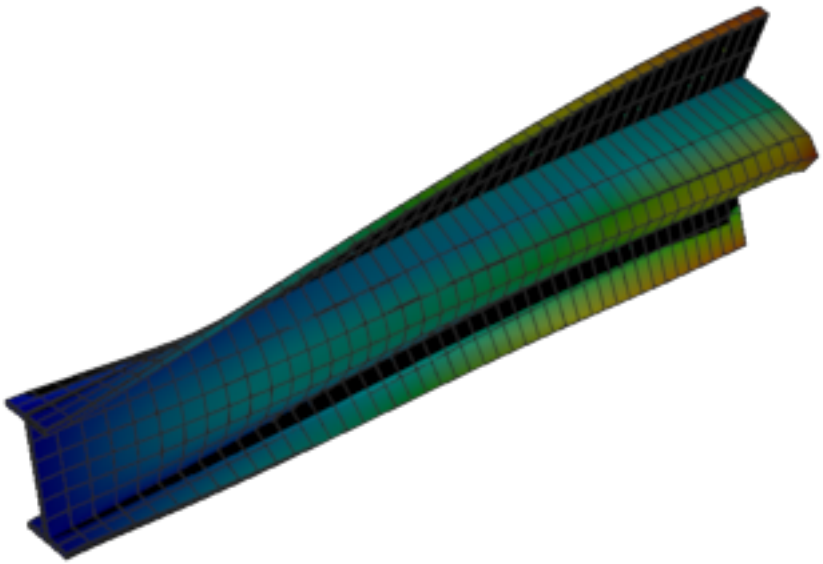


Figure 477: Calculix — mode 6, 131.365 Hz

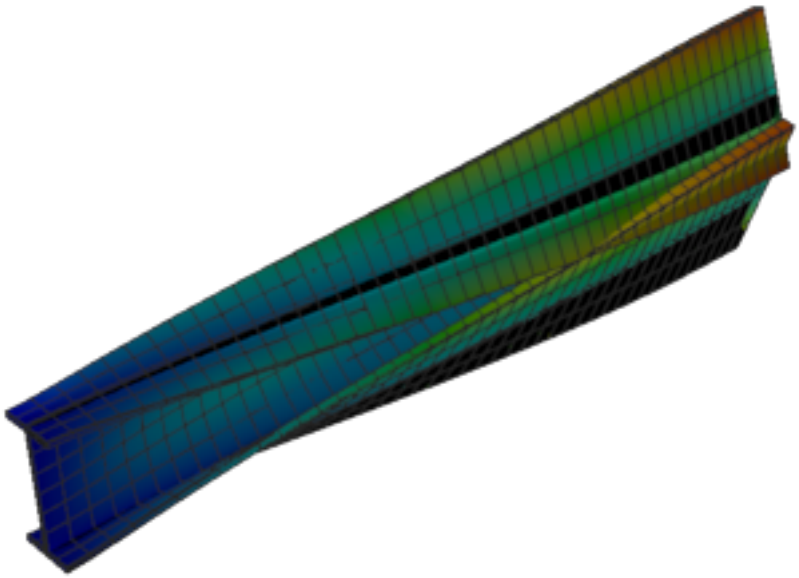


Figure 478: Code_Aster — mode 6, 131.365 Hz

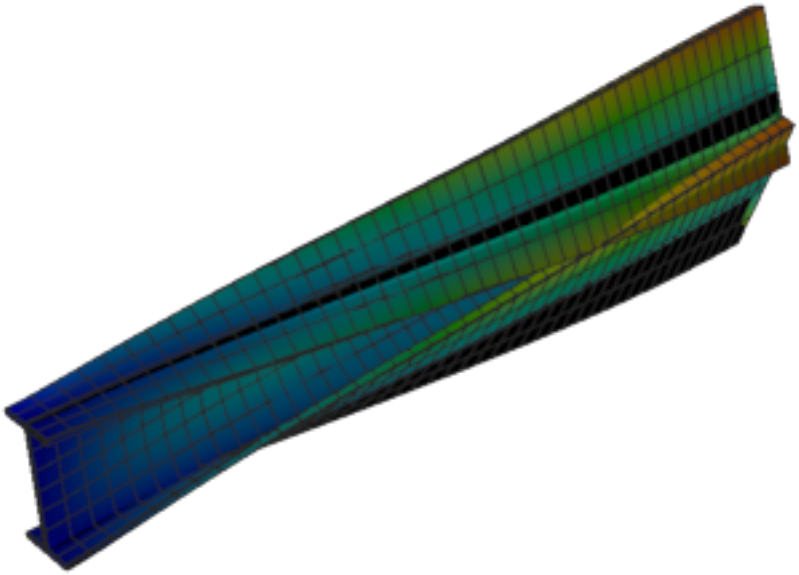


Figure 479: Sesam — mode 6, 130.865 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	131.370	+0.10	681273	2.05e-24	0.01	1.33e-19	1.43e-12	-0.103	-3.65e-10
Calculix	131.365	+0.09	681274	8.04e-24	0.01	6.41e-19	2.84e-12	0.103	8.01e-10
Code_Aster	131.365	+0.09	681273	1.58e-23	0.01	9.32e-21	-3.98e-12	-0.103	-9.65e-11
Sesam	130.865	-0.29	676090	4.18e-25	0.01	8.04e-22	6.46e-13	-0.096	2.84e-11

Mode 7

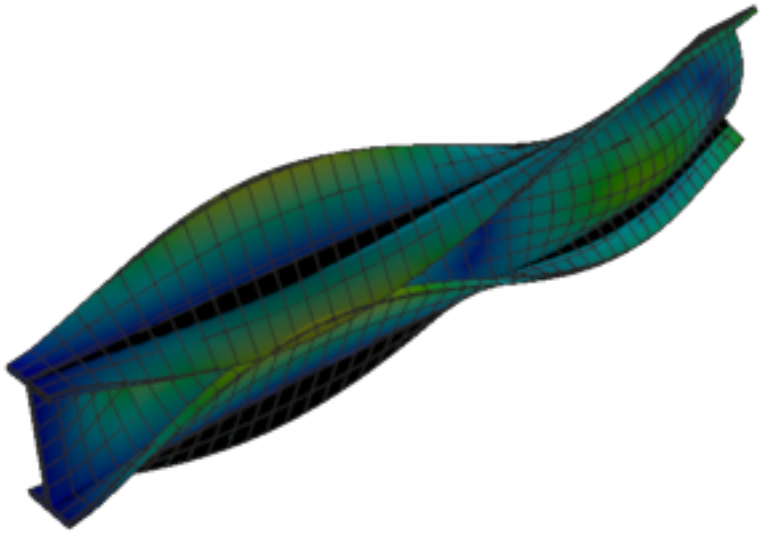


Figure 480: Abaqus — mode 7, 149.958 Hz

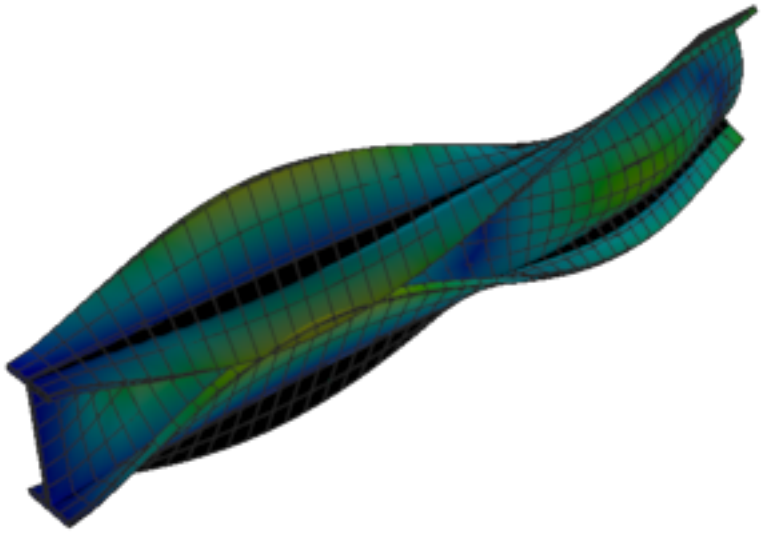


Figure 481: Calculix — mode 7, 149.958 Hz

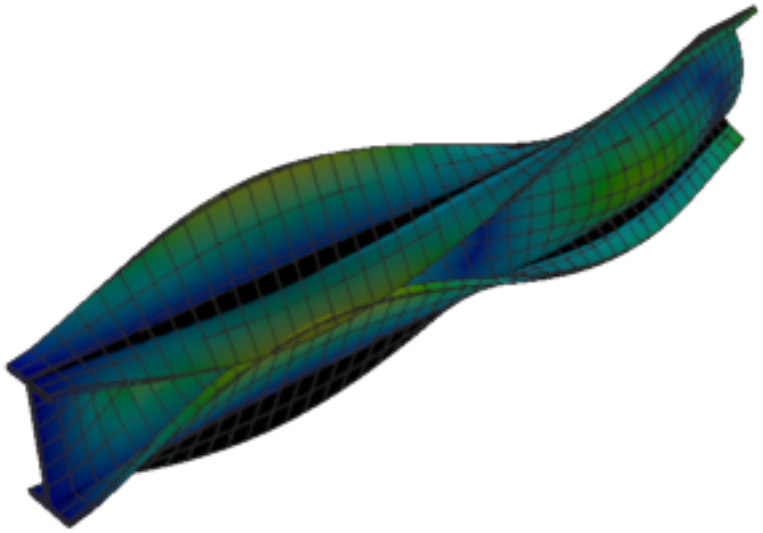


Figure 482: Code_Aster — mode 7, 149.958 Hz

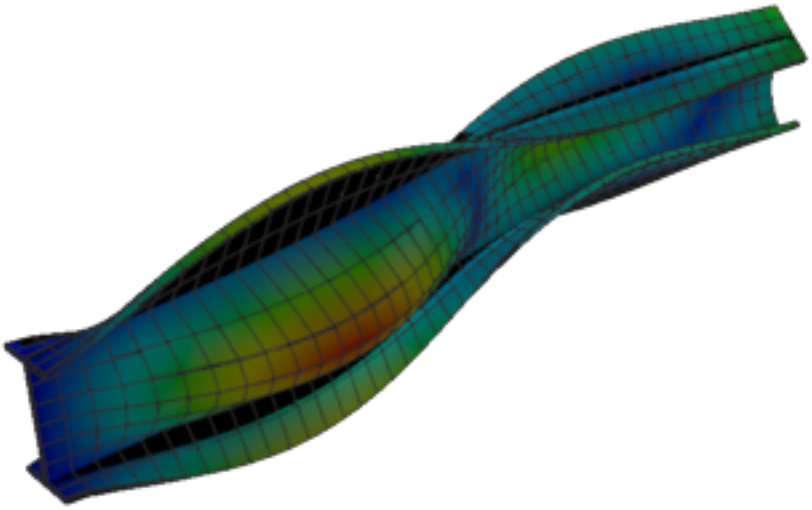


Figure 483: Sesam — mode 7, 148.496 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	149.960	+0.25	887766	2.76e-24	3.31	8.83e-21	1.66e-12	1.818	9.40e-11
Calculix	149.958	+0.24	887766	1.74e-27	3.31	4.35e-20	-4.17e-14	1.818	2.08e-10
Code_Aster	149.958	+0.24	887766	1.23e-24	3.31	2.03e-22	1.11e-12	1.818	1.43e-11
Sesam	148.496	-0.73	870539	8.20e-26	3.38	3.30e-22	2.86e-13	-1.839	1.82e-11

Mode 8

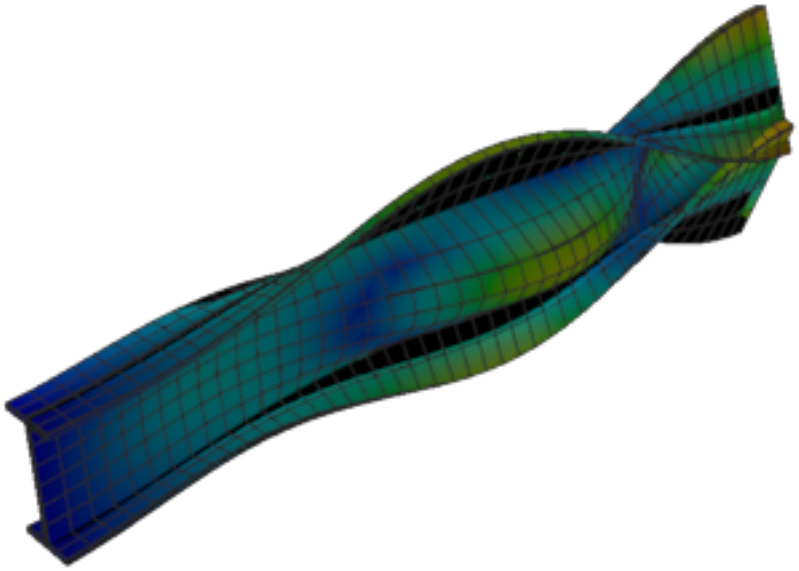


Figure 484: Abaqus — mode 8, 173.925 Hz

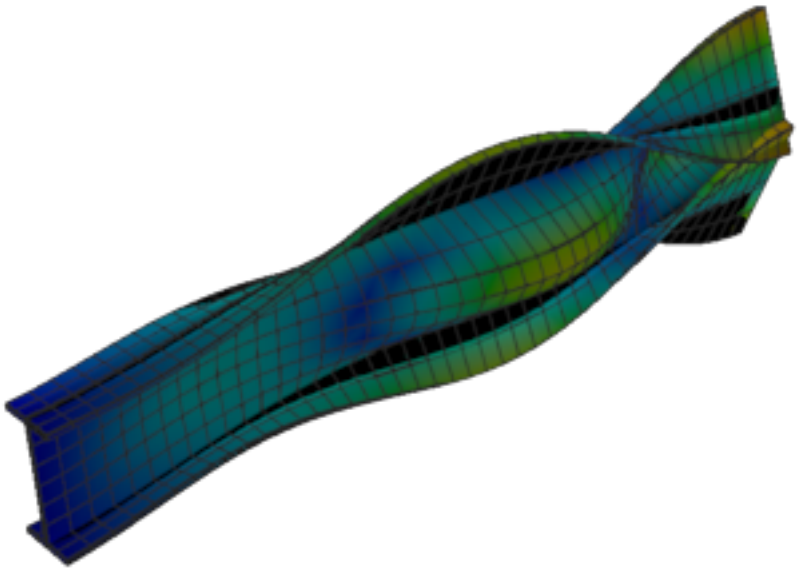


Figure 485: Calculix — mode 8, 173.925 Hz

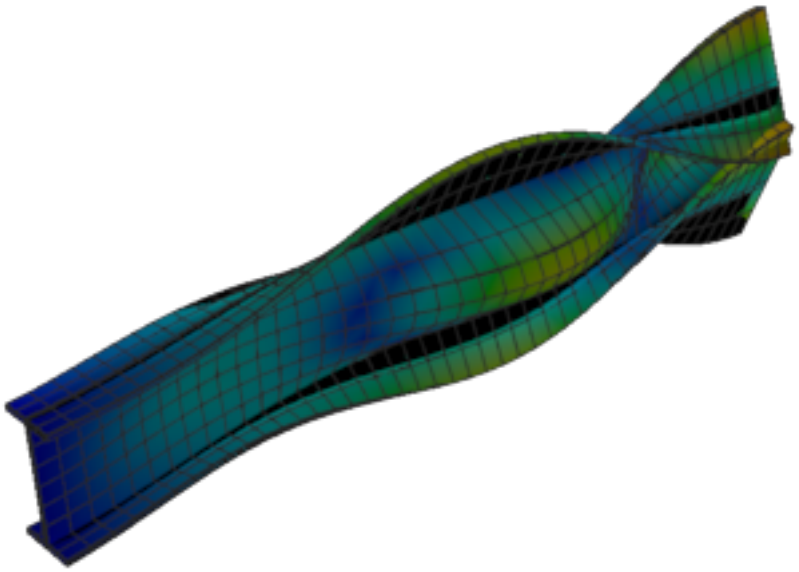


Figure 486: Code_Aster — mode 8, 173.925 Hz

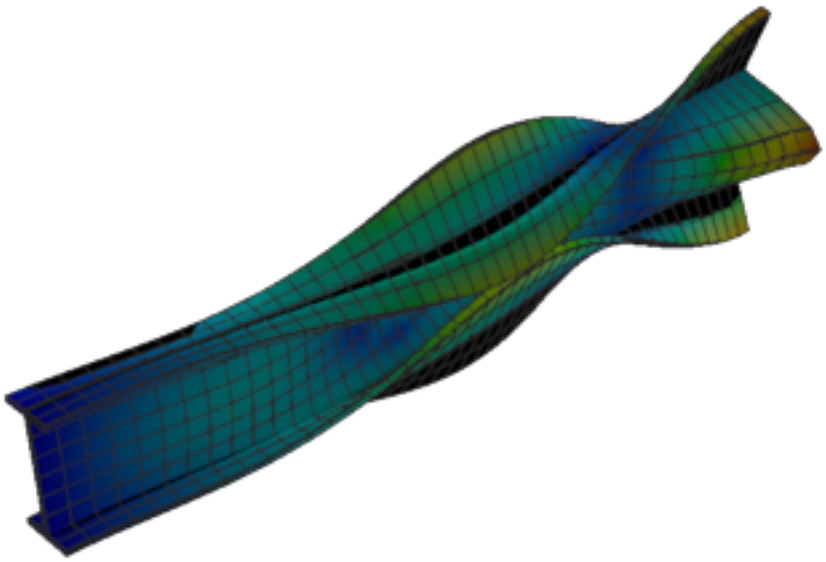


Figure 487: Sesam — mode 8, 172.243 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	173.920	+0.24	1.19422e+06	4.36e-26	5.42	1.76e-20	2.09e-13	2.328	-1.33e-10
Calculix	173.925	+0.24	1.19422e+06	4.25e-25	5.42	4.87e-20	-6.52e-13	2.328	-2.21e-10
Code_Aster	173.925	+0.24	1.19422e+06	2.37e-25	5.42	2.61e-23	-4.87e-13	2.328	-5.11e-12
Sesam	172.243	-0.73	1.17124e+06	1.58e-26	5.31	8.78e-22	-1.26e-13	-2.305	-2.96e-11

Mode 9

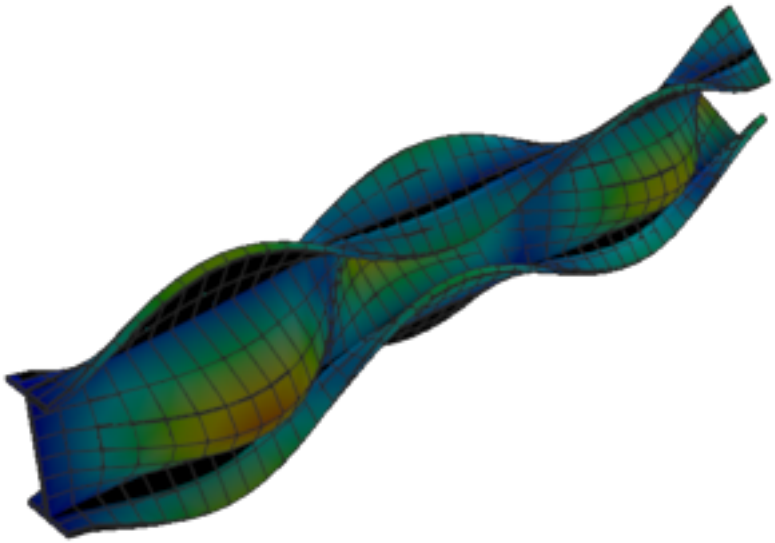


Figure 488: Abaqus — mode 9, 205.292 Hz

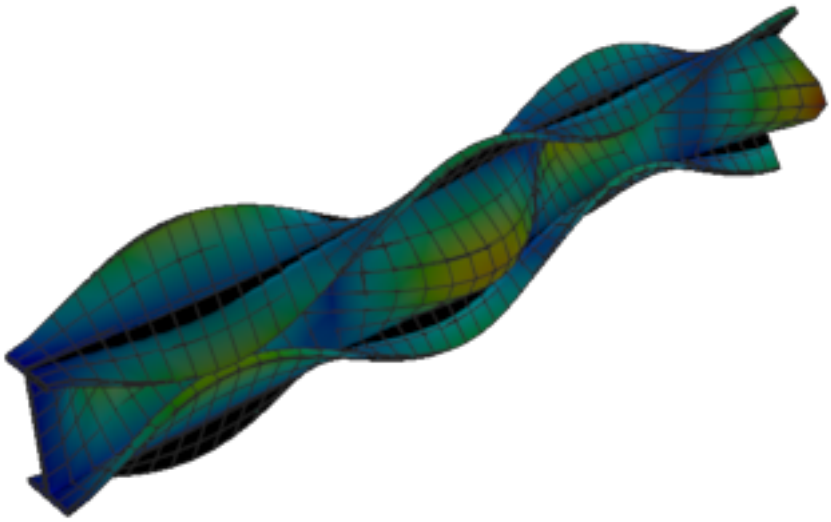


Figure 489: Calculix — mode 9, 205.292 Hz

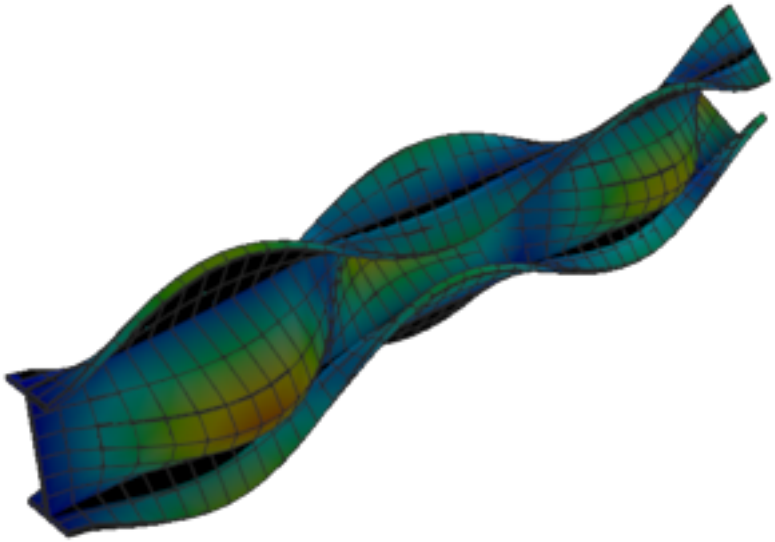


Figure 490: Code_Aster — mode 9, 205.292 Hz

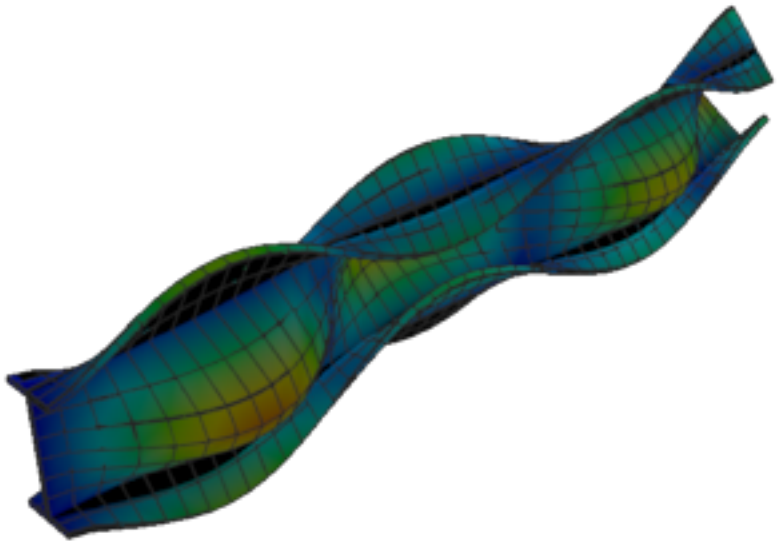


Figure 491: Sesam — mode 9, 201.904 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	205.290	+0.41	1.66381e+06	4.78e-27	0.52	2.91e-22	6.91e-14	-0.719	1.71e-11
Calculix	205.292	+0.41	1.6638e+06	5.77e-25	0.52	1.64e-21	-7.59e-13	0.720	4.04e-11
Code_Aster	205.292	+0.41	1.66381e+06	7.17e-24	0.52	1.15e-22	2.68e-12	-0.719	-1.07e-11
Sesam	201.904	-1.24	1.60935e+06	5.31e-25	0.56	3.60e-22	-7.28e-13	-0.746	1.90e-11

Mode 10

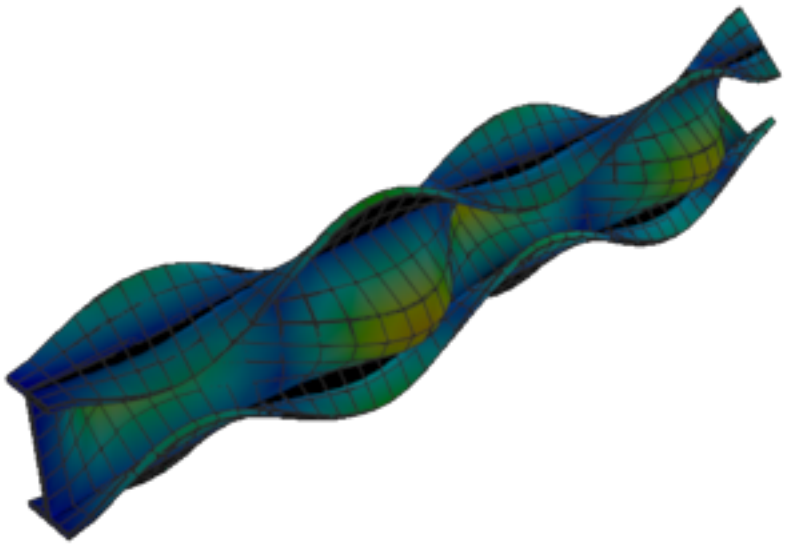


Figure 492: Abaqus — mode 10, 251.347 Hz

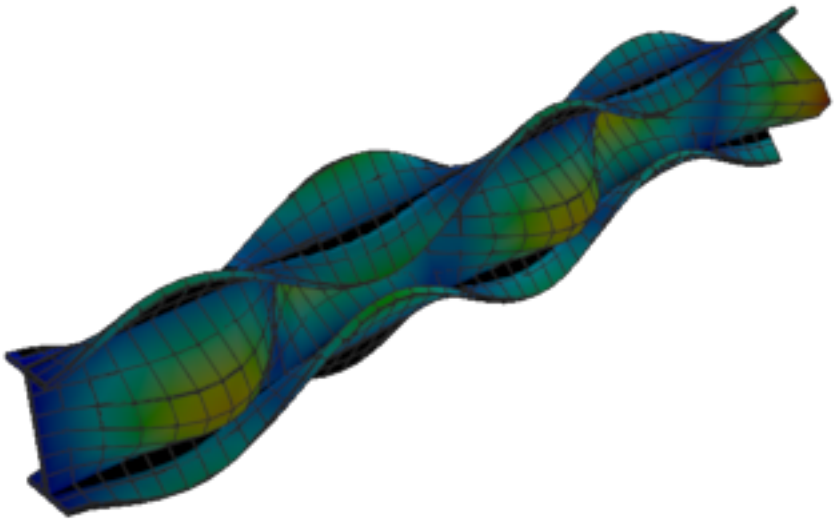


Figure 493: Calculix — mode 10, 251.347 Hz

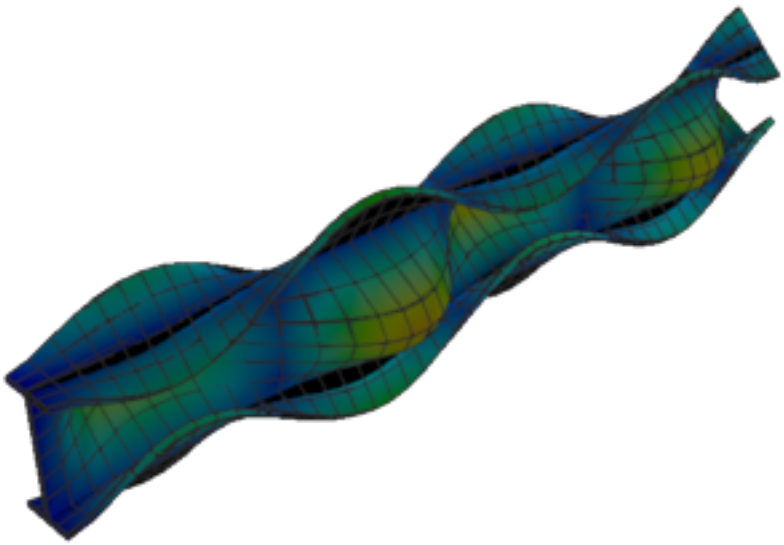


Figure 494: Code_Aster — mode 10, 251.347 Hz

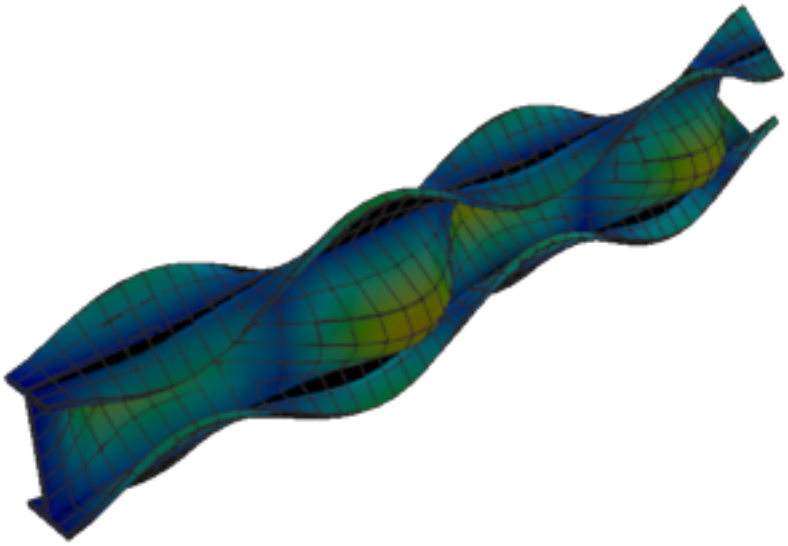


Figure 495: Sesam — mode 10, 246.353 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	251.350	+0.50	2.49407e+06	1.08e-21	1.60	3.13e-19	3.29e-11	1.267	-5.59e-10
Calculix	251.347	+0.50	2.49406e+06	1.19e-25	1.61	7.95e-19	-3.45e-13	-1.267	8.92e-10
Code_Aster	251.347	+0.50	2.49407e+06	7.51e-25	1.60	1.30e-18	-8.67e-13	1.267	-1.14e-09
Sesam	246.353	-1.50	2.39594e+06	1.32e-18	1.47	7.64e-19	-1.15e-09	1.211	8.74e-10

Mode 11

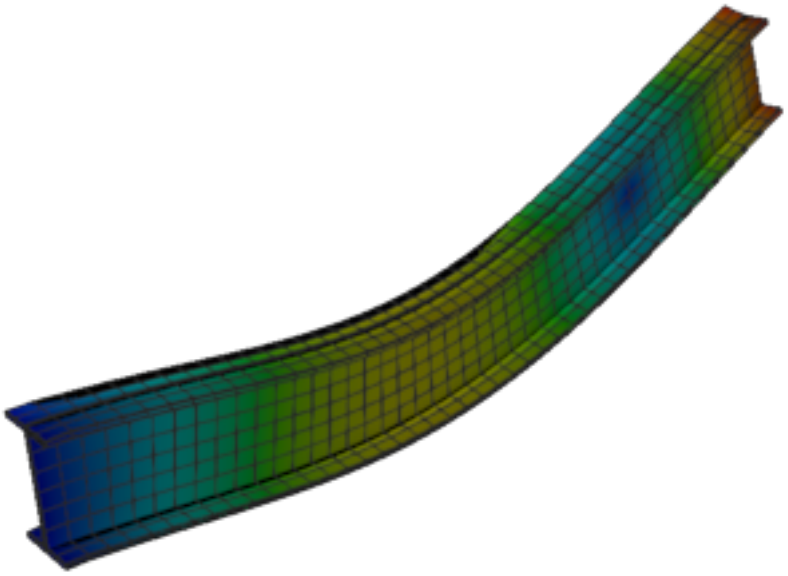


Figure 496: Abaqus — mode 11, 252.917 Hz

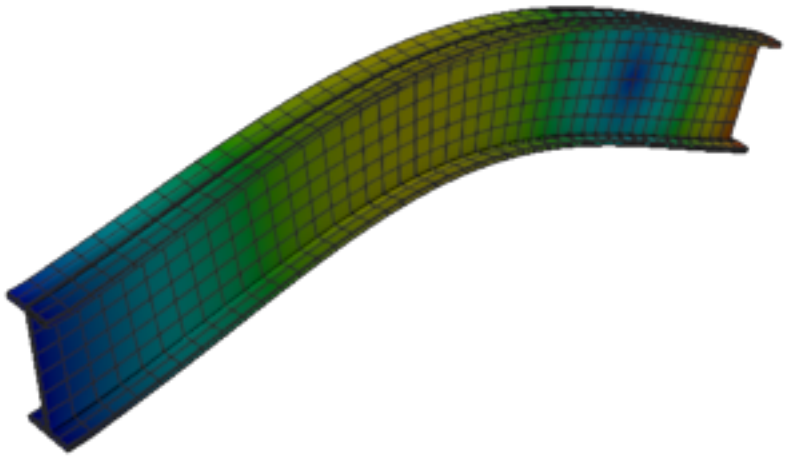


Figure 497: Calculix — mode 11, 252.917 Hz

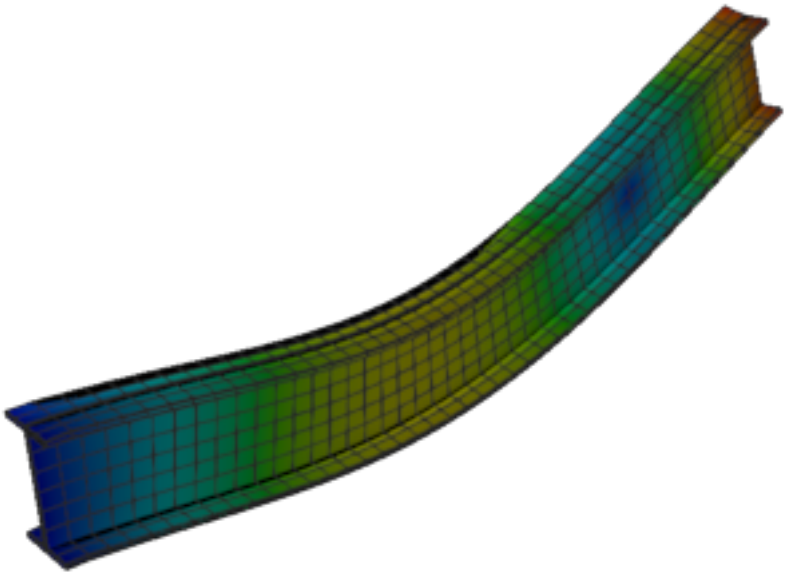


Figure 498: Code_Aster — mode 11, 252.917 Hz

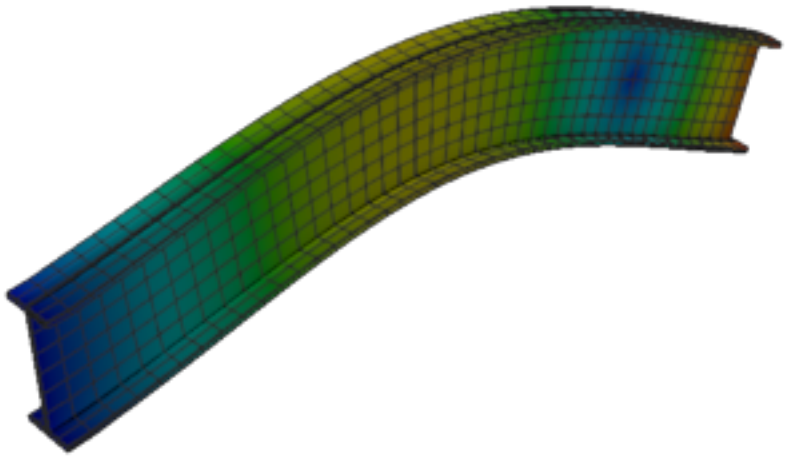


Figure 499: Sesam — mode 11, 252.742 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.920	+0.02	2.52532e+06	4.22e-22	4.70e-20	40.25	2.05e-11	2.17e-10	-6.344
Calculix	252.917	+0.02	2.52532e+06	2.59e-23	3.32e-20	40.25	-5.09e-12	1.82e-10	6.345
Code_Aster	252.917	+0.02	2.52532e+06	9.19e-24	5.59e-20	40.25	-3.03e-12	-2.37e-10	-6.344
Sesam	252.742	-0.05	2.52182e+06	2.26e-15	2.81e-16	40.28	-4.75e-08	1.68e-08	6.346

Solid, 2nd order, HEX, reduced integration

Mode 1

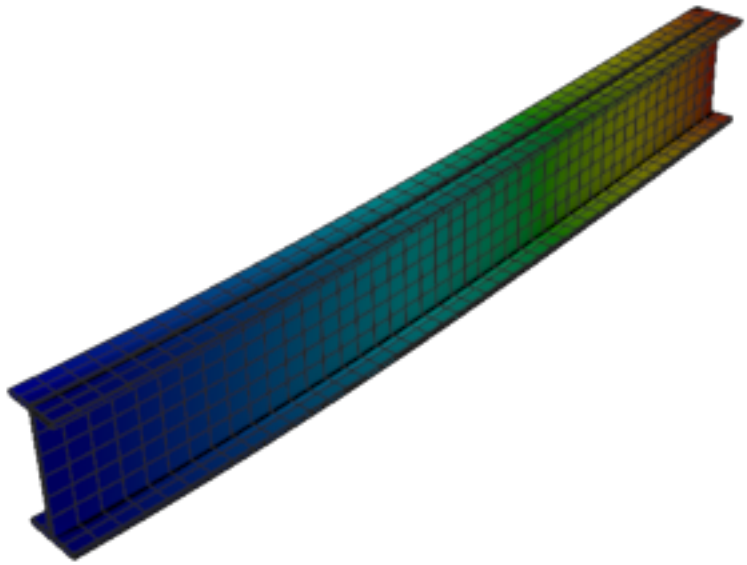


Figure 500: Abaqus — mode 1, 12.943 Hz

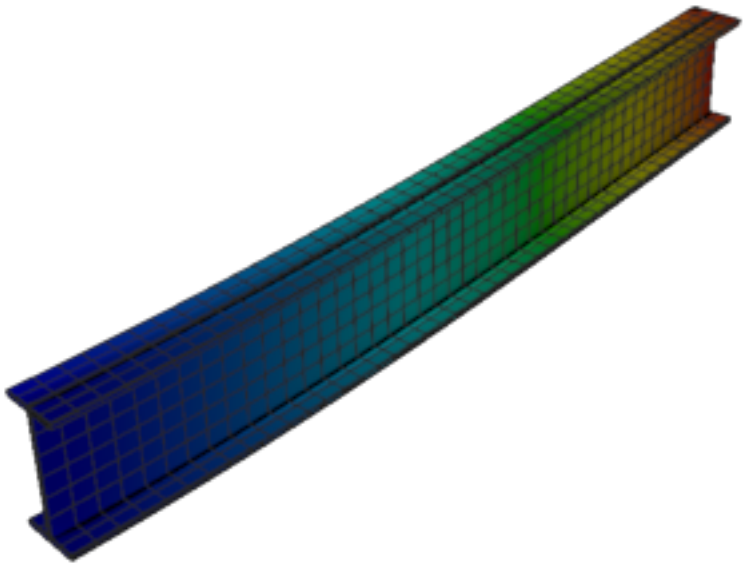


Figure 501: Calculix — mode 1, 12.943 Hz

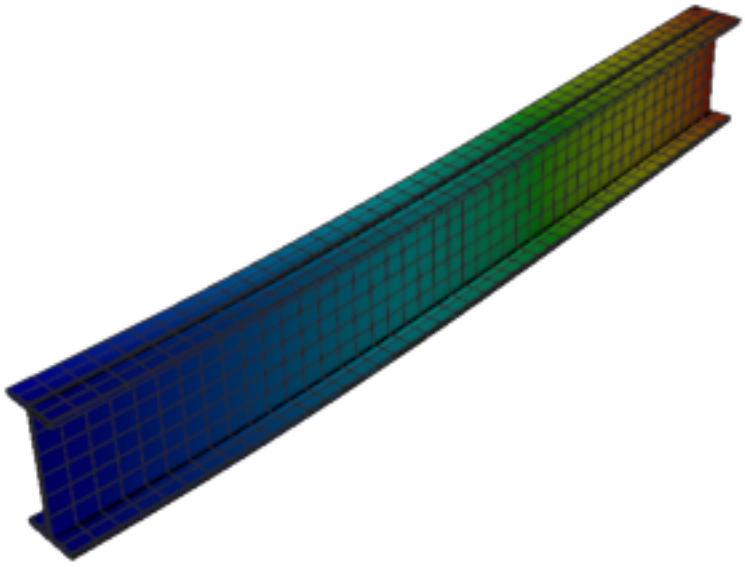


Figure 502: Code_Aster — mode 1, 12.943 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.943	1.29e-03	6613.2	5.11e-26	116.56	2.93e-20	-2.26e-13	10.796	1.71e-10
Calculix	12.943	-6.37e-04	6613.22	1.66e-24	116.56	6.19e-20	1.29e-12	10.796	2.49e-10
Code_Aster	12.943	-6.57e-04	6613.22	1.19e-25	116.56	5.87e-21	-3.45e-13	10.796	-7.66e-11

Mode 2

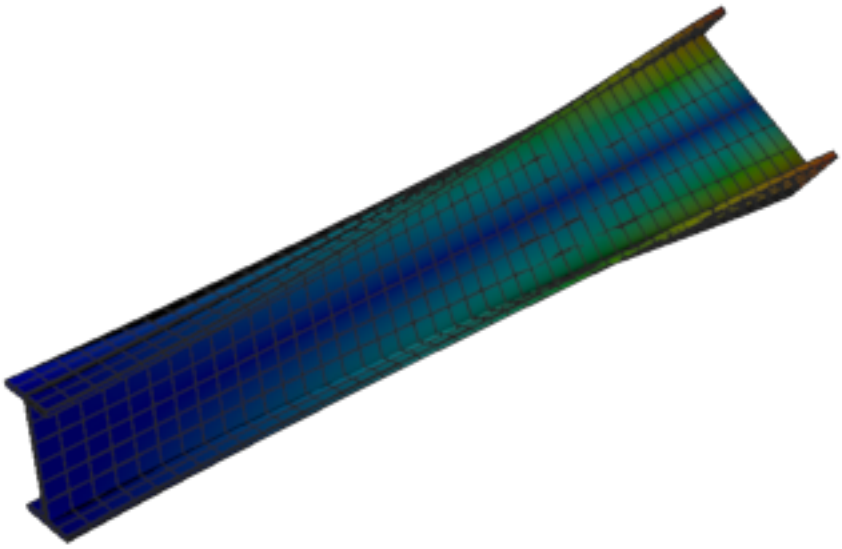


Figure 503: Abaqus — mode 2, 20.029 Hz

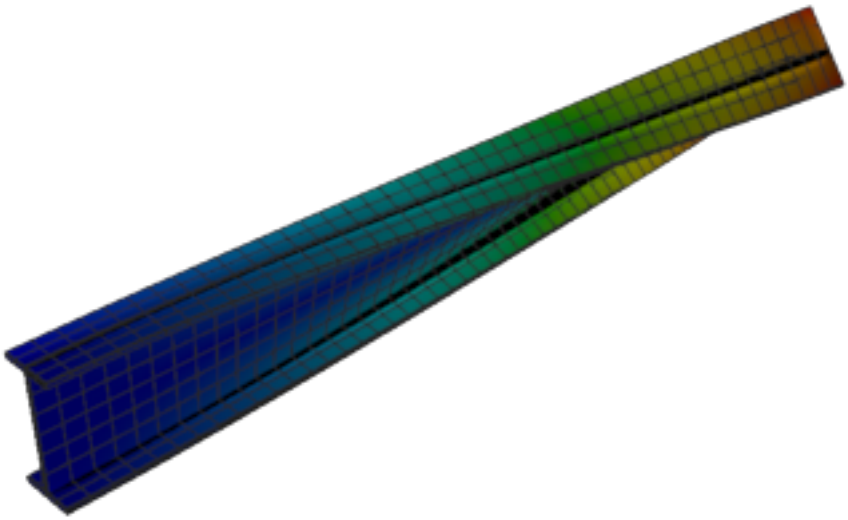


Figure 504: Calculix — mode 2, 20.029 Hz

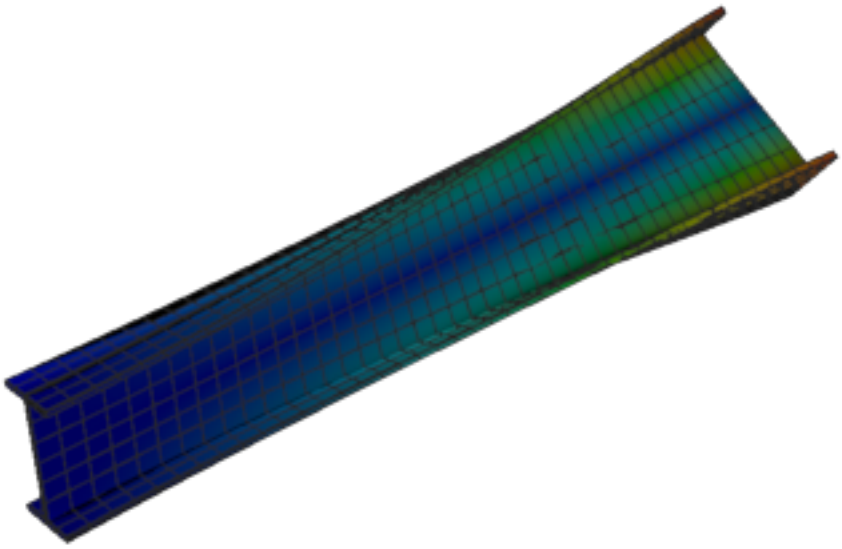


Figure 505: Code_Aster — mode 2, 20.029 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.029	8.03e-04	15837	2.00e-26	8.47e-16	7.79e-20	-1.41e-13	2.91e-08	-2.79e-10
Calculix	20.029	-3.95e-04	15836.8	1.31e-25	7.93e-16	7.72e-19	-3.62e-13	-2.82e-08	8.79e-10
Code_Aster	20.029	-4.08e-04	15836.8	5.43e-25	1.48e-17	1.39e-19	-7.37e-13	3.85e-09	3.73e-10

Mode 3

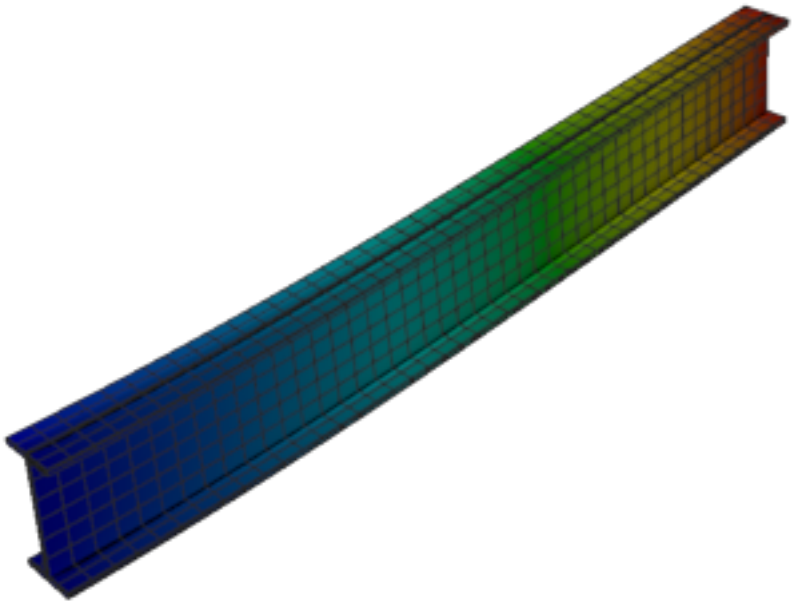


Figure 506: Abaqus — mode 3, 50.498 Hz

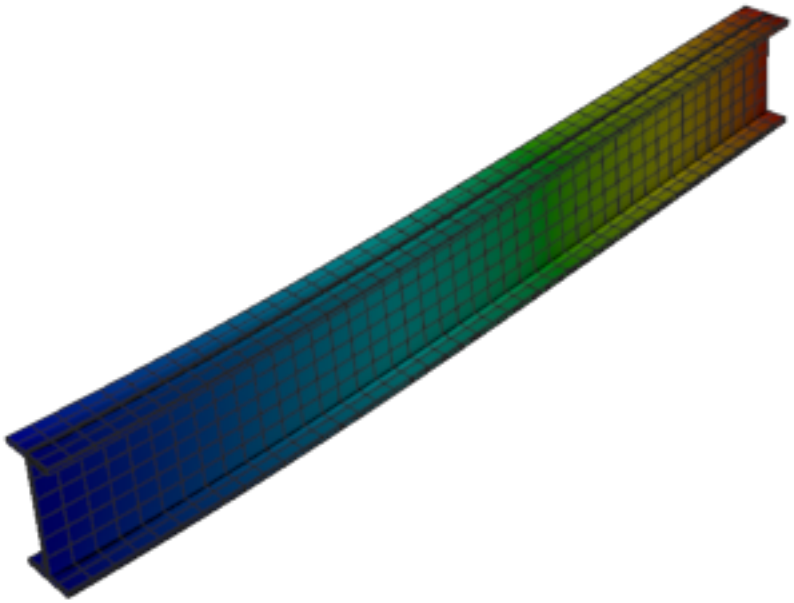


Figure 507: Calculix — mode 3, 50.498 Hz

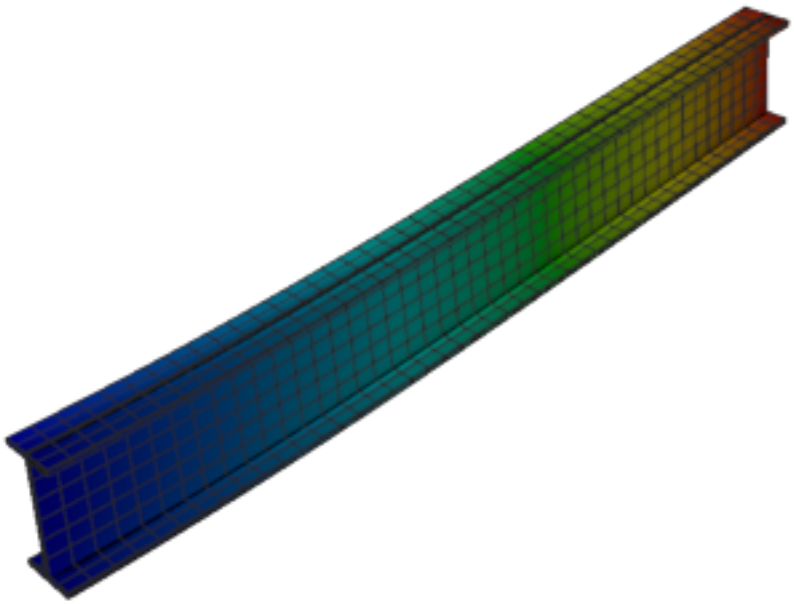


Figure 508: Code_Aster — mode 3, 50.498 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.498	3.96e-04	100671	1.64e-24	3.78e-20	118.41	-1.28e-12	-1.94e-10	10.882
Calculix	50.498	-1.98e-04	100671	4.96e-24	8.00e-20	118.41	-2.23e-12	-2.83e-10	10.882
Code_Aster	50.498	-1.98e-04	100671	2.17e-24	1.42e-21	118.41	-1.47e-12	3.77e-11	10.882

Mode 4

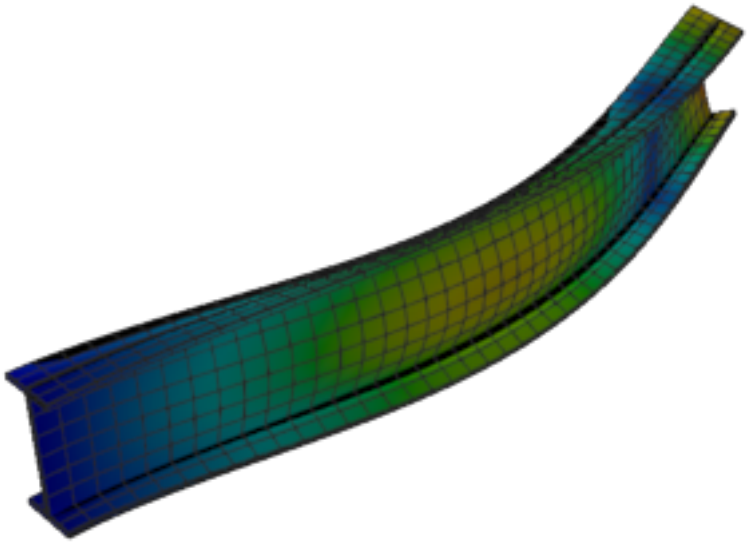


Figure 509: Abaqus — mode 4, 76.687 Hz

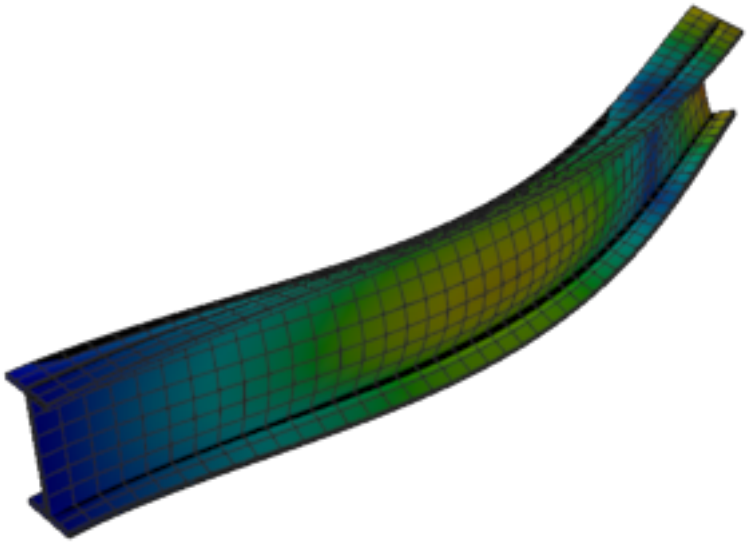


Figure 510: Calculix — mode 4, 76.688 Hz

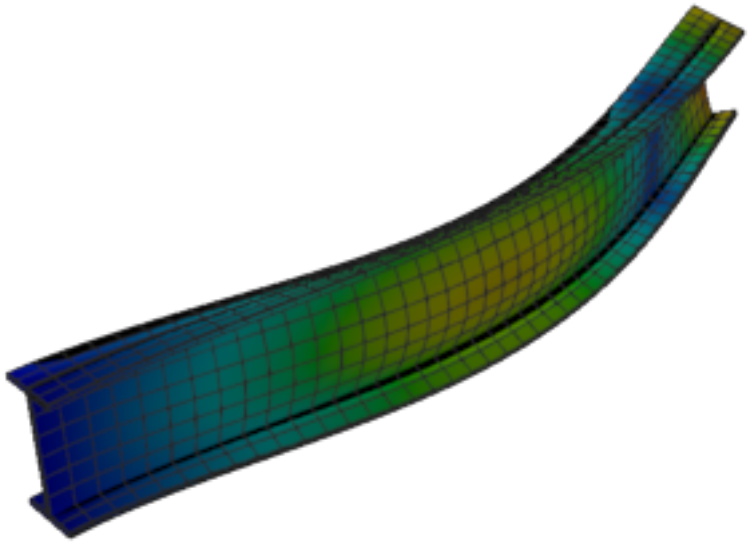


Figure 511: Code_Aster — mode 4, 76.687 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	76.687	-4.40e-04	232171	2.11e-25	35.90	8.07e-22	-4.60e-13	-5.992	-2.84e-11
Calculix	76.688	2.77e-04	232172	1.40e-24	35.90	3.37e-21	1.18e-12	-5.992	-5.81e-11
Code_Aster	76.687	1.64e-04	232171	6.43e-25	35.90	1.04e-20	-8.02e-13	-5.992	-1.02e-10

Mode 5

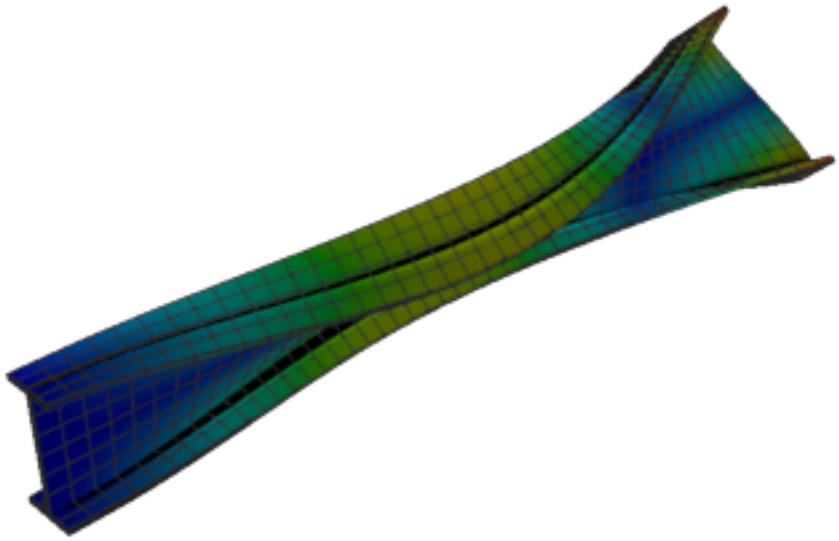


Figure 512: Abaqus — mode 5, 97.391 Hz

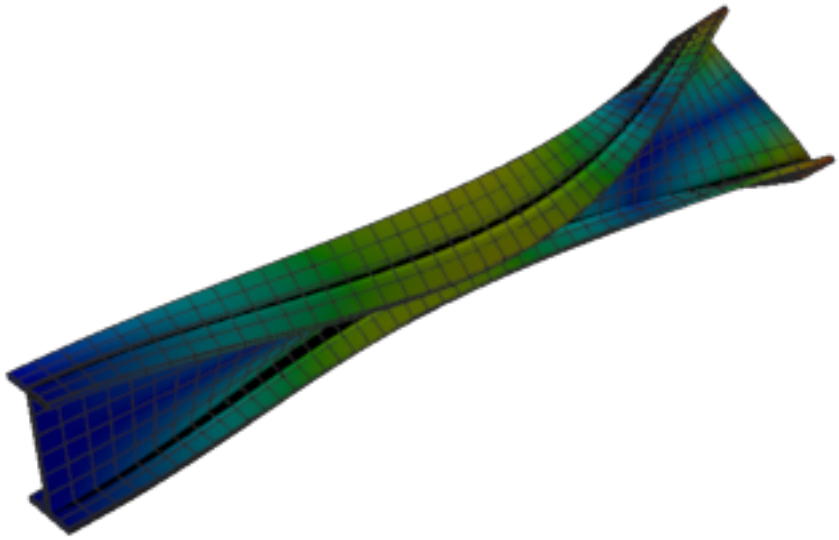


Figure 513: Calculix — mode 5, 97.391 Hz

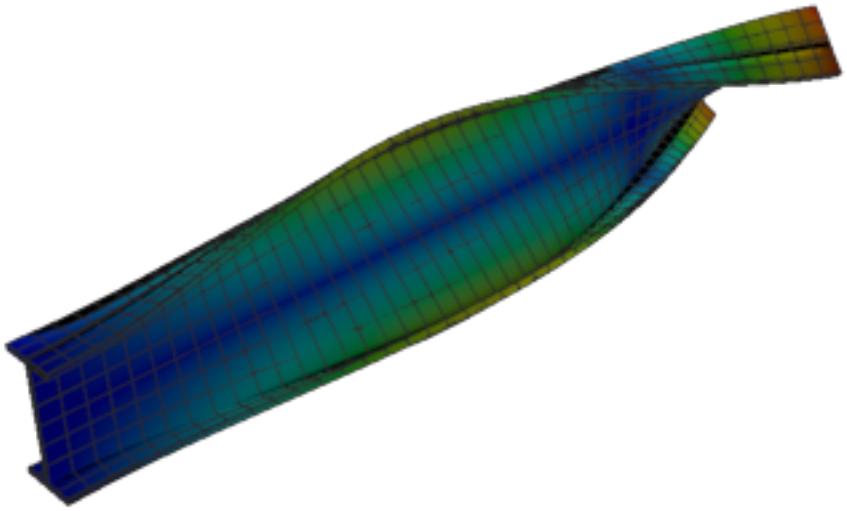


Figure 514: Code_Aster — mode 5, 97.391 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.391	-1.71e-04	374455	1.68e-25	8.25e-19	2.52e-24	4.10e-13	-9.08e-10	-1.59e-12
Calculix	97.391	1.06e-04	374455	5.59e-25	2.56e-19	9.90e-21	-7.47e-13	-5.06e-10	9.95e-11
Code_Aster	97.391	6.57e-05	374455	1.30e-24	1.42e-18	4.98e-21	-1.14e-12	-1.19e-09	7.06e-11

Mode 6

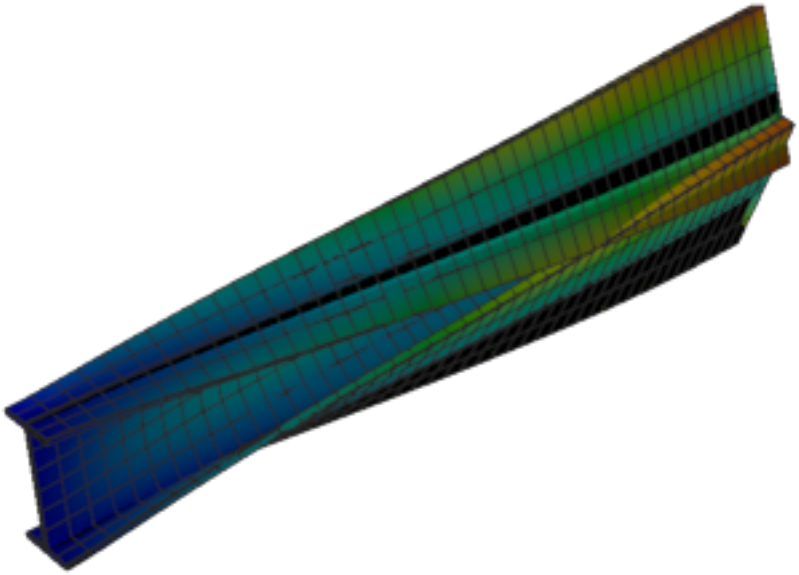


Figure 515: Abaqus — mode 6, 130.809 Hz

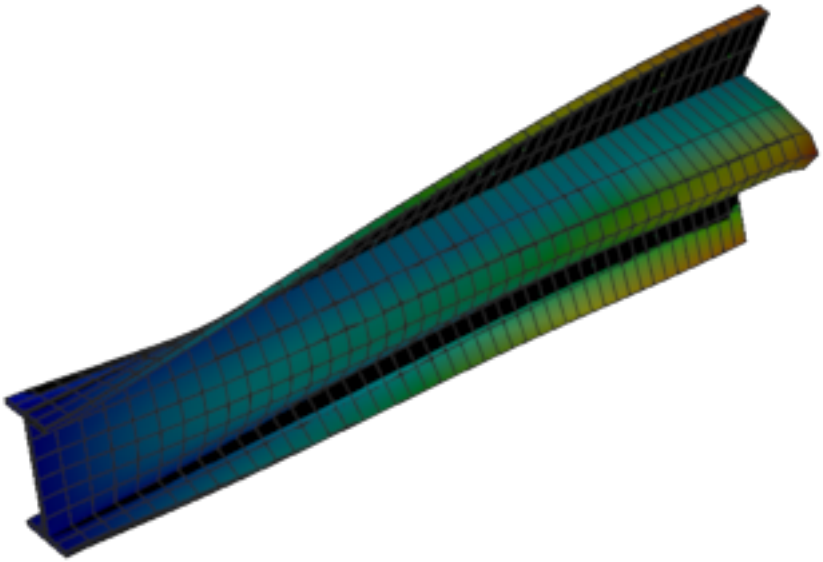


Figure 516: Calculix — mode 6, 130.814 Hz

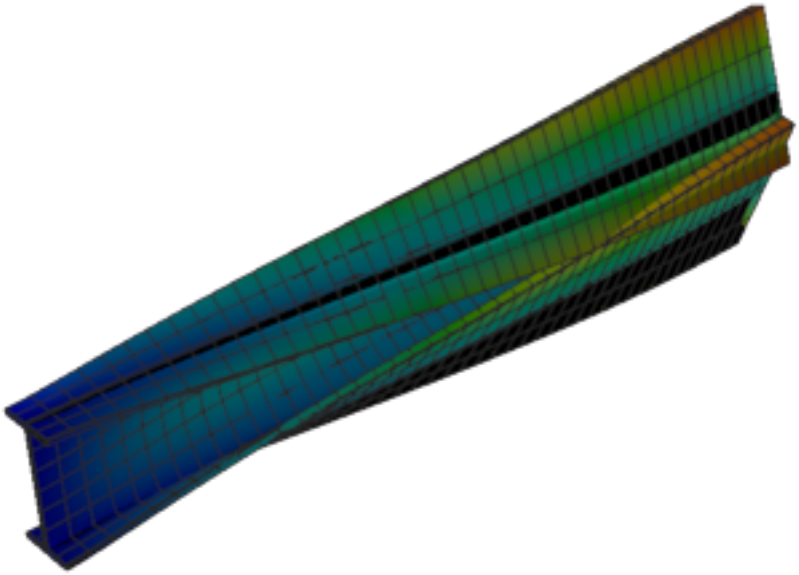


Figure 517: Code_Aster — mode 6, 130.809 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	130.810	-7.26e-04	675515	5.64e-26	0.01	6.07e-21	-2.37e-13	-0.097	7.79e-11
Calculix	130.814	2.18e-03	675564	4.82e-25	0.01	4.46e-21	-6.94e-13	0.096	6.68e-11
Code_Aster	130.809	-1.45e-03	675515	3.85e-25	0.01	6.15e-21	6.20e-13	-0.097	-7.84e-11

Mode 7

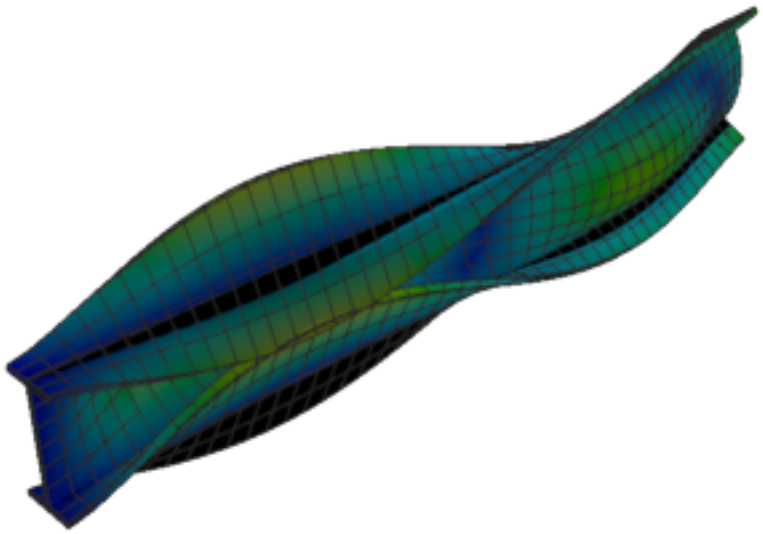


Figure 518: Abaqus — mode 7, 148.474 Hz

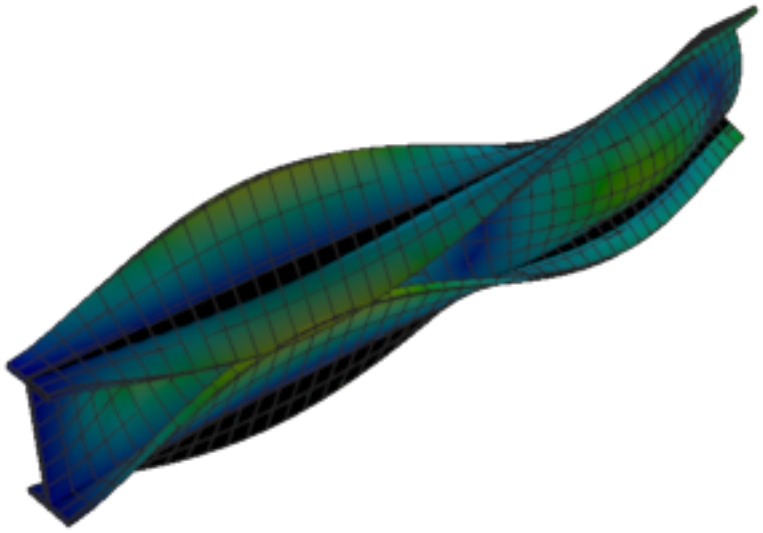


Figure 519: Calculix — mode 7, 148.478 Hz

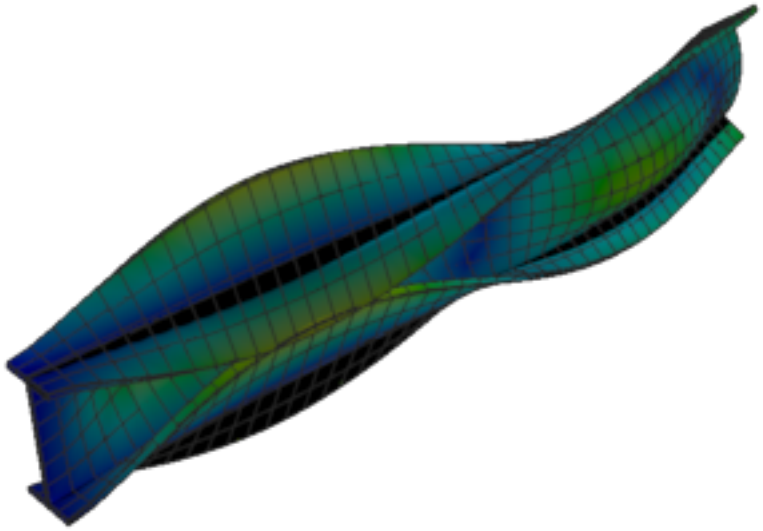


Figure 520: Code_Aster — mode 7, 148.474 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	148.470	-2.76e-03	870287	1.93e-24	3.39	2.12e-22	-1.39e-12	1.840	-1.46e-11
Calculix	148.478	2.63e-03	870330	4.10e-25	3.39	8.55e-22	6.41e-13	1.840	2.92e-11
Code_Aster	148.474	1.37e-04	870287	1.52e-25	3.39	1.11e-22	-3.90e-13	1.840	1.05e-11

Mode 8

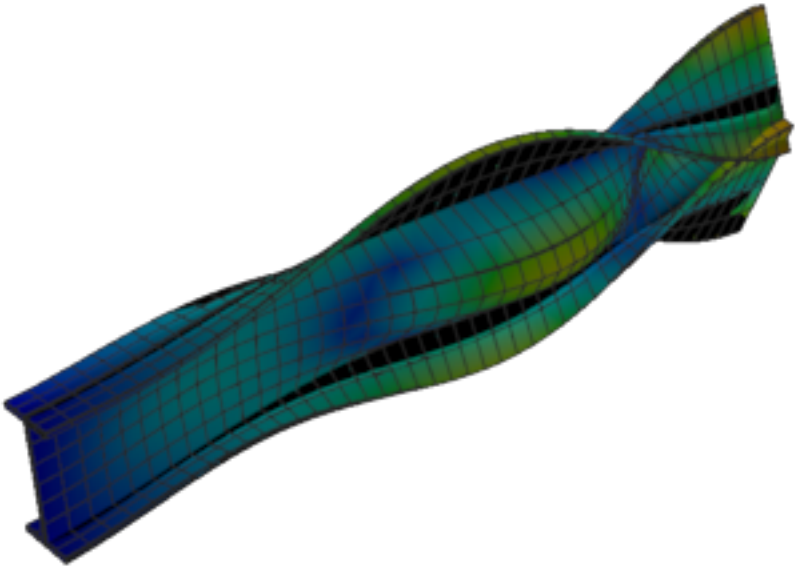


Figure 521: Abaqus — mode 8, 172.171 Hz

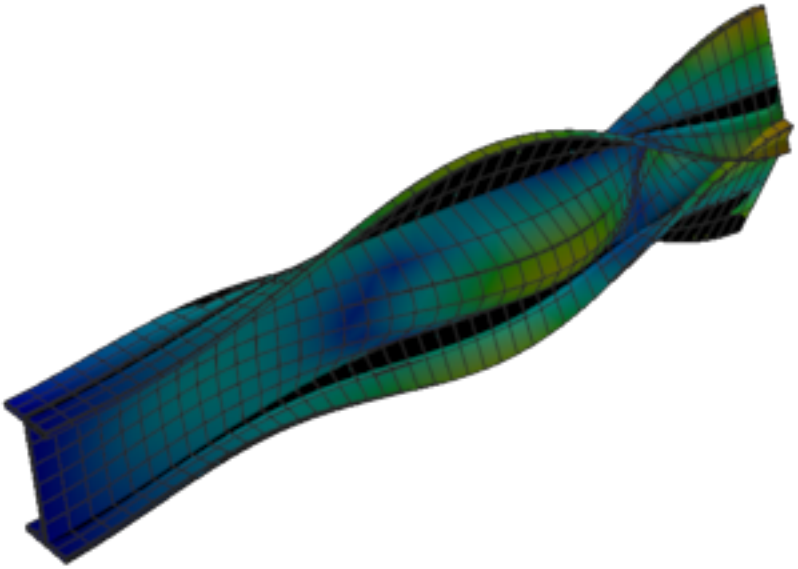


Figure 522: Calculix — mode 8, 172.177 Hz

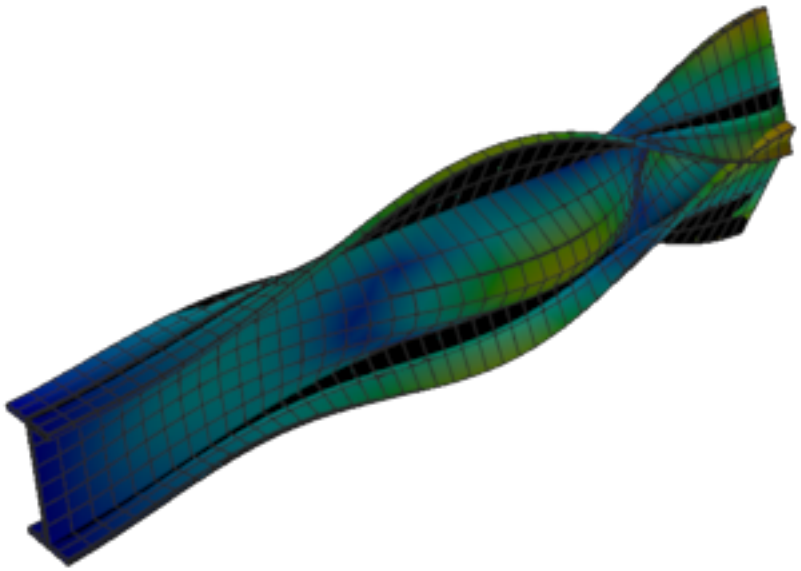


Figure 523: Code_Aster — mode 8, 172.171 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	172.170	-1.58e-03	1.17026e+06	1.27e-25	5.31	9.73e-21	3.57e-13	2.304	9.86e-11
Calculix	172.177	2.37e-03	1.17033e+06	1.37e-24	5.31	1.97e-22	1.17e-12	2.304	1.40e-11
Code_Aster	172.171	-7.89e-04	1.17026e+06	2.56e-24	5.31	1.73e-20	-1.60e-12	2.304	-1.32e-10

Mode 9

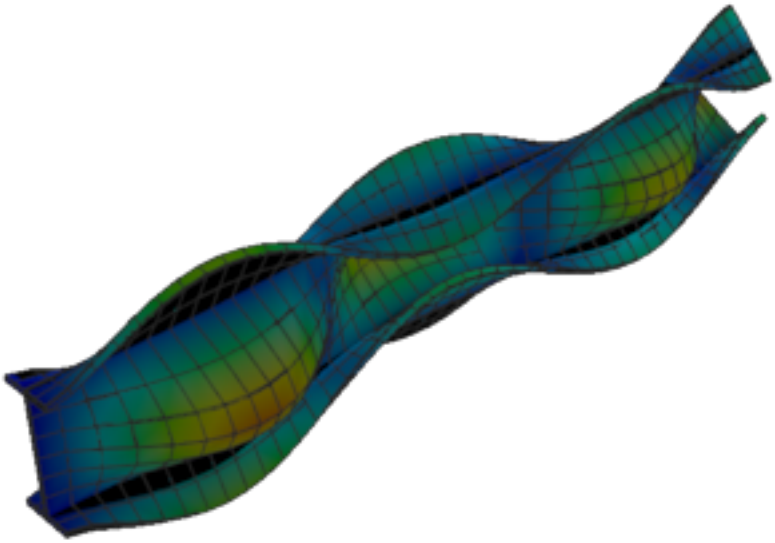


Figure 524: Abaqus — mode 9, 201.839 Hz

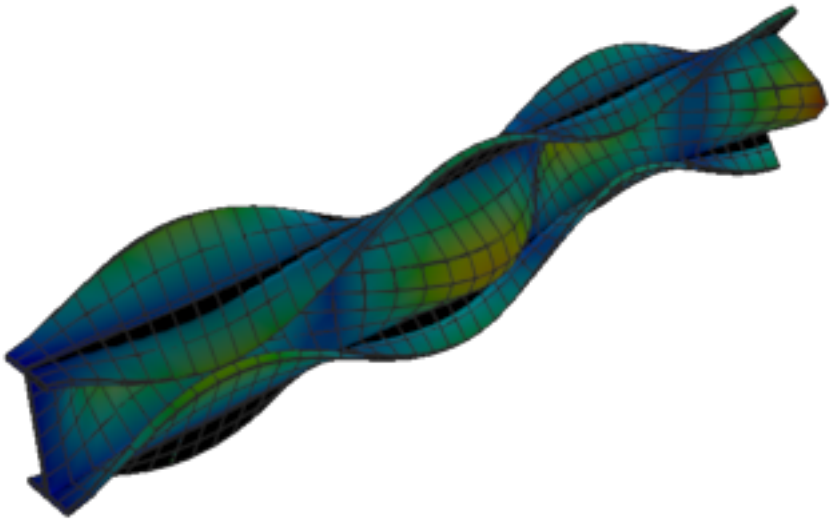


Figure 525: Calculix — mode 9, 201.846 Hz

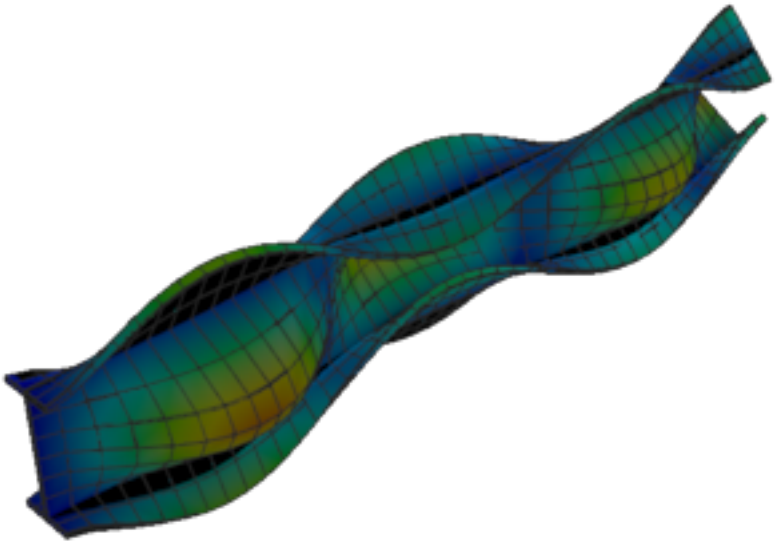


Figure 526: Code_Aster — mode 9, 201.839 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	201.840	-8.14e-04	1.60831e+06	9.86e-25	0.56	1.17e-21	-9.93e-13	-0.747	3.42e-11
Calculix	201.846	2.01e-03	1.60842e+06	3.90e-25	0.56	7.43e-22	-6.24e-13	0.747	2.73e-11
Code_Aster	201.839	-1.20e-03	1.60831e+06	3.60e-24	0.56	4.60e-21	-1.90e-12	-0.747	-6.78e-11

Mode 10

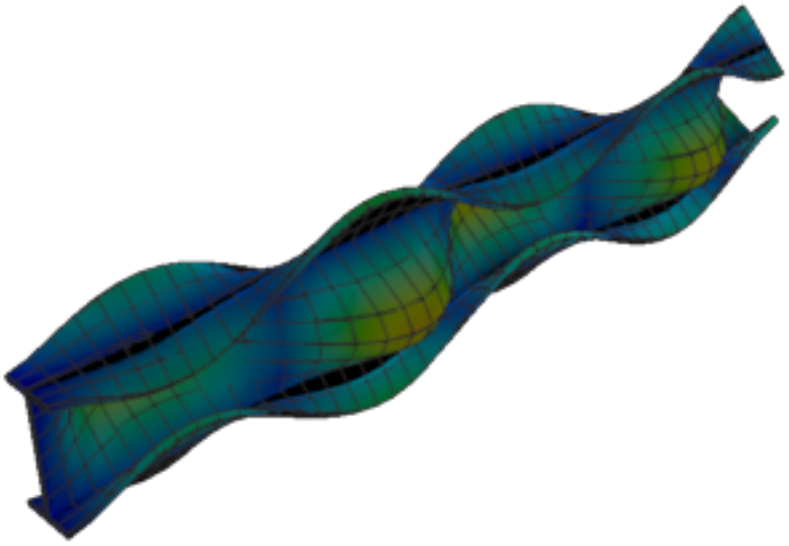


Figure 527: Abaqus — mode 10, 246.165 Hz

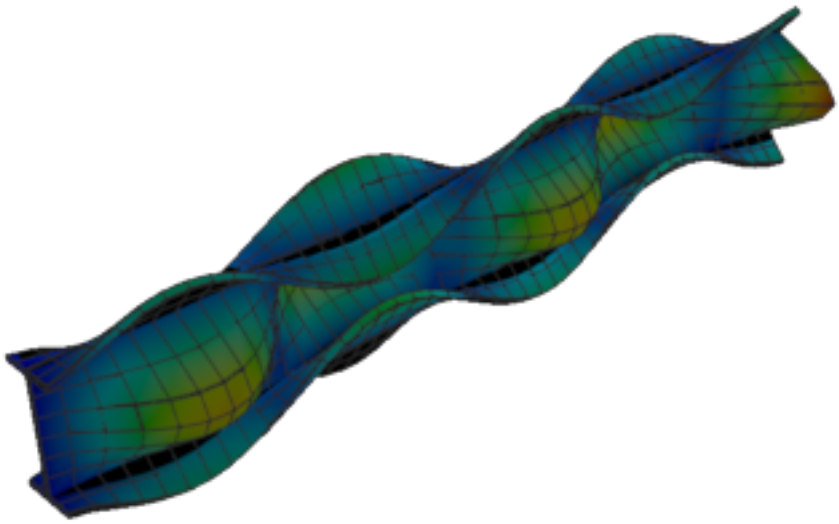


Figure 528: Calculix — mode 10, 246.177 Hz

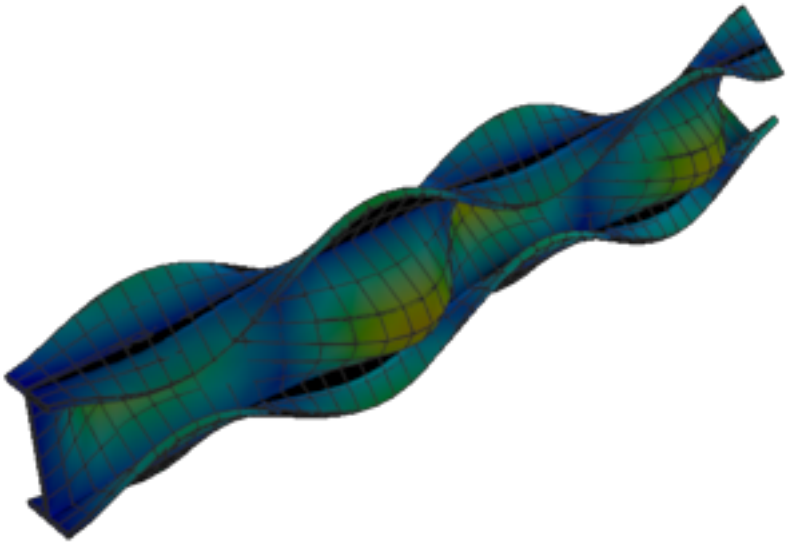


Figure 529: Code_Aster — mode 10, 246.165 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	246.170	-2.59e-04	2.39228e+06	3.32e-21	1.47	1.28e-24	5.76e-11	1.212	-1.13e-12
Calculix	246.177	2.54e-03	2.39251e+06	4.40e-29	1.47	7.89e-21	-6.63e-15	-1.213	8.88e-11
Code_Aster	246.165	-2.28e-03	2.39228e+06	4.39e-25	1.47	1.15e-19	6.62e-13	1.212	-3.38e-10

Mode 11

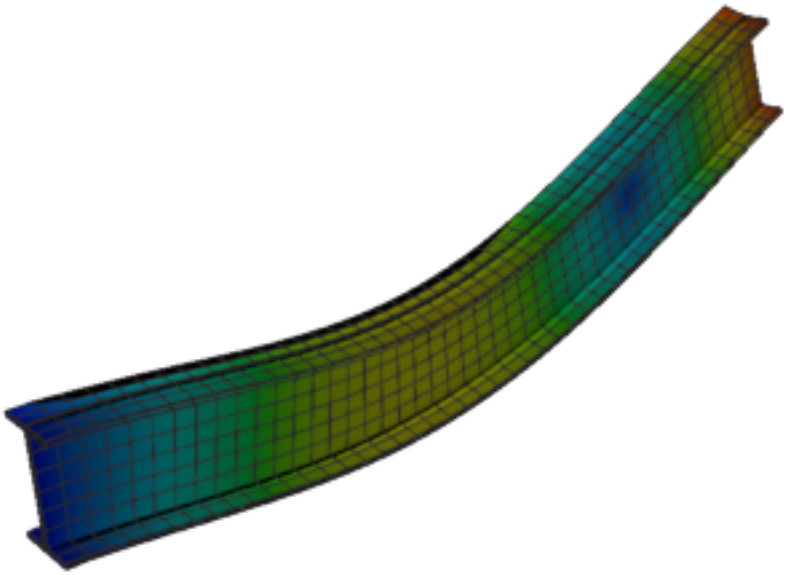


Figure 530: Abaqus — mode 11, 252.573 Hz

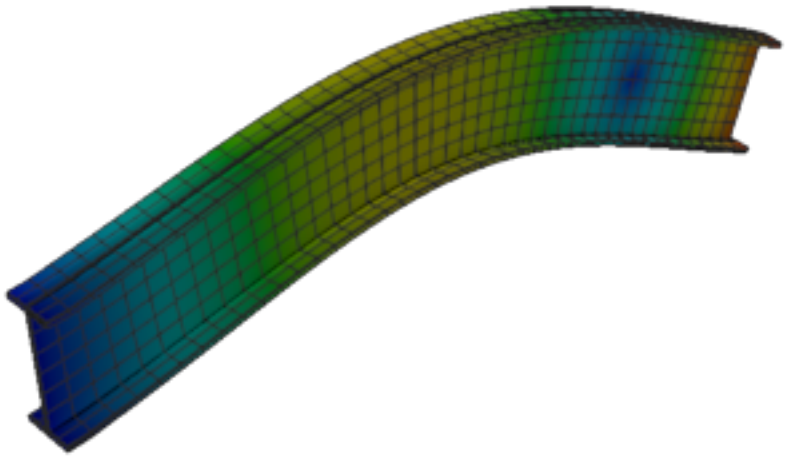


Figure 531: Calculix — mode 11, 252.573 Hz

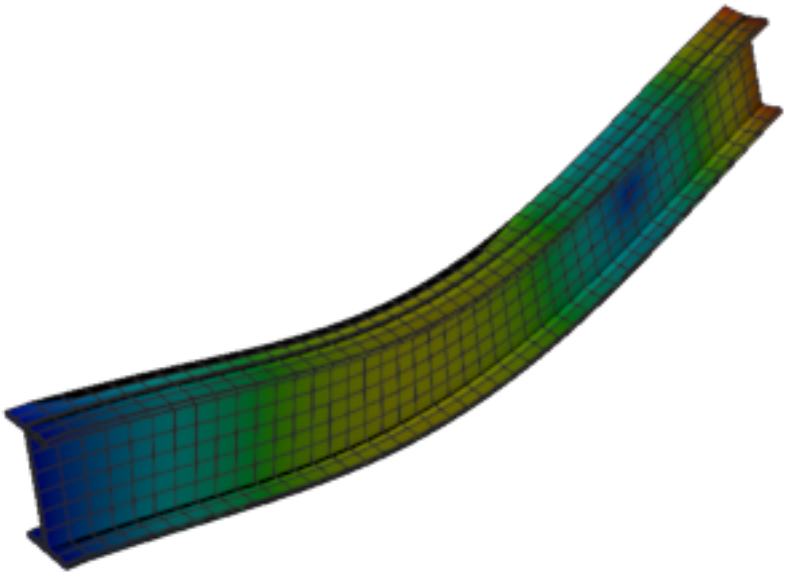


Figure 532: Code_Aster — mode 11, 252.573 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.570	-8.59e-04	2.51846e+06	4.57e-21	2.25e-19	40.26	6.76e-11	4.75e-10	-6.345
Calculix	252.573	4.48e-04	2.51846e+06	1.29e-24	5.04e-23	40.26	1.14e-12	-7.10e-12	6.345
Code_Aster	252.573	4.11e-04	2.51846e+06	1.85e-24	3.66e-21	40.26	-1.36e-12	-6.05e-11	-6.345

Plate eigenvalue analysis

The plain and the stiffened strip, at the eigen case’s 0.0625 m seed.

Shell, 1st order, plain

Mode 1

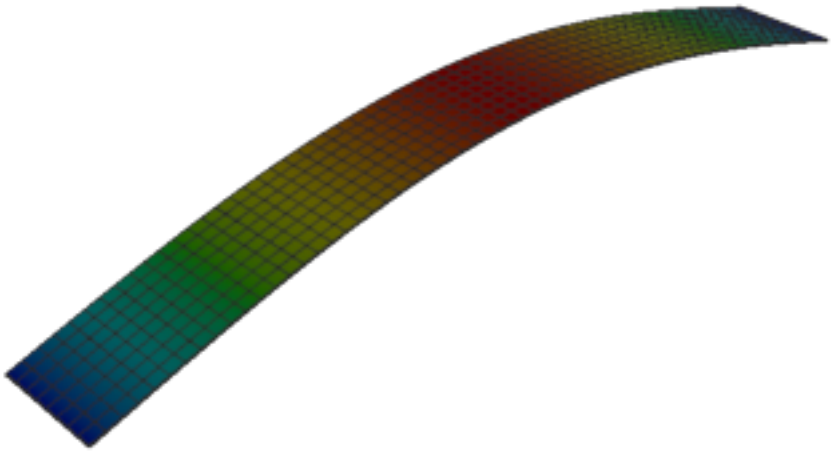


Figure 533: Abaqus — mode 1, 1.537 Hz

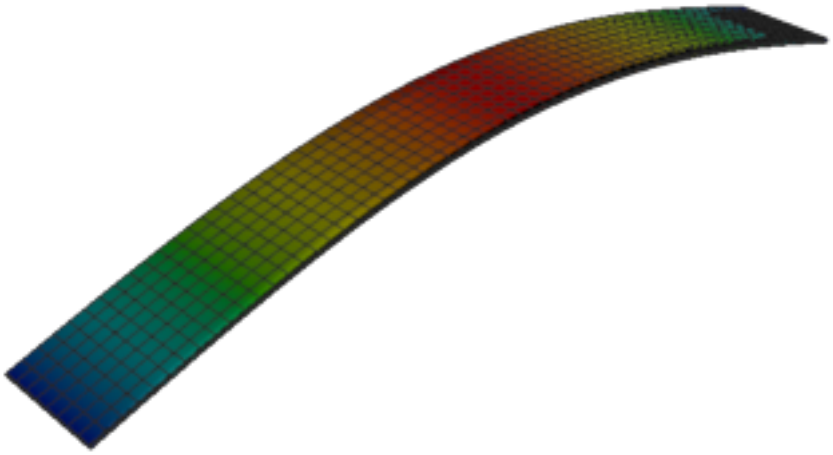


Figure 534: Calculix — mode 1, 1.537 Hz

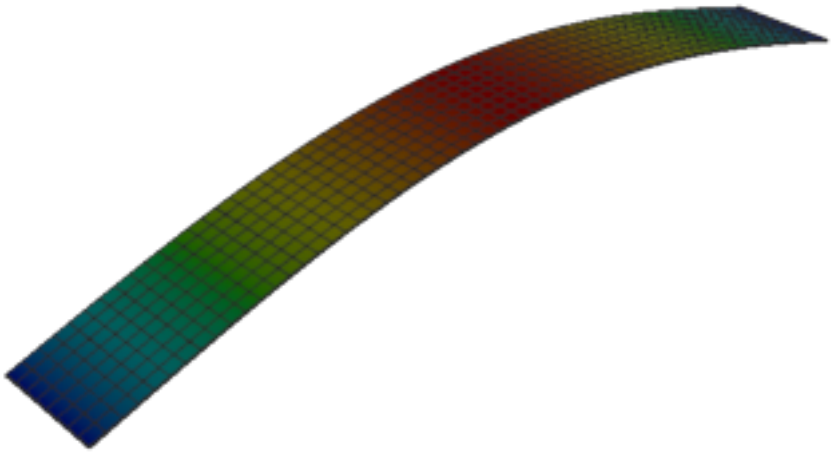


Figure 535: Code_Aster — mode 1, 1.537 Hz

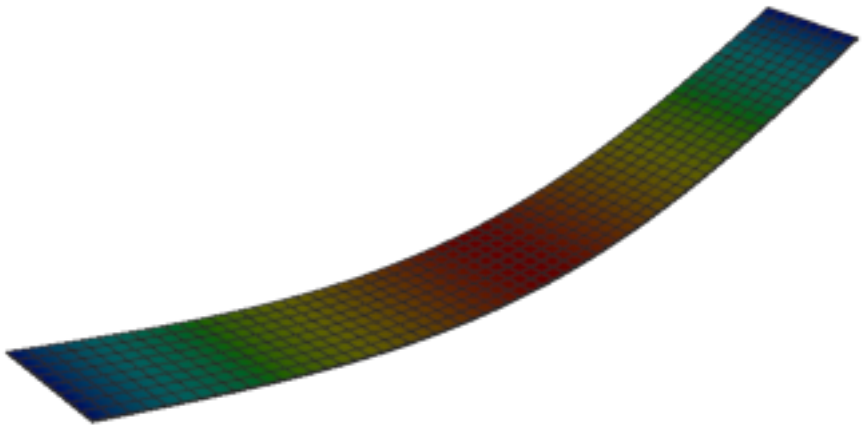


Figure 536: Sesam — mode 1, 1.537 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	1.537	+0.03	93.269	6.40e-31	3.03e-35	127.26	-8.00e-16	5.50e-18	11.281
Calculix	1.537	+0.01	93.2317	3.65e-09	1.07e-29	127.22	6.04e-05	-3.26e-15	11.279
Code_Aster	1.537	2.03e-07	93.2152	2.24e-49	1.19e-57	127.26	-4.74e-25	-3.46e-29	11.281
Sesam	1.537	-4.98e-06	93.2152	1.43e-40	1.48e-45	127.17	-1.19e-20	3.85e-23	-11.277
Closed form	1.537								

Mode 2

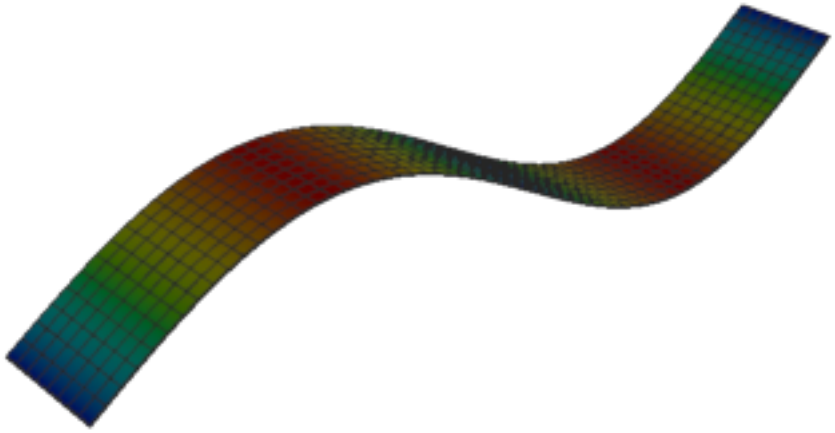


Figure 537: Abaqus — mode 2, 6.154 Hz

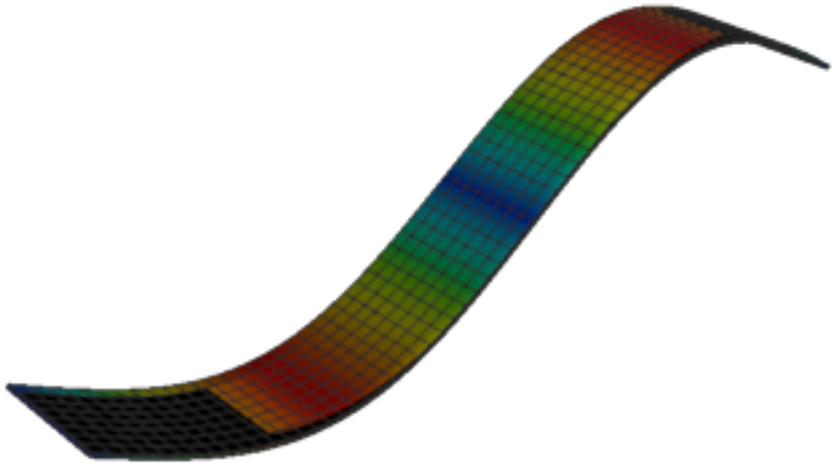


Figure 538: Calculix — mode 2, 6.149 Hz

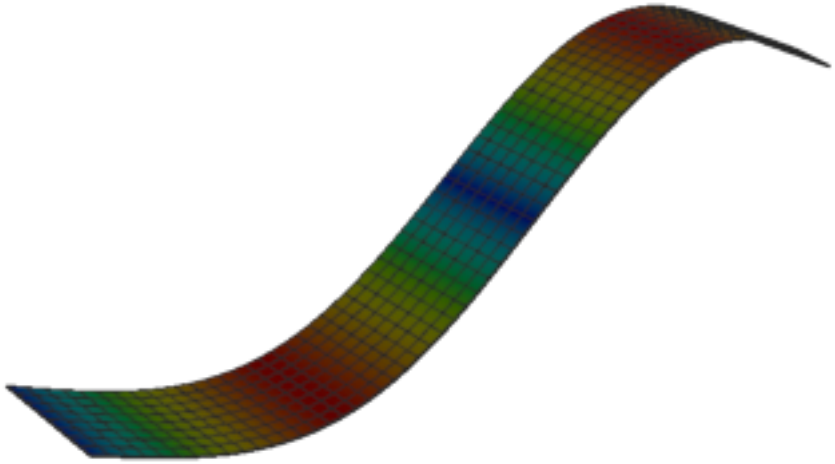


Figure 539: Code_Aster — mode 2, 6.146 Hz

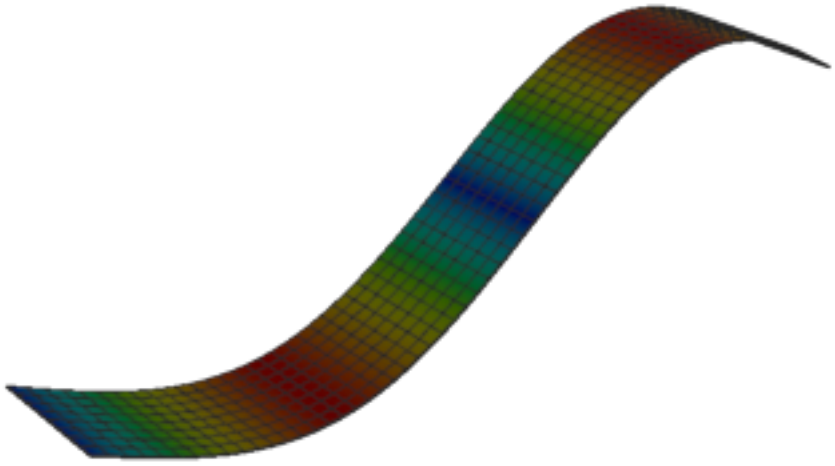


Figure 540: Sesam — mode 2, 6.146 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	6.154	+0.12	1494.9	5.42e-31	1.34e-33	5.32e-21	7.36e-16	-3.66e-17	7.29e-11
Calculix	6.149	+0.04	1492.5	1.46e-08	9.33e-31	4.33e-15	-1.21e-04	9.66e-16	-6.58e-08
Code_Aster	6.146	4.29e-06	1491.44	1.65e-44	1.01e-52	1.29e-22	-1.29e-22	-1.00e-26	1.13e-11
Sesam	6.146	1.13e-05	1491.44	1.40e-34	3.63e-39	9.71e-23	1.18e-17	-6.03e-20	-9.85e-12
Closed form	6.146								

Mode 3

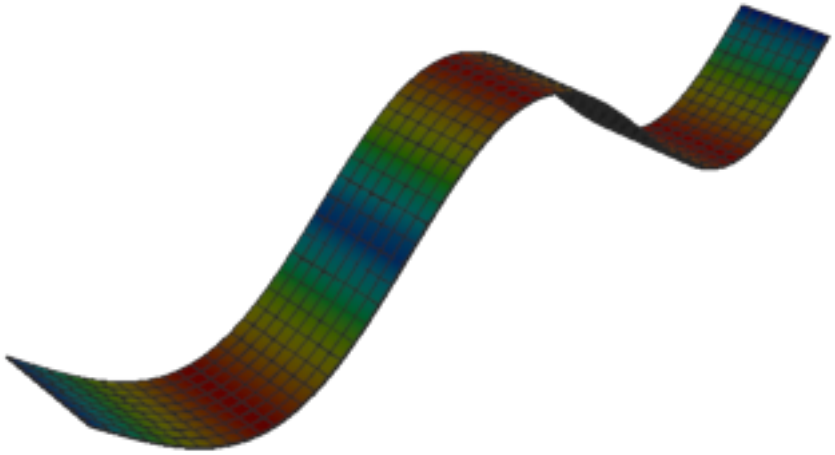


Figure 541: Abaqus — mode 3, 13.866 Hz

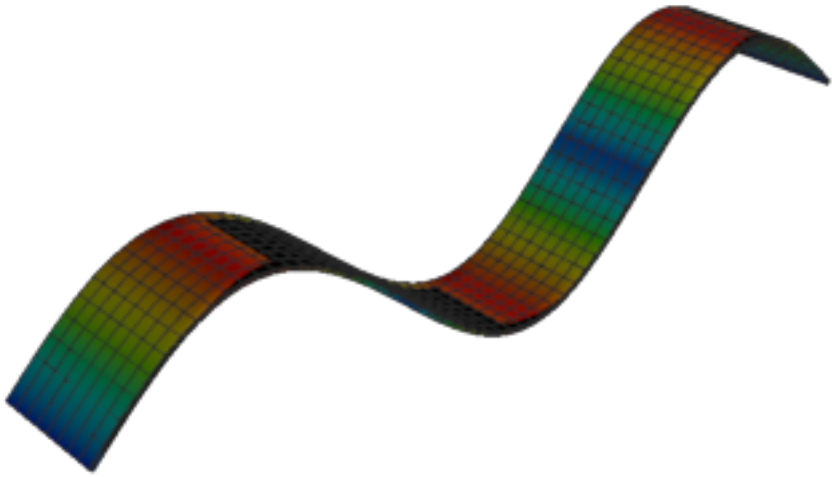


Figure 542: Calculix — mode 3, 13.840 Hz

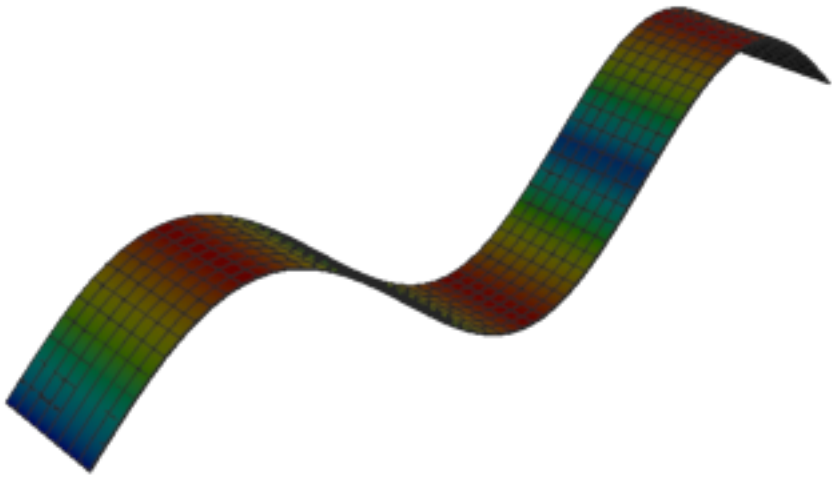


Figure 543: Code_Aster — mode 3, 13.829 Hz

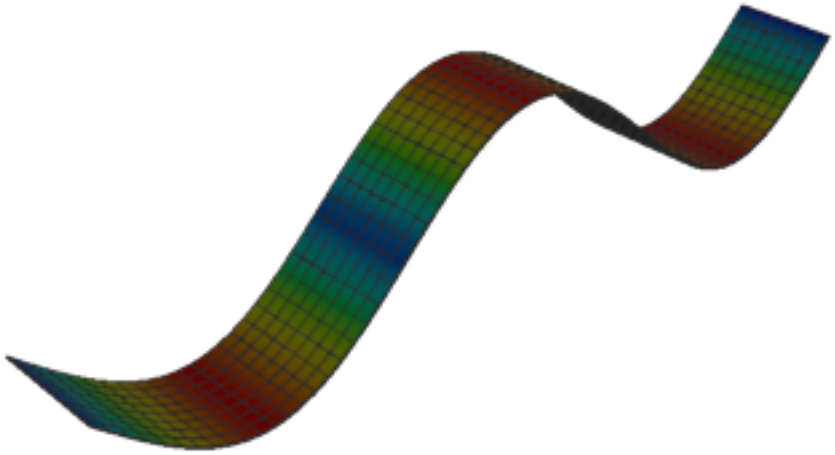


Figure 544: Sesam — mode 3, 13.829 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	13.866	+0.26	7589.9	4.43e-30	5.80e-32	14.14	2.10e-15	-2.41e-16	-3.760
Calculix	13.840	+0.08	7562.44	3.27e-08	4.89e-31	14.11	1.81e-04	6.99e-16	3.756
Code_Aster	13.829	2.17e-05	7550.43	2.99e-41	9.78e-50	14.14	-5.47e-21	-3.13e-25	3.760
Sesam	13.829	6.01e-05	7550.43	1.99e-35	2.24e-40	14.05	-4.46e-18	1.50e-20	-3.748
Closed form	13.829								

Mode 4

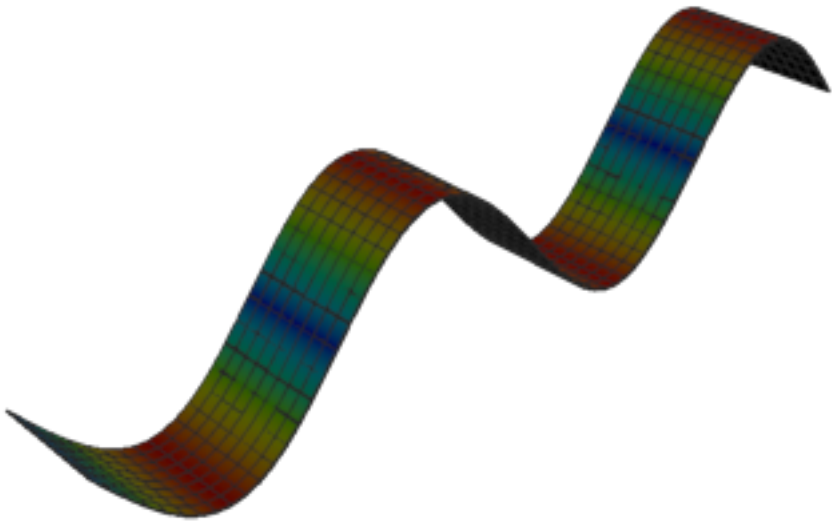


Figure 545: Abaqus — mode 4, 24.700 Hz

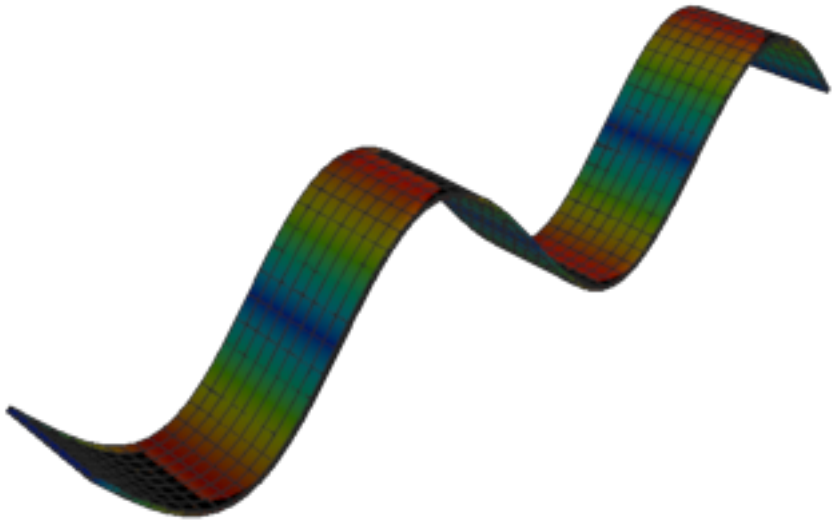


Figure 546: Calculix — mode 4, 24.621 Hz

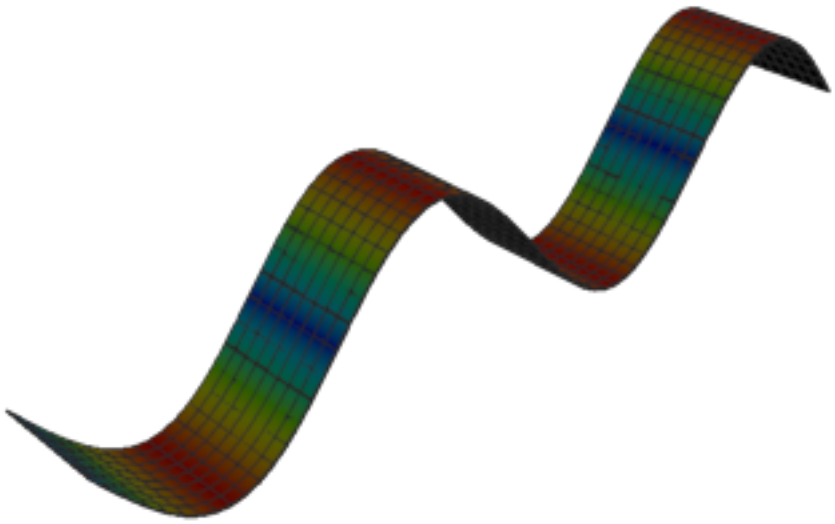


Figure 547: Code_Aster — mode 4, 24.586 Hz

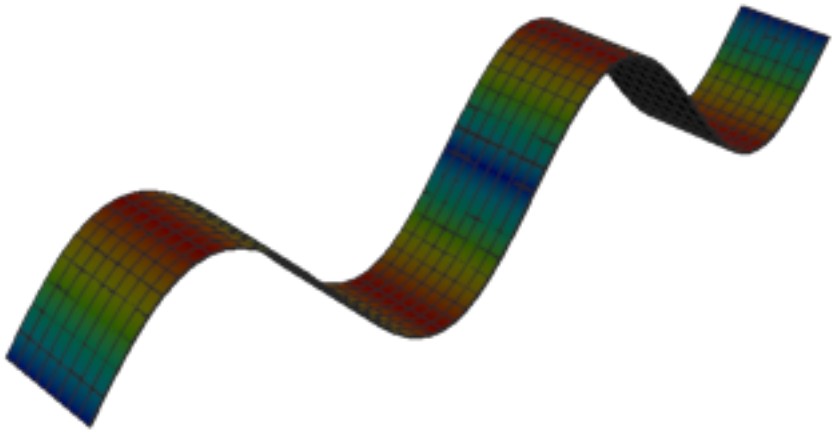


Figure 548: Sesam — mode 4, 24.586 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	24.700	+0.46	24086	1.04e-28	7.36e-32	1.83e-23	-1.02e-14	2.71e-16	4.27e-12
Calculix	24.621	+0.14	23930.6	5.80e-08	9.05e-31	5.81e-21	-2.41e-04	-9.51e-16	-7.62e-11
Code_Aster	24.586	6.86e-05	23863.1	5.65e-40	4.68e-49	3.83e-24	-2.38e-20	6.84e-25	1.96e-12
Sesam	24.586	1.01e-04	23863.1	9.91e-32	6.38e-36	1.14e-23	-3.15e-16	2.53e-18	3.37e-12
Closed form	24.586								

Mode 5

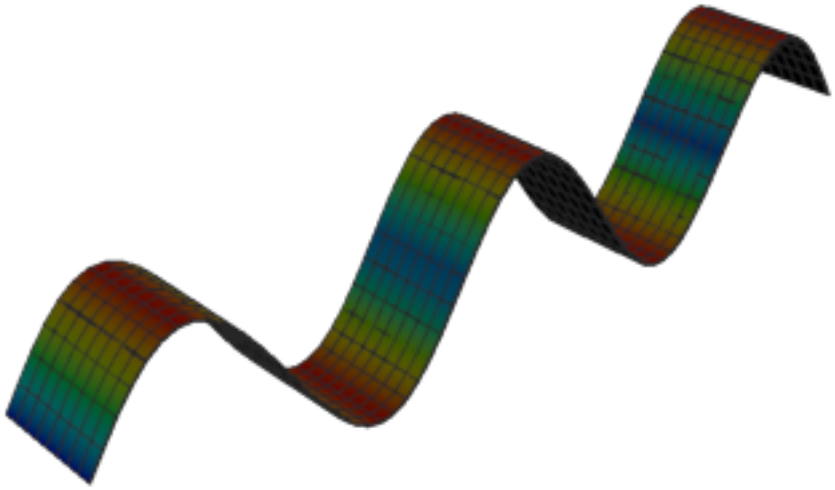


Figure 549: Abaqus — mode 5, 38.695 Hz

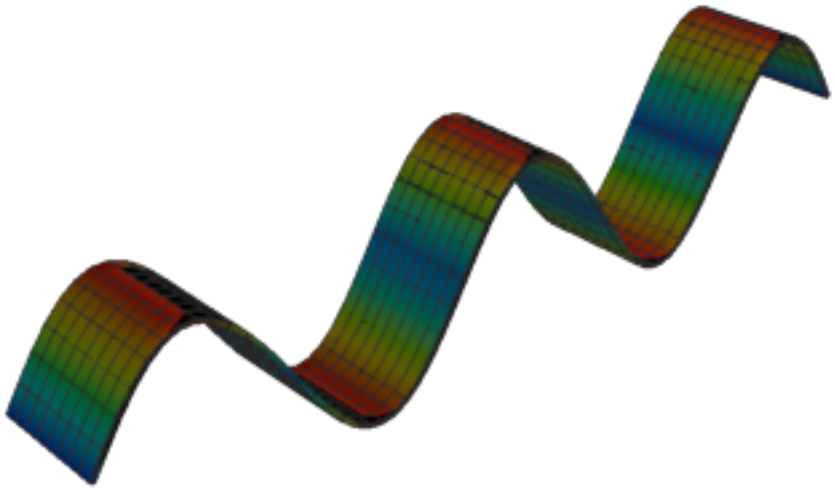


Figure 550: Calculix — mode 5, 38.500 Hz

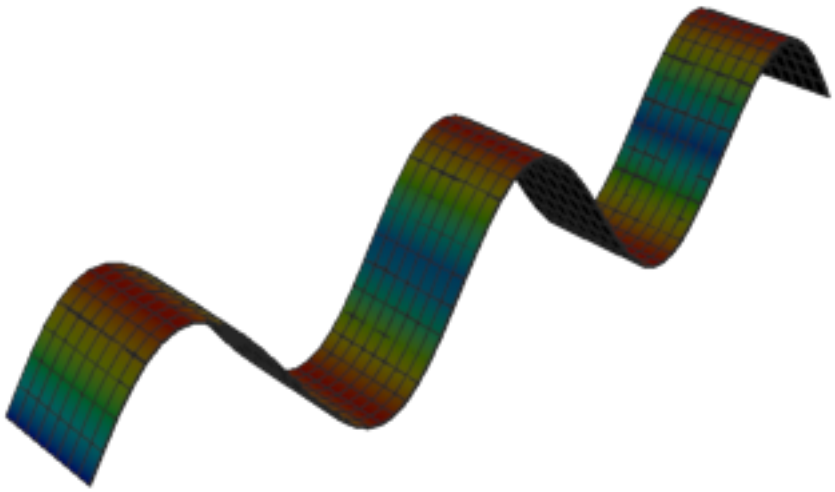


Figure 551: Code_Aster — mode 5, 38.415 Hz

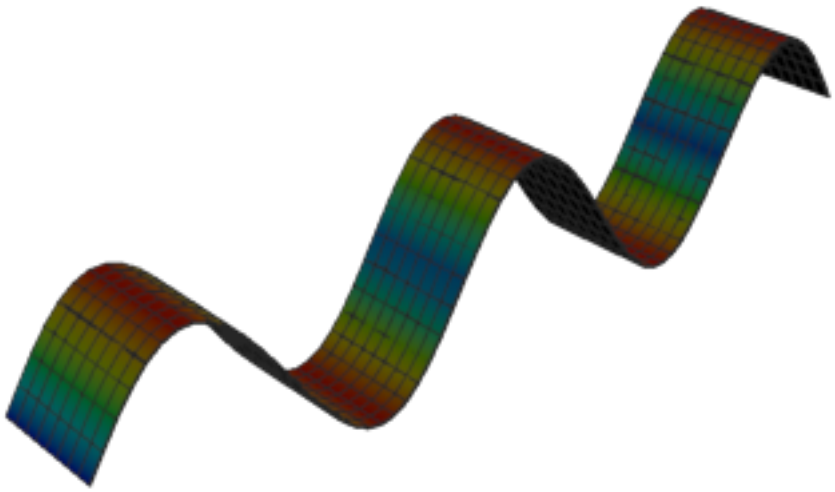


Figure 552: Sesam — mode 5, 38.415 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	38.695	+0.73	59111	1.02e-28	1.84e-30	5.09	1.01e-14	-1.36e-15	2.256
Calculix	38.500	+0.22	58517.4	9.02e-08	1.18e-30	5.06	3.00e-04	1.09e-15	2.249
Code_Aster	38.415	1.67e-04	58259.7	6.94e-39	7.45e-45	5.09	-8.33e-20	8.63e-23	2.256
Sesam	38.415	2.42e-04	58259.8	1.79e-26	3.30e-27	5.00	1.34e-13	5.75e-14	2.236
Closed form	38.415								

Mode 6

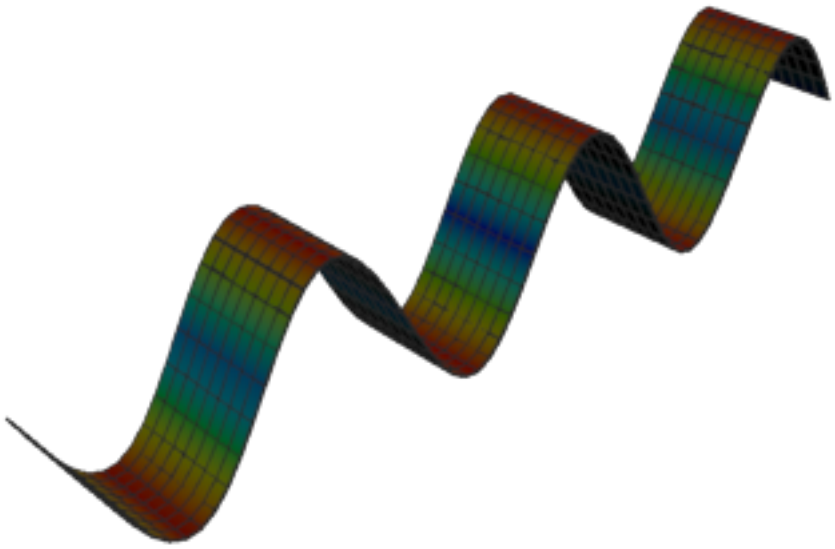


Figure 553: Abaqus — mode 6, 55.900 Hz

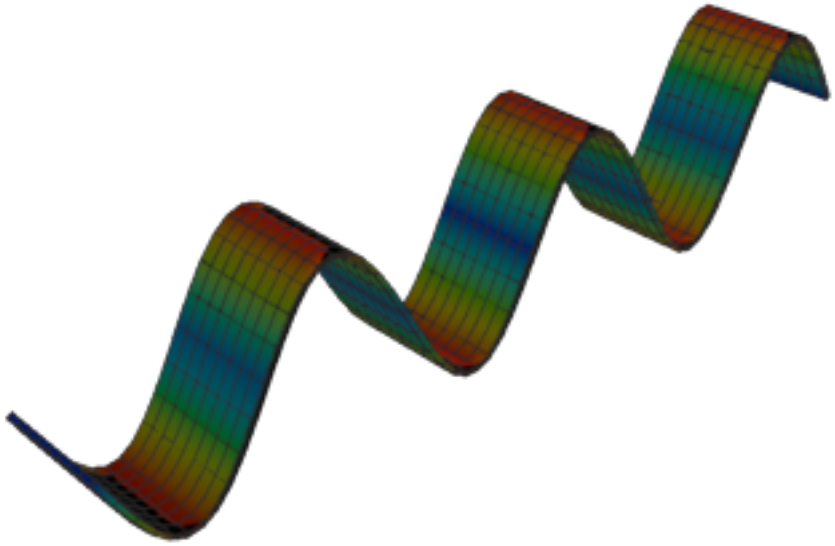


Figure 554: Calculix — mode 6, 55.494 Hz

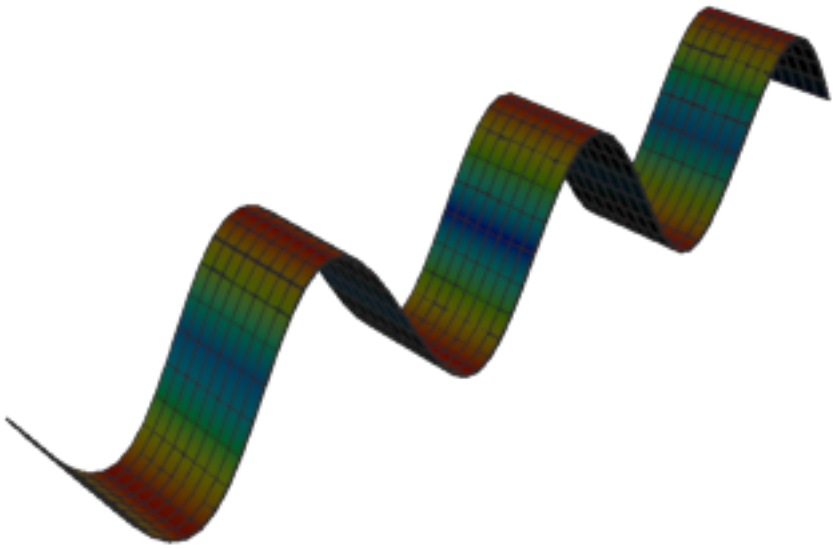


Figure 555: Code_Aster — mode 6, 55.318 Hz

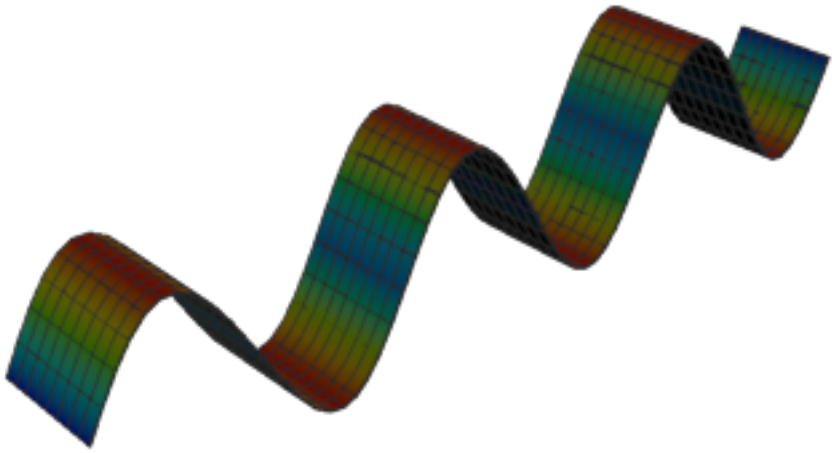


Figure 556: Sesam — mode 6, 55.318 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	55.900	+1.05	123361	4.94e-20	7.80e-26	3.09e-20	-2.22e-10	-2.79e-13	-1.76e-10
Calculix	55.494	+0.32	121578	1.29e-07	1.46e-30	2.42e-19	-3.60e-04	1.21e-15	4.92e-10
Code_Aster	55.318	3.46e-04	120808	1.66e-37	2.85e-42	5.43e-26	-4.07e-19	-1.69e-21	2.33e-13
Sesam	55.318	5.30e-04	120808	1.43e-17	2.77e-18	6.63e-16	3.78e-09	1.66e-09	-2.58e-08
Closed form	55.318								

Shell, 1st order, stiffened

Mode 1

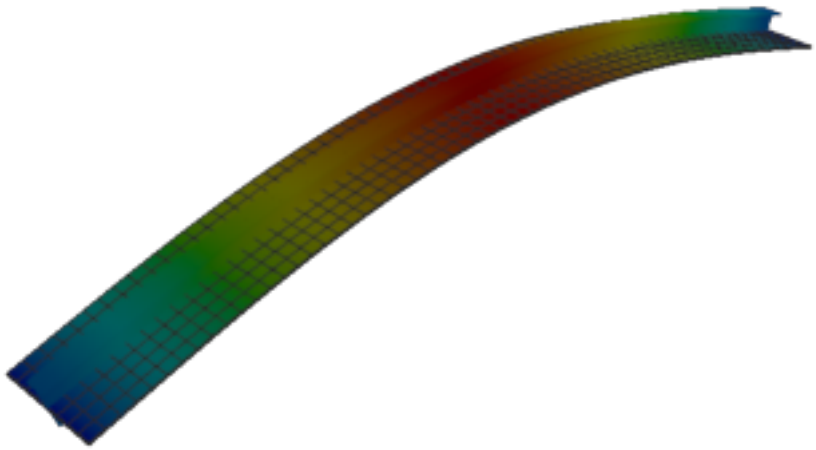


Figure 557: Abaqus — mode 1, 21.242 Hz

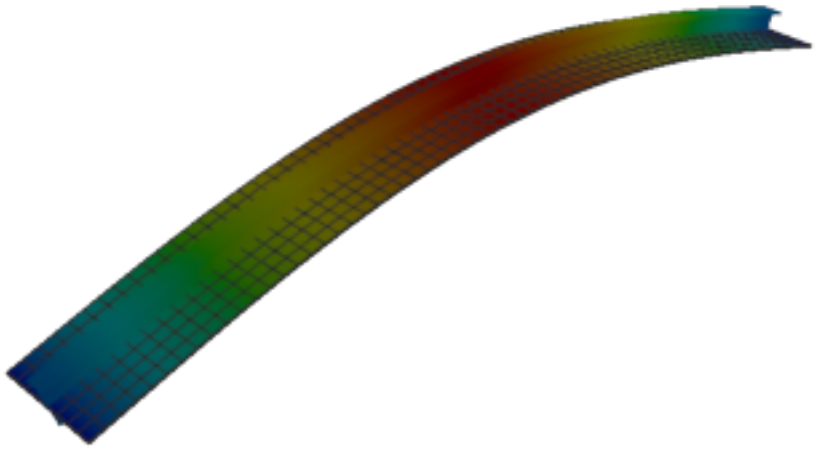


Figure 558: Code_Aster — mode 1, 19.298 Hz

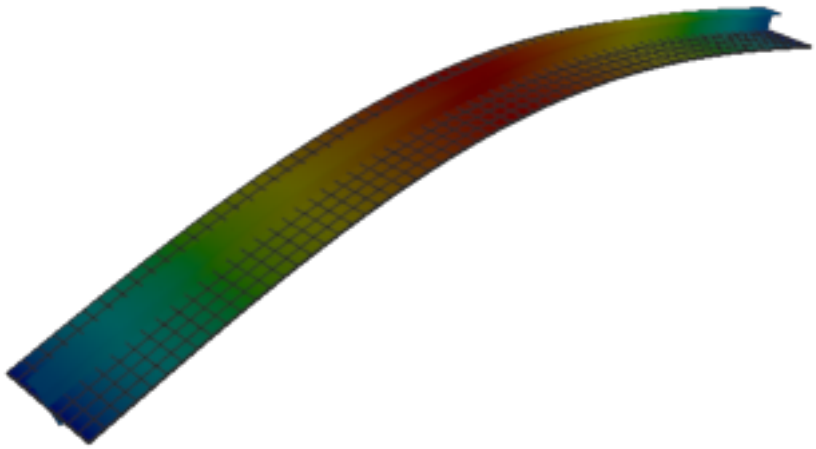


Figure 559: Sesam — mode 1, 19.157 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	21.242	+6.75	17813	0.05	9.45e-26	195.80	-0.214	-3.07e-13	13.993
Code_Aster	19.298	-3.02	14702.3	1.78e-39	5.54e-47	196.08	4.22e-20	7.44e-24	14.003
Sesam	19.157	-3.73	14488	2.36e-35	2.32e-35	195.94	4.86e-18	4.82e-18	13.998

Mode 2

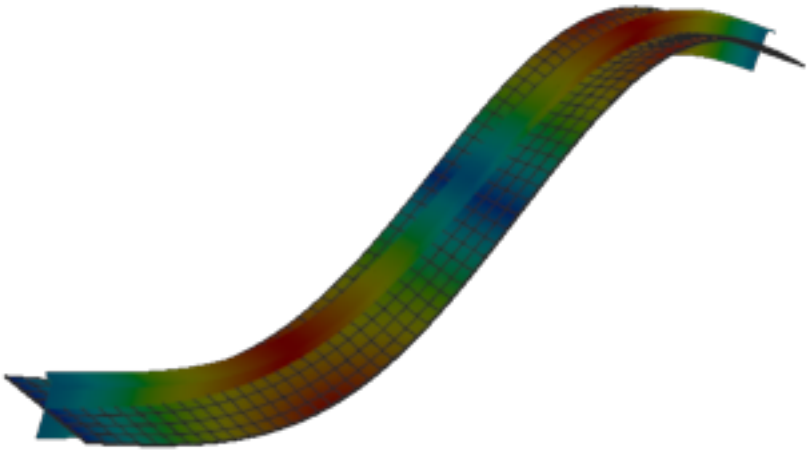


Figure 560: Abaqus — mode 2, 80.469 Hz

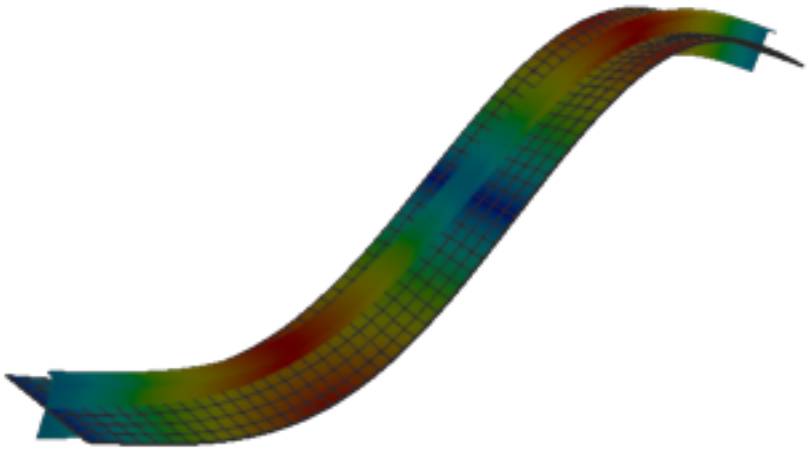


Figure 561: Code_Aster — mode 2, 75.261 Hz

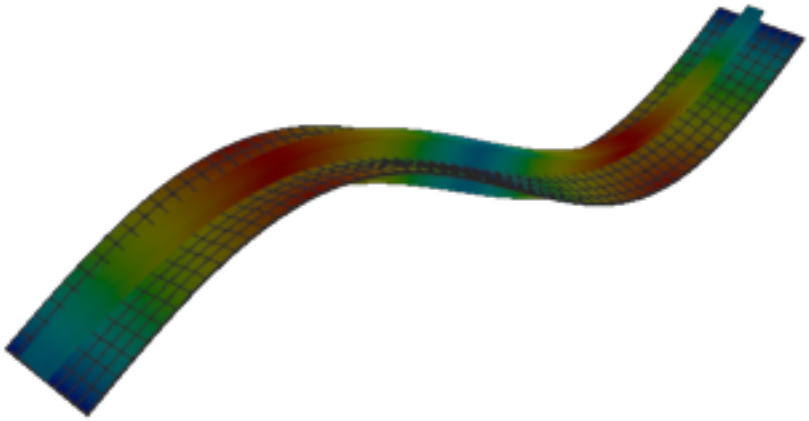


Figure 562: Sesam — mode 2, 73.241 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	80.469	+5.43	255632	0.16	1.91e-25	3.42e-05	0.403	4.37e-13	0.006
Code_Aster	75.261	-1.39	223612	6.65e-36	1.89e-42	2.29e-23	-2.58e-18	-1.38e-21	4.78e-12
Sesam	73.241	-4.04	211769	1.97e-32	2.23e-33	1.18e-23	1.40e-16	4.72e-17	-3.44e-12

Mode 3

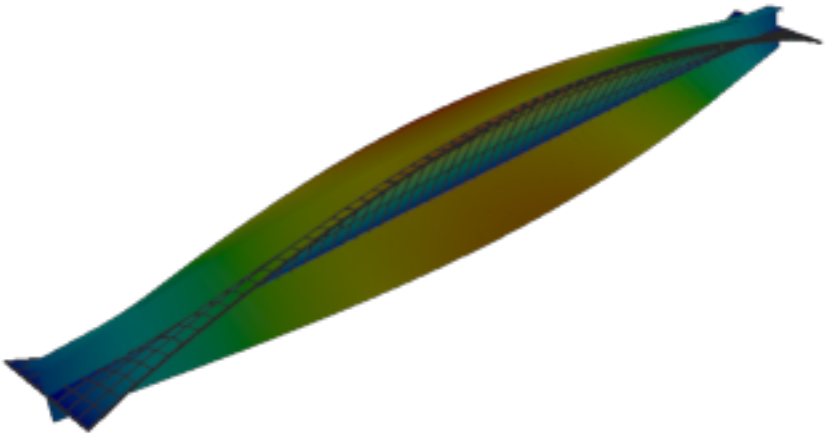


Figure 563: Abaqus — mode 3, 93.088 Hz

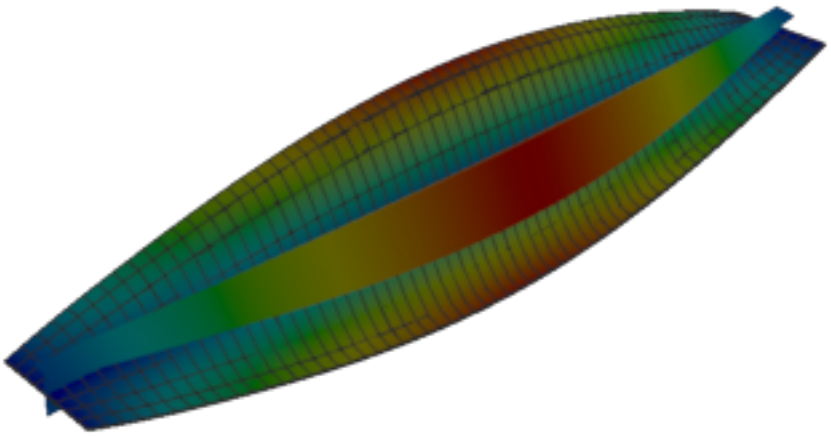


Figure 564: Code_Aster — mode 3, 92.494 Hz

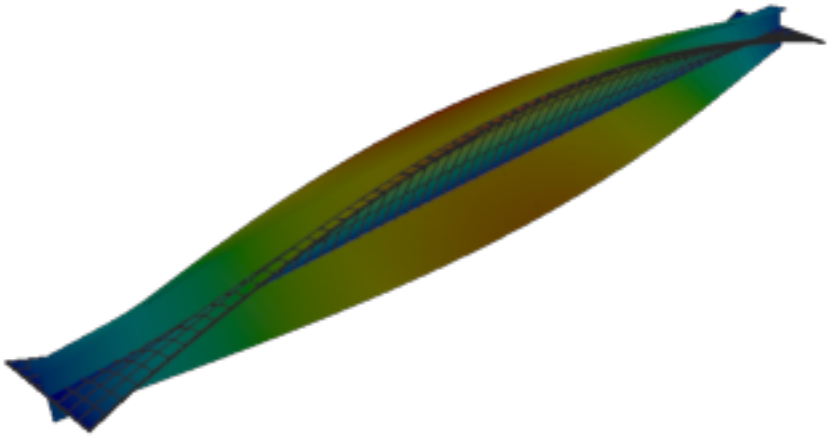


Figure 565: Sesam — mode 3, 92.540 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	93.088	+0.41	342097	2.75e-24	3.95	1.03e-23	1.66e-12	1.988	3.22e-12
Code_Aster	92.494	-0.23	337741	4.55e-35	3.26e-41	2.73e-26	6.75e-18	-5.71e-21	-1.65e-13
Sesam	92.540	-0.18	338080	2.00e-31	1.99e-07	1.09e-25	4.48e-16	4.46e-04	-3.29e-13

Mode 4

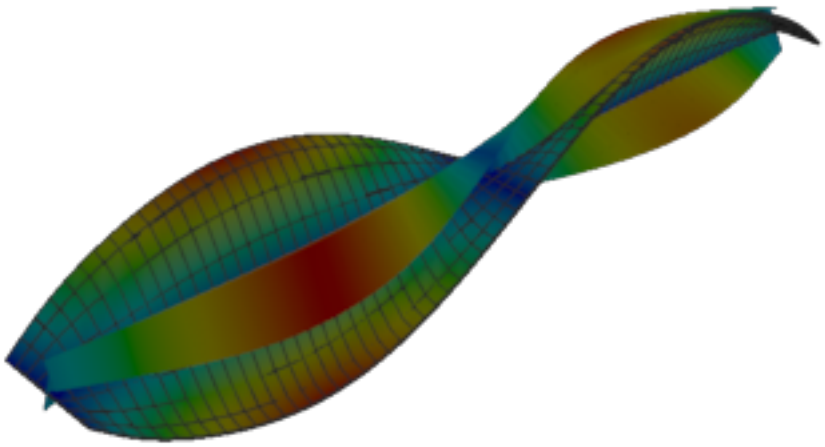


Figure 566: Abaqus — mode 4, 106.618 Hz

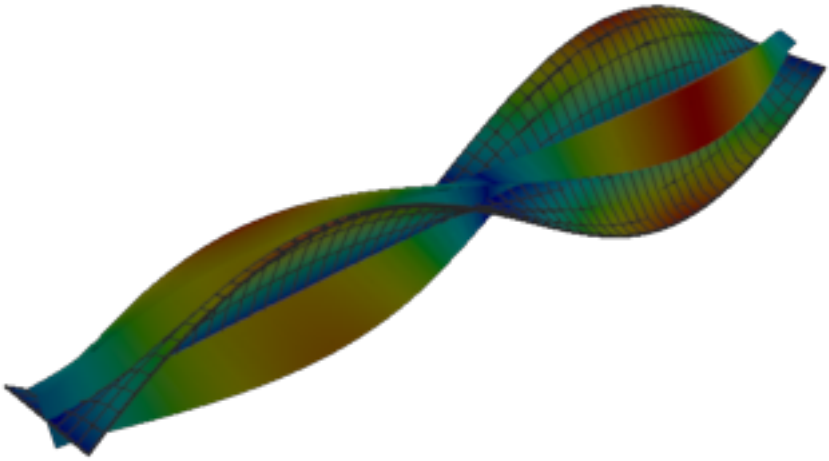


Figure 567: Code_Aster — mode 4, 100.176 Hz

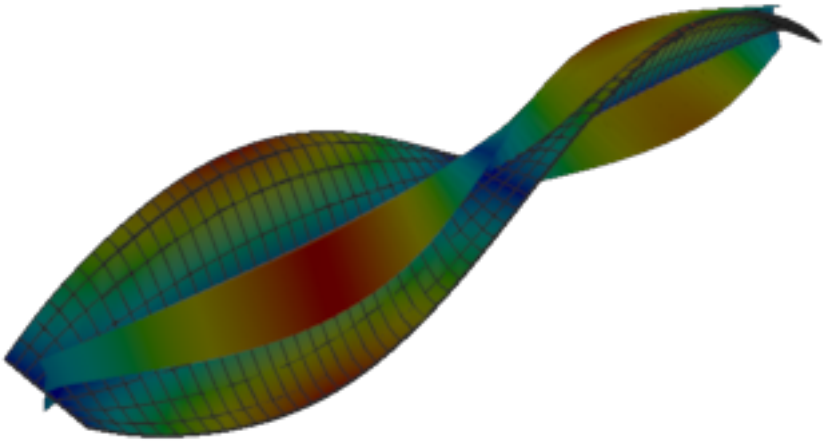


Figure 568: Sesam — mode 4, 100.875 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	106.620	+3.96	448765	1.14e-22	2.35e-08	5.91e-22	1.07e-11	1.53e-04	-2.43e-11
Code_Aster	100.176	-2.32	396174	2.37e-35	2.54e-40	5.09e-26	-4.87e-18	1.59e-20	-2.26e-13
Sesam	100.875	-1.64	401723	1.35e-30	9.30e-10	9.03e-26	-1.16e-15	3.05e-05	3.01e-13

Mode 5

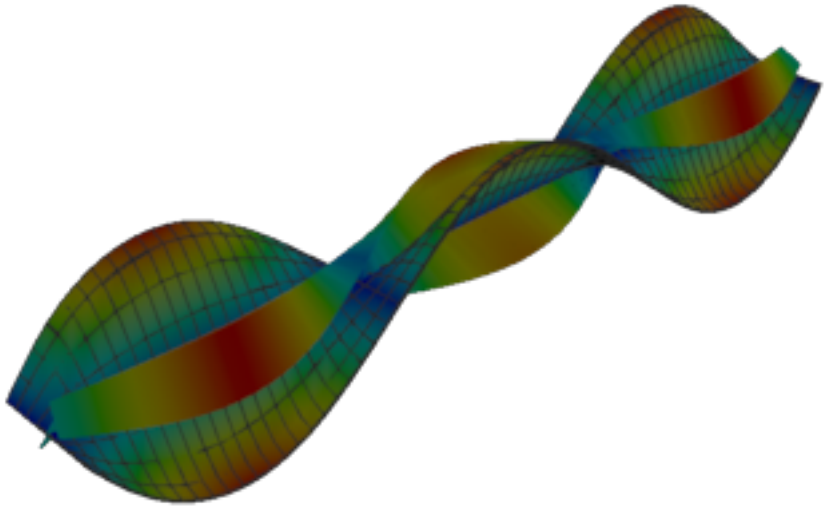


Figure 569: Abaqus — mode 5, 128.504 Hz

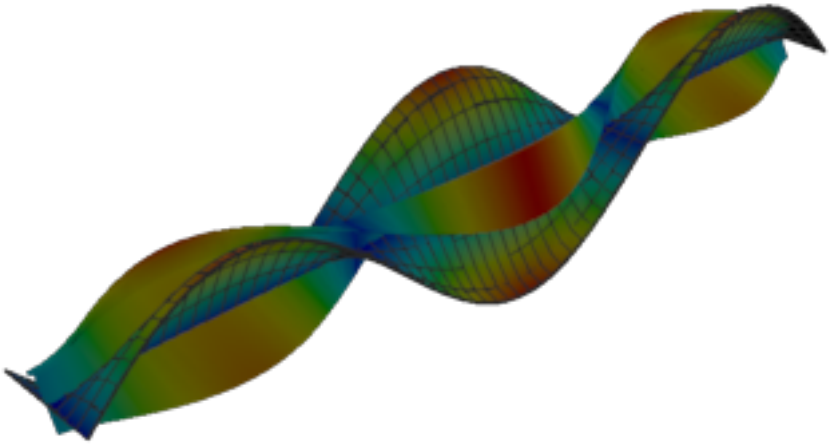


Figure 570: Code_Aster — mode 5, 112.038 Hz

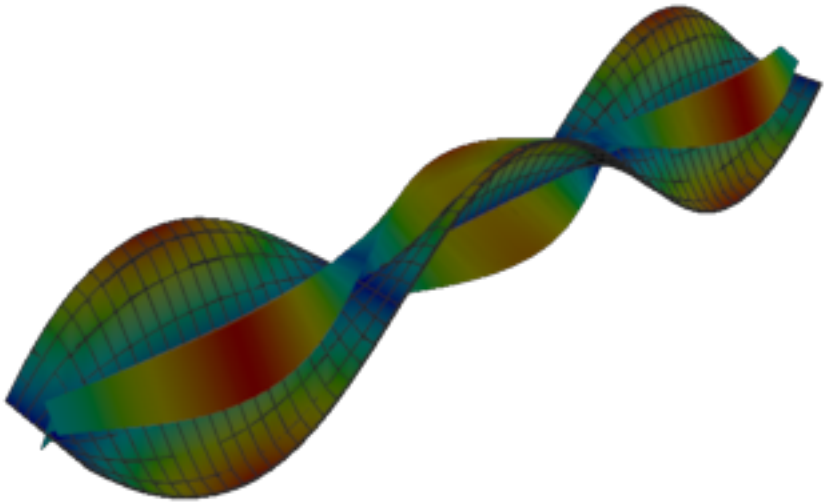


Figure 571: Sesam — mode 5, 114.982 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	128.500	+8.43	651919	5.66e-20	0.39	1.44e-19	2.38e-10	-0.624	3.79e-10
Code_Aster	112.038	-5.46	495550	7.84e-34	1.81e-41	8.00e-27	2.80e-17	4.26e-21	-8.94e-14
Sesam	114.982	-2.97	521937	2.16e-24	4.46e-07	8.33e-24	-1.47e-12	-0.001	2.89e-12

Mode 6

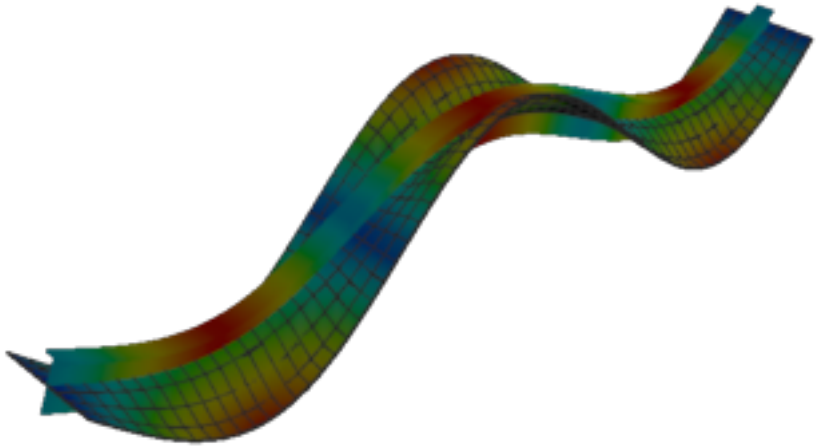


Figure 572: Abaqus — mode 6, 155.525 Hz

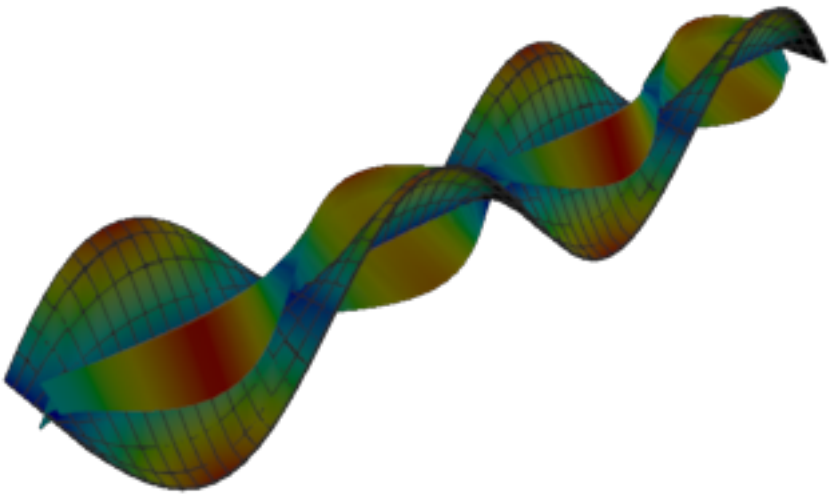


Figure 573: Code_Aster — mode 6, 127.230 Hz

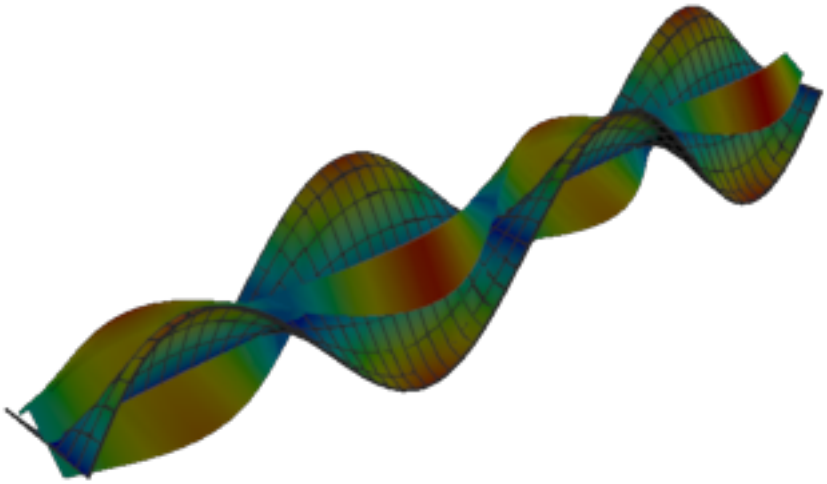


Figure 574: Sesam — mode 6, 134.472 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	155.520	+11.83	954899	0.26	3.11e-16	20.18	0.507	1.76e-08	-4.492
Code_Aster	127.230	-8.52	639054	7.78e-34	1.87e-39	8.70e-26	-2.79e-17	4.33e-20	-2.95e-13
Sesam	134.472	-3.31	713882	4.52e-18	4.40e-11	2.57e-17	-2.13e-09	6.63e-06	5.07e-09

Plate static analysis

The deflected strip under 1 kPa, plain and stiffened, at the same 0.0625 m seed. The other seeds deform into the same shape; their convergence is in the summary table.

Shell, 1st order, plain, seed 0.0625 m

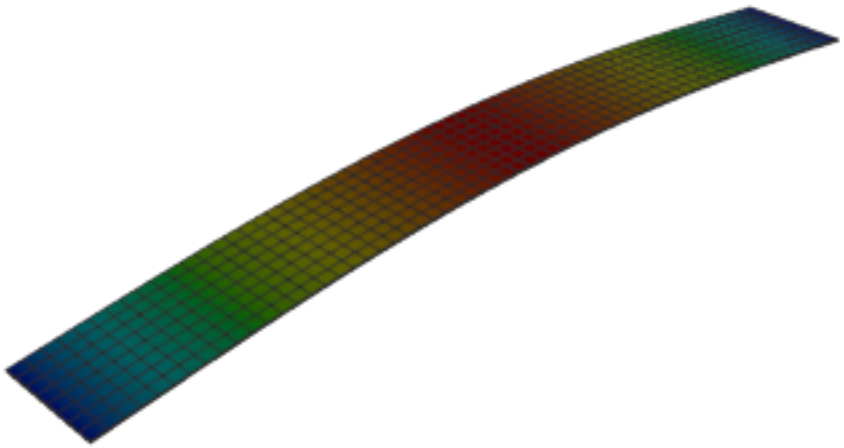


Figure 575: Abaqus — displacement

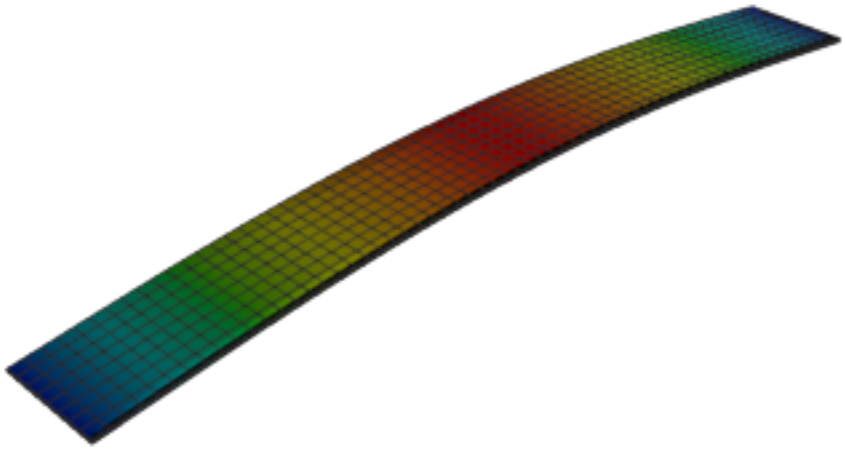


Figure 576: Calculix — displacement

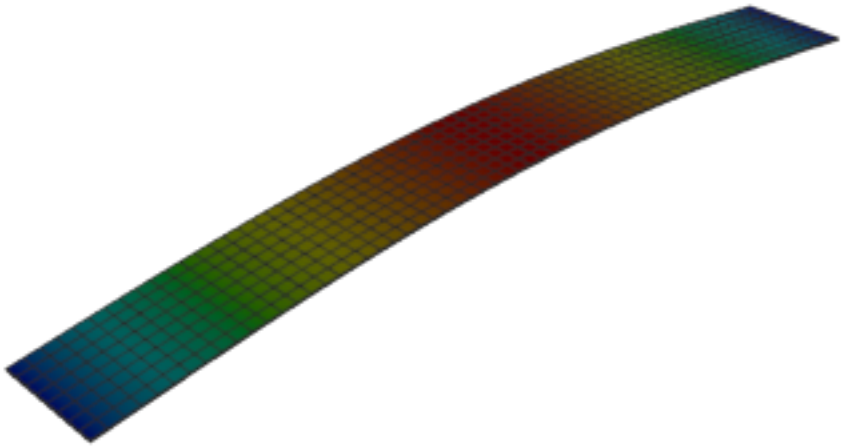


Figure 577: Code_Aster — displacement

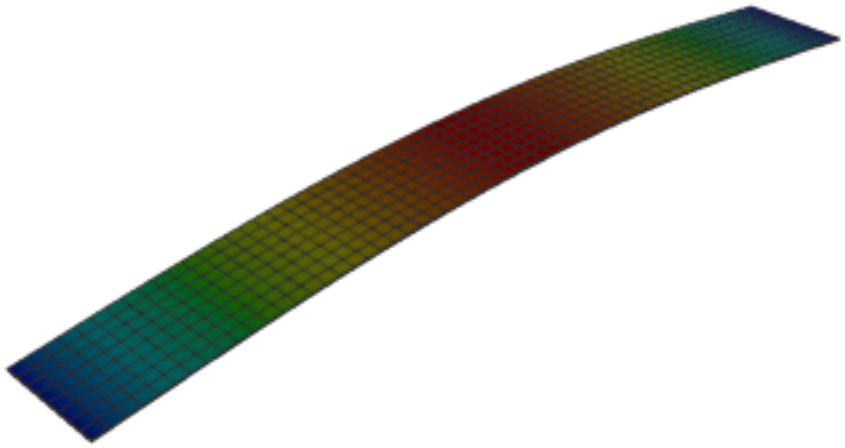


Figure 578: Sesam — displacement

Solver	Mid-span u_z [mm]	Δ vs closed form [%]	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	173.2686	-0.04	173.2686	5.59e-17	4.12e-29
Calculix	173.2680	-0.04	173.2680	0.6932	2.94e-10
Code_Aster	173.2995	-0.02	173.2995	0.0000	0.0000
Sesam	173.2995	-0.02	173.2995	0.0000	0.0000
Closed form	173.3333				

Shell, 1st order, stiffened, seed 0.0625 m

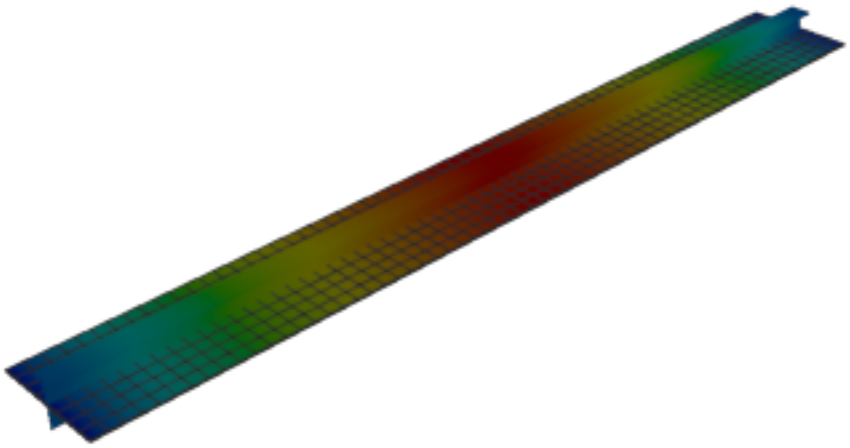


Figure 579: Abaqus — displacement

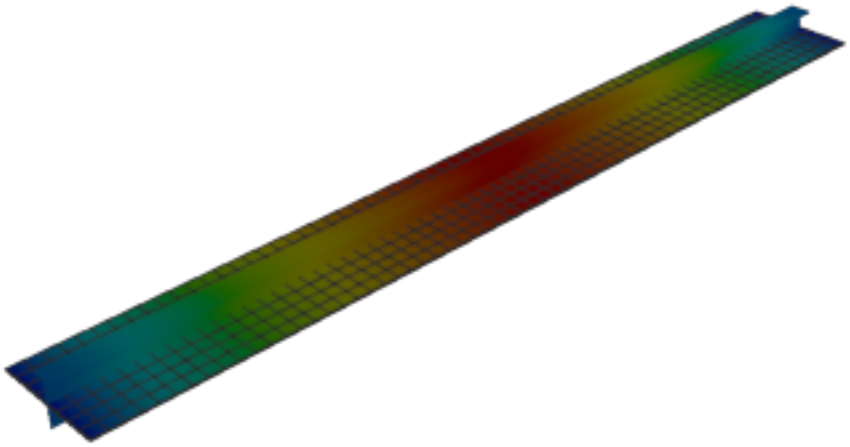


Figure 580: Code_Aster — displacement

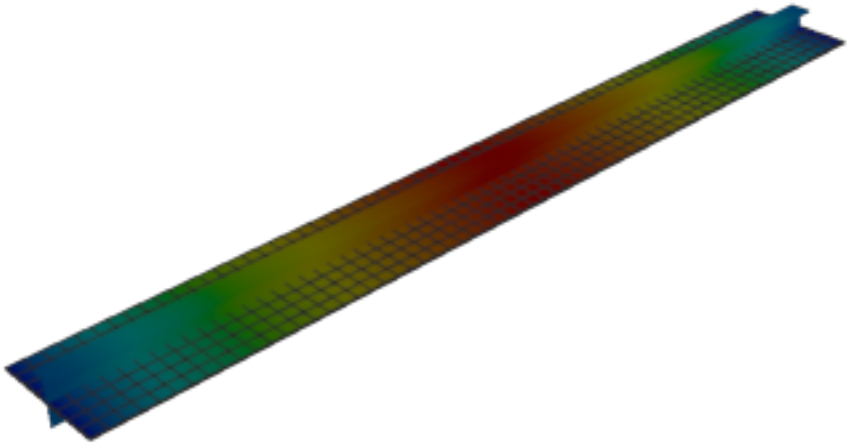


Figure 581: Sesam — displacement

Solver	Mid-span u_z [mm]	Δ vs mean [%]	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	0.5855	-12.93	0.5934	0.0118	2.12e-05
Code_Aster	0.7107	+5.70	0.7191	0.0000	0.0000
Sesam	0.7210	+7.23	0.7294	1.18e-17	4.08e-17